

Custom Compiler™ Custom Router User Guide

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Contents

New in This Release	16
Related Products, Publications, and Trademarks	16
Conventions	16
Customer Support	17
Statement on Inclusivity and Diversity	18

1. About the Custom Router	19
Custom Router Description	19
Custom Router Key Features	19
Routing Methods	21
Route Object Properties	22
Custom Router User Interface	22

2. Getting Started	26
Opening the Layout Editor and Enabling the Custom Router	26
Net-Based Routing Workflow	28
Color Routing Workflow	31
Device-Group Routing Workflow	34

3. Preparing for Routing	36
Preparing for Routing Workflow	37
Checking the Technology	37
Checking the Constraints in the Technology	37
Specifying the Routing Layers	38
Preparing and Reviewing the Design	41
Setting Up Tracks for Routing	41
Setting Up for Crossing Constraint Group Regions	42
Controlling Route Placement	42
Checking Routing Components	43
Optimizing Routing Results	43

Contents

Assigning Objects to Nets	43
Routing Across the Hierarchy	44
Creating Abstract Views to Facilitate Routing	44
Turning On Flight Lines	44
Creating Blockages for Center Routing	44
Routing Designs With Power/Ground Nets	46
Reducing the Number of Short Circuits	46
Preventing Artificial Short Circuits	48
Minimizing CPU and Memory Usage	49
Preference Syntax	49
Requirements	49
Examples	49
4. Setting Routing Constraints	51
About Routing Constraints	51
Before You Start	52
Creating Routing Constraints	53
Viewing and Modifying Constraint Parameters	55
Deleting Routing Constraints	57
5. Topology Routing	58
About Topology Routing	59
Route Topology Components	59
Methods for Creating Route Topologies	62
Topology Viewing Options	63
Workflow for Topology Routing	63
Before You Start	64
Setting Routing Constraints	64
Preparing for Topology Creation	65
Setting Electromigration Constraints	65
Setting Resistance Constraints	65
Setting Auto Router Options	66
Displaying the Topology Toolbar	66
Creating Trunk-Style Topologies	67
Creating Freeform Topologies	70

Contents

Creating Route Topologies Automatically	73
About the Automatic Topology Creation Method	73
Before You Start Automatic Topology Creation	73
Starting Automatic Topology Creation	78
Viewing the Topology Generation Report	79
Creating Topologies from Existing Routes	80
Setting Topology Segment Properties	81
Setting Topology Viewing Options	83
Editing Topologies	87
Moving Topologies	87
Moving Starting or Ending Locations	89
Merging Freeform Topology Segments	90
Merging Trunks	91
Converting Freeform Segments to Trunks	92
Converting Trunk Topologies to Freeform Topologies	92
Splitting Topologies	93
Adding Trunk Tapoffs	94
Deleting Trunk Tapoffs	94
Deleting Topologies	94
Deleting Topologies Using the Topology Assistant	95
Deleting Selected Topologies on the Layout Canvas	96
Grouping Nets That Have Topologies	96
Estimating Route Congestion	99
Enabling or Disabling Topology-Based Shielding	99
Setting Default Topology-Based Shielding Behavior	99
Enabling or Disabling Shielding for All Topologies on a Net	100
Enabling or Disabling Topology Segment Shielding	100
Enabling or Disabling Shielding for Trunk Topology Tapoffs	101
Generating Routes Based on Topologies	102
Importing and Exporting Topologies	104
Importing Route Topologies	104
Exporting Route Topologies	105
Reusing Existing Topologies in Different Designs	106
About Topology Files	107
Modifying Topology Files for Reuse	108
Topology Toolbar Reference	108

Topology Assistant Reference	111
Topology Assistant Toolbar	112
Display Topology Section	112
Filter Section	113
Nets Table	113
Topology Assistant Shortcut Menu	114
Topology Shortcut Menu Commands	114
6. Defining a Routing Area	117
About the Routing Area	117
Before You Start	119
Displaying the Routing Area Toolbar	119
Defining a Rectangular Area	120
Defining a Polygonal Area	121
Setting the Viewport as the Routing Area	123
Using Tcl Preferences to Define the Routing Area	124
7. Auto Routing	125
About the Auto Router	125
Enabling the Auto Router	126
Checking for Constraint Compliance	126
About the Data Checker	127
Running the Data Checker	128
Checking for Design Errors	129
About the Preroute Checker	129
Design Errors	130
Design Warnings	132
Technology Errors	134
Running the Preroute Checker	134
Viewing and Resolving Design Errors	136
Starting the Auto Router	137
Before You Start	137
Procedure	138
Managing Option Sets	139
Opening the Auto Route Dialog Box	140

Contents

Creating Option Sets	140
Modifying Option Sets	141
Renaming Option Sets	141
Deleting Option Sets	142
Saving Option Sets to a Directory	142
Reloading Options Sets from a Directory	142
Auto Router Behavior	143
Unrouted Connections	143
Prerouted Connections	143
Partial Dangling Connections	144
Auto Route Toolbar and Dialog Box Reference	145
Basic Options	145
General Tab	147
Layer Tab	149
Via Tab	152
Bus/Diff Pair Tab	155
Shield Tab	157
Matching Tab	162
Lengthening Style	163
Lengthen Within	167
Advanced Tab	167
8. Interactive Routing	170
About the Interactive Router	171
Enabling the Interactive Router	172
Customizing Your Work Preferences	174
Interactive Router Keyboard Shortcuts	174
Displaying and Hiding the Keyboard Shortcut Reference	176
Reconfiguring Keyboard Shortcuts	176
Checking for Design Errors	178
About the Preroute Checker	178
Design Errors	179
Design Warnings	181
Technology Errors	183
Running the Preroute Checker	183
Viewing and Resolving Design Errors	185
Applying and Ignoring Rules During Routing	185

Contents

Follow-the-Cursor Routing	188
Before You Start	188
Follow-the-Cursor Routing Flow	188
Assigning and Locking Colors Manually	191
Bending Routes	192
Dropping Vias Manually	193
Adding Vias Manually	194
Routing Around or Through Obstacles	195
Switching to Point-to-Point Mode	196
Welding	196
Point-to-Point Routing	197
Bus Routing	200
About Interactive Bus Routing	201
Bus Routes and Leader Bits	201
Bit Spacing	202
Corner Styles	203
Relative Routing	204
Routing Bus Bits	205
Adjusting Route Widths Automatically	208
Dropping Vias Automatically	211
Replicating the Last Route	213
Cloning Routes	213
Adding Strap Connections	217
Managing Option Sets	221
Opening the Create Route Dialog Box	222
Creating Option Sets	223
Modifying Option Sets	224
Renaming Option Sets	225
Deleting Option Sets	225
Saving Option Sets to a Directory	225
Reloading Options Sets from a Directory	226
Route Toolbar and Create Route Dialog Box Reference	226
Route Toolbar	226
Create Route Dialog Box	231
Common Options	231
General Tab	233
Layer Tab	237
Via Tab	239

Cloning/Welding Tab	242
Bus Tab	243
Add Via Dialog Box	246
Shortcut Menu Commands (FTC and P2P)	247
Shortcut Menu Commands (Interactive Bus Routing)	250
Auto Welding Mode	252
Options	253
9. Pattern Routing	256
About Pattern Routing	257
Pattern Route Assistant	257
Pattern Route Structures	258
Pattern Route Components	259
Workflow for Pattern Routing	261
Preparing for EM-Aware Pattern Routing	263
Adding Custom Patterns in the Pattern Catalog	263
Creating the Custom Pattern File	265
Importing Custom Patterns Into the Pattern Catalog	266
Importing Custom Patterns Using the Pattern Route Assistant	266
Setting the Path to the Custom Pattern File	269
Saving Modified Route Patterns to the Catalog	269
Exporting Patterns From the Pattern Catalog	271
Displaying the Pattern Route Assistant	272
Setting Up Pattern Routing	272
Selecting a Figure Group	273
Choosing Route Patterns	275
Filtering Route Patterns in the Catalog	276
Choosing a Pattern for a Net	277
Choosing the Same Pattern for Multiple Nets	277
Grouping Nets for Symmetric Routing	278
Copying a Pattern to One or More Nets	279
Deleting a Route Pattern	281
Selecting Different Route Patterns for a Net	282
Previewing Route Patterns	283
Adjusting Pattern Components on the Layout Canvas	285
Moving Pattern Components	285
Editing Components on the Layout Canvas	287
Adding and Removing Trunks	289

Contents

Adding and Removing Spines	290
Adjusting Pattern Component Attributes in the Assistant	291
Setting the Layer and Width Attributes	293
Specifying the Position Attribute	294
Extending Tapoffs, Trunks, and Spines	299
Adding Spines	304
Specifying the Number of Wire Stripes	306
Setting Cluster Attributes	307
Running the Group Planner	310
Specifying Group Settings	312
Creating Patterned Routes	318
Creating and Extracting Route Patterns	319
Workflow for Creating and Extracting Route Patterns	320
Before You Start	320
Pattern Routing Guidelines	320
Limitation	320
Extracting Route Patterns	321
Saving and Viewing Pattern Route Constraints	323
Deleting Routes in Patterns	325
Pattern Route Assistant Reference	326
Toolbar	327
Group	328
Net and Pattern	328
Pattern Control	328
Group Settings	331
Trunk Spacing	331
Update Group	332
Other Group Settings	334
Assistant Menu	335
Starter Route Patterns	337
Route Pattern Description	339
Via Options Dialog Box	340
10. Trunk Routing (Manual Method)	342
About the Generate Trunk Command	342
Before You Start Trunk Routing	344
Adding Trunk Routes	346
Trunk Toolbar and Trunk Generator Dialog Box Reference	348

General Options	350
Track Options	352
Instance Options	353
Extension Side	353
Shield Options	355
Trim Length Options	357
11. Trunk Routing (Automatic Method)	359
About Automatic Trunk Routing	359
Constraints	360
Top-Level Trunks	360
Routing Area	360
Top-Level Trunk Layer	360
Top-Level Trunk Placement	362
-startSide	362
-align	362
Examples	362
12. Power Routing	365
About Power Routing	365
Power Structure Components	365
Power Mesh Creation Commands	367
Before You Start	367
13. Hierarchical Routing	368
About Hierarchical Routing	368
Bottom-to-Top Routing	368
Routing Order	371
Before You Start	373
Routing Using rt::routeHierarchy	373
Viewing Hierarchical Routing Reports	374
Viewing Preroute Reports	374
Viewing Postroute Reports	375
14. Adding Post-Route Shielding	376

Contents

About Post-Route Shielding	376
Before You Add Post-Route Shielding	377
Adding Shielding	377
Enabling the Add Shield Command	377
Setting Options in the Add Shield Toolbar	378
Setting Options in the Add Shield Dialog Box	379
Starting the Shielding Operation	380
Adding Point-to-Point (P2P) Shielding	380
Before You Start	380
Enabling the P2P Shield Command	380
Adding Shielding	381
Setting Options in the P2P Shield Toolbar	382
Setting Options in the P2P Shield Dialog Box	384
Deleting Shielding	386
Shielding Styles and Behavior	386
Coaxial Shielding	387
Parallel Shielding	388
Tandem Shielding	388
Differential Pair and Bus Shielding	389
Group Shielding	390
Shared Shielding	392
Maximizing Shared Shielding	395
Shield Vias	396

15. Viewing and Modifying Route Attributes	398
About Routes and Route Shapes	398
Viewing and Editing Route Shape Properties	399

16. Deleting Routes	403
Ways You Can Delete Routes	403
Before You Start	404
Deleting Routes Created by the Custom Router	404
Deleting Routes Created by Any Method	405

17. Checking Route Congestion	406
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Contents

Rule Filter	406
About the Congestion Checker	408
Congestion Calculations	408
Global Routing Cells	408
Analysis Results	409
Starting the Congestion Checker	410
Viewing the Results on the Canvas	412
Ways to Modify the Display Settings	412
Modifying Display Settings	413
Displaying Congestion Statistics in Tooltips	416
Viewing the Congestion Report in the Console	418
18. Custom Rules	419
About Custom Rules	419
Overview of Rule Selection Canvas	420
Custom Rules Dialog box	422
Using Rules Mode	422
User-Defined Rule Modes	424
Rules View	429
Rule Description	432
19. Mesh Routing	434
Generating Mesh Routes	434
About Mesh Route Generation	434
Mesh Route Dialog Box Reference	436
Mesh Route Dialog Box	436
Recording Transaction History	446
Avoiding Custom Pullback From Being Impaired	446
Computing Full Design Rule and Half Design Rule	449
Creating an Instance	451
Creating Mesh Route With Different Intra-Group and Inter-Group Pitch	452
Storing Mesh Route Creation Information	454
About Mesh Route Creation Information	454
Deleting Mesh Routes	456
About Mesh Route Deletion	456

	Delete Custom Route Shapes Toolbar	457
	Delete Mesh Toolbar	458
20.	Analyzing and Troubleshooting Routing Results	460
	Analyzing Routing Results Workflow	460
	Viewing the Routing Summary	461
	Preroute Information	461
	Routing Stage Information	463
	Post-Route Summary	464
	Viewing Detailed Net Information	466
	Creating a Net Report Using the Design Navigator	466
	Creating a Net Report Using le::reportNets	468
	Troubleshooting Unrouted Nets	470
	Checking and Fixing Track-Related Issues	471
	Before You Start	471
	Checking and Reporting Track-Related Issues	471
	Fixing Track-Related Issues	472
	Examples	473
	Entire Design	473
	Specific Objects	473
	Specific Checks	473
	Instances	473
	Running the Postroute Checker	473
A.	Routing Constraint Reference	475
	Antenna Defaults Constraint	477
	Bus Constraint	478
	Color Matching Constraint	481
	Differential Group Constraint	483
	Driver Taper Halo Constraint	485
	Grids Constraint	485
	Hier Net Spacing Constraint	487
	Inter Group Spacing Constraint	489
	Layers Constraint	489
	Length Constraint	492

Net Priority Constraint	493
Net Shielding Constraint	494
Pattern Route Constraint	501
Shield Overhang Constraint	502
Taper Halo Constraint	504
Via Reliability Constraint	505
Vias Constraint	505
Wire Matching Constraint	507

B. Interactive Routing Cursor Movement and Behavior	509
Basic Interactive Cursor Movements	509
Interactions with Obstructions	511
Routing Over and Beyond Objects	513
Routing Around Obstacles in Trackless Designs	514
Adding Points Along the Route	515
Adding Vias	516
Unroutable Objects	518
Jogs and Via Costs in Routing	518
Self-Intersecting Routing Patterns and Same-Net Obstructions	522

C. Troubleshooting Error Messages	524
Where the Error Messages Are Displayed	524
Console	524
Text Viewer	525
Types of Error Messages	526
Custom Router Error Messages	527
GCR Error Messages	527
IROUTE Error Messages	560
PATTERN_ROUTE_ASSISTANT Error Messages	563

About This User Guide

This user guide explains how to use the Custom Router component of the Custom Compiler™ Layout Editor to add routes to integrated circuit designs.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Synopsys® Custom Compiler™

Synopsys Fusion Compiler™

Synopsys IC Compiler™ II

Synopsys IC Compiler™

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
<code>Courier</code>	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

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Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

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If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

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Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

About the Custom Router

This chapter provides an overview of the Custom Router.

This chapter contains the following topics:

- [Custom Router Description](#)
- [Custom Router Key Features](#)
- [Routing Methods](#)
- [Route Object Properties](#)
- [Custom Router User Interface](#)

Custom Router Description

The Custom Router is a shape-based router that creates interconnects (or routes) automatically for digital and mixed-signal nets at all levels of the chip design. The Custom Router integrates design and electrical rules as part of the routing process to provide a complete routing solution. For digital IC designs, you can preroute sensitive nets using a set of custom routing options and continue with the Fusion Compiler or IC Compiler II tool to complete the physical implementation.

For information about the rules or constraints you can apply, see [Setting Routing Constraints](#). For information about routing components in digital IC designs, see About Analog and Digital Co-designing for Fusion Compiler co-designing flows or About Analog and Digital Co-designing for IC Compiler II co-designing flows.

Custom Router Key Features

The Custom Router offers a number of features that creates routes quickly and efficiently while meeting special routing requirements. These features include:

- Topology routing
- Preroute checks for design-rule compliance and design errors

- Global routing
- User-guided routing assistance
- Support for designs with non-default rules (spacing, layer, grid, via cuts)
- Multi-cut and bar vias
- Bus routing
- Differential pair and triplet routing
- Corridor routing
- Color routing
- Local interconnect routing using contactless metal layers (Interactive Router only)
- Route pattern selection, modification, and reuse
- Route guides
- Pin tapering
- Shielding (automatically during routing, and post-route)
- Track- and grid-adherence during routing and shielding
- Support for gridded tracks (start and stop points on tracks)
- Routing within a subsection of a design
- Assisted device interactive routing
- Support for standard via variants

Routing Methods

The Custom Router lets you create routes automatically using one of the following methods:

- **Auto Routing:** Routes multiple nets simultaneously. The router automatically adds wires to placed components. You do not need to interact with the router during the auto-routing process. For more information, see [About the Auto Router](#).
- **Interactive Routing:** Routes single nets interactively. Based on your guidance, the router creates the interconnects automatically. For more information, see [About the Interactive Router](#).
- **Pattern Routing:** Reuses exiting route patterns in other regions of the same design to reduce labor-intensive repetition. For more information, see [Pattern Routing](#).
- **Topology Routing:** Prescribes electrically aware pathways that the Auto Router must follow. You can import and export the topologies for backup and recovery. The exported topologies can also be reused in different designs. See [Topology Routing](#).

Note:

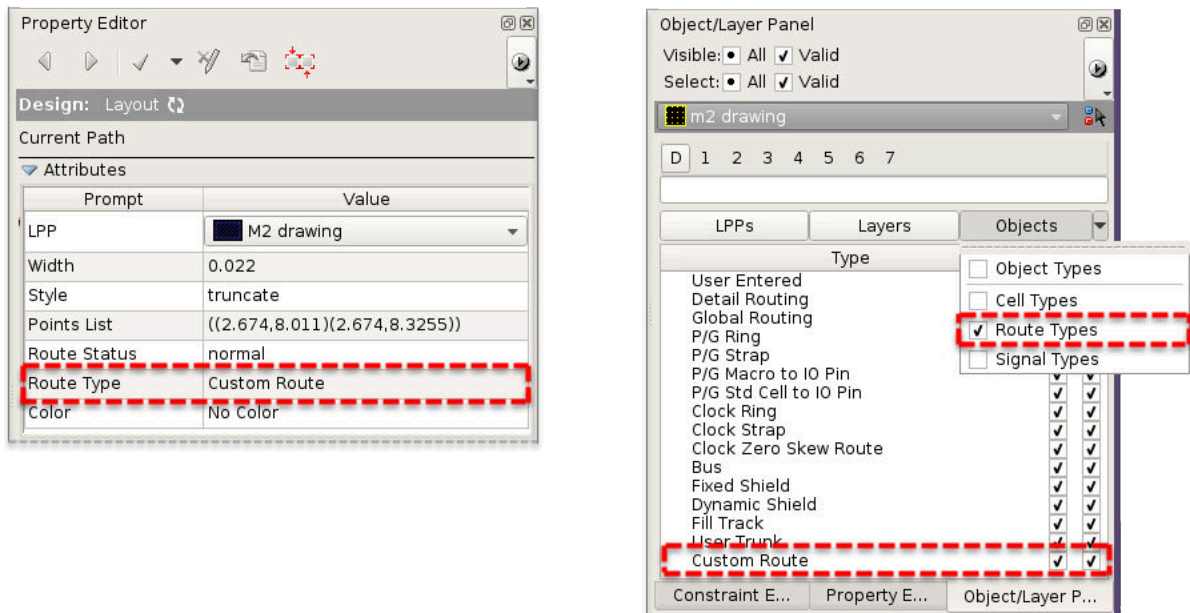
Except when performing differential-pair routing, the Auto Router and Interactive Router support orthogonal routing only.

To route packaging designs, see [About the Packaging Auto Router](#).

Route Object Properties

Shapes that are created by the Custom Router have the **Custom Route** Route Type. The **Custom Route** property is used by the [Delete Custom Route Shapes](#) command to identify the shapes to delete. You can also use it to identify shapes that are added by the Custom Router in Fusion Compiler or IC Compiler II co-design processes.

You can view the Route Type property using the Property Editor. You can also control the Custom Route objects' visibility and selectability using the Object/Layer Panel.



Custom Router User Interface

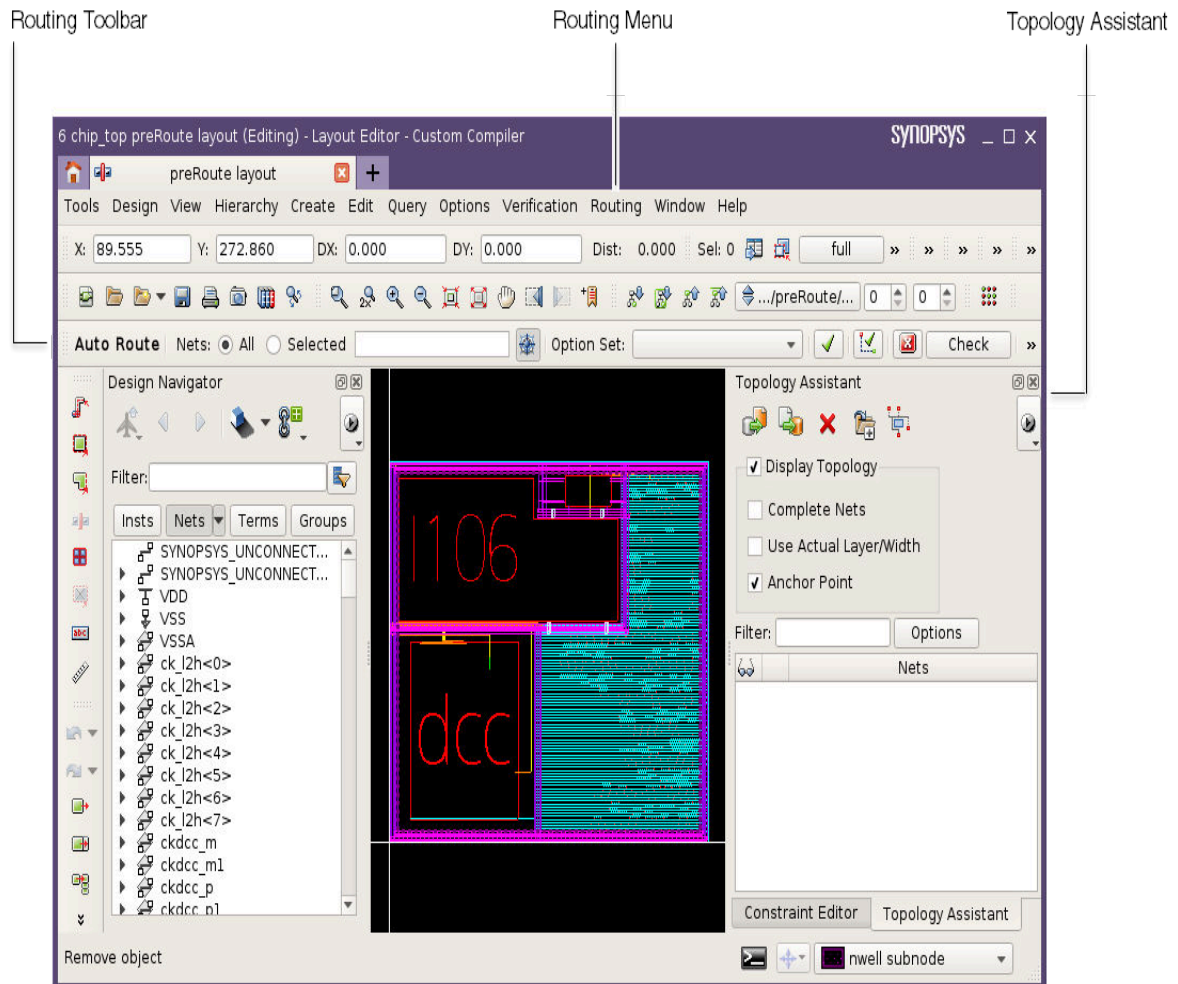
The Custom Router is an integrated component of the Custom Compiler Layout Editor. The following figure shows the Layout Editor with the Custom Router is enabled. Notice the following:

- The **Routing** menu appears on the menu bar.
- Based on the **Routing** menu command you choose, the corresponding router toolbar appears.
- Other features are available for routing tasks. For example, you can select nets in the Design Navigator, shown on the left side of the window by default. You can also add design-specific routing constraints in the Constraint Editor, shown on the right side of the window by default.

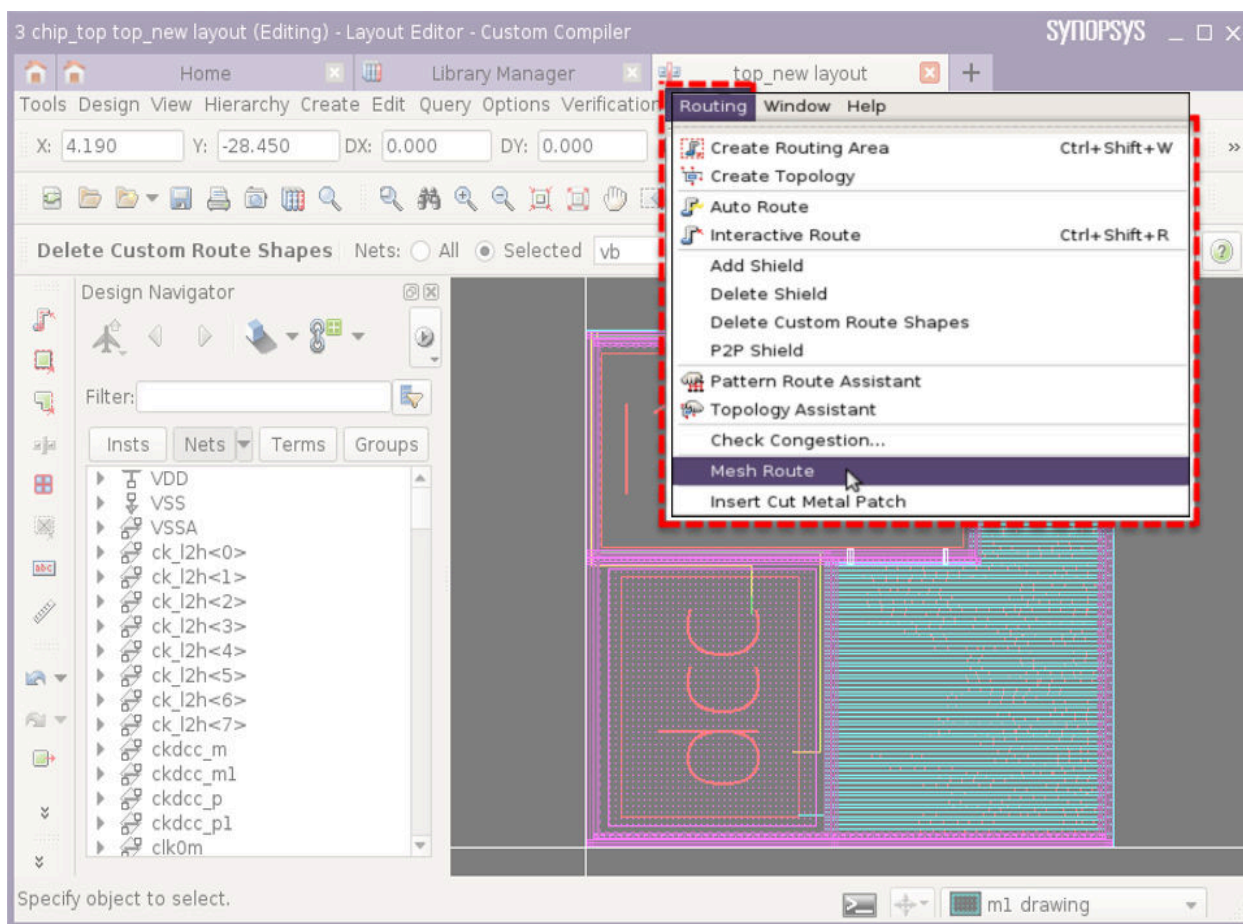
For instructions on how to enable the Custom Router and the **Routing** menu, see [Getting Started](#). For information about the Custom Compiler Layout Editor, including the Layout Editor tools, assistants, and toolbars, see [About the Layout Editor](#).

Note:

Instead of using the graphical user interface, you can write Tcl scripts to batch process routing requests.



You can display routing toolbars using the commands in the **Routing** menu.



Command	Description
Create Routing Area	Displays the Routing Area toolbar so that you can specify an area to route only the components within that area. See Defining a Routing Area .
Create Topology	Displays the Topology toolbar so that you can define route topologies to prescribe pathways that the Auto Router must follow. The topologies capture net-based routing intents and are saved with the design. See Topology Routing .
Auto Route	Displays the Auto Route toolbar so that you can use information in the technology and design to automatically add wires on one or more nets. For information, see Auto Routing .
Interactive Route	Displays the Route toolbar so that you can create routes for single nets interactively. See Interactive Routing or Interactive Routing .
Add Shield	Displays the Add Shield toolbar so that you can add post-route shielding. See Adding Post-Route Shielding .

Command	Description
Delete Shield	Displays the Delete Shield toolbar so that you can delete shielding. See Adding Post-Route Shielding .
Delete Custom Route Shapes	Displays the Delete Custom Route Shapes toolbar so that you can delete routes that were added by the Custom Router. See Deleting Routes .
P2P Shield	Displays the P2P Shield toolbar so that you can manually add partial, point-to-point shielding for a routed net. See Adding Post-Route Shielding .
Pattern Route Assistant	Displays the Pattern Route Assistant so that you can choose a route pattern and create routes based on that pattern. See Pattern Routing .
Topology Assistant	Displays the Topology Assistant so that you can set topology viewing options. See Setting Topology Viewing Options .
Check Congestion	Displays the Check Congestion dialog box so that you can predict and manage route congestion in your designs. See Checking Route Congestion .
Mesh Route	Displays the Mesh Route dialog box so that you can create a power mesh on multiple layers and nets and generate mesh-like shapes. See Generating Mesh Routes .

Most of the menu commands have associated shortcut keys. For more information, see [Layout Editor Shortcut Keys](#).

2

Getting Started

This chapter explains how to start the Custom Compiler Layout Editor and enable the Custom Router. This chapter also provides a high-level view of the routing workflows.

This chapter contains the following topics:

- [Opening the Layout Editor and Enabling the Custom Router](#)
- [Net-Based Routing Workflow](#)
- [Color Routing Workflow](#)
- [Device-Group Routing Workflow](#)

For the topology routing workflow, see [Topology Routing](#).

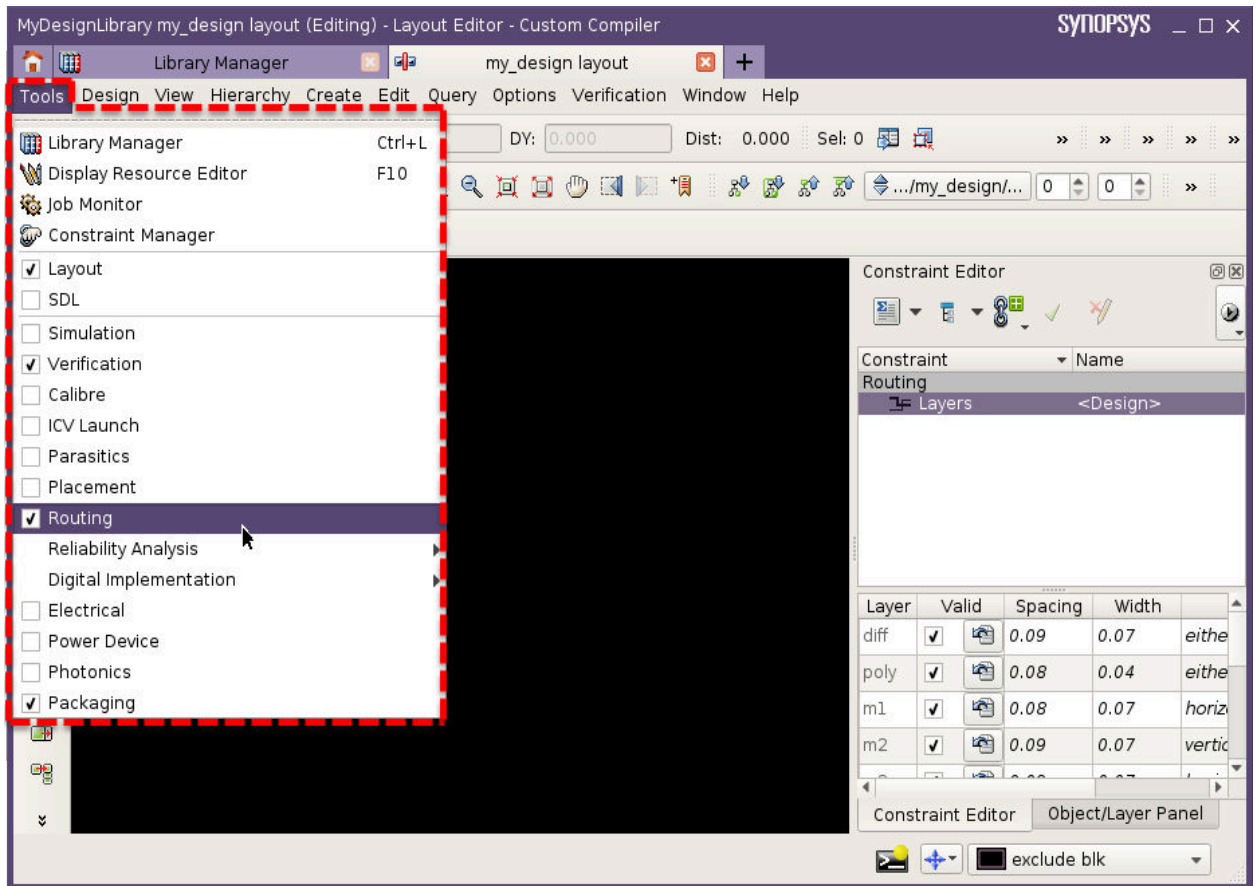
Opening the Layout Editor and Enabling the Custom Router

To start the Layout Editor and enable the Custom Router:

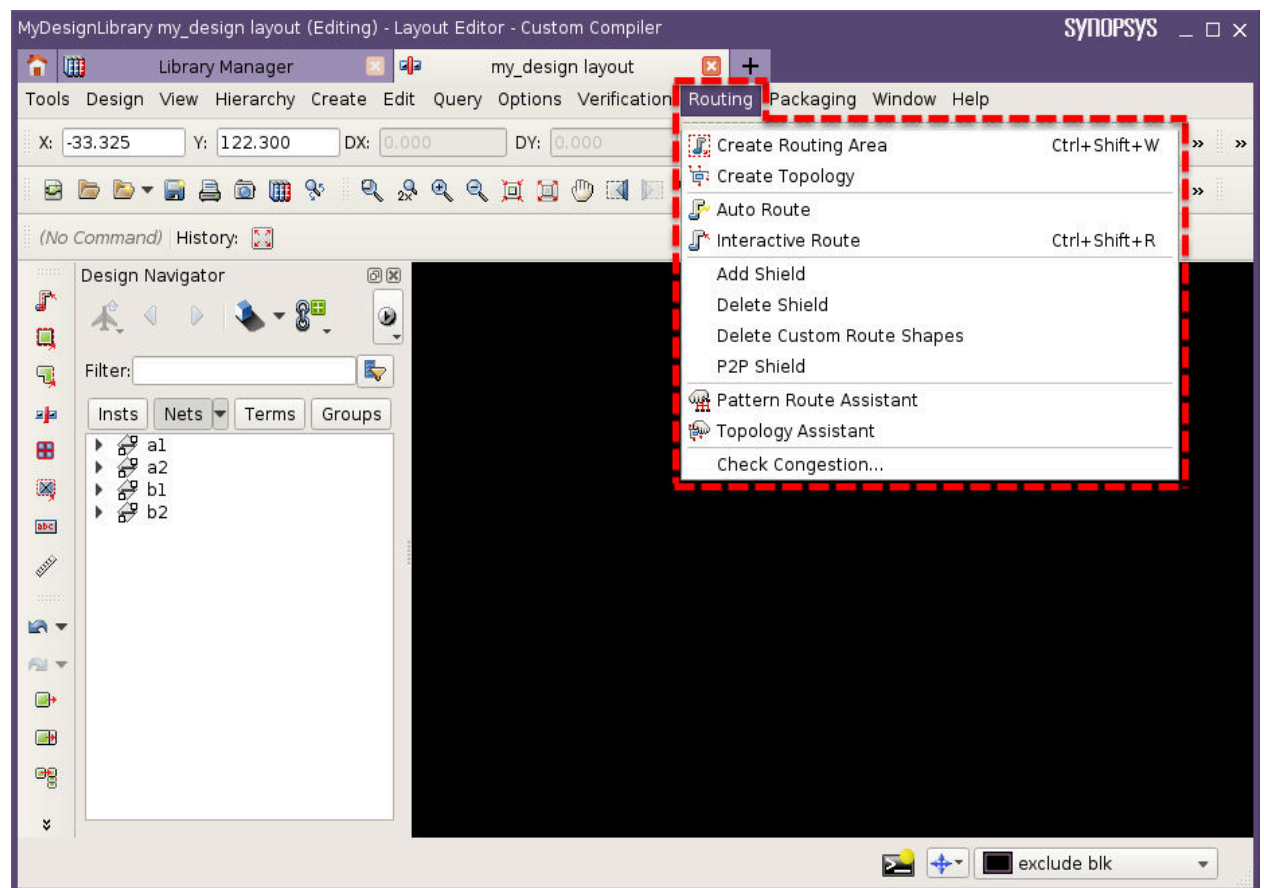
1. Start a Custom Compiler session. See [Starting Sessions](#).
2. Open the design layout. See [Opening Designs](#). Opening a design layout automatically opens the Layout Editor.
3. To use the new Interactive Router, make sure the `rtUse202003IR` Tcl preference is set to `false` (default). To use the legacy Interactive Router, set the `rtUse202003IR` Tcl preference to `true`.
4. In the Layout Editor, choose **Tools > Routing**.

Chapter 2: Getting Started

Opening the Layout Editor and Enabling the Custom Router



The **Routing** menu appears on the menu bar.

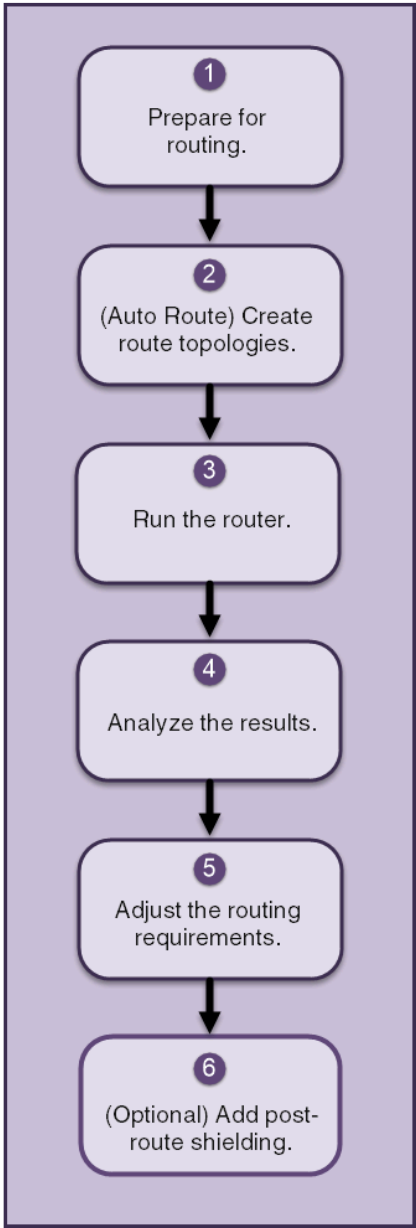


Net-Based Routing Workflow

This topic describes the net-based workflow. See also the [Color Routing Workflow](#).

Net-based routing connects shapes that are on the nets that you select in the Design Navigator. The following figure shows the high-level net-based routing workflow. Click a step in the workflow to see the detailed description of the step.

The workflow can vary, depending on your design, the default constraints used, and the routing results. Typically, steps 2 through 4 are repeated until the routes are optimized. In addition, the workflow implies an overall iterative process that involves making incremental adjustments to the layout and routing after checking resistance and electromigration.



The following table describes each step in the net-based routing workflow.

Step	Description
1 Prepare for routing.	Prepare for routing as follows: • Check the technology for the necessary process-level constraints and specify routing constraints. • Review the design for connection information. • Check and add design constraints. See Setting Routing Constraints. • Assess pin blockages and run the Data Checker. For information, see Preparing for Routing.
2 (Auto Route) Create route topologies.	Create route topologies to guide auto routing. See Topology Routing .
3 Run the router.	Run the router as follows: • In the Layout Editor, turn on the Connectivity Engine. • Turn on the DRD feature. • In the Design Navigator, display the flight lines, clear existing flight lines, and then select the nets. For instructions, see Displaying and Hiding Flight Lines. • Define a subsection (or area) for routing. You can check the routing within the specified area without having to load the data of the entire design. See Defining a Routing Area. • Use one of the following routing methods: auto routing, or interactive routing. See About the Auto Router or Interactive Routing. • (Optional) Reuse existing route patterns within the design. For information, see Pattern Routing. Note that the Auto Router can add shielding during the routing process. If you created route topologies, you can start auto routing from the Topology toolbar. See Workflow for Topology Routing.
4 Analyze results.	Check the routing results. Look for unrouted nets and constraints that are not met. See Analyzing and Troubleshooting Routing Results. Check for route congestion issues. See Checking Route Congestion.
5 Adjust routing requirements.	To refine the routing results: • Add or change routing constraints. For instructions, see Setting Routing Constraints. • Change the routing parameters. For example, the Auto Router adds shielding during the routing process. You can turn off shielding or specify the shielding type when setting up the Auto Route options. For information, see Auto Routing.
Repeat steps 3 through 5.	Iteratively edit the routes and re-run the router until you have addressed all errors and optimized the routes.

Step		Description
6	(Optional) Add shielding.	To add shielding after routing, use the Routing > Add Shield command. For information, see Adding Post-Route Shielding .

If you want to choose a route pattern and create routes, see [Pattern Routing](#). For routing packaging designs, see Workflow for Routing Packaging Designs.

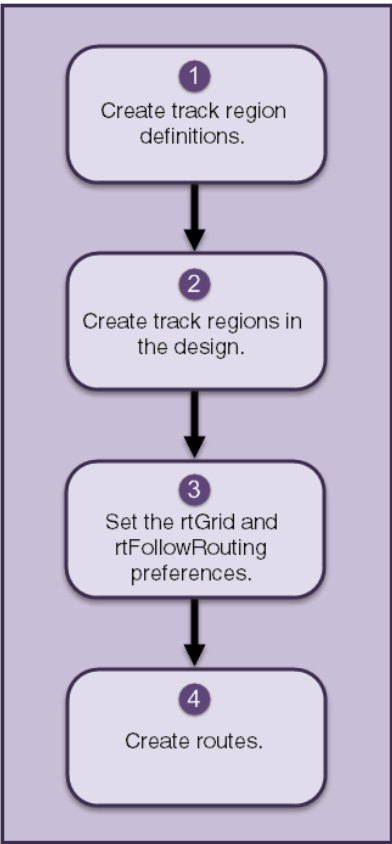
After routing, you can check for electrical violations (electromigration, resistance, capacitance) and shield coverage. See In-Design Electrical Checking and Reporting.

Color Routing Workflow

This topic describes the color routing workflow. See also the [Net-Based Routing Workflow](#).

In advanced node designs that contain FinFET components, multiple patterning is used to improve resolution as shape density increases. The technique impacts every aspect of the design process, including routing.

The following figure shows the workflow for color routing. Click a step in the workflow to see the detailed description of the step.



The following table describes each step in the color routing workflow.

Step		Description
1	Create track region definitions.	Make sure the technology database contains track region definitions. Make sure the track region definitions specify track colors. Routes inherit the color of the tracks.
2	Create track regions in the design.	Use the Track Region assistant to create track regions in the design. Make sure the tracks and track regions that you want to use are Active. The router will only snap routes to active tracks. To facilitate the design process, you can make the tracks and track regions Visible so that you can see them on the canvas. For more information about creating track regions, see the <i>Custom Compiler Environment User Guide</i>.

Step		Description
3	Set the track adherence mode.	(Auto Router) When ensuring that routes snap to the tracks during auto routing, specify the track adherence mode. See the Layer Tab in the Auto Route dialog box. (Interactive Router) To ensure that routes snap to the tracks during interactive routing, set the Layer Tab in the Create Route dialog box. For information, see the <i>Custom Compiler Tcl Reference Manual</i>.
4	Create routes.	For net-based routing, see Net-Based Routing Workflow .

During the routing operation, you have the option of snapping routes to active tracks. If the tracks have color, the routes will inherit the track color. However, when preroute pins have a color that is different from the track, the route will inherit the color of the preroute pins when the route connects. See [Color Propagation](#).

Note:

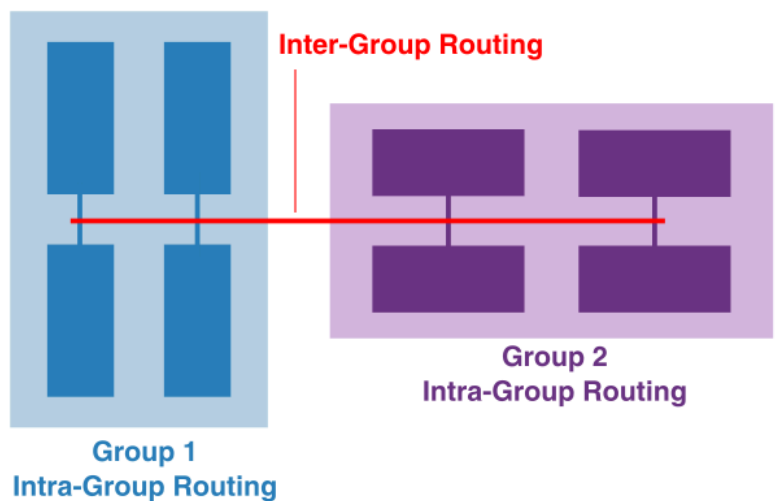
Color stitching is not supported.

If you choose to add shielding, parallel shielding is added to colored routes. For information about adding shielding, see [Adding Post-Route Shielding](#).

Device-Group Routing Workflow

This topic describes the workflow for device-group routing. Device group routing consists of the following:

- **Intra-group routing:** Connecting devices within a figure group.
- **Inter-group routing:** Connecting devices in different figure groups.



Step	Description	
1	Connect devices within the figure group.	Use Pattern Routing to connect devices within a figure group. These intra-device connections typically form fishbone or backbone structures.
2	Connect devices in different figure groups.	Use Topology Routing to plan and automatically connect devices in different figure groups at the higher levels of the design hierarchy.

3

Preparing for Routing

This chapter explains how to check for the readiness of your design before you start the routing operation.

This chapter contains the following topics:

- [Preparing for Routing Workflow](#)
- [Checking the Technology](#)
- [Preparing and Reviewing the Design](#)
- [Optimizing Routing Results](#)
- [Minimizing CPU and Memory Usage](#)

Preparing for Routing Workflow

Before you start routing:

1. Check the technology that is associated with the design. Make sure it contains the necessary process-level constraints. See [Checking the Technology](#).
2. Check the constraints that have been added to the design. You can use the Constraint Editor to view the constraints and adjust the constraint parameters. For information, see [Setting Routing Constraints](#).
3. Prepare and review the design. If you want to route on tracks, make sure you create tracks in your design. You should also check the design to ensure routing components exist. Assess the number of pin blockages in the design before spending time creating actual routes. See [Preparing and Reviewing the Design](#).
4. Specify the necessary connection settings to ensure successful routing. See [Optimizing Routing Results](#).
5. To minimize CPU and memory usage when routing very large designs, you can use the `rtImportPartialLayers` Tcl preference to specify the routing layers of interest. See [Minimizing CPU and Memory Usage](#).

Checking the Technology

Because the Custom Router creates routes based on process constraints in the technology, you must make sure the necessary constraints are correctly defined in the technology.

- [Checking the Constraints in the Technology](#)
- [Specifying the Routing Layers](#)

Checking the Constraints in the Technology

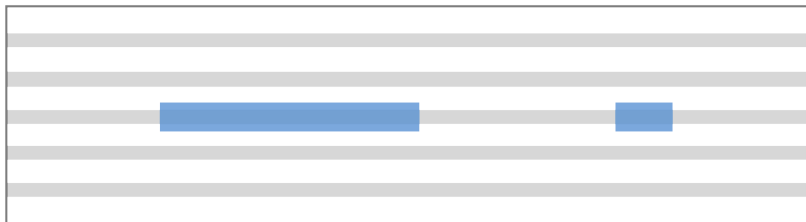
Make sure the following are correctly defined:

- Layer definitions (`validRoutingLayers`)
- Minimum spacing, width of the routing layers, and other constraints in the foundry group
- Via definitions (`viaDefs` and `validRoutingVias`)
- (Optional) Layer functions
- (Optional) Routing direction

- Track region and track pattern definitions
- (Optional) Grid pitch and offset (or by default, the router will use the manufacturing grid pitch and offset)

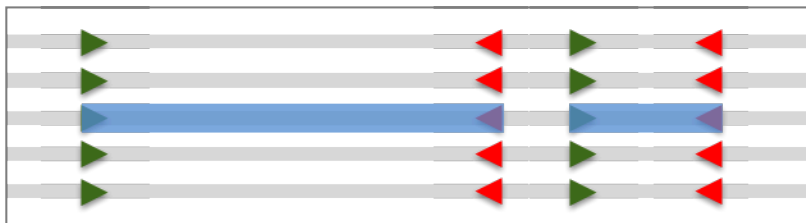
If you want the routes to start and end at specific locations on the tracks, be sure to define grids containing permissible starting and stopping points in the track region or track pattern. For track regions, specify the `-grid*` arguments when defining `tracks()`. For track patterns, specify the `-grid*` arguments when using the `le::createTrack` and `le::createTrackPattern` Tcl commands.

No Grid



- Tracks
- Interconnects/Routes
- Start Points
- Stop Points

With Grid



When you use the Interactive Router, you can create local interconnects. To do this, make sure you:

- Add the `leConnectingLayer` constraint to the foundry constraint group. In addition, define the connecting layers in the `leConnectingLayer` constraint.
- Define the local interconnect routing layers in the `functions` section of the technology database. The layers should have the `contactlessMetal` function.

You can use the Constraint Manager Tool to view and manage OA constraints defined in the technology database.

Specifying the Routing Layers

In some technologies, multiple metal layers that have the same mask number have different layer numbers. The different layer numbers permit the layers to be visually

distinguishable on the canvas (for example, signal routes from power and ground routes), even though they will be physically manufactured on the same layer.

Check the technology file and see if multiple layers on the same mask have different layer numbers. If so, you can use the `rtUserRoutingLayers` Tcl preference to let the router know which layers to use for routing. If you do not specify the routing layers, the router will use the metal layers with the lowest layer numbers as the default routing layers.

The following example shows a technology file where:

- `metal1`, `vdd1`, and `vss1` are assigned mask number 10, but they have different layer numbers: 10, 20, and 29, respectively.
- `metal2`, `vdd2`, and `vss2` are assigned mask number 13, but they have different layer numbers: 13, 21, and 28, respectively.

```
layerDefinitions(
  techLayers(
    ; ( layerName  layerNum  abbreviation )
    ; ( -----  -
    ( NWELL        1      NWELL        )
    ( DIFF         2      DIFF         )
    ( POLY         3      POLY         )
    ( CONT         7      CONT         )
    ( METAL1       10     METAL1       )
    ( METAL1A      20     METAL1A      )
    ( METAL1B      30     METAL1B      )
    ( VIA1         11     VIA1         )
    ( METAL2       12     METAL2       )
    ( METAL2A      22     METAL2A      )
    ( METAL2B      32     METAL2B      )
    ...

  layerRules(
    functions(
      ; ( layer      function      [maskNumber] )
      ; ( -----
      ( POLY        poly         5         )
      ( CONT        cut          6         )
      ( METAL1      metal        10        )
      ( METAL1A     metal        10        )
      ( METAL1B     metal        10        )
      ( VIA1        cut          11        )
      ( METAL2      metal        12        )
      ( METAL2A     metal        12        )
      ( METAL2B     metal        12        )
      ...
```

If you want to use `metal1` and `metal2` as the routing layers, then set the preference as follows:

```
rtUserRoutingLayers {metal1, metal2}
```

In this example, if you do not set the preference, the router will still use `metal1` and `metal2` as the routing layers, because they have the lowest layer numbers on masks 10

and 13, respectively. However, the following example shows that `vdd1` has a lower layer number than `metal1`. In this case, if the preference is not set, the router will use `vdd1` and `metal2`, respectively.

```
layerDefinitions(
  techLayers(
    ; ( layerName  layerNum  abbreviation )
    ; ( -----  - - - - -  - - - - - )
    ( NWELL        1      NWELL        )
    ( DIFF         2      DIFF         )
    ( POLY         3      POLY         )
    ( CONT         7      CONT         )
    ( METAL1       10     METAL1       )
    ( METAL1A      9      METAL1A      )
    ( METAL1B     30     METAL1B      )
    ( VIA1        11     VIA1         )
    ( METAL2       12     METAL2       )
    ( METAL2A     22     METAL2A      )
    ( METAL2B     32     METAL2B      )
    ...
  )
  layerRules(
    functions(
      ; ( layer      function  [maskNumber] )
      ; ( -----  - - - - -  - - - - - )
      ( POLY        poly      5          )
      ( CONT        cut       6          )
      ( METAL1      metal     10         )
      ( METAL1A     metal     10         )
      ( METAL1B     metal     10         )
      ( VIA1        cut       11         )
      ( METAL2      metal     12         )
      ( METAL2A     metal     12         )
      ( METAL2B     metal     12         )
      ...
    )
  )
)
```

The layers specified by `rtUserRoutingLayers`:

- Are automatically selected in the **Layer** tab of the **Auto Route** dialog box. For example, if you specified `metal1` and `metal2` using the preference, **metal1** and **metal2** will be the only layers selected in the **Layer** tab.
- Override the specifications in the `validRoutingLayers` constraint in the technology library and any specifications in the Constraint Editor.

When setting the preference, you can only specify metal layers. You cannot specify non-metal layers, such as poly layers. In addition, the Tcl preference setting is applicable at the design level only. You cannot apply it to specific nets or net groups.

For more information about the `rtUserRoutingLayers` Tcl preference, see the *Custom Compiler Tcl Reference Manual*.

Preparing and Reviewing the Design

Before starting the Auto Router, you should prepare and review the design. This topic describes what you should look for and how to prepare for routing:

- [Setting Up Tracks for Routing](#)
- [Setting Up for Crossing Constraint Group Regions](#)
- [Controlling Route Placement](#)
- [Checking Routing Components](#)

Setting Up Tracks for Routing

If you want routes to adhere (or snap) to tracks, make sure routing layers have track regions or track patterns. In addition, the tracks must be active.

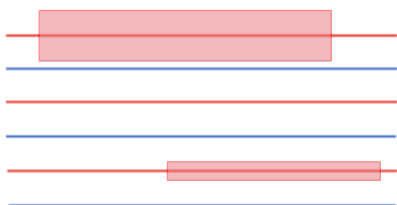
You can create, edit, and activate tracks using the track region PCell options of the **Create > Rectangle** command. Alternatively, you can use the Track Region Assistant or the Track Pattern Assistant. If both track regions and track patterns exist, track regions will be used.

If you are using the Track Region Assistant or the Track Pattern Assistant, make sure the track definitions exist in the technology database. If you want gridded tracks, make sure the track grids are defined in the technology database. See [Checking the Technology](#).

Routes inherit their colors from the tracks. However, you can set the `rtFlipTrackColorForNdrRouting` Tcl preference to `true` to permit route colors to change when necessary to prevent spacing violations.

When snapping to default track types, if a route does not meet the layer's `minWidth` requirement, the route can automatically change color so that it can be placed closer to neighboring routes.

`rtFlipTrackColorForNdrRouting = false`



`rtFlipTrackColorForNdrRouting = true`



Setting Up for Crossing Constraint Group Regions

Region-based constraints are foundry or alternate-foundry rules that apply to a specific region (physical layer or layer-purpose pair). When creating routes, you should be aware of the compatibility of the rules across adjacent regions and how they impact route placement.

On a per layer basis, if a region constraint exists for a given routing rule (and prime parameters), the corresponding foundry rule is ignored (similarly for other precedence scenarios)

Before you start, use the `leTechFoundryConstraintGroups` Tcl preference to determine the order of the constraint groups. See [Crossing Constraint Group Regions](#).

To set up for routing across constraint group regions:

1. In the **Auto Route** or **Create Route** dialog boxes, click the **Layer** tab.
2. Select the layer associated with the routing region containing the region-based constraint group.
3. Click the **Via** tab.
4. Select the foundry or alternate-foundry vias.

During auto routing, if the route cannot cross into another region, the connection will be left open, and an error message lets you know about the open connection.

During interactive routing, if it is not possible to cross into another boundary because of rule violations, the pointer will display an error message to let you know.

Controlling Route Placement

In addition to using tracks, you can add the following components to control route placement:

- Blockages
- Route corridors
- Topologies

For more information, see [Ways to Control Route Placement](#).

Checking Routing Components

To ensure successful routing, make sure:

- The design contains the necessary components for routing. For example, the design should contain at least two pins.
- Look for and resolve pin blockages.

Optimizing Routing Results

This topic describes the following:

- [Assigning Objects to Nets](#)
- [Routing Across the Hierarchy](#)
- [Creating Abstract Views to Facilitate Routing](#)
- [Turning On Flight Lines](#)
- [Creating Blockages for Center Routing](#)
- [Routing Designs With Power/Ground Nets](#)
- [Reducing the Number of Short Circuits](#)
- [Preventing Artificial Short Circuits](#)

Assigning Objects to Nets

The Auto Router relies on connectivity information in the design to create routes. Make sure pins, shapes, figures, and instance terminals are assigned net information. The Auto Router treats non-netted objects as obstructions. For designs that do not have connectivity information, you can write scripts to add it. If you need assistance, contact Synopsys Technical Support Center.

The Interactive Router does not require connectivity information. It can add routes between non-netted objects. It cannot add route between non-netted objects and netted objects.

Routing Across the Hierarchy

The Custom Router adds routes only at the top (0x) level of the design. To permit the router to connect to objects on lower levels:

1. In the Layout Editor, choose **Options > Design**. The **Layout Design Options** dialog box opens.
2. In the Connectivity Tab, select **Connectivity Engine**.
3. Set the **Depth** as follows:
 - 0: The Custom Router connects to 0x netted shapes or to 1x Inst terms. (Default)
 - > 0: The Custom Router connects to contiguous metal below 0x (top-level) pins within the Connectivity Engine depth. If you want the Custom Router to see all the shapes in the design, set it to the total number of levels in the design (typically 1 for ICC designs and 32 for block level designs). To limit the levels, set the option to a higher level.

Creating Abstract Views to Facilitate Routing

If the design is large or complex, you can create an abstract view of the design to limit the hierarchy depth and reduce the routing time. See [Creating Abstract Views From Designs](#).

Turning On Flight Lines

Turn on **Flight Lines** for the nets that need to be routed. Flight lines identify required connections between physical objects on the same net. See [Understanding Flight Lines](#). To turn on Flight Lines, see [Displaying and Hiding Flight Lines](#).

Creating Blockages for Center Routing

Channels are route passageways formed by the space between the following:

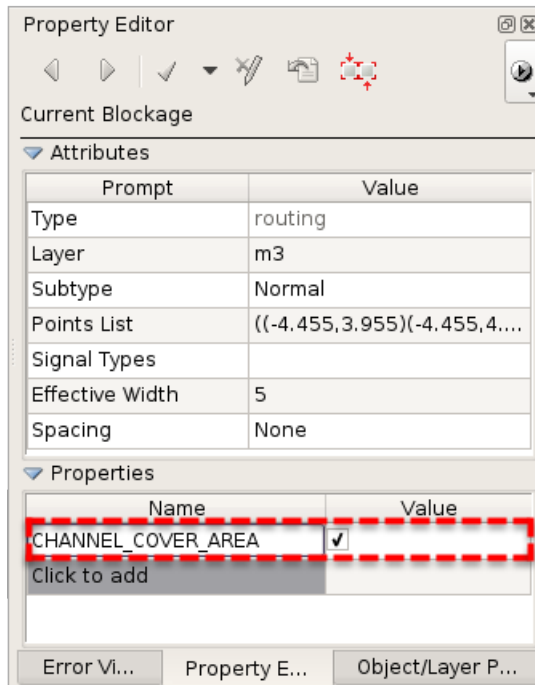
- A macro and a boundary
- Two adjacent macros
- A macro and a group of standard cells
- A group of standard cells and a boundary
- Two groups of standard cells

Some applications require routes to be centered in the channels. In designs where channel-forming objects are on different hierarchy levels, the router might not know which target objects to use when determining the center of channels.

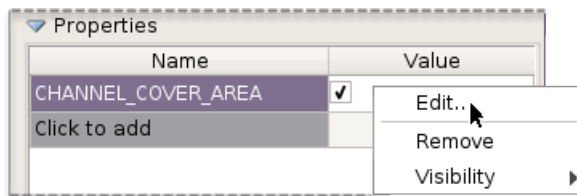
To aid the router, you can create top-level blockage objects that have the `CHANNEL_COVER_AREA` property. The router will use these special blockage objects to determine the center of channels.

To create a special blockage object:

1. In the design layout, create a blockage at the top level.
2. Display the Property Editor.
3. On the layout canvas, select the blockage object. The properties of the blockage appears in the Property Editor.
4. In the Property Editor, add `CHANNEL_COVER_AREA`.



5. Right-click the **Value** column and choose **Edit**.



6. In the **Edit Property** dialog box that appears, set the following and click **OK**:

- **Type:** boolean
- **Value:** ☒

Routing Designs With Power/Ground Nets

If your design contains 0x shapes on power/ground nets, the Custom Router will route all the power/ground pins of the selected nets.

If your design does not contains 0x shapes on power/ground nets, you can reduce the overall route runtime by setting the `rtMaxPinNumRoutePGLimit` Tcl preference. During the routing operation, the router determines the total number of pins on the power/ground signal type nets and takes the following actions:

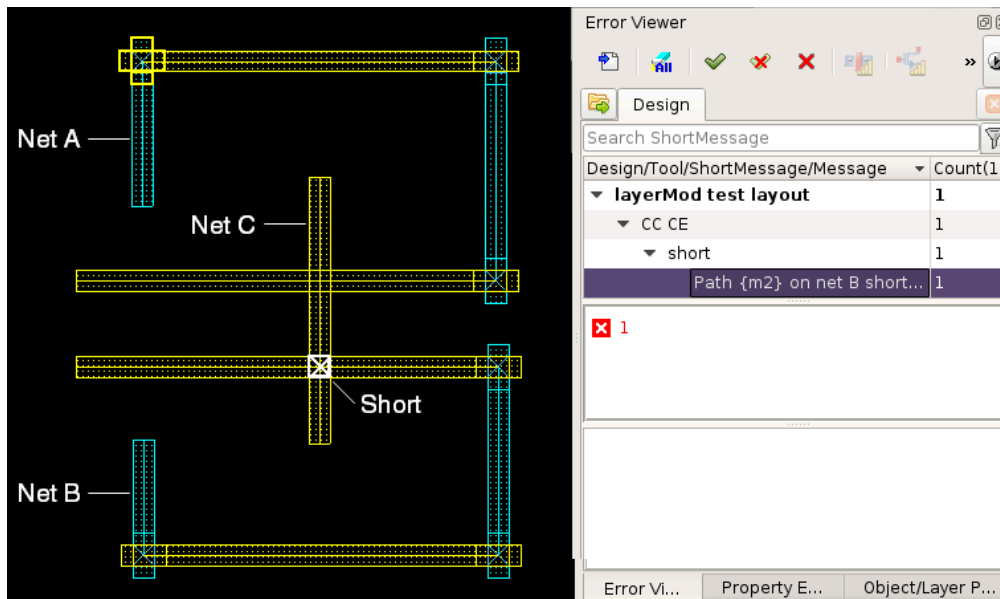
Number of pins is...	Route the power/ground nets?
<code><= rtMaxPinNumRoutePGLimit</code>	Yes
<code>> rtMaxPinNumRoutePGLimit</code>	No A warning message is displayed in the Console.

The default preference value is 30. To ensure that the router does not route the power/ground nets regardless of the number of pins, set `rtMaxPinNumRoutePGLimit` to 0.

Reducing the Number of Short Circuits

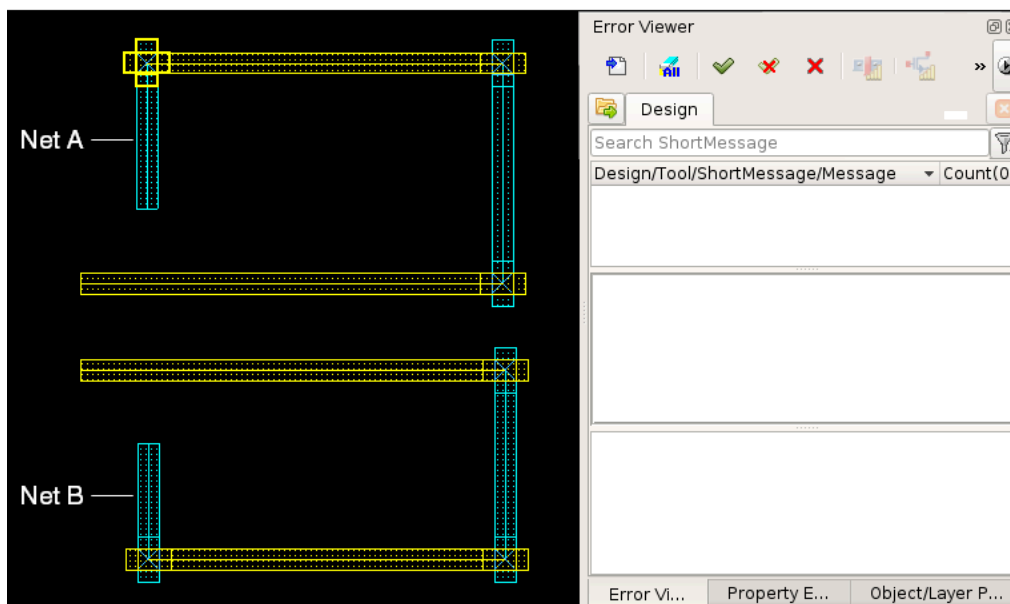
A short circuit occurs when path segments on the same layer but belonging to different nets touch, causing the nets to connect. The following example shows that nets A, B, and

Net C are all on the same layer. Adding the net C path as shown connects nets A and B, thus causing a short circuit.



To route the design with the minimum number of short circuits, set the `rtDeleteShortWires` Tcl preference to `true`. During the route operation, the router will remove paths that will cause short circuits. For example, setting the preference to `true` removes the net C path, thus preventing a short circuit at that location.

`rtDeleteShortWires = true`



Preventing Artificial Short Circuits

In cases where a resistor has two pins that connect to different nets, you can add the `leAbstract` Boolean property in the Property Editor and set it to true to stop the router from tracing connectivity. Doing so prevents artificial short circuiting and ensures correct routing.

You can add the `leAbstract` property at the cell or instance level. When adding at the cell level, all instances referring to it will force the router to stop tracing connectivity.

Minimizing CPU and Memory Usage

Before routing very large Custom Compiler, Fusion Compiler, or IC Compiler II designs, you can set the `rtImportPartialLayers` Tcl preference to specify the layers of interest. When you start routing, the Custom Router loads only the data from those layers, minimizing CPU and memory usage during the routing operation.

This topic describes the following:

- [Preference Syntax](#)
- [Requirements](#)
- [Examples](#)

Preference Syntax

```
rtImportPartialLayers -value "<layer_1> <layer_2> <layer_n>"
```

The layers you specify do not have to be contiguous. For example, you can specify m1, m3, and m4. The router will ignore invalid layers.

Requirements

Make sure you:

- Set the preference before you choose **Routing > Auto Route** or **Routing > Interactive Route**.
- Specify only the following types of layers:
 - Metal
 - Poly
 - Diffusion
 - Contactless metal
- Include pin layers so that the router can connect to the pins.

Examples

In the following example, the preference setting will cause the router to load the data for layers m2, m3, and m4 before creating routes.

```
db::setPrefValue rtImportPartialLayers -value "m2 m3 m4" -scope  
[de::getActiveContext]
```

In the following example, the preference setting will cause the router to load the data for layers m4 and m6 before creating routes.

```
db::setPrefValue rtImportPartialLayers -value "m4 m6" -scope  
[de::getActiveContext]
```

4

Setting Routing Constraints

This chapter describes the constraints supported by the Custom Router, and explains how you can add and modify design-specific constraints (intents) in the Constraint Editor.

This chapter contains the following topics:

- [About Routing Constraints](#)
- [Before You Start](#)
- [Creating Routing Constraints](#)
- [Viewing and Modifying Constraint Parameters](#)
- [Deleting Routing Constraints](#)

For instructions on how to view, import, and export constraints in the Constraint Editor, see the *Custom Compiler Environment User Guide*.

About Routing Constraints

By default, the software uses the process-level constraint values in the technology database. However, you can create design-specific constraints (or intents) to further restrict placement and routing. Design-specific constraints can be applied to a group of nets or an individual net.

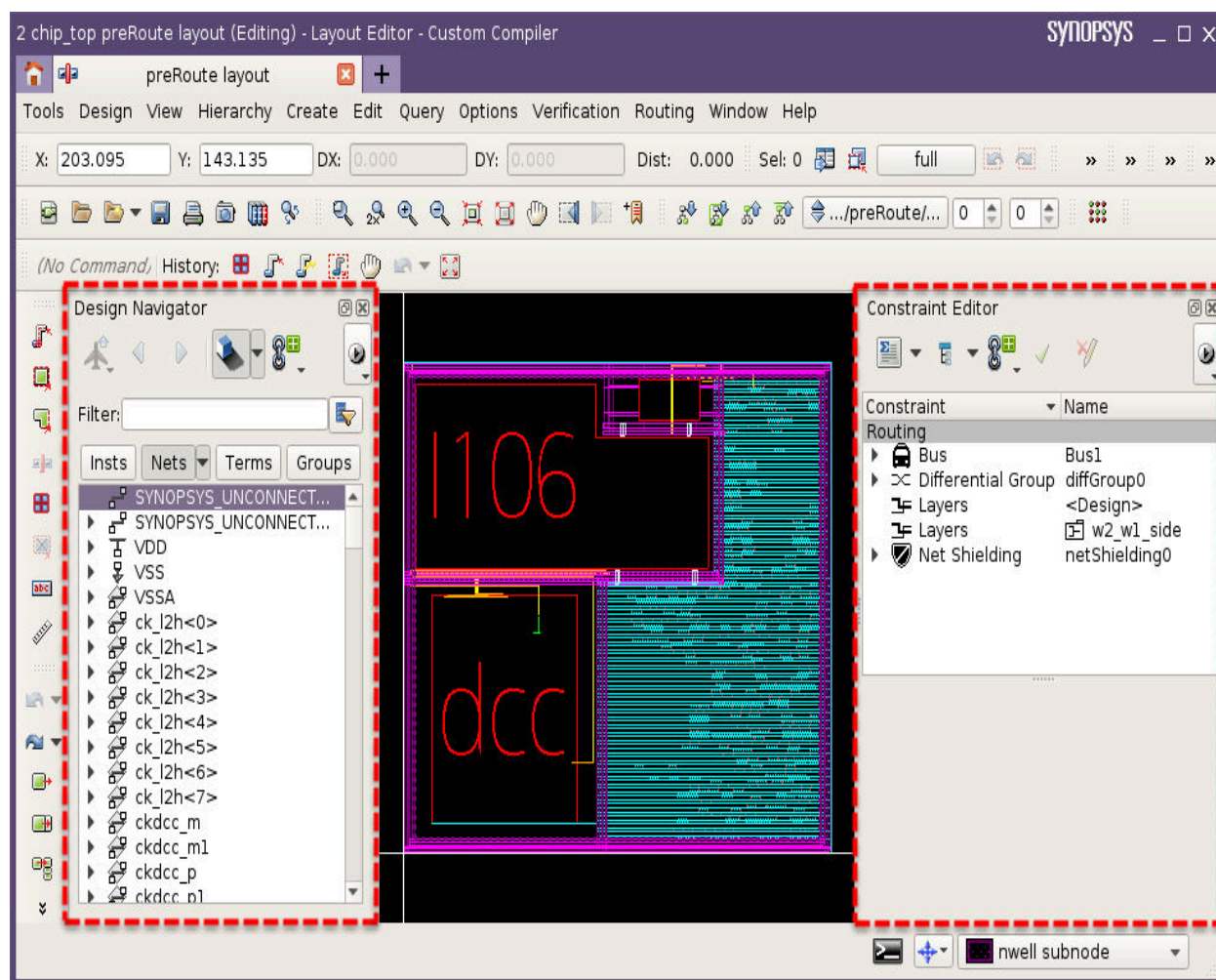
Routing constraints are design-specific constraints (intents) that control routing operations and results. For example, you can specify valid routing layers, the minimum width of each routing layer, preferred direction for each layer, and shielding styles.

You use the Constraint Editor to create, view, and modify Routing constraints. For process constraints that are applied to all designs, review the technology database.

Before You Start

Make sure you display the following assistants:

Assistant	Description
Design Navigator	Allows you to select the nets to apply the constraints. Alternatively, you can select shapes on the canvas, and the nets associated with the shapes are automatically selected.
Constraint Editor	Allows you to create constraints for the selected nets, modify constraint parameters, and delete existing constraints.

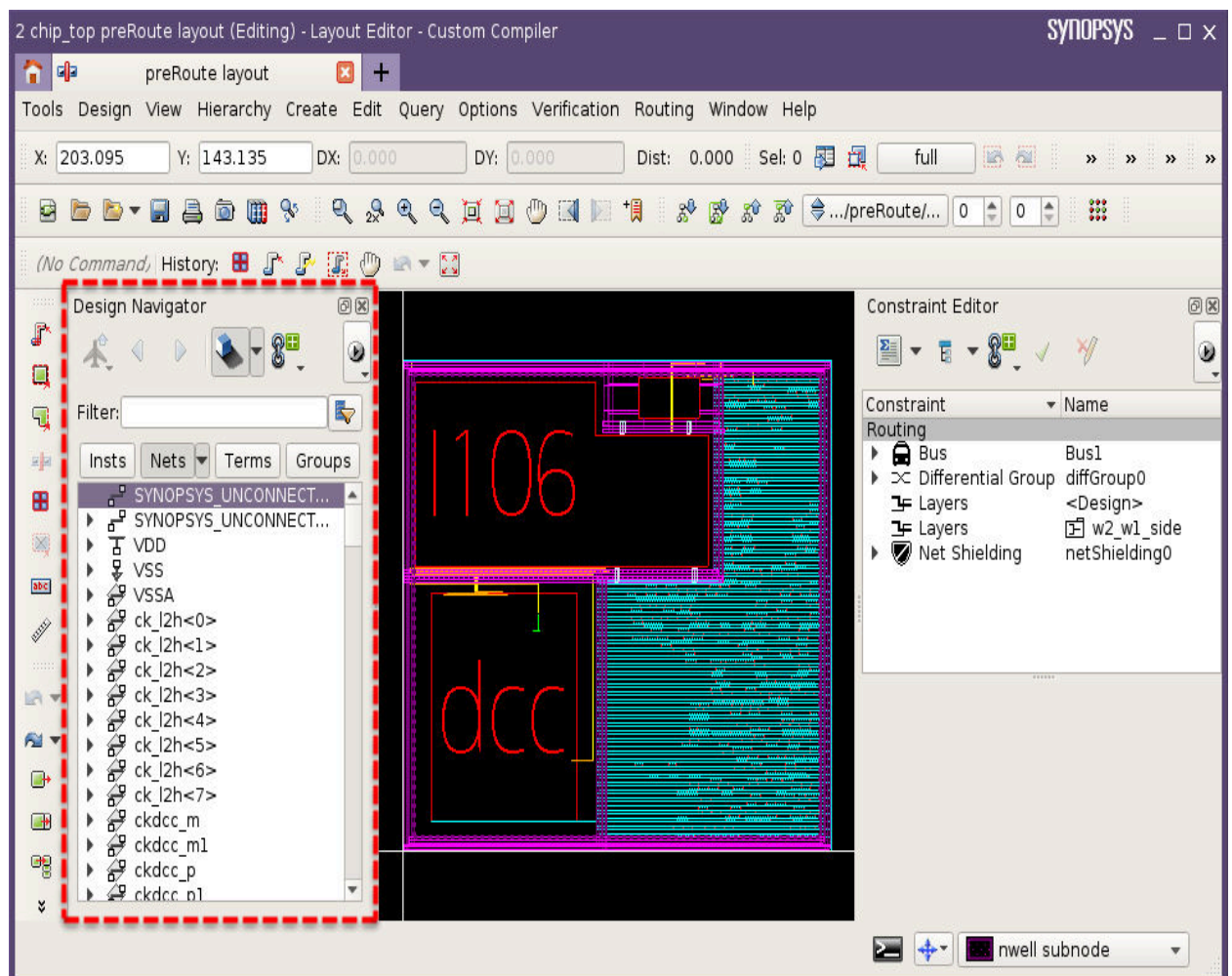


Creating Routing Constraints

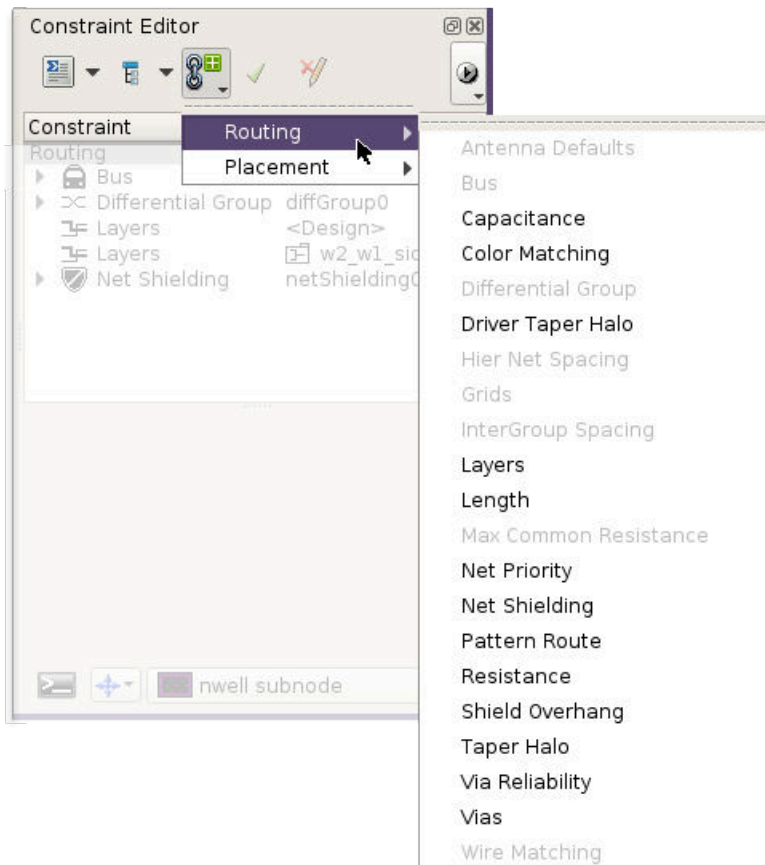
By default, the router automatically applies the constraint values in the technology during routing. However, you can create design-specific constraints (intents) to further restrict routing.

To create a new routing constraint:

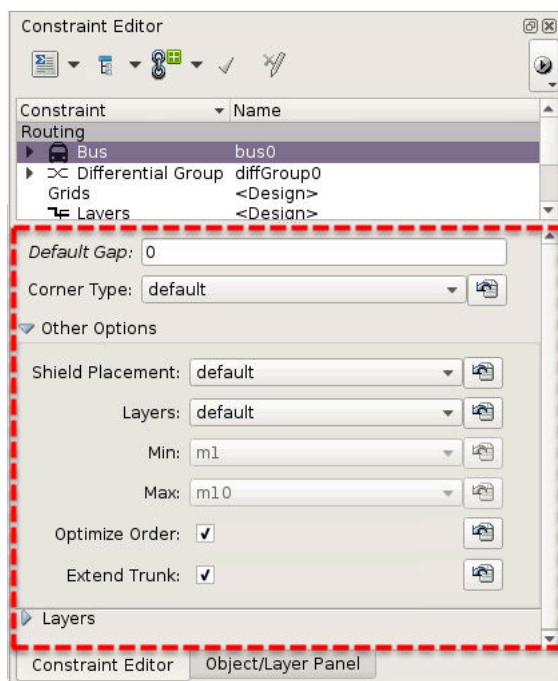
1. Make sure the Design Navigator and Constraint Editor are open in the Layout Editor. For instructions, see [Before You Start](#).
2. In the Design Navigator, select the nets or net groups to which you want to apply the new constraint.



3. In the Constraint Editor, click **Create Constraint** , choose **Routing**, and then select the constraint you want to create.



The newly added constraint appears in the Constraint list. The area below the Constraint list refreshes and displays the parameters for the selected constraint.



For the descriptions of the constraint parameters, see [Routing Constraint Reference](#).

Viewing and Modifying Constraint Parameters

By default, the constraint parameter values are set to the values in the technology library. You can override the values in the technology library by supplying the desired values in the Constraint Editor. Some design-specific constraints (intents) also supplement the process rules.

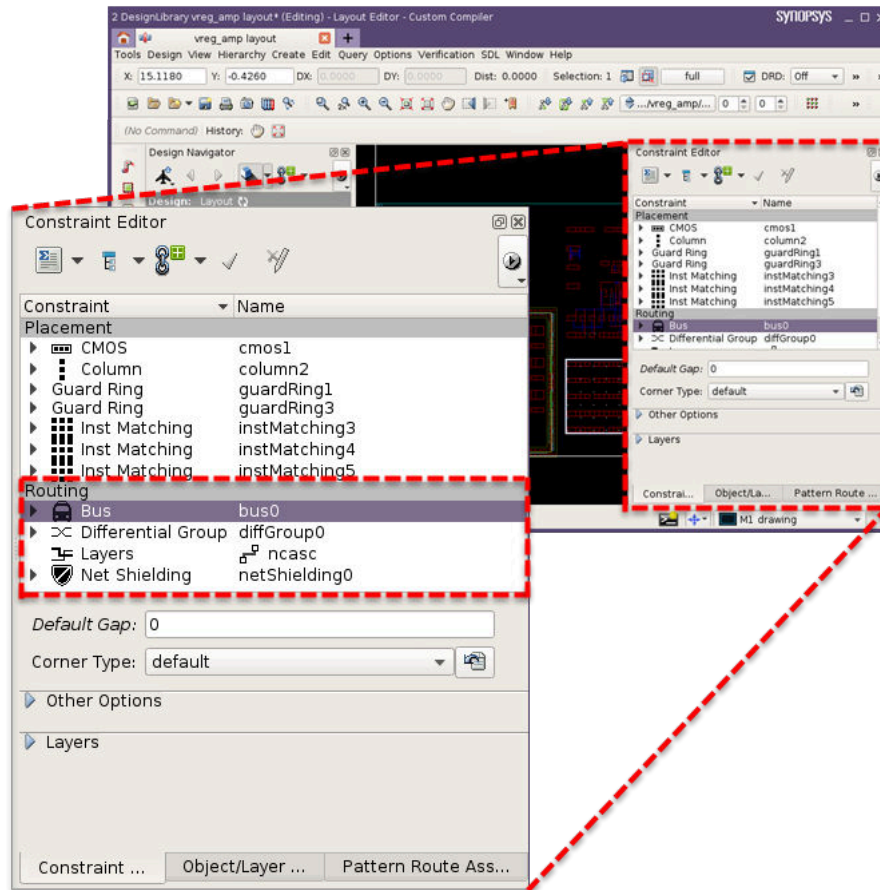
Note:

The constraints you set can override process rules to further restrict routing. You cannot set constraints to violate process rules.

This topic describes how to view and modify the constraint parameters using the Constraint Editor. To modify constraint values in the technology library, review the technology database.

To view or modify the constraint parameters:

1. In the Layout Editor, display the Constraint Editor. For instructions, see [Before You Start](#).
2. In the Constraint Editor, locate the **Routing** section, and then choose the constraint you want to view or modify. The constraint parameters appear.

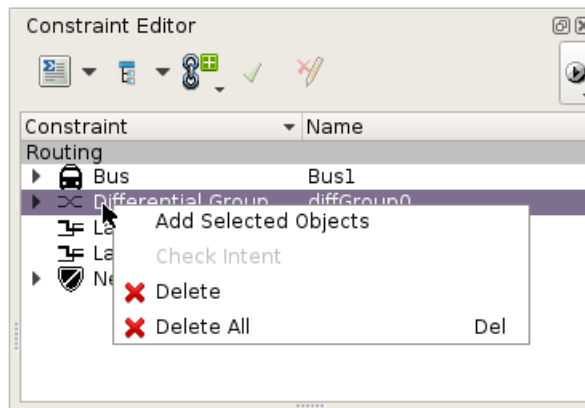


3. View or modify the constraint parameters. For the description of the parameters, see [Routing Constraint Reference](#).
4. Notice that after you change a parameter value, a  button appears next to the changed value. To reset a parameter back to its default value, click .
5. When you are finished modifying the constraint:
 - Click **Apply Changes** () to save the modifications.
 - Click **Discard Changes** () to revert back to the previously saved settings.

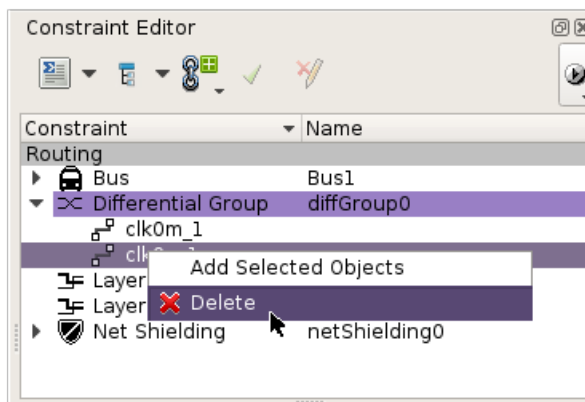
Deleting Routing Constraints

To delete an existing constraint or to remove a constraint from a net:

1. Make sure the Constraint Editor is open in the Layout Editor. For instructions, see [Before You Start](#).
2. To delete a constraint:
 - a. In the Constraint Editor, right-click the constraint in the **Constraint** list, and then choose **Delete**. If you want to delete all of the constraints, click **Delete All**.
 - b. Click **Apply Changes** () to save the modifications.



3. To remove a constraint from a net:
 - a. In the Constraint Editor, right-click the net under the constraint, and then choose **Delete**.
 - b. Click **Apply Changes** () to save the modifications.



5

Topology Routing

This chapter describes how to create and edit route topologies, set layout and electrically aware constraints, and create routes based on the topology guidance.

This chapter contains the following topics:

- [About Topology Routing](#)
- [Workflow for Topology Routing](#)
- [Before You Start](#)
- [Displaying the Topology Toolbar](#)
- [Creating Trunk-Style Topologies](#)
- [Creating Freeform Topologies](#)
- [Creating Route Topologies Automatically](#)
- [Creating Topologies from Existing Routes](#)
- [Setting Topology Segment Properties](#)
- [Setting Topology Viewing Options](#)
- [Editing Topologies](#)
- [Deleting Topologies](#)
- [Grouping Nets That Have Topologies](#)
- [Estimating Route Congestion](#)
- [Enabling or Disabling Topology-Based Shielding](#)
- [Generating Routes Based on Topologies](#)
- [Importing and Exporting Topologies](#)
- [Reusing Existing Topologies in Different Designs](#)
- [Topology Toolbar Reference](#)

- [Topology Assistant Reference](#)
- [Topology Shortcut Menu Commands](#)

About Topology Routing

Controlling route placement can help you optimize the manufacturability and performance of your design. One of the ways you can control route placement is to plan and create route topologies.

You can define route topologies to prescribe pathways that the Auto Router must follow. The topologies capture net-based routing intents and are saved with the design.

This topic describes the following:

- [Route Topology Components](#)
- [Methods for Creating Route Topologies](#)
- [Topology Viewing Options](#)

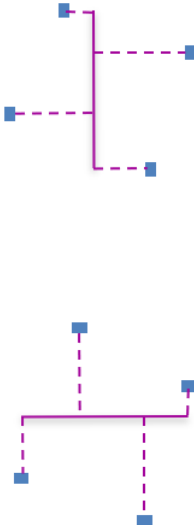
Route Topology Components

Each route topology consists of the following:

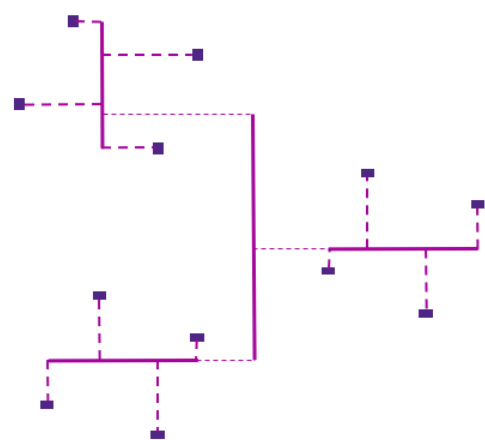
- Topology style
- Topology objects
- Topology properties

Topology Style

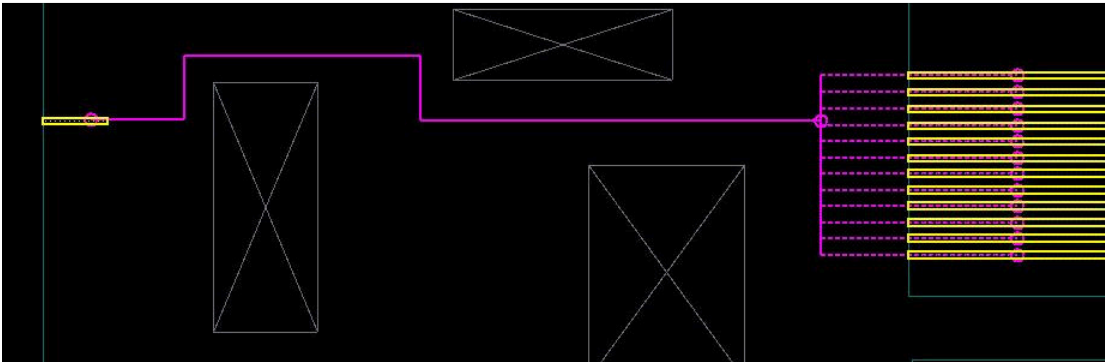
The two route topology styles you can create are:

Topology Style	Structure	Description
Trunk (vertical or horizontal)		Adds fishbone or backbone route guidelines between selected pins or topology segments. See Creating Trunk-Style Topologies .
Freeform		Adds route guidelines from pins or topology segments. Use this style to create loop or mesh patterns. You can also use this style to guide routes around obstacles (). See Creating Freeform Topologies .

You can combine the two styles in one design. In addition, you can connect route topologies across multiple hierarchy levels.



Combined Topology Styles



Topology Objects

Topologies consist of the following objects:

-  **Pin**
-  **Guideline**
-  **Tapoff**
-  **Anchor Point**

Option	Description
Guideline	Lines that indicate where the routes will be placed.

Option	Description
Anchor Point	Shapes that indicate where the topology connects to pins or other topologies. Tcl preference: <code>rtTopologyDisplayAnchorPoint</code>

Because route topologies are a set of constraints, topology objects are not OA objects. You can only create or edit them using the available topology commands.

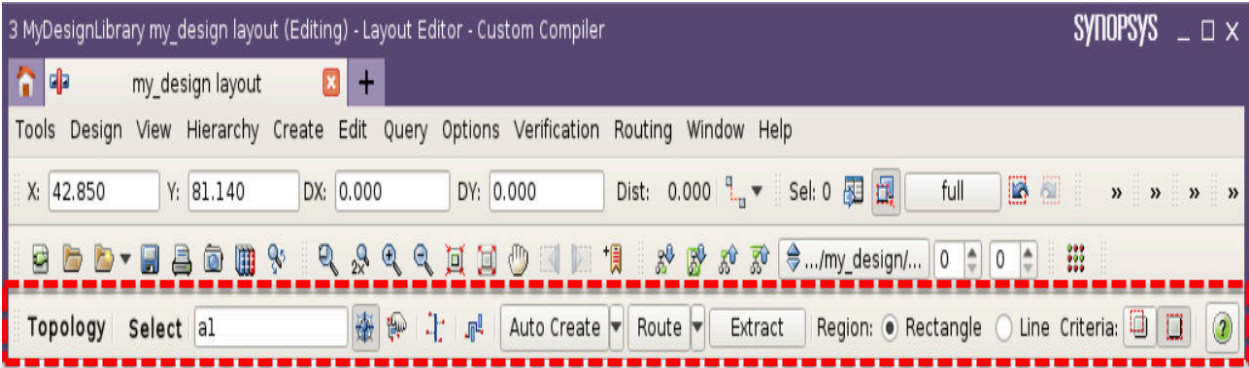
Topology Properties

Topology properties consists of layer, width, shield, and number of stripes information. The properties are used when you generate routes based on the topologies. See [Setting Topology Segment Properties](#).

The topology properties are associated with nets and saved with the design. As long as they remain intact, detailed routes can be regenerated from the topology, even if you modify the design.

Methods for Creating Route Topologies

You use the **Routing > Create Topology** command to create route topologies.



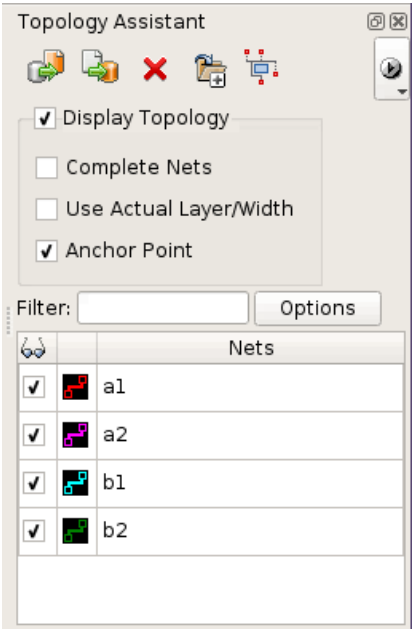
The command provides two ways to create route topologies:

Method	Description
Manual	Manually create topologies in one of the styles: Creating Trunk-Style Topologies Creating Freeform Topologies
Automatic	Automatically generate topologies for selected nets. See Creating Route Topologies Automatically .

Regardless of the method you use, routes created from the topologies adhere to tracks defined in the design.

Topology Viewing Options

You use the Topology Assistant to view the list of all nets with topologies. You can also use the assistant to set topology viewing options: show or hide topology guidelines, change topology guideline colors, show or hide anchor points, and so on. See [Setting Topology Viewing Options](#).



Workflow for Topology Routing

The basic workflow for topology routing is as follows:

Step	See...
1	Prepare for topology routing. Before You Start
2	Display the Topology toolbar. Displaying the Topology Toolbar
3	Create topology. • Creating Trunk-Style Topologies • Creating Freeform Topologies • Creating Route Topologies Automatically • Creating Topologies from Existing Routes • Importing Route Topologies

Step		See...
4	Set topology routing properties	Setting Topology Segment Properties
5	Edit topologies.	Editing Topologies
6	(Optional) Delete topologies.	Deleting Topologies
7	(Optional) Grouping nets that have topologies.	Grouping Nets That Have Topologies
8	Estimate route congestion.	Estimating Route Congestion
9	Add shielding to topologies.	Enabling or Disabling Topology-Based Shielding
9	Generate routes based on topologies.	Generating Routes Based on Topologies
10	(Optional) Export topologies.	Exporting Route Topologies

Before You Start

To optimize the results of topology routing, be sure to review and complete any prerequisite tasks. For example, to ensure electrical compliance, you would need to define the relevant electrical constraints.

This topic describes tasks to be completed before you start creating route topologies:

- [Setting Routing Constraints](#)
- [Preparing for Topology Creation](#)
- [Setting Electromigration Constraints](#)
- [Setting Resistance Constraints](#)
- [Setting Auto Router Options](#)

Setting Routing Constraints

Set the following routing constraints, which the topology inherits based on scope:

- [Layers Constraint](#): Specifies minimum width and spacing constraints per layer.
- [Via Reliability Constraint](#): Specifies the minimum number of via cuts for per layer.

After you create the topology, you can edit the constraints on specific segments. See [Setting Topology Segment Properties](#). If you selected **Shield** when setting the topology segment properties, make sure you set the [Net Shielding Constraint](#).

Preparing for Topology Creation

Before you create a route topology:

- Review the design and plan the route topology.
- Display the Design Navigator so that you can see the list of nets you will be working with.
- In the Design Navigator, turn on the **Shadow Mode** so that you can better see the pins on selected nets. See [Placing Design Areas in Shadow](#).
- Display the flight lines so that you can clearly see the connections on the selected nets. See [Displaying and Hiding Flight Lines](#).

For inter-device group routing, set the `rtInterGroupATCConnectionMethod Tcl` preference to specify the layer to use:

Preference Value	Description
topLayer	Connects only the trunks and spines that are on the topmost layer.
topHVLayers	Connects only the trunks and spines that are on the topmost horizontal and topmost vertical layer.
trunkAndSpine	Connects the trunks and spines on all layers.

Setting Electromigration Constraints

To ensure that the routes generated from the topology are compliant with electromigration design rules, make sure all instance terminals connected to the net are back-annotated in the layout with AC average current data. See [Annotating Layouts with Current Data](#).

Before generating routes based on the topologies, you can turn on and off electrical checks. See [Generating Routes Based on Topologies](#).

Setting Resistance Constraints

To ensure that the routes generated from the topology are compliant with resistance constraints, set the Resistance constraint with maximum pin-to-pin resistance information in the Constraint Editor Assistant or use the `cm::createResistance Tcl` command. To set

the Resistance constraint in the Constraint Editor, see [Defining Pin-to-Pin Resistance Constraints](#).

To set sheet resistance constraints, see [Calculating Resistance Using Sheet Resistance](#) and set the options in the **Electrical Options** dialog box. Depending on the **Flow Settings** you select on the **Main** tab, you need to specify the relevant Nxtgrd file on the **Nxtgrd** tab and the TLUPlus file on the **TLU+** tab. See [Setting EM, Resistance, and Capacitance Options](#).

Before generating routes based on the topologies, you can turn on and off electrical checks. See [Generating Routes Based on Topologies](#).

Setting Auto Router Options

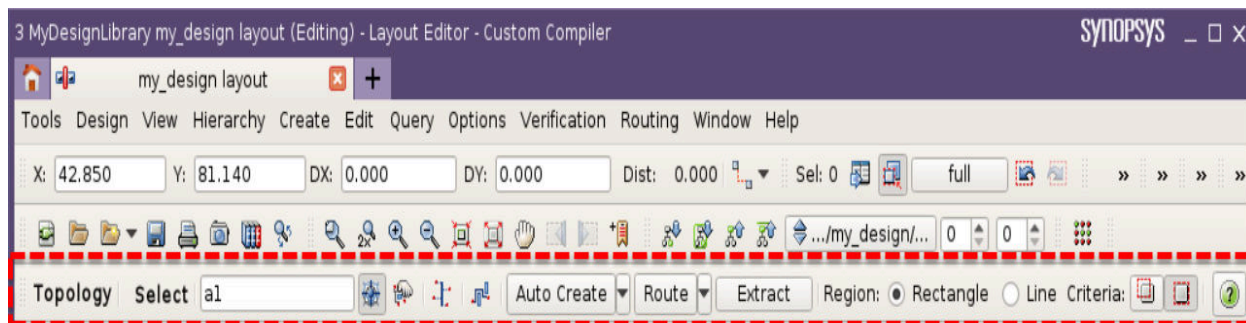
Because the Topology **Route** command uses the Auto Router, make sure you set the Auto Router options to optimize the results. See [Auto Route Toolbar and Dialog Box Reference](#).

Displaying the Topology Toolbar

You use the Custom Router **Topology** toolbar to create route topologies.

To display the **Topology** toolbar:

1. Open the design layout.
2. In the Layout Editor, choose **Tools > Routing**. The **Routing** menu appears on the Layout Editor menu bar.
3. Choose **Routing > Create Topology**. The **Topology** toolbar appears.

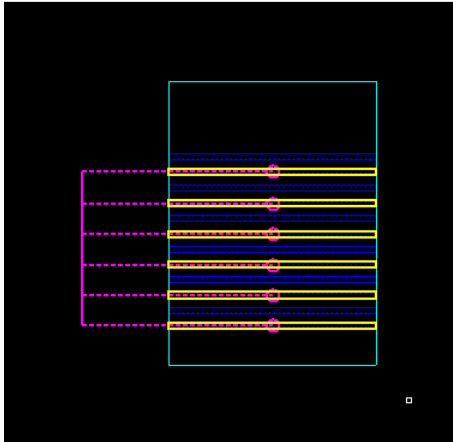


For a quick reference of the toolbar options, see [Topology Toolbar Reference](#).

Creating Trunk-Style Topologies

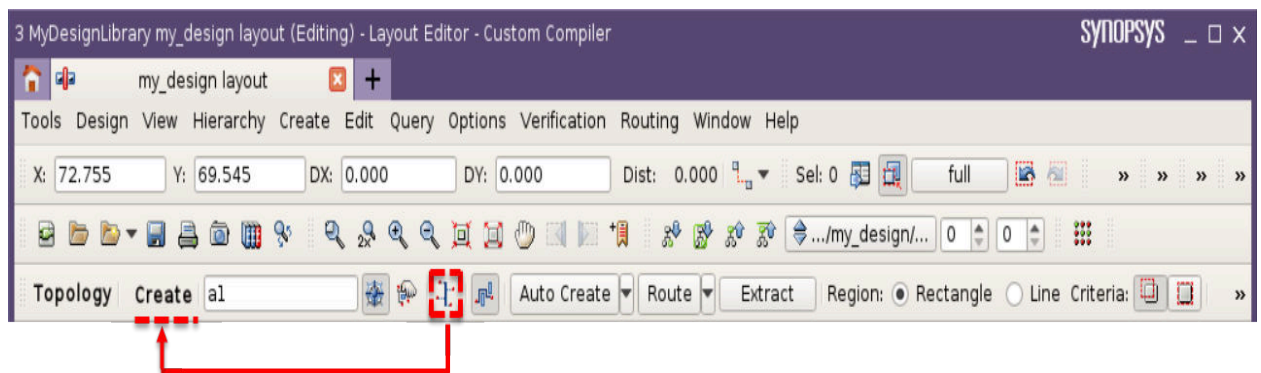
Trunk-style topologies have fishbone or backbone structures. You select pins, other topology components, or existing routes in the design, and the command automatically creates a trunk and tapoffs to connect the pins. You can adjust the trunk position, rotate the trunk, and add or remove tapoffs.

Backbone trunk-style topology



To create trunk-style topologies:

1. In the Design Navigator, select the net.
2. On the **Topology** toolbar, click **Trunk**. The status area on the toolbar changes to **Create**.

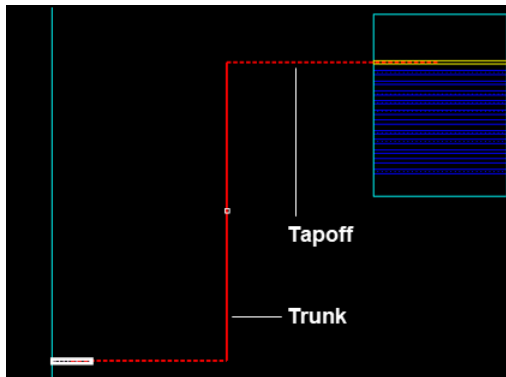


3. On the layout canvas, select at least two pins using one of the following methods:

Selection method	Description
Area-select multiple pins	On the Topology toolbar: a. Select Region: Rectangle. b. Click one of the following: : Selects objects within the rectangular area. : Select objects within the rectangular area or intersecting the rectangular boundary. Draw the rectangle on the canvas.
Line-select multiple pins	On the Topology toolbar, select Region: Line , and then draw a line across the pins you want to select on the canvas.
Cherry-pick multiple pins	Shift+click to select pins. Ctrl+click to remove pins from the selection.

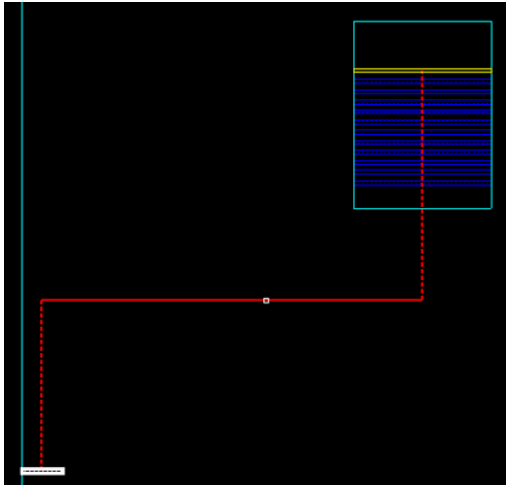
As soon as you select two or more pins, the topology guidelines (trunk and tapoffs) appear on the canvas. If the trunk overlaps with pins, no tapoffs are created.

In the following example, the guidelines appear in red. To change the color (display packet) of the guidelines, see [Setting Topology Viewing Options](#).



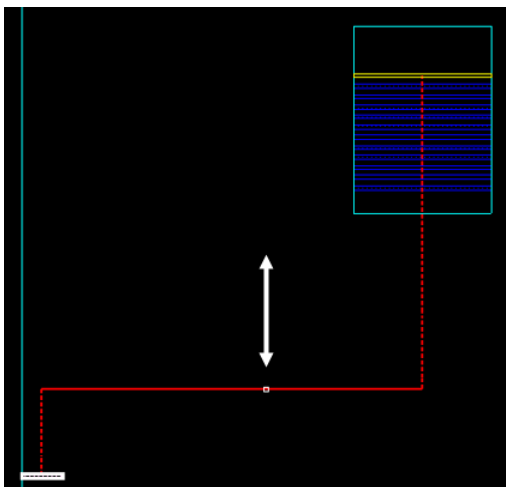
4. To rotate the trunk, press R.

The newly created trunk rotates.



If you want to rotate trunks after you create them, use the **Merge to Horizontal Trunk** and **Merge to Vertical Trunk** commands. See [Merging Trunks](#).

5. To adjust the trunk position, move the pointer up and down for vertical trunks, or left and right for horizontal trunks.



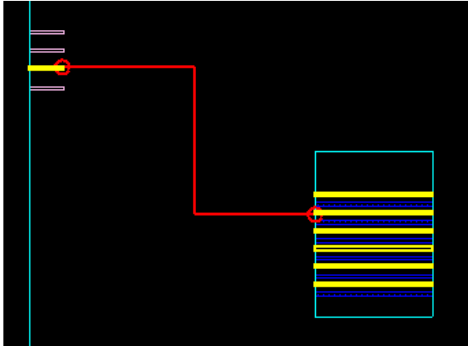
6. When you are satisfied with the trunk position, click the canvas to commit the topology.
7. Press Esc to return to the Select mode. Press Esc again to exit the **Topology** command.

To edit the trunk topology, see [Editing Topologies](#). To change the properties of a topology segment, such as the layer, width, shielding, and wire stripes, see [Setting Topology Segment Properties](#).

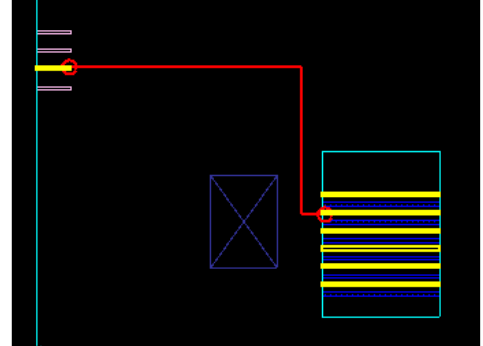
Creating Freeform Topologies

In freeform topologies, you can start from and end at pins, existing topology line, or existing routes. Freeform topologies lets you have manual control over placement, especially around obstacles.

No blockage



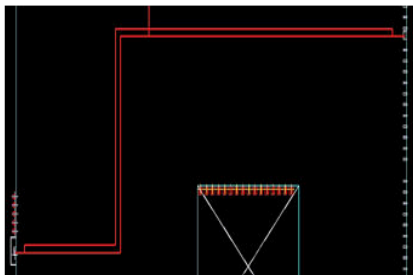
Topology guidance around blockage



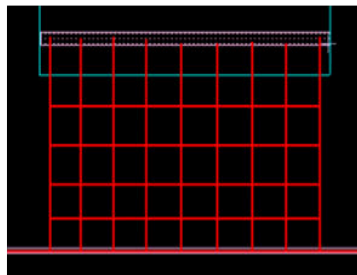
Using the freeform style, a variety of route topology patterns are possible:

Example Topology Patterns

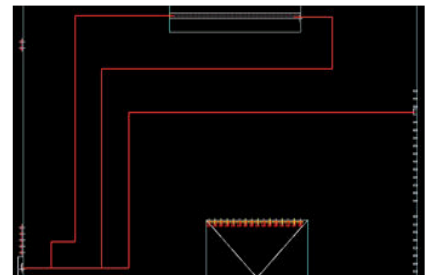
Parallel



Mesh

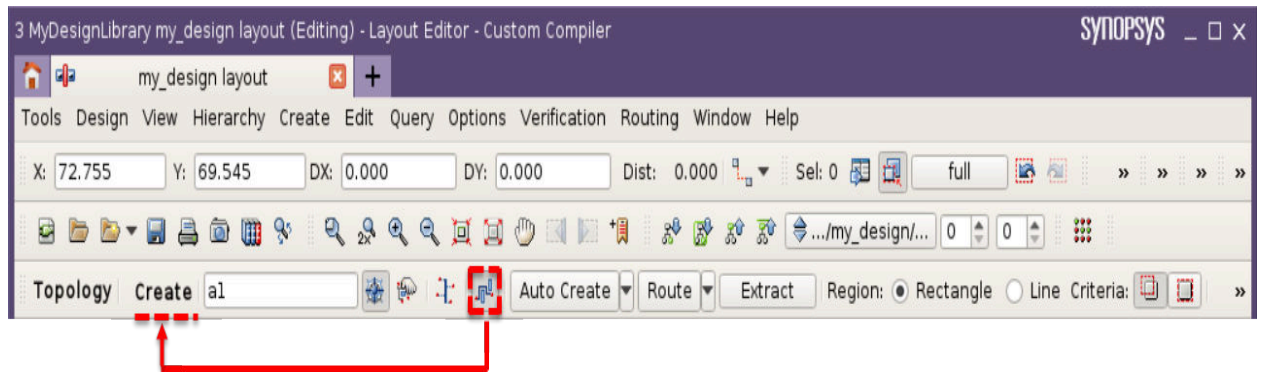


Star



To create freeform topologies:

1. In the Design Navigator, select the net.
2. On the **Topology** toolbar, click **Freeform**. The status area on the toolbar changes to **Create**.



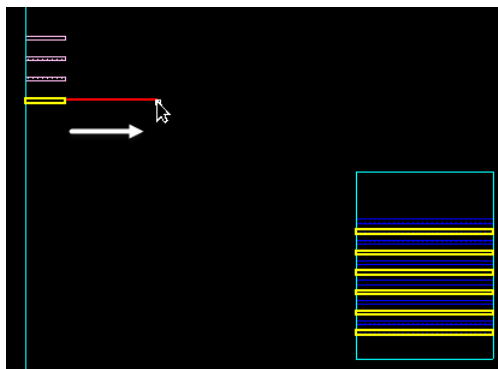
3. On the layout canvas, select the starting pin or a topology segment.

If the area is crowded and you are unable to easily select the pin or segment:

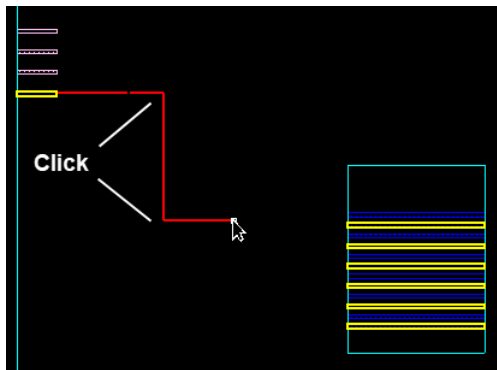
- a. Move the pointer to the starting pin or segment.
- b. Press 1 to cycle through the objects beneath the pointer until the shape or segment of interest is active.
- c. Click the canvas to select the active shape or segment.

4. Move the pointer away from the pin or topology segment. A topology line appears on the canvas.

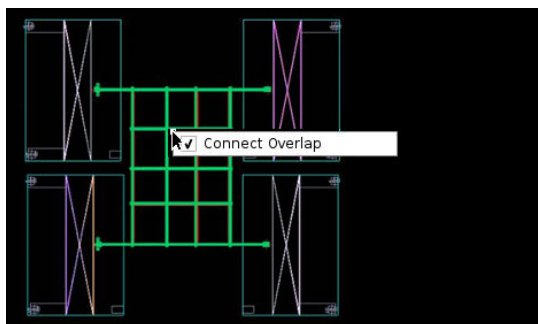
In the following example, a red line appears. To change the color (display packet) of topology lines, see [Setting Topology Viewing Options](#).



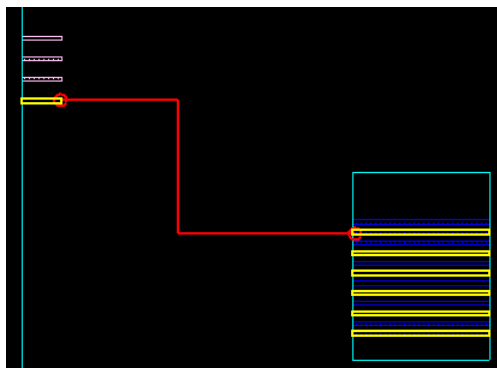
5. To bend or add route points, click the locations on the canvas where you want to bend or add route points.



6. To add vias and connect to a pin or another topology at the intersection, right-click and select **Connect Overlap**. Use this option for creating loop or mesh topologies.



7. Click the ending pin, topology segment, or existing route to finish and commit the topology.



8. Press Esc to return to the Select mode. Press Esc again to exit the **Topology** command.

To edit the freeform topology, see [Editing Topologies](#). To change the properties of a topology segment, such as the layer, width, shielding, wire stripes, see [Setting Topology Segment Properties](#).

Creating Route Topologies Automatically

The **Auto Create** command generates freeform topologies automatically for the selected nets.

This topic describes the following:

- [About the Automatic Topology Creation Method](#)
- [Before You Start Automatic Topology Creation](#)
- [Starting Automatic Topology Creation](#)
- [Viewing the Topology Generation Report](#)

About the Automatic Topology Creation Method

You can use the **Auto Create** command to generate freeform topologies automatically for selected nets. Automating the task can reduce the overall time it takes to route a design.

The **Auto Create** command uses the valid routing layer for the associated topology segment and the appropriate width based on the constraints. In addition, the command creates shielding for the topologies by default if net shielding constraints are defined for the nets. See [Enabling or Disabling Topology-Based Shielding](#).

If the nets you selected already have associated route topologies:

- The command will skip the nets with existing topologies.
- If a selected net has partial topologies, the command will attempt to complete them.

If you turn on the option to create topologies for nets that connect to standard cells, be sure to review the results and check for the following:

- Connection or pin access issues
- Unroutes and DRC errors

Before You Start Automatic Topology Creation

Be sure to complete the prerequisites for automatic topology creation.

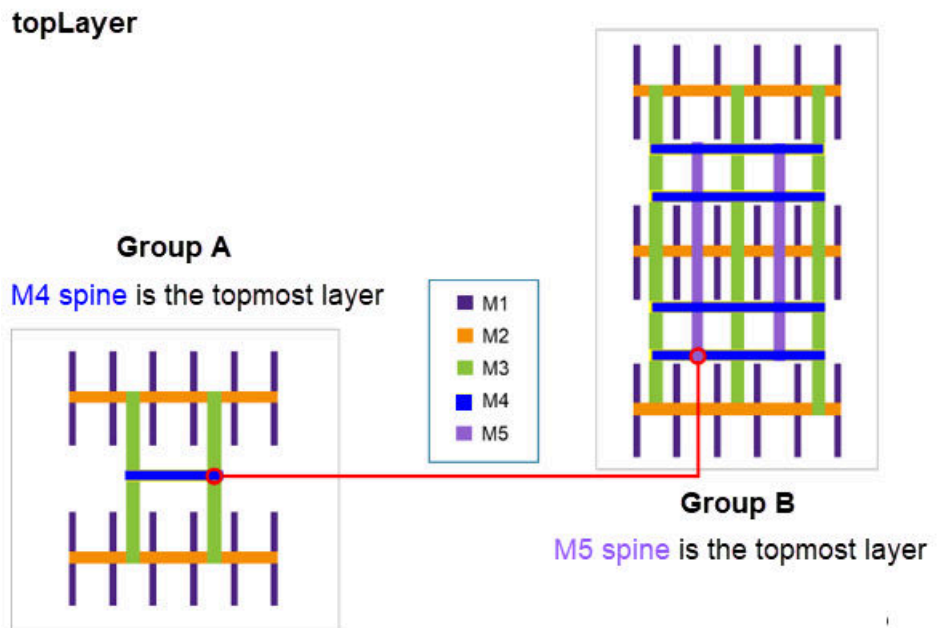
To ensure successful results:

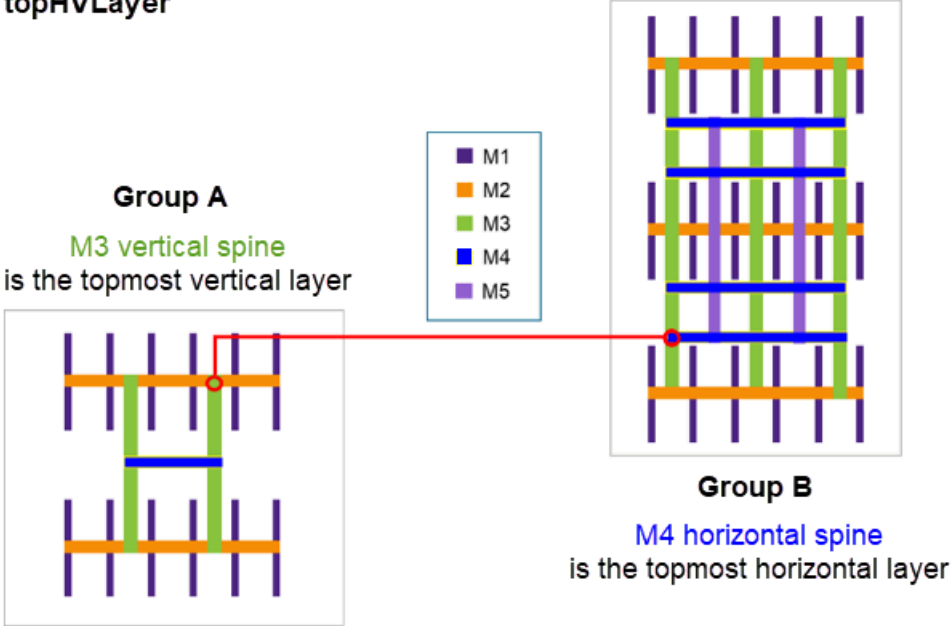
- Review existing topologies in the design. Remove topologies that do not meet your requirements and let the **Auto Create** command generate another solution.
- Because the command will not generate wrong-way route guidance, change the layer setting to use specific layers after the topology is generated. See [Setting Topology Segment Properties](#).
- If you want to add loop or mesh topologies, use Freeform Topology and the **Connect Overlap** option to create them. See [Creating Freeform Topologies](#).

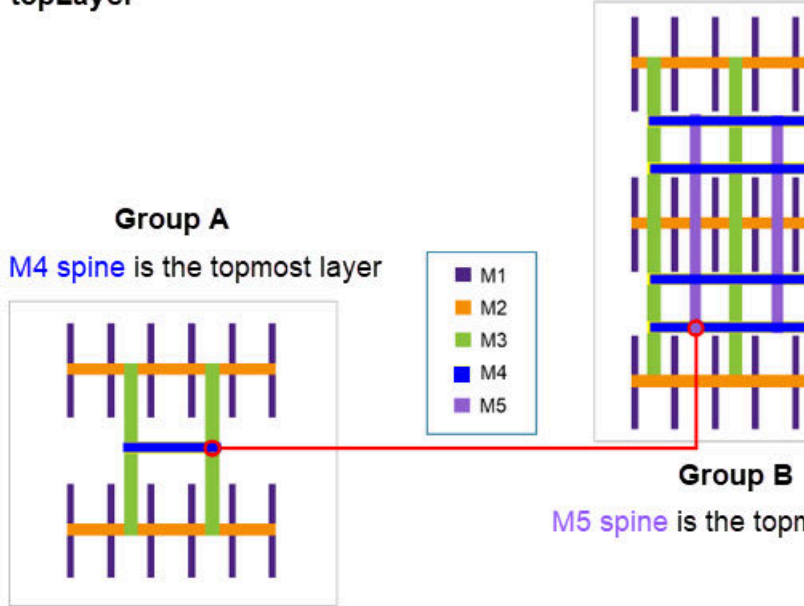
Because **Auto Create** does not support bus route guidance, you will need to create bus routes manually.

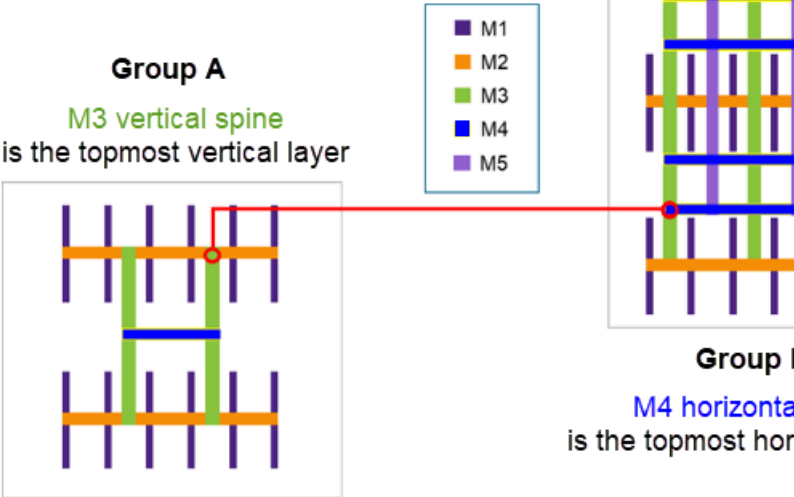
To specify the trunk and spine layers that can be used to connect device groups, set the `rtInterGroupATCConnectionMethod` Tcl preference:

Preference value	Description
<code>topLayer</code>	Only the trunks and spines on the topmost layer in one group can connect to trunks and spines on the topmost layer in another group. In the example, the M4 spine in group A can connect to the M5 spine in group B because they are both on the topmost layers in their groups.



Preference value	Description
<code>topHVLayers</code>	(Default) Only the trunks and spines on the two topmost layers in one group can connect to trunks and spines on the two topmost layers in another group. The two topmost layers consist of a horizontal trunk or spine on one layer and a vertical trunk or spine on another layer. In the example, the M3 spine is the topmost vertical layer in group A, and the M4 spine is the topmost horizontal layer in group B. So, these two spines can connect. topHVLayer  Group A M3 vertical spine is the topmost vertical layer Group B M4 horizontal spine is the topmost horizontal layer Although not shown, the M4 spine is the topmost horizontal layer in group A, and the M5 spine is the topmost vertical layer in group B. So, these two spines can also connect.
<code>anyTrunkAndSpine</code>	The trunks or spines on any layer in one group can connect to the trunks or spines on any layer in another group.

Preference value	Description
topLayer	Only the trunks and spines on the topmost layer in one group can connect to trunks and spines on the topmost layer in another group. In the example, the M4 spine in group A can connect to the M5 spine in group B because they are both on the topmost layers in their groups. topLayer  Group A M4 spine is the topmost layer Group B M5 spine is the topmost layer Legend: M1 (purple), M2 (orange), M3 (green), M4 (blue), M5 (purple)

Preference value	Description
<code>topHVLayers</code>	(Default) Only the trunks and spines on the two topmost layers in one group can connect to trunks and spines on the two topmost layers in another group. The two topmost layers consist of a horizontal trunk or spine on one layer and a vertical trunk or spine on another layer. In the example, the M3 spine is the topmost vertical layer in group A, and the M4 spine is the topmost horizontal layer in group B. So, these two spines can connect. topHVLayer  Although not shown, the M4 spine is the topmost horizontal layer in group A, and the M5 spine is the topmost vertical layer in group B. So, these two spines can also connect.
<code>anyTrunkAndSpine</code>	The trunks or spines on any layer in one group can connect to the trunks or spines on any layer in another group.

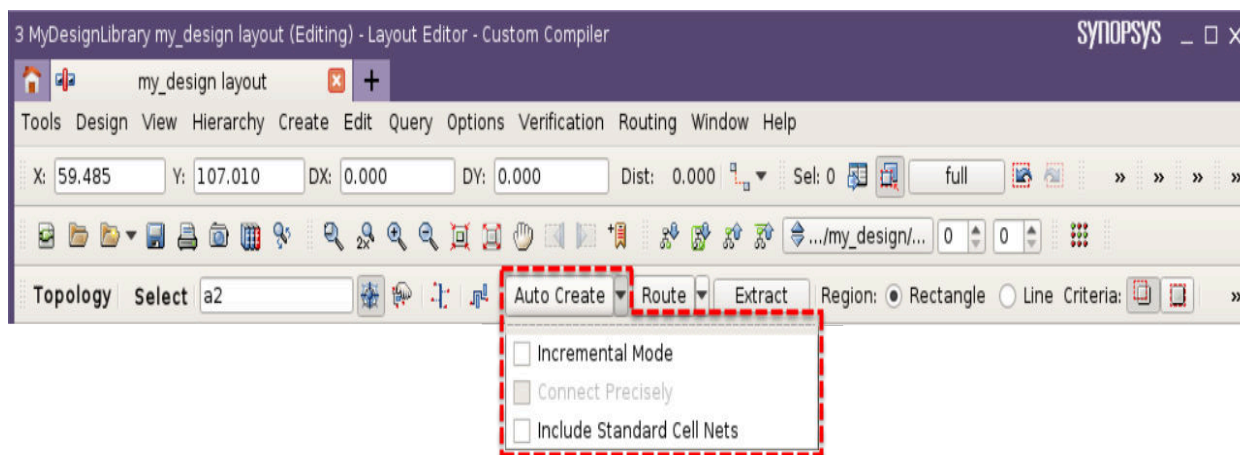
See Also

- [Device-Group Routing Workflow](#)

Starting Automatic Topology Creation

To create route topologies automatically:

1. In the Design Navigator, select one or more nets.
2. On the **Topology** toolbar, click **Auto Create**, and select the applicable options:



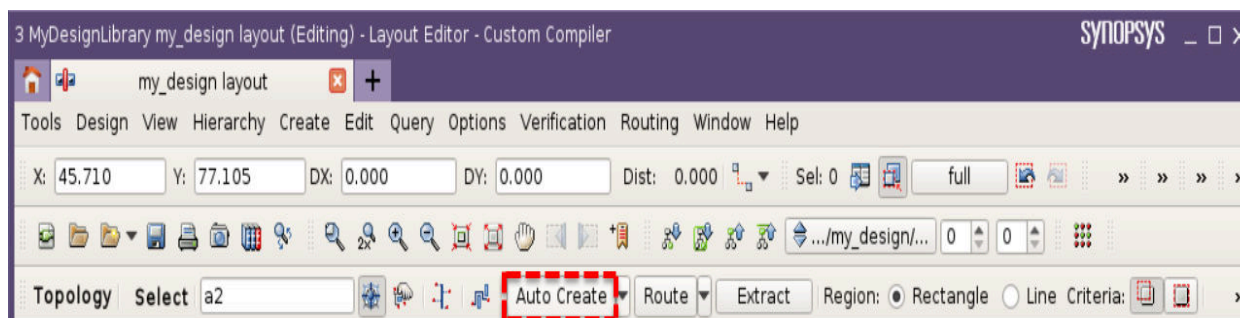
- **Incremental Mode:** Creates partial topologies and lets you complete the topology for the entire net incrementally. Select this option if the net is large and complex.

Tcl preference: `rtTopologyIncrementalRoute`

- **Connect Precisely:** Connects selected pins or existing topologies using the freeform structure. See [Route Topology Components](#). This option is only available if you select **Incremental Mode**.
- **Include Standard Cell Nets:** Creates topologies for nets that connect to standard cell pins.

Tcl preference: `rtTopologyIncludeStdCellNets`

3. On the **Topology** toolbar, click **Auto Create**.



Route topologies are created for the selected nets.

4. Review the generated topologies and adjust them as needed. See [Editing Topologies](#).

Viewing the Topology Generation Report

After route topologies are generated, a summary report appears in the Console. The report presents the following information:

Report Section	Description
Auto Create Topology Report	• Nets required: Number of nets selected for automatic topology generation. • Skipped: Number of nets that were skipped because they do not support the topology operation. • Attempted: Number of nets for which topology generation were attempted. • Success: Number of nets for which topologies were successfully generated. • Fail: Number of nets that do not have topologies because the operation failed.
Skipped Nets	List of nets that were skipped and the reasons for skipping them.
Fail Nets	List of nets for which the automatic topology creation failed.

The following example summary report shows 11 nets selected for automatic topology generation. **Auto Create** skipped 2 nets because one net was already fully routed, and the other net is a supply net. **Auto Create** made attempts to create topologies for 9 nets total. 8 topologies were created successfully, but the attempt failed for the in<3> net.

```
*****Auto Create Topology Report*****
* Nets Required Skipped Attempted Success Fail *
*          11      2          9      8      1 *
*****
* Skipped Nets:
* net160: Is already fully routed.
* in<5>: Supply net.
*****
* Fail Nets:
* in<3>
*****
```

Creating Topologies from Existing Routes

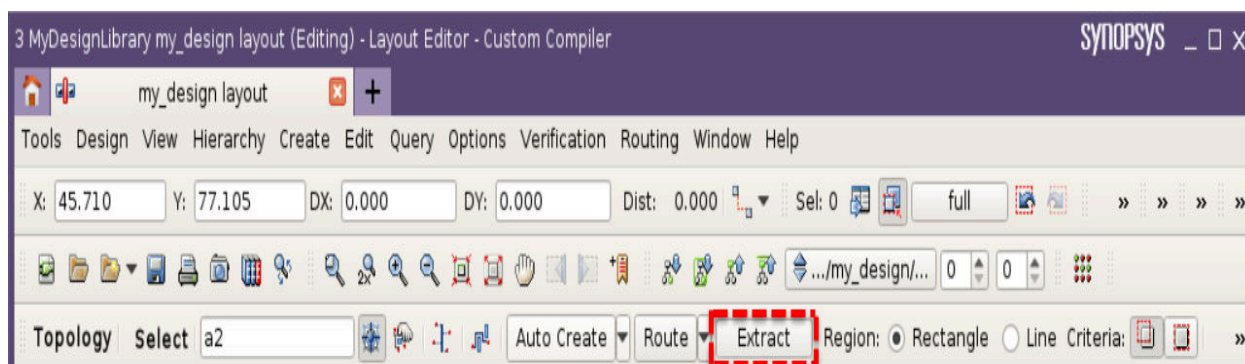
You can create new topologies by extracting them from existing routes. This method can be especially useful for layout ECOs. For example:

- In placement ECOs, you can extract topologies from existing routes, delete the routes, move blocks, and regenerate new routes from the topologies.
- In route ECOs, you can extract topologies from existing routes, delete the existing routes of specific nets, adjust the topologies, and regenerate the routes for those nets from the modified topologies.

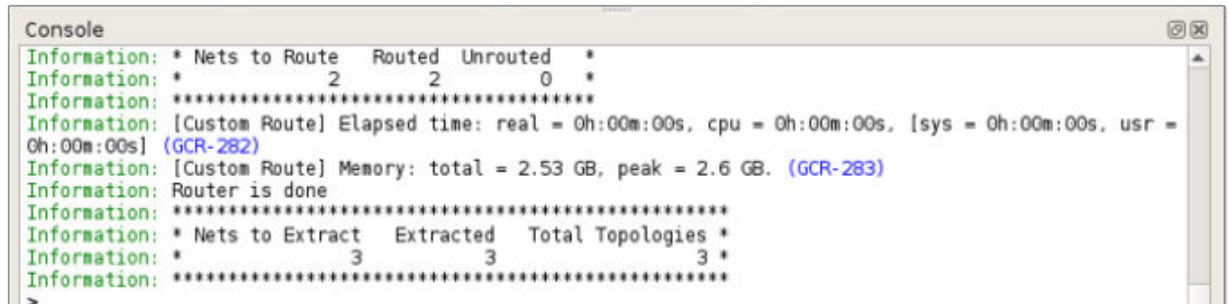
The extraction creates freeform topologies. If the existing routes were generated from trunk topologies, the extraction process creates the equivalent freeform topologies.

To extract topologies from existing routes:

1. In the Layout Editor, display the Design Navigator.
2. On the Design Navigator **Nets** tab, select the nets.
3. On the **Topology** toolbar, click **Extract**. The topology for the selected nets are extracted.



In the Console, a report shows the number of nets that were selected and the number of topologies that were extracted. Typically, the number of nets and the number of topologies extracted are the same.

A screenshot of a console window titled "Console" with a standard Windows-style title bar (minimize, maximize, close buttons). The window contains green text on a black background. The text shows routing statistics: "Information: * Nets to Route Routed Unrouted *", "Information: * 2 2 0 *", "Information: *****", "Information: [Custom Route] Elapsed time: real = 0h:00m:00s, cpu = 0h:00m:00s, [sys = 0h:00m:00s, usr = 0h:00m:00s] (GCR-282)", "Information: [Custom Route] Memory: total = 2.53 GB, peak = 2.6 GB. (GCR-283)", "Information: Router is done", "Information: *****", "Information: * Nets to Extract Extracted Total Topologies *", "Information: * 3 3 3 *", "Information: *****", and a prompt ">".

```
Console
Information: * Nets to Route  Routed  Unrouted  *
Information: *          2      2      0      *
Information: *****
Information: [Custom Route] Elapsed time: real = 0h:00m:00s, cpu = 0h:00m:00s, [sys = 0h:00m:00s, usr =
0h:00m:00s] (GCR-282)
Information: [Custom Route] Memory: total = 2.53 GB, peak = 2.6 GB. (GCR-283)
Information: Router is done
Information: *****
Information: * Nets to Extract  Extracted  Total Topologies *
Information: *          3      3      3      *
Information: *****
>
```

Setting Topology Segment Properties

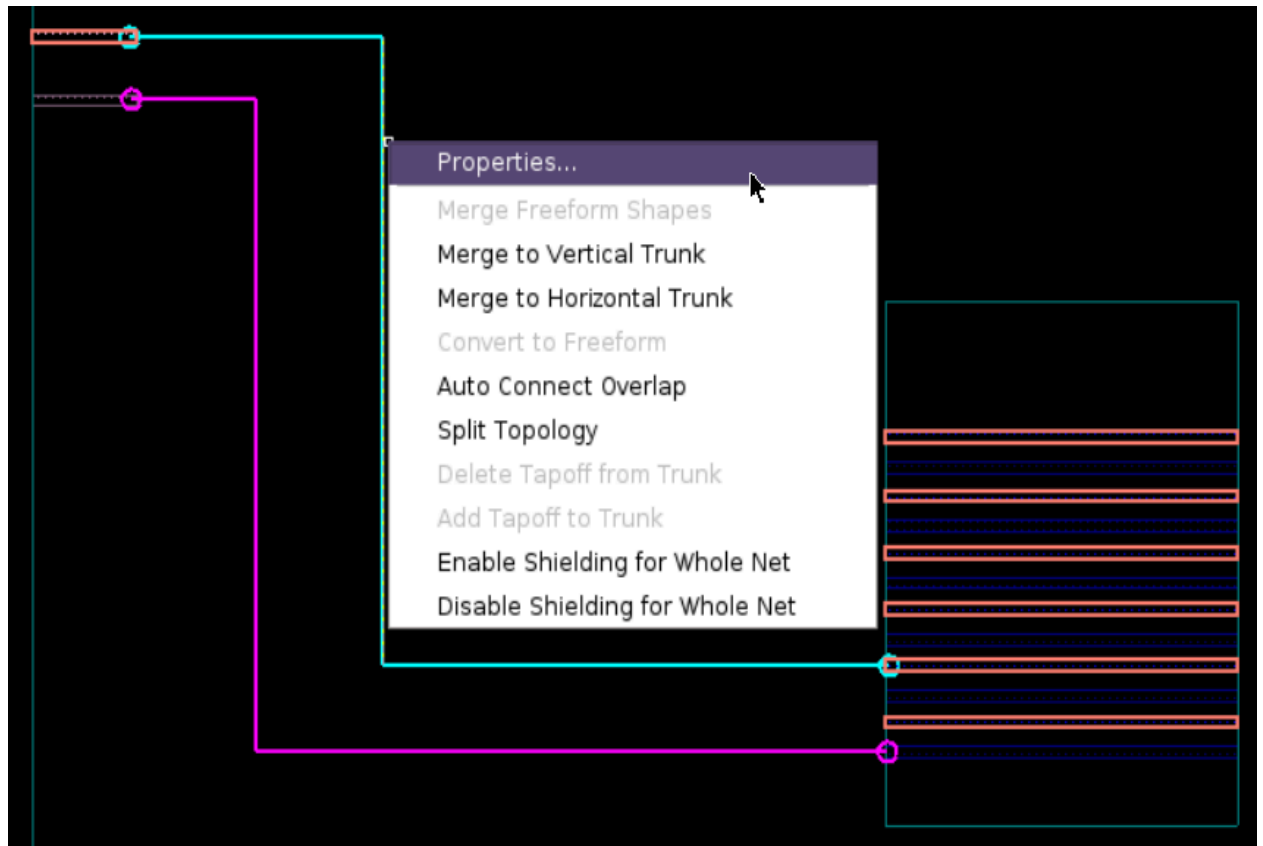
After you create route topologies, you can associate the following topology routing properties with trunk and freeform topology segments: layer, width, and shielding. The Auto Router will use the information during routing.

Note:

You cannot set the properties for trunk topology tapoff segments.

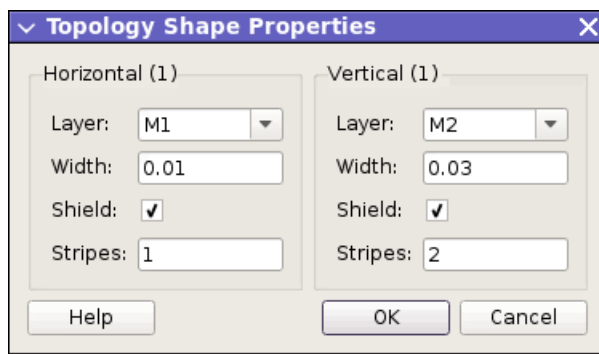
To set topology routing properties:

1. On the canvas, select one or more topology segments.
2. Right-click and choose **Properties**.



The **Topology Shape Properties** dialog box opens. The topology properties are displayed by orientation (Horizontal and Vertical). The number of topology segments selected are also shown.

In the following example, one horizontal and one vertical segment were selected.



3. Set the following options for each segment:

Property	Description
Layer	The layer to use for the route segment. This setting overrides the Layer setting in the Auto Route dialog box. If you do not specify a layer, the Auto Router will use the default layer and width defined by the technology or design-specific layer constraint.
Width	The width of the route segment. This setting overrides the layer Width setting in the Auto Route dialog box. Set the Layer property before setting the Width property. Setting Width to 0 uses the default width, which is one of the following: ◦ Largest of the following width values: technology <code>minWidth</code>, or non-default rule (NDR) minimum width. ◦ If you specify Stripes, Width is the minimum sum of the stripe widths. ◦ If the topology segment is snapping to routing tracks, Width inherits the track width.
Shield	Applies the Net Shielding Constraint to the route segment. For other shielding methods, see Enabling or Disabling Topology-Based Shielding.
Stripes	Creates the specified number of stripes in the route segment. For example, if you specify 2 Stripes, the Topology Router will create a route segment that consists of two parallel stripes. The maximum number of stripes you can specify is controlled by the <code>rtMaxTopologyStripeNum</code> Tcl preference. Stripes does not apply to segments that connect to pins.

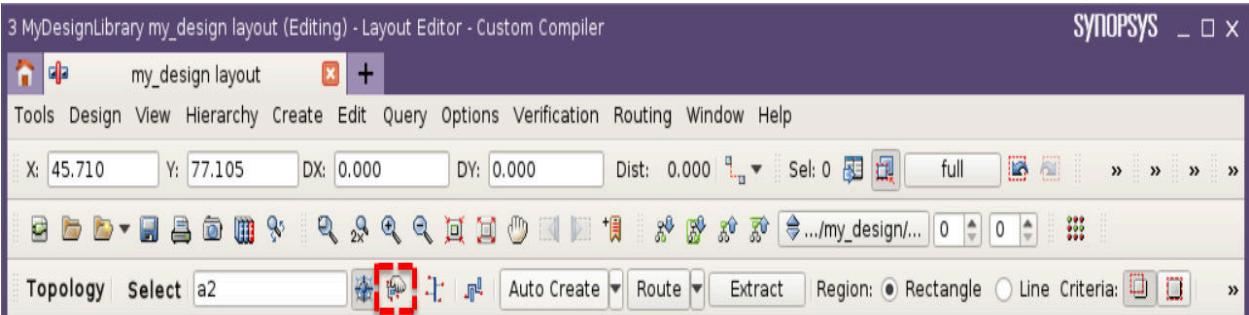
4. Click **OK** to save the properties.

Setting Topology Viewing Options

You use the Topology Assistant to show or hide topology guidelines. When the topology guidelines are displayed, you can use the assistant to change the colors.

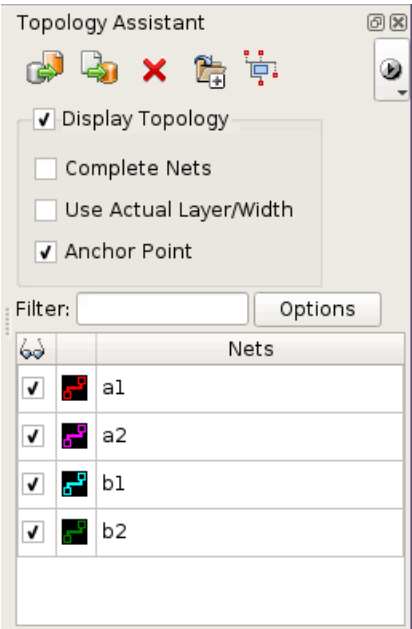
To set the topology viewing options:

1. On the **Topology** toolbar, click the **Topology Assistant** button.



Alternatively, in the Layout Editor, choose **Window > Assistants > Topology Assistant**.

The Topology Assistant appears on the right side of the Layout Editor window.



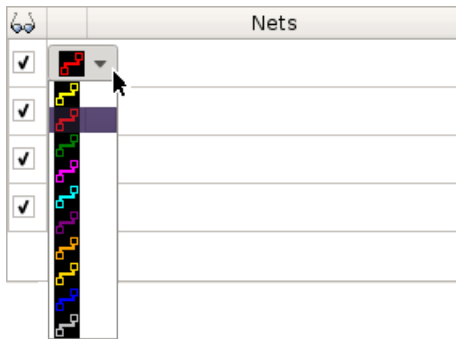
2. Set the following **Display Topology** options:

Option	Description
Display Topology	Displays or hides the topology guidelines on the layout canvas for open nets only. Tcl preference: rtTopologyDisplay

Option	Description
Complete Nets	Displays the topology guidelines for both open and complete nets. Tcl preference: <code>rtTopologyDisplayCompleteNet</code>
Use Actual Layer/Width	Draws topology using the display packet and width of the assigned layer. If the topology does not have an assigned layer, the topology guideline will still be drawn. Tcl preference: <code>rtTopologyUseActualLayer</code>
Anchor Point	Displays or hides the topology anchor points. See Route Topology Components . Tcl preference: <code>rtTopologyDisplayAnchorPoint</code>

- In the **Nets** table, select **Visibility**  if you want to display the specific net topologies. To hide the topologies, clear the **Visibility**  check box.
- To change a topology guideline color, click the guideline color in the **Nets** table, and choose a different color.

By default, the topology guideline colors are derived from the flight line display packets. You can override the default by selecting a different color in the Topology Assistant.



- To return to the default guideline colors, select the nets, click , and choose **Reset Topology Color**.

By default, when flight lines are turned on in the Design Navigator, topology guidelines are displayed using the same display packets. For example, if net_A flight lines are red, the topology guidelines for net_A are also red.

If flight lines are turned off, the topology guidelines will cycle through the color list per net. To change the list of colors, in the Display Resource Editor, find the following packet names and change the packet properties:

Packet Name	Description
routeTapoff	Color and line style of the tapoff component.
routeTrunk	Color and line style of the trunk or free-style guidelines.

Editing Topologies

After you create route topologies, you can edit them. For example, you can move segments, merge segments and trunks, convert topology styles, add trunk tapoffs, and delete trunk tapoffs.

This topic describes the following:

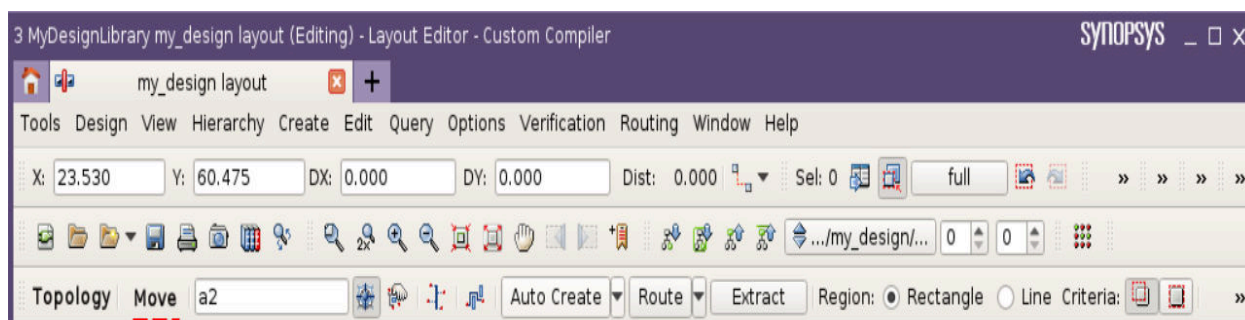
- [Moving Topologies](#)
- [Moving Starting or Ending Locations](#)
- [Merging Freeform Topology Segments](#)
- [Merging Trunks](#)
- [Converting Freeform Segments to Trunks](#)
- [Converting Trunk Topologies to Freeform Topologies](#)
- [Splitting Topologies](#)
- [Adding Trunk Tapoffs](#)
- [Deleting Trunk Tapoffs](#)

Moving Topologies

You can move an entire route topology, a component of the topology, or multiple topologies together. The topology components can be on different nets.

To move a topology:

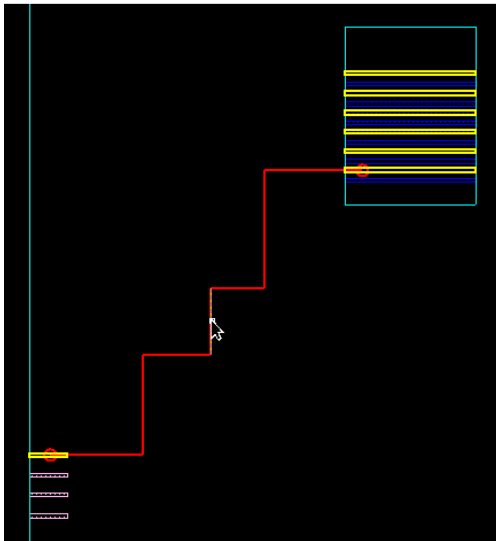
1. With the **Topology** command still active, press M. On the **Topology** toolbar, the status changes to **Move**.



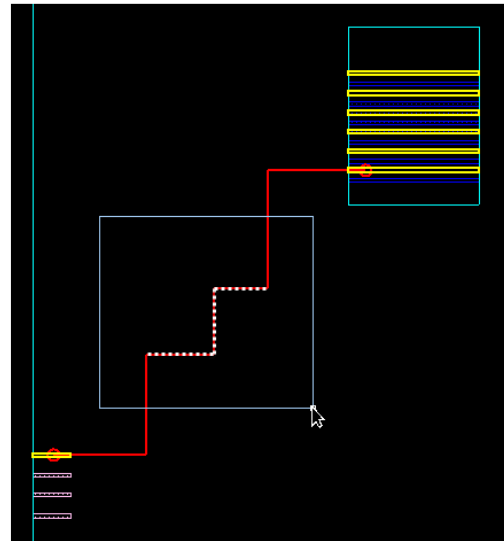
2. On the canvas, use one of the following methods to select the topology components you want to move.

- Select a single topology segment.
- Area-select multiple topology segments.
- Right-click the canvas and choose **Select All Topology Shapes** to select all topology shapes in the design.

Select a topology segment



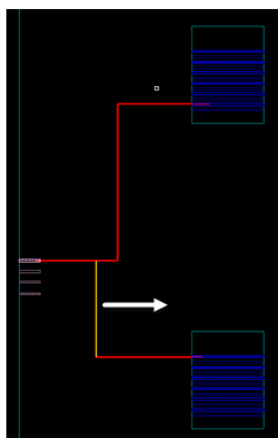
Area-select multiple segments



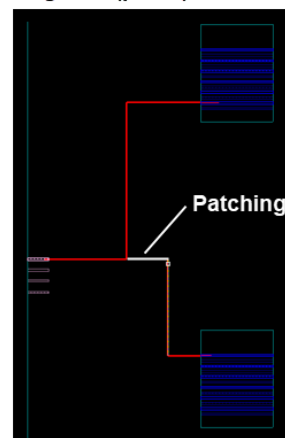
3. Move the selected topology components to the target location. The topology Move command operates similarly to the Layout Editor Move command and supports preselection and postselection. See Preselecting and Postselecting Objects.

During the move operation, segments might be extended or created (patched).

Move right



Segment (patch) created



4. When the topology or topology component is at its target location, click the canvas to commit the modification.
5. Press Esc to return to the Select mode. Press Esc again to exit the **Topology** command.

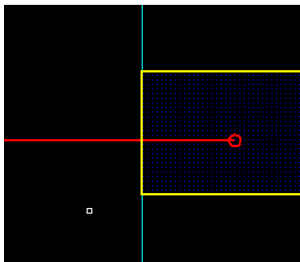
Moving Starting or Ending Locations

You can change the starting or ending locations of the topology.

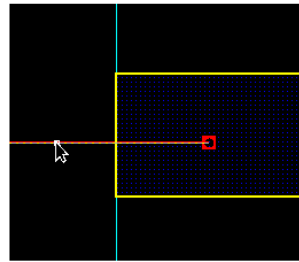
To move the starting or ending locations of the route topology:

1. On the canvas, select the segment that connects to the starting or ending topology shape. If the shape is on a pin, another topology shape, or existing route, a square highlighting shape appears.

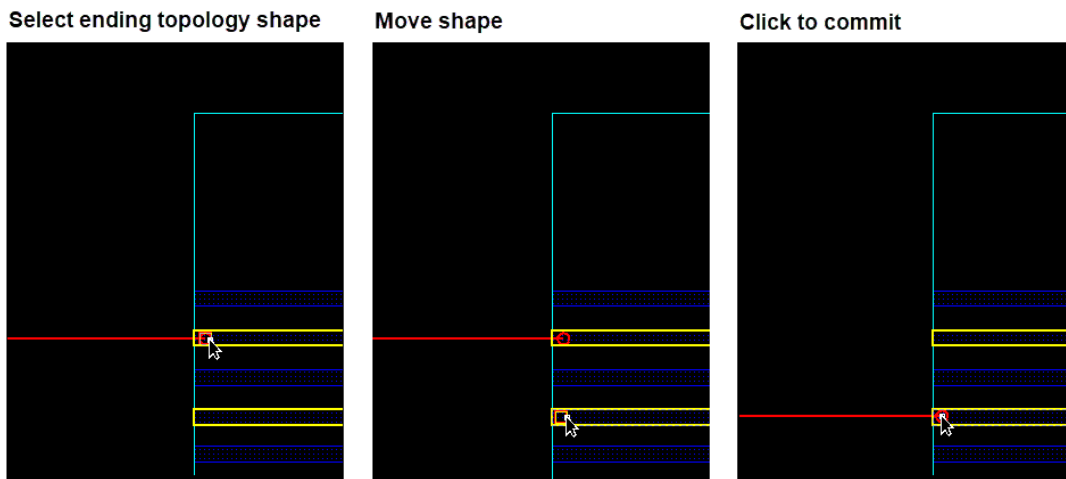
Before selecting segment



After selecting segment



2. Select the square shape.
3. Press M, click the square shape, move it to a same-net target pin, topology segment, or existing route and click to commit the change. The topology line moves and reconnects to the starting or ending shape.



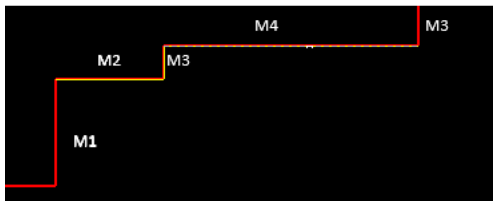
4. Press Esc to return to the Select mode.

Merging Freeform Topology Segments

You can merge segments of a freeform topology. Doing so can reduce the length of the resulting routes or simplify the route path, thus improving design performance.

In the following example, merging the two horizontal guidelines (M2 and M4) reduces the number of layers that must be traversed between the M1 and M3 vertical guidelines.

Original topology



Merge M2 and M4



For the merge operation to be successful, the following requirements must be met:

- The components must be from the same freeform topology.
- The topology components must be continuous.
- You must select at least two components for merging.

When merging freeform topologies:

- If the merging components have the same topology properties, the merged shapes will still have the same topology properties.
- If the merging components have different topology properties, the best fitting rule for the merged topology will be used.

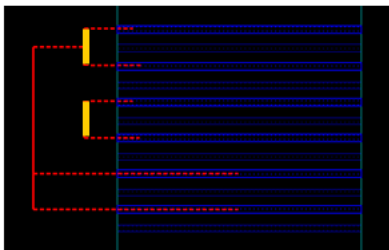
To merge components within freeform topologies:

1. On the canvas, select the two components that you want to merge.
2. Right-click and choose **Merge Freeform Shapes** The selected components are merge.

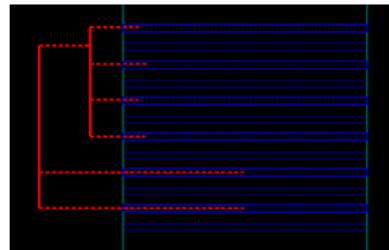
Merging Trunks

You can use the **Merge to Horizontal Trunk** and **Merge to Vertical Trunk** commands to merge multiple trunks into a single trunk. The merged trunk inherits the properties from one of the trunks. You can also use the commands to rotate single trunks.

Before



After



To merge or rotate trunks:

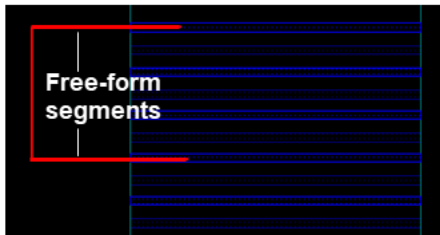
1. On the canvas, select the trunks you want to merge or rotate.
2. Right-click and choose Merge to **Merge to Horizontal Trunk** or **Merge to Vertical Trunk**.

Converting Freeform Segments to Trunks

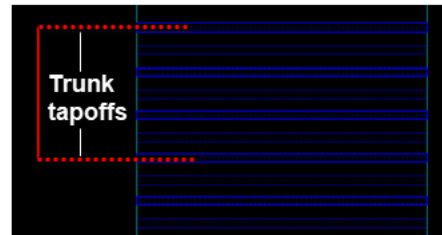
You can use the **Merge to Horizontal Trunk** and **Merge to Vertical Trunk** commands to convert selected freeform topologies into a trunk topology.

- If a freeform topology is selected, the entire topology is converted to a trunk-style topology.
- The trunk properties are determined by the selected topology.
- If the topology is connected to other topologies, the connections will be retained after the merge.

Before



After



To convert freeform topologies to trunk-style topologies:

1. On the canvas, select the topology segments you want to merge.
2. Right-click and choose Merge to **Merge to Horizontal Trunk** or **Merge to Vertical Trunk**.

Converting Trunk Topologies to Freeform Topologies

The **Convert to Freeform** command converts selected trunk topology segments to freeform topology segments. This command is useful when you have an existing trunk topology structure, and you want to apply different shielding to individual tapoff segments.

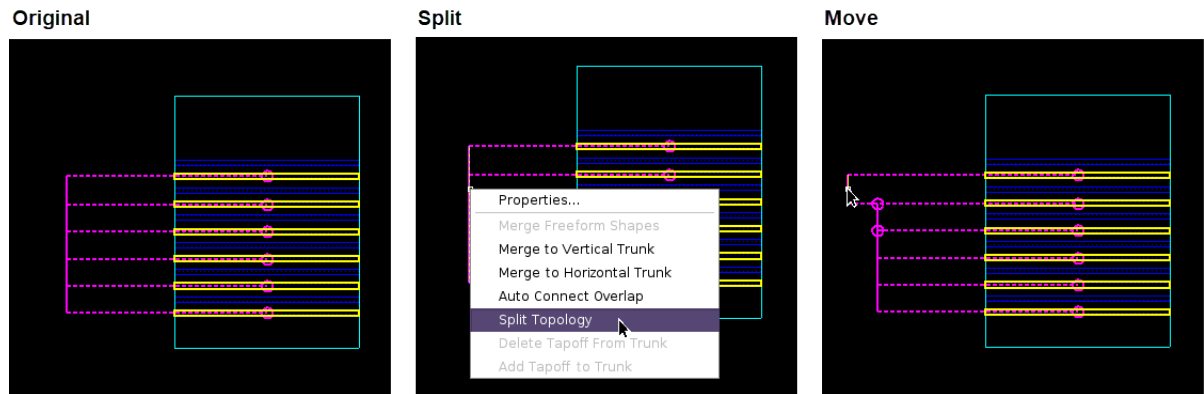
To convert trunk-style topologies to freeform topology segments:

1. On the canvas, select the trunk-style topology segments you want to convert:
 - Select a trunk-style segment to convert the segment and all connected tapoffs.
 - Select individual tapoffs to convert only the selected tapoff segments.
2. Right-click and choose **Convert to Freeform**.

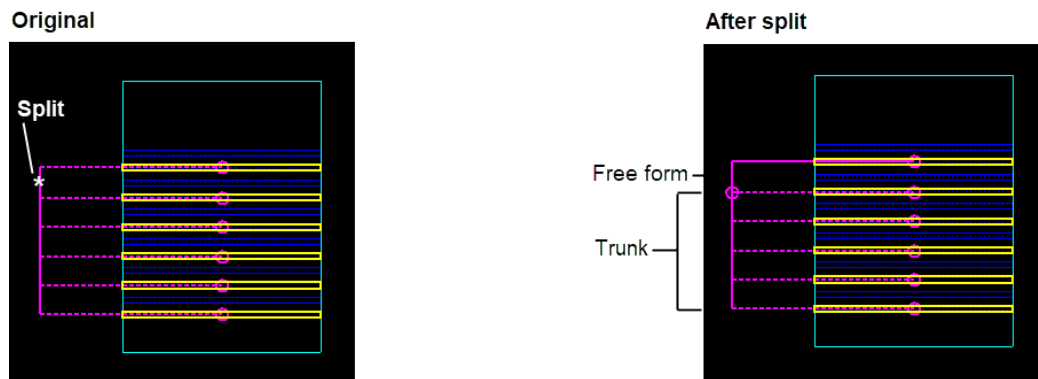
Splitting Topologies

You can use the **Split Topology** command to split a topology segment or trunk into multiple segments. Splitting topology segments allows you to change the geometry of the topology.

After splitting a topology, you can move the independent segments or components to manipulate the structure.



Depending on where the split occurs, the resulting topology can consist of trunk and freeform styles. The following example shows that the split caused a pin to be isolated. Because trunk-style topologies require two or more pins, the single-pin isolation produced a freeform style in the resulting topology.



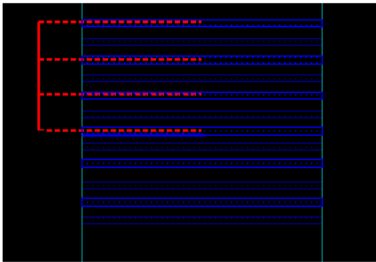
To split a topology segment:

1. On the canvas, select the segment you want to split.
2. Position the pointer at the location where you want the split to occur, right-click, and choose **Split Topology**.

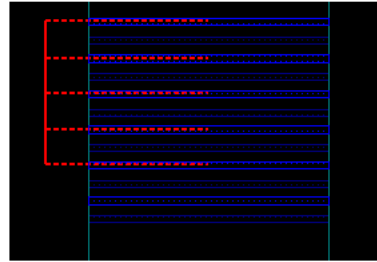
Adding Trunk Tapoffs

You can use the **Add Tapoffs** command to extend an existing trunk topology segment and add an additional tapoff.

Before



After



To add a tapoff to a trunk:

1. On the canvas, select the trunk and the pin you want to connect to.
2. Right-click, and choose **Add Tapoff**.

A tapoff is added to the topology.

Deleting Trunk Tapoffs

You can use the **Delete Tapoffs** command to extend an existing trunk topology segment and add an additional tapoff.

To delete a tapoff from a trunk:

1. On the canvas, select the tapoff you want to delete.
2. Right-click, and choose **Delete Tapoff**.

Deleting Topologies

You can delete specific topologies, or you can delete all the topologies in the design.

Use one of the following methods to delete route topologies:

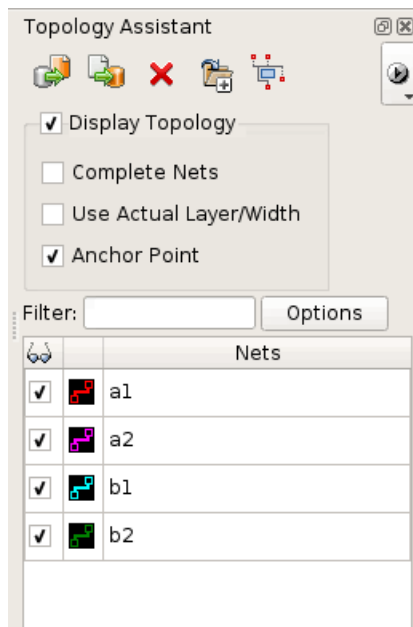
- [Deleting Topologies Using the Topology Assistant](#)
- [Deleting Selected Topologies on the Layout Canvas](#)

Deleting Topologies Using the Topology Assistant

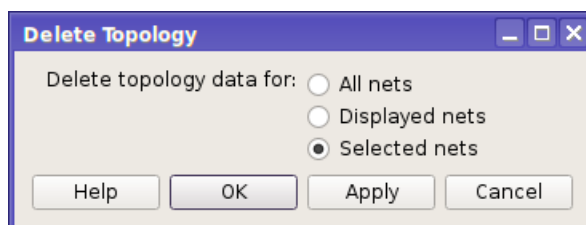
The Topology Assistant lists all nets that have topologies in the design. You can use the assistant to delete, import, or export topologies.

To delete topologies using the Topology Assistant:

1. In the Layout Editor, choose **Window > Assistants > Topology Assistant**. The Topology Assistant appears on the right side of the Layout Editor window.



2. In the Topology Assistant, select the nets that have the topologies you want to delete.
3. Click **Delete Topology** . The **Delete Topology** dialog box opens.



4. Select one of the following:
 - **All nets:** Deletes all topologies in the design.
 - **Displayed nets:** Deletes the topologies that are currently displayed.
 - **Selected nets:** Deletes the topologies of the selected nets.
5. Click **OK**.

Deleting Selected Topologies on the Layout Canvas

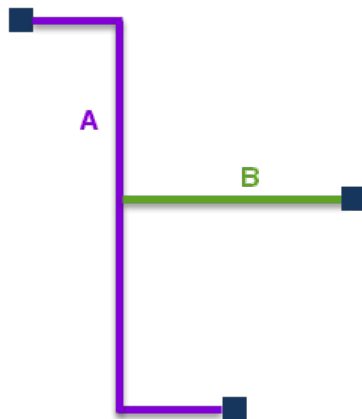
Instead of using the Topology Assistant to delete topologies, you can delete topologies directly from the layout canvas.

To delete selected topologies on the layout canvas:

1. On the layout canvas, select the topologies you want to delete.
2. Press Delete.

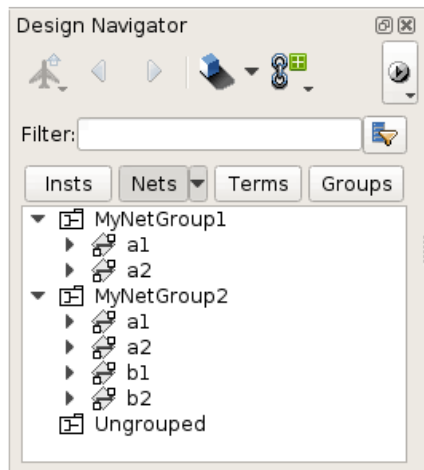
The selected topologies are deleted. If other topologies are dependent on the deleted topology, the other topologies are deleted also.

In the following example, topology B depends topology A. If you delete A, then B is also deleted. However, if you delete B, A remains.



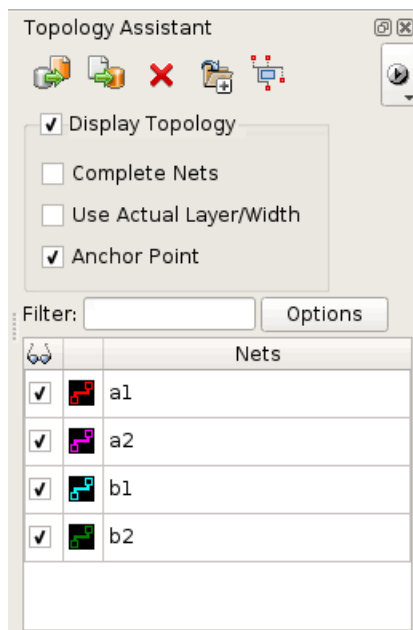
Grouping Nets That Have Topologies

You can group nets that have topologies so that they appear together in the Design Navigator. Grouping the nets allows you to easily locate topologies of interest.

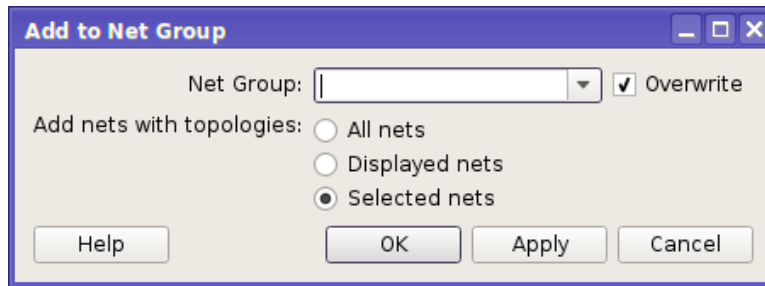


To group nets that have topologies:

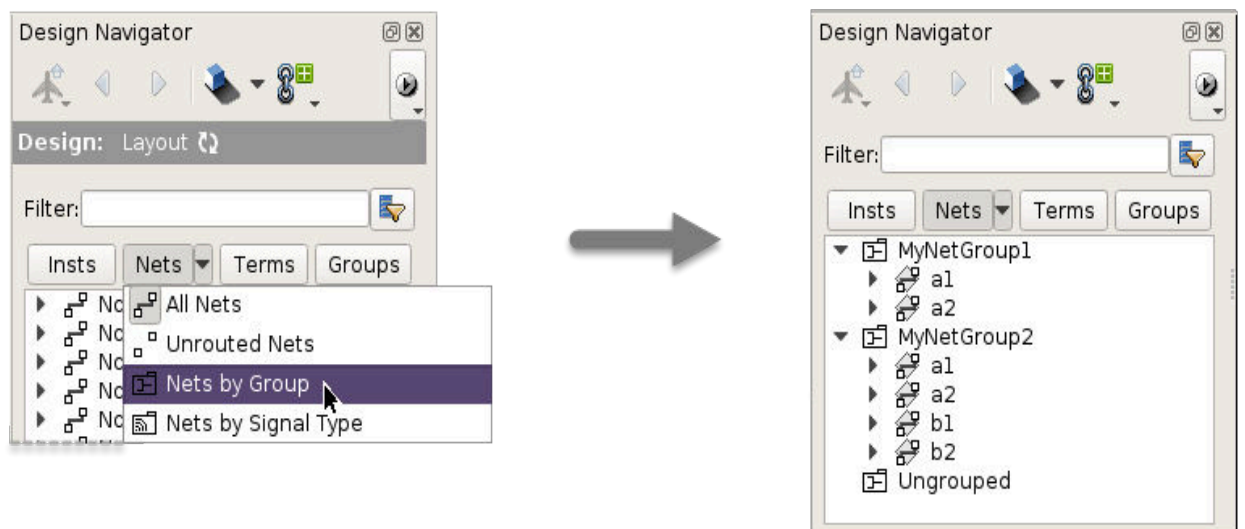
1. In the Layout Editor, choose **Window > Assistants > Topology Assistant**. The Topology Assistant appears on the right side of the Layout Editor window.



2. In the Topology Assistant, select the nets that you want to group, and click **Add to Net Group** . The **Add to Net Group** dialog box opens.



3. In the **Add to Net Group** dialog box:
 - a. Choose an existing **Net Group**, or type a **Net Group** name to create a new net group.
 - b. Select the nets to want to add to the net group:
 - **All nets**: Adds all nets that have topologies to the net group.
 - **Displayed nets**: Adds the nets that are currently displayed to the net group.
 - **Selected nets**: Adds the selected nets to the net group.
4. (Existing net group only) Select **Overwrite** if you want to overwrite the existing nets in the net group. If you do not select the option, the specified nets will be appended to the existing list.
5. Click **OK**.
6. To view the grouped nets, in the Design Navigator, click **Nets**, and choose **Nets by Group**. The Nets tab refreshes and displays the net groups.



Estimating Route Congestion

Because the Congestion Checker considers route topologies, you can use the checker to estimate potential route congestions in the design. For instructions, see [Checking Route Congestion](#).

Enabling or Disabling Topology-Based Shielding

By default, topology-based shielding is turned on for trunk-style, freeform, and automatic topology creation so that the Auto Router adds shielding during route creation. You have the option of disabling or re-enabling topology-based shielding.

This topic describes the ways in which you can enable or disable topology-based shielding:

- [Setting Default Topology-Based Shielding Behavior](#)
- [Enabling or Disabling Shielding for All Topologies on a Net](#)
- [Enabling or Disabling Topology Segment Shielding](#)
- [Enabling or Disabling Shielding for Trunk Topology Tapoffs](#)

Setting Default Topology-Based Shielding Behavior

Because topology-based shielding is turned on by default, shielding is added for every segment in each topology you create. You can control the default behavior using the `rtTopologyShieldDefault` Tcl preference.

To set the default topology-based shielding behavior:

- In the Custom Compiler Console Tab or Console Assistant, type the following, and then press Enter:

```
> db::setPrefValue rtTopologyShieldDefault -value <true|false> -scope [de::getActiveContext]
```

where:

- `true`: Turns on the **Shield** property for all segments of a topology by default.
- `false`: Turns off the **Shield** property for all segments of a topology by default.

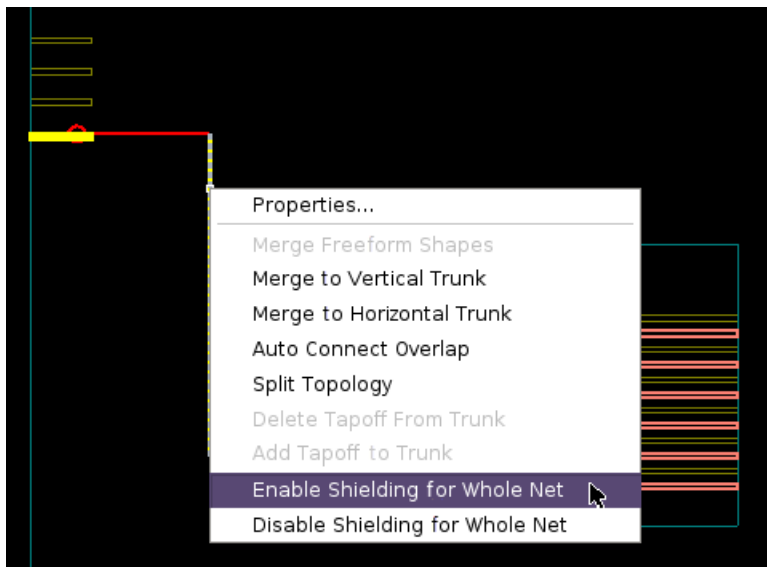
For information about the **Shield** property, see [Setting Topology Segment Properties](#).

Enabling or Disabling Shielding for All Topologies on a Net

Instead of editing shield settings for individual topology segments, you can enable or disable shielding for all topologies on a net.

To enable or disable shielding for all topologies on a net:

1. On the canvas, select one of the topology segments on the net.
2. Right-click and choose **Enable Shielding for Whole Net** or **Disable Shielding for Whole Net**.



If the command is unavailable, check to see if the shielding constraints are defined for the net or nets.

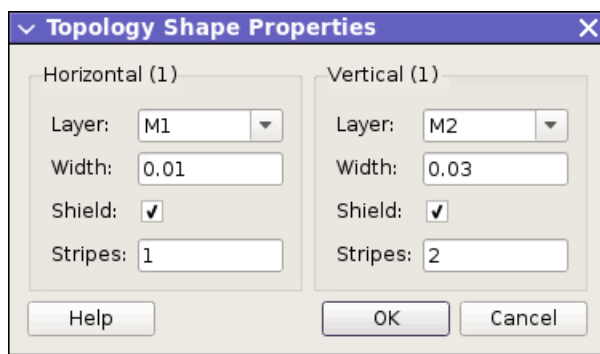
Enabling or Disabling Topology Segment Shielding

After a topology is created, you have the option of editing segment properties, including whether shielding should be created for the selected segments during routing. You use this method to add shielding to freeform and trunk-style topology segments.

You cannot use this method to add shielding to trunk-style tapoffs. To add shielding to trunk-style tapoffs, see [Enabling or Disabling Shielding for Trunk Topology Tapoffs](#).

To enable or disable the shield settings for selected topology segments:

1. Follow the instructions in [Setting Topology Segment Properties](#).
2. In the **Topology Shape Properties** dialog box, select or clear the **Shield** check box.

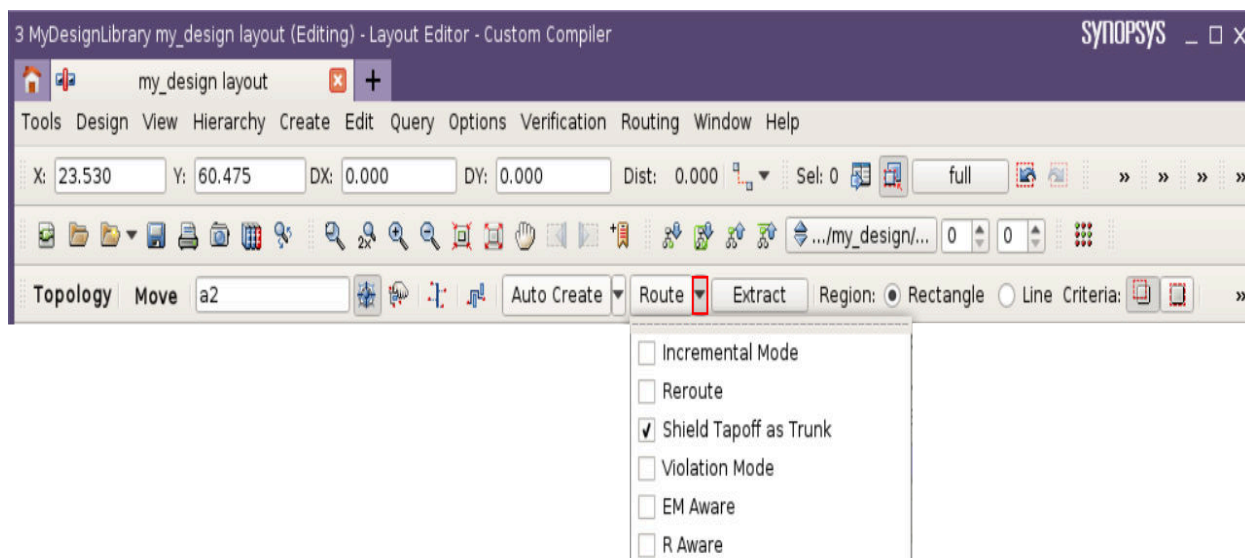


Enabling or Disabling Shielding for Trunk Topology Tapoffs

Enabling shielding on all nets and on topology segments does not add shielding to tapoffs in trunk-style topologies. Instead, you enable trunk tapoff shielding through the **Route** menu on the **Topology** toolbar.

To enable or disable the shielding for trunk topology tapoffs:

- On the **Topology** toolbar, choose **Route > Shield Tapoff as Trunk**.



Generating Routes Based on Topologies

After you specify the route topology settings, you can generate the routes.

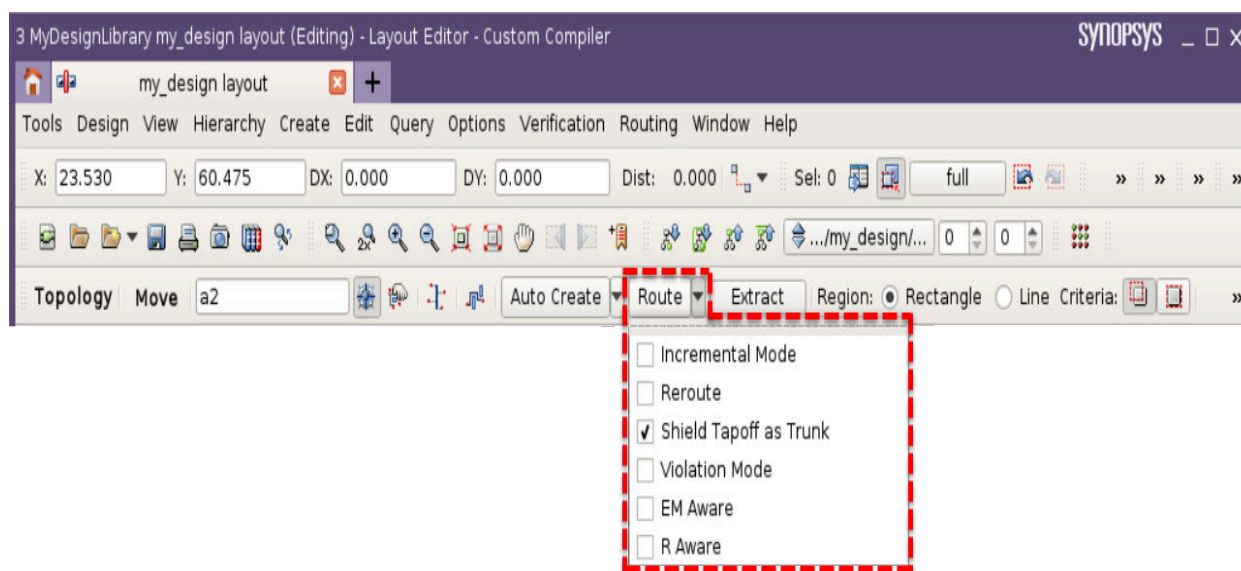
[Before You Start](#), review and complete the prerequisites for this task.

To generate routes for one or more nets:

1. On the Design Navigator **Nets** tab or the [Topology Assistant Nets](#) table, select one or more nets.

If you are routing a single net, instead of selecting the net from the Design Navigator, you can type the net name on the **Topology** toolbar.

2. On the **Topology** toolbar, click **Route**, and select the application options:



- **Incremental Mode:** Creates partial routes and lets you complete the route for the entire net incrementally. The Auto Router will not create routes for pins that do not have topologies. Select this option if the net is large and complex.

Tcl preference: `rtTopologyIncrementalRoute`

- **Reroute:** Removes existing route shapes and shielding shapes before creating new routes. This option becomes unavailable when you select **Incremental Mode**.

Tcl preference: `rtTopologyReroute`

- **Shield Tapoff as Trunk:** Adds shielding to trunk tapoff segments, applying the same shield settings used by the corresponding trunk. (To add shielding to tapoffs

in freeform topologies, set individual segment properties. See [Setting Topology Segment Properties](#).)

Tcl preference: `rtTopologyShieldTapoffAsTrunk`

- **Violation Mode:** Creates the routes even if design rule violations occur.

Tcl preference: `rtTopologyViolationMode`

- **EM Aware:** Runs the Electrical Reporter to check for electromigration (EM) violations. If EM violations are detected, the width of the route segments will be increased to avoid the violations.

Tcl preference: `rtTopologyEMAware`

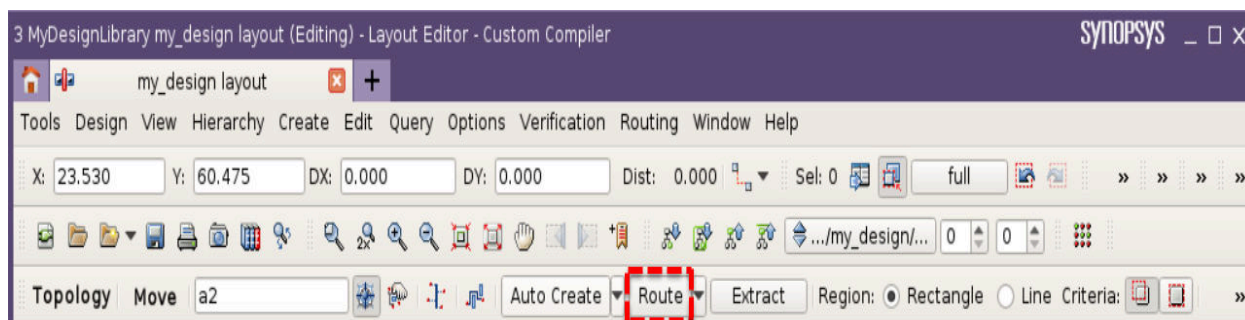
- **R Aware:** Runs the Electrical Reporter to check for pin-to-pin maximum resistance violations. If resistance constraint violations are detected, the width of the route segments will be increased to comply with the constraint.

Tcl preference: `rtTopologyRAware`

You can choose which flow to set in the **Main** tab and then specify a `.nxtgrd` file in the **Nxtgrd** tab and a **TLU+** file in the **TLU+** tab. See the following topics for more information:

- To set pin-to-pin resistance constraint, see [Defining Pin-to-Pin Resistance Constraints](#).
- To set `sheetResistance`, see [Setting EM, Resistance, and Capacitance Options](#)

3. On the **Topology** toolbar, click **Route**.



The Auto Router initializes and creates routes based on the topologies in the design. The Auto Router options and the electrical constraints you set are applied during routing.

Importing and Exporting Topologies

You can import and export route topologies for backup and recovery. You can also export existing topologies, modify the net and terminal names, and apply the topologies to different nets in another design.

This topic describes the following:

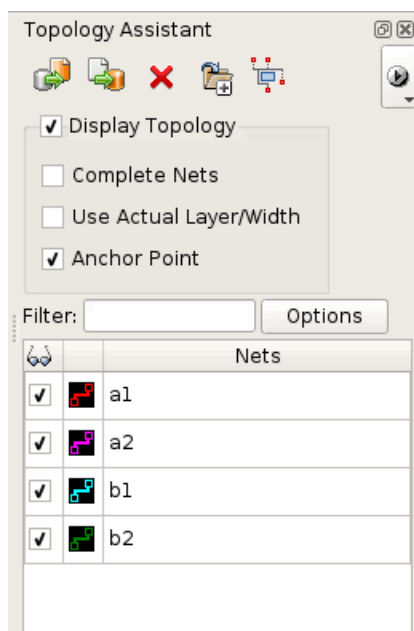
- [Importing Route Topologies](#)
- [Exporting Route Topologies](#)

See [Modifying Topology Files for Reuse](#) if you want to reuse the exported topologies in different designs.

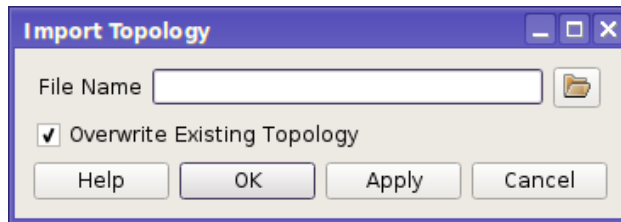
Importing Route Topologies

To import route topologies from a file:

1. In the Layout Editor, choose **Window > Assistants > Topology Assistant**. The Topology Assistant appears on the right side of the Layout Editor window.



2. On the Topology Assistant toolbar, click **Import Topology** . The Import Topology dialog box opens.

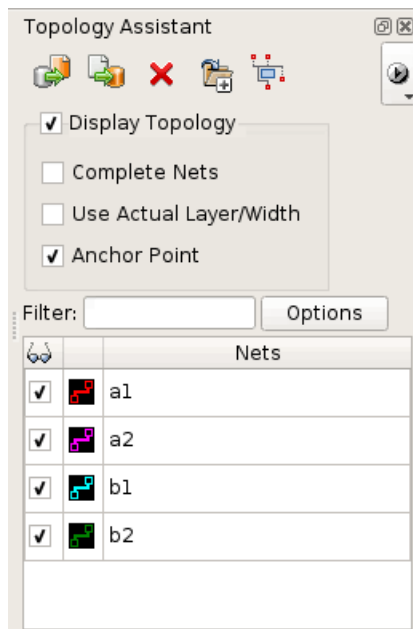


3. In the **Import Topology** dialog box:
 - a. In the **File Name** box, specify the topology file.
 - b. Select **Overwrite Existing Topology** if you want to overwrite existing topologies in the design.
 - c. Click **OK**.

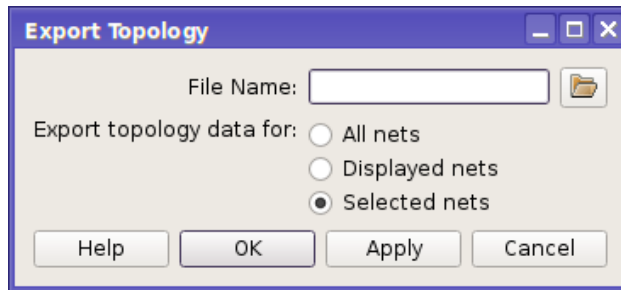
Exporting Route Topologies

To export route topologies to a file:

1. In the Layout Editor, choose **Window > Assistants > Topology Assistant**. The Topology Assistant appears on the right side of the Layout Editor window.



2. On the Topology Assistant toolbar, click **Export Topology** . The Import Topology dialog box opens.



3. In the **Export Topology** dialog box:
 - a. In the **File Name** box, specify the name of the topology file and where you want to store it.
 - b. Select the topology data to export:
 - **All nets**: Exports all nets that have topology data.
 - **Displayed nets**: Exports currently displayed nets.
 - **Selected nets**: Exports selected nets.
 - c. Click **OK**. The topology information is exported. See [Modifying Topology Files for Reuse](#).

Reusing Existing Topologies in Different Designs

After exporting the route topology information, you can modify the net, terminal, and pin information so that the same topologies can be applied to different nets and pins in the same or different design.

This topic describes the following:

- [About Topology Files](#)
- [Modifying Topology Files for Reuse](#)

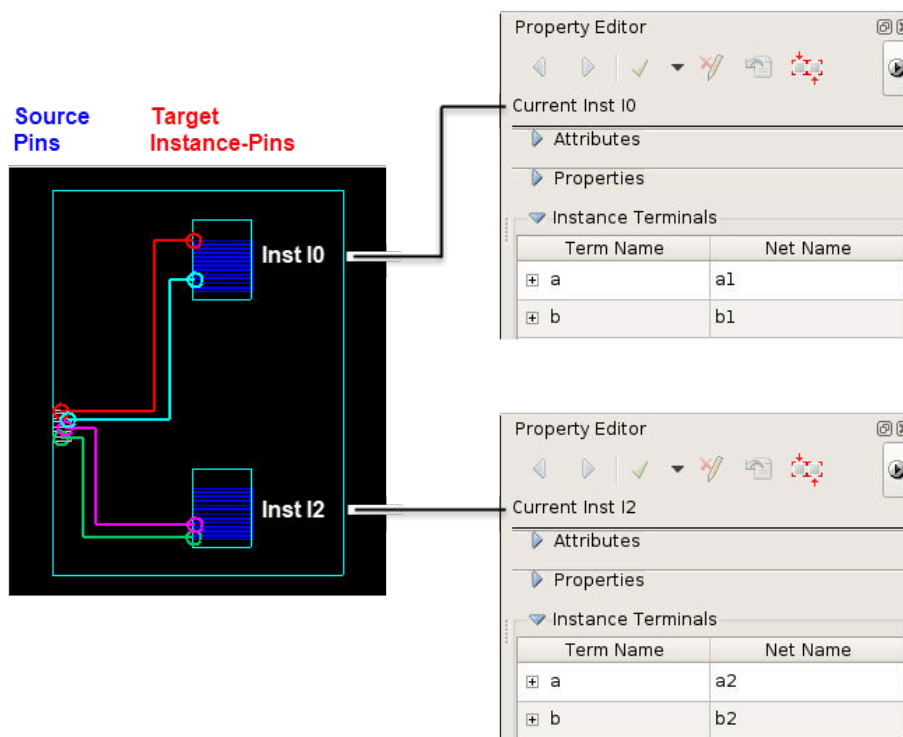
About Topology Files

A topology file is created when you export topologies. The file contains:

- Names of the nets that you selected for export.
- Tcl commands used to create the topologies. The commands indicate:
 - The topology style: freeform or trunk.
 - The source and target terminals or pins.
 - The direction and offset of the topology components.

For details, see `rt::createFreeFormTopology` and `rt::createTrunkTopology`.

In the following example, topologies were created for four nets: a1, b1, a2, and b2. Each topology connects a pair of pins. You can view information about the nets, terminals, and pins in the Property Editor.



The exported file for this example is as follows. In this example, all four topologies are freeform.

Source Pins

Target Instance-Pins

```
# Net: a1
set topology0 [rt::createFreeFormTopology -source [list
a1-a1 0 0.558000 0.050000] -target [list I0-a 0 0.014344
0.417500] -pattern {freeform horizontal 21915/46477
vertical 39855/40108 horizontal 8930/46477 ;} -net {a1}
-design [ed] ]
# Net: a2
set topology1 [rt::createFreeFormTopology -source [list
a2-a2 0 0.784250 0.682500] -target [list I2-a_mj4 0
0.040456 0.067500] -pattern {freeform horizontal
7005/46477 vertical -22505/22659 horizontal 23290/46477 ;}
-net {a2} -design [ed] ]
# Net: b1
set topology2 [rt::createFreeFormTopology -source [list
b1-b1 0 0.921750 0.565000] -target [list I0-b_mj3 0
0.034571 0.572500] -pattern {freeform horizontal
23885/46477 vertical 32655/33052 horizontal 5780/46477 ;}
-net {b1} -design [ed] ]
# Net: b2
set topology3 [rt::createFreeFormTopology -source [list
b2-b2 0 0.489250 0.020000] -target [list I2-b_mj4 0
0.023170 0.197500] -pattern {freeform horizontal
5430/46477 vertical -22980/23451 horizontal 25810/46477 ;}
-net {b2} -design [ed] ]
```

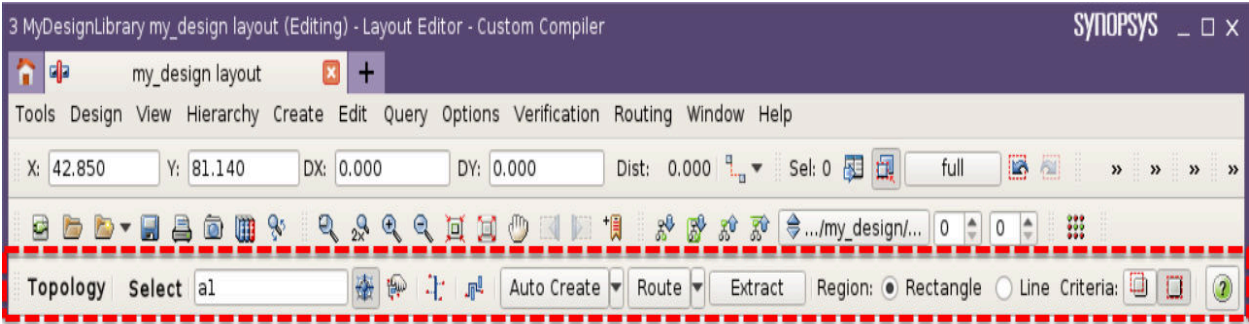
Modifying Topology Files for Reuse

To reuse the same topologies for different nets in the same design or in another design:

1. [Export the topologies](#) from the design.
2. Open the exported topology file in a text editor.
3. Change the net names, instance names, and pin names. See [About Topology Files](#).
4. [Import the modified topology file](#) in the same or different design.

Topology Toolbar Reference

This topic presents quick reference of the **Topology** toolbar. For instructions on using the Topology feature, see [Methods for Creating Route Topologies](#).



Option	Description
Select/Create/Move	Indicates the topology operation mode: • Select: You are selecting pins or topology components. • Create: You are creating Trunk or Freeform topologies. • Move: You are moving selected components. Press Esc at any time to return to the default mode (Select).
Net box	Lets you specify the names of the nets you selected. If you selected nets in the Design Navigator, the selected nets appear in the box. Tcl preference: rtTopologyNet
	Displays or hides the Design Navigator so that you can select the nets.
	Displays or hides the Topology Assistant .
	Creates trunk-style topology. See Route Topology Components .
	Lets you create freeform style topology. See Route Topology Components .
Auto Create	Automatically creates route topologies. Before you start, you can select the following options: • Incremental Mode: Creates partial topologies and lets you complete the topology for the entire net incrementally. Select this option if the net is large and complex. Tcl preference: rtTopologyIncrementalRoute • Connect Precisely: Connects selected pins or existing topologies using the freeform structure. See Route Topology Components. This option is only available if you select Incremental Mode. • Include Standard Cell Nets: Creates topologies for nets that connect to standard cell pins. Tcl preference: rtTopologyIncludeStdCellNets See Creating Route Topologies Automatically .

Option	Description
	Calls the Auto Router to create routes based on the topologies in the design. Before you start routing, you can select the following options: • Incremental Mode: Creates partial routes and lets you complete the route for the entire net incrementally. The Auto Router will not create routes for pins that do not have topologies. Select this option if the net is large and complex. Tcl preference: <code>rtTopologyIncrementalRoute</code> • Reroute: Removes existing route shapes and shielding shapes before creating new routes. This option becomes unavailable when you select Incremental Mode. Tcl preference: <code>rtTopologyReroute</code> • Shield Tapoff as Trunk: Adds shielding to trunk tapoff segments, applying the same shield settings used by the corresponding trunk. (To add shielding to tapoffs in freeform topologies, set individual segment properties. See Setting Topology Segment Properties.) Tcl preference: <code>rtTopologyShieldTapoffAsTrunk</code> • Violation Mode: Creates the routes even if design rule violations occur. Tcl preference: <code>rtTopologyViolationMode</code> • EM Aware: Runs the Electrical Reporter to check for electromigration (EM) violations. If EM violations are detected, the width of the route segments will be increased to avoid the violations. Tcl preference: <code>rtTopologyEMAware</code> • R Aware: Runs the Electrical Reporter to check for pin-to-pin maximum resistance violations. If resistance constraint violations are detected, the width of the route segments will be increased to comply with the constraint. Tcl preference: <code>rtTopologyRAware</code> You can choose which flow to set in the Main tab and then specify a <code>.nxtgrd</code> file in the Nxtgrd tab and a TLUPlus file in the TLU+ tab. See the following topics for more information: ◦ To set pin-to-pin resistance constraint, see Defining Pin-to-Pin Resistance Constraints. ◦ To set sheetResistance, see Setting EM, Resistance, and Capacitance Options
	Extracts topology from existing routes.
Region	Lets you select pins using area-selection (Rectangle) or line-selection (Line). If you select Rectangle, you can select the following options: • : Selects objects within the rectangular area. • : Select objects within the rectangular area or intersecting the rectangular boundary.

See Also

- [Topology Assistant Reference](#)
- [Topology Shortcut Menu Commands](#)

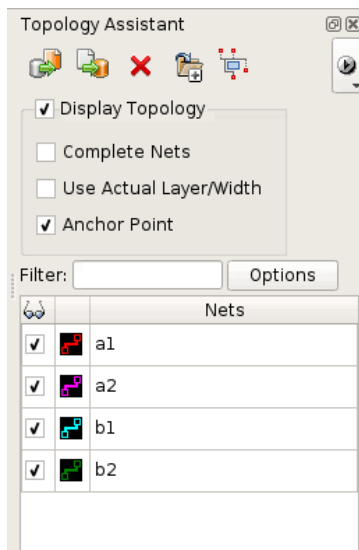
Topology Assistant Reference

This topic presents quick reference of the Topology Assistant. For instructions on using the Topology feature, see [Methods for Creating Route Topologies](#).

The Topology Assistant consists of the following:

- [Topology Assistant Toolbar](#)
- [Display Topology Section](#)
- [Filter Section](#)
- [Nets Table](#)
- [Topology Assistant Shortcut Menu](#)

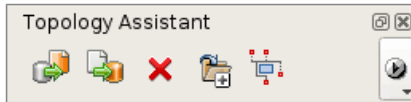
Several [Topology Assistant Shortcut Menu](#) commands are also available.



See also:

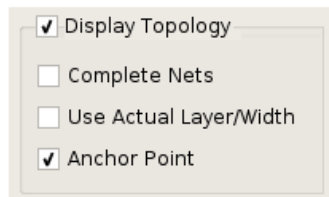
- [Topology Toolbar Reference](#)
- [Topology Shortcut Menu Commands](#)

Topology Assistant Toolbar



Option	Description
	Exports route topologies to a file. Tcl preference: <code>rtTopologyImportExportFilePath</code>
	Imports route topologies from a file. Tcl preference: <code>rtTopologyImportExportFilePath</code>
	Deletes topologies of selected nets. Tcl preference: <code>rtTopologyDeleteMode</code>
	Groups selected nets that have topologies. Tcl preference: <code>rtTopologyAddToNetGroupMode</code>
	Displays or hides the Topology toolbar.

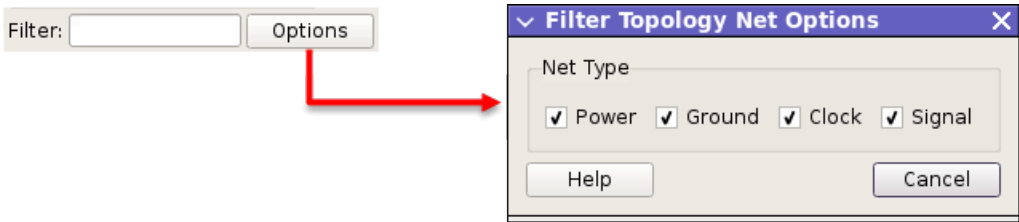
Display Topology Section



Option	Description
Display Topology	Displays or hides the topology guidelines on the layout canvas for open nets only. Tcl preference: <code>rtTopologyDisplay</code>
Complete Nets	Displays the topology guidelines for both open and complete nets. Tcl preference: <code>rtTopologyDisplayCompleteNet</code>
Use Actual Layer/Width	Draws topology using the display packet and width of the assigned layer. If the topology does not have an assigned layer, the topology guideline will still be drawn. Tcl preference: <code>rtTopologyUseActualLayer</code>

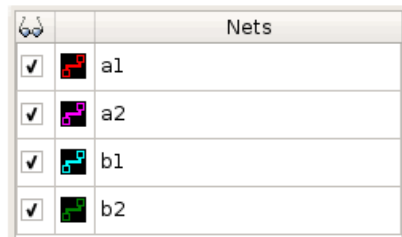
Option	Description
Anchor Point	Displays or hides the topology anchor points. See Route Topology Components . Tcl preference: rtTopologyDisplayAnchorPoint

Filter Section



Option	Description
Filter	Lets you specify the filter criteria for the net names.
Options	Opens the Filter Topology Net Options dialog box. The dialog box lets you select the Net Type to display.

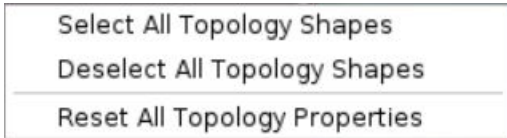
Nets Table



Option	Description
	Shows or hides selected nets.
	Specifies the topology guideline color.
Nets	Names of nets that have topologies.

Topology Assistant Shortcut Menu

The following shortcut menu commands appear when you right-click selected nets in the Topology Assistant:



Command	Description
Select All Topology Shapes	Selects all topology shapes on the net.
Deselect All Topology Shapes	Deselects all topology shapes on the net.
Reset All Topology Properties	For the selected topology segments on the net, resets the Layer , Width , and Stripe properties to the default values, and resets the Shield property to the value defined by the <code>rtTopologyShieldDefault</code> Tcl preference. See Setting Topology Segment Properties .

Topology Shortcut Menu Commands

When creating route topologies, you can right-click the canvas and choose commands to view segment properties, turn on shielding, and modify the topology. For example, you can add or delete tapoffs, convert freeform topologies to trunk-style topologies, or convert trunk-style topologies to freeform topologies.

The shortcut commands that are available during topology creation and editing are:

Properties...
Select All Topology Shapes
Reset Properties
Merge Freeform Shapes
Merge to Vertical Trunk
Merge to Horizontal Trunk
Convert to Freeform
Auto Connect Overlap
Split Topology
Delete Tapoff From Trunk
Add Tapoff to Trunk
Enable Shielding for Whole Net
Disable Shielding for Whole Net

Command	Description
Properties	Opens the Topology Shape Properties dialog box. See Setting Topology Segment Properties .
Select All Topology Shapes	Selects all topology shapes in the design.
Reset Properties	For the selected topology segments, resets the Layer , Width , and Stripe properties to the default values, and resets the Shield property to the value defined by the <code>rtTopologyShieldDefault</code> Tcl preference. See Setting Topology Segment Properties .
Merge Freeform Shapes	Merges segments of a freeform topology. See Merging Freeform Topology Segments .
Merge to Vertical Trunk	Merges multiple trunks into a single vertical trunk. See Merging Trunks .
Merge to Horizontal Trunk	Merges multiple trunks into a single horizontal trunk. See Merging Trunks .
Convert to Freeform	Converts selected trunk topology segments to freeform topology segments. See Converting Trunk Topologies to Freeform Topologies .
Auto Connect Overlap	Connects the topology automatically to crossing topologies.
Split Topology	Splits a topology segment or trunk into multiple segments. See Splitting Topologies .

Command	Description
Delete Tapoff From Trunk	Deletes trunk tapoffs. See Deleting Trunk Tapoffs .
Add Tapoff to Trunk	Adds tapoffs to trunk topology. See Adding Trunk Tapoffs .
Enable Shielding for Whole Net	Turns on topology-based shielding for all topologies on a net. See Enabling or Disabling Shielding for All Topologies on a Net .
Disable Shielding for Whole Net	Turns off topology-based shielding for all topologies on a net. See Enabling or Disabling Shielding for All Topologies on a Net .

See Also

- [Topology Toolbar Reference](#)
- [Topology Assistant Reference](#)

6

Defining a Routing Area

This chapter describes how to define a subsection of the design for routing.

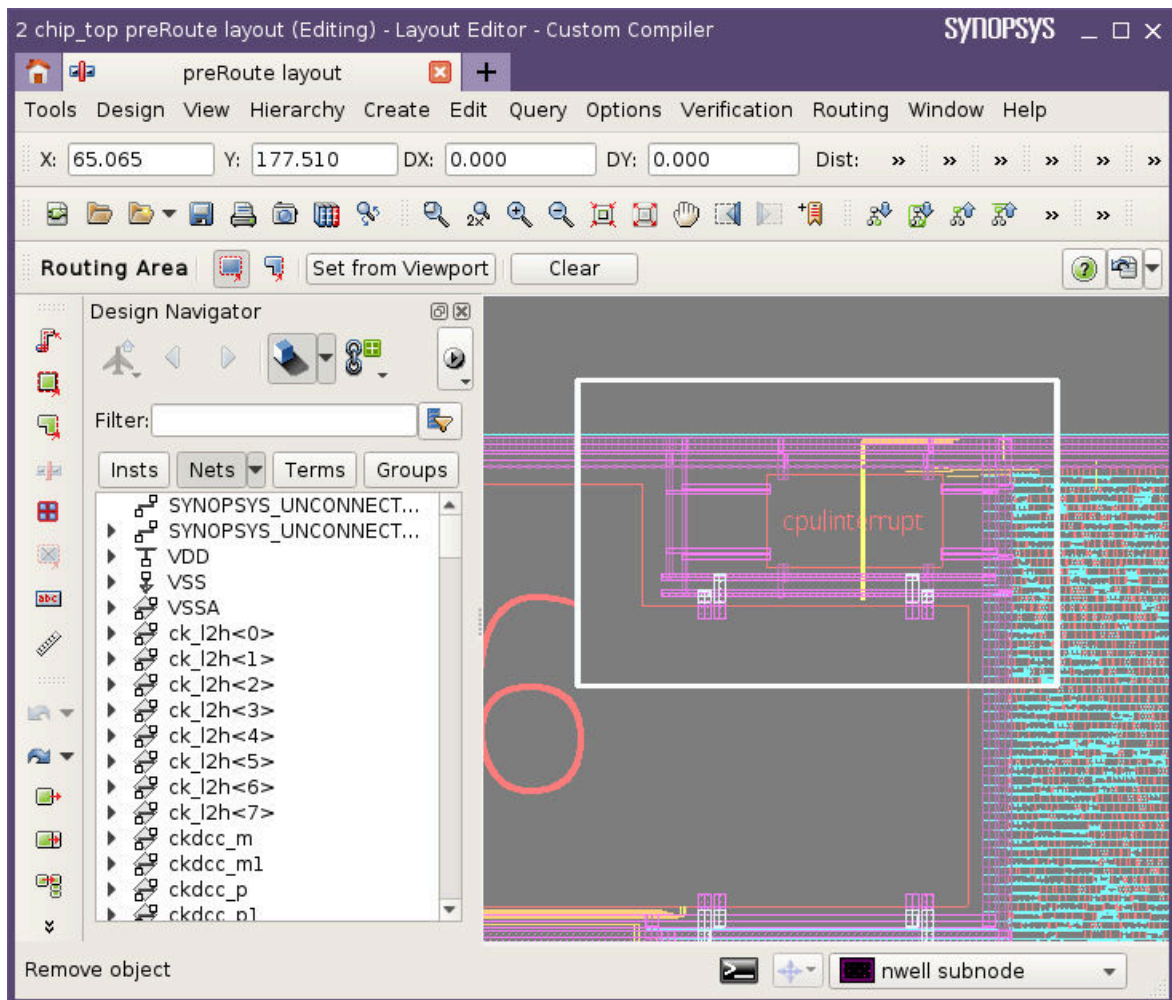
This chapter contains the following topics:

- [About the Routing Area](#)
- [Before You Start](#)
- [Displaying the Routing Area Toolbar](#)
- [Defining a Rectangular Area](#)
- [Defining a Polygonal Area](#)
- [Setting the Viewport as the Routing Area](#)
- [Using Tcl Preferences to Define the Routing Area](#)

About the Routing Area

You can define an area of a design and route only the components within that area. Use this feature if your design is large, you only need to create or modify routes within a specific area, and you want to minimize the data loading and processing time.

In the following example, a routing area is defined in a design. In the example, a white rectangle encloses the defined routing area. Note that you can create a polygonal area instead. When you start the routing operation, the router loads and processes only the data for the components within the defined area.



When defining a routing area, consider the following:

- All source and destination points of selected nets have to be within the defined area. The router will not be able to complete the routing if a source or destination point is missing.
- You cannot create routes between components that are inside of the defined area and components that are outside of the defined area.

If you are in the follow-the-cursor interactive routing mode, and you click on a component that is outside of the defined area, the router will create a route to the nearest valid component within the defined area.

If you are in the point-to-point (P2P) interactive routing mode, and you click on a component outside of the defined area, the router will ignore the routing request and display a warning message in the Console.

- You can create and change a defined area at any time. The router will reload the associated data when you start the Auto Router or Interactive Router.

Before You Start

Make sure:

- Your design is open in the Layout Editor.
- The Custom Router is enabled.
- The region of interest is displayed in the viewport.

For instructions, see [Opening the Layout Editor and Enabling the Custom Router](#).

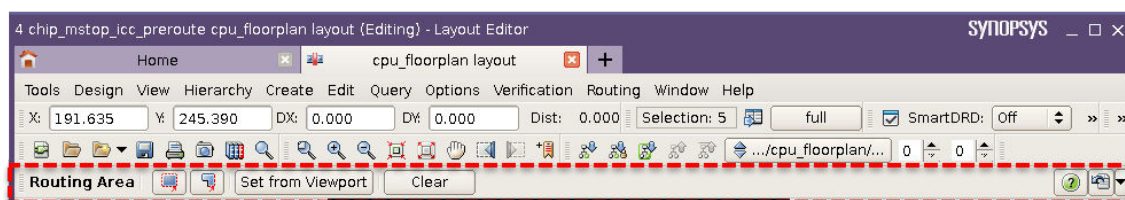
Displaying the Routing Area Toolbar

You use the **Routing Area** toolbar to define routing areas.

To display the **Routing Area** toolbar:

- In the Layout Editor, choose **Routing > Create Routing Area**.

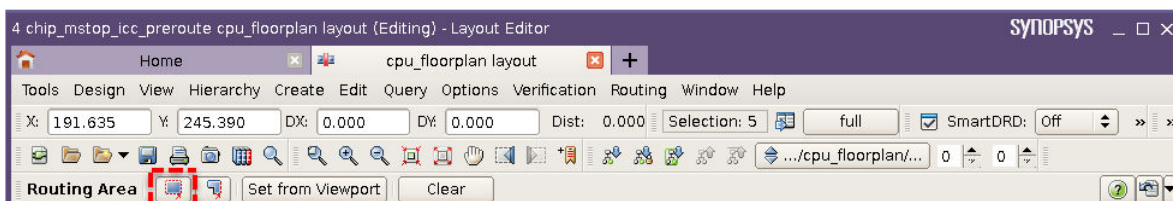
The **Routing Area** toolbar appears.



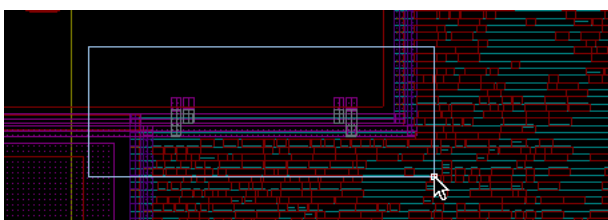
Defining a Rectangular Area

To define a rectangular area:

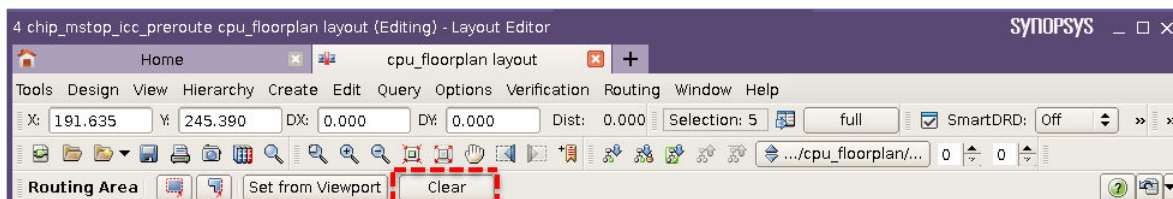
1. On the **Routing Area** toolbar, click the **Set Routing Area Shape Style to Rectangle** button.



2. On the canvas, draw a rectangle. To do this, place the cursor on the canvas, click and drag diagonally in the desired direction, and then release the mouse button. A white rectangle appears on the canvas.

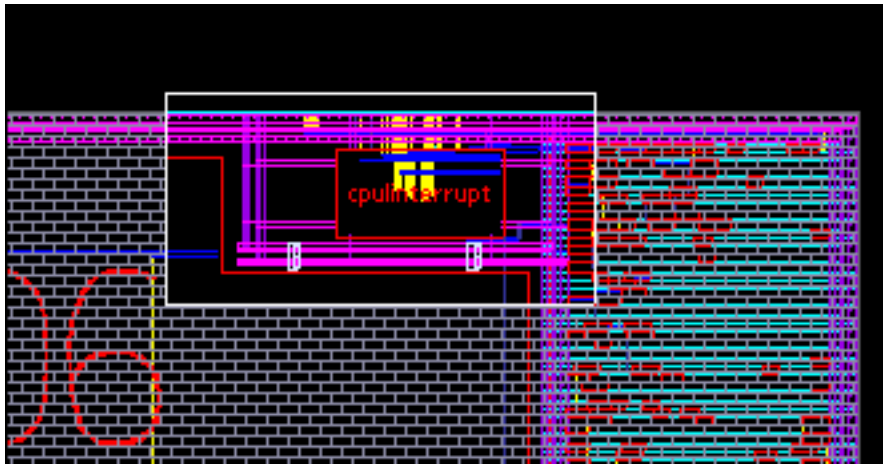


3. If you want to change the routing area, on the **Routing Area** toolbar, click **Clear**, and then repeat step 2.



To create routes within the defined area, see [Auto Routing](#) or [Interactive Routing](#).

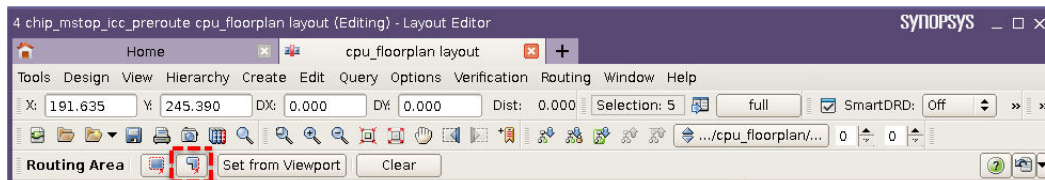
During the routing operation, the router adds a patterned mask over the region outside of the defined routing area. The mask is a visual aid that helps you to distinguish the defined area from the rest of the design. The mask is not part of the design and is removed after the routing operation is finished.



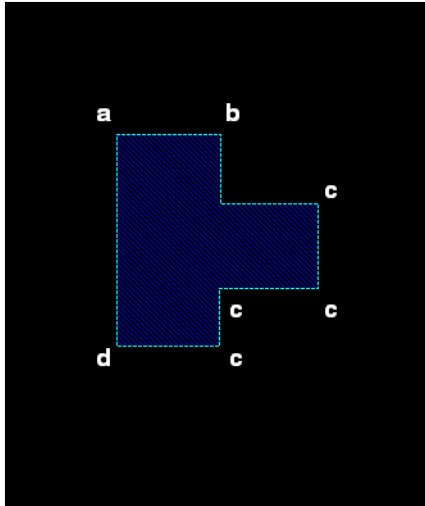
Defining a Polygonal Area

To define a polygonal area:

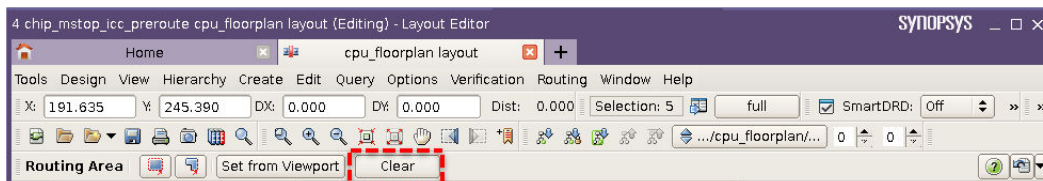
1. On the **Routing Area** toolbar, click the **Set Routing Area Shape Style to Polygon** button.



2. On the canvas, draw a polygon as follows:
 - a. Click the canvas where you want to create the first vertex of the polygon.
 - b. Click the canvas where you want to create the next vertex.
 - c. Repeat b to add more vertices.
 - d. When you are ready to create the final vertex, double-click the canvas or press Enter.

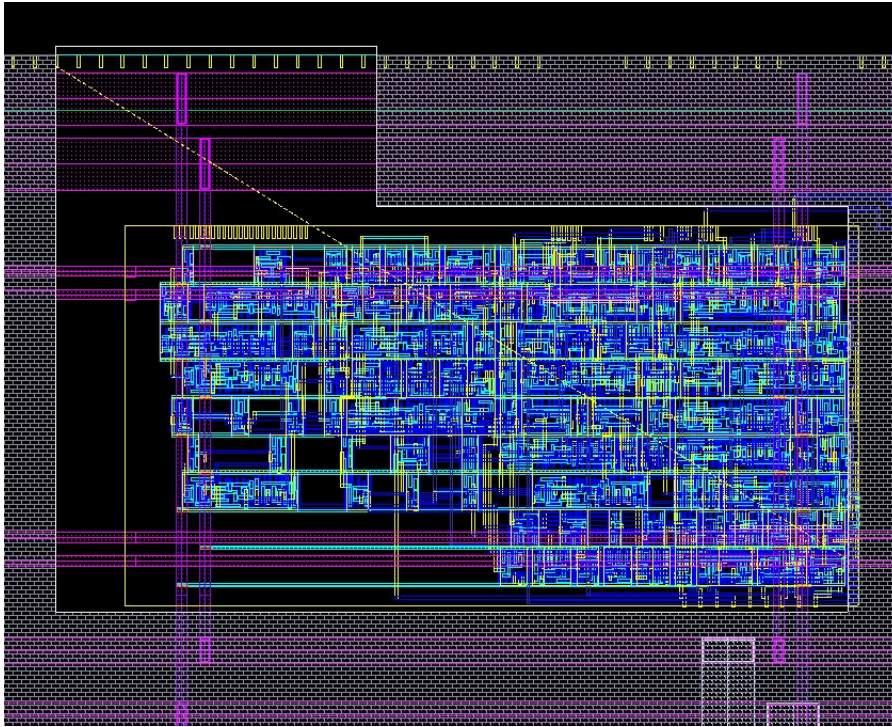


3. If you want to change the routing area, on the **Routing Area** toolbar, click **Clear**, and then repeat step 2.



To create routes within the defined area, see [Auto Routing](#) or [Interactive Routing](#).

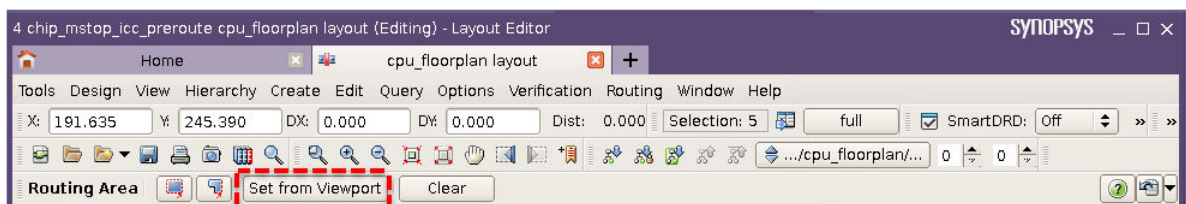
During the routing operation, the router adds a patterned mask over the region outside of the defined routing area. The mask is a visual aid that helps you to distinguish the defined area from the rest of the design. The mask is not part of the design and is removed after the routing operation is finished.



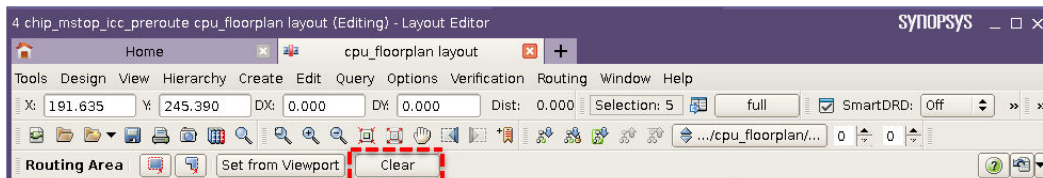
Setting the Viewport as the Routing Area

To set the viewport as the routing area:

1. Zoom in and pan so that the area of interest fits the viewport.
2. On the **Routing Area** toolbar, click **Set from Viewport**.



3. If you want to change the routing area, on the **Routing Area** toolbar, click **Clear**, and then repeat step 2.



To create routes within the defined area, see [Auto Routing](#) or [Interactive Routing](#).

Using Tcl Preferences to Define the Routing Area

You can also use the `rtRoutingArea` Tcl preference to define the polygonal area. In the following example, the `rtRoutingArea` preference defines a polygon, and the `-value` argument specifies the coordinates of the vertices.

```
db::setPrefValue rtRoutingArea -scope [de::getActiveContext]  
  
-value {{144.16 175.085} {148.455 175.085} {148.455 178.32} {147.165  
178.32} {147.165 176.545} {145.52 176.545} {145.52 175.91} {144.16  
175.91}}
```


7

Auto Routing

This chapter describes how to use the Auto Router.

This chapter contains the following topics:

- [About the Auto Router](#)
- [Enabling the Auto Router](#)
- [Checking for Constraint Compliance](#)
- [Checking for Design Errors](#)
- [Starting the Auto Router](#)
- [Managing Option Sets](#)
- [Auto Router Behavior](#)
- [Auto Route Toolbar and Dialog Box Reference](#)

About the Auto Router

The Auto Router uses information that is available in the technology and the design to automatically add wires on one or more nets at the block level. During the routing operation, the Auto Router adds paths and vias. Optionally, it can add differential pair routing, bus and river routing, coaxial routing, shielding, and other routing features you specify. You do not need to interact with the router during the auto-routing operation. Individual, bus, and differential-pair routes are automatically snapped to active tracks if you turn on this option. The same option will also automatically snap corresponding parallel, jumpers for parallel, and bus shielding to active tracks during auto routing.

The Auto Router works with prerouted mixed-signal Fusion Compiler or IC Compiler designs and respects corridors that were created in these designs. A bus or diff-pair must contain the same set of corridors. Within a bus or differential pair, the corridors are accumulative. The capacity at any section of corridors should be large enough for bus wires and vias passing through.

Enabling the Auto Router

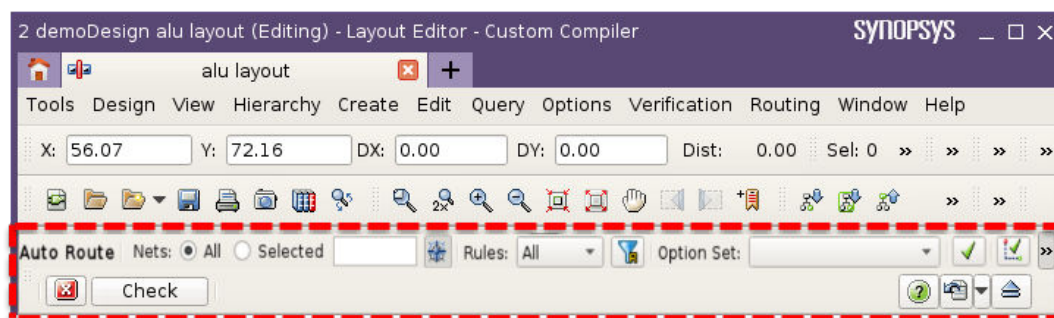
Caution:

Before you start, make sure you review the information in [Preparing for Routing](#).
If you want to route a subsection (area) of the design, define the area. See [Defining a Routing Area](#).

You must enable the Auto Router so that you can set the routing options and start the routing operation.

To enable the Auto Router:

1. In the Layout Editor, make sure the Custom Router is enabled. For instructions, see [Getting Started](#).
2. Choose **Routing > Auto Route**. The **Auto Route** toolbar appears.



Checking for Constraint Compliance

Before you start the Auto Router, you should run the Data Checker to make sure the design complies with technology and design-specific constraints. Doing so ensures that the Auto Router will run successfully.

This topic describes the following:

- [About the Data Checker](#)
- [Running the Data Checker](#)

To troubleshoot Data Checker errors, see [Troubleshooting Error Messages](#).

About the Data Checker

The Data Checker is a Custom Router feature that checks the integrity of the design database and verifies the design rules. The Data Checker looks for errors in the following:

- Design

The Data Checker looks for missed PR boundaries for any block, invalid placement of a pin or instance, unroutable pins, missing pins in a net, blocked pins, and cell views that have no reference technology.

- Technology

The Data Checker searches the technology for missing rules. Also, because poly layers and metal layers are treated as routable layers, the Data Checker displays errors if it encounters one or more of the following conditions:

- No routable layer is turned on.
- No routable layer is defined in the technology.
- No minimum width, minimum spacing, or manufacture grid is defined for any routable layer.
- No via is defined for any routable layer.

- Design-specific constraints

The Data Checker looks for:

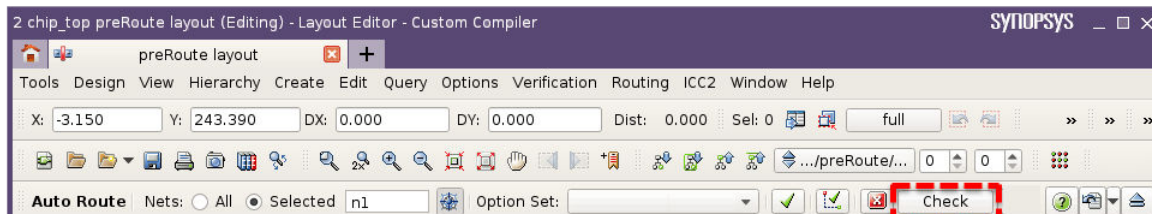
- Design-specific constraints that are set incorrectly. The Data Checker looks at the routing layers, routing directions, routing grids, non-default rule width and spacing, shielding width and gap, differential pair and triplet pin topology, and net matching values.
- Shielding constraint has width or spacing value that is less than the technology minimum width or spacing value.
- Non-default rule has width or spacing value that is less than the technology minimum width or spacing value.
- A routing grid value you set is less than the minimum value for the routing grid (the minimum value is the sum of the layer minimum width plus minimum spacing).

Results of the data check are displayed in the Text Viewer.

Running the Data Checker

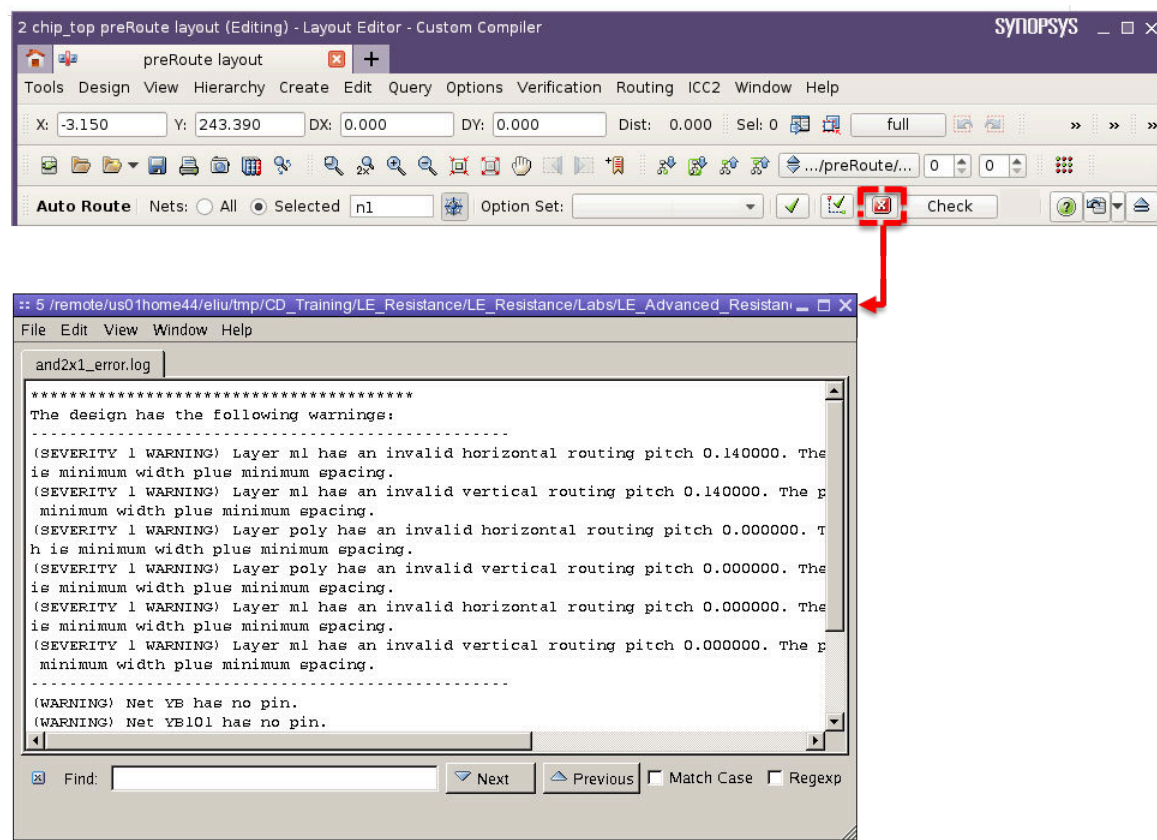
To run the Data Checker:

- On the **Auto Route** toolbar, click **Check**.



To view the warning and error messages:

- Click  on the **Auto Route** toolbar. The Text Viewer appears.



Resolve any errors before running the Auto Router. See [Troubleshooting Error Messages](#).

Checking for Design Errors

This topic describes the following:

- [About the Preroute Checker](#)
- [Running the Preroute Checker](#)
- [Viewing and Resolving Design Errors](#)

About the Preroute Checker

Before you start the Auto Router, you can run the Preroute Checker to make sure the design is free of design-level errors. Doing so ensures that the Auto Router will run successfully.

The Preroute Checker is a Custom Router feature that checks the design for the following:

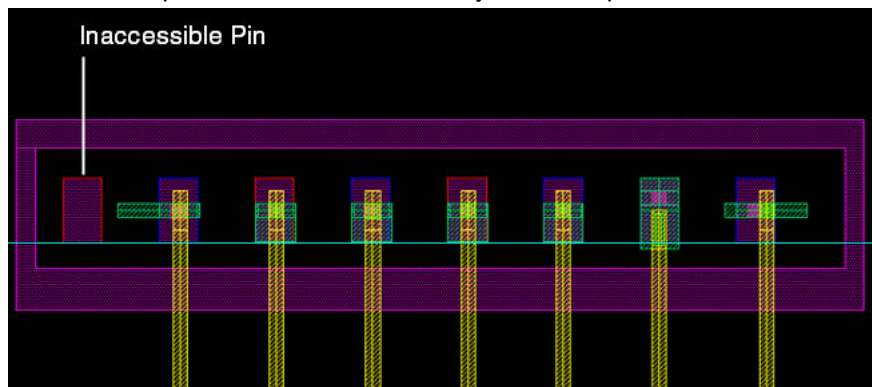
- [Design Errors](#)
- [Design Warnings](#)
- [Technology Errors](#)

Error markers are added in the design and listed in the Error Viewer so that you can easily locate and resolve the errors. In addition, a summary report is displayed in the Console and log file. See [Viewing and Resolving Design Errors](#).

Design Errors

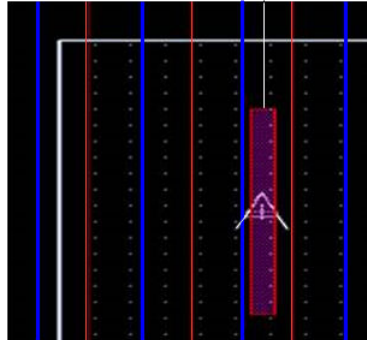
The Custom Router will display error messages for the following design errors:

Type of Design Error	Description
Pin Accessibility	Pins that are not accessibility during routing: • Pins that are not accessible through vias • Pins that are completely blocked • Pin widths that are not within the <code>allowedWidthRange</code> Layer constraint In the following example, the Preroute Checker will display an error for the left-most pin because it is blocked by other shapes.



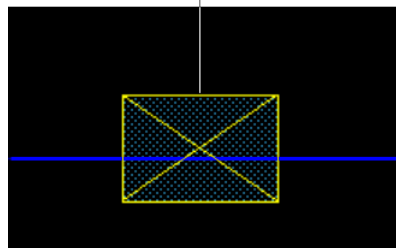
Type of Design Error	Description
Off Track Pin	(Designs that have routing tracks) Pins that are not on routing tracks or have widths different from the tracks. In the following example, the Preroute Checker will display an error for the pin that is placed between the blue and red track.

Off-Track Pin



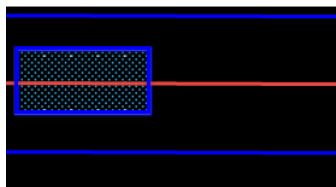
In the following example, the Preroute Checker will display an error for the pin that is on the blue track, but the center of the pin is not on the track.

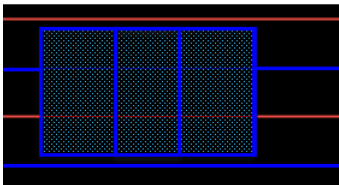
Off-Track Pin



Pin and Track Color	(Designs that have colored routing tracks) Pin colors that do not match track colors. In the following example, the Preroute Checker will display an error for the blue pin that is placed on the red track.
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Blue Pin on Red Track

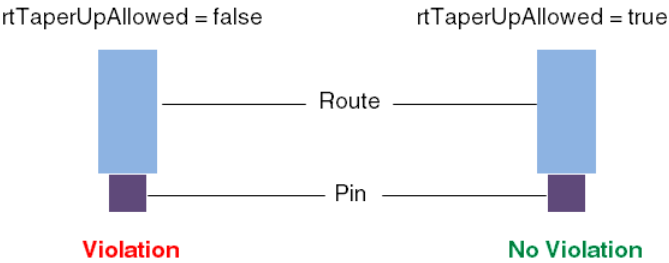
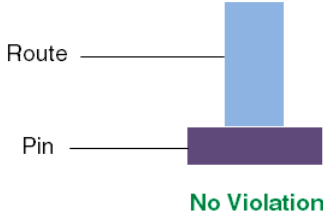


Type of Design Error	Description
Pin Shorts	Short circuits caused by preroute pin shapes that are positioned incorrectly. In the following example, the Preroute Checker will display an error because the two overlapping blue pins are causing short circuitry. Short - Overlapping Pins 

Design Warnings

The Custom Router will display warning messages for the following:

Type of Design Issue	Description
Single Pin in Routing Area	Pins that require connections to other pins that are outside of the defined routing area.
Track Region and Layer Direction Mismatch	(Designs that have routing tracks) The tracks in the track region do not match the routing layer direction. For example, the Preroute Checker will display an error message if the tracks in a track region are vertical, but the direction specified for the routing layer is horizontal.

Type of Design Issue	Description
Pin Size	Pins that have widths that are: • Not within <code>allowedWidthRange</code> and might result in reroutes. For example, the pin width is 0.12, and <code>allowedWidthRange</code> is 0.03 to 0.10. • Narrower than the route layer width constraint, and <code>rtTaperUpAllowed</code> is false. Pin Width < Route Layer Width <code>rtTaperUpAllowed = false</code> <code>rtTaperUpAllowed = true</code>  Pin Width > Route Layer Width <code>rtTaperUpAllowed = true or false</code>  Unnetted Shape Shapes that do not have net assignments. Label on Shapes Without Pins Shapes that have labels, and the label text should match net names in the design, but: • The shapes do not have pin or net associations. • The net names do not match the label text. Pin Off Grid Points (Designs that have routing tracks) Both ends of pins are not on grid points. If only one end of the pin is on the grid point, the Auto Router will extend the other end to prevent violation. Nets Without Shapes Nets that do not have any shapes associated with them.

Technology Errors

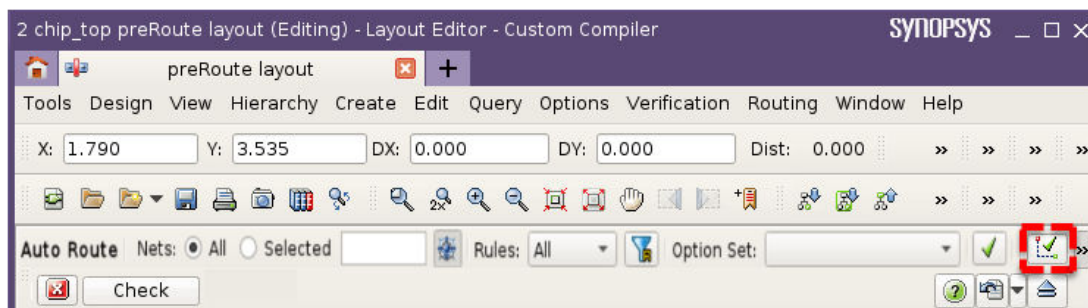
The Custom Router will display error messages for the following errors:

Type of Design Issue	Description
Missing Via Definitions	Missing via definitions between metal layers, starting from M0.
Missing Width Constraints	Missing <code>minWidth</code> constraint for metal, contactless, and poly layers.
Missing Spacing Constraints	Missing spacing constraint for metal, contactless, and poly layers.

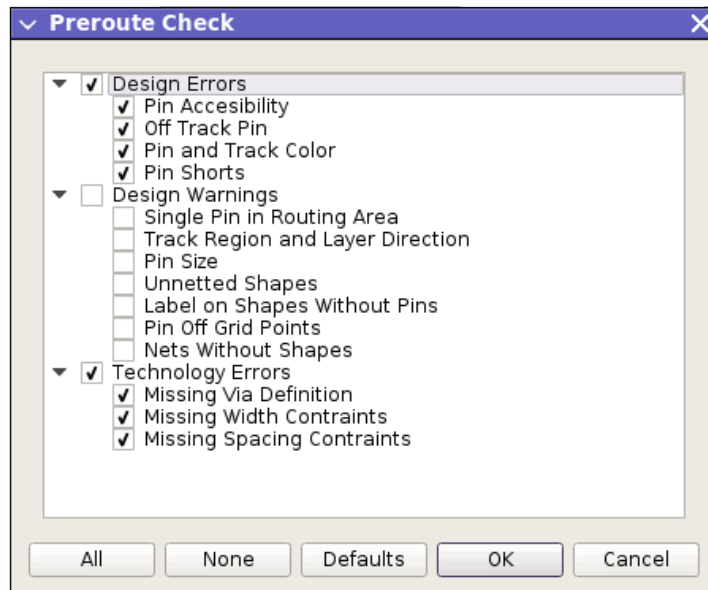
Running the Preroute Checker

To run the Preroute Checker:

1. In the Design Navigator, select the nets you want to check. Alternatively, you can [define the area](#) in which you will run the checker.
2. On the **Auto Route** toolbar, click **Preroute Check**.

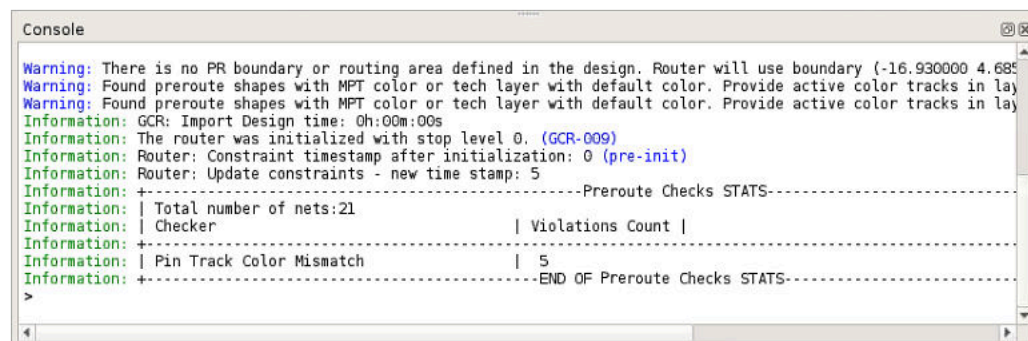


3. In the **Preroute Check** dialog box, select the checks you want to run, and click **OK**.



4. Click **OK**. The Preroute Checker starts checking the design.

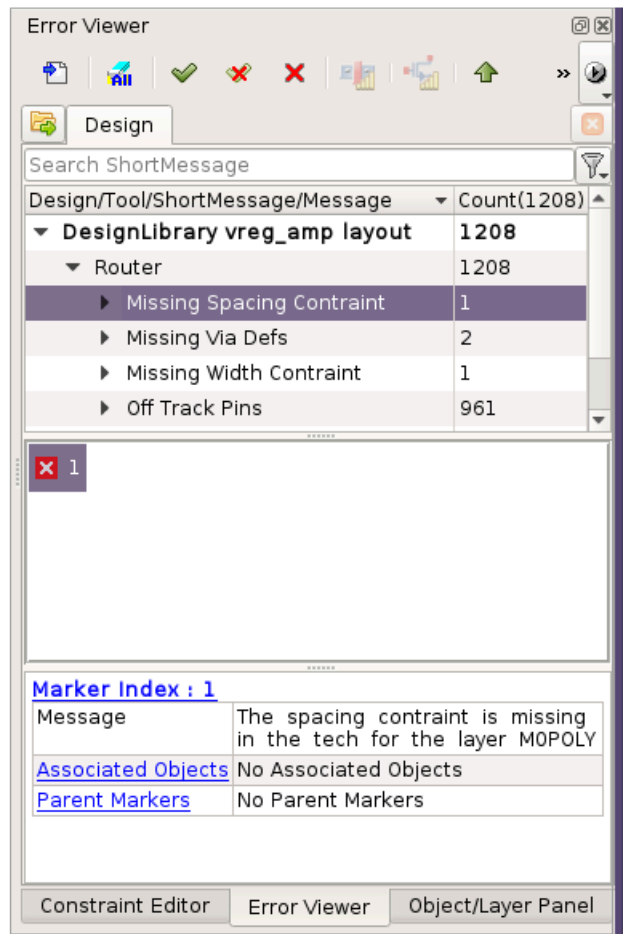
When the checker is finished, error markers appear on the canvas. A summary is available in the Console and log file.



Viewing and Resolving Design Errors

To view and resolve the errors:

1. Display the Error Viewer. The list of errors appears in the viewer.



2. Select an error in the Error Viewer. The viewport zooms in to the location of the error.
3. Resolve the error.

The Error Viewer explains the causes of the errors. See also [Troubleshooting Error Messages](#).

Starting the Auto Router

The Auto Router uses information that is available in the technology and the design to automatically add wires on one or more nets at the block level. You use the **Routing > > Auto Route** command to automatically add routes in your design.

This topic describes the following:

- [Before You Start](#)
- [Procedure](#)

For topology-based routing, see [Topology Routing](#).

Before You Start

To ensure that the auto routing process and results are successful, you must complete some preparations, such as turning on the Connectivity Engine and the DRD feature.

Before you start the Auto Router:

1. Make sure you have completed all the preparations. See:
 - [Net-Based Routing Workflow](#)
 - [Preparing for Routing](#)
 - [Setting Routing Constraints](#)
2. If the design contains pseudo parallel connections, set the `Routing Tcl Tcl` preference to `true` so that the Auto Router does not add routes for those connections. (When the preference is true, the Connectivity Engine treats pseudo parallel nets as complete or already routed, and the [Post-Route Summary](#) in the Console will show the nets as already routed.)
3. In the Layout Editor, turn on the Connectivity Engine.

The Connectivity Engine helps to determine whether the Auto Router will create routes based on the following:

Connectivity Engine determines that net is ...	Net has associated topology?	Auto Router will...
Complete	No	Not create routes
Complete	Yes	Route the net
Incomplete	No	Route the net

Connectivity Engine determines that net is ...	Net has associated topology?	Auto Router will...
Incomplete	Yes	Route the net

If the Connectivity Engine is turned off, the Auto Router will create routes.

4. Turn on the DRD feature.
5. In the Design Navigator, display the flight lines, clear existing flight lines, and then select the nets. See [Displaying and Hiding Flight Lines](#).
6. If you want to route a subsection (area) of the design, define the area first before you start routing. See [Defining a Routing Area](#).
7. (Co-design flows) If you want the Auto Router to treat weak pin connections as strong or must-join connections, set the `leWeakConnectionModeOverride` Tcl preference. See also [How Fusion Compiler Must-Join Pins are Treated](#) and [How IC Compiler II Must-Join Pins are Treated](#).

Procedure

After you complete the routing preparations, you can start the Auto Router.

Note:

Routing occurs in real time. Therefore, you cannot edit or modify a design during the routing operation. However, you can interrupt a routing operation and save the current routed state.

When a single net from a river, bus, or differential group is selected for routing, the whole group is routed.

To start the Auto Router:

1. Display the **Auto Route** toolbar. See [Enabling the Auto Router](#).
2. On the **Auto Route** toolbar, do one of the following:
 - Choose the **Option Set** you want to use. To create option sets, see [Managing Option Sets](#).
 - Click **Eject**  to open the **Auto Route** dialog box and set the auto routing options. See [Auto Route Toolbar and Dialog Box Reference](#).

3. Start the Auto Router using one of the following methods:

- On the **Auto Route** toolbar, click **Route** ()
- If you are in the **Auto Route** dialog box, click **Apply**.

The router initializes, loads the data, runs the Data Checker, and then starts the routing operation. Routes are created based on the order of the nets in the Design Navigator.

The Auto Router displays a progress bar. A message in the Console lets you know that the Auto Router is busy.

4. (Optional) To interrupt the routing operation and save the current routing state, press Ctrl+C or click **Stop** in the progress dialog box.

When the operation finishes, a completion message appears in the Console. If error messages appear in the Console during routing, see [Troubleshooting Error Messages](#).

You can view and edit the route shape properties in the Property Editor. See [Viewing and Modifying Route Attributes](#). For instructions on viewing and editing via shape properties, see [Modifying Via Attributes and Properties](#).

Managing Option Sets

After you set the options on the **Auto Route** toolbar or dialog box, you can create option sets and recall the settings later within the same design session. Doing so allows you to experiment with different settings without losing them.

You can also save the options to a directory, permitting them to be shared across design sessions or with other users.

This topic describes the following:

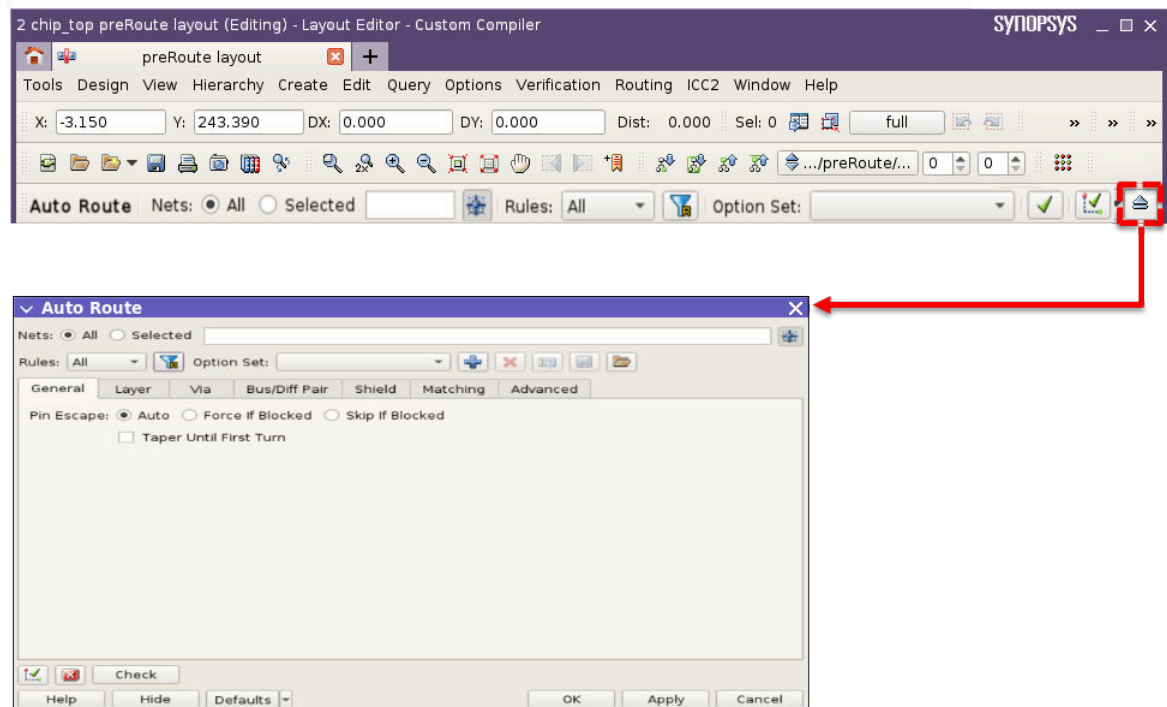
- [Opening the Auto Route Dialog Box](#)
- [Creating Option Sets](#)
- [Modifying Option Sets](#)
- [Renaming Option Sets](#)
- [Deleting Option Sets](#)
- [Saving Option Sets to a Directory](#)
- [Reloading Options Sets from a Directory](#)

Opening the Auto Route Dialog Box

You create, modify, and save auto routing options in the **Auto Route** dialog box.

To open the **Auto Route** dialog box:

- On the **Auto Route** toolbar, click **Eject** .

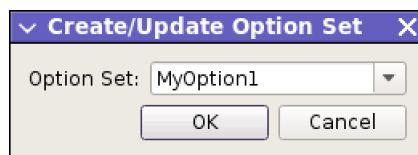


Creating Option Sets

Option sets are retained within the same design session only. If you want to recall the options in a later session, see [Managing Option Sets](#).

To create a new option set:

1. In the **Option Sets** area, click . The **Create/Update Option Set** dialog box opens.



2. Type a name in the **Option Set** box.

3. Click **OK** to save the option set to memory and close the dialog box. To save the option set permanently to a file, see [Managing Option Sets](#).

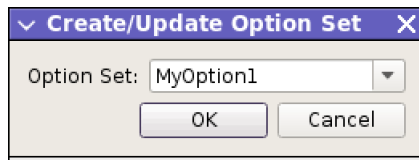
Caution:

If you do not save the option set to a file, you lose your changes when you exit the software.

Modifying Option Sets

To modify an existing option set:

1. In the **Option Set** area, choose the option set that you want to modify.
2. Change the auto route settings. See [Auto Route Toolbar and Dialog Box Reference](#).
3. In the **Option Sets** area, click . The **Create/Update Option Set** dialog box opens.

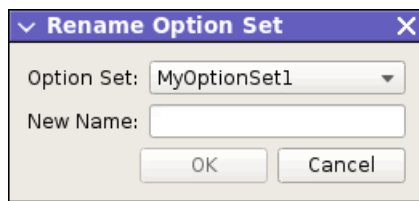


4. Click **OK** to save the changes and close the dialog box.

Renaming Option Sets

To rename an existing option set:

1. In the **Option Set** area, click . The **Rename Option Set** dialog box opens.

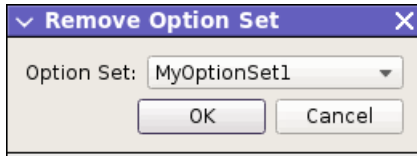


2. Choose the **Option Set** you want to rename.
3. Type a **New Name** for the option set.
4. Click **OK** to save the change and close the dialog box.

Deleting Option Sets

To delete an existing option set:

1. In the **Option Set** area, click . The **Remove Option Set** dialog box opens.



2. Choose the **Option Set** you want to delete.
3. Click **OK** to delete the option set and close the dialog box.

Saving Option Sets to a Directory

To save all existing options to a directory:

1. In the **Option Set** area, click . The **Save Option Sets** dialog box opens.
2. Select the directory you want to use.
3. Click **Choose**. The existing option sets are saved in the directory. Each option set is saved as an XML file.

Reloading Options Sets from a Directory

To load the option sets from a directory:

1. In the **Option Set** area, click . The **Load Option Sets** dialog box opens.
2. Select the directory you want to load.
3. Click **Choose**. The option set names appear in the Option Set list.

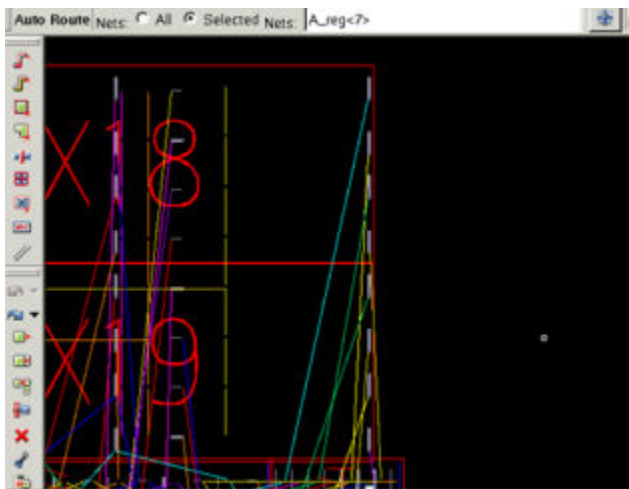
Auto Router Behavior

A Custom Compiler layout can contain a combination of unrouted connections, prerouted connections, or partially routed connections. The Custom Router behavior depends on the connection types, as described in the following sections.

- [Unrouted Connections](#)
- [Prerouted Connections](#)
- [Partial Dangling Connections](#)

Unrouted Connections

By default, the Custom Router routes all selected unrouted (or unconnected) terminals at the top level. If you do not set constraints in the Constraint Editor, the router will use the rules in the technology to create interconnects based on a minimum spanning tree or closest target. The following example shows unrouted connections in a design with flight lines enabled to demonstrate this concept.



To set design-specific constraints, see [Setting Routing Constraints](#). To set process-level constraints, edit the technology database.

Prerouted Connections

The Custom Router automatically derives connectivity for any pin-to-pin prerouted connection that it detects. Completed connections can be ripped up and rerouted, unless their route status is set to locked or fixed.

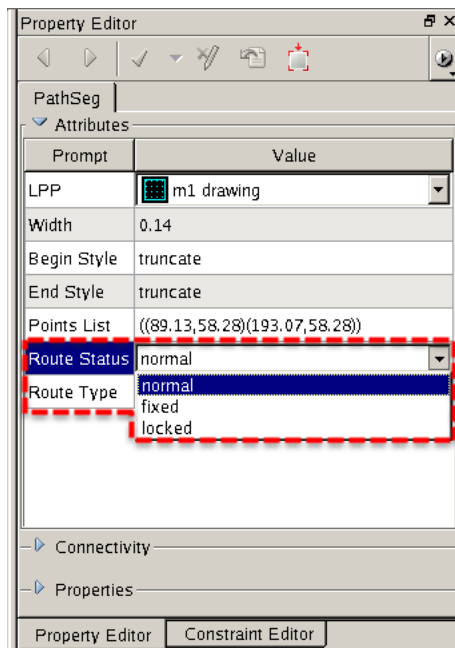
A route status can be normal, locked, or fixed.

- When the status is set to locked or fixed on all shapes of a given net, then the tool cannot modify the connection.
- If a connection has a route status of normal, then the tool can modify the connection.

Note:

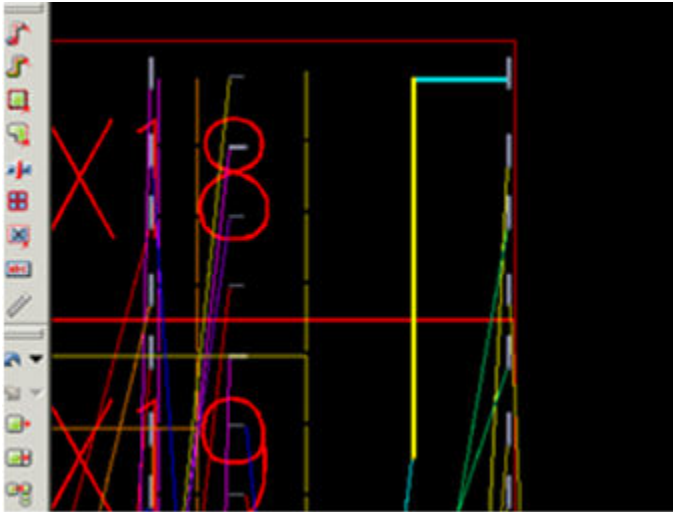
The router will not modify the route status on a routed connection.

Use the Property Editor to view or set the Route Status.



Partial Dangling Connections

You can create any partial dangling route connected to any terminal pin. By default, the tool can connect anywhere along a dangling path to complete an interconnect. However, if there is a floating netted route, the tool can route to it but not modify it. Whether the router routes to a dangling end or the body of a floating wire depends on the proximity to the pin.



Auto Route Toolbar and Dialog Box Reference

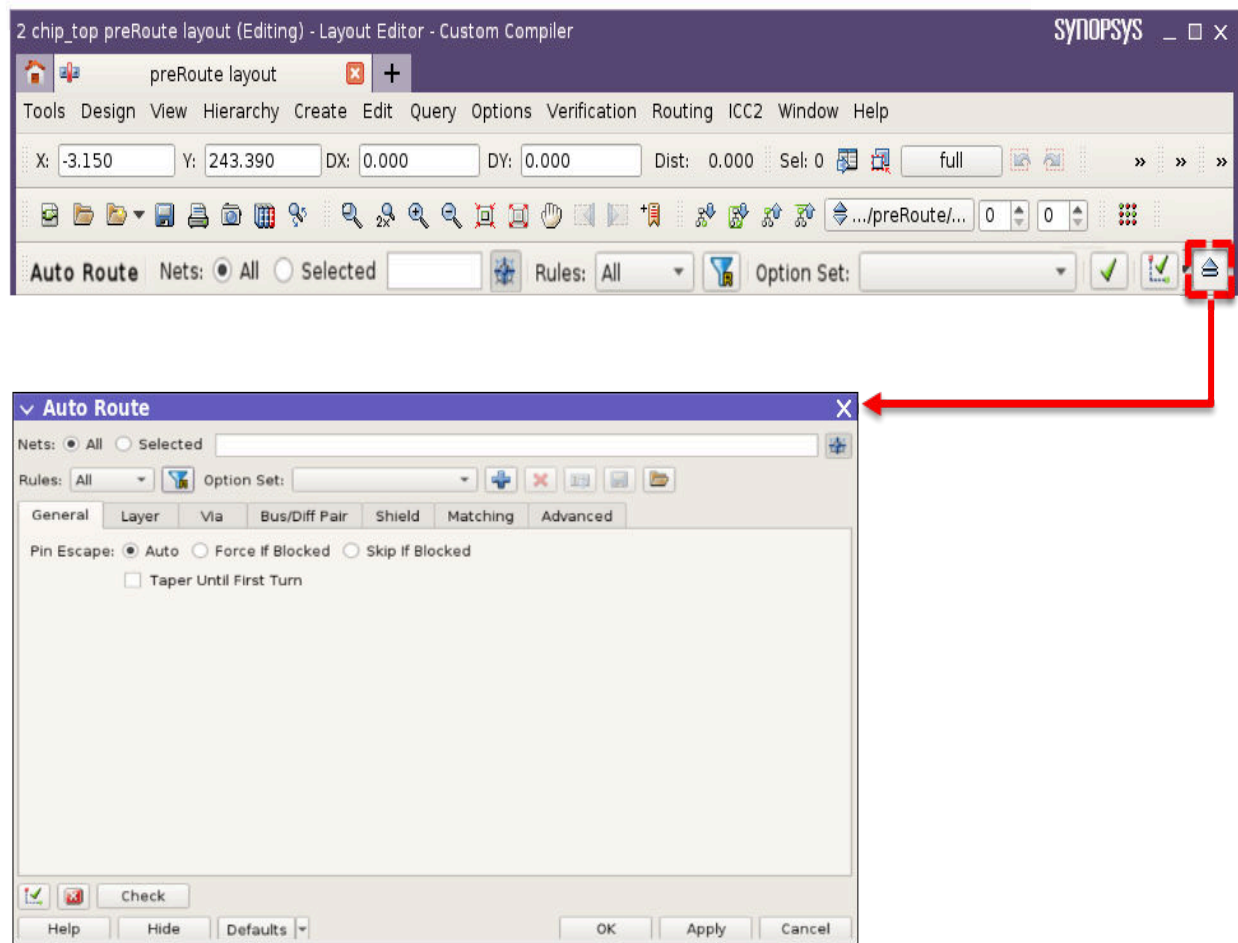
This topic provides a quick reference of the **Auto Route** toolbar and dialog box. For instructions on creating routes automatically, see [Starting the Auto Router](#).

The toolbar and dialog box contain the following:

- [Basic Options](#)
- [General Tab](#)
- [Layer Tab](#)
- [Via Tab](#)
- [Bus/Diff Pair Tab](#)
- [Shield Tab](#)
- [Matching Tab](#)
- [Advanced Tab](#)

Basic Options

In the **Auto Route** dialog box, the basic options are the options that appear in both the toolbar and dialog box. In the dialog box, the common options appear above and below the tabs.

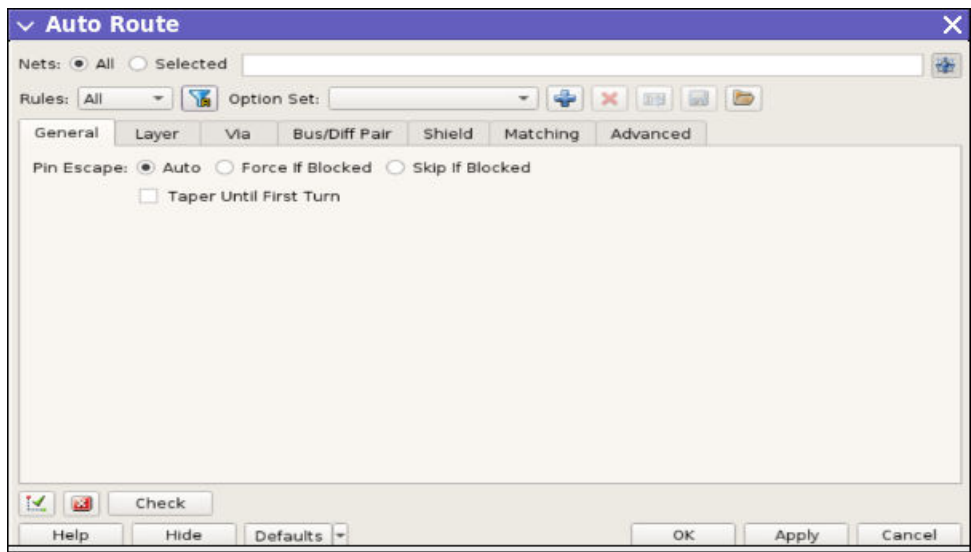


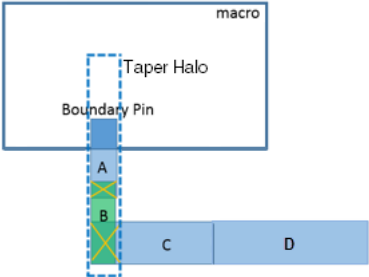
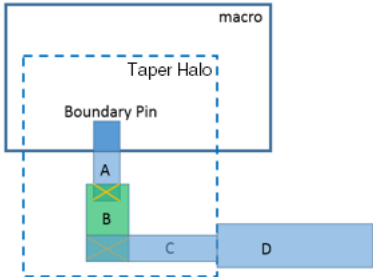
Option	Description
Nets	Specifies the nets you want to route: • All: Routes all nets. • Selected: Routes only the selected nets. Selected net names appear in the box. You can select nets in the Design Navigator, or you can type the net names in the Nets box. See Net-Based Routing Workflow or Color Routing Workflow .
	Displays or hides the Design Navigator.
Option Set	Displays the chosen set of options. See Managing Option Sets .
	Starts the auto routing based on the options and specifications you provided.
	Runs the Preroute Checker. See Checking for Design Errors .

Option	Description
	Opens a text viewer, which displays any errors or warnings found during data checking or an auto route operation.
	Starts the Data Checker. See Checking for Constraint Compliance .
 (toolbar) Help (dialog box)	Displays the Custom Compiler help.
 (toolbar) Defaults (dialog box)	Creates new default options and values, or resets to the selected default options and values. Click the button, and then select one of the following: • Make Defaults for: Uses the current information to create a new set of defaults for cellview, cell, library, or user. • Save: Saves your current preferences as an XML file. • Load from: Loads from the default cellview, cell, library, or user. • Load: Lets you load a previously saved preference file. • Purge from: Lets you delete the default options and values from the cellview, cell, library, or user.
 (toolbar) Hide (dialog box)	Opens or hides the Auto Route dialog box. See Auto Route Toolbar and Dialog Box Reference .

General Tab

In the **Auto Route** dialog box, the **General** tab specifies pin escaping and route tapering options.



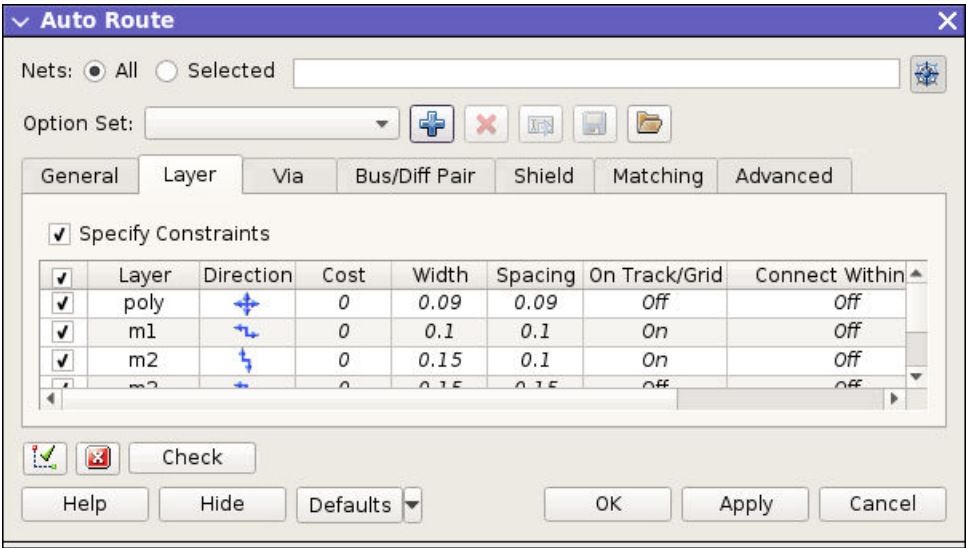
Option	Description
Pin Escape	Specifies how to route to pins when access is blocked. Select one of the following: • Auto: Routes to blocked pins when pin escape succeeds. • Force If Blocked: Relaxes rules at blocked pins to permit connections. • Skip If Blocked: Routes close to blocked pins. Adds markers between the pins and the dangling routes. You might want to use this option to route most of the net automatically, leaving a short segment to route interactively using either the Interactive Route or Create Interconnect command. Tcl preference: <code>rtPinEscapeControl</code>
Taper Until First Turn	Tapers the route between the boundary pin and the first route turn. ☐ Taper Until First Turn ☒ Taper Until First Turn  Tcl preference: <code>rtTaperUntilFirstTurn</code>

Layer Tab

In the **Auto Route** dialog box, the **Layer** tab options specify the valid routing layers and routing directions.

You have the option of populating the **Layer** tab with information from the Constraint Editor. Although you can edit the settings and then run an auto routing operation in the current session, the changes are not written back to the Constraint Editor. Also, if you close the **Auto Route** dialog box, your changes are deleted from memory.

The **Layer** tab contains the following options:



Option	Description
Specify Constraints	Overrides the design-specific constraints in the Constraint Editor. When you select this option, additional columns appear in the table: Direction , Cost , Width , Spacing . The values you enter for these columns override the corresponding values in the Constraint Editor. See Setting Routing Constraints about the Constraint Editor. Tcl preference: rtUseDesignConstraint

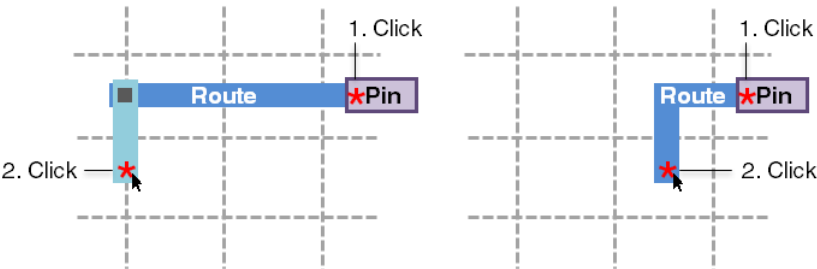
Layer table:

Option	Description
Layer	Lists the routing layers. Select the layers that you want to use for routing.

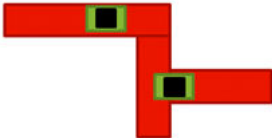
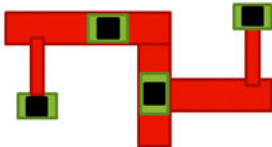
Option	Description
Direction	Specifies the preferred direction: : The router determines the direction. : The preferred direction is vertical. : The preferred direction is horizontal. : The direction will be vertical only. : The direction will be horizontal only. Tcl preference: <code>rtLayerDirection</code>
Cost	Specifies the layer's cost, or routing bias. Type a value between 0 and 100, where: 0 = Costs the least, thus the layer will be used more than layers that cost more. 100 = Costs the most, thus the layer will be used less frequently than layers that cost less. To ensure that the router uses all routing layers equally, specify 0. To make the router use certain layers more than others, specify larger values. For example, if you want the router to place more wires on m8 than m10, specify 0 for m8 and specify 50 for m10. Tcl preference: <code>rtLayerCostAR</code>
Width	Specifies the minimum layer width. Tcl preference: <code>rtLayerWidth</code>
Spacing	Specifies the minimum layer spacing. Tcl preference: <code>rtLayerSpacing</code>

Option	Description
On Track/Grid	Specifies whether the route and corresponding parallel shields, jumpers for parallel shields, and bus shields will snap to the wire tracks or the routing grid. Choose one of the following: • On: Snaps routes and shields to active tracks. If the layer does not have active tracks, the routes and shields will snap to the routing grid. If you choose this option, the Cost value is set to 4 automatically, and you cannot change it. When you choose On, routes will not connect to off-track pins. • Off: Does not snap routes and shields to active tracks or grids. Choose this option for gridless routing. • Relax: Snaps routes and shields to active tracks. However, the routes can move off tracks and grids to connect to off-track or off-grid pins. After connecting to off-track pins, the routes will remain off-track until they need to change layer and turn.

On Track/Grid = Relax



If tracks and routing grids are not defined and you choose **On**, the Auto Router uses an internally determined grid. The grid pitch is route width + minSpacing between shapes. The grid offset is 0 or pitch/2, whichever permits more pins on the grid.

Option	Description
Connect Within Pin	(Select Options: Pin to display this column.) Specifies the position of vias when they are connected to pins. For each layer, select one of the following: • Off: No pin connection requirements. Off is the default selection. • ViaAllPins: A via connected to a pin must be contained completely within the pin shape. You can still route wires to and from the pin. • ViaWireAllPins: A via or wire connecting to a pin must be contained completely within the pin shape. • ViaStandardCellPins: A via connecting to a standard cell pin must be contained completely within the pin. You can still route wires to and from the pin. Select this mode for mixed macro and standard cell routing to control different pin access. • ViaWireStandardCellPins: A via or wire connecting to a standard cell pin must be contained completely within the pin shape. Select this mode for mixed macro and standard cell routing to control different pin access. The following example shows two vias on a pin layer. The vias are contained completely within the pins. The example is the result of selecting either ViaWireAllPins or ViaWireStandardCellPins.  The following example shows four vias and their positions on a pin layer. The two center vias are completely within the pins. The two outer vias are connected to the pins but are not completely enclosed. The example is the result of selecting either ViaAllPins or ViaStandardCellPins.  Tcl preference: <code>rtConnectWithinPin</code>

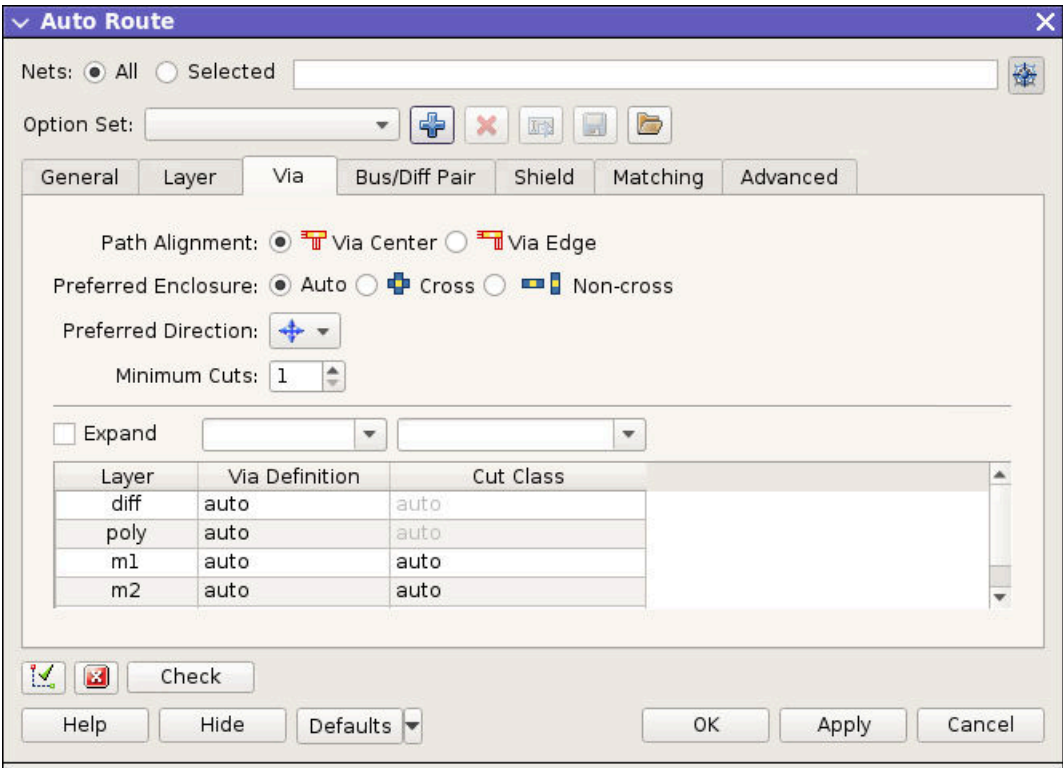
Via Tab

In the **Auto Route** dialog box, the **Via** tab options specify how vias are placed during routing.

Note:

The via options you set in the Custom Router is universal. The via settings you specify for the Auto Router is also applied when you use the Interactive Router and the Pattern Route Assistant.

The **Via** tab contains the following options:



Option	Description
Path Alignment	Aligns the connecting path and via using one of the following methods: • Via Center: Aligns the connecting path’s centerline with the center of the via. • Via Edge: Aligns the edge of the connecting path with the edge of the via. Tcl preferences: <code>rtAllowOffsetVia</code>
Preferred Enclosure	Specifies how the via’s metal layers are positioned: • Auto: The router determines the positions. • Cross: The layers are perpendicular. • Non-cross: The layers are parallel. Tcl preferences: <code>rtViaPreferredEnclosure</code>

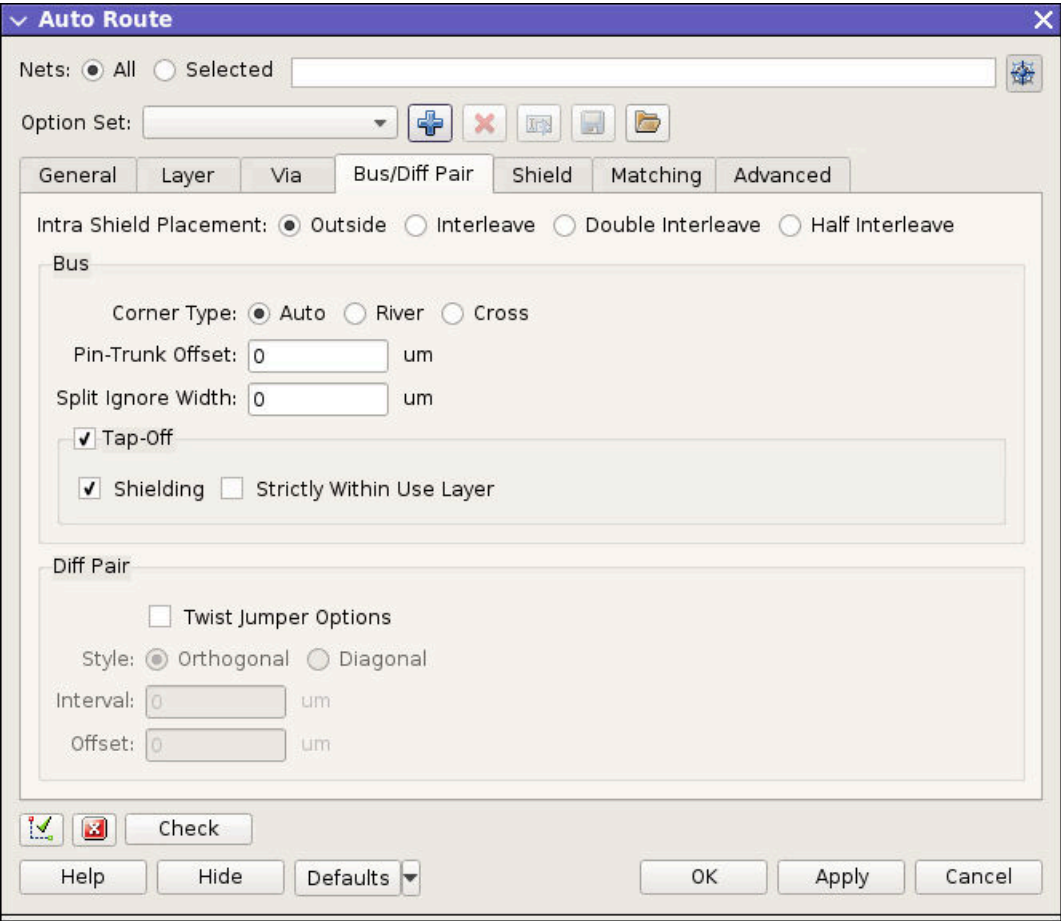
Option	Description
Preferred Direction	Specifies the via orientation: : The router determines the orientation. : The preferred orientation is vertical. : The preferred orientation is horizontal. : The orientation will be vertical only. : The orientation will be horizontal only. Tcl preferences: <code>rtPrefBarDoubleCutViaOrient</code>, <code>rtStrictFollowPrefBarDoubleCutViaOrient</code>
Minimum Cuts	The minimum number of cuts in the via. Tcl preferences: <code>rtMinimumCut</code>

Layer table:

Option	Description
Expand	Displays or hides additional columns in the layer table: Direction, Layer1 Enc, and Layer2 Enc. Tcl preferences: <code>rtUseExpandedViaSetting</code>
Layer	Lists the layers.
Via Definition	Specifies the via definition to use when adding the via on the specific layer.
Cut Class	Specifies the cut class to use when adding the via.
Direction	Specifies the via direction (or orientation). : The router determines the orientation. : The preferred orientation is vertical. : The preferred orientation is horizontal. : The orientation will be vertical only. : The orientation will be horizontal only.
Layer1/2 Enc	Amount to extend layer1 and layer2 beyond the cut edges in the X and Y direction.

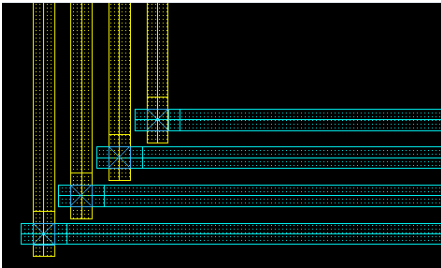
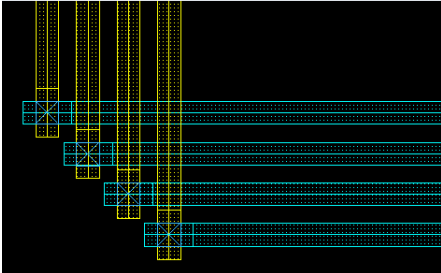
Bus/Diff Pair Tab

In the **Auto Route** dialog box, the **Bus/Diff Pair** tab options specify how bus bits and differential pairs are routed and shielded.



Option	Description
Intra Shield Placement	The method in which the bus trunk is to be shielded: • Outside: Places shields on the outside of the trunk. • Interleave: Inserts shared shielding between two adjacent bits. • Double Interleave: Shields each bit individually. • Half Interleave: Inserts shared shielding every two bits. Tcl preference: <code>rtIntraShieldPlacement</code>

Bus Options:

Option	Description
Corner Type	Specifies how the bits are routed as they turn: • Auto: Automatically determine optimal corner type for bit/pin alignment to reduce routing congestion. • River: Routes the bits so that they do not overlap at the corner.  • Cross: Routes the bits so that they overlap at the corner.  Tcl preference: <code>rtBusCornerType</code>
Pin-Trunk Offset	Specifies the offset for generated bus virtual pin. Tcl preferences: <code>rtPinTrunkOffset</code>
Split Ignore Width	Specifies the wire width threshold of any obstructing power grid or route which triggers the bus trunk to be split into more than one section. Tcl preferences: <code>rtBusSplitIgnoreWidth</code>
Tap-Off	Connects pins to bus. Tcl preferences: <code>rtBusTapoff</code> You can select the following tapoff options: • Shielding: Adds shielding on the tapoffs. If you do not select the option, only the bus trunk has shielding. • Strictly Within Use Layer: Performs bus tap-off using the valid routing layers only. Tcl preferences: <code>rtBusTapoffWithinUseLayer</code>, <code>rtBusTapoffWithinUseLayer</code>

Diff Pair Options:

Option	Description
Twist Jumper Options	Twists the nets in the group. Tcl preference: <code>rtDiffPairTwisted</code>
Style	Specifies the style of the twist: • Orthogonal: Applies 90-degree twists to the wires. • Diagonal: Applies 45-degree twists to the wires.
Interval	Specifies the distance between each twist.
Offset	Specifies the distance between the first twist and the connecting pins.

Shield Tab

In the **Auto Route** dialog box, the **Shield** tab options specify how shielding is applied during routing.

For instructions on how to add shielding, see [Creating Routing Constraints](#).

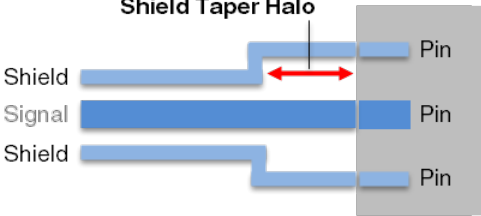
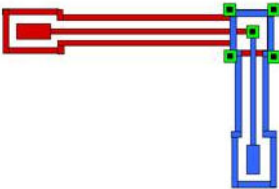
The **Shield** tab contains the following options:

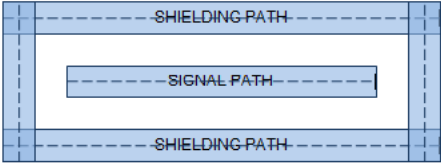
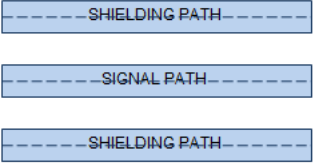
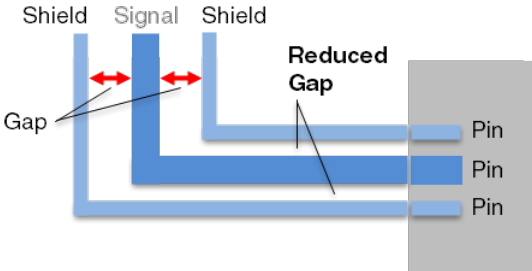
The screenshot shows the 'Auto Route' dialog box with the 'Shield' tab selected. The 'Nets' section has 'All' selected. The 'Option Set' is empty. The 'Shield' tab contains the following options:

- Shield Nets:** [Empty text box]
- Taper Halo:** [0]
- Enclose:** ☒ Pin ☒ Via
- ☒ Allow Wrong Way ☐ Connect to Aligned Pin
- ☒ Sharing ☐ Use Via Array for Coaxial
- Jumper Spacing (um):**
 - Tandem:** [0] **Parallel:** [0] **Coaxial:** [0]
- Minimum Length (um):**
 - Signal Segment:** [0] **Connected Shield:** [0]
- ☐ **Connect to Supply**
 - Min Shield Length:** [0] um
 - ☐ **Connect at Mesh Overlap**
 - Layer Offset:** ☐ Above ☒ Either ☐ Below
 - ☐ **Connect with Route**
 - Tie Spacing:** [0] um
 - Search Distance:** [0] um
- ☐ **Report Coverage**

At the bottom, there are icons for a checkmark, a red X, and a 'Check' button. Below these are 'Help', 'Hide', and 'Defaults' buttons. At the very bottom are 'OK', 'Apply', and 'Cancel' buttons.

Option	Description
Shield Nets	Specifies the net to use as shield. The router will use this value only if the information is not provided in the Net Shielding Constraint . Tcl preferences: rtShieldNet, rtShieldSecondNet

Option	Description
Taper Halo	Specifies the distance between the pin and the location at which the shield must taper. Default: 0.0 (tapers the shield between the pin and the first turn.) Shield Taper Halo  Tcl preference: <code>rtShieldTaperHalo</code>
Enclose: Pin	When selected, extends the parallel shield line to wrap around a pin if there is enough room.  Tcl preference: <code>rtShieldEnclosePin</code>
Enclose: Via	When selected, the parallel shield line is extended to wrap around a via. Tcl preference: <code>rtShieldEncloseVia</code>

Option	Description
Allow Wrong Way	Adds shielding paths in one of the following ways: - Select the option to add shielding paths in either of the two directions across the routing layers.  - Clear the check box so that all shielding paths are oriented in the same direction across all routing layers.  If a layer permits bi-directional routing, the router can create shielding paths in both directions, but the segments will have open ends. Tcl preference: <code>rtAllowWrongWayShield</code>
Connect to Aligned Pin	Reduces the distance between the shield and the signal to connect to the shield pins.  Tcl preference: <code>rtConnectShieldToAlignedPin</code>

Option	Description
Sharing	Permits shield to be shared among signal nets in the same group that have parallel and split coaxial shield styles. Shield Sharing does not apply to other shield styles. For more information, see Shared Shielding. The Shield Sharing option is also available in the Add Shields dialog box and the P2P Shielding Options dialog box. Enabling or disabling Shield Sharing in one of the dialog boxes automatically enables or disables the option in other dialog box. For more information, see Setting Options in the Add Shield Dialog Box and Setting Options in the P2P Shield Dialog Box. To apply shield sharing to a specific group of nets, use the Sharing setting in the Net Shielding constraint. For information about the Constraint Editor, see Setting Routing Constraints. To route wires close to power straps and use the power straps as shared shielding, set the <code>rtUsePowerStrapAsShield</code> preference to <code>true</code>. This preference is applicable for auto routing only.
Use Via Array for Coaxial	Uses the via array to tie solid coaxial shields instead of jumper vias. Tcl preference: <code>rtShieldUseViaArrayForCoaxial</code>
Jumper Spacing	Specifies the spacing between consecutive jumper pairs that connect different types of shields: Tandem, Parallel, and Coaxial. If the value is not set or is 0, no jumper or jumper via is added. For the descriptions of the different shield types, see Shielding Styles and Behavior. Tcl preferences: <code>rtShieldTandemJumper</code>, <code>rtShieldParallelJumper</code>, <code>rtShieldCoaxialJumper</code>
Minimum Length (um)	Specifies the minimum length of signal segments and associated shield segments. If a signal or connected shield segment is shorter than the specified value, the router will not let you create a shield segment for it. • Signal Segment: The minimum length of the signal segment. The default value is 0 um. The Minimum Segment control in the shield constraint can also specify the minimum signal shield length. Note that point-to-point (P2P) shielding can shield part of a signal segment. Therefore, minimum signal segment has no impact on P2P shielding. • Connected Shield: The minimum length of connected shield segments. The default value is 0 um. Tcl preferences: <code>rtShieldMinSignalSeg</code>, <code>rtMinConnectedShieldLength</code>
Connect to Supply	Lets you specify requirements for how shields will connect to the nearest power mesh. Tcl preference: <code>rtConnectShield</code> • Min Shield Length (um): Specifies the minimum shield length required to connect to the power supply. Tcl preference: <code>rtMinShieldSupply</code>

Option	Description
	• Connect at Mesh Overlap: Creates multiple shield connections to the power mesh for a shield. Creates a shield via at each intersection of a shield path segment and a power strap. If a shield does not intersect with any power strap, the default four-pass connecting process is followed. Tcl preference: <code>rtShieldConnectAtMeshOverlap</code> • Layer Offset: Specifies the power mesh layer to connect to the shield: Above, Either above or below, or Below the shield layer. The options are available only if you enable Connect At Mesh Overlap. Tcl preference: <code>rtShieldConnectLayerOffset</code> • Connect with Route: Connects shields to ground. This option creates long ties that might block future routing. Select this option only if you must connect shields to ground using the Auto Router. Note that these shields are marked as dynamic shields in ICC designs. Tcl preference: <code>rtShieldConnectWithRoute</code> • Tie Spacing (um): The distance, in micrometers, between wire-tie connections along a shield path segment. Tcl preference: <code>rtShieldTieSpacing</code> • Search Distance (um): The distance that the Auto Router uses when searching for targets to tie. Tcl preference: <code>rtShieldTiePowerSearch</code>
Report Coverage	Creates a shield coverage report at the end of the auto-route run. The coverage can be larger than 100% if you selected Enclose Pin or Enclose Via , and any shielding added in the back of the via is considered extra shielding.

If you want to connect shielding to VSS instance pins, you can set the `rtConnectShieldToAlignedPins` preference to `true`.

Shield vias are automatically added when you add shielding. By default, the router will use the corresponding signal's via definitions for shield via creation. If, for some reason, you want to use different via definitions when adding [Shield Vias](#), you can do one of the following:

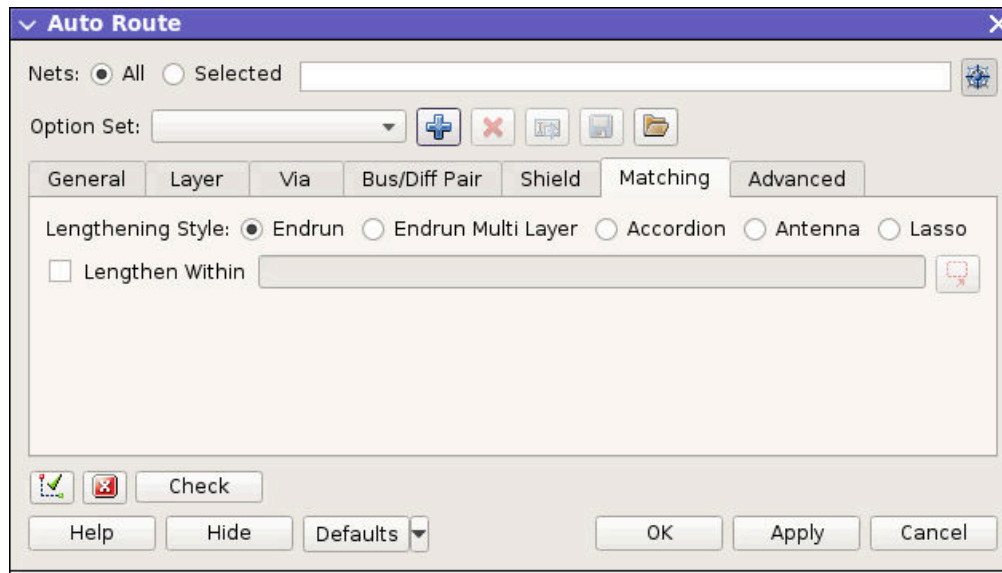
- Choose the via definitions to use in the [Net Shielding Constraint](#).
- Set the `rtSyncShieldViasWithSignalVias` Tcl preference to `false`.

Matching Tab

In the **Auto Route** dialog box, the **Matching** tab lets you specify how to match wire lengths so that you can minimize clock skews and timing violations.

In the **Matching** tab, you can specify the following:

- **Lengthening Style**: The style to use when extending the wire length.
- **Lengthen Within**: The area within which the wire should be lengthened.



Lengthening Style

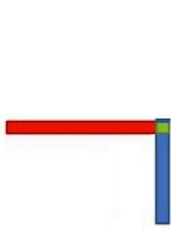
The **Lengthening Style** option specifies the style to use when extending the wire lengths. This section describes the style selections available in the **Matching** tab. Unless noted, you can use the options in multi-corner advanced-node designs.

To use the `rtRcRoutingStyle` Tcl preference, see the *Custom Compiler Tcl Reference Guide*.

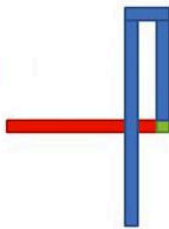
In the **Matching** tab, select one of the following:

Option	Description
Endrun	Extends the wire length either horizontally or vertically with a single-layer loop. The short segment at the top of the loop is in the non-preferred routing direction and the loop path can overlap with the main path. The router creates two types of loops: • <i>Endrun</i>: Adds length when the wire turns 90° between two pins. • <i>Side run</i>: Adds length to a straight-running wire between two pins. In the following two examples, the wires are lengthened in the vertical direction.

Before



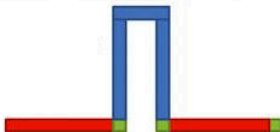
After (Endrun)



Before



After (Side Run)

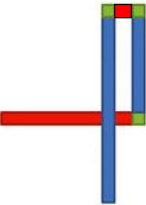


To further optimize length matching, you can set the `rtAllowCreateSiderunSmallLoop` Tcl preference to true.

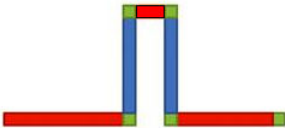
Endrun Multi Layer

Extends the wire length using the endrun style. However, instead of a single-layer loop, the endrun loop uses two or more layers.

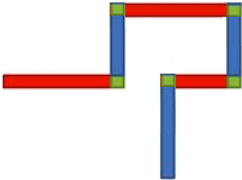
Endrun

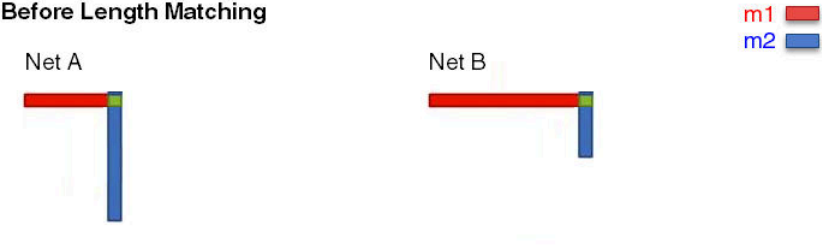
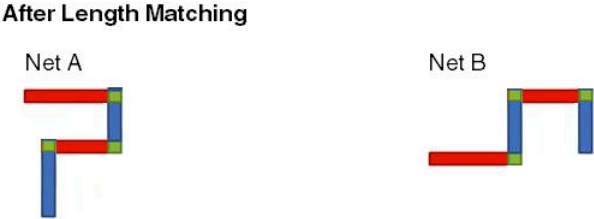
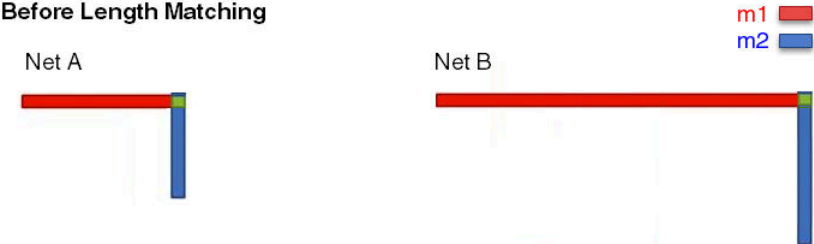
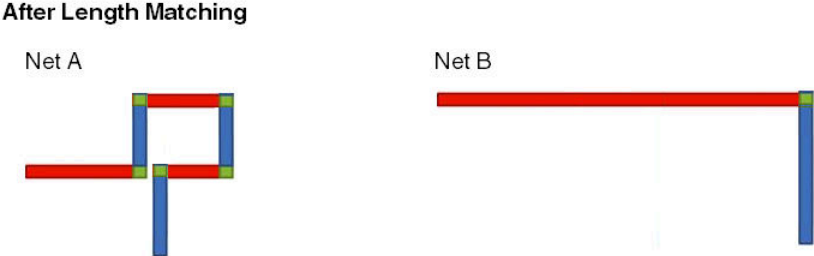


Side Run



To further optimize length matching, you can set the `rtAllowCreateSiderunSmallLoop` Tcl preference to true.

Option	Description
Accordion	Extends the wire length with multiple same-layer loops. 
Antenna	Extends individual wire segments in the preferred routing direction to increase capacitance without affecting wire resistance. Use this style in older node designs, where clock skews can be addressed by manipulating the wire capacitance only. 
Lasso	Extends the wire length horizontally and vertically with a two-layer lasso-like loop. The loop can consist of three or four path segments that do not overlap. The lasso length matching is only applicable if the nets you are matching use the same routing layers. Before After (Lasso)  

Option	Description
	You can use the lasso style to match length per layer. Multiple lasso loops are permitted per net. The following example shows how the wires on net A and net B are matched. The m1 wire on net A is extended to match the m1 wire on net B. The m2 wire on net B is extended to match the m2 wire on net A. Before Length Matching  After Length Matching  You can also use the lasso style to match the total wire length. In this case, only one lasso loop is added to one of the nets, the lasso loop is square-shaped, and the lengths matching is not based on layers. Before Length Matching  After Length Matching 

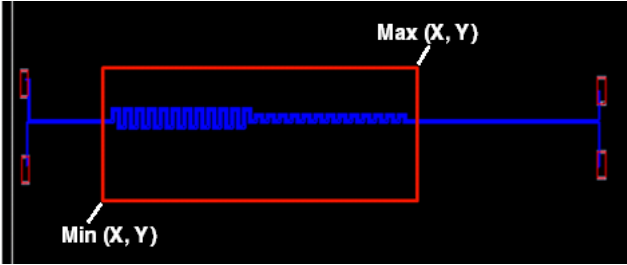
Lengthen Within

The **Lengthen Within** option lets you specify the area within which the router can lengthen wires. You must specify the area before routing.

If you want to use the `rtUseMatchBox` or `rtMatchBox` Tcl preference, see the *Custom Compiler Tcl Reference Guide*.

To lengthen wires within a specified area:

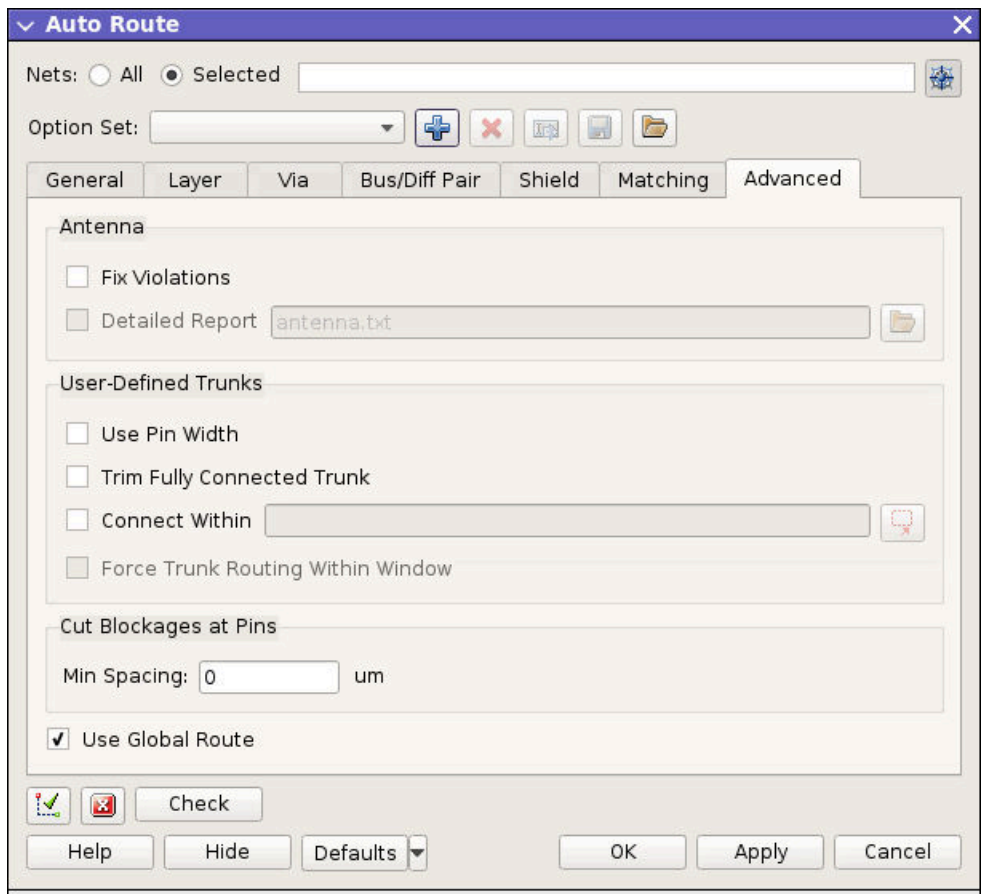
1. Set the Wire Matching constraints. For instructions, see [Wire Matching Constraint](#).
2. In the **Matching** tab, select **Lengthen Within**, and then use one of the following methods to define the wire-lengthening area:

Option	Description
Min X, Y Max X, Y	Type the absolute coordinates in the Min X, Y , Max X , and Y boxes. The Min X, Y coordinates specify the lower left corner of the boundary box. The Max X, Y coordinates specify the upper right corner of the boundary box. 
	Click the button, and then draw a rectangle on the canvas. The coordinate values automatically appear in the Min X, Y , Max X , and Y boxes.

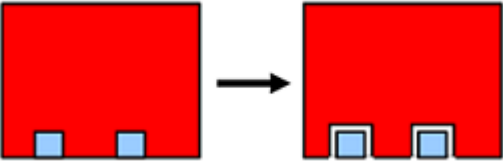
3. In the **Auto Route** dialog box, click **Apply**. The router applies the Wire Matching constraints and adds wire length only in a minimum-length constrained net or adds length for a matched group of nets.

Advanced Tab

The **Advanced** tab options allow you to fix antenna violations, determine how user-defined trunks are handled, and enable global routing.



Option	Description
Antenna: Fix Violations	Fixes any process antenna violations on metal and cut layers. When disabled, the router ignores the process antenna violations. Tcl preference: <code>rtAntennaFixViolations</code>
Antenna: Detailed Report	Generates a detailed report showing all nets affected by the antenna fix operation, in addition to the summary report. - When disabled, the Auto Router generates only the summary report, which appears in the console and log file. - When enabled, the Auto Router generates both the summary report, which appears in the console and log file, and a detailed report in the file specified in the File Name box. The report file is in the text format. By default, it is named <code>antenna.txt</code>, and it is stored in the current working directory. To change the file name, type a new name in the box. Click  to specify a new file location. Tcl preference: <code>rtAntennaDetailedReport</code>

Option	Description
User Defined Trunks	Specifies the following options for objects whose route type is User Trunk: • Use Pin Width: Routes a pin to a trunk using the pin width instead of the width defined in the technology or constraints you specified in the Constraint Editor. • Trim Fully Connected Trunk: Removes overhanging extensions beyond the intersection of trunks and connecting segments. • Connect Within: Lets you define an area within which all wires must connect to corresponding pins using the shortest possible routes. Type the absolute coordinates of the area in the Min X, Y, Max X, and Y boxes. The Min X, Y coordinates specify the lower left corner of the boundary box. The Max X, Y coordinates specify the upper right corner of the boundary box. Alternatively, click , and then draw a rectangle on the canvas. The coordinate values are automatically entered in the Min X, Y, Max X, and Y boxes. The boundary of the defined area is displayed in the design when the routing command is active. • Force trunk routing Within window: Forces the routing within the area you specified. This option is available only if you select Connect Within. Tcl preferences: <code>rtTrunkUsePinWidth</code>, <code>rtTrimConnectedTrunk</code>, <code>rtUseTrunkBox</code>, <code>rtForceInsideTrunkBox</code>
Cut Blockages At Pins: Min Spacing	Creates cutouts in pin blockages so that the router can access the pins during routing. You can specify the distance between the pins and the cutout, in micrometers (um). The following examples show a spacing specification of 0 um (left) and a value > 0 um (right).  Tcl preference: <code>rtCutBlockagesAtPinsMinSpacing</code>
Use Global Route	Approximates the course of the routes before detailed routes are added. Doing so can reduce the number of rip-ups and reroutes, thus improving overall routing performance and results, especially in block-level routing. The option is turned on by default. Tcl preference: <code>rtUseGlobalRoute</code>

8

Interactive Routing

The Interactive Router allows you to manually guide the route, and the router creates the interconnects automatically. You can interactively route one or more nets.

This chapter provides instructions on using the new Interactive Router.

- [About the Interactive Router](#)
- [Enabling the Interactive Router](#)
- [Customizing Your Work Preferences](#)
- [Checking for Design Errors](#)
- [Applying and Ignoring Rules During Routing](#)
- [Follow-the-Cursor Routing](#)
- [Point-to-Point Routing](#)
- [Bus Routing](#)
- [Adjusting Route Widths Automatically](#)
- [Dropping Vias Automatically](#)
- [Replicating the Last Route](#)
- [Cloning Routes](#)
- [Adding Strap Connections](#)
- [Managing Option Sets](#)
- [Route Toolbar and Create Route Dialog Box Reference](#)

For instructions on using the Pre-2020.03 Interactive Router, see the [Custom Compiler Pre-2020.03 Interactive Router User Guide](#).

About the Interactive Router

The Interactive Router lets you route single nets and multiple nets interactively. You manually determine the route, and the router creates the interconnects automatically. When obstacles are encountered, the router gives you manual control, providing additional flexibility to work around complex topologies. Potential design rule violations are displayed on the canvas as you route, allowing you to avoid or fix the errors as they occur.

The Interactive Router supports the following modes:

- **Follow-the-Cursor:** The router creates the interconnect automatically between points as you click on the canvas. See [Follow-the-Cursor Routing](#).
- **Point-to-Point (P2P):** The router creates the interconnect automatically after you specify a startpoint and an endpoint on the canvas. See [Point-to-Point Routing](#).
- **Strap:** The router automatically connects multiple shapes with a single click. See [Adding Strap Connections](#).
- **Bus:** The router creates routes for multiple nets simultaneously. See [Bus Routing](#).
- **Cloning and Welding:** The **Auto Cloning** mode can assist you with device-level routing by automatically replicating routes within and between rows of devices. The **Welding** mode automatically branches (taps off) from the main interconnect (trunk) to connect to nearby pins on the same net. See [Cloning Routes](#) and [Follow-the-Cursor Routing](#).

During the routing operation, the Interactive Router:

- Automatically creates DRC aware routes.
- Complies with non-default rules (NDRs).
- Gives you the option of manually routing around obstructions.
- Automatically inserts vias or lets you manually insert vias. You have the option of selecting and positioning the via.
- Lets you add routes on contactless metal layers for local interconnect routing.
- Automatically tapers the starting or ending paths.
- Automatically snaps individual routes and differential-pair routes to active tracks if you turn on this option.
- Routes unnetted shapes.

Enabling the Interactive Router

Important:

Before you start, make sure you review the information in [Preparing for Routing](#). If you want to route a subsection (area) of the design, define the area. For instructions, see [Defining a Routing Area](#).

You must enable the Interactive Router so that you can set the routing options and start the routing operation.

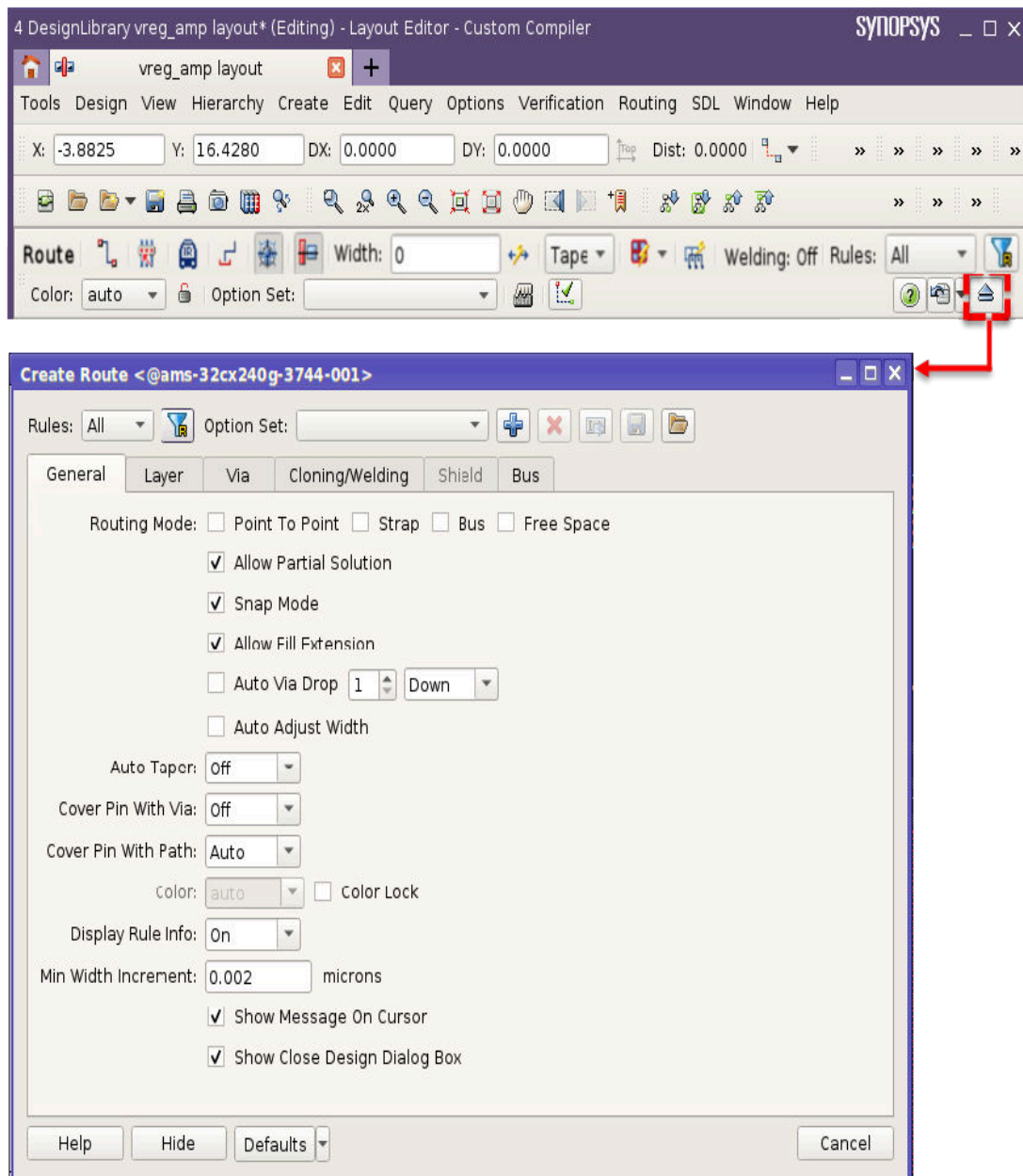
To enable the Interactive Router:

1. In the Layout Editor, make sure the Custom Router is enabled. For instructions, see [Getting Started](#). Make sure the `rtUse202003IR` Tcl preference is `false` (default).
2. Choose **Routing > Interactive Route**.

In the background, the Interactive Router initializes and loads the data. When the initialization and data loading are finished, the **Route** toolbar appears.

3. To display the **Create Route** dialog box, click **Eject**  on the **Route** toolbar.

To hide the dialog box and return to the toolbar, click **Hide** in the dialog box.



Customizing Your Work Preferences

This topic describes the following:

- [Interactive Router Keyboard Shortcuts](#)
- [Displaying and Hiding the Keyboard Shortcut Reference](#)
- [Reconfiguring Keyboard Shortcuts](#)

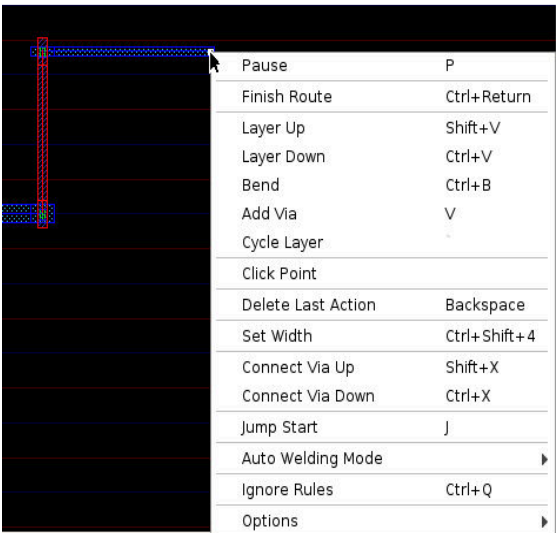
Interactive Router Keyboard Shortcuts

The Interactive Router has its own default set of keyboard shortcuts that you can use to quickly access routing commands and options. Depending on your work habits, you might want to:

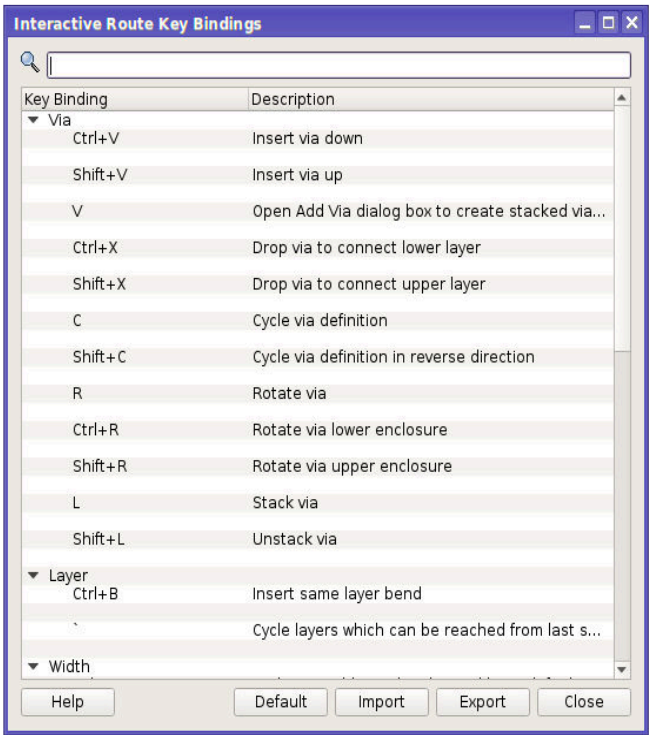
- Right-click on the layout canvas as you route to view the available commands you can use. See [Shortcut Menu Commands \(FTC and P2P\)](#).
- Keep the list of keyboard shortcuts on your desktop and refer to it while you are routing.

The **Interactive Route Key Bindings** dialog box provides a reference of the keyboard shortcuts. The dialog box also lets you customize key bindings.

View Available Commands During Routing



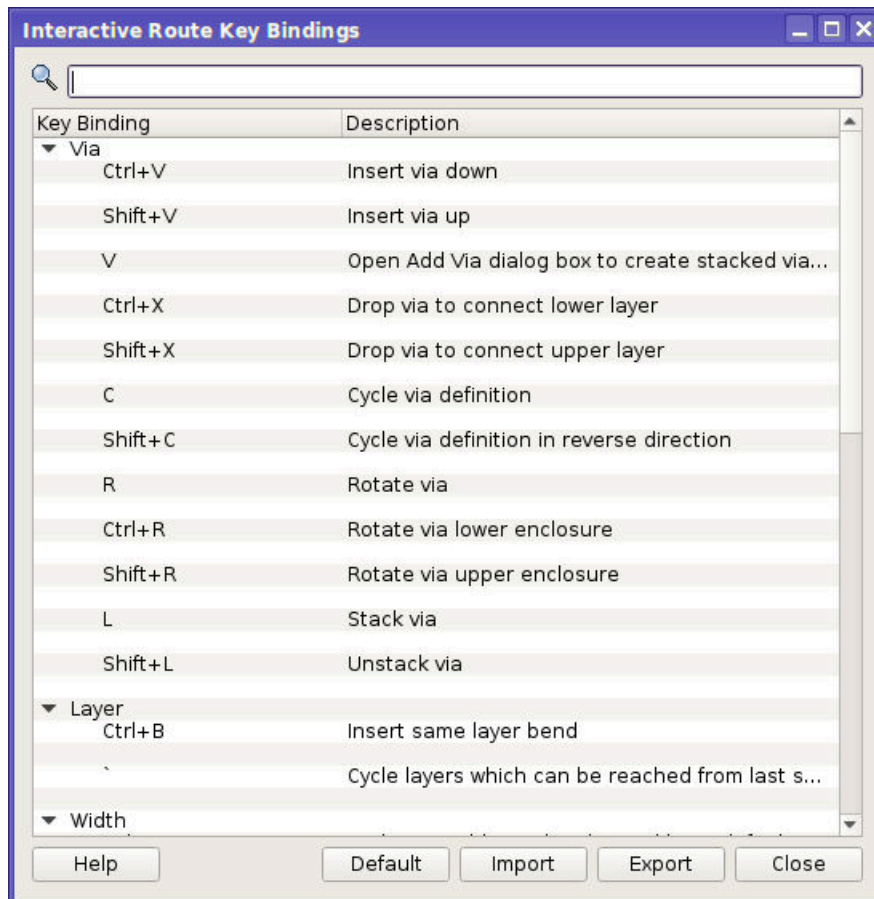
View Available Keyboard Shortcut Reference



Displaying and Hiding the Keyboard Shortcut Reference

To display or hide the Interactive Router keyboard shortcut reference:

1. On the **Route** toolbar, click **Key Binding** . The **Interactive Router Key Bindings** dialog box opens.



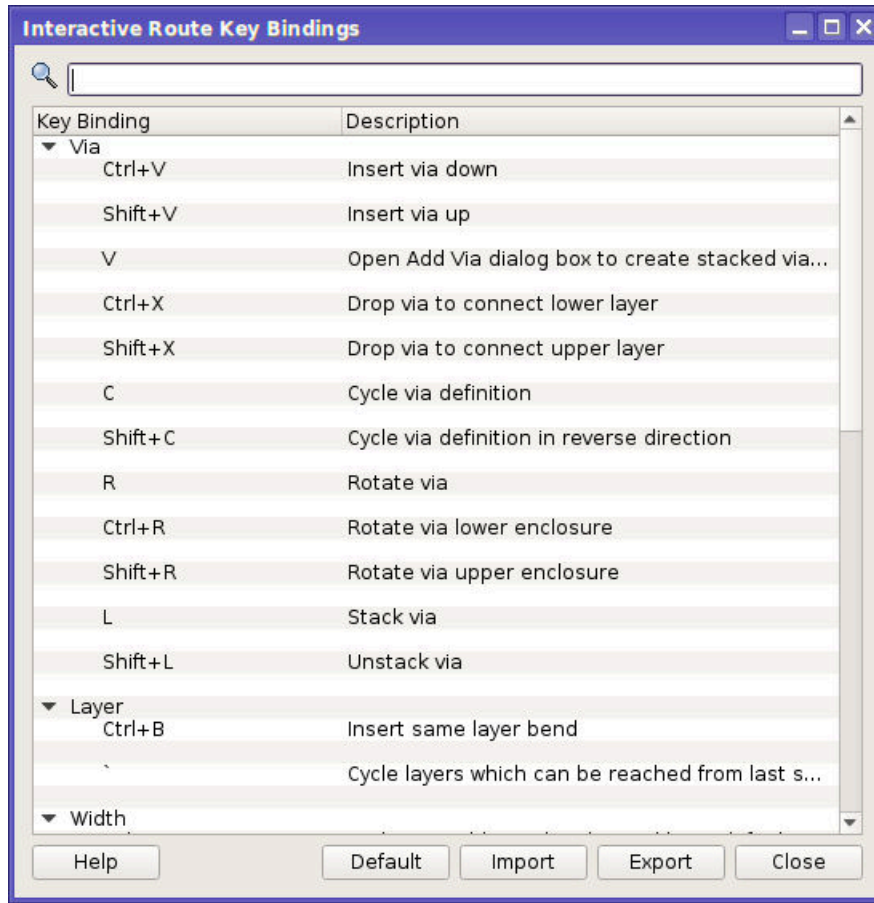
2. When you are finished and want to hide the dialog box, click **Close**.

Reconfiguring Keyboard Shortcuts

Because every designer can have a different work style, you can change the shortcut key assignments to better fit your preferences. This section describes how to reconfigure the Interactive Router keyboard shortcuts. For information about selecting or configuring the global Custom Compiler keyboard shortcuts, see Customizing Keyboard Shortcuts.

To reconfigure the Interactive Router keyboard shortcuts:

1. On the **Route** toolbar, click **Key Binding** . The **Interactive Router Key Bindings** dialog box opens.



2. Click **Export** and specify the location for the exported file.
3. Click **Save**. The key binding file is exported.
4. Open the exported file in a text editor and modify the key bindings.
5. In the **Interactive Router Key Bindings** dialog box, click **Import** and select the file you modified.
6. Click **OK**. The modified key bindings are imported, and you can view the modifications in the dialog box. In addition, you can start to use the modified bindings.

Checking for Design Errors

This topic describes the following:

- [About the Preroute Checker](#)
- [Running the Preroute Checker](#)
- [Viewing and Resolving Design Errors](#)

About the Preroute Checker

Before you start the Interactive Router, you can run the Preroute Checker to make sure the design is free of design-level errors. Doing so ensures that the Interactive Router will run successfully.

The Preroute Checker is a Custom Router feature that checks the design for the following:

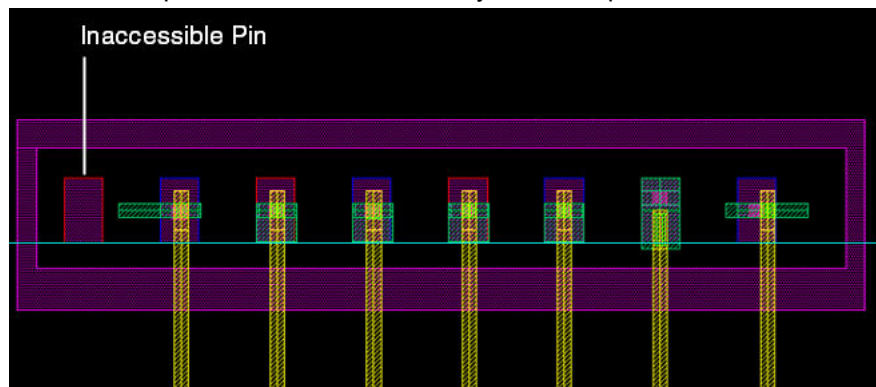
- [Design Errors](#)
- [Design Warnings](#)
- [Technology Errors](#)

Error markers are added in the design and listed in the Error Viewer so that you can easily locate and resolve the errors. In addition, a summary report is displayed in the Console and log file. See [Viewing and Resolving Design Errors](#).

Design Errors

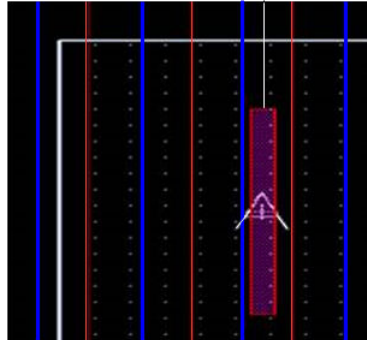
The Custom Router will display error messages for the following design errors:

Type of Design Error	Description
Pin Accessibility	Pins that are not accessibility during routing: • Pins that are not accessible through vias • Pins that are completely blocked • Pin widths that are not within the <code>allowedWidthRange</code> Layer constraint In the following example, the Preroute Checker will display an error for the left-most pin because it is blocked by other shapes.



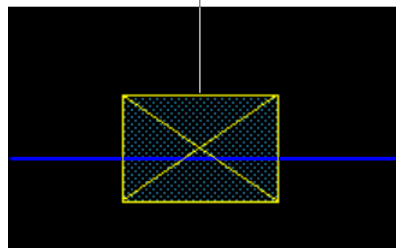
Type of Design Error	Description
Off Track Pin	(Designs that have routing tracks) Pins that are not on routing tracks or have widths different from the tracks. In the following example, the Preroute Checker will display an error for the pin that is placed between the blue and red track.

Off-Track Pin



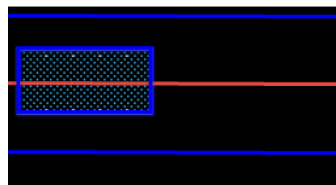
In the following example, the Preroute Checker will display an error for the pin that is on the blue track, but the center of the pin is not on the track.

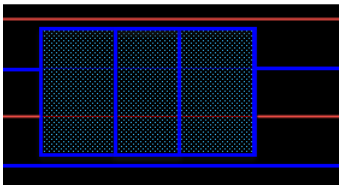
Off-Track Pin



Pin and Track Color	(Designs that have colored routing tracks) Pin colors that do not match track colors. In the following example, the Preroute Checker will display an error for the blue pin that is placed on the red track.
---------------------	---

Blue Pin on Red Track

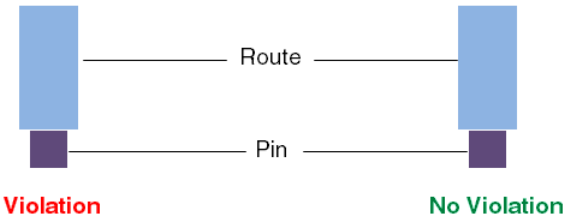


Type of Design Error	Description
Pin Shorts	Short circuits caused by preroute pin shapes that are positioned incorrectly. In the following example, the Preroute Checker will display an error because the two overlapping blue pins are causing short circuitry. Short - Overlapping Pins 

Design Warnings

The Custom Router will display warning messages for the following:

Type of Design Issue	Description
Single Pin in Routing Area	Pins that require connections to other pins that are outside of the defined routing area.
Track Region and Layer Direction Mismatch	(Designs that have routing tracks) The tracks in the track region do not match the routing layer direction. For example, the Preroute Checker will display an error message if the tracks in a track region are vertical, but the direction specified for the routing layer is horizontal.

Type of Design Issue	Description
Pin Size	Pins that have widths that are: - Not within <code>allowedWidthRange</code> and might result in reroutes. For example, the pin width is 0.12, and <code>allowedWidthRange</code> is 0.03 to 0.10. - Narrower than the route layer width constraint, and <code>rtTaperUpAllowed</code> is false. Pin Width < Route Layer Width <code>rtTaperUpAllowed = false</code> <code>rtTaperUpAllowed = true</code>  Pin Width > Route Layer Width <code>rtTaperUpAllowed = true or false</code> 
Unnetted Shape	Shapes that do not have net assignments.
Label on Shapes Without Pins	Shapes that have labels, and the label text should match net names in the design, but: - The shapes do not have pin or net associations. - The net names do not match the label text.
Pin Off Grid Points	(Designs that have routing tracks) Both ends of pins are not on grid points. If only one end of the pin is on the grid point, the Interactive Router will extend the other end to prevent violation.
Nets Without Shapes	Nets that do not have any shapes associated with them.

Technology Errors

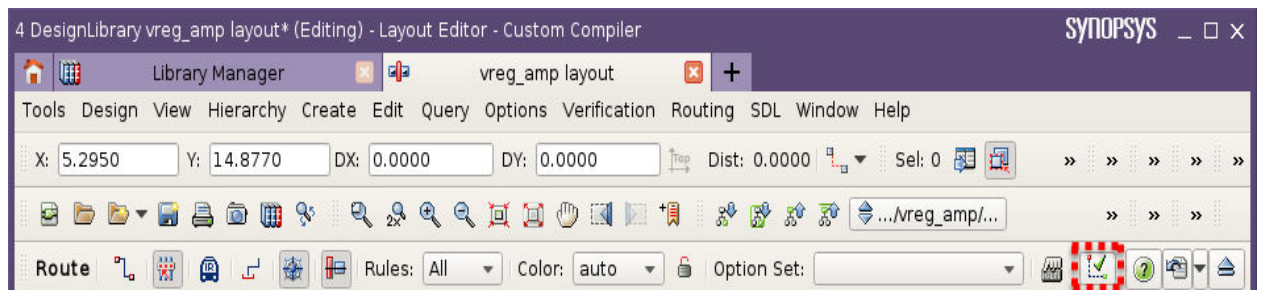
The Custom Router will display error messages for the following errors:

Type of Design Issue	Description
Missing Via Definitions	Missing via definitions between metal layers, starting from M0.
Missing Width Constraints	Missing <code>minWidth</code> constraint for metal, contactless, and poly layers.
Missing Spacing Constraints	Missing spacing constraint for metal, contactless, and poly layers.

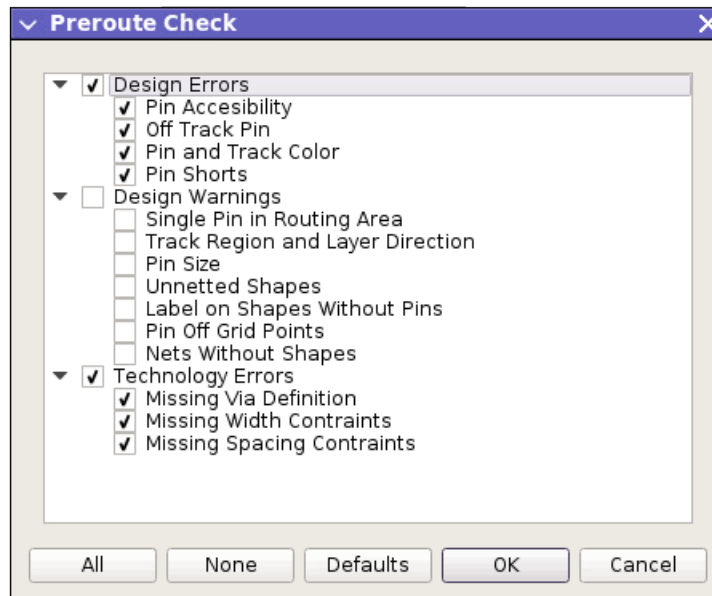
Running the Preroute Checker

To run the Preroute Checker:

1. In the Design Navigator, select the nets you want to check. Alternatively, you can [define the area](#) in which you will run the checker.
2. On the **Route** toolbar, click **Preroute Check**.

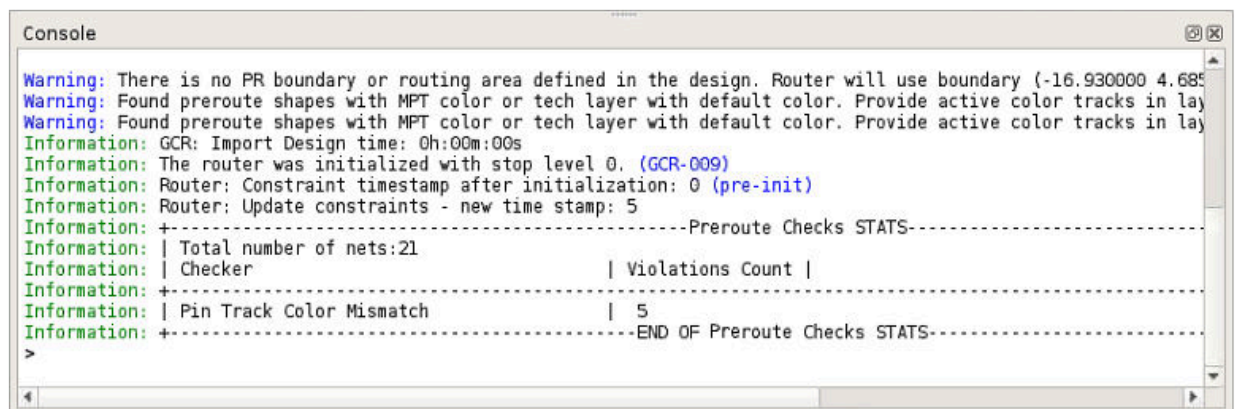


3. In the **Preroute Check** dialog box, select the checks you want to run, and click **OK**.



4. Click **OK**. The Preroute Checker starts checking the design.

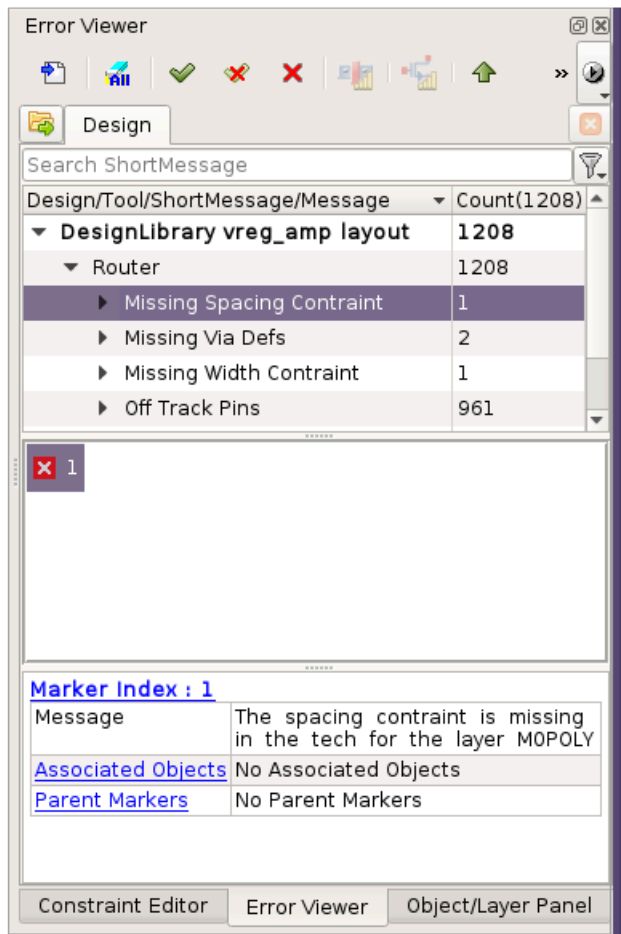
When the checker is finished, error markers appear on the canvas. A summary is available in the Console and log file.



Viewing and Resolving Design Errors

To view and resolve the errors:

1. Display the Error Viewer. The list of errors appears in the viewer.



2. Select an error in the Error Viewer. The viewport zooms in to the location of the error.
3. Resolve the error.

The Error Viewer explains the causes of the errors. See also [Troubleshooting Error Messages](#).

Applying and Ignoring Rules During Routing

To minimize corrections downstream, you can apply design rules during interactive routing. You can also ignore design rules on the last segment if you want to connect unnetted and

netted routes during follow-the-cursor interactive routing. See [Follow-the-Cursor Routing Flow](#).

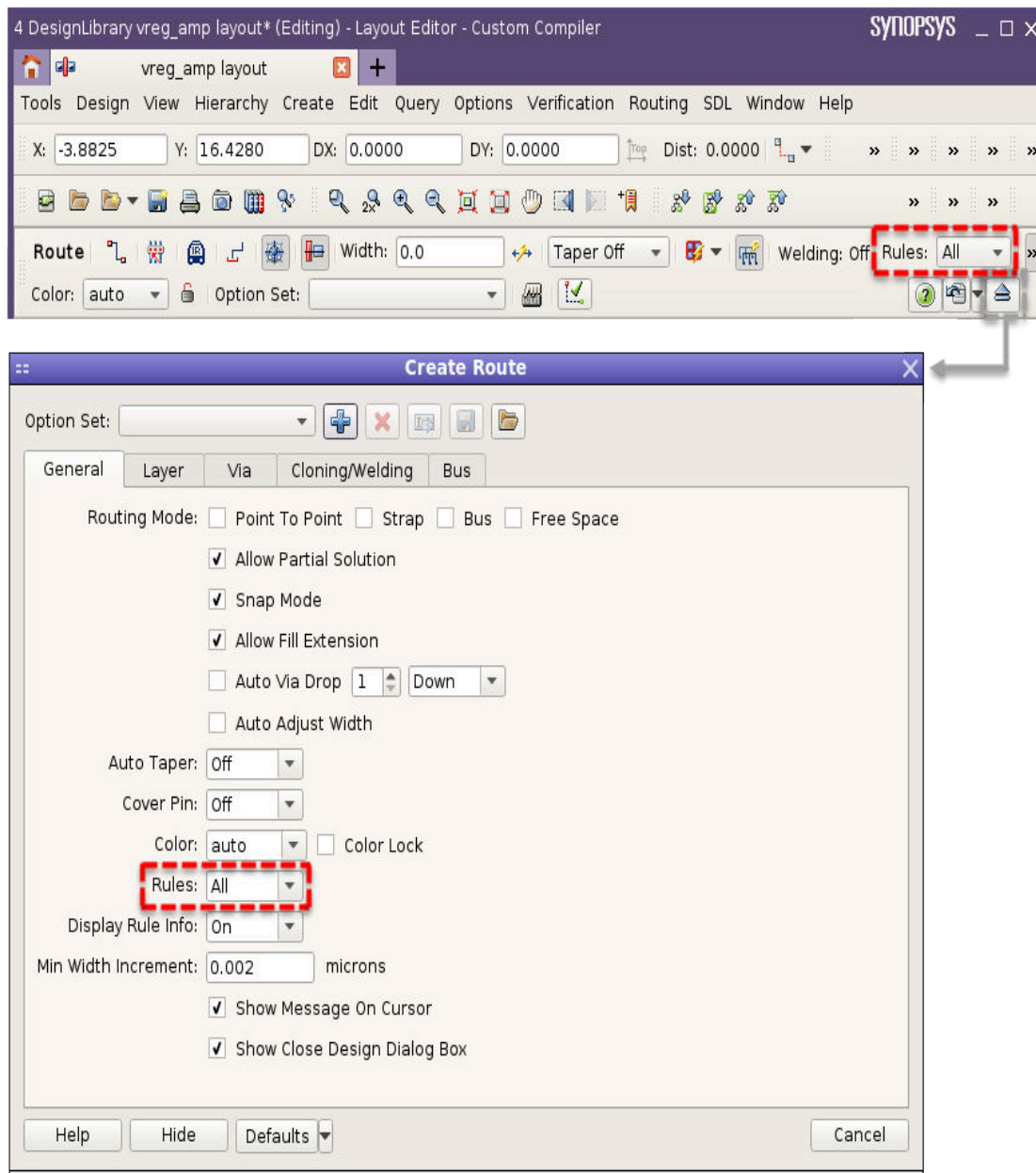
To apply or ignore design rules during interactive routing:

- On the **Route** toolbar or in the **Create Route** dialog box, choose the **Rules** to apply:
 - **All**: Applies all design rules during routing.
 - **LVS**: Applies only layout-versus-schematic (LVS) comparison rules during routing.
 - **None**: Does not apply any design rules during routing.

If you choose **All** or **LVS**, you have the option of pressing Ctrl+Q to ignore the rules for the current segment during routing. (See [Interactive Router Keyboard Shortcuts](#).) Doing so allows you to draw the route segment despite the rule violation, such as a blockage, and return later to resolve the issue (for example, remove the blockage). To resume to apply the rules during routing, press Ctrl+Q again.

Chapter 8: Interactive Routing

Applying and Ignoring Rules During Routing



Follow-the-Cursor Routing

This topic describes the following:

- [Before You Start](#)
- [Follow-the-Cursor Routing Flow](#)
- [Assigning and Locking Colors Manually](#)
- [Bending Routes](#)
- [Dropping Vias Manually](#)
- [Adding Vias Manually](#)
- [Routing Around or Through Obstacles](#)
- [Switching to Point-to-Point Mode](#)
- [Welding](#)

Before You Start

Before you start:

- Review the tasks to be completed before you start routing. See [Getting Started](#).
- Enable the Interactive Router and display either the **Route** toolbar or the **Create Route** dialog box. See [Getting Started](#).
- Set up your keyboard shortcuts to best fit your work style. See [Customizing Your Work Preferences](#).

Follow-the-Cursor Routing Flow

To create a route using the follow-the-cursor method:

1. Make sure you are in the Follow-the-Cursor mode. On the **Route** toolbar:

- To start routing from a pin, turn off **Point to Point**  and **Free Space FTC** .



- To start from anywhere on the canvas, turn on **Free Space FTC** .



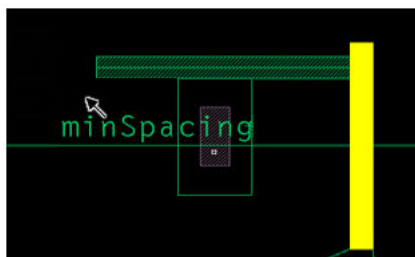
2. Set the routing options. See [Route Toolbar and Create Route Dialog Box Reference](#).
3. (Optional) To display the keyboard shortcuts, on the **Route** toolbar, click **Key Binding** , or press H. You might need to refer to the shortcuts during routing. See [Interactive Router Keyboard Shortcuts](#).
4. Start routing using one of the following methods:
 - (Follow-the-Cursor) To start from a pin, click the pin on the canvas. The pin's net and layer information are automatically used for routing.
 - (Free Space FTC) To start from anywhere on the canvas, select the net in the Design Navigator, select the layer-purpose pair in the Object/Layer Panel, and click the location on the canvas. If you selected **Tap Layer**, the starting path inherits the layer information from the starting shape or pin.

If multiple shapes are under the pointer, the active object is selected. Information about the active object is displayed on the Layout Editor Object Info toolbar. See Object Info Toolbar.

5. If you are manually adjusting the path width, use Ctrl+scroll up to increase the width, and Ctrl+scroll down to decrease the width.
6. Move the pointer on the canvas to create the route.

A path is drawn on the canvas. If you are routing with tracks, the path is snapped to the nearest track.

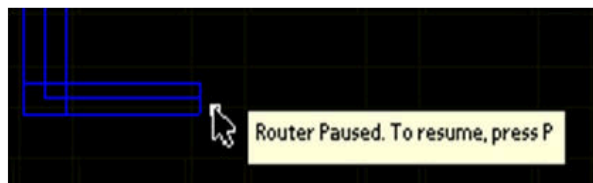
If the path violates design rules, a message appears on the canvas to let you know the specific rule that is violated. You can fix the violation before continuing. In the following example, the `minSpacing` rule is violated. To fix the violation, move the route away from the rectangular obstacle.



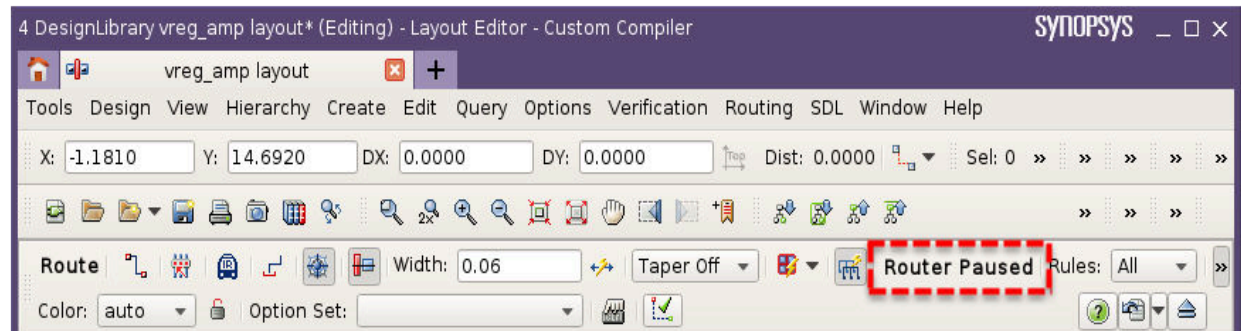
7. See the following for additional routing operations:
 - [Assigning and Locking Colors Manually](#) (in designs that have colored layer definitions in the technology database)
 - [Bending Routes](#)

- Adding vias using one of the following methods:
 - [Dropping Vias Manually](#)
 - [Adding Vias Manually](#)
 - [Dropping Vias Automatically](#)
 - [Routing Around or Through Obstacles](#)
 - [Switching to Point-to-Point Mode](#)
 - [Welding](#)
8. If you make a mistake during the follow-the-cursor mode, press Backspace to remove the last path added.
9. To pause the routing operation:
- a. At the location where you want to pause or make adjustments, right-click the canvas and choose **Pause**. Depending on the `rtDisplayMessageOnCursor` Tcl preference setting, the **Router Paused** message appears on the layout canvas at the pointer or on the **Route** toolbar.

`rtDisplayMessageOnCursor = true`



`rtDisplayMessageOnCursor = false`



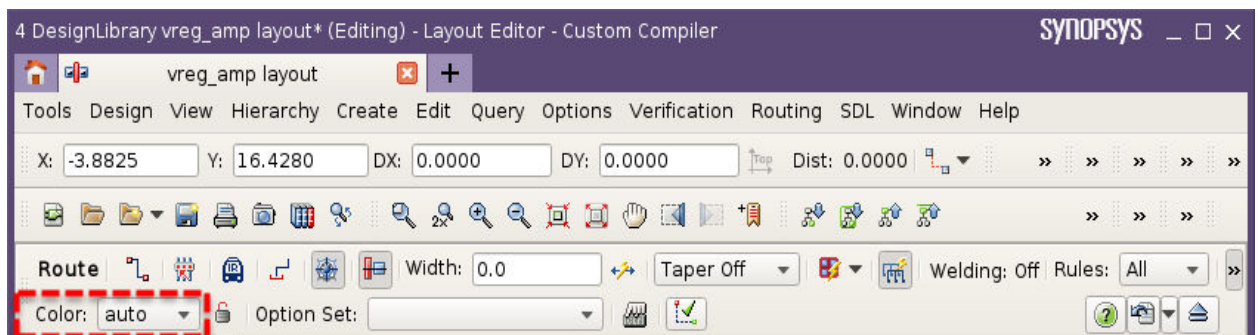
- b. If you want to use one of the shortcut menu commands to make adjustments, choose the command. See [Shortcut Menu Commands \(FTC and P2P\)](#).
 - c. To resuming routing, right-click the canvas and choose **Resume**.
10. To connect an unnetted route to a netted object, right-click and choose **Ignore Rules** when adding the last path segment so that the connection does not create a DRC error.
- You do not need to use Ignore Rules when connecting a netted route to an unnetted object.
11. To complete the route, do one of the following:
- Click the final compatible pin or wire.
 - Ctrl+click the final compatible pin or wire to end the route and start the next route from the same pin or wire.
 - Double-click any place in the design, including open space.
 - Press Enter at any place in the design, including open space.
12. Press Esc to exit the follow-the-cursor routing command.

You can view and edit the route shape properties in the Property Editor. For instructions on viewing and editing via shape properties, see [Modifying Via Attributes and Properties](#).

If error messages appear in the Console during routing, see [Troubleshooting Error Messages](#).

Assigning and Locking Colors Manually

In designs that contain color track definitions in the technology database, you can assign colors to path segments using the **Color** and **Color Lock**  settings on the **Route** toolbar. The **Color** setting applies to the current path segment and subsequent segments until you change the selection. The **Color Lock** setting is applied globally to all routes in the design.



The **Color** selections are:

- **auto:** (Default) Automatically assigns the color of the track to the path based on the Color Propagation Rules. For example, if you place the path on a red track, the path will be assigned the red color. If no propagation can be applied, then the shape is assigned the default layer color.

In the point-to-point routing mode, **auto** is the only applicable Color selection.

- **<colorName>:** Assigns the selected color, <colorName> to the path segment. The available color names are defined in the technology file.

When a path segment has a color, it can only snap to tracks that has the same color. For example, if a path segment is red, it can only snap to red tracks.

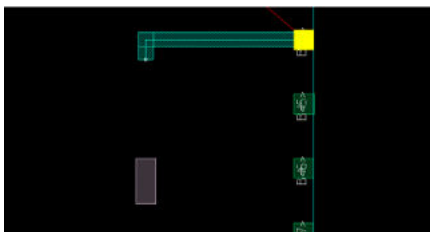
If the selected color is invalid, the path segment inherits the track color as if the color selection is **auto**. For example, if you select green, but the layer can only be red or blue, the path segment inherits the blue color when placed on a blue track.

- **same:** Assumes that nearby shapes have the same color and applies the `multiPatternSpacing` constraint.
- **No Color:** Does not apply any color to the route segment. Select this value if you want to assign colors later using a different utility.

If a layer has two colors, and you selected color1, the router will route on color1 tracks only and will ignore color2 tracks. If `leColorName` is set to auto, then IR is free to choose any track. If `leColorName` is set to No Color, then IR is free to choose any track. If `leColorName` is set to same, then IR is free to choose any track and will apply same color spacing wrt to all the existing shapes irrespective of their colors.

Bending Routes

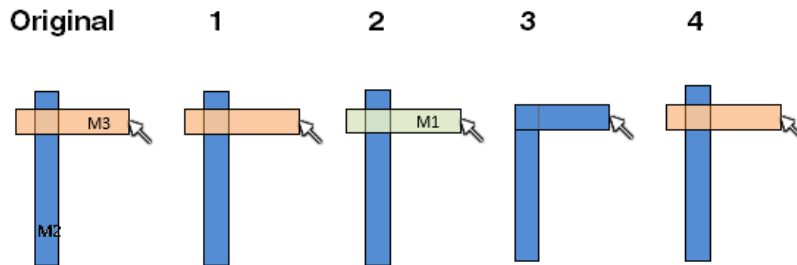
To add same-layer bends, press Ctrl+B. A bend is added on the current layer.



The bend location changes as you move the pointer. The bend commits when you click the canvas or press Spacebar.

To lock or change the layer of the path after a bend:

- Press L once to lock the current layer so that it cannot be changed.
- Press L a second time to cycle to the next layer below the current layer.
- Press L a third time to cycle to the current layer, forming a same-layer bend.
- Press L a fourth time to cycle back to the original solution.



The router will only place routes on visible layers. For example, if M1 is not visible, when you create a bend from M2, the perpendicular path will not be placed on M1. To change the visibility of a layer, see [Making LPP Sets Visible and Selectable](#).

Dropping Vias Manually

You drop vias to:

- Add vias and connect to layers above or below the routing layer.
- Continue routing on the current routing layer.

To drop one or more vias manually, use one of the following keyboard shortcuts when the pointer is over the target shape:

- Ctrl+X: Drops one or more vias to connect to the target layer below the current layer.
- Shift+X: Drops one or more vias to connect to the target layer above the current layer.

For example, if you are currently routing on m1, when you reach the target m5 shape, press Shift+X to drop vias to connect to the m5 shape, and then continue to route on m1.

If you want the vias to be added automatically during routing, turn on **Auto Via Drop**  on the **Route** toolbar. See [Dropping Vias Automatically](#).

Adding Vias Manually

You add vias to:

- Add vias and connect to layers above or below the routing layer.
- Continue routing on the new target routing layer.

To add one or more vias manually and continue routing from the new target layer, use one of the following keyboard shortcuts:

- Ctrl+V: Adds a via below the current layer.
- Shift+V: Adds a via above the current layer.
- V: Opens the [Add Via Dialog Box](#) and lets you specify the connecting layers, via definitions, via stacking, and number of cuts.

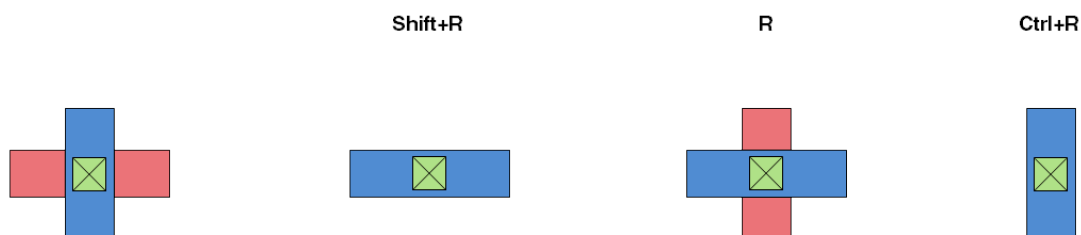
For example, if you are currently routing on m1, you can press Shift+V to add vias to connect to an m5 shape, and then continue to route from m5.

If Auto Adjust Width is turned on, the width of the connecting path is adjusted to be rule-compliant.

To select the via definition, press C to cycle through the available via definitions until the desired via appears on the canvas. Click to commit the via.

To rotate the via, use one of the following keyboard shortcuts:

- R: Rotates the entire via.
- Ctrl+R: Rotates only the metal layer below the cuts.
- Shift+R: Rotates only the metal layer above the cuts.



When connecting paths that are several layers apart, stacked vias are automatically added. To remove the stacking, press Shift+S. To add stacked vias, press S.

The via position changes as you move the pointer. The via position commits when you click the canvas or press Spacebar.

Note:

In the previous release, when you pressed M, the tool exited the interactive routing and invoked the `Move` command.

Now, when you press M, the tool mirrors custom Via.

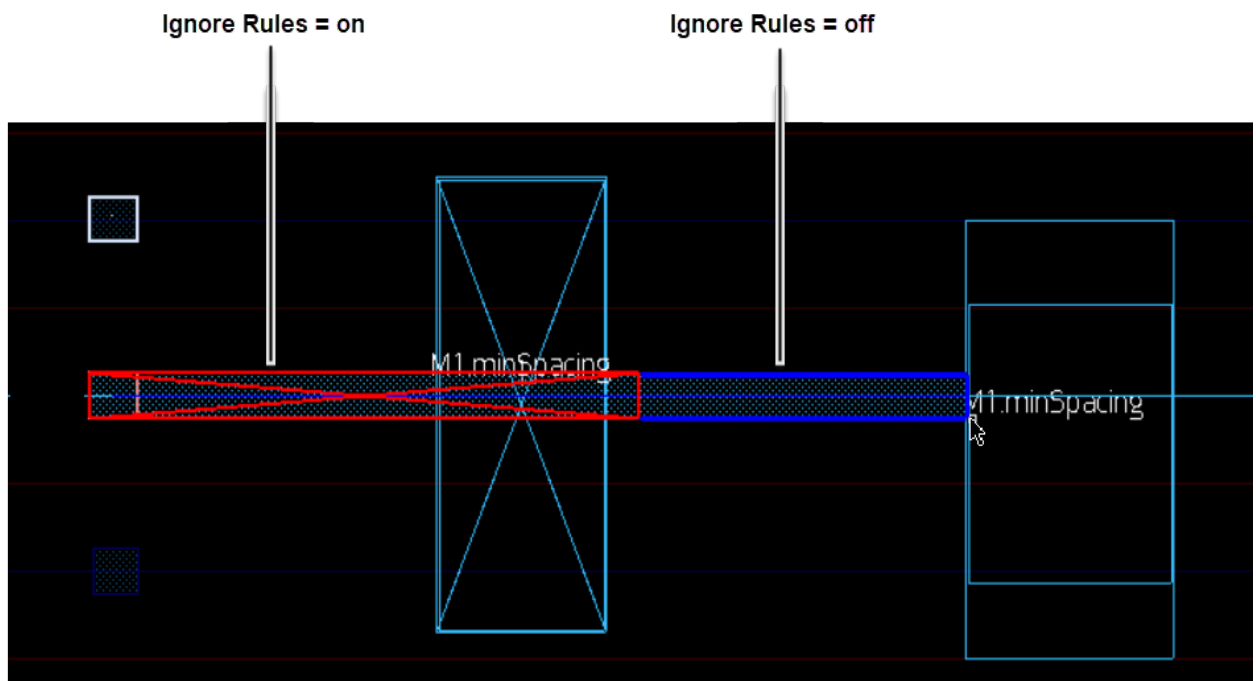
Routing Around or Through Obstacles

Click the canvas to add route points as you work around obstacles.

If you want the route through an obstacle:

1. As the route approaches the obstacle, press `Ctrl+Q` to turn on the Ignore Rules mode.
2. Continue to route through the obstacle.
3. After the route clears the opposite edge of the obstacle, press `Ctrl+Q` to turn off the Ignore Rules mode.

The route segment created during the Ignore Rules mode and the obstacles that are routed through have X indicators to provide a visual reminder. You can return to this location and adjust the route or the obstacle later.



Switching to Point-to-Point Mode

To turn on the point-to-point mode and allow the router to create routes automatically between two points:

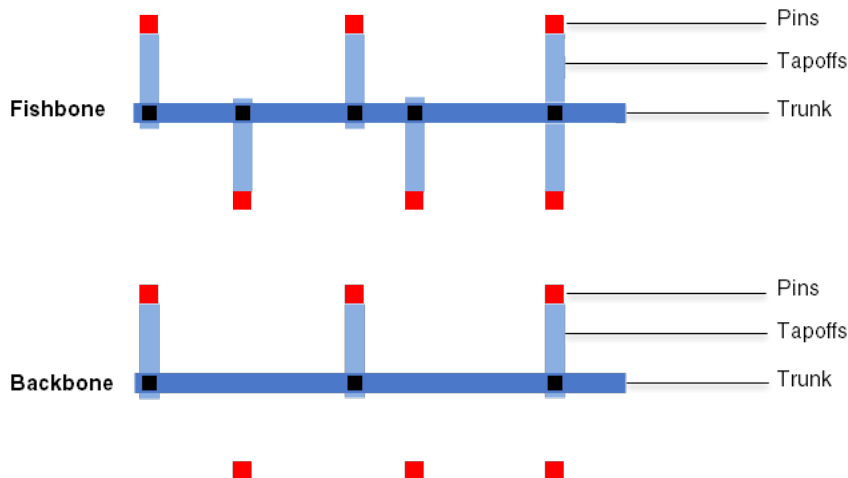
1. Position the pointer at the starting location on the canvas, and press J to turn on the Jump (point-to-point) mode. The route stops following the pointer and flight lines appear on the canvas.
 2. (Optional) In the Object/Layer Panel, select the layer to use at the end point.
 3. On the canvas, click the end location or target object. An optimal route is created, based on the cost, layer, direction, and other options you specified in the **Create Route** dialog box.
 4. To exit the Jump mode, press Backspace. You can continue routing using the Follow-the-Cursor mode.
-

Welding

To turn on the **Welding** mode, press E.

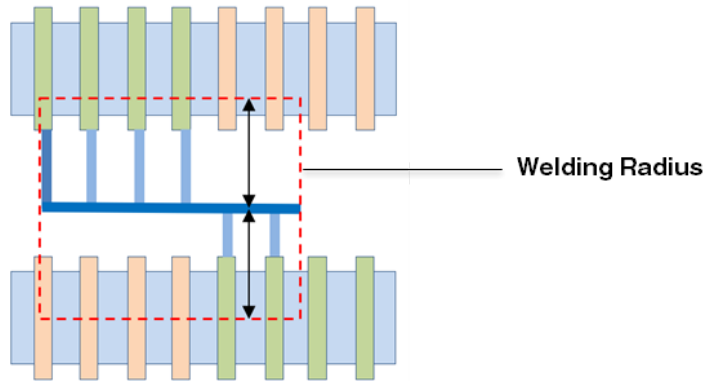
Welding automatically branches (taps off) from the main interconnect (trunk) to connect to nearby pins on the same net.

Pressing E once creates Backbone structures. Pressing E a second time creates Fishbone structures.



The router considers adding tapoff connections to objects that are within a certain distance from the trunk in the fishbone mode, called Welding Radius. A rectangle is drawn around the trunk to indicate the distance. To increase the distance, press Shift+Control+ScrollUp.

To decrease the distance radius, press Shift+Control+ScrollDown. You can also set the Welding Radius in the **Create Route** dialog box. See [Route Toolbar](#).



By default, pins on all the layer are considered for welding. If you want to exclude pins on specific layers, make sure they are not selected in the **Create Route** dialog box. See [Route Toolbar](#).

Point-to-Point Routing

This topic describes how to use the graphical user interface for point-to-point routing. To run point-to-point routing in batch mode, use the `rt::createP2PRoute` Tcl command.

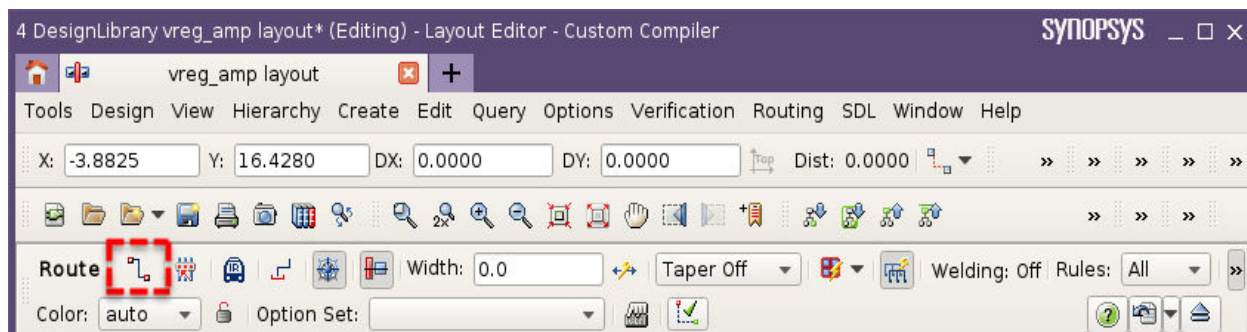
Before you start:

- Review the tasks to be completed before you start routing. See [Getting Started](#).
- Enable the Interactive Router and display either the **Route** toolbar or the **Create Route** dialog box. See [Getting Started](#).
- Set up your keyboard shortcuts to best fit your work style. See [Customizing Your Work Preferences](#).

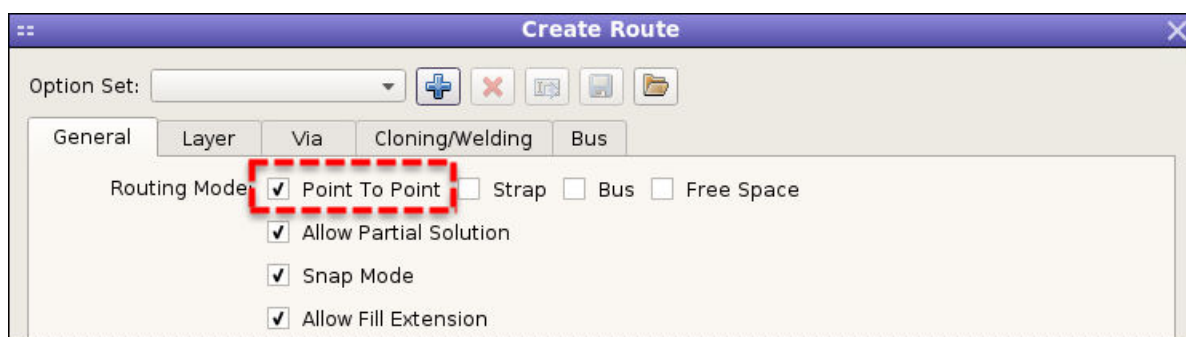
To create a route using the point-to-point method:

1. Turn on the Point-to-Point mode using one of the following methods:

- On the **Route** toolbar, click **Point to Point** .



- In the **Create Route** dialog box, select **Point to Point**.



If you are routing a design that contain color track definitions in the technology database, the only **Color** selection available is **auto**. See [Route Toolbar and Create Route Dialog Box Reference](#).

2. Set other routing options. See [Route Toolbar](#).

3. Start routing using one of the following methods:

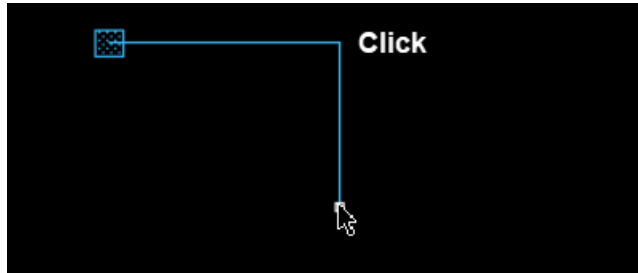
- To start from a pin, click the pin on the canvas. The pin's net and layer information is automatically used for routing.
- To start from any location, select the net in the Design Navigator, select the layer-purpose pair in the Object/Layer Panel, and click the location on the canvas.

If multiple shapes are under the pointer, the active object is selected. Information about the active object is displayed on the Layout Editor Object Info toolbar. See [Object Info Toolbar](#).

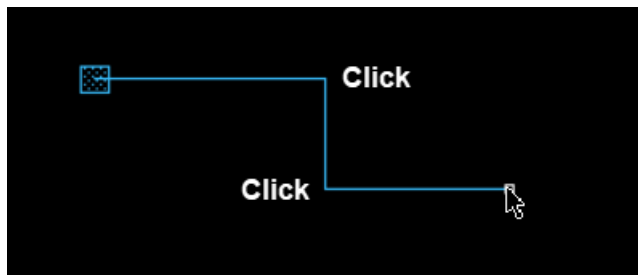
4. Continue clicking to specify intermediate guide points.

Examples:

- To create an L-shaped turn, click once on the canvas and continue in the perpendicular direction.

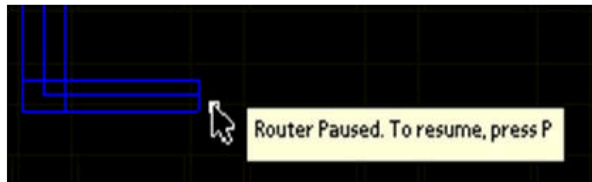


- To create a Z-shaped route, click two times on the canvas.

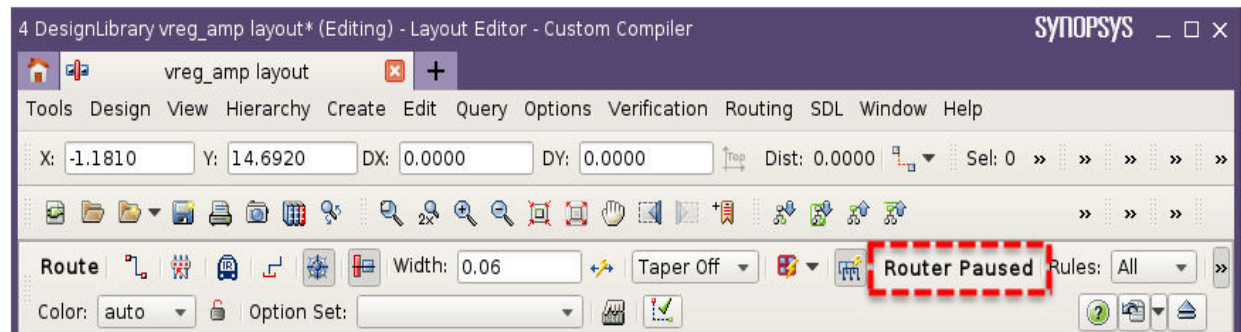


5. If you make a mistake, you can press Backspace to remove previously entered points.
6. To pause the routing operation:
 - a. At the location where you want to pause or make adjustments, right-click the canvas and choose **Pause**. Depending on the `rtDisplayMessageOnCursor Tcl` preference setting, the **Router Paused** message appears on the layout canvas at the pointer or on the **Route** toolbar.

rtDisplayMessageOnCursor = true



rtDisplayMessageOnCursor = false



- b. If you want to use one of the shortcut menu commands to make adjustments, choose the command. See [Shortcut Menu Commands \(FTC and P2P\)](#).
- c. To resuming routing, right-click the canvas and choose **Resume**.
7. To complete the route, click a compatible pin or wire.
8. Press Esc to exit the point-to-point routing command.

Note:

If the route fails, review the console messages. You might need to provide more precise point-to-point guidance.

If error messages appear in the Console during routing, see [Troubleshooting Error Messages](#).

Bus Routing

You can use the Interactive Router to route multiple nets simultaneously. The interactive bus routing method checks for rule violations as you route. Doing so lets you avoid the violations and reduce costly iterations in your design process.

This topic describes the following:

- [About Interactive Bus Routing](#)
- [Routing Bus Bits](#)

About Interactive Bus Routing

The Interactive Router **Bus FTC** mode provides features to facilitate multi-net routing. Understanding the bus routing concepts and how the features work can produce optimal results.

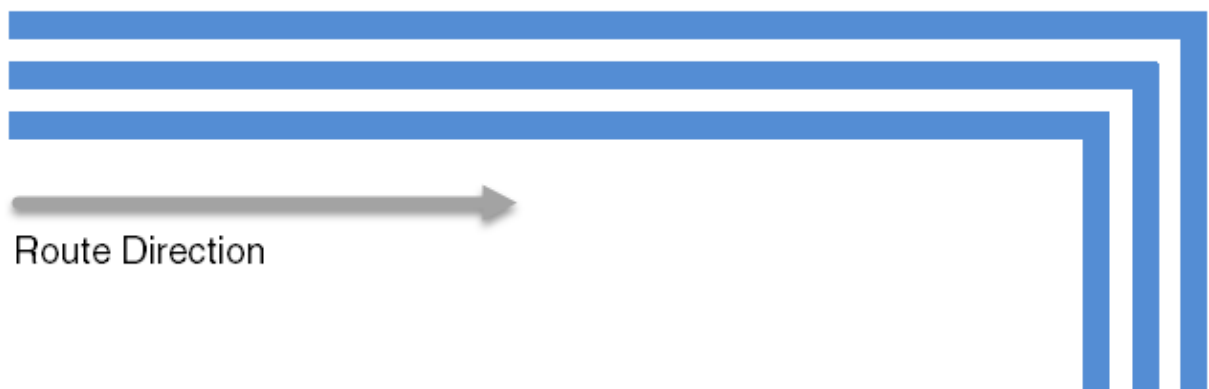
This topic describes the following:

- [Bus Routes and Leader Bits](#)
- [Bit Spacing](#)
- [Corner Styles](#)
- [Relative Routing](#)

Bus Routes and Leader Bits

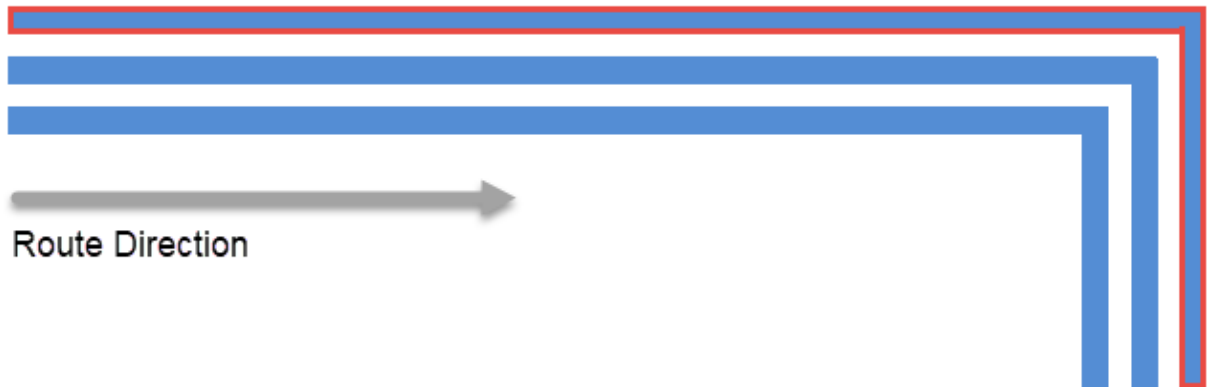
A bus is a group of nets that you route simultaneously. Individual routes in a bus are called bits.

Bus with Three Bits



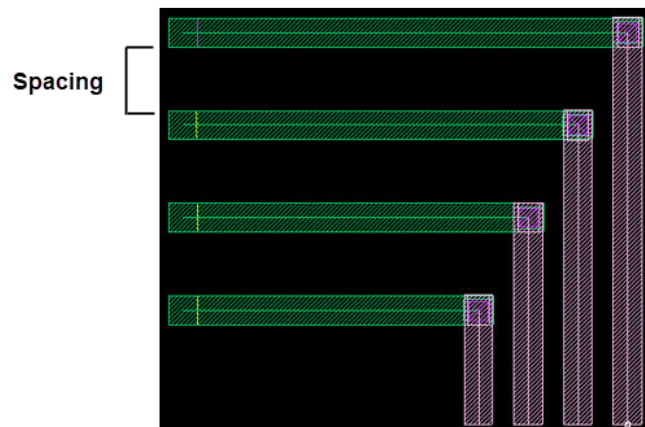
When you select the nets to route together, you specify a *leader net* or *leader bit*. The remaining nets become the *follower nets* or *follower bits*. During routing, you guide the leader bit on the layout canvas, and the follower bits try to remain as close to the leader bit as possible while copying the topology and direction of the leader bit.

Bus with Leader Bit



Bit Spacing

Bit spacing is the edge-to-edge distance between bits.

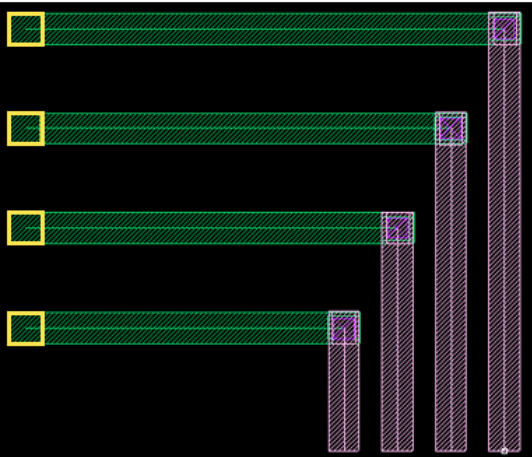


The bit spacing applied depends on the following:

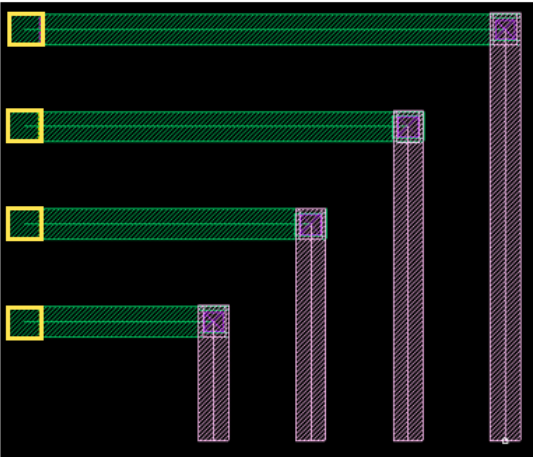
- The **Spacing** value you specify on the **Route** toolbar or **Create Route** dialog box.
- If you do not specify a **Spacing** value:
 - `minSpacing`, if the **Rules** option is **None** or **LVS**.
 - Calculation based on all rules (including `minSpacing`) if the **Rules** option is **All**.

You can also specify that the bus bit spacing match the spacing of the starting pins using the **Keep Pin Spacing** option (`rtBusKeepPinSpacing`).

Default Spacing

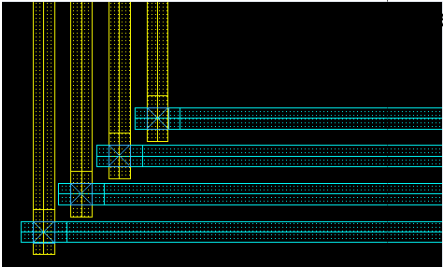
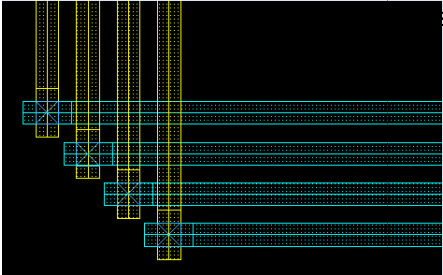


Keep Pin Spacing



Corner Styles

Two corner styles are possible when creating bends in bus routing:

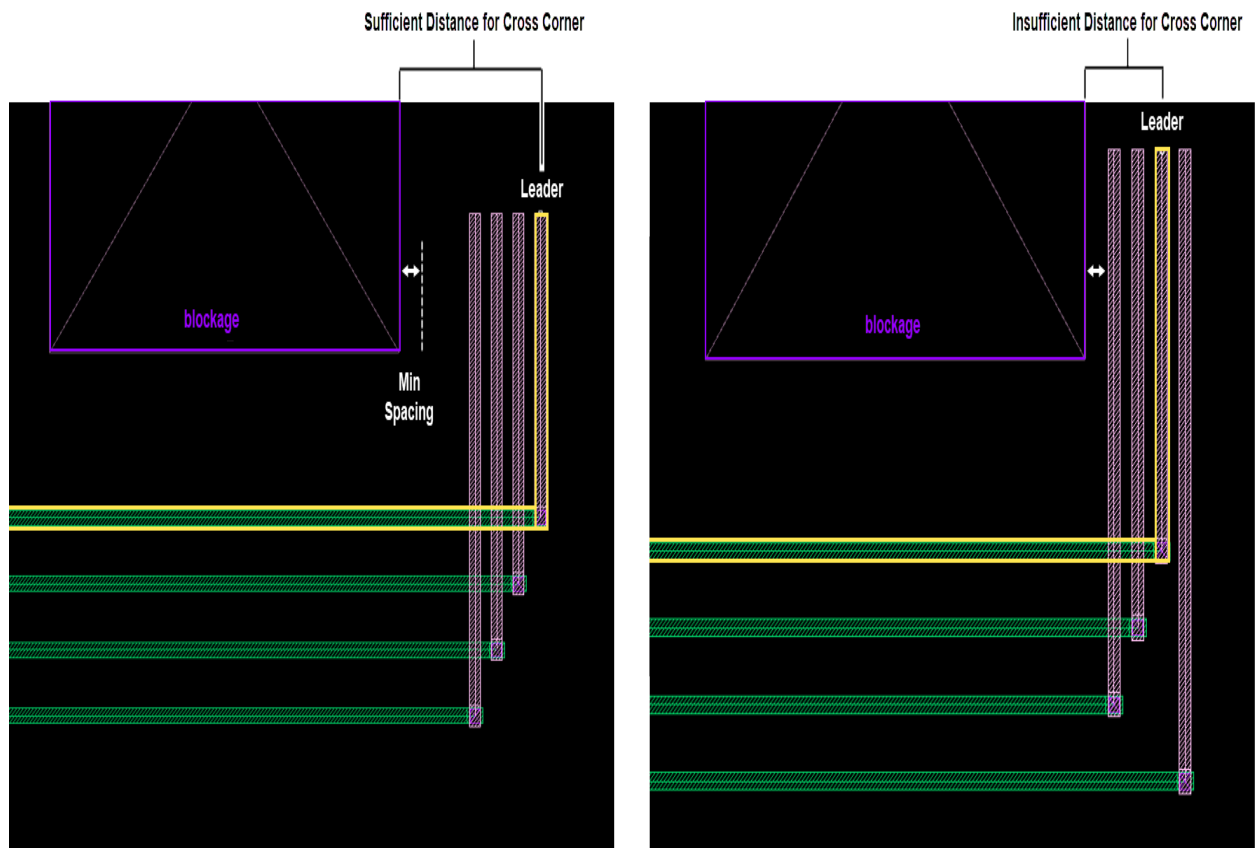
Style	Name	Description
	River	Routes the bits so that they do not overlap at the corner.
	Cross	Routes the bits so that they overlap at the corner.

By default, the Interactive Router automatically determines the best style to use. However, you can explicitly specify the River style or the Cross style.

Regardless of the setting, the Interactive Router tries to apply the specified style whenever possible. However, to prevent design-rule violations, the Interactive Router might use a different style or combine styles.

In the following example, the Cross corner style is selected.

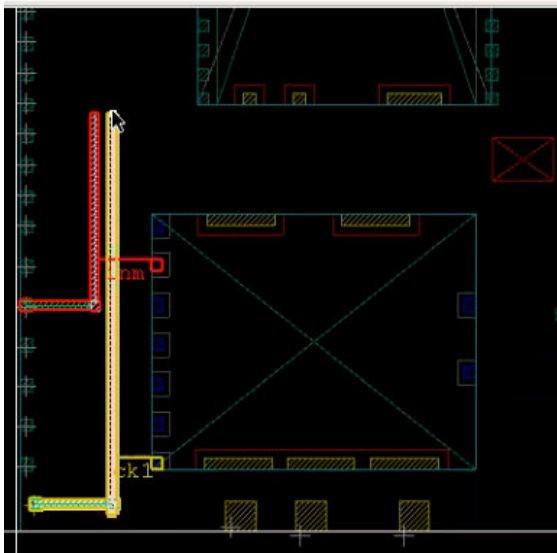
- If you provide sufficient distance between the leader bit and the obstacle so that all follower bits can fit between them, the Cross style is used.
- If you place the leader bit too close to the obstacle so that the fourth bit cannot fit next to the obstacle, the fourth bit is moved to the right of the leader bit.



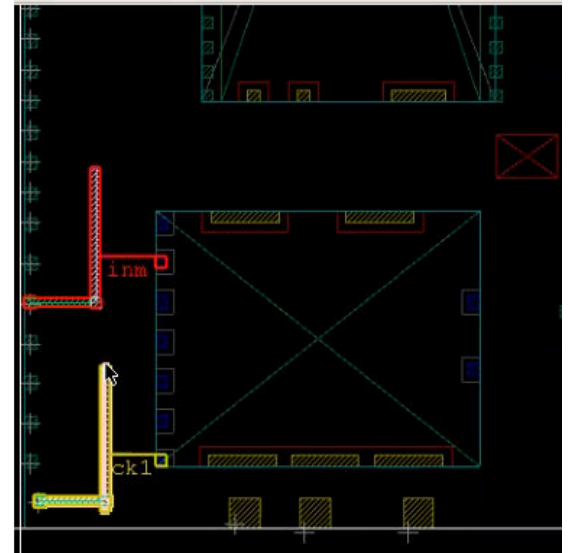
Relative Routing

During bus routing, the follower bits try to remain as close as possible to the leader bit so that the ends of the bits are aligned. However, you can route bits relative to the pin location so that after bending, the ends of the bits can cascade and not align. Use the relative mode when the starting pins and the ending pins do not align spatially.

Remain Close to Leader Bit



Remain Relative to Starting Pin Locations

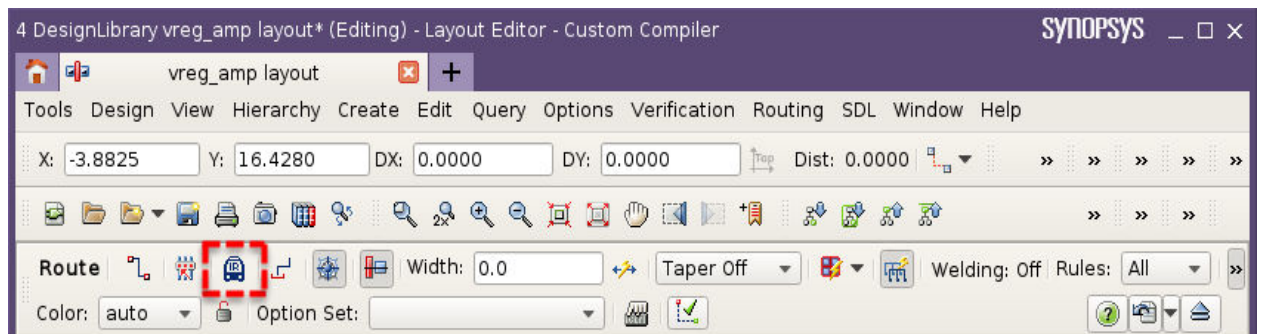


Routing Bus Bits

To start interactive bus routing, turn on the **Bus FTC** mode on the **Route** toolbar, and set the options. You can preset some options before you start. You can also change these settings during the routing operation and apply the new settings.

To route multiple nets:

1. On the **Route** toolbar, click **Bus FTC**  to turn on the bus routing mode.

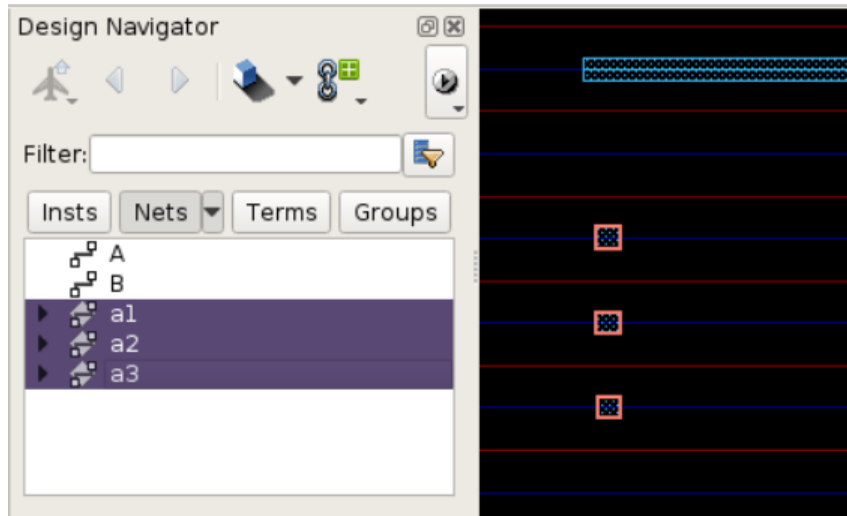


2. Set the bus routing options using one of the following:
 - [Route toolbar](#)
 - [Create Route dialog box](#)

3. Select the nets using one of the following methods:

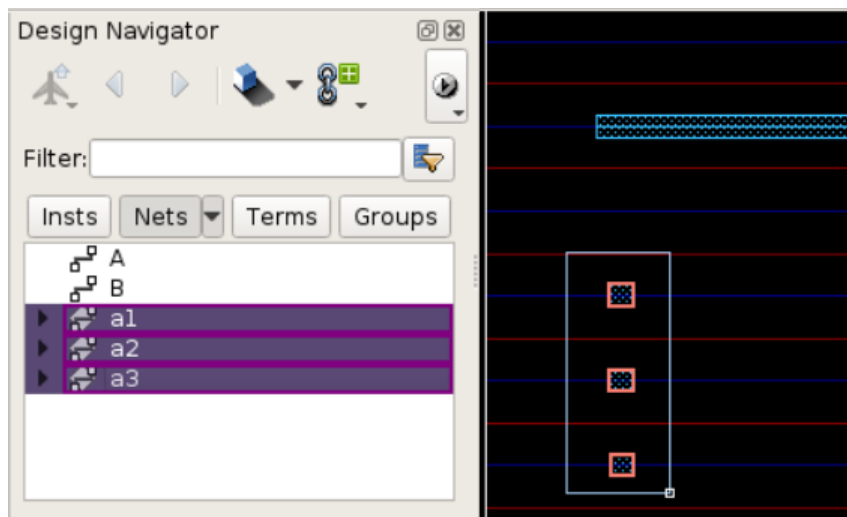
- In the Design Navigator, select the nets.

On the layout canvas, the starting shapes are highlighted.



- On the layout canvas, area-select the starting shapes. The starting shapes of the nets (such as pins) must be on the same layer, and they should align partially or fully.

The pins on the layout canvas are highlighted. In addition, the Design Navigator shows that the corresponding nets are selected.

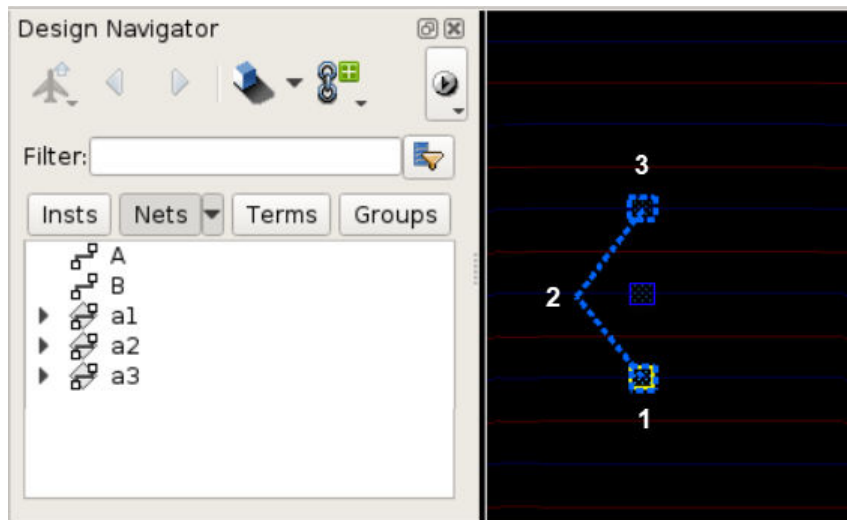


- On the layout canvas, make sure no net is selected in the Design Navigator, and select the starting shapes individually. Alternatively, right-click the first shape you want to select and choose **Select Bus**.

To select non-contiguous shapes, click the empty space near the shapes you want to skip.

As you make the selections, a blue dotted outline appears around the selected shapes. A blue dotted line follows the movement of the pointer.

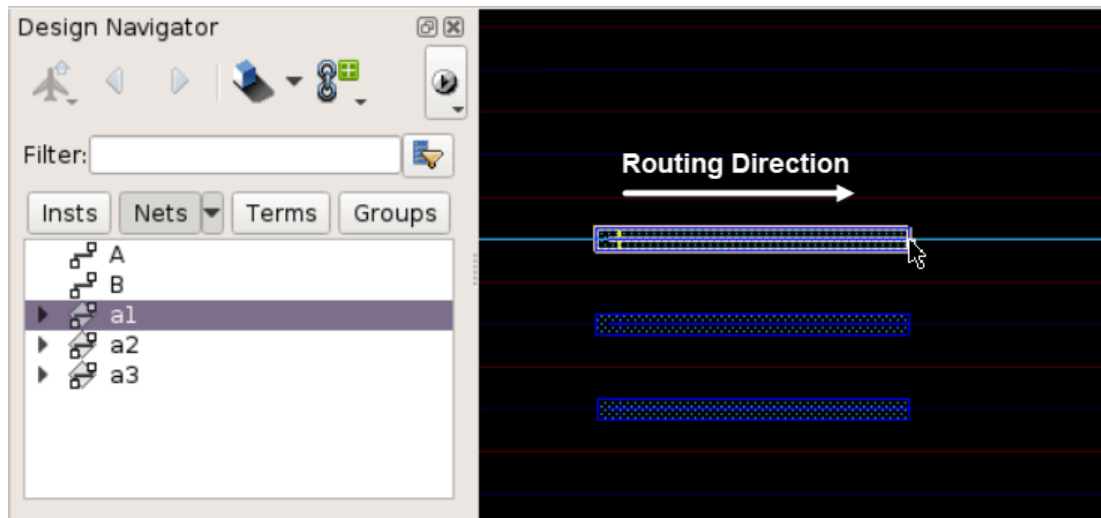
When you are finished, double-click the last shape.



4. Select one of the starting shapes so that it becomes the leading bit. The remaining bits automatically become follower bits.

During routing, the topology of the leading bit is copied and applied to the follower bits. See [Bus Routes and Leader Bits](#).

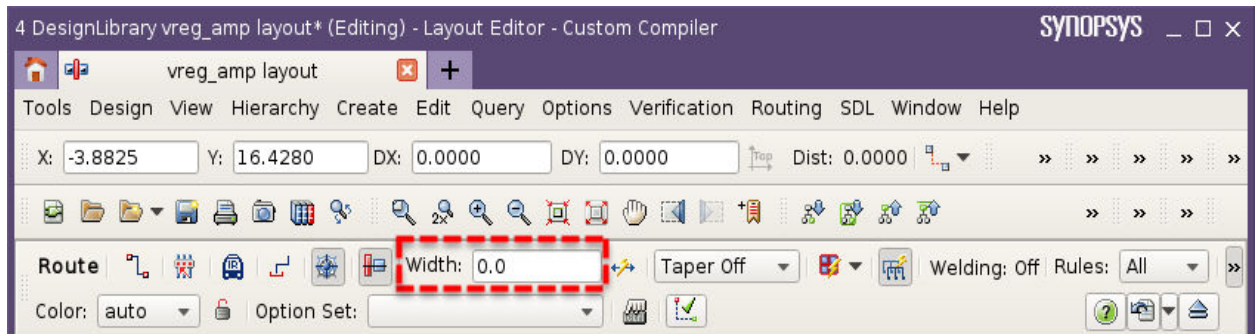
5. Start routing by moving the pointer away from the starting shapes.



6. For additional routing instructions, such as adding vias or routing through obstacles, see the procedure from step 2 in [Follow-the-Cursor Routing](#).
7. (Optional) Change the following on the **Route** toolbar or in the **Create Route** dialog box during the routing operation:
 - **Spacing**: Type a new value to change the spacing between bits.
 - **Corner Type**: Ctrl+3 to cycle through and apply another style. See the [Bus Tab](#) for the available corner styles.
8. To connect to interconnects:
 - a. Right-click and choose **Select Bus Target**.
 - b. Click the target interconnect.
9. When you are finished, press Enter to complete and commit the route.

Adjusting Route Widths Automatically

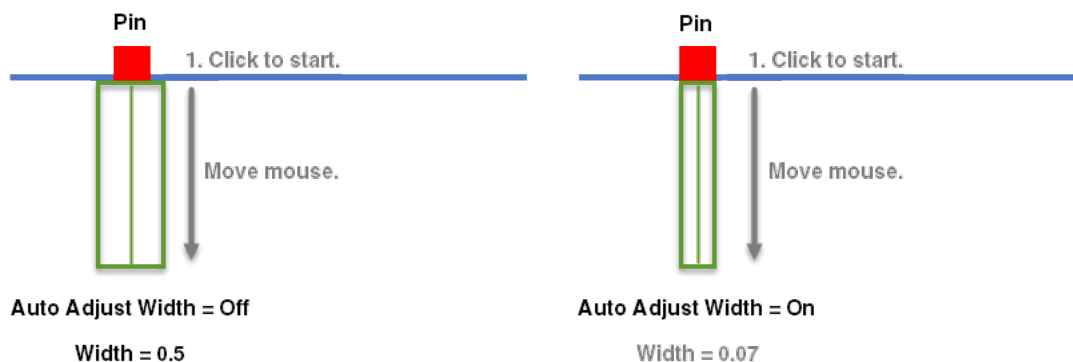
Width and **Auto Adjust Width**  are options that you can specify on the **Route** toolbar for follow-the-cursor and point-to-point interactive routing. For the description of the options, see [Route Toolbar](#).



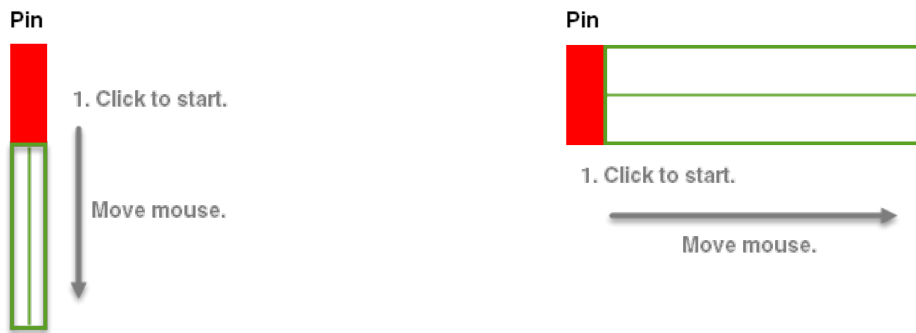
You can use **Width** and **Auto Adjust Width**  together when creating routes. When you start routing, **Auto Adjust Width** automatically applies the width that matches the starting pin geometry.

In the following example, the pin width is 0.07 in both scenarios.

- In the first scenario (left), **Auto Adjust Width** is turned off, and the **Width** value is set to 0.5. When you click the pin to start routing, the first route segment has a width of 0.5, which is wider than the pin width (0.07).
- In the second scenario (right), **Auto Adjust Width** is turned on. The router automatically adjusts the route width to match that of the pin. The **Width** box becomes unavailable for editing and displays the width value applied: 0.07.



When **Auto Adjust Width** is turned on, the width of the first route segment is determined by the dimension of the pin and the direction of the route, as shown in the following examples.



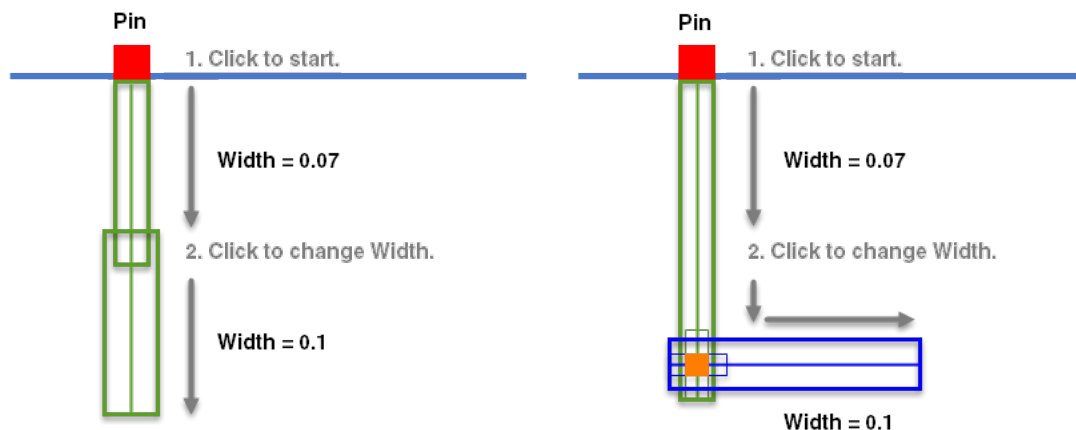
To override the automatically adjusted starting width, you can press Tab to cycle through predetermined width settings. For information about the Tab width settings, see [Route Toolbar](#).

After the first route segment is placed, you can change the route width, regardless of the **Auto Adjust Width** setting. For example, you might want to widen the route to reduce resistance.

To change the route width, click the route at the location where you want the change to occur, and then type the new value in the **Width** box. Alternatively, you can press Ctrl +Shift+4 to cycle through the predetermined width settings.

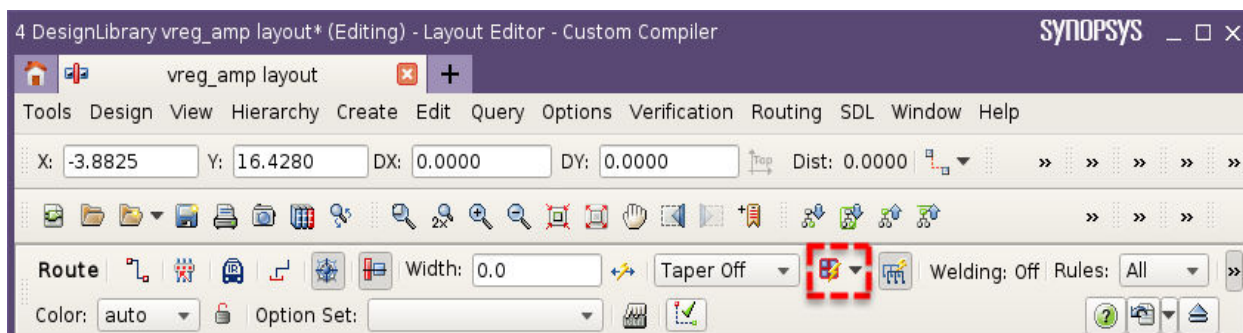
The new route width is applied as follows:

- If you continue the route in the same direction, the new width is applied from the most recently committed point (click) on the current layer, as shown in the first scenario (left).
- If you change the route direction, the new width is applied when you start the new route direction, as shown in the second scenario (right).



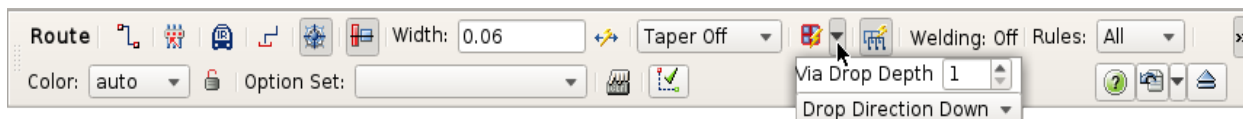
Dropping Vias Automatically

Auto Via Drop is an option that you can select on the **Route** toolbar for follow-the-cursor and point-to-point interactive routing.



Auto Via Drop can assist you with power grid and device-level routing by automatically adding vias along routes to connect to same-net wires or pins that are below the routing wire. By default, the via connects to same-net shapes one layer below the routing wire.

- To connect beyond one layer, set the **Via Drop Depth**.
- To set the direction of the via drop, choose the **Drop Direction: Down, Up, or Both** (both up and down).

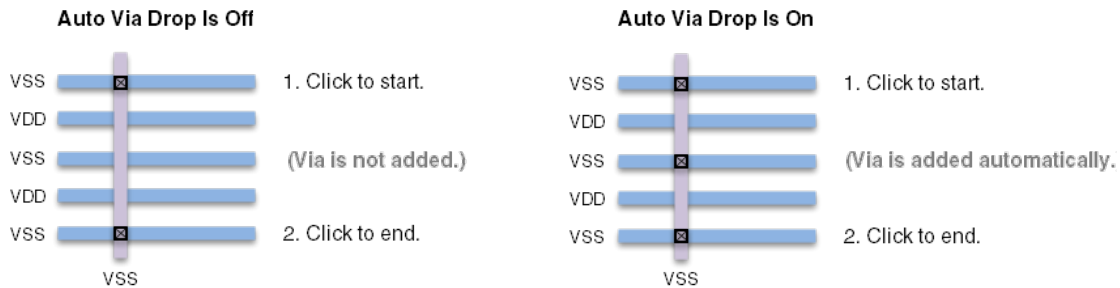


During Auto Via Drop operations:

- The router adds stacked vias when **Via Drop Depth** is > 1 .
- The router adds vias to connect to wires, pre-wires, or pins.
- The router only adds vias if the vias can be fully enclosed by the connecting wires or pins.
- The vias can be single-cut vias or via arrays. Single vias include cut class specified vias.
- Multiple vias can be added along the path being interactively routed.

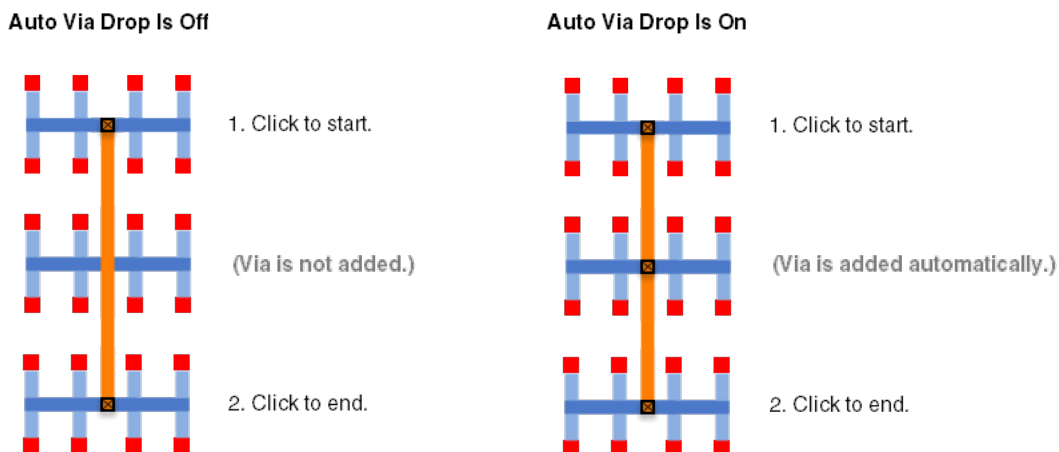
The following power grid examples show the difference between routing without **Auto Via Drop** and routing with **Auto Via Drop**.

- **Auto Via Drop is turned off:** When you create routes in a power grid, vias are added only when you right-click along the route and use the **Via Up**, **Via Down**, and **Add Via** commands. In the example, vias are added manually at the two locations shown.
- **Auto Via Drop is turned on:** As you create routes in the power grid, vias are automatically added to connect to the wire or pin that is on the layer below. In the example, vias are automatically added at every VSS-to-VSS and VDD-to-VDD connection as the routes are created across the grid.



The following auto welding examples show the difference between routing without **Auto Via Drop** and routing with **Auto Via Drop**. For information about auto welding, see [Follow-the-Cursor Routing](#).

- **Auto Via Drop is turned off:** When you route manually to connect the fishbone structures shown, vias are added only at the starting and ending locations.
- **Auto Via Drop is turned on:** When you route manually to connect the fishbone structures shown, a via is automatically added to connect to the wire (or trunk) in the second fishbone structure.



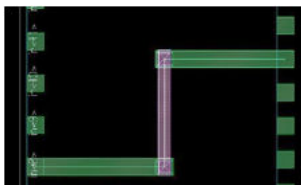
To add routes and drop vias manually, see [Follow-the-Cursor Routing](#) and [Dropping Vias Manually](#). See also [Adding Vias Manually](#).

Replicating the Last Route

To save time, you can use the Redo Route command to replicate the last route you created to connect a different set of pins.

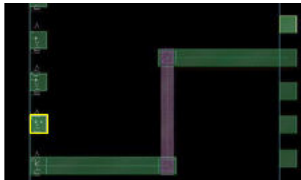
To replicate a route:

1. Create a route.

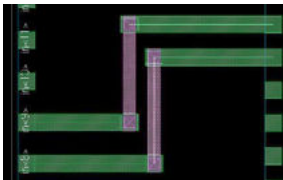


2. Select another pin that will be the starting pin for the replication.

The starting pin layer must be the same as the starting pin layer of the original route.

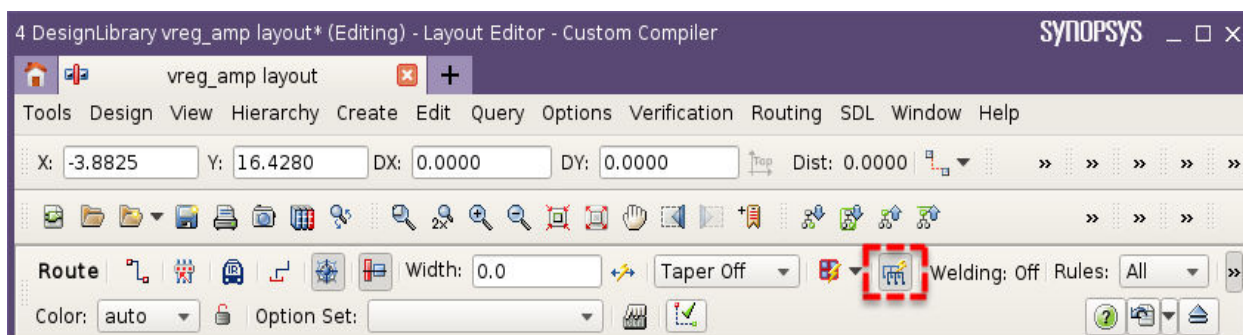


3. Press Ctrl+D. The route you created in step 1 is copied. The new connections are based on the starting pin you selected in step 2.



Cloning Routes

Auto Cloning is an option available on the **Route** toolbar for follow-the-cursor and point-to-point interactive routing. Click the button on the toolbar to turn on the feature. You can set the associated options in the **Create Route** dialog box. See [Route Toolbar](#).

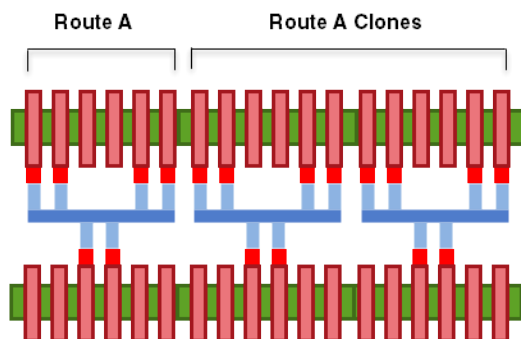


Auto Cloning can assist you with device-level routing by automatically replicating routes within and between rows of devices. **Auto Cloning** is applied to the current segment and is typically used with **Welding** and **Auto Via Drop**.

When the routes of two clones (or master and clone) touch, they are merged into a single route. This enables rapid creation of complex tree-like patterns, as shown below.

During interactive routing, the router looks for matching patterns of devices and pins on the same net. If it finds device and pin patterns that match, the router will replicate the route for those devices. The clone shapes must be on the same net, on the same layer, and have the same dimensions as the corresponding original shapes. When the routes of two clones touch, the clones are merged and become a single route.

In the following example, two clones are created within the same row.



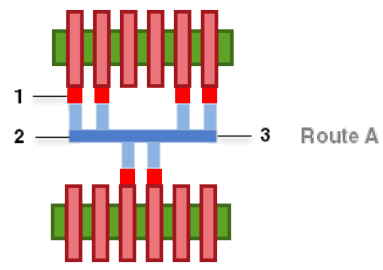
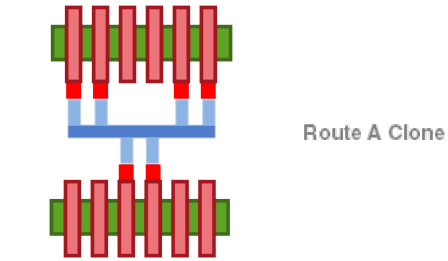
Vias are added automatically at the pins. If you select the **Auto Via Drop** option, vias are also added along wires that are one layer above or below the welding wire. For more information, see [Dropping Vias Automatically](#).

Before you start interactive routing with the **Auto Cloning** option, consider the following:

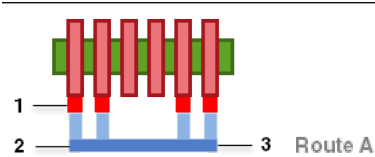
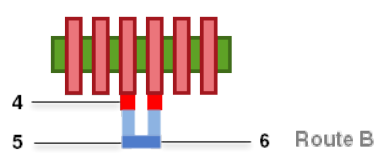
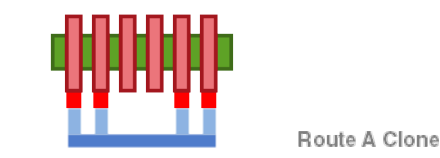
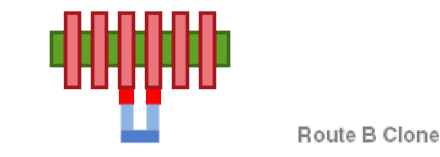
- Synopsys recommends that you use the **Auto Cloning** option while performing follow-the-cursor interactive routing. Follow-the-cursor routing lets you preview potential welding routes and make adjustments as you move the cursor on the canvas, before you commit the route. Because point-to-point routing does not offer the same level of preview, you would have to reroute to make any changes.

- If you want to confine the cloning operations within a smaller region in the design, define a routing area first. For information about defining a routing area, see [Defining a Routing Area](#).
- The router will not clone existing routes. Cloning can only occur when you create new routes.
- Cloning routes can start at a pin, from any location on a trunk, or from an existing route.
- The maximum number of clones per route is determined by the `rtMaxCloning` Tcl preference. The default setting is 64.

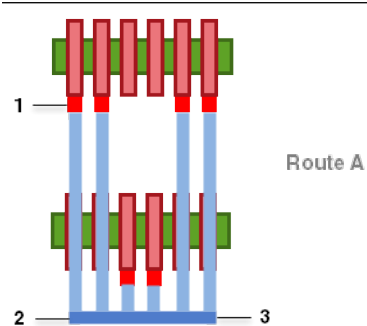
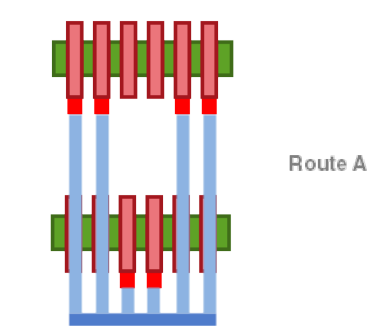
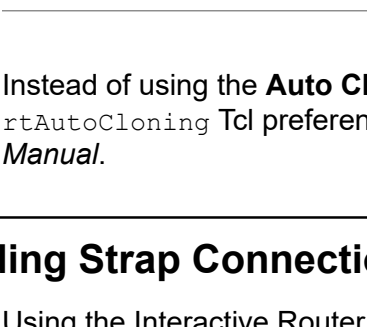
In the following **Auto Cloning** example, four devices are placed as shown, and the **Fishbone Welding** is chosen. All of the red pins are on the same net. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing. 2 Route downward, and then click midway across the first channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins on the same net. Because the third and fourth devices are identical to the first two devices and have the same pin positions, the router clones the current route in the third channel.
	3 Below the last pin, press Enter or double-click to finish the route. The cloned route also changes the route direction and finishes at the last pin.

In the following **Auto Cloning** example, four devices are placed as shown, and the **Backbone Welding** is chosen. All of the red pins are on the same net. To route the red pins, you would:

Step	
	1, 4 Click the first pin to start follow-the-cursor interactive routing.
	2, 5 Route downward, and then click midway across the channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins on the same net. Because the third and fourth devices are identical to the first two devices and have the same pin positions, the router clones the current routes in the third and fourth channels.
	3, 6 Below the last pin, press Enter or double-click to finish the route. The cloned routes also change the route direction and finish at the last pin.
	

In the following **Auto Cloning** example, four devices are placed as shown, and the **Backbone Welding** is chosen. All of the red pins are on the same net. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing.
	2 Route downward, and then click midway across the second channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins on the same net. Because there are matching rows of devices and connections, the router clones the current route and connections as shown.
	3 Below the last pin, press Enter or double-click to finish the route.

Instead of using the **Auto Cloning** option on the **Route** toolbar, you can use the `rtAutoCloning` Tcl preference. For information, see the *Custom Compiler Tcl Reference Manual*.

Adding Strap Connections

Using the Interactive Router **Strap** mode, you can connect multiple same-layer, same-net shapes with a single click. If you turned on **Rules** and specified **Color**, the strapping operation applies those specifications. This topic provides instructions on creating strap connections.

Before you start, make sure

- The design contains routing tracks. The **Strap** mode icon  does not appear on the **Route** toolbar if your design does not have routing tracks.
- If you only want to connect a subset of shapes in the design, determine the target region of the design before you create the straps. If you do not specify a region, the Interactive Router **Strap** mode searches for same-layer and same-net shapes in the entire design.

To add strap connections to multiple shapes:

1. (Optional) Define the target region within which the Interactive Router should look for same-net shapes. See [Defining a Routing Area](#).

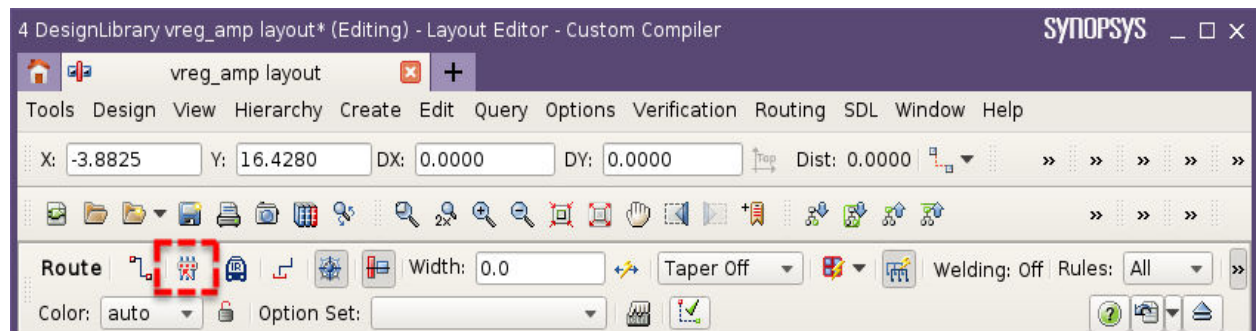
If you do not specify a region, the Interactive Router searches for same-layer and same-net shapes in the entire design.

2. In the **Object/Layer** Panel, select the layer-purpose pair (LPP) of the strap shapes with which you want connect the strap, so the selected layer is the active layer.

For example, if the shapes you want to connect are on Mx-drawing and you want to connect Mx-drawing with M2-strap, select M2-drawing.

3. Turn on the **Strap** mode using one of the following methods:

- On the **Route** toolbar, click **Strap** .



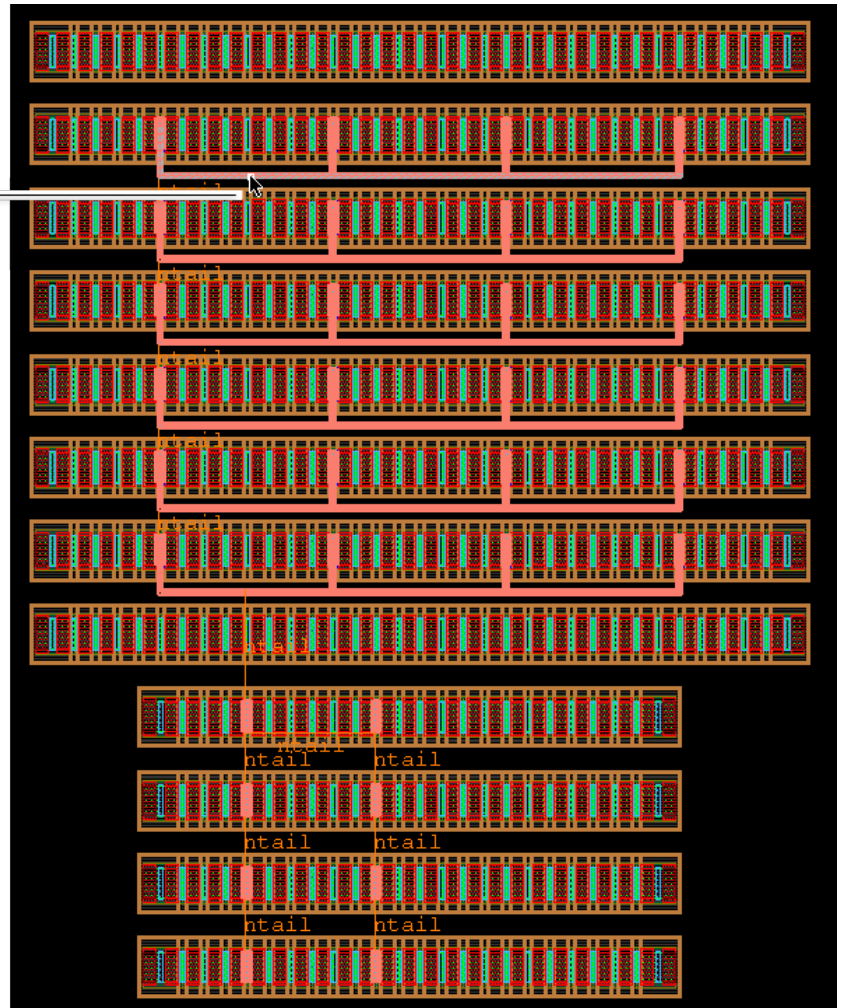
- In the **Create Route** dialog box, select **Strap**.



If you do not see the **Strap** mode button or option, check your design and make sure it contains routing tracks.

4. (Optional) Set other routing options. See [Route Toolbar](#).
5. On the layout canvas, click one of the shapes that you want to connect.

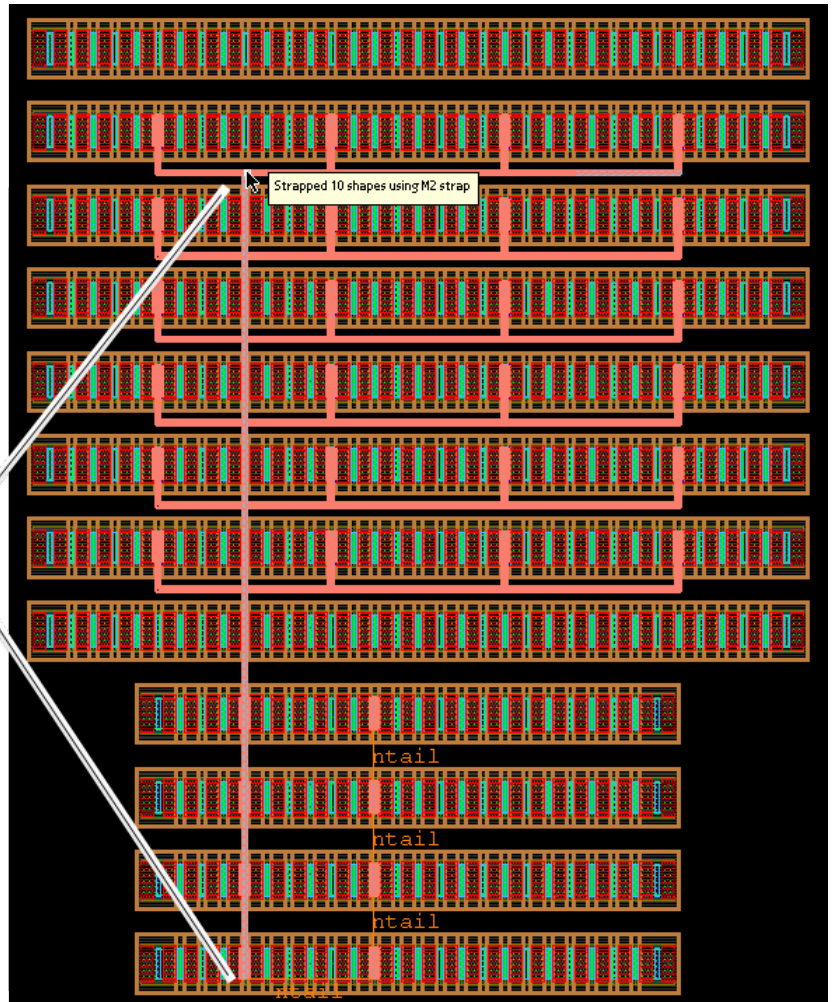
Click one of the shapes you
want to connect



The strapping route is automatically created on the nearest track to connect the multiple shapes, in the preferred direction of the active layer. Vias are also automatically added if needed. A tooltip lets you know the number of shapes connected during the operation.

The Interactive Router **Strap** mode drops vias to the same-net shapes only on one layer above or one layer below the strap layer.

Strapping route
connects same-net
shapes in the design.



6. Press **Esc** to exit the routing command.

Managing Option Sets

After you set the options on the **Route** toolbar or **Create Route** dialog box, you can create option sets and recall the settings later within the same design session. Doing so allows you to experiment with different settings without losing them.

You can also save the options to a directory, permitting them to be shared across design sessions or with other users.

This topic describes the following:

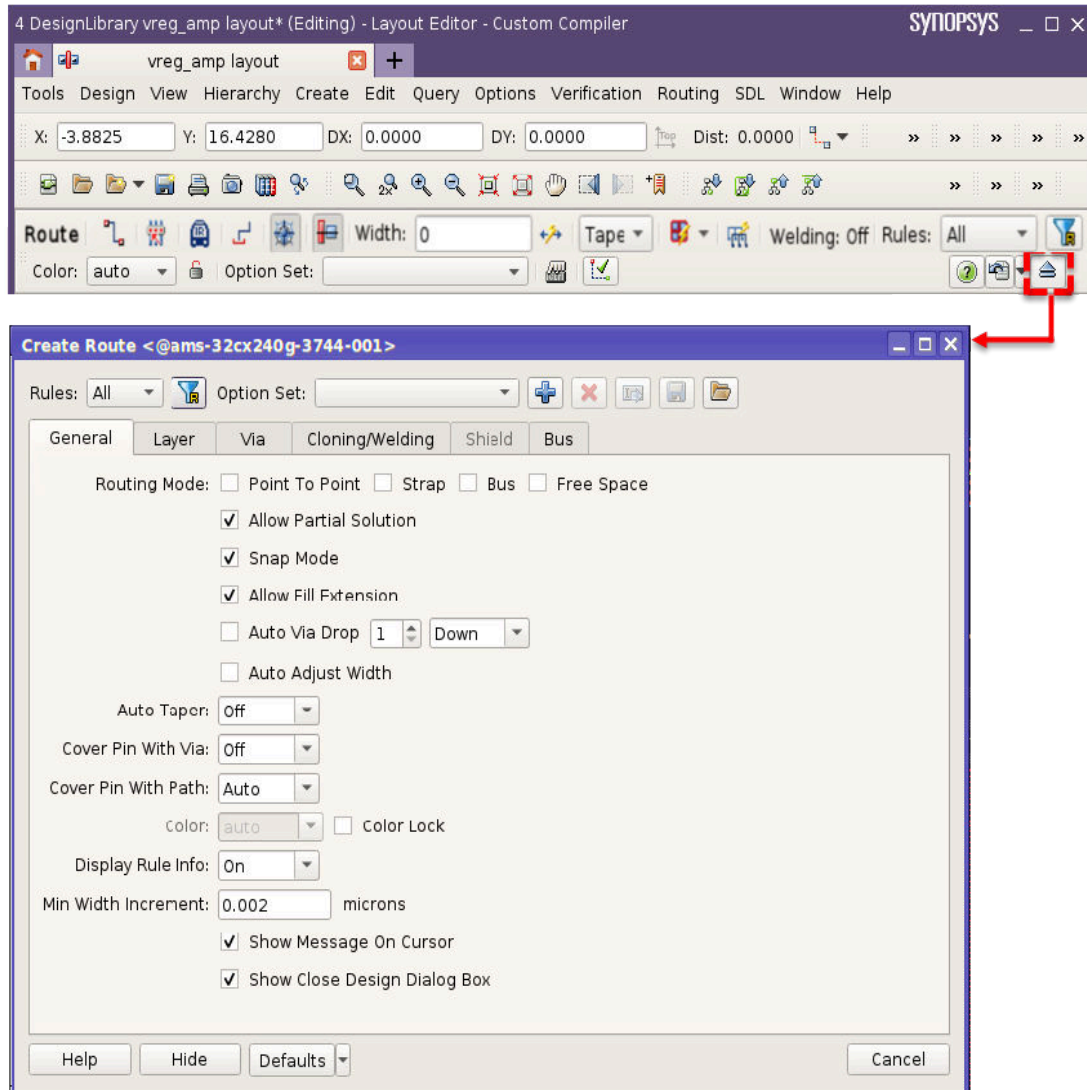
- [Opening the Create Route Dialog Box](#)
- [Creating Option Sets](#)
- [Modifying Option Sets](#)
- [Renaming Option Sets](#)
- [Deleting Option Sets](#)
- [Saving Option Sets to a Directory](#)
- [Reloading Options Sets from a Directory](#)

Opening the Create Route Dialog Box

You create, modify, and save interactive routing options in the **Create Route** dialog box.

To open the **Create Route** dialog box:

- On the **Route** toolbar, click **Eject** .

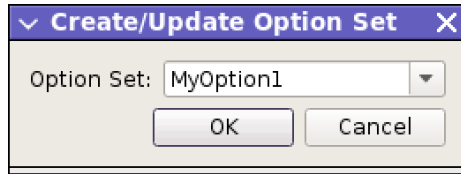


Creating Option Sets

Option sets are retained within the same design session only. If you want to recall the options in a later session, see [Saving Option Sets to a Directory](#).

To create a new option set:

1. In the **Option Sets** area, click . The **Create/Update Option Set** dialog box opens.



2. Type a name in the **Option Set** box.
3. Click **OK** to save the option set to memory and close the dialog box. To save the option set permanently to a file, see [Saving Option Sets to a Directory](#).

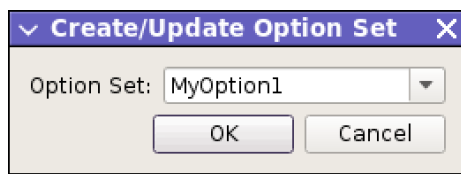
Caution:

If you do not save the option set to a file, you lose your changes when you exit the software.

Modifying Option Sets

To modify an existing option set:

1. In the **Option Set** area, choose the option set that you want to modify.
2. Change the interactive route settings. See [Route Toolbar](#).
3. In the **Option Sets** area, click . The **Create/Update Option Set** dialog box opens.

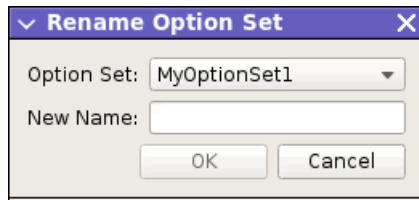


4. Click **OK** to save the changes and close the dialog box.

Renaming Option Sets

To rename an existing option set:

1. In the **Option Set** area, click . The **Rename Option Set** dialog box opens.

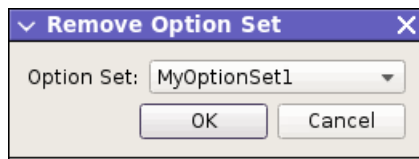


2. Choose the **Option Set** you want to rename.
3. Type a **New Name** for the option set.
4. Click **OK** to save the change and close the dialog box.

Deleting Option Sets

To delete an existing option set:

1. In the **Option Set** area, click . The **Remove Option Set** dialog box opens.



2. Choose the **Option Set** you want to delete.
3. Click **OK** to delete the option set and close the dialog box.

Saving Option Sets to a Directory

To save all existing options to a directory:

1. In the **Option Set** area, click . The **Save Option Sets** dialog box opens.
2. Select the directory you want to use.
3. Click **Choose**. The existing option sets are saved in the directory. Each option set is saved as an XML file.

Reloading Options Sets from a Directory

To load the option sets from a directory:

1. In the **Option Set** area, click . The **Load Option Sets** dialog box opens.
2. Select the directory you want to load.
3. Click **Choose**. The option set names appear in the Option Set list.

Route Toolbar and Create Route Dialog Box Reference

The **Route** toolbar and the corresponding **Create Route** dialog box contain settings that determine the interactive routing methods and specify route features that meet your design requirements.

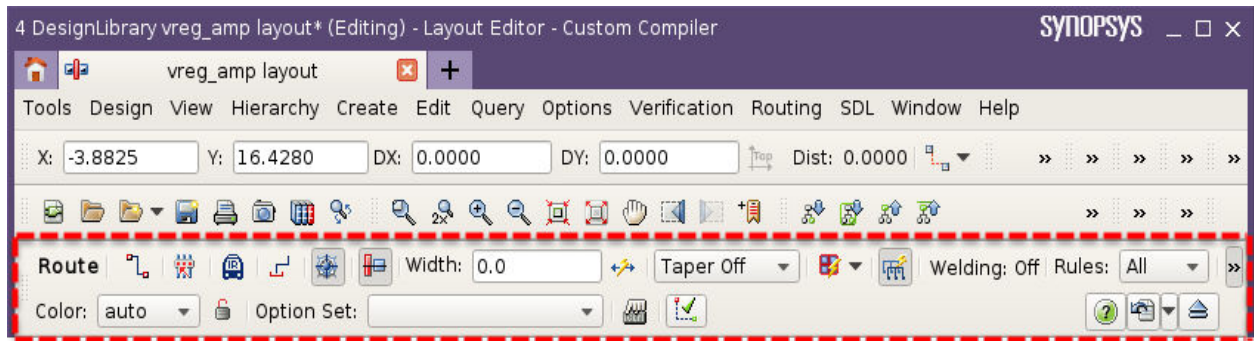
This topic provides a quick reference of the following:

- [Route Toolbar](#)
- [Create Route Dialog Box](#)
- [Add Via Dialog Box](#)
- [Shortcut Menu Commands \(FTC and P2P\)](#)
- [Shortcut Menu Commands \(Interactive Bus Routing\)](#)
- [Auto Welding Mode](#)
- [Options](#)

Route Toolbar

The **Route** toolbar provides quick access to frequently used Interactive Router options.

This topic provides a quick reference of the **Route** toolbar. For instructions on creating routes interactively, see [Follow-the-Cursor Routing](#) and [Point-to-Point Routing](#).



Option	Description
	Specifies the interactive routing method: : Point-to-point method. See Point-to-Point Routing. Tcl preference: <code>rtlRP2P</code> : Strap mode. See Adding Strap Connections. Tcl preference: <code>rtlRStrap</code> : Bus routing method. See Bus Routing. Tcl preference: <code>rtlRBus</code> : Free-space follow-the-cursor method. Use this method to start routing from anywhere on the canvas. See Follow-the-Cursor Routing. Tcl preference: <code>rtlRFreeSpace</code> To start follow-the-cursor routing from a pin, turn off  and .
	Displays or hides the Design Navigator so that you can select the nets. See Net-Based Routing Workflow or Color Routing Workflow.
Design Navigator	
Net	(Follow the Cursor only) Specifies the net to be routed.
	Snaps bends, vias, and the pointer to the center of shapes on the active net. Turn on the option when aligning the path segment to shapes on lower layers.
Snap	Note: Routing tracks always has priority.
Tap Layer	(Free Space FTC only) Turns on tapping so that the starting path inherits the layer information from the starting shape or pin. If you do not select the option, the starting layer will be from the active layer-purpose pair (LPP).

Option	Description
Width	Allows you to specify the width of the route, or displays the current width of the route. The Width box becomes unavailable for the first route segment when you turn on Auto Adjust Width. In this case, you can view the width that is applied. (Auto Adjust Width is off) Type a width value to apply to all layers. (Auto Adjust Width is on) View the width to be applied to the first route segment. After the first route segment, you can override the value and type the desired width in the Width box.] When Auto Adjust Width is on, the width value displayed follows the order of precedence: • The width value previously set in the session. This value will apply to all layers. • An NDR width value associated with the net and relevant layers. Routing width can vary between layers. • A design constraint width value. Routing width can vary between layers. • Foundry rules width. Routing width can vary between layers. For more information, see Adjusting Route Widths Automatically. Tcl preference: <code>leWidth</code>
 Auto Adjust Width	Automatically applies the width that matches the width of the connecting shape and ignores the Width value for the first route segment. After the first route segment is placed, you can change the Width value, or you can press Tab to cycle through the following predetermined widths (values appear in the Width box): • The starting route width, which is based on the pin geometry and route direction. • The greater of the following: the non-default rule (NDR) width value associated with the net, or the current layer's default routing width. For more information, see Adjusting Route Widths Automatically. Tcl preference: <code>leInterconnectAdjustWidth</code>
Spacing (Bus routing only)	Centerline-to-centerline distance between bits. Type one of the following values: • 0 (default): Uses the <code>minSpacing</code> value. • > 0: Uses the value you specify. When you specify a custom value, the Interactive Router uses the Rules setting (<code>rtBusKeepPinSpacing</code>) to determine the spacing value applied: • All: Applies the greater of the custom value or <code>minSpacing</code> value. • None or LVS: Applies the custom spacing. See also Bit Spacing. Tcl preference: <code>leBusSpacing</code>

Option	Description
Taper	Specifies whether the router should perform tapering adjustments (up or down) when connecting to or from a terminal or wire. Auto tapering determines the routing width of the first and last path segment of the routed interconnect based on the dimensions of the level 1 pins and level 0 netted rectangles. • Off: Disables automatic tapering. • Start: Tapers the first path segment. • End: Tapers the last path segment. • Both: Tapers the width of the first and last path segment. Tcl preference: <code>rtAutoTaperMode</code>
Auto Via Drop	Automatically adds vias to the same-net shapes located below the current wires being routed. Click the arrow to specify the number of layers for the operation and the drop direction. This option is typically used with the Auto Welding Mode and Auto Cloning options. For more information, see Dropping Vias Automatically. Tcl preferences: <code>rtAutoViaDrop</code>, <code>rtAutoViaDropDepth</code>, <code>rtAutoViaDropDirection</code>
	Turns on or off route cloning. See Cloning Routes.
Cloning	
Welding	Displays the state of the Welding mode: Off, Backbone, or Fishbone. To turn on the Welding mode and to set the welding style, see Follow-the-Cursor Routing.
Rules	Specifies the set of rules to be checked during interactive routing: • All: Checks all design rules during routing. • LVS: Checks only layout-versus-schematic comparison rules during routing. • None: Does not check any design rules during routing. Tcl preference: <code>rtEnableRules</code>

Option	Description
Color	Assigns a color to the current route segment: • auto: (Default) Automatically assigns the color of the track to the path based on the Color Propagation Rules. For example, if you place the path on a red track, the path will be assigned the red color. If no propagation can be applied, then the shape is assigned the default layer color. In the point-to-point routing mode, auto is the only applicable Color selection. • <colorName>: Assigns the selected color, <colorName> to the path segment. The available color names are defined in the technology file. When a path segment has a color, it can only snap to tracks that has the same color. For example, if a path segment is red, it can only snap to red tracks. If the selected color is invalid, the path segment inherits the track color as if the color selection is auto. For example, if you select green, but the layer can only be red or blue, the path segment inherits the blue color when placed on a blue track. • same: Assumes that nearby shapes have the same color and applies the <code>multiPatternSpacing</code> constraint. • No Color: Does not apply any color to the route segment. Select this value if you want to assign colors later using a different utility. The Color option is available only if colors are defined in the technology database, and at least one of the layers has a color defined, and you are currently routing on the colored layer. Tcl preference: <code>leColorName</code>
 /  (Color Lock/Unlock)	Locks or unlocks the assigned colors. To prevent modifications, set to Lock. Locking and unlocking a shape's color changes its ability to propagate its color to or inherit color from overlapping shapes. For more information, see Color Propagation. The lock setting applies to the entire route, not just the route segment. The Color option is available only if colors are defined in the technology database, and at least one of the layers has a color defined, and you are currently routing on the colored layer. Tcl preference: <code>leCreateLockColor</code>
Option Set	Displays the chosen set of options. See Managing Option Sets .
 Key Binding	Displays or hides the Interactive Router Key Bindings dialog box. See Customizing Your Work Preferences .
 Preroute Check	Runs the preroute checker. See Checking for Design Errors .
 Help	Displays the Custom Compiler help.

Option	Description
	Opens the Create Route Dialog Box .
Eject	

See Also

- [Create Route Dialog Box](#)

Create Route Dialog Box

The **Create Route** dialog box provides the same options that are on the **Route** toolbar. Additional settings are available in the dialog box for advanced interactive routing features.

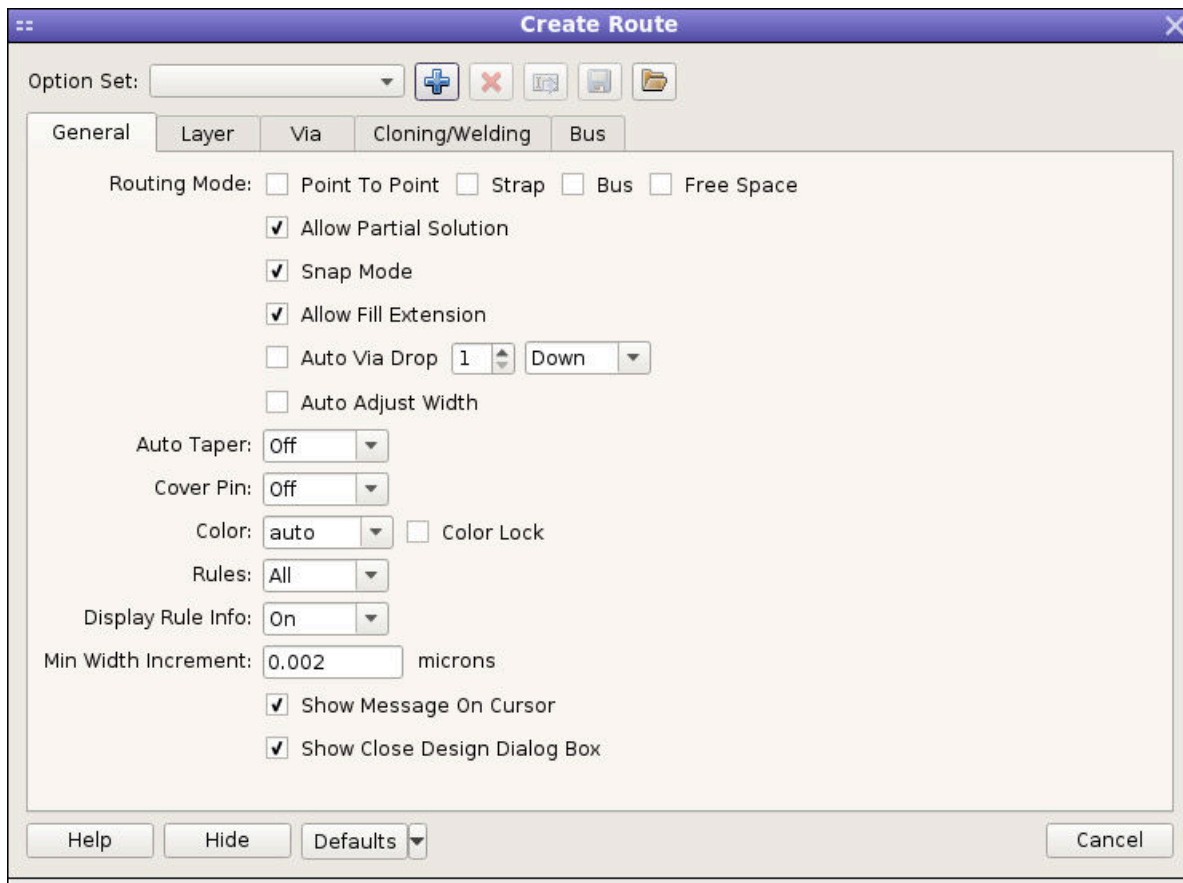
This topic provides a quick reference of the **Create Route** dialog box. For instructions on creating routes interactively, see [Follow-the-Cursor Routing](#) and [Point-to-Point Routing](#).

The **Create Route** dialog box contains the following:

- [Common Options](#)
- [General Tab](#)
- [Layer Tab](#)
- [Via Tab](#)
- [Cloning/Welding Tab](#)
- [Bus Tab](#)

Common Options

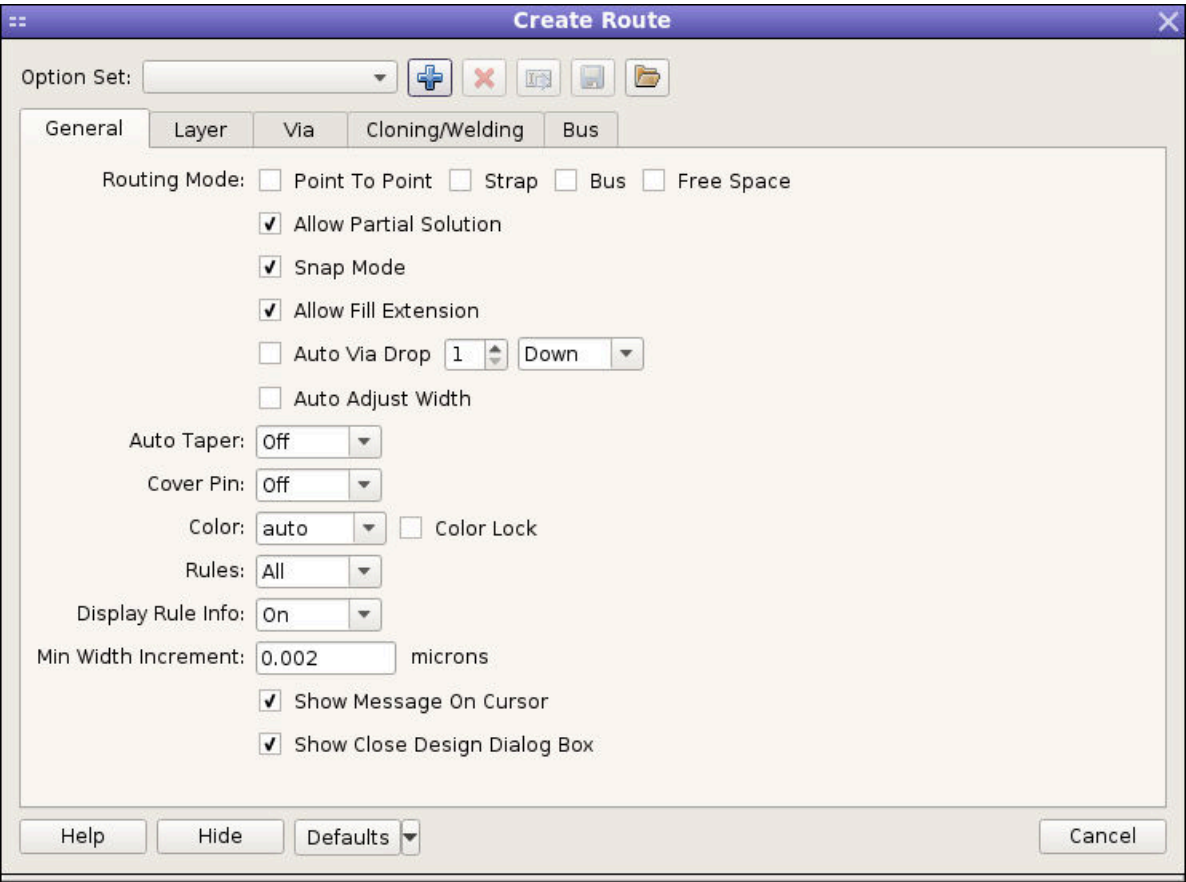
At the top and bottom of the **Create Route** dialog box, a set of common options can be set regardless of the tab you are on.



Option	Description
Option Set	Displays the chosen set of options. Click the buttons next to Option Set to create new, delete, rename, save, and load option sets. See Managing Option Sets .
Help	Displays the Custom Compiler help.
Hide	Hides the Create Route dialog box and returns to the Route toolbar. See Route Toolbar .

General Tab

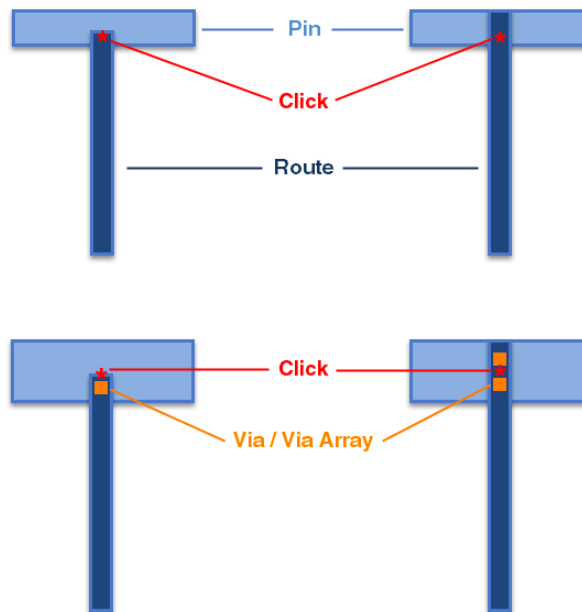
The **General** tab contains options that specify the interactive routing method and other settings that define route shapes and placement.



Option	Description
Routing Mode	Specifies the interactive routing method: • ☒ Point to Point: Point-to-point method. See Point-to-Point Routing. • ☒ Strap: Strap mode. See Adding Strap Connections. Tcl preference: <code>rtIRStrap</code>. • ☐ Point to Point ☐ Free Space: Follow-the-cursor method. See Follow-the-Cursor Routing. • ☒ Bus: Interactive bus routing method. See Bus Routing. • ☒ Free Space: Free Space method. See Follow-the-Cursor Routing. Tcl preference: <code>rtIRP2P</code>

Option	Description
Allow Partial Solution	(Point-to-Point routing only) Routes to a location near the target when the target is not accessible. Tcl preference: <code>rtAllowPartialSolution</code>
Snap Mode	Aligns the path to the center of existing same-net shapes. Tcl preference: <code>rtSnapMode</code>
Allow Fill Extension	Extends shapes during routing to ensure rule compliance. Tcl preference: <code>rtAllowFillExtension</code>
Auto Via Drop	Automatically adds vias to the same-net shapes located below the current wires being routed. Click the arrow to specify the number of layers for the operation and the drop direction. This option is typically used with the Auto Welding Mode and Auto Cloning options. For more information, see Dropping Vias Automatically . Tcl preferences: <code>rtAutoViaDrop</code> , <code>rtAutoViaDropDepth</code> , <code>rtAutoViaDropDirection</code>
Auto Via Drop Depth	Specifies the layer depth and drop direction when automatically adding vias. Tcl preference: <code>rtAutoViaDropDepth</code> , <code>rtAutoViaDropDirection</code>
Auto Adjust Width	Automatically applies the width that matches the width of the connecting shape and ignores the Width value for the first route segment. After the first route segment is placed, you can change the Width value, or you can press Tab to cycle through the following predetermined widths (values appear in the Width box): • The starting route width, which is based on the pin geometry and route direction. • The greater of the following: the non-default rule (NDR) width value associated with the net, or the current layer's default routing width For more information, see Adjusting Route Widths Automatically . Tcl preference: <code>leInterconnectAdjustWidth</code>
Auto Taper	Specifies whether the router should perform tapering adjustments (up or down) when connecting to or from a terminal or wire. Auto tapering determines the routing width of the first and last path segment of the routed interconnect based on the dimensions of the level 1 pins and level 0 netted rectangles. • Off: Disables automatic tapering. • Start: Tapers the first path segment. • End: Tapers the last path segment. • Both: Tapers the width of the first and last path segment. Tcl preference: <code>rtAutoTaperMode</code>

Option	Description
Cover Pin	Extends the route to cover the entire length of the pin or rectangular shape, regardless of where you clicked within the pin or rectangular shape boundary when starting or ending the route. If vias are required to connect to the pins, extending the route can result in via arrays. Specify one of the following: • Off: Does not extend the route. • Both: Extends the route at both the starting and ending shapes. • Start: Extends the route at the starting shape only. • End: Extends the route at the ending shape only. <code>rtCoverPin = off</code> <code>rtCoverPin = start, end, both</code>



Option	Description
Color	Assigns a color to the current route segment: • auto: (Default) Automatically assigns the color of the track to the path based on the Color Propagation Rules. For example, if you place the path on a red track, the path will be assigned the red color. If no propagation can be applied, then the shape is assigned the default layer color. In the point-to-point routing mode, auto is the only applicable Color selection. • <colorName>: Assigns the selected color, <colorName> to the path segment. The available color names are defined in the technology file. When a path segment has a color, it can only snap to tracks that has the same color. For example, if a path segment is red, it can only snap to red tracks. If the selected color is invalid, the path segment inherits the track color as if the color selection is auto. For example, if you select green, but the layer can only be red or blue, the path segment inherits the blue color when placed on a blue track. • same: Assumes that nearby shapes have the same color and applies the <code>multiPatternSpacing</code> constraint. • No Color: Does not apply any color to the route segment. Select this value if you want to assign colors later using a different utility. The Color option is available only if colors are defined in the technology database, and at least one of the layers has a color defined, and you are currently routing on the colored layer. Tcl preference: <code>leColorName</code>
Color Lock	Locks or unlocks the assigned colors. To prevent modifications, set to Lock. Locking and unlocking a shape's color changes its ability to propagate its color to or inherit color from overlapping shapes. For more information, see Color Propagation. The lock setting applies to the entire route, not just the route segment. The Color option is available only if colors are defined in the technology database, and at least one of the layers has a color defined, and you are currently routing on the colored layer. Tcl preference: <code>leCreateLockColor</code>
Rules	Specifies the set of rules to be checked during interactive routing: • All: Checks all design rules during routing. • LVS: Checks only layout-versus-schematic comparison rules during routing. • None: Does not check any design rules during routing. Tcl preference: <code>rtEnableRules</code>
Display Rule Info	Displays rule violation information during routing. Select one of the following: • on: Displays rule names, and one or more rectangular boxes to visually indicate the areas where violations will occur. • off: Does not display rule names or visual indicators. • box: Displays one or more rectangular boxes to visually indicate the areas where violations will occur. Does not display rule names. Tcl preference: <code>rtDisplayRuleInfo</code>

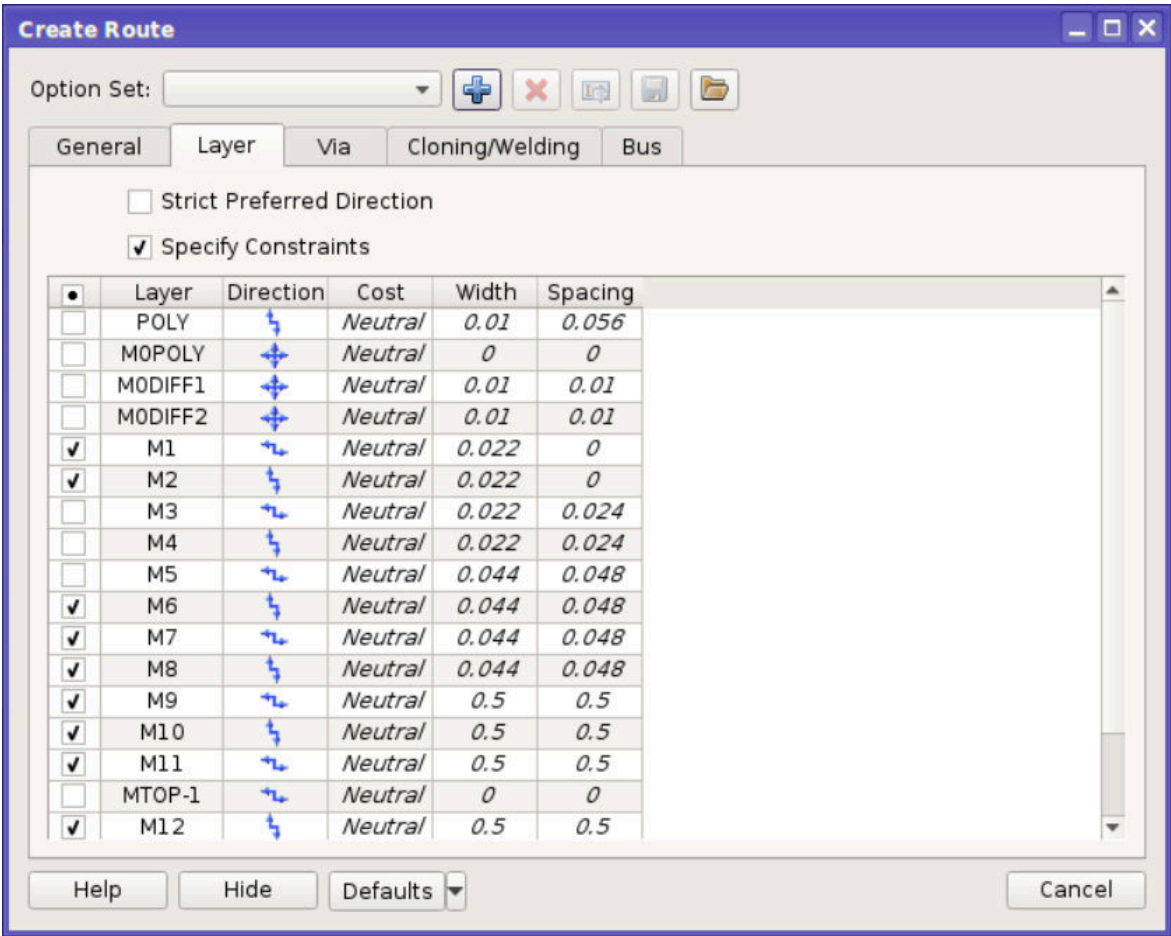
Option	Description
Min Width Increment	Specifies the width increment when using Ctrl+scroll. The default value is two times the manufacturing grid. Tcl preference: <code>rtMinimumWidthIncrement</code>
Show Message on Cursor	Shows warning messages next to the pointer (cursor) when the pointer is on a shape that might need adjustments. Regardless of whether you turn on this option, warning messages are always displayed in the Console.
Show Close Design Dialog Box	Opens the Close Design dialog box and lets you save or discard the modifications before closing the design.

Layer Tab

You use the **Layer** tab options to specify the valid routing layers and routing directions.

By default, the **Layer** tab fields are populated with information from the Constraint Editor. Although you can edit the settings and then run an interactive routing operation in the current session, the changes are not written back to the Constraint Editor. Also, if you close the **Create Route** dialog box, your changes are deleted from memory.

The **Layer** tab contains the following options:



Option	Description
Strict Preferred Direction	Creates only preferred-direction routes. Select the option if you want to prevent the router from creating wrong-way routes. For example, in the technology database, the m1 layer definition permits only vertical routes. If you select Strict Preferred Direction, you can only add vertical routes on m1. If you do not select Strict Preferred Direction, you can add vertical and horizontal (wrong-way) routes on m1. Tcl preference: <code>rtStrictPreferredDirection</code>
Specify Constraints	Overrides the design-specific constraints in the Constraint Editor. When you select this option, additional columns appear in the table: Direction, Width, Spacing. The values you enter for these columns override the corresponding values in the Constraint Editor. See Setting Routing Constraints about the Constraint Editor. Tcl preference: <code>rtUseDesignConstraint</code>

Layer table:

Option	Description
Layer	Lists the routing layers. Select the layers that you want to use for routing.
Direction	Specifies the preferred direction: : The router determines the direction. : The preferred direction is vertical. : The preferred direction is horizontal. : The direction will be vertical only. : The direction will be horizontal only. Tcl preference: <code>rtLayerDirection</code>
Cost	Specifies the restrictions to route on the layer: Low: The cost to route on this layer is less than the cost to route on layers that are neutral. Neutral: No routing cost bias exists on the layer. High: The cost to route on this layer is greater than the cost to route on layers that are neutral. The router will use a low-cost layer freely. As the cost increases, the layer usage becomes more restricted.
Width	Specifies the minimum layer width. Tcl preference: <code>rtLayerWidth</code>
Spacing	Specifies the minimum layer spacing. Tcl preference: <code>rtLayerSpacing</code>

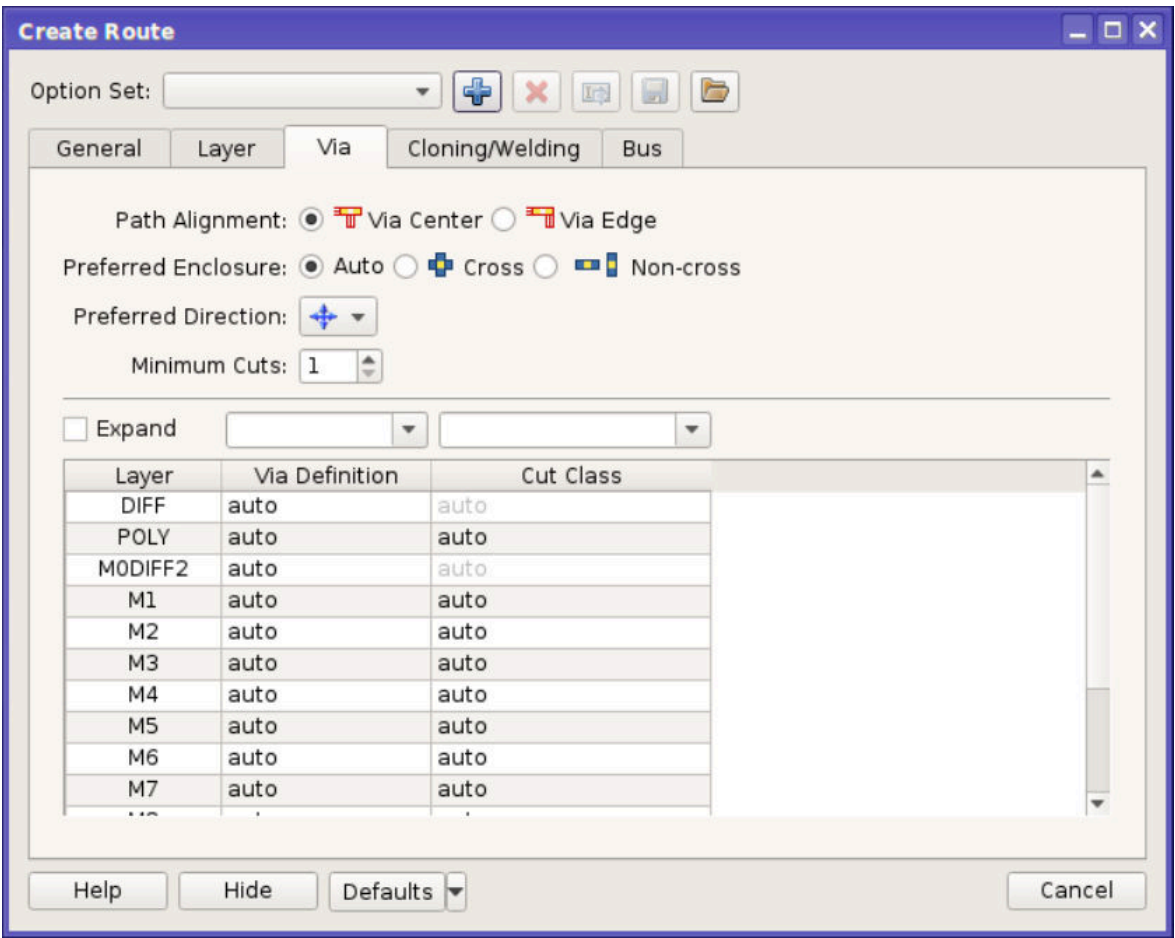
Via Tab

The **Via** tab lets you specify how vias are placed during interactive routing.

Note:

The via options you set in the Custom Router is universal. The via settings you specify for the Interactive Router is also applied when you use the Auto Router and the Pattern Route Assistant.

The **Via** tab contains the following options:



Option	Description
Path Alignment	Aligns the connecting path and via using one of the following methods: • Via Center: Aligns the connecting path’s centerline with the center of the via. • Via Edge: Aligns the edge of the connecting path with the edge of the via. Tcl preferences: <code>rtAllowOffsetVia</code>
Preferred Enclosure	Specifies how the via’s metal layers are positioned: • Auto: The router determines the positions. • Cross: The layers are perpendicular. • Non-cross: The layers are parallel. Tcl preferences: <code>rtViaPreferredEnclosure</code>

Option	Description
Preferred Direction	Specifies the via orientation: : The router determines the orientation. : The preferred orientation is vertical. : The preferred orientation is horizontal. : The orientation will be vertical only. : The orientation will be horizontal only. Tcl preferences: <code>rtPrefBarDoubleCutViaOrient</code>, <code>rtStrictFollowPrefBarDoubleCutViaOrient</code>
Minimum Cuts	The minimum number of cuts in the via. Tcl preferences: <code>rtMinimumCut</code>

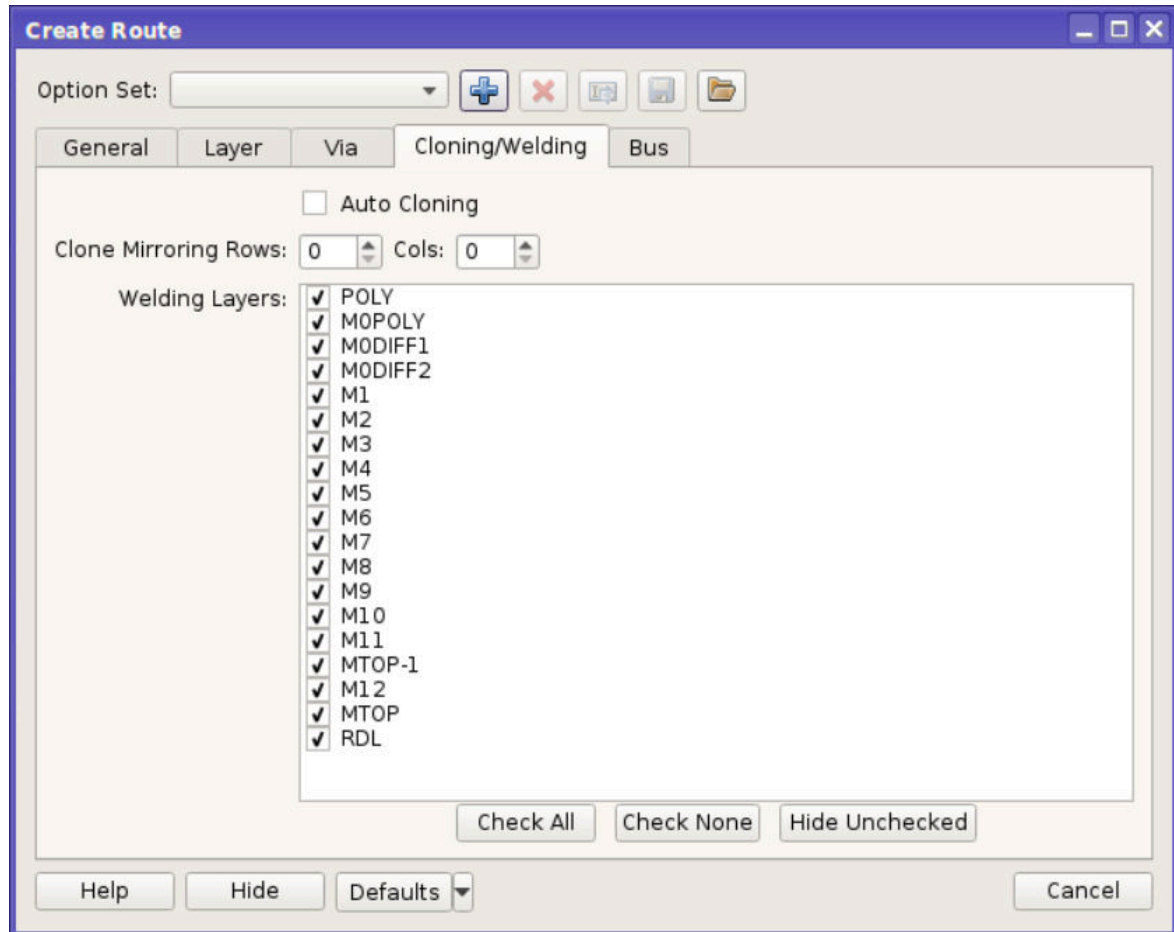
Layer table:

Option	Description
Expand	Displays or hides additional columns in the layer table: Direction, Layer1 Enc, and Layer2 Enc. Tcl preferences: <code>rtUseExpandedViaSetting</code>
Layer	Lists the layers.
Via Definition	Specifies the via definition to use when adding the via on the specific layer.
Cut Class	Specifies the cut class to use when adding the via.
Direction	Specifies the via direction (or orientation). : The router determines the orientation. : The preferred orientation is vertical. : The preferred orientation is horizontal. : The orientation will be vertical only. : The orientation will be horizontal only.
Layer1/2 Enc	Amount to extend layer1 and layer2 beyond the cut edges in the X and Y direction.

Cloning/Welding Tab

The **Cloning/Welding** tab settings specify whether cloning is run automatically, the cloning structure, and the welding layers.

The **Cloning/Welding** tab contains the following options:



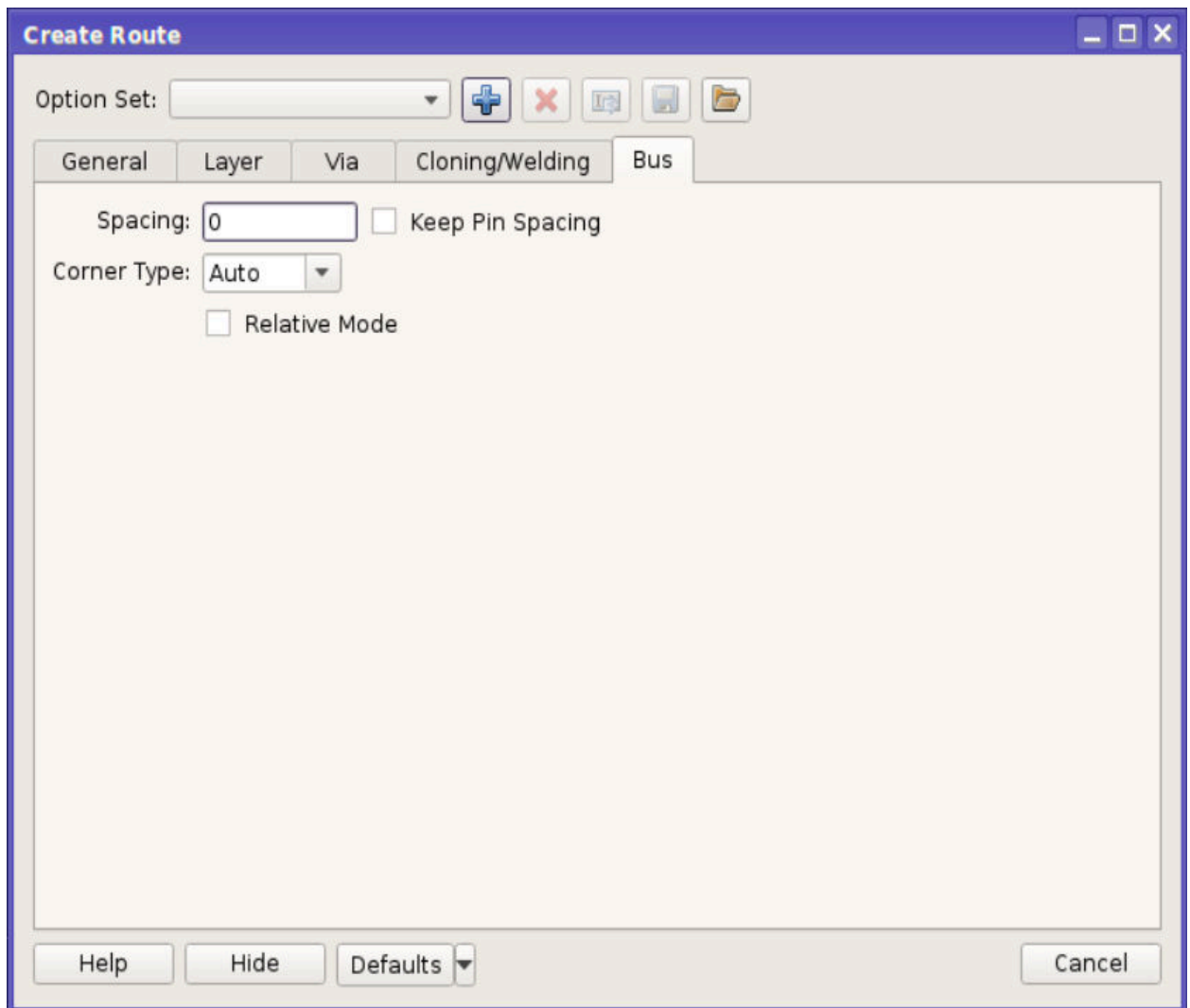
Option	Description
Auto Cloning	Automatically replicates routes and connections between rows of devices. This option is typically used with the Auto Via Drop and Auto Welding Mode options. For more information, see Cloning Routes . Tcl preference: <code>rtAutoCloning</code>

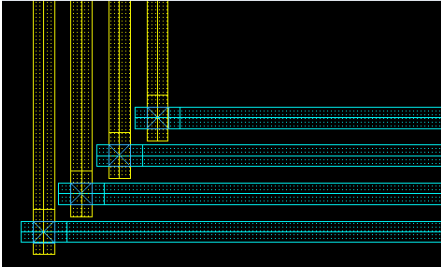
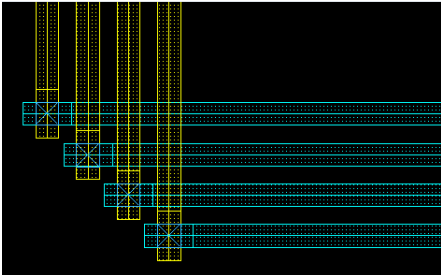
Option	Description
Clone Mirroring Rows, Cols	Creates clones that mirror the original route pattern. • Rows: The cloned routes are flipped vertically about the x-axis after the specified number of rows. • Cols: The cloned routes are flipped horizontally about the y-axis after the specified number of columns. Tcl preference: <code>rtCloneMirrorRows</code>, <code>rtCloneMirrorCols</code>
Welding Layers	Specifies the layers to be considered during welding operations. Click Check All to select all layers in the list. Click Check None to exclude all layers in the list. Click Hide Unchecked to hide the excluded layers.

Bus Tab

The **Bus** tab settings specify bit spacing, bit pattern at corners, and whether to use the Relative Mode.

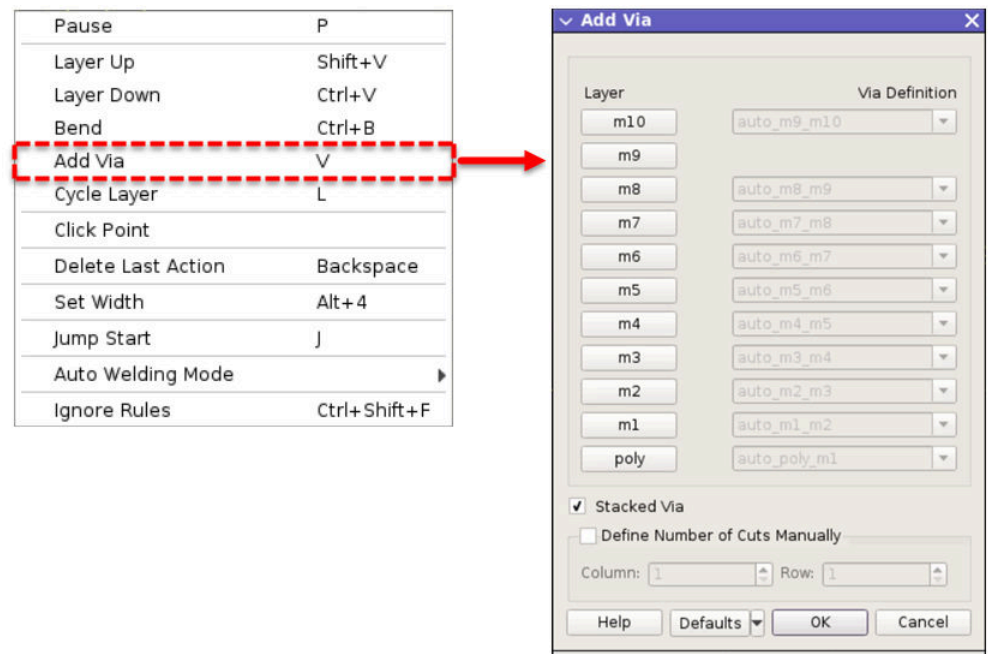
The **Bus** tab contains the following options:



Option	Description
Spacing	Centerline-to-centerline distance between bits. Type one of the following values: • 0 (default): Uses the <code>minSpacing</code> value. • > 0: Uses the value you specify. When you specify a custom value, the Interactive Router uses the Rules setting (<code>rtBusKeepPinSpacing</code>) to determine the spacing value applied: • All: Applies the greater of the custom value or <code>minSpacing</code> value. • None or LVS: Applies the custom spacing. See also Bit Spacing. Tcl preference: <code>leBusSpacing</code>
Keep Pin Spacing	Uses pin spacing as bus bit spacing. When you select this option, the Spacing option is grayed out. See also Bit Spacing. Tcl preference: <code>rtBusKeepPinSpacing</code>
Corner Type	Specifies how the bus bits are routed at corners. • auto: Automatically determines the optimal corner type for bit or pin alignment to reduce routing congestion. • river: Routes the bits so that they do not overlap at the corner.  • cross: Routes the bits so that they overlap at the corner.  Tcl preference: <code>rtBusCornerType</code>
Relative Mode	Routes bits relative to the pin location so that after bending, the ends of the bits can cascade and not align. See also Relative Routing. Tcl preference: <code>rtBusRelativeMode</code>

Add Via Dialog Box

While using the Interactive Router in the **Follow the Cursor** mode, you can right-click and add vias. You can specify the via definitions and the corresponding layers in real-time in the **Add Via** dialog box.



The **Add Via** dialog box contains the following options:

Option	Description
Layer	The possible layers the router can reach from the current follow-the-cursor routing layer. To insert a via to a new routing layer, the user must select, by clicking on the corresponding button, a different layer than the current one. If the layer above or below the current layer is not a valid routing layer, the layer will not be displayed in the interface.
Via Definition	By default, the default via per adjacent layer will be chosen. Users may choose a different standard Via Definition for each layer to move to. Modifications to the Via Definition value will be remembered in the current design session for subsequent Add Via actions but will not apply to Via Up/Down actions which will continue to use the defined standard default via.
Stacked Via	Stacks the specified vias.

Option	Description
Define Number of Cuts Manually	Specifies how to calculate the number of via rows and columns. When disabled, the router calculates number based on the current routing. When enabled, you can manually enter the number of via rows and columns.
Column/Row	Defines the number of cut rows and cut columns that the router applies to the parameterized via currently selected to be inserted on the cursor.

See Also

- [Follow-the-Cursor Routing](#)
- [Interactive Router Shortcut Menu](#)

Shortcut Menu Commands (FTC and P2P)

During interactive follow-the-cursor and point-to-point routing, you can right-click the canvas and choose commands to pause routing, route to another layer, add points along the route, change route width, turn on or off automatic welding, add or rotate via, and so on.

The shortcut commands that are available during interactive follow-the-cursor and point-to-point routing are:

Follow the Cursor

Pause	P
Finish Route	Ctrl+Return
Layer Up	Shift+V
Layer Down	Ctrl+V
Bend	Ctrl+B
Add Via	V
Cycle Layer	~
Click Point	
Delete Last Action	Backspace
Set Width	Ctrl+Shift+4
Connect Via Up	Shift+X
Connect Via Down	Ctrl+X
Jump Start	J
Auto Welding Mode	
Ignore Rules	Ctrl+Q
Options	

☒ Off

☐ Backbone

☐ Fishbone

Commit Route, Restart

Redo RouteCtrl+D

Cycle ViaDef NextC

Cycle ViaDef PreviousShift+C

Rotate ViaR

Rotate Via Lower EnclosureCtrl+R

Rotate Via Upper EnclosureShift+R

Stack ViaL

UnStack ViaShift+L

Cycle WidthTab

Increase Width

Decrease Width

Increase WeldRadius

Decrease WeldRadius

Point to Point

Pause	P
Finish Route	Ctrl+Return
Add Point	
Commit On Unnetted Shape	

Command	Description	Default Shortcut Keys
Pause/Resume	Pauses or resumes the interactive routing.	P
Finish Route	Completes and commits the route.	Ctrl+Return

Command	Description	Default Shortcut Keys
Layer Up (follow-the-cursor)	Continues routing on the adjacent layer above the current layer.	Shift+V
Layer Down (follow-the-cursor)	Continues routing on the adjacent layer below the current layer.	Ctrl+V
Bend (follow-the-cursor)	Adds a bend on the same layer.	Ctrl+B
Add Via (follow-the-cursor)	Lets you select a specific standard via for a given layer transition. See Follow-the-Cursor Routing .	V
Cycle Layer (follow-the-cursor)	Locks or changes the layer of the path after a bend. See Follow-the-Cursor Routing .	`
Click Point / Add Point	Adds an intermediate point (follow-the-cursor mode) or guide point (point-to-point mode).	Click
Delete Last Action (follow-the-cursor)	Deletes the last-entered point.	Backspace
Set Width (follow-the-cursor)	Opens the Set Width dialog box and lets you specify the path width. See Adjusting Route Widths Automatically .	Ctrl+Shift+4
Connect Via Up (follow-the-cursor)	Adds one or more vias to connect to the metal layer above the current layer. If the metal layer is multiple layers away, stacked vias are added.	Shift+X
Connect Via Down (follow-the-cursor)	Adds one or more vias to connect to the metal layer below the current layer. If the metal layer is multiple layers away, stacked vias are added.	Ctrl+X
Jump Start (follow-the-cursor)	Starts point-to-point routing while you are in the follow-the-cursor mode. See Follow-the-Cursor Routing .	J
Auto Welding Mode (follow-the-cursor)	Turns on or off the Welding mode. When Welding is on, you can select the welding style: Backbone or Fishbone. See Auto Welding Mode options and Follow-the-Cursor Routing .	E (not shown)
Ignore Rules (follow-the-cursor)	Lets you route through obstacles. See Follow-the-Cursor Routing .	Ctrl+Q
Commit on Unnetted Shape	Commits the routes created for unnetted shapes.	–
Options	See Options .	–

Shortcut Menu Commands (Interactive Bus Routing)

During interactive bus routing, you can right-click the canvas and choose commands to pause routing, route to another layer, add points along the route, change route width, add via, and so on.

The shortcut commands that are available during interactive bus routing are:

Interactive Bus Routing

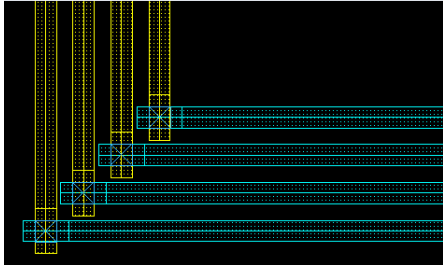
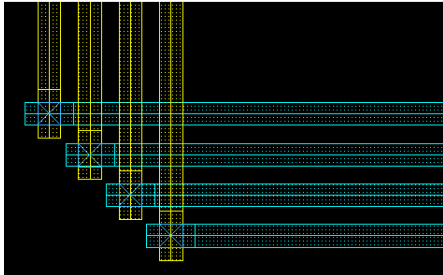
Pause	K
Select Bus Target	
Finish Route	Ctrl+Return
Layer Up	Shift+V
Layer Down	Ctrl+V
Bend	Ctrl+B
Add Via	U
Cycle Layer	Ctrl+P
Click Point	
Delete Last Action	Backspace
Set Width	Ctrl+Shift+4
Jump Start	J
Bus Corner	▶
Ignore Rules	Ctrl+Q
Options	▶

☒ Auto
☐ River
☐ Cross

Commit Route, Restart

Redo Route	Ctrl+D
Cycle ViaDef Next	C
Cycle ViaDef Previous	Shift+C
Rotate Via	R
Rotate Via Lower Enclosure	Ctrl+R
Rotate Via Upper Enclosure	Shift+R
Stack Via	L
UnStack Via	Shift+L
Cycle Width	Tab
Increase Width	
Decrease Width	

Command	Description	Default Shortcut Keys
Pause/Resume	Pauses or resumes the interactive routing.	P
Select Bus / Select Bus Target	Turns on the selection mode so that you can select the shapes (for example, pins) and initiate bus routing.	
Finish Route	Completes and commits the route.	Ctrl+Return
Layer Up	Continues routing on the adjacent layer above the current layer.	Shift+V
Layer Down	Continues routing on the adjacent layer below the current layer.	Ctrl+V
Bend	Adds a bend on the same layer.	Ctrl+B
Add Via	Lets you select a specific standard via for a given layer transition. See Follow-the-Cursor Routing .	V
Cycle Layer	Locks or changes the layer of the path after a bend. See Follow-the-Cursor Routing .	`
Click Point / Add Point	Adds an intermediate point (follow-the-cursor mode) or guide point (point-to-point mode).	Click
Delete Last Action	Deletes the last-entered point.	Backspace
Set Width	Opens the Set Width dialog box and lets you specify the path width. See Adjusting Route Widths Automatically .	Ctrl+Shift+4
Jump Start	Starts point-to-point routing while you are in the follow-the-cursor mode. See Follow-the-Cursor Routing . You can use Jump Start to suspend the current action, skip to a new location on the canvas, and continue to route using the point-to-point mode from the new location.	J

Command	Description	Default Shortcut Keys
Bus Corner	Specifies how the bus bits are routed at corners. • auto: Automatically determines the optimal corner type for bit or pin alignment to reduce routing congestion. • river: Routes the bits so that they do not overlap at the corner.  • cross: Routes the bits so that they overlap at the corner.  Tcl preference: <code>rtBusCornerType</code>	–
Ignore Rules	Lets you route through obstacles. See Follow-the-Cursor Routing .	Ctrl+Q
Options	See Options .	–

Auto Welding Mode

From the [Interactive Router shortcut menu](#), you can turn on or off the **Auto Welding Mode**. When **Auto Welding Mode** is turned on, you can specify the welding style.

Interactive Router Shortcut Menu

Pause	P
Finish Route	Ctrl+Return
Layer Up	Shift+V
Layer Down	Ctrl+V
Bend	Ctrl+B
Add Via	V
Cycle Layer	~
Click Point	
Delete Last Action	Backspace
Set Width	Ctrl+Shift+4
Connect Via Up	Shift+X
Connect Via Down	Ctrl+X
Jump Start	J
Auto Welding Mode	▶
Ignore Rules	Ctrl+Q
Options	▶

☒ Off
☐ Backbone
☐ Fishbone

Option	Description
Off	Does not automatically connect to pins in neighboring devices or routes.
Backbone	Automatically connects to pins in neighboring devices or routes on one side of the trunk only.
Fishbone	Automatically connects to pins in neighboring devices or routes on both sides of the trunk.

Options

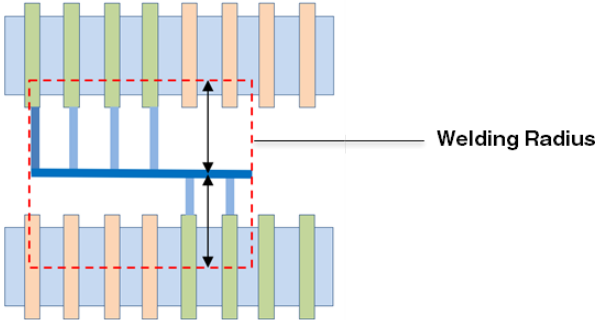
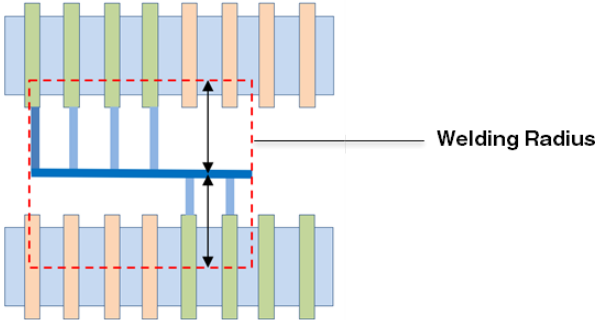
From the [Interactive Router shortcut menu](#), you set options to cycle via definitions, rotate via enclosures, stack or unstack vias, change the route width, and commit the route.

Interactive Router Shortcut Menu

Pause	P
Finish Route	Ctrl+Return
Layer Up	Shift+V
Layer Down	Ctrl+V
Bend	Ctrl+B
Add Via	V
Cycle Layer	~
Click Point	
Delete Last Action	Backspace
Set Width	Ctrl+ Shift+ 4
Connect Via Up	Shift+X
Connect Via Down	Ctrl+X
Jump Start	J
Auto Welding Mode	▶
Ignore Rules	Ctrl+Q
Options	▶

Commit Route, Restart	
Redo Route	Ctrl+D
Cycle ViaDef Next	C
Cycle ViaDef Previous	Shift+C
Rotate Via	R
Rotate Via Lower Enclosure	Ctrl+R
Rotate Via Upper Enclosure	Shift+R
Stack Via	L
UnStack Via	Shift+L
Cycle Width	Tab
Increase Width	
Decrease Width	
Increase WeldRadius	
Decrease WeldRadius	

Option	Description
Commit Route, Restart	Saves the current routes you completed and starts routing again..
Redo Route	Recreates the routes that you just deleted.
Cycle ViaDef Next	Cycles to the next via definition or cut class.
Cycle ViaDef Previous	Cycles to the previous via definition or cut class.

Option	Description
Rotate Via	Rotates the entire via. See Adding Vias Manually .
Rotate Via Lower Enclosure	Rotates only the metal layer below the cuts. See Adding Vias Manually .
Rotate Via Upper Enclosure	Rotates only the metal layer below the cuts. See Adding Vias Manually .
Stack Via	Arrange the vias so that they are vertically stacked.
Unstack Via	Disassemble the stacked via.
Cycle Width	Cycles through the valid route widths.
Increase Width	Widens the route width.
Decrease Width	Reduces the route width.
Increase Weld Radius	Increases the distance to be considered for adding tapoffs between the route trunk and devices.  This option does not apply to bus routing.
Decrease Weld Radius	Decreases the distance to be considered for adding tapoffs between the route trunk and devices.  This option does not apply to bus routing.

9

Pattern Routing

This chapter explains how to use the Pattern Route Assistant to automate complex and repetitive routing tasks.

This chapter contains the following:

- [About Pattern Routing](#)
- [Workflow for Pattern Routing](#)
- [Preparing for EM-Aware Pattern Routing](#)
- [Adding Custom Patterns in the Pattern Catalog](#)
- [Displaying the Pattern Route Assistant](#)
- [Setting Up Pattern Routing](#)
- [Creating Patterned Routes](#)
- [Creating and Extracting Route Patterns](#)
- [Saving and Viewing Pattern Route Constraints](#)
- [Deleting Routes in Patterns](#)
- [Pattern Route Assistant Reference](#)

About Pattern Routing

This topic describes the following:

- [Pattern Route Assistant](#)
- [Pattern Route Structures](#)
- [Pattern Route Components](#)

Pattern Route Assistant

 [Show me](#) how to use the Pattern Route Assistant.

The Pattern Route Assistant routes devices that are created by the Symbolic Editor **Matching** mode, and is ideal for routing multiple fanout nets that require repetitive routing patterns. Instead of adding routes manually, you can use the assistant to automate complicated and repetitive routing tasks. The Pattern Route Assistant is tightly coupled with the Symbolic Editor to ensure matched placement of route patterns for analog device-level layouts.

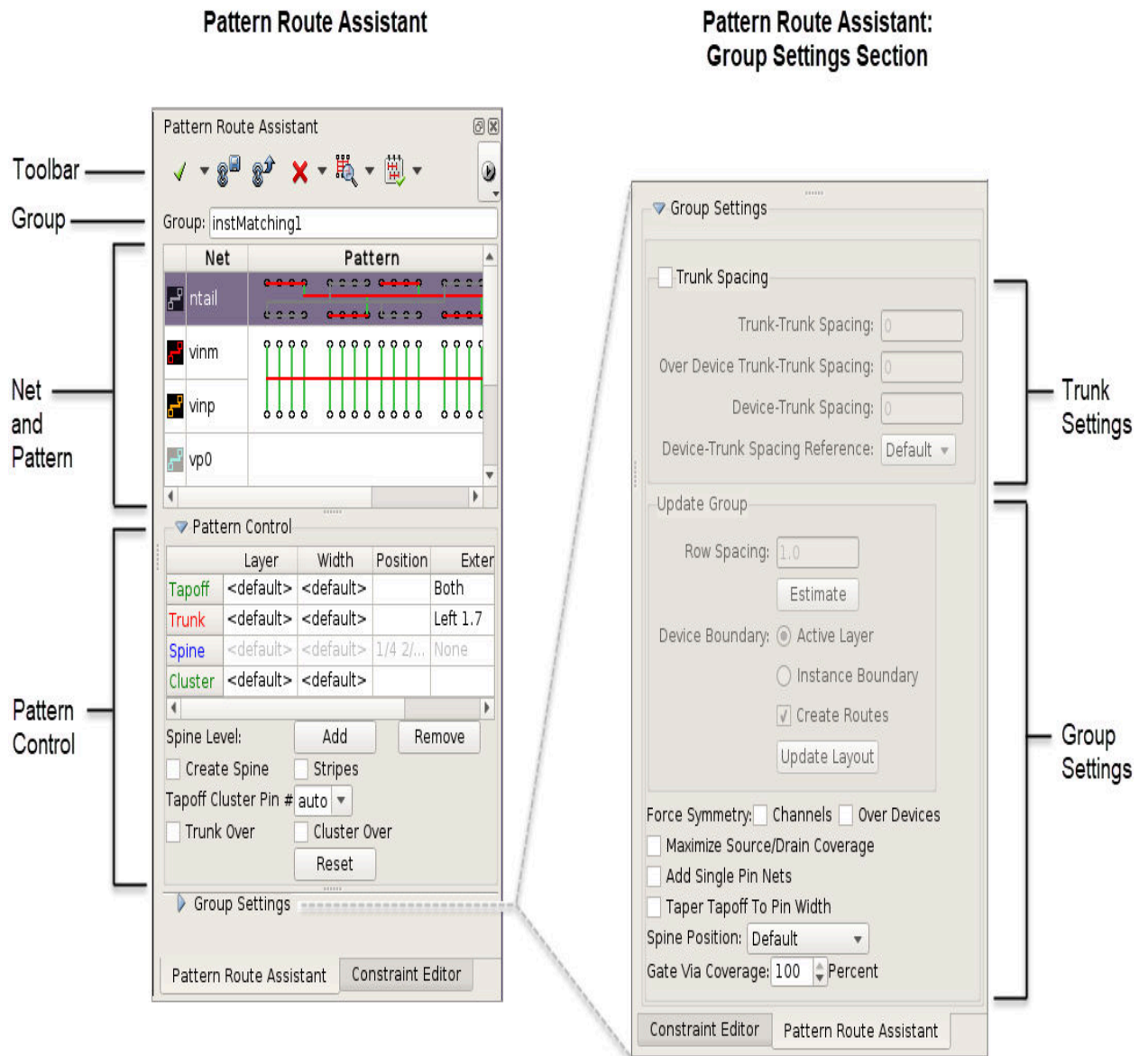
The Pattern Route Assistant provides a set of commonly used route patterns from which you can choose. The patterns can be applied to both FinFET and planar process nodes. If your design has special requirements, you can add custom patterns to the catalog. See [Adding Custom Patterns in the Pattern Catalog](#).

The assistant also has several options that let you refine the chosen pattern. For example, you can modify route attributes to adjust the tapoffs, trunks, spines, and routing layers.

When choosing a route pattern for multi-net routing, you can use the assistant's Group Planner to automatically determine the best topology, layer, and position for to each net in a figure group. You can also let the assistant automatically adjust the rows to ensure sufficient spacing between them.

Route patterns are stored as **InstMatching Pattern Route** constraints and can be reused as templates. See [Saving and Viewing Pattern Route Constraints](#).

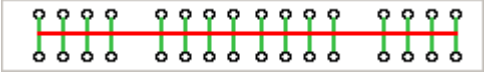
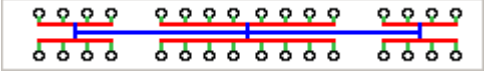
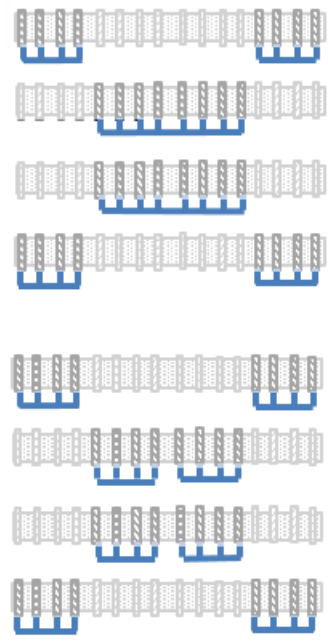
The following diagram shows the components of the Pattern Route Assistant. For the quick reference, see [Pattern Route Assistant Reference](#).



Pattern Route Structures

The Pattern Route Assistant provides a catalog of factory-supplied starter route patterns that cover basic and common scenarios. These patterns exhibit one of the route structures described in this topic.

Route patterns are defined by a descriptive language called Route Pattern Description Reference.

Structure	Description
Backbone	 Structures that connect to pins on only one side of the trunk.
Fishbone	 Structures that connect to pins on both sides of the trunk.
Intermittent	 Structures that connect or group consecutive pins within a channel.
Match	 Structures where the smallest intermittent pin-trunk group is matched within the route pattern.

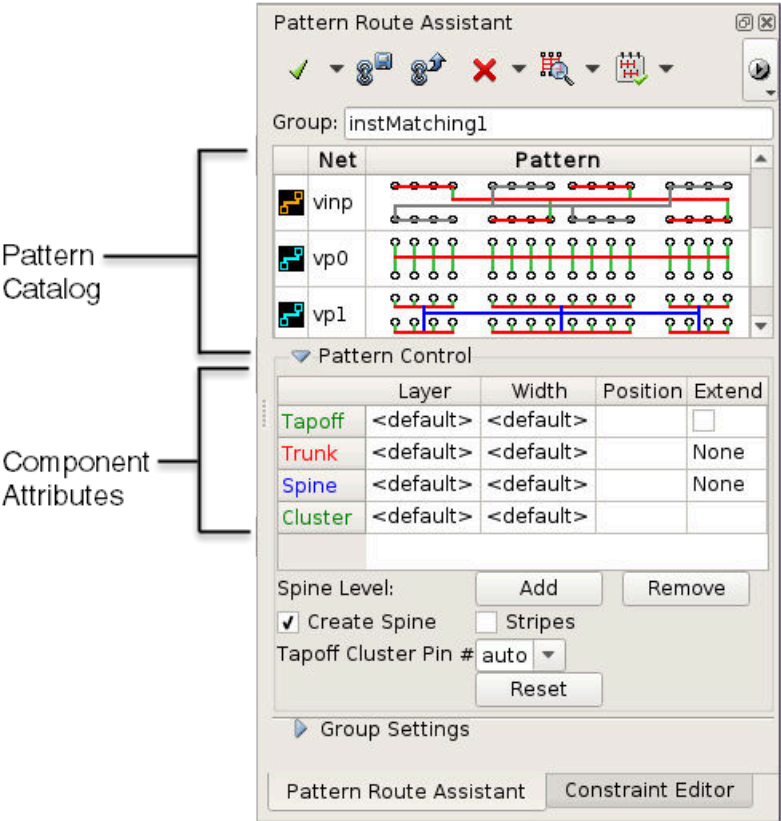
See Also

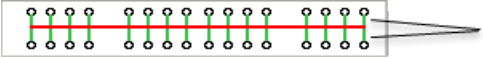
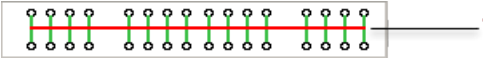
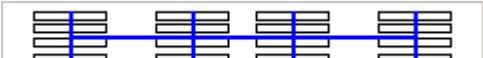
- [Route Pattern Description](#)

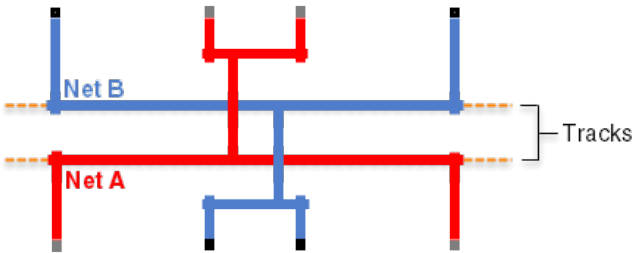
Pattern Route Components

Each backbone and fishbone route pattern structure consists of components that you can adjust to refine default route patterns. See [Adjusting Pattern Component Attributes in the Assistant](#).

The following table lists the route pattern components. In the assistant, the pattern components in the catalog and the attributes table are color-matched.



Route Component	Description
Tapoff	Route segment that branches from a trunk and connects to a pin. Examples: 
Trunk	Any horizontal path. Trunks can connect directly to pins. Examples: 
Spine	Route segment that connects trunks and other spines. Examples:  A spine-only pattern treats pins as trunks and adds spines to connect the trunks: 

Route Component	Description
Cluster	Same-net pins that are connected in groups to reduce the number of routes in channels. Doing so allows routes on different nets to co-exist in the same channel.  With a less congested channel, routes from different nets can co-exist in the same channel. In the following example, routes for net A and net B can be routed in the same channel after some of the pins are clustered. 

Workflow for Pattern Routing

The basic workflow for pattern routing is as follows:

Step		See...
1	Prepare for EM-aware pattern routing.	Preparing for EM-Aware Pattern Routing
2	(Optional) Add custom patterns in the pattern catalog.	Adding Custom Patterns in the Pattern Catalog
3	Display the Pattern Route Assistant.	Displaying the Pattern Route Assistant

Step	See...
4	Set up and run pattern routing. If you want to apply new route patterns, see: 1.Setting Up Pattern Routing 2.Previewing Route Patterns 3.Adjusting Pattern Components on the Layout Canvas and Adjusting Pattern Component Attributes in the Assistant 4.Running the Group Planner 5.Specifying Group Settings 6.Creating Patterned Routes If you want to extract a pattern from a net and apply the same pattern to another net, see Creating and Extracting Route Patterns.
5	(Optional) Preserve a modified route pattern for reuse later. Saving and Viewing Pattern Route Constraints
6	(Optional) Delete route patterns. Deleting Routes in Patterns

Preparing for EM-Aware Pattern Routing

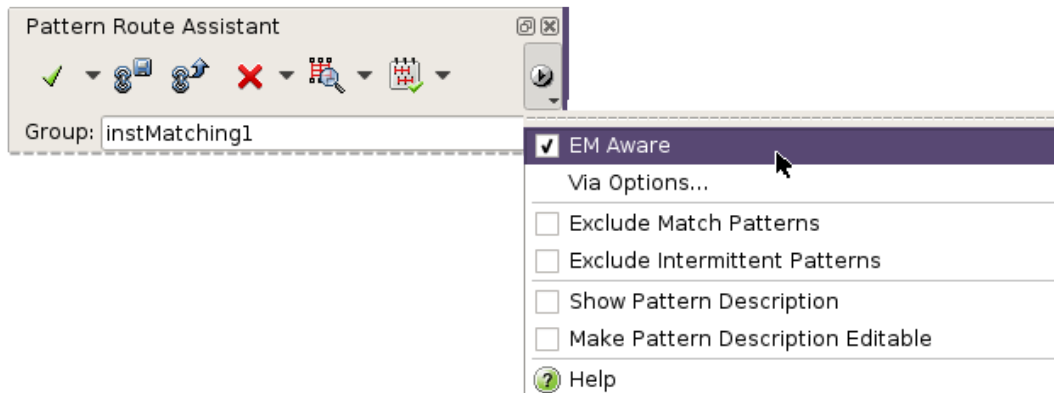
Electromigration is the gradual erosion of wire material caused by long-term exposure to high current density. It can cause reliability issues, especially as the number of design components increases, and the clearance between the components decreases.

To assess electromigration risks, you can run electromigration (EM) analysis. Instead of waiting until you complete the layout, you can automate the EM analysis as part of the net or group pattern routing process. The Pattern Route Assistant will run the EM Reporter before pattern route generation, and use the report to change wire widths or stripe wires to resolve EM issues.

To prepare for EM-aware pattern routing:

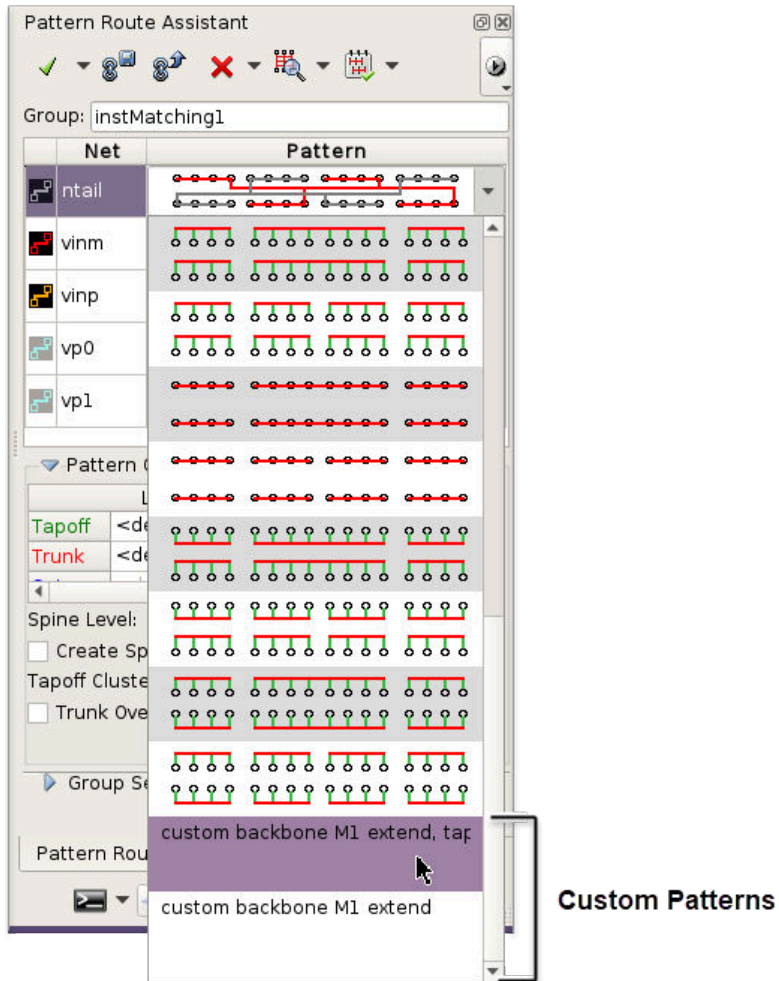
1. Add current values to instance terminals and pins in the layout using one or both of the following methods:
 - Run a simulation, capture the current data, and annotate the current values in the layout. See [Workflow for Dynamic EM Analysis](#) or [Workflow for Static EM Analysis](#).
 - Select instance terminals and pins in the layout and edit the current values in the Property Editor.
2. Set up for EM reporting. See [Reporting Electromigration](#).

When setting up for pattern routing, make sure you select **EM Aware** in the Pattern Route Assistant menu.



Adding Custom Patterns in the Pattern Catalog

Although factory-supplied starter route patterns cover basic and common route scenarios, you can add custom patterns to meet special requirements. The custom patterns you add can be displayed in the Pattern Route Assistant.



This topic describes the following:

- [Creating the Custom Pattern File](#)
- [Importing Custom Patterns Into the Pattern Catalog](#)
- [Saving Modified Route Patterns to the Catalog](#)
- [Exporting Patterns From the Pattern Catalog](#)

See Also

- [Starter Route Patterns](#)
- [Choosing Route Patterns](#)

Creating the Custom Pattern File

In addition to the factory-supplied starter route patterns, you can add custom patterns to meet special design requirements. The custom patterns have text descriptions using the Route Pattern Description syntax and are stored in a custom pattern file, an ASCII file.

To create a custom pattern file:

1. Create an ASCII file using a text editor.
2. In the newly created file, add the following two lines for each custom route pattern:

```
<pattern_name>
```

```
<route_pattern_description>
```

Custom Pattern Information	Description
pattern_name	The name that appears in the pattern catalog. You should provide a name that summarizes the pattern description.
route_pattern_description	The text description of the pattern. For syntax and detailed description, see the Route Pattern Description Reference.

The following example shows a custom pattern file containing the descriptions of two patterns.

Custom Pattern ASCII File

```
custom backbone M1 extend, tapoff poly
backbone track 2/3 layer metall width 0.3 extend tapoff layer poly; no spine;

custom backbone M1 extend
backbone track 2/3 layer metall width 0.3 extend; no spine;
```

3. Save the file.

See Also

- [Route Pattern Description Reference](#)
- [Importing Custom Patterns Into the Pattern Catalog](#)

- [Saving Modified Route Patterns to the Catalog](#)
- [Choosing Route Patterns](#)

Importing Custom Patterns Into the Pattern Catalog

After you create custom patterns, you can import them into the pattern catalog so that they are displayed in the Pattern Route Assistant.

Use one of the following methods to import the custom patterns:

- [Importing Custom Patterns Using the Pattern Route Assistant](#)
- [Setting the Path to the Custom Pattern File](#)

See Also

- [Choosing Route Patterns](#)
- [Route Pattern Description Reference](#)

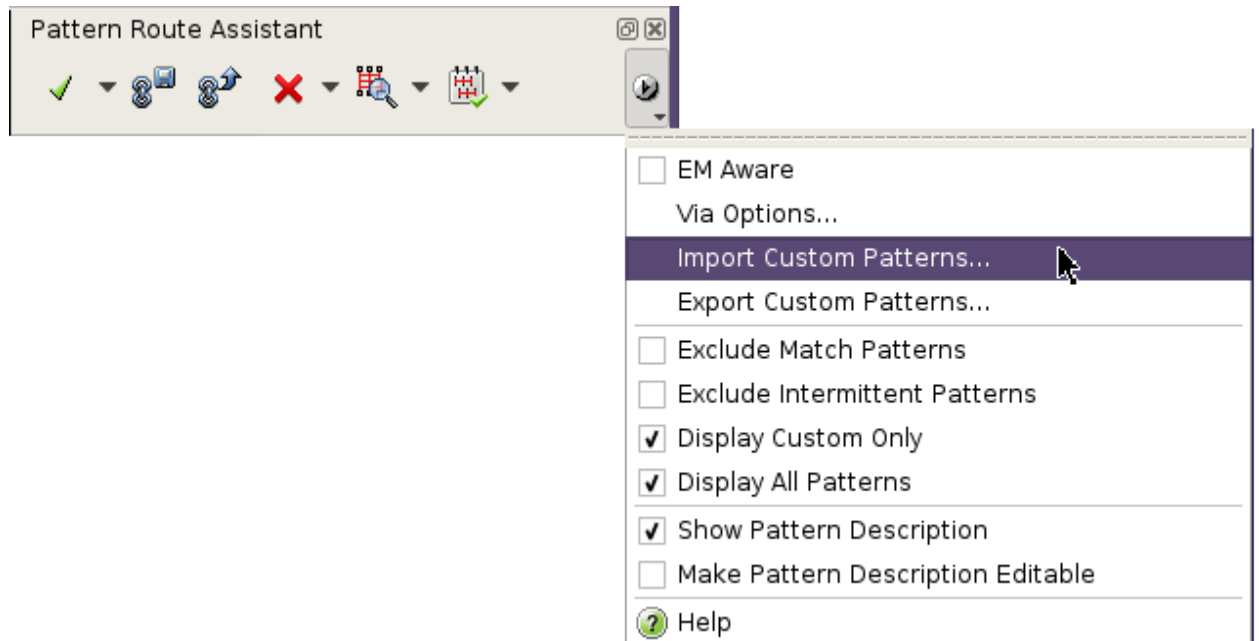
Importing Custom Patterns Using the Pattern Route Assistant

You can use the Pattern Route Assistant to import newly created custom patterns.

Before you start, make sure you add the new pattern descriptions in the custom pattern file. See [Creating the Custom Pattern File](#).

To import new custom patterns:

1. [Display the Pattern Route Assistant](#).
2. From the Pattern Route Assistant menu, choose **Import Custom Patterns**.



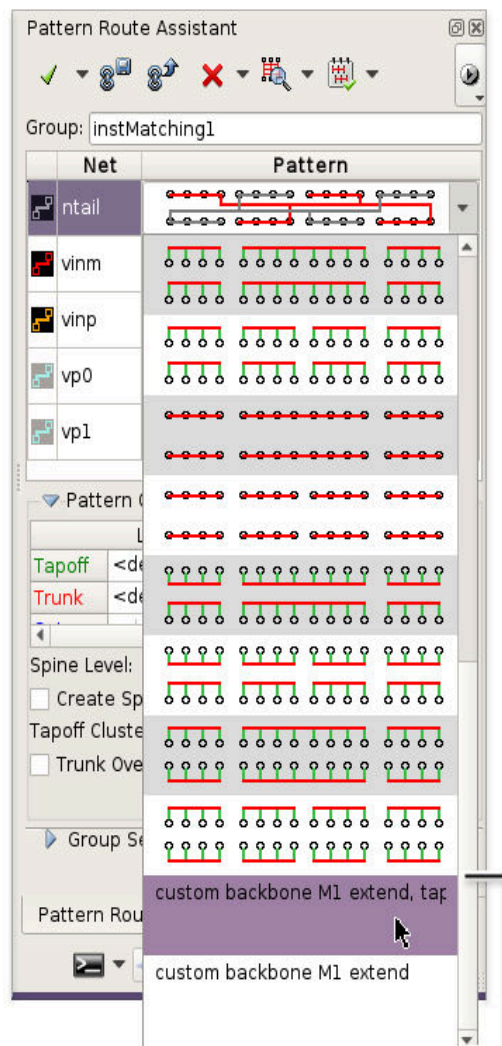
3. In the **Import Custom Patterns** dialog box, locate and select the [custom pattern file](#) you created, and click **OK**.

The custom pattern names appear in the pattern catalog.

Custom Pattern ASCII File

```
custom backbone M1 extend, tapoff poly
backbone track 2/3 layer metall1 width 0.3 extend tapoff layer poly; no spine;

custom backbone M1 extend
backbone track 2/3 layer metall1 width 0.3 extend; no spine;
```



Custom Patterns

Setting the Path to the Custom Pattern File

Instead of using the Pattern Route Assistant to import the custom patterns, you can set the `rtPatternRouteAssistantCustomPatternFile` Tcl preference.

You must set the preference before you open the design. Setting the Tcl preference:

- Lets the Pattern Route Assistant know where to find the file and the custom patterns.
- Lists the custom patterns in the pattern catalog.

To set the Tcl preference:

► Type the following in the Custom Compiler Console:

```
db::setPrefValue rtPatternRouteAssistantCustomPatternFile -value  
~myDirectory/custom_pattern.txt
```

where:

- `myDirectory` is the name of the directory where the custom pattern file is stored.
- `custom_pattern.txt` is the name of the custom pattern file.

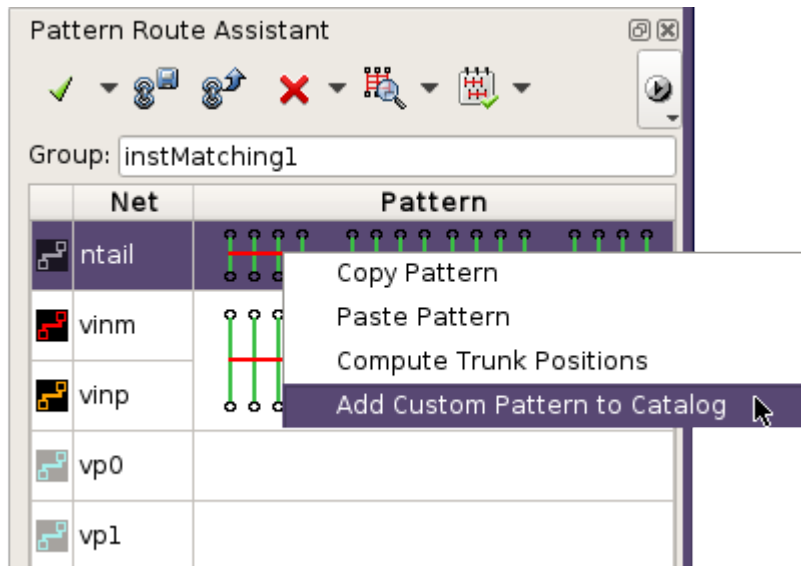
When you open the design and display the Pattern Route Assistant, custom patterns are displayed in the pattern catalog.

Saving Modified Route Patterns to the Catalog

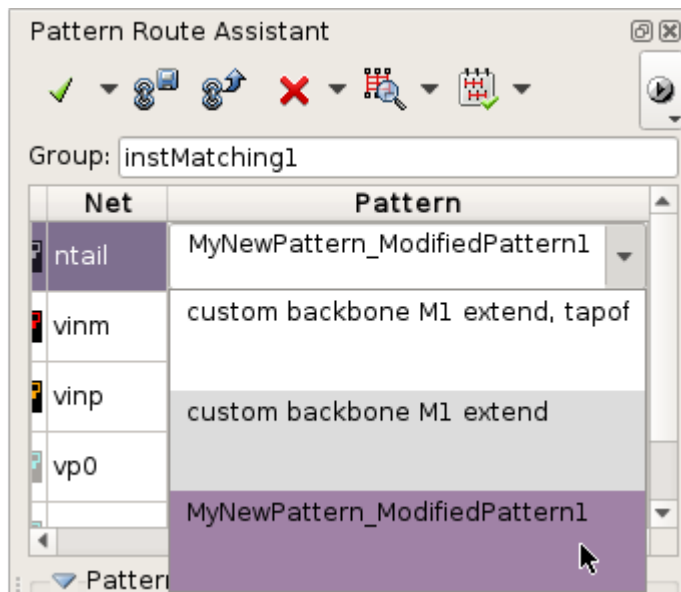
After you modify existing route patterns, you can save the modified patterns as a new patterns in the catalog.

To save modified patterns as new patterns in the catalog:

- After you finish making modifications to an existing pattern, right-click the pattern in the catalog and choose **Add Custom Pattern to Catalog**.



The newly added pattern appears in the catalog.



If you want to share the new pattern, you can export the patterns in the catalog. See [Exporting Patterns From the Pattern Catalog](#).

See Also

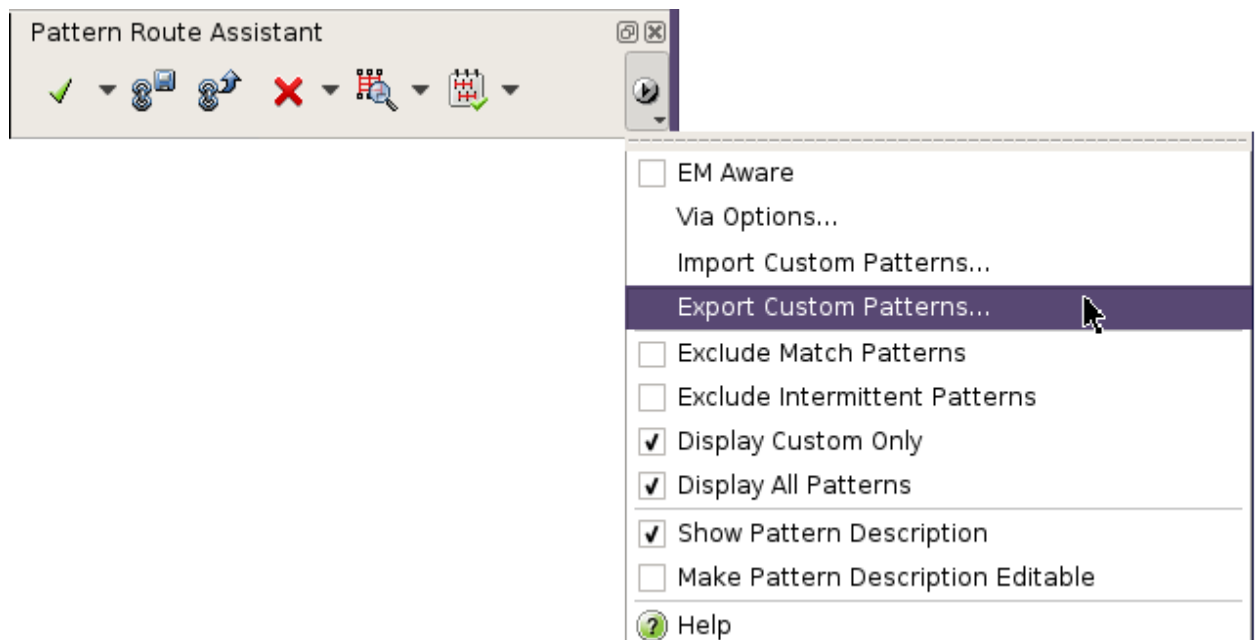
- [Choosing Route Patterns](#)
- [Adjusting Pattern Components on the Layout Canvas](#)
- [Adjusting Pattern Component Attributes in the Assistant](#)
- [Creating and Extracting Route Patterns](#)

Exporting Patterns From the Pattern Catalog

After you create new custom patterns by modifying existing patterns, you can export the patterns in the catalog to share with other designers.

To export the patterns in the catalog:

1. [Display the Pattern Route Assistant.](#)
2. From the Pattern Route Assistant menu, choose **Export Custom Patterns**.



3. In the **Export Custom Patterns** dialog box, select the folder, specify the name of the exported file, and click **OK**.

See Also

- [Importing Custom Patterns Into the Pattern Catalog](#)
- [Creating the Custom Pattern File](#)

- [Saving Modified Route Patterns to the Catalog](#)
- [Adjusting Pattern Components on the Layout Canvas](#)
- [Adjusting Pattern Component Attributes in the Assistant](#)

Displaying the Pattern Route Assistant

To display the Pattern Route Assistant, use one of the following methods:

- Choose **Routing > Pattern Route Assistant**.
- Choose **Window > Assistants > Pattern Route Assistant**.

Setting Up Pattern Routing

Before applying route patterns in your design, you need to select and adjust the pattern, specify group settings, and check the planner to ensure the most optimal patterns are applied to the nets.

This topic describes the following:

Step		See...
1	Select a figure group.	Selecting a Figure Group
2	Choose route patterns for the nets in the figure group.	• Choosing Route Patterns • Selecting Different Route Patterns for a Net
3	Preview the route pattern.	Previewing Route Patterns
3	(Optional) Adjust the route components to optimize the results.	• Adjusting Pattern Components on the Layout Canvas • Adjusting Pattern Component Attributes in the Assistant
4*	Run the group planner. If you have critical nets, specify the group settings for those nets first, and then run the group planner for the remaining nets.	Running the Group Planner
5*	Specify group settings.	Specifying Group Settings

* Steps 4 and 5 can be run iteratively to plan route patterns incrementally. For example, you can use the incremental method to determine the routes for critical nets first, and then plan the remaining routes around them.

Selecting a Figure Group

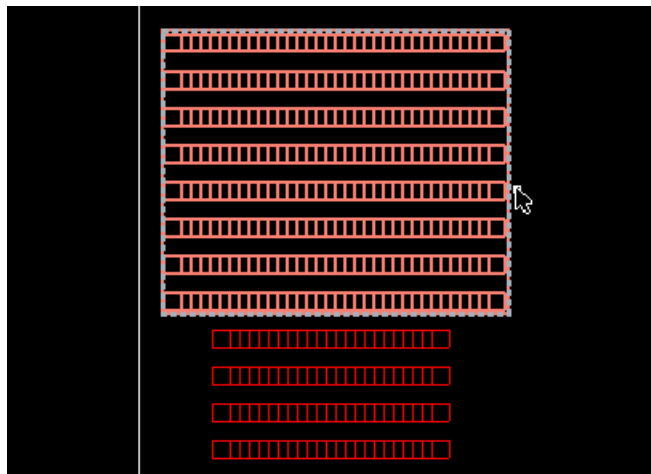
Pattern routes are typically applied to devices in a figure group. Therefore, the first step when creating pattern routes is to select the figure group.

Note:

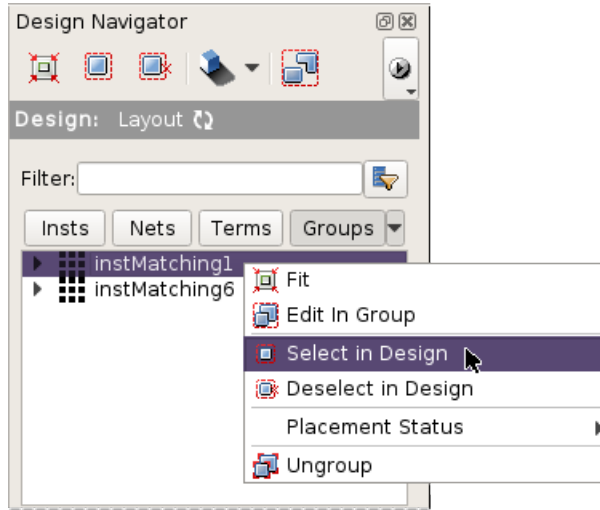
You must select a figure group to display the list of nets in the Pattern Route Assistant.

To select a figure group:

- Use one of the following methods:
 - On the layout canvas, select the figure group.

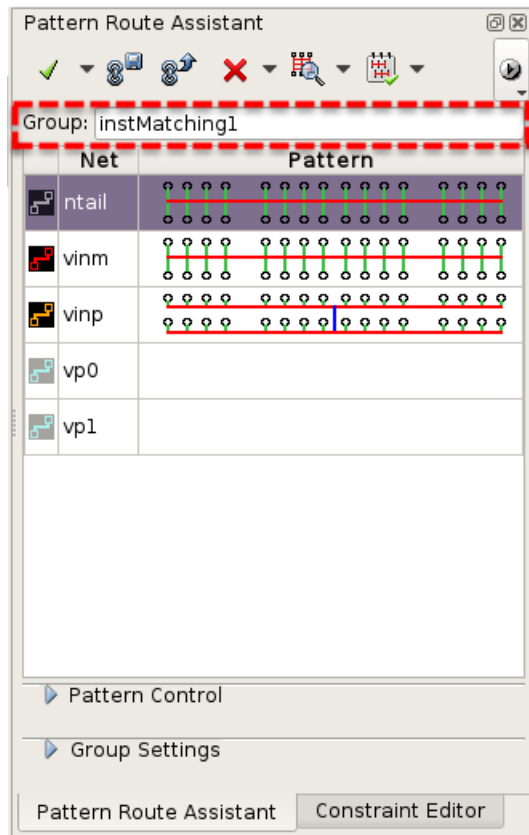


- In the Design Navigator, in either the **Design: Schematic** mode or the **Design: Layout** mode, click the **Groups** tab, right-click the group name, and choose **Select in Design**.



In the Pattern Route Assistant:

- The name of the group you selected appears in the **Group** box.
- The list of nets in the group is displayed below the **Group** box.



Choosing Route Patterns

After you select a figure group, you can choose route patterns for the nets. You can choose route patterns in the pattern catalog.

This topic describes the following:

- [Filtering Route Patterns in the Catalog](#)
- [Choosing a Pattern for a Net](#)
- [Choosing the Same Pattern for Multiple Nets](#)
- [Grouping Nets for Symmetric Routing](#)

- [Copying a Pattern to One or More Nets](#)
- [Deleting a Route Pattern](#)

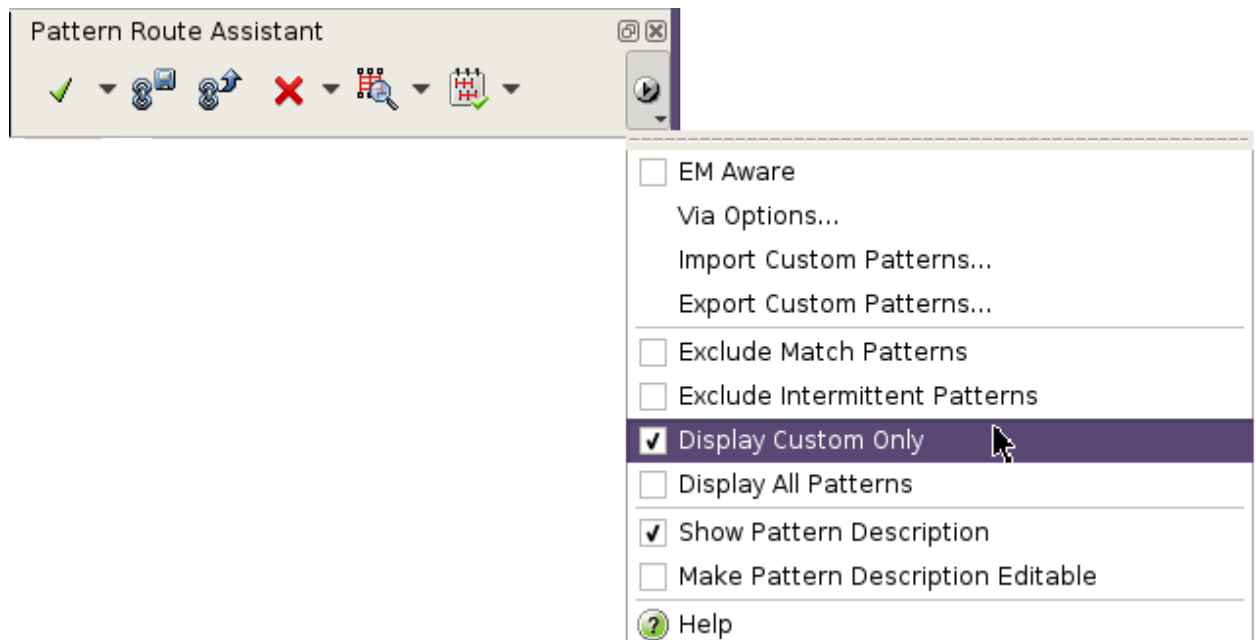
For information about the route patterns in the catalog, see [Starter Route Patterns](#) and [Adding Custom Patterns in the Pattern Catalog](#).

Filtering Route Patterns in the Catalog

If the list of patterns in the catalog is long and you only want to select a custom pattern, you can filter the list to display only the custom patterns.

To display only the custom patterns:

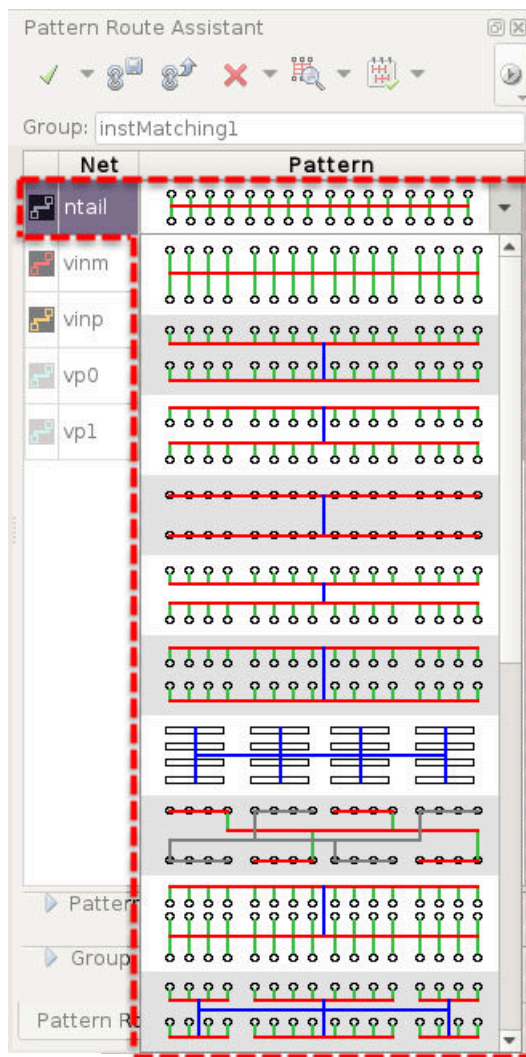
- From the Pattern Route Assistant menu, choose **Display Custom Only**.



Choosing a Pattern for a Net

To choose a pattern for a single net:

1. In the Pattern Route Assistant, select a net. The selected net becomes highlighted.
2. Choose a pattern from the catalog.

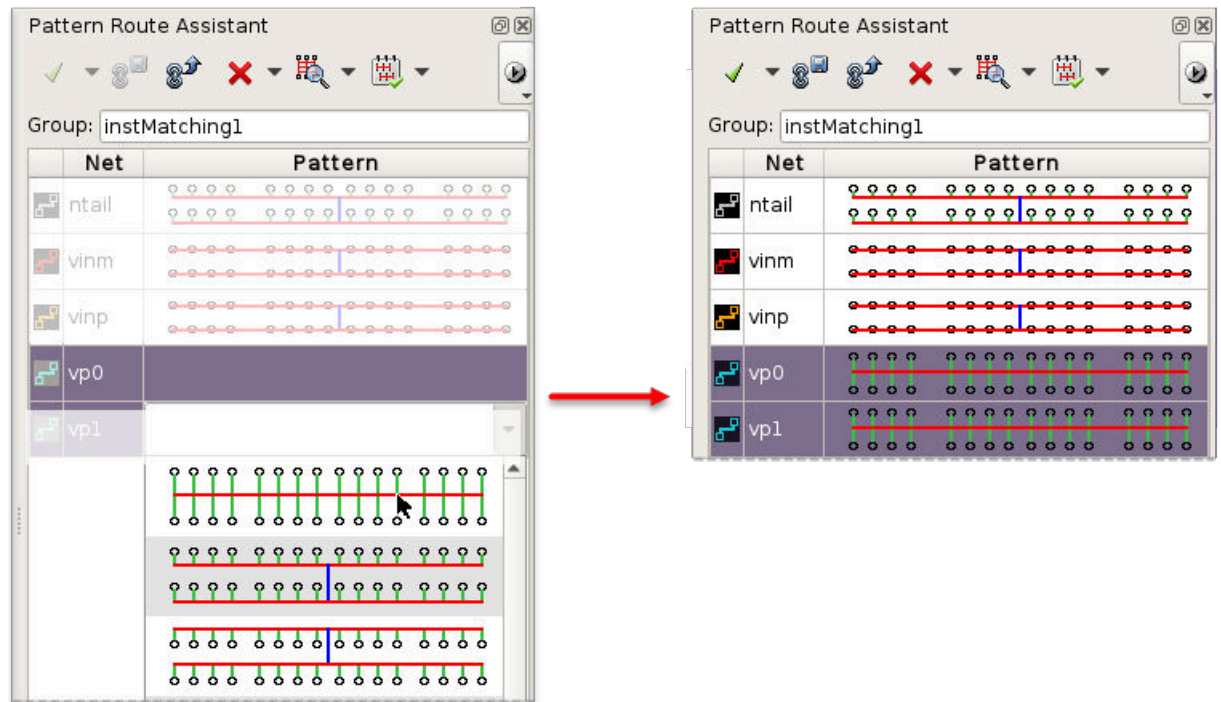


Choosing the Same Pattern for Multiple Nets

You can choose the same pattern and assign it to multiple nets. To do this, you must first select the nets, and then choose a pattern to assign it to the selected nets.

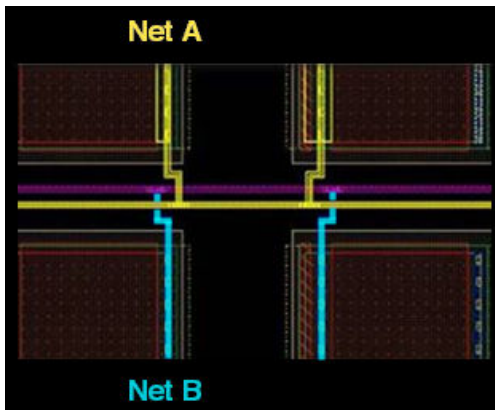
To choose the same pattern for multiple nets:

1. In the Pattern Route Assistant, select multiple nets. The selected nets become highlighted.
2. Choose a pattern for one of the nets in the group. The same pattern is assigned to all the nets in the group.



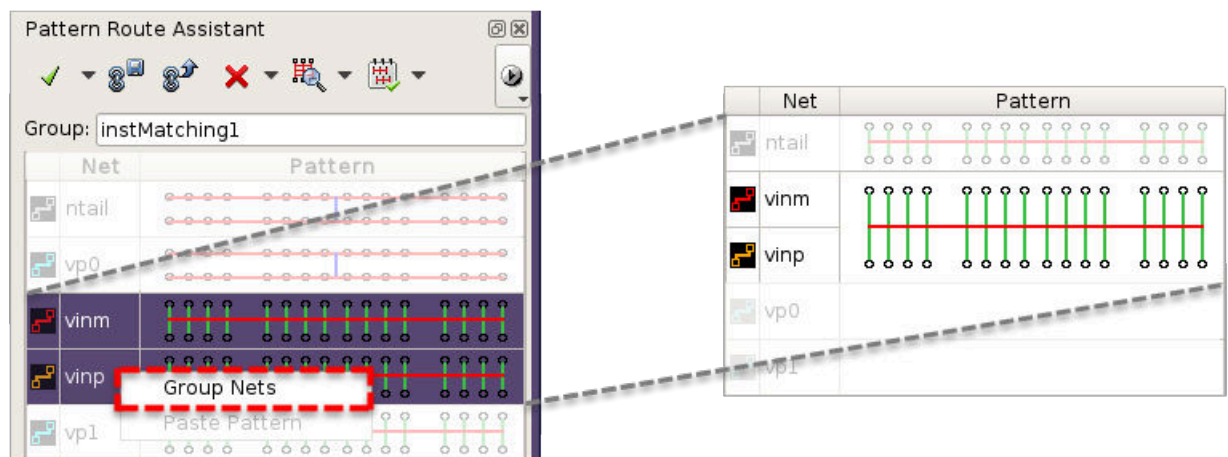
Grouping Nets for Symmetric Routing

To ensure two routes exhibit the same electrical characteristics, you can select two nets and apply a symmetric pattern. If jogs are added, they are also symmetric. In the following example, nets A and B are grouped so that the resulting routes are symmetric.



To group two nets:

1. In the Pattern Route Assistant, select two nets. The selected nets become highlighted.
2. Right-click the selected nets and choose **Group Nets**.
3. Choose a pattern for one of the nets. The same pattern is assigned to the other net. When the nets are routed, the routes and any jogs added will become symmetrical.

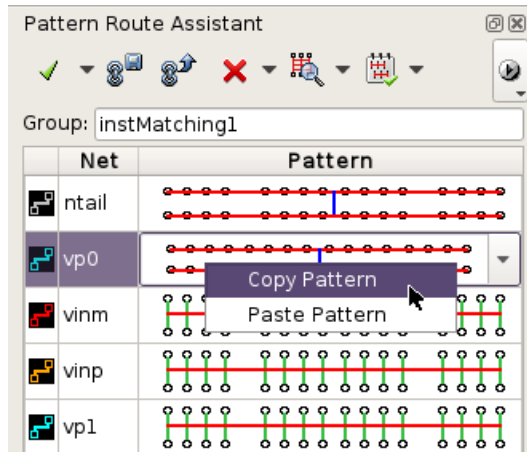


Copying a Pattern to One or More Nets

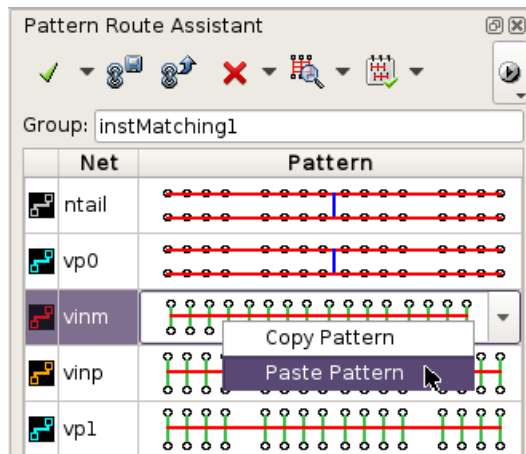
You can copy a route pattern from one net to other nets in the same design.

To copy a pattern to other nets:

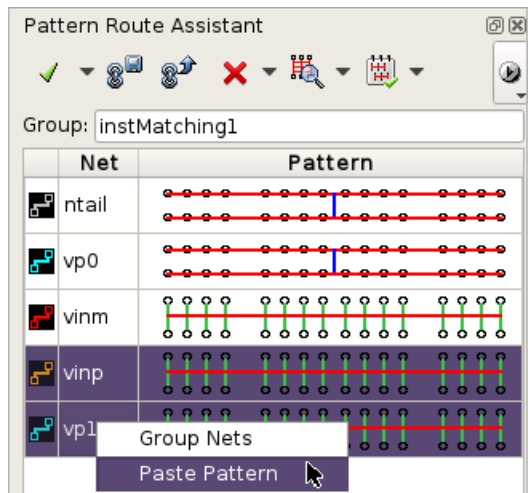
1. In the Pattern Route Assistant, right-click a pattern in the **Net** list and choose **Copy Pattern**.



2. Paste the chosen pattern to one or more nets.
 - To paste the pattern to a single net, right-click the net and choose **Paste Pattern**.



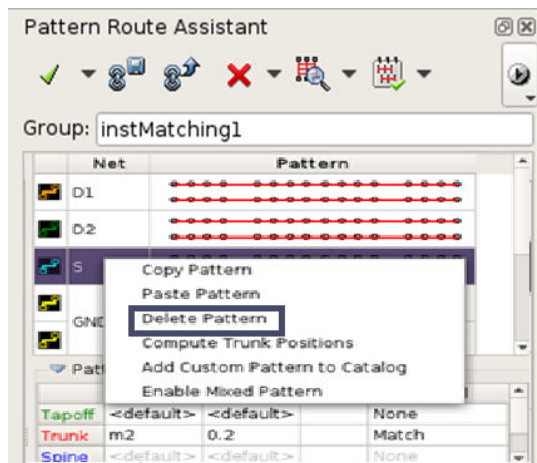
- To paste the pattern to multiple nets, select multiple nets, right-click the nets, and choose **Paste Pattern**.



Deleting a Route Pattern

You can delete a route pattern from one net or multiple nets.

- In the **Pattern Route Assistant**, right-click a pattern in the **Net** list and choose **Delete Pattern**.



To delete route patterns from multiple nets, select multiple nets, right-click the nets, and choose **Delete Pattern**.

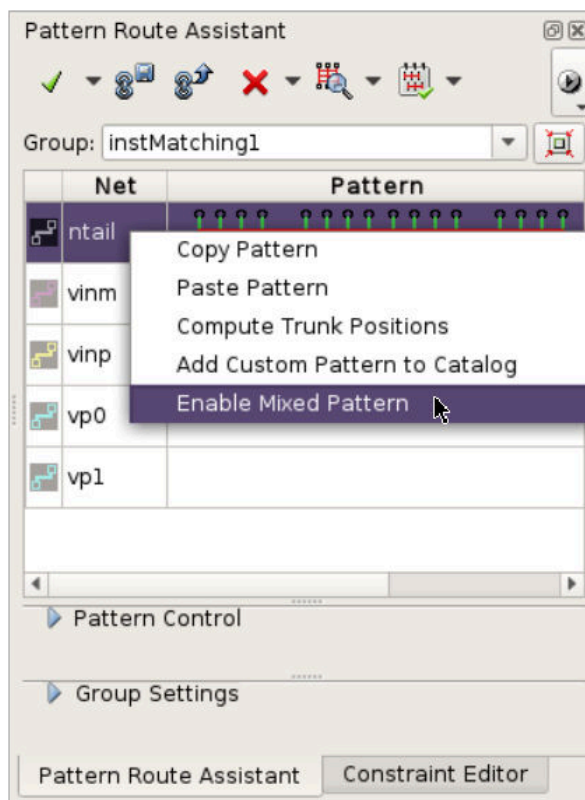
Selecting Different Route Patterns for a Net

You can select different route patterns for routing gate and source/drain pins that are on the same net. This topic describes how to select different patterns for a net.

To select different route patterns for a net:

1. In the pattern catalog section, right-click the net name and choose **Enable Mixed Pattern**.

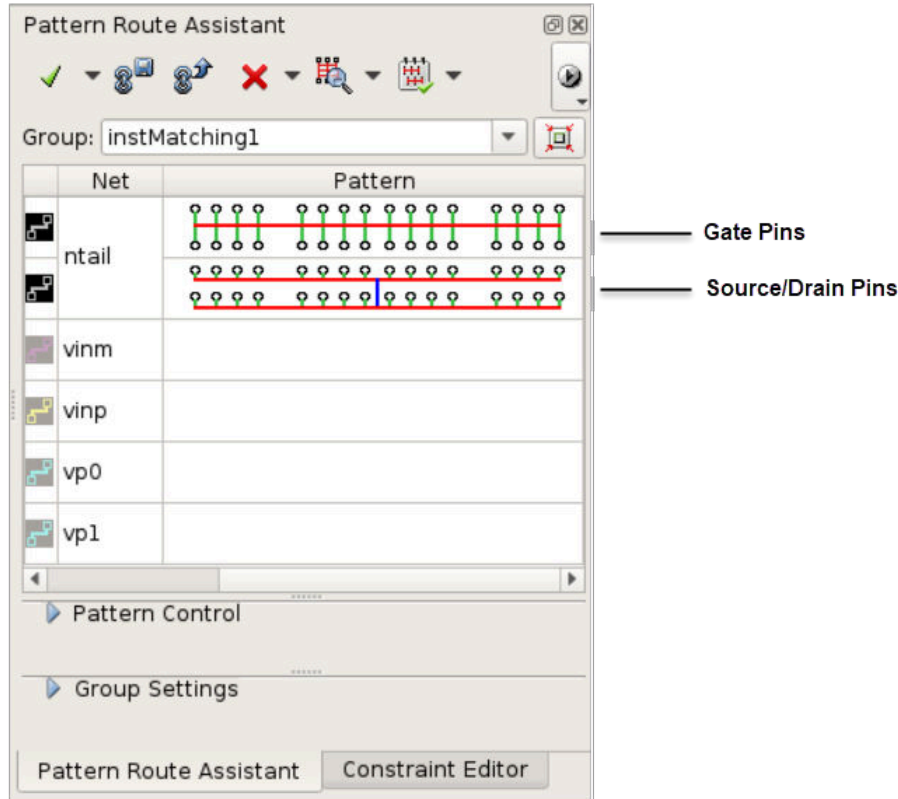
Two rows appear next to the net name.



2. Select a pattern for each row.

Pattern in...	Is for...
First row	Gate pin routing
Second row	Source/drain pin routing

In the following example, the ntail net has two different route patterns. The pattern specified in the first row is for gate pin routing, and the pattern specified in the second row is for source/drain pin routing.



3. To return the net to a single route pattern, right-click the net and choose **Disable Mixed Pattern**.

Previewing Route Patterns

Before creating the route pattern, you can preview the results so that you can adjust the trunk and spine positions directly on the layout canvas. This topic describes how to display a preview of the route pattern on the layout canvas. To make adjustments, see [Adjusting Pattern Components on the Layout Canvas](#) and [Adjusting Pattern Component Attributes in the Assistant](#).

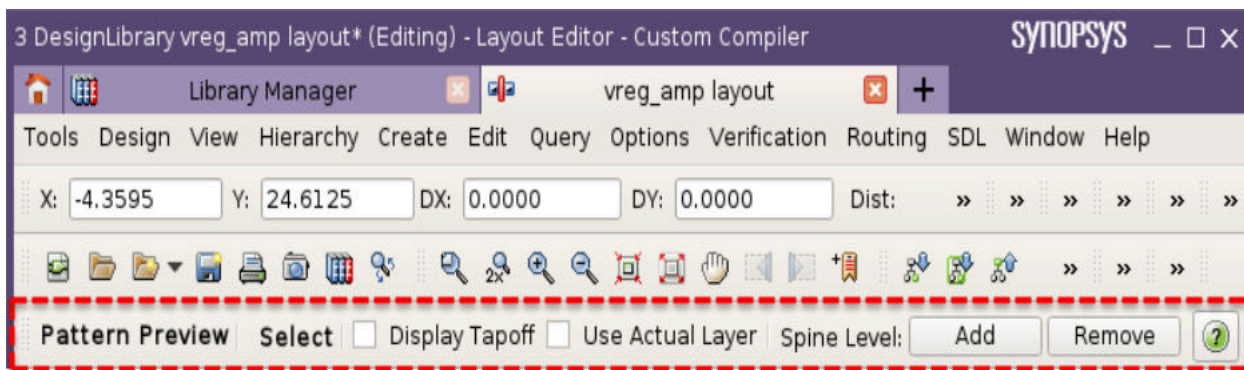
Before you start:

- Make sure the Connectivity Engine is turned on. See [Enabling the Connectivity Engine](#).
- Display the flight lines. See [Displaying and Hiding Flight Lines](#).

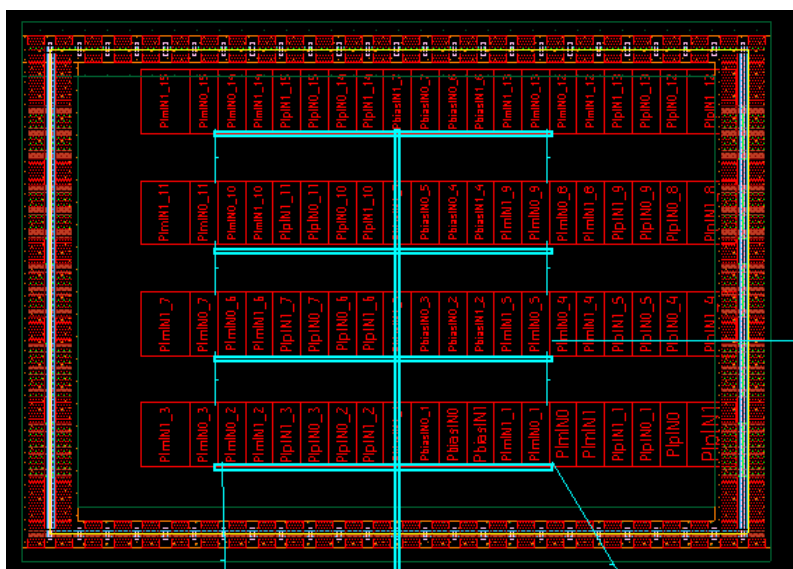
To preview the route pattern on the layout canvas:

1. On the Pattern Route Assistant toolbar, click **Preview Pattern** .

The **Pattern Preview** toolbar appears.



On the layout canvas, the preview trunks and spines appear. The flight line color is used to render the preview routes.



2. On the **Pattern Preview** toolbar, select the options you want to use:
 - **Display Tapoff**: Displays the tapoff segments.
 - **Use Actual Layer**: Displays the preview routes on the layers that will actually be used during routing.

The layout canvas updates to display the tapoffs and actual layer that will be used for routing.

3. Review the preview and assess whether modifications are required.

See Also

- [Adding and Removing Spines](#)
- [Adjusting Pattern Components on the Layout Canvas](#)
- [Adjusting Pattern Component Attributes in the Assistant](#)

Adjusting Pattern Components on the Layout Canvas

After choosing the most applicable route pattern from the pattern catalog, you can refine the components by adjusting their attributes.

You can adjust the attributes using one of the following methods:

- (Recommended) Moving and editing components directly on the layout canvas. The changes you make on the canvas are automatically synchronized with the component attributes in the Pattern Route Assistant.

Manipulating objects on the layout canvas is easier than to guess what values you should use when adjusting the components.

- Modifying the attribute values in the **Pattern Control** section of the Pattern Route Assistant. The changes you make in the assistant are automatically displayed on the layout canvas.

This topic describes how to adjust the route pattern components on the layout canvas. To modify the attributes using the **Pattern Control** section in the Pattern Route Assistant, see [Adjusting Pattern Component Attributes in the Assistant](#).

On the layout canvas, you can adjust the route pattern as follows:

- [Moving Pattern Components](#)
- [Editing Components on the Layout Canvas](#)
- [Adding and Removing Trunks](#)
- [Adding and Removing Spines](#)

Moving Pattern Components

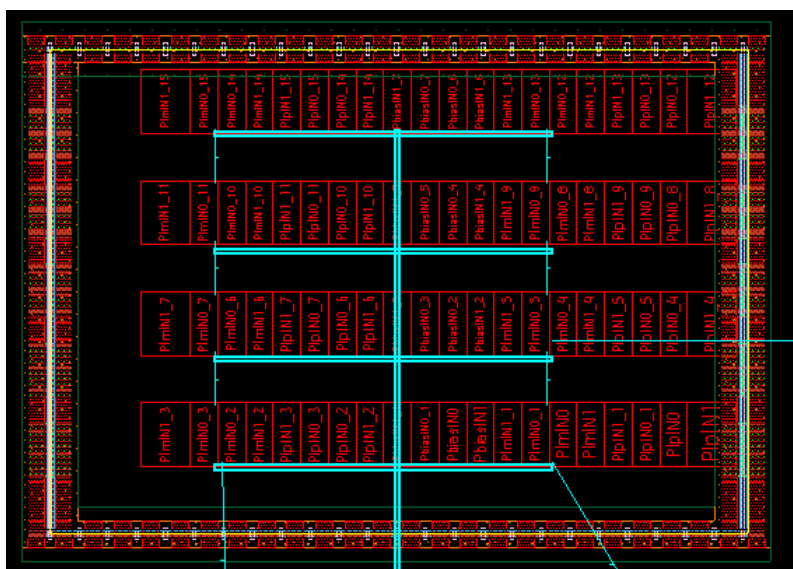
While previewing the route patterns, you can move trunks and spines to adjust their positions in the pattern. You can make these adjustments directly on the layout canvas.

Trunks and spines can be moved to available track locations only. If you want to move a trunk or spine to a location in the channel that currently does not have any tracks, set the `rtPatternRouteIncreaseTrackByFactor` Tcl preference before moving the route pattern components to increase the number of available tracks. For example, if the existing number of tracks is 10, setting the preference value to 2 increases the number of tracks to 20, and setting preference value to 3 increases the number of tracks to 30.

To adjust the route pattern on the layout canvas:

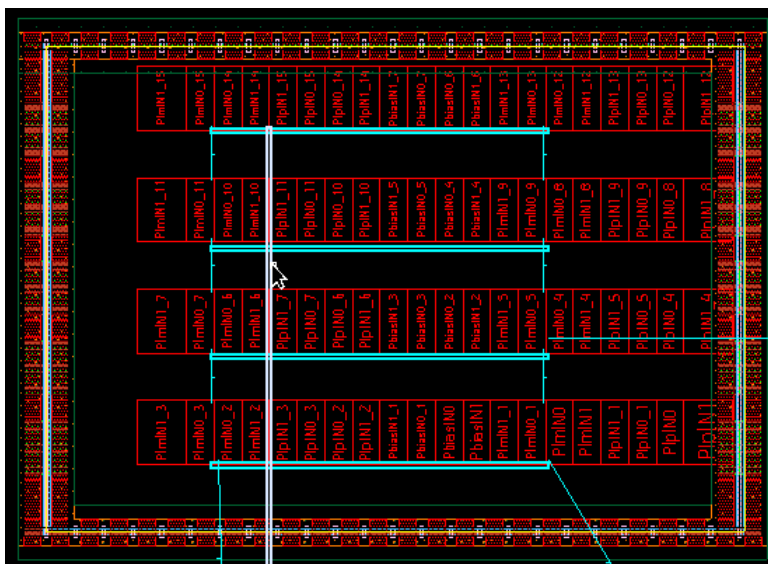
1. Turn on the route pattern preview mode. See [Previewing Route Patterns](#).

On the layout canvas, the preview trunks and spines appear.



2. To adjust a trunk or spine on the layout canvas, use one of the following methods:
 - Preselect method:
 - a. Press M to enable the **Move** command.
 - b. On the layout canvas, select the path you want move.
 - c. Move the path to the new location.
 - d. Click the canvas to commit the move.
 - Postselect method:
 - a. On the layout canvas, select the path you want move.
 - b. Press M to enable the **Move** command.

- c. Click the path on the canvas to start the **Move** command, and move it to the new location.
- d. Click the canvas to commit the move.



3. Check the component attributes in the **Pattern Control** section of the Pattern Route Assistant to review the updated values, such as the **Trunk Position** and **Spine Position**. See [Adjusting Pattern Component Attributes in the Assistant](#).
4. (Optional) To undo the last modification, press U.

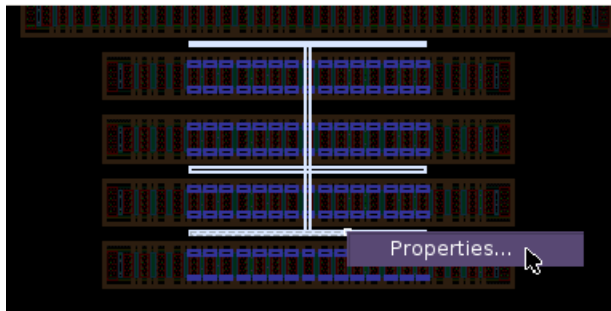
You can press U repeatedly until the action that you want to undo is undone, or until you return to the previously saved state.

Editing Components on the Layout Canvas

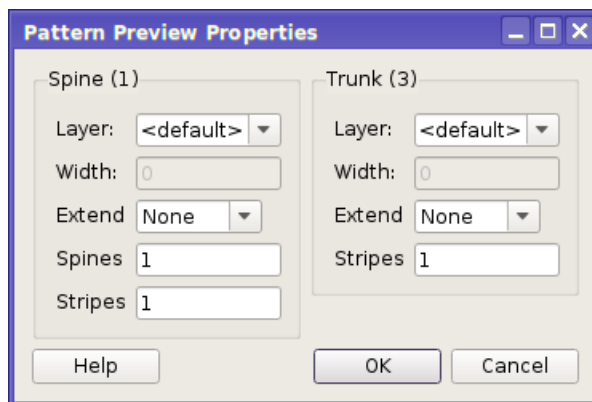
You can edit the Layer, Width, Extend, Spines, and Stripes attributes from the layout canvas.

To edit the pattern attributes from the layout canvas:

1. Turn on the route pattern preview mode. See [Previewing Route Patterns](#).
2. On the layout canvas, select the paths you want to adjust.
3. Right-click the selected paths, and choose **Properties**.



The **Pattern Preview Properties** dialog box opens. The dialog box displays a section for each component type you selected. In the following example, one spine and 3 trunks were selected.



4. For each component type, choose the **Layer** to use for the route component. This setting overrides the **Layer** setting in the Auto Route dialog box.

If you do not specify a layer, the Auto Router will use the default layer and width defined by the technology or design-specific layer constraint.

5. Set the **Width** of the component.

This setting overrides the layer **Width** setting in the Auto Route dialog box.

Set the Layer property before setting the Width property.

Setting Width to 0 uses the default width, which is one of the following:

- Largest of the following width values: technology `minWidth`, or non-default rule (NDR) minimum width.
- If you specify **Stripes**, **Width** is the minimum sum of the stripe widths.
- If the topology segment is snapping to routing tracks, **Width** inherits the track width.

6. Choose whether to lengthen tapoffs, trunks, or spines, and the method of the extension. For detailed descriptions of the choices (**None**, **Trunks**, **Both**, **Top**, **Bottom**), see [Adjusting Pattern Component Attributes in the Assistant](#).
7. Increase or decrease the number of **Spines**.

For example, if you change the number of Spines to 3, the preview updates and displays 3 spines.
8. Increase or decrease the number of wire **Stripes** in trunks and spines.
9. When you are finished, click **OK**. The route pattern updates.
10. (Optional) To undo the last modification, press U.

You can press U repeatedly until the action that you want to undo is undone, or until you return to the previously saved state.

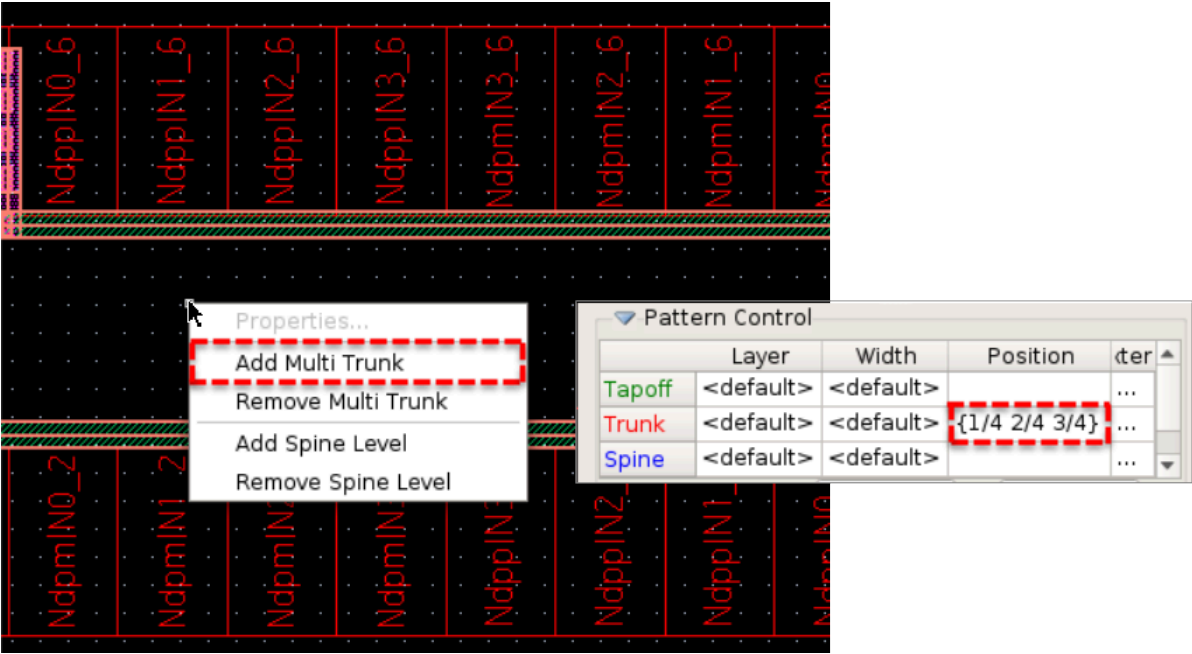
Adding and Removing Trunks

While previewing the route patterns, you can add new trunks in a channel. The trunks in the channel must all be on the same net.

To add or remove trunks in a channel:

1. Turn on **Preview Pattern**. See [Previewing Route Patterns](#).
2. In the device channel where you want to add trunks, right-click and choose **Add Multi Trunk**. Repeat this step to add additional trunks in the channel.

In the Pattern Route Assistant **Pattern Control** section, fraction values appear in the Trunk **Position** column. For information about the values, see [Specifying the Position Attribute](#).



3. To remove a trunk from the channel, right-click and choose **Remove Multi Trunk**.

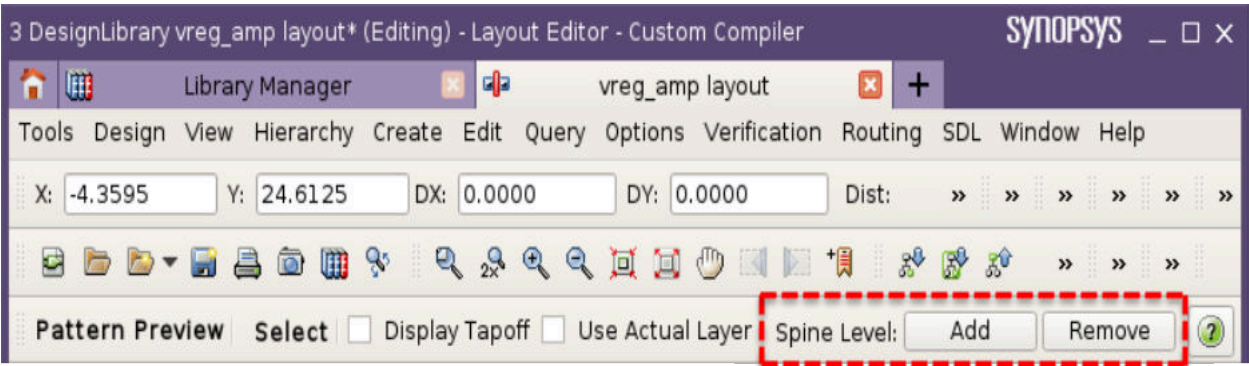
One of the trunks in the channel is removed, and the Position values are updated in the **Pattern Control** table.

Adding and Removing Spines

While previewing the route patterns, you can add new spines to connect existing spines. You can also remove the most recently added spine.

To add or remove spines, use one of the following methods:

- On the **Pattern Preview** toolbar, locate **Spine Level**, and click **Add** or **Remove**.



As you add or remove spines from the toolbar, the changes are also reflected in the Pattern Route Assistant **Pattern Control** section. See [Adjusting Pattern Component Attributes in the Assistant](#).

- On the layout canvas, at the location where you want to add or remove the spine, right-click and choose **Add Spine Level** or **Remove Spine Level**.



Adjusting Pattern Component Attributes in the Assistant

After choosing the most applicable route pattern from the pattern catalog, you can refine the components by adjusting their attributes.

You can adjust the attributes using one of the following methods:

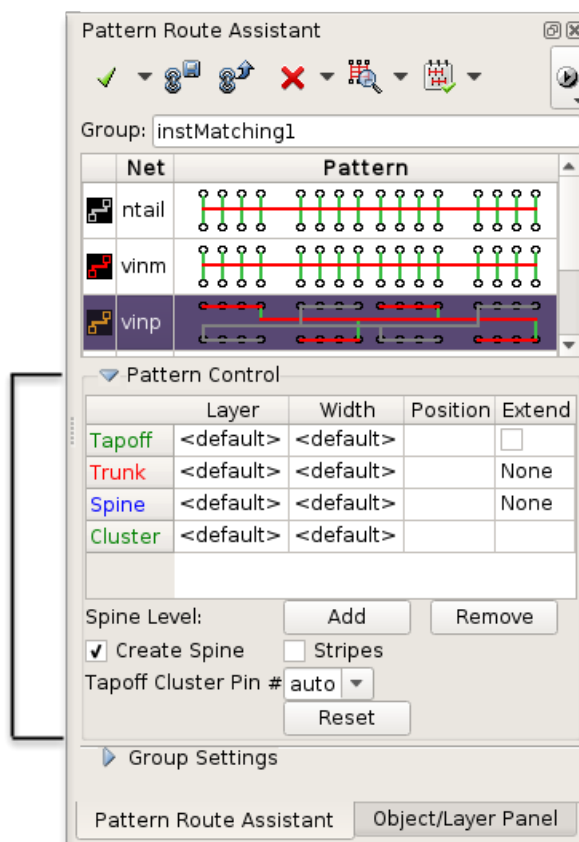
- (Recommended) Moving and editing components directly on the layout canvas. The changes you make on the canvas are automatically synchronized with the component attributes in the Pattern Route Assistant.

Manipulating objects on the layout canvas is easier than to guess what values you should use when adjusting the components. Therefore, this is the recommended method.

- Modifying the attribute values in the **Pattern Control** section of the Pattern Route Assistant. The changes you make in the assistant are automatically displayed on the layout canvas.

This topic describes how to adjust the pattern component attributes using the **Pattern Control** section in the Pattern Route Assistant.

- [Setting the Layer and Width Attributes](#)
- [Specifying the Position Attribute](#)
- [Extending Tapoffs, Trunks, and Spines](#)
- [Adding Spines](#)
- [Specifying the Number of Wire Stripes](#)
- [Setting Cluster Attributes](#)



To move and edit the components on the layout canvas, see [Adjusting Pattern Components on the Layout Canvas](#).

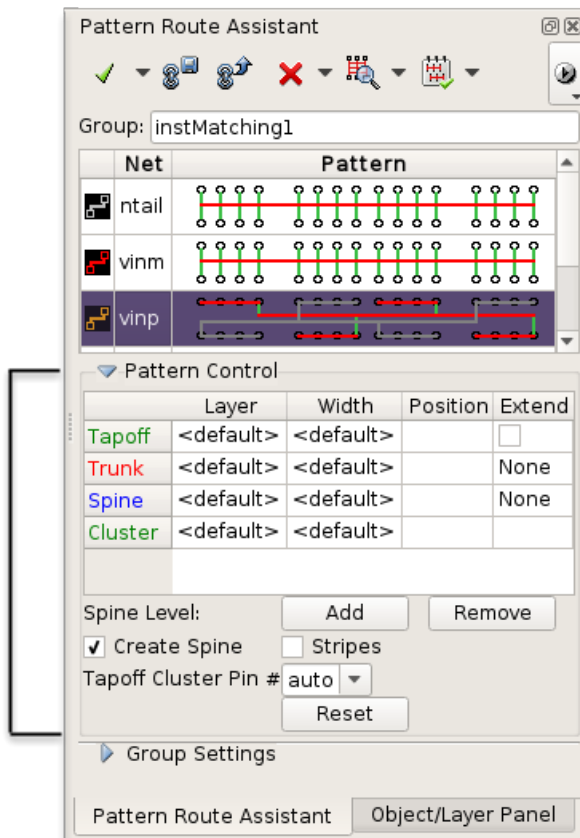
Note:

The **Cluster** attribute only appears if you selected a net that is assigned a cluster pattern. See [Pattern Route Components](#).

The customized pattern is available for the current design only.

Setting the Layer and Width Attributes

In the Pattern Route Assistant, you can set the **Layer** and **Width** attributes for tapoff, trunk, spine, or cluster connections. The attributes are in the **Pattern Control** section of the assistant.



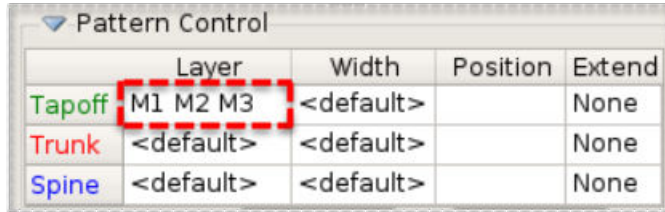
Note:

The **Cluster** attribute only appears if you selected a net that is assigned a cluster pattern. See [Pattern Route Components](#).

To set the Layer and Width attributes:

1. In the Pattern Route Assistant, select a net or a group of nets that have the same pattern.
2. In the **Pattern Control** section, choose the **Layer** of the tapoff, trunk, spine, or cluster connection. The list of layers is derived from the technology database.

If you want to stack the routes, type the names of the layers to use for the stacking. For example, if you want to use layers M1, M2, and M3 for stacked routing, type M1 M2 M3 in the **Layer** box. Use a character space between the layer names.



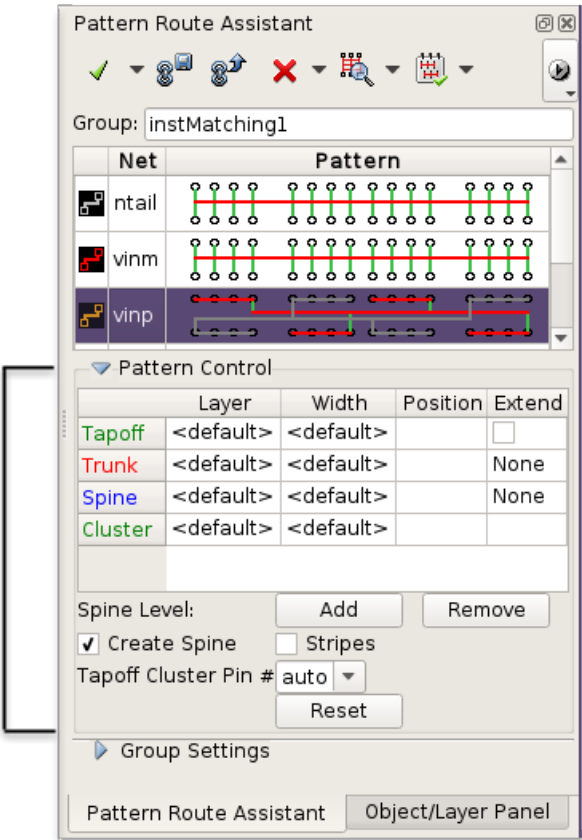
3. Choose the **Width** of the tapoff, trunk, spine, or cluster connection. The list contains supported widths for the selected layer, which are derived from the technology database.

All stacked layers must use the same width.

Specifying the Position Attribute

In the Pattern Route Assistant, you can specify the **Position** of the tapoff, trunk, spine, or cluster connections. The attribute is in the **Pattern Control** section of the assistant.

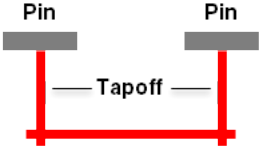
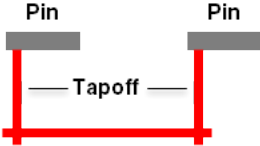
Although you can type a value to set the position, you might want to visually adjust the positions on the layout canvas first. See [Previewing Route Patterns](#).



Note:

The **Cluster** attribute only appears if you selected a net that is assigned a cluster pattern. See [Pattern Route Components](#).

The positions of the connecting objects that you specify are relative to the location of the selected nets and to the connecting objects themselves. You express the positions using fractions.

Connecting Object	Position Description
Pins	By default, tapoffs are centered at the pins (1/2). You can change its position by changing the position ratio. For example, you can specify 1/4 to move the tapoff to the left, as shown. You specify one value for pin positions. Tapoff Position = 1/2 (Default)  The diagram shows two gray rectangular pins. A red horizontal line connects the bottom of each pin. A vertical red line, labeled 'Tapoff', is positioned exactly in the middle of the space between the two pins. Tapoff Position = 1/4  The diagram shows two gray rectangular pins. A red horizontal line connects the bottom of each pin. A vertical red line, labeled 'Tapoff', is positioned closer to the left pin than the right pin, representing a 1/4 position ratio.

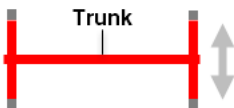
Connecting Object	Position Description
-------------------	----------------------

Trunks

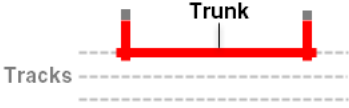
Single trunk in a channel

In fishbone structures, the trunk is centered in the channel by default. In backbone structures, the trunk is snapped to the track closest to the pins by default.

Trunk Position = 1/2
(Fishbone Default)

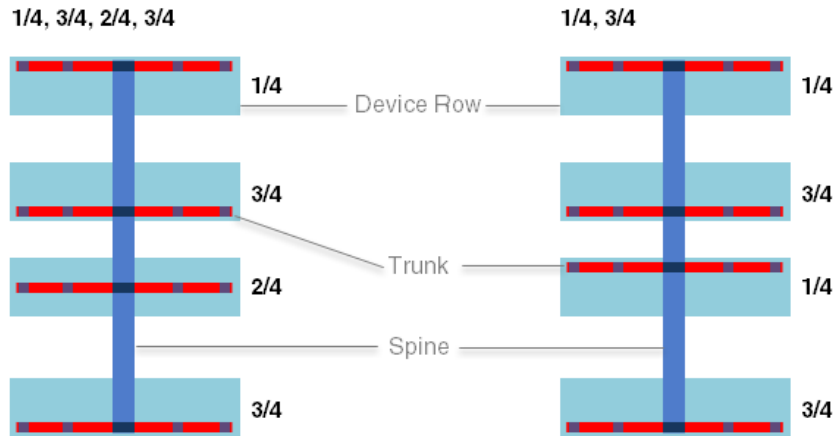


Trunk Position
(Backbone Default)



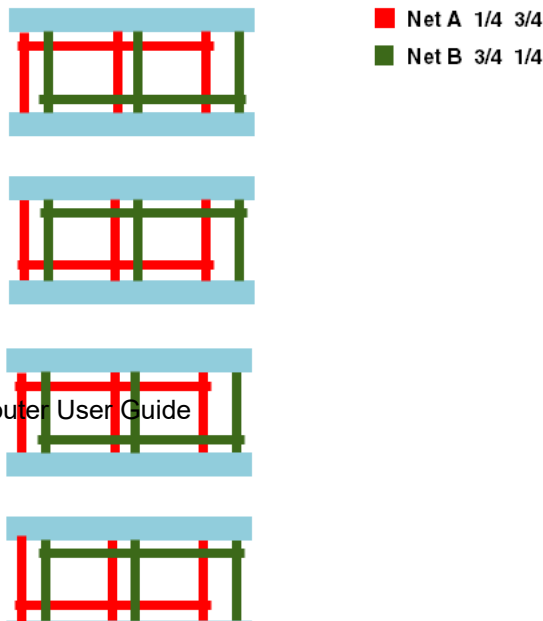
You can specify more than one trunk position. You must add a character space between position values.

For example, suppose a route pattern contains four trunks that are over devices. You can specify four different trunk positions: 1/4 3/4 2/4 3/4. If you only specify two trunk positions (1/4 3/4), the positions will be repeated.



One example application for specifying repeating trunk positions is to create symmetric or matched routing patterns.

Symmetric Routing Pattern



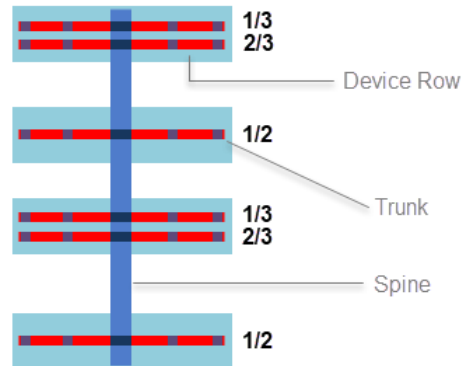
Connecting Object	Position Description
-------------------	----------------------

Multiple same-net trunks in a channel

If you want to place multiple same-net trunks in the same channel, use the following **Position** syntax: $\{1/3\ 2/3\}\ \{1/2\}$.

In the example, two trunks are placed in the first channel at positions 1/3 and 2/3, and one trunk is placed in the second channel at position 1/2.

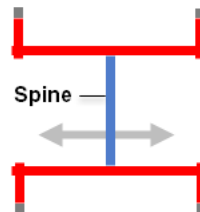
$\{1/3, 2/3\}\ \{1/2\}$



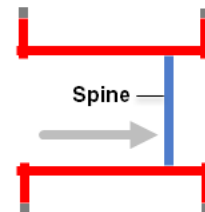
Spines

By default, the spine is centered within the row spacing.

Spine Position = 1/2
(Default)

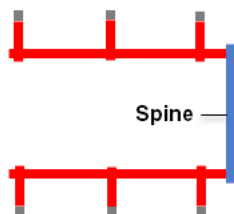


Spine Position = 2/3

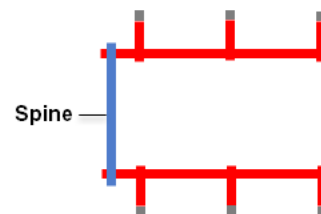


To place a spine to the right beyond the row, specify a fraction > 1 . To place a spine to the left beyond the row, specify a negative fraction.

Spine Position = 5/4



Spine Position = -1/4



To specify the position of the connecting objects:

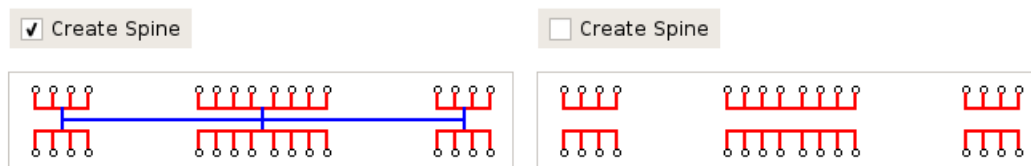
1. In the Pattern Route Assistant, select a net or a group of nets that have the same pattern.
2. Set the **Position** of the tapoff, trunk, or spine.

Type one or more values in the **Position** box:

- Fractional value between 0 and 1: Positions the tapoffs, trunks, or spines at the pin or within the row. For example, 1/2 centers trunks within their rows.
- Value < 0: Positions spines to the left of the row.
- Value > 1: Positions spines to the right of the row.

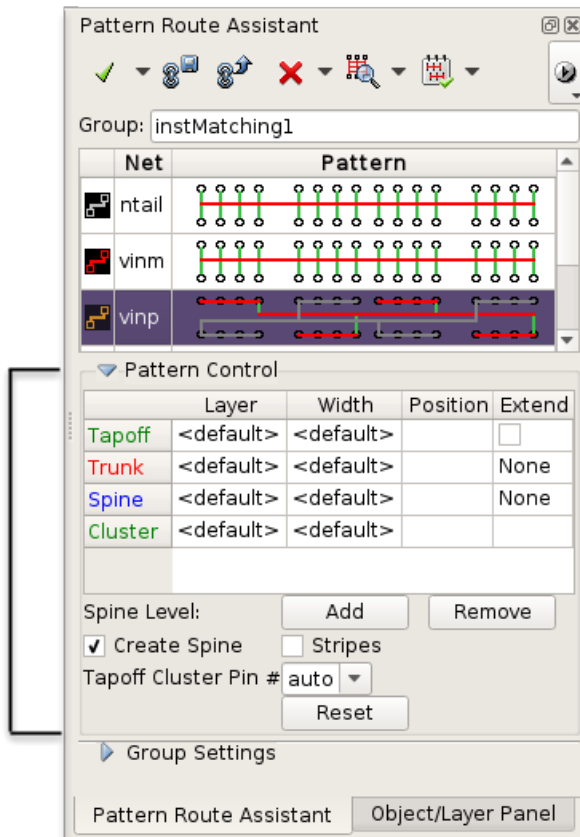
If you specify multiple values for trunks and spines, add a character space between position values.

3. Select **Create Spine** to add spines in the route patterns. Clear the check box if you do not want to add spines in the route pattern.



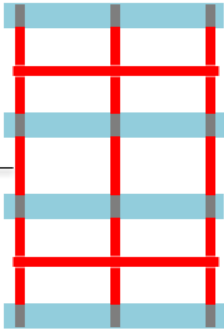
Extending Tapoffs, Trunks, and Spines

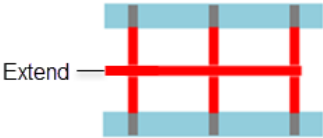
You can lengthen the tapoff, trunk, or spine using the **Extend** option in the **Pattern Control** section of the Pattern Route Assistant. The **Extend** attribute does not apply to **Cluster** patterns.

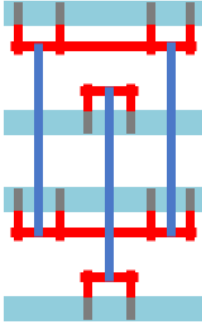
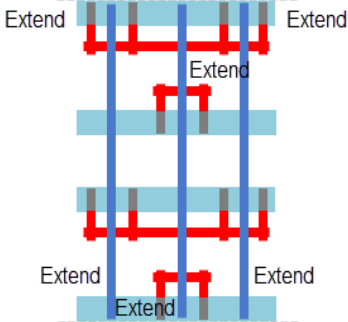
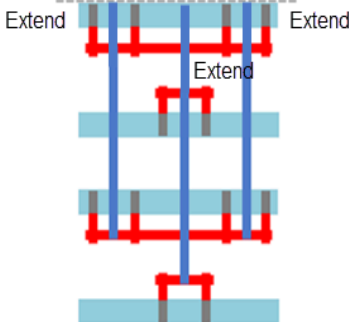
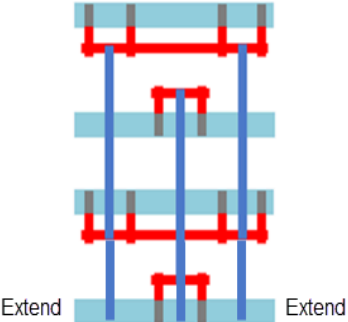


To lengthen or extend tapoffs, trunks, or spines:

1. In the Pattern Route Assistant, select a net or a group of nets that have the same pattern.
2. In the **Pattern Control** section, choose the **Extend** option to lengthen the tapoff, trunk, or spine.

When extending...	Choose...	
Tapoffs	Extend: None	Extend: Both
		
		Extend — 
	Extend: None	Extend: Match
		
		Extend 

When extending...	Choose...	
Trunks	Extend: None	Extend: Both
		
	Extend: Left	Extend: Right
		
	Extend: None	Extend: Match
		
	Extend: None	Extend: Pinedge
		

When extending...	Choose...	
Spines	Extend: None 	Extend: Both 
	Extend: Top 	Extend: Bottom 

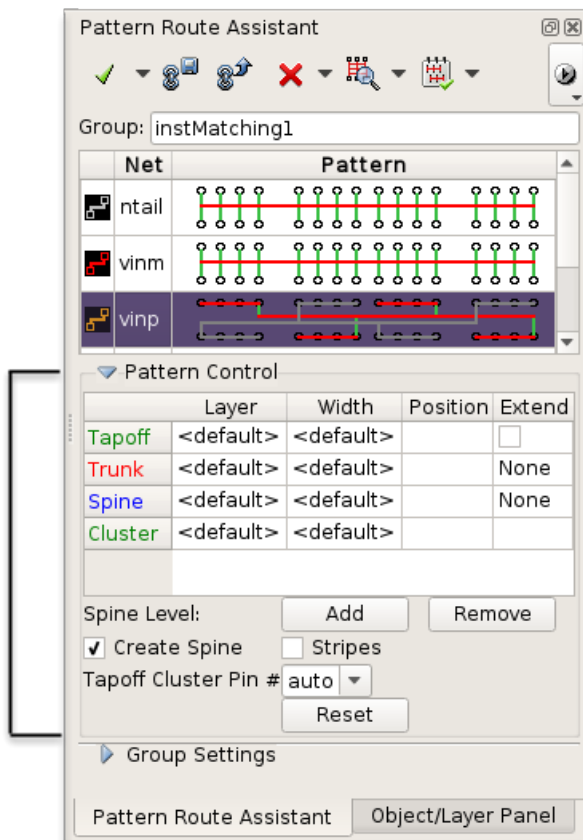
- To explicitly specify a length value, type the value (in um) next to the **Extend** selection.

You cannot explicitly specify an extension value for **Match**.

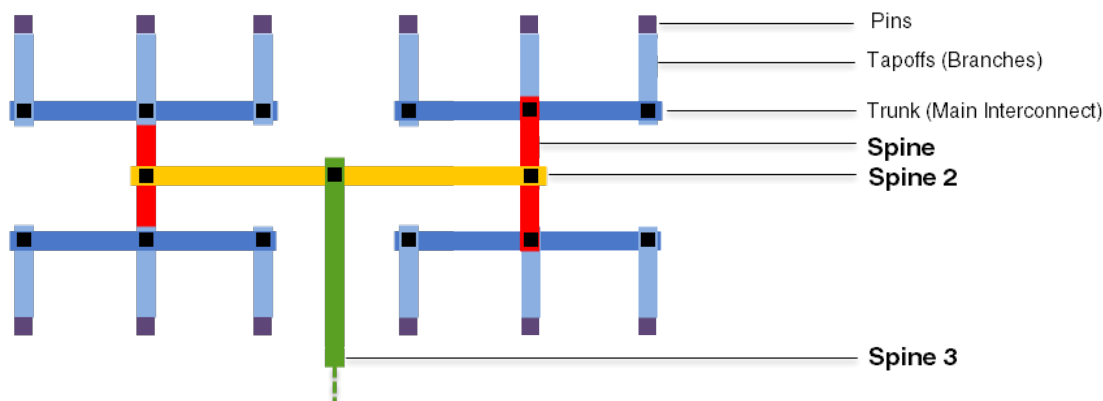
Pattern Control			
	Position	Extend	Stripes
Tapoff		Both	
Trunk		Left 1.7	1
Spine	1/4 2/4 3/4	None	3
Spine 2	1/2	None	1

Adding Spines

In the Pattern Route Assistant, you can add spines to selected nets. The **Create Spine** and **Spine Level: Add** options are in the **Pattern Control** section of the assistant.



Adding another level of spine connects existing spines. In the following example, Spine connects trunks, Spine 2 connects Spines, Spine 3 connects Spine 2s, and so on.



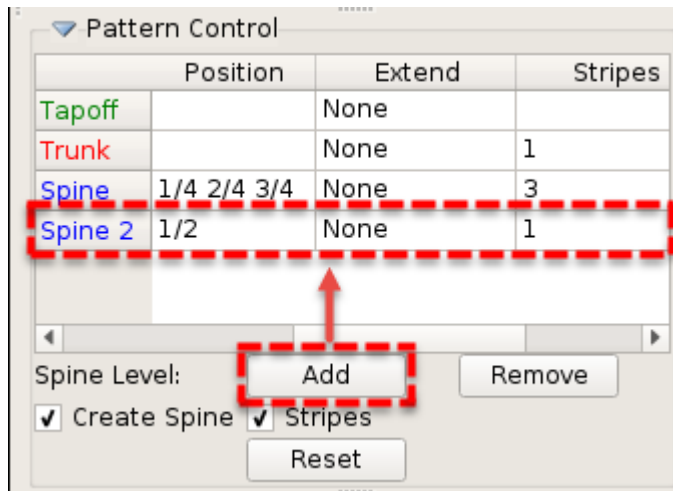
To add spines:

1. In the Pattern Route Assistant, select a net or a group of nets that have the same pattern.
2. In the **Pattern Control** section, select **Create Spine** to add spines to connect trunks or other spines.



3. To insert new spines that connect existing spines, click **Spine Level: Add**.

A new **Spine n** row appears in the attributes table.



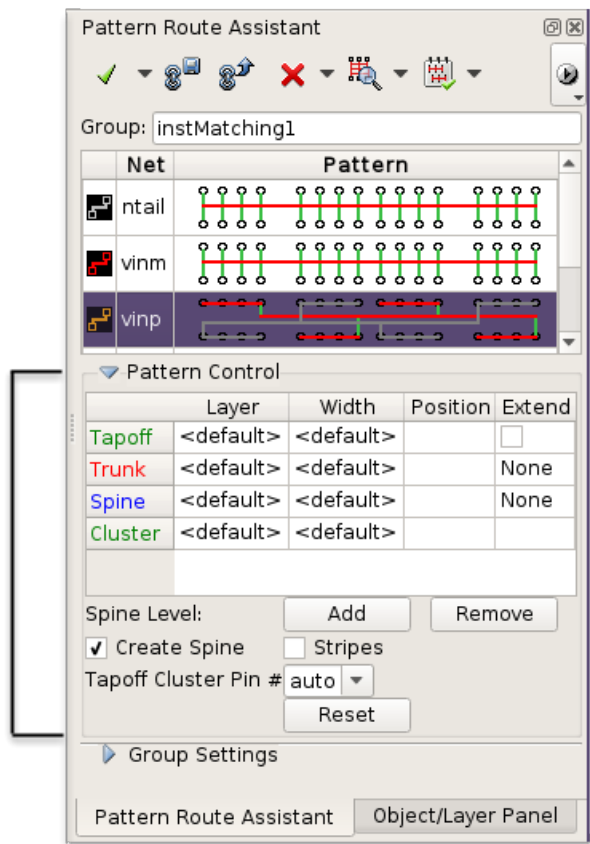
4. Specify the attributes for the newly added spine: **Layer**, **Width**, **Position**, **Extend**, and **Stripes**.
5. To delete the most recently added **Spine n** from the attributes table, click **Spine Level: Remove**.

See Also

- [Setting the Layer and Width Attributes](#)
- [Specifying the Position Attribute](#)
- [Extending Tapoffs, Trunks, and Spines](#)
- [Specifying the Number of Wire Stripes](#)

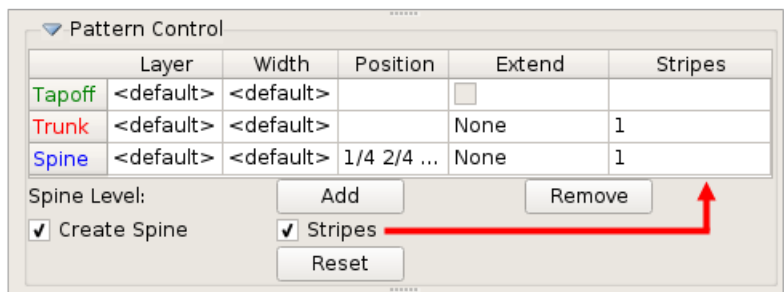
Specifying the Number of Wire Stripes

In the Pattern Route Assistant, you can specify the number of wire stripes in trunks and spines. The **Stripes** attribute is in the **Pattern Control** section of the assistant.



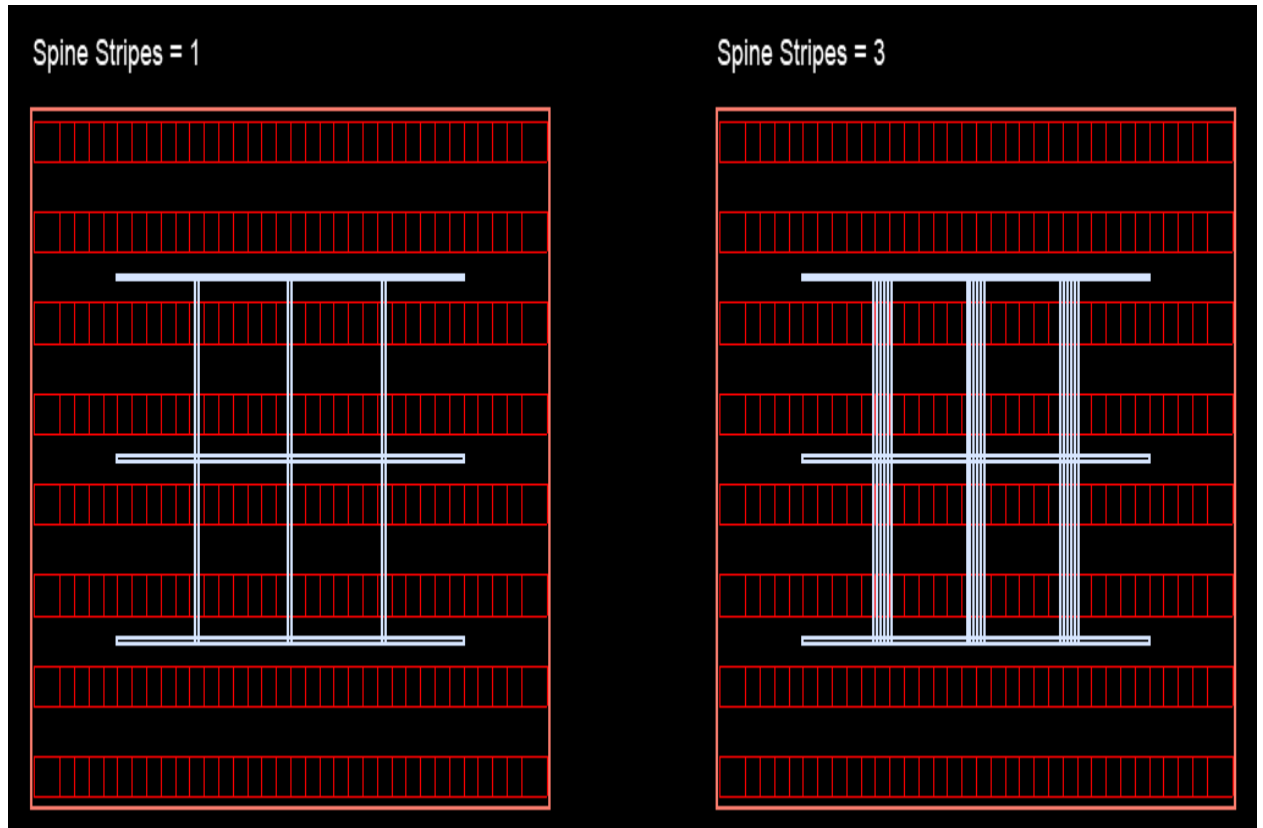
To specify the number of wire stripes:

1. In the Pattern Route Assistant, select a net or a group of nets that have the same pattern.
2. In the **Pattern Control** section, select **Stripes** to display the Stripes column.



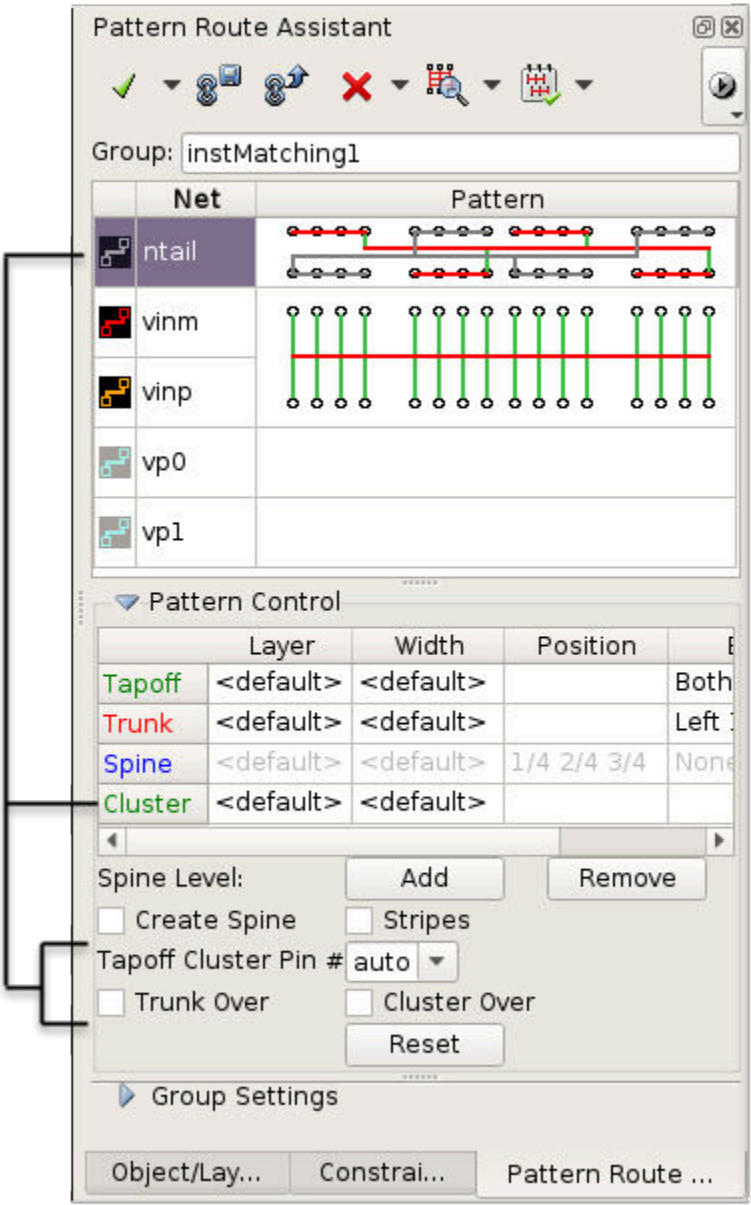
3. In the **Stripes** column, type the number of wire stripes in trunk or spine.

In the following example, the number of spine stripes is changed from 1 to 3.



Setting Cluster Attributes

In the Pattern Route Assistant, you can set attributes to refine cluster patterns. The attributes are in the **Pattern Control** section of the assistant.



To set cluster pattern attributes:

1. In the Pattern Route Assistant, select a net or a group of nets that have the cluster pattern.
2. In the **Pattern Control** section, choose the **Tapoff Cluster Pin #**, the number of pins in a cluster:
 - **auto**: (Default) Allows the router to determine the number of pins that are grouped in a cluster. Different clusters might contain different number of pins.
 - **<integer>**: Specifies the number of pins in the clusters. The router will try to group the same number of pins in every cluster.

If the number of neighboring pins exceeds the value you specify, the router will create multiple clusters and group any remaining pins.

In the following example, # Pins in Tapoff Cluster is 3. The router will create as many 3-pin clusters as possible.

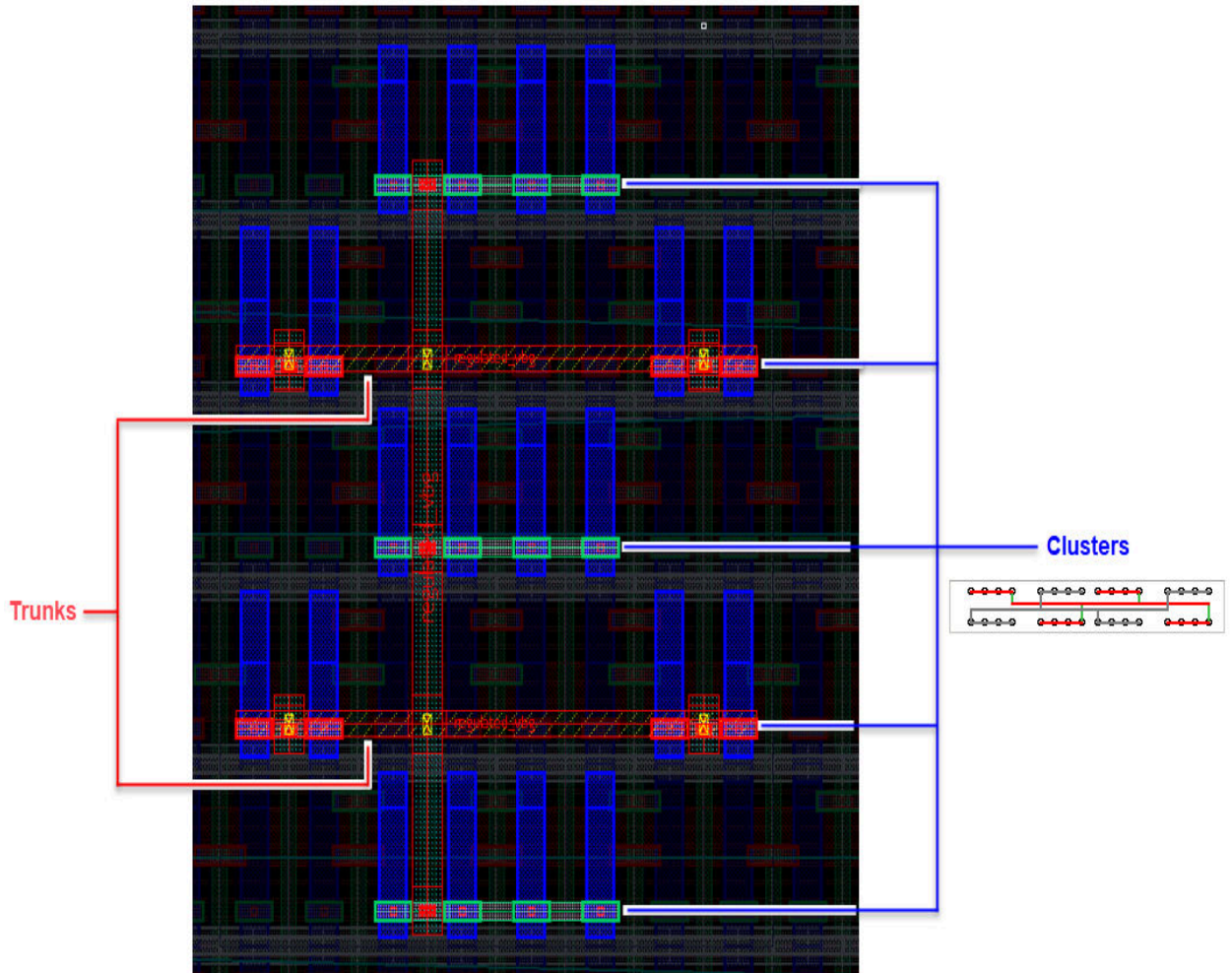


If the number of neighboring pins is less than the value you specify, the router will still group the pins.

In the following example, the # Pins in Tapoff Cluster is 10. The router will still create a cluster of 9 pins.



3. Select **Trunk Over** if you want to connect two or more clusters using trunks that are on a different layer, thus eliminating the need for tapoffs.



4. Select **Cluster Over** if you want to connect over the pins without routing inside the channel.

Running the Group Planner

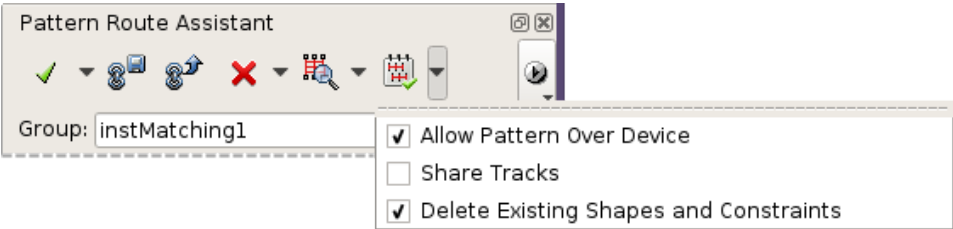
The Group Planner uses the **Pattern Control** settings, the **Group Settings**, and existing Pattern Route constraint to determine the best route pattern for each net in the figure group.

If you have not yet specified the **Group Settings**, the default options are used to create a rough starting pattern. After running the Group Planner, you can refine the pattern by adjusting the **Pattern Control** and **Group Settings**. See [Specifying Group Settings](#).

If you have critical nets, see [Specifying Group Settings](#) to determine the placement of the critical routes first, and then use the Group Planner to plan new routes around them.

To run the Group Planner:

1. On the Pattern Route Assistant toolbar, display the **Group Planner** menu, and then select the options you want to use.



Option	Description
Allow Pattern Over Device	Permits routes to be placed on top of the devices if necessary. Clear the check box if you want the routes to be placed in channels only. Tcl preference: <code>rtPatternRouteAssistantAllowPatternOverDevice</code>
Share Tracks	Permits trunks within the same channel to share tracks to optimize track and channel use. ☒ Share Tracks ☐ Share Tracks 
Delete Existing Shapes and Constraints	Deletes existing route shapes, including vias, and associated constraints before determining the route patterns. Clear the check box if you want to retain existing routes and vias and add new routes to unconnected pins. Clear the check box if you want plan route patterns incrementally. Tcl preference: <code>rtPatternRouteAssistantRemoveRouteAndConstraintBeforePlanning</code>

2. Click **Group Planner** .

After a few seconds, a report appears in the Console. The report shows the number of nets planned, the row spacing and the pattern used for each net.

```
*****
* Nets to Plan Estimated Row Spacing Time *
*      5      0.698  0:00:00 *
*****
* Nets Pattern
* ntail backbone over track 1 / 3 layer M3 extend;
* vp0 backbone over track 2 / 3 layer M3 extend;
* vinm backbone track 1 / 5 layer M3 extend mirror;
* vinp backbone track 2 / 5 layer M3 extend mirror;
* vpl fishbone track 3 / 5 layer M3 extend;
*****
```

3. If you need to make adjustments, see [Adjusting Pattern Component Attributes in the Assistant](#) and [Specifying Group Settings](#).

After running the Group Planner and reviewing the computed routes, you have the option of making any necessary adjustments to refine the results. The **Group Settings** let you modify trunk spacing, row spacing, and other placement settings. To adjust pattern components, see [Specifying Group Settings](#) and [Adjusting Pattern Component Attributes in the Assistant](#).

Specifying Group Settings

The **Group Settings** let you specify trunk spacing, row spacing, and other device-placement options. Together with the **Pattern Control** settings, they determine the resulting route pattern.

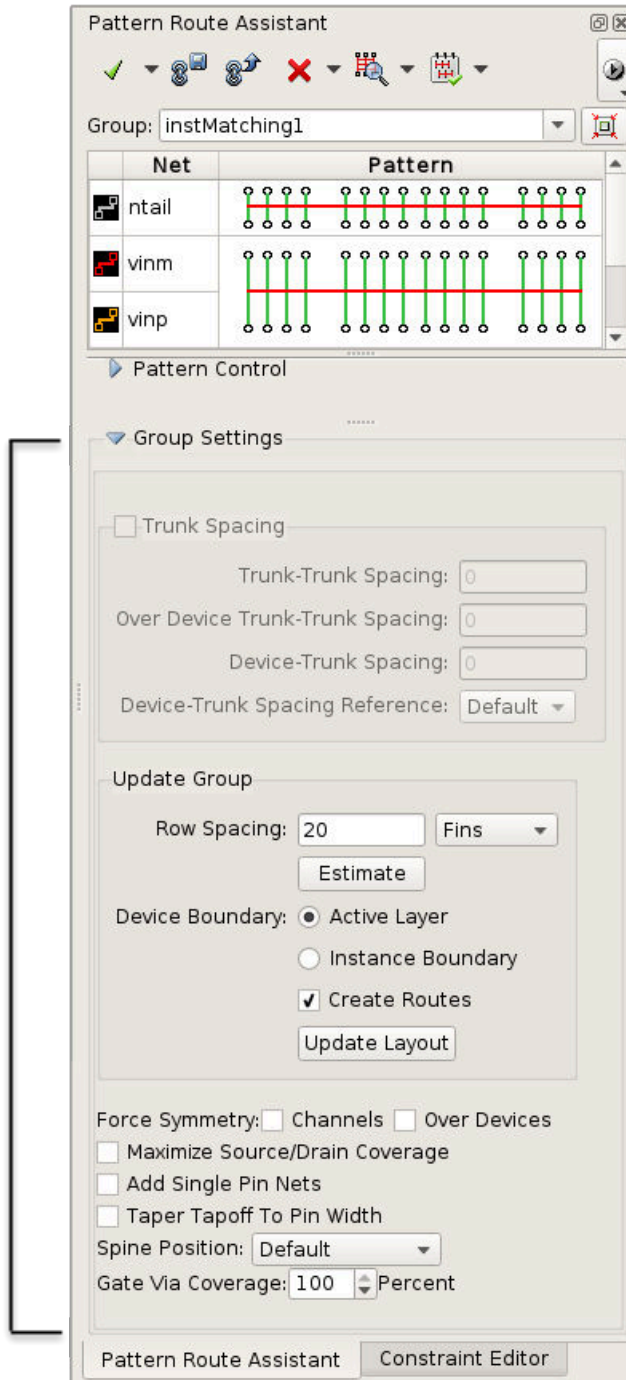
You can use the **Group Settings** with the Group Planner iteratively to plan route patterns incrementally. For example, you can use the Group Settings to determine the placement of critical routes first, and then use the Group Planner to plan new routes around them. See [Running the Group Planner](#).

Caution:

Always run the Group Planner after you modify the Trunk Spacing values.

Note:

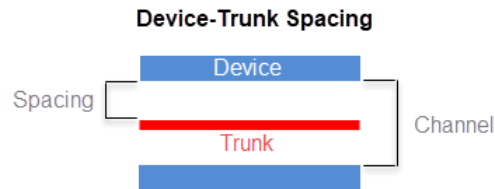
The Trunk Spacing values are saved for the currently selected group.



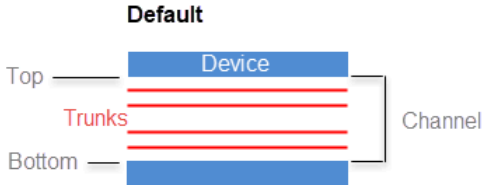
To specify the **Group Settings**:

1. To adjust the trunk spacing, select or specify the following:

Option	Description
Trunk Spacing	Lets you control the spacing between adjacent trunks, and the spacing between devices and trunks. Tcl preference: <code>rtPatternRouteAssistantTrunkTrunkSpacingEnable</code>
Trunk-Trunk Spacing	Edge-to-edge distance between adjacent trunks in channels only. Tcl preference: <code>rtPatternRouteAssistantTrunkTrunkSpacing</code>
Over Device Trunk-Trunk Spacing	Edge-to-edge distance between adjacent trunks that are placed on top of the devices instead of in channels. Tcl preference: <code>rtPatternRouteAssistantOverDeviceTrunkTrunkSpacing</code>
Device-Trunk Spacing	Edge-to-edge distance between the device and the adjacent trunk in the channel.



Tcl preference:
`rtPatternRouteAssistantTrunkDeviceSpacing`

Option	Description
Device-Trunk Spacing Reference	Side of the device channel from which Device-Trunk Spacing is determined: ◦ Default: Both the top and bottom sides of the channel. ◦ Top: Top side of the device channel. ◦ Bottom: Bottom side of the device channel. Device-Trunk Spacing Reference Default  Top Top — Trunks Bottom — Channel Top Top — Device Bottom Bottom — Device Tcl preference: rtPatternRouteAssistantTrunkDevices pacingRef

2. To adjust the distance between adjacent rows:
- In the **Update Group** section, click **Estimate** to calculate the estimated row spacing required.
 - If necessary, edit the calculated **Row Spacing** value, and choose the unit of measure (**fins** or **microns**).

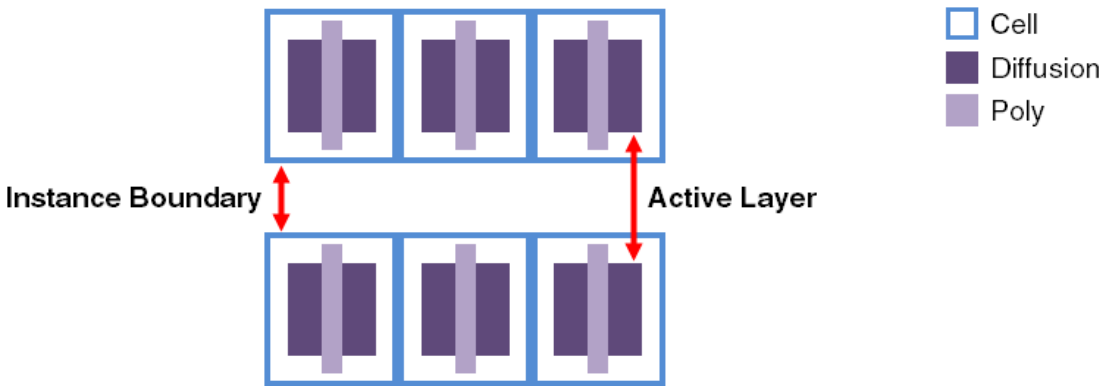
You can specify more than one row spacing value. If the number of rows is greater than the number of row spacing values, the values are repeated. For example, if a figure group contains five rows, and you specify two row spacing values (5 7), then the values will repeat: 5 7 5 7 5. When specifying multiple values, add a character space between the values.

The row spacing should accommodate the routes without increasing the design footprint significantly.

The Row Spacing value in the Route Pattern Assistant is synchronized with the **Different Row Spacing** specification in the Symbolic Editor. See the Creation Tab in the **Symbolic Design Options** dialog box.

3. To specify the channel definition (space between cell and device rows) to use for pattern routing, select the **Device Boundary** option:

Device Boundary Option	Description
Instance Boundary	The channel is between cell boundaries.
Active Layer	The channel is between poly-diffusion overlaps.



The channel definition you select can impact the tracks that are available for routing. If you select Active Layer, the channel is wider, and tracks that are within the instance boundary are also available for routing.

Note:

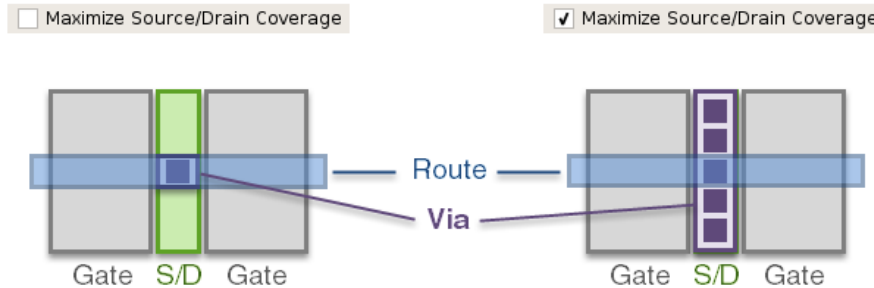
The Device Boundary selection in the Pattern Route Assistant is synchronized with the **Device Boundary** selection (`lxDeviceBoundary`) in the Symbolic Editor. See the Creation Tab in the **Symbolic Design Options** dialog box.

4. Select **Force Symmetry** to consider the number of trunks to be routed in an area and ensure that routes are distributed evenly in **Channels**, **Over Devices**, or both.

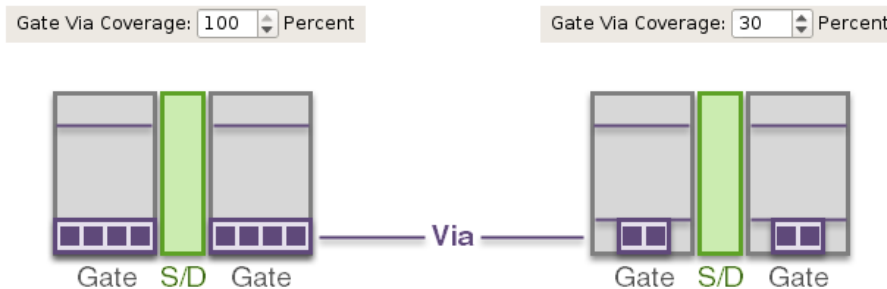
If the option is off, routes might not be distributed evenly. For example, the first route in the channel is always at the center. If a second route is added, it will be placed above or below the center route, resulting in an uneven pattern in the channel.

5. Select **Maximize Source/Drain Coverage** to maximize the cut array (number of cuts) in vias that connect routes to source/drain pins in devices. If you are concerned

that routes might block horizontal connections, clear the check box to reduce the via coverage.



- To specify the percent of gate pin area to use for vias, set the **Gate Via Coverage** option. If you want the via to fully cover the gate pin, set the option to 100%. Setting a value less than 100% leaves space for other connections. Via coverage always starts from the center of the gate and expands outward.



- Select **Add Single Pin Nets** to include nets that have only one pin.
If you have single-pin nets, you can select this option and use the **Extend** option to extend trunks to connect to the single pins. If you do not select the option, single-pin nets will not be routed.
- Select **Taper Tapoff to Pin Width** to adjust the width of the route to match the width of the pin.
- Choose the **Spine Position** to determine how spines are placed if no track regions exist in the design for the spine layer:
 - Default:** Considers all possible spine positions and honors the layer pitch requirements.
 - Gate Only:** Considers only the spine positions that overlap with the gates on poly layers.

- **SD Only:** Considers only the spine positions that overlap with source and drain pins.
- **Terminal Aware:** Considers spine positions that overlap with gates, source pins, and drain pins first, and then evaluates the spacing between terminals.

Tcl preference: `rtPatternRouteAssistantSpinePosition`

10. Select **Create Routes** if you want the assistant to save the route pattern constraints and create the routes based on the route plans when you click **Update Layout**.

If you do not select this option, you can click  (**Apply Pattern**) to create the routes after you click **Update Layout**. See [Creating Patterned Routes](#).

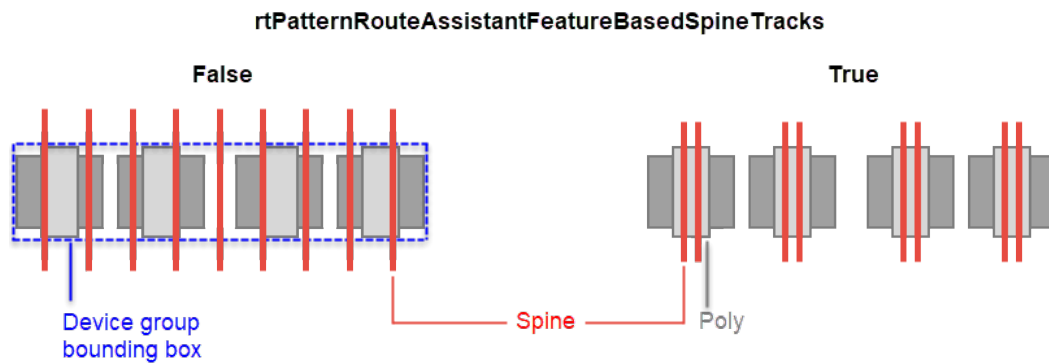
After adjusting the running the **Group Settings** for critical nets, you can use the Group Planner to generated planned routes for the remaining nets. See [Running the Group Planner](#).

Creating Patterned Routes

After you finish choosing route patterns, adjust the pattern components, and adjust the group settings, you can create the patterned routes.

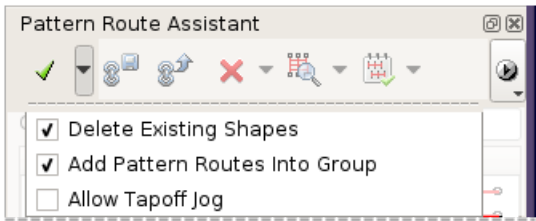
If you want to create the routes during **Update Layout**, see [Specifying Group Settings](#).

Before you start, if you want to ensure that spines are always distributed evenly within poly shape boundaries, set the `rtPatternRouteAssistantFeatureBasedSpineTracks` Tcl preference to `true`. When the preference is false, the spines are distributed evenly within group device bounding boxes.



To route the nets using the chosen patterns:

1. In the Pattern Route Assistant, select one or more nets you want to route.
2. On the Pattern Route Assistant toolbar, display the **Apply Pattern**  menu and select the options:



Option	Description
Delete Existing Shapes	Deletes all existing shapes before adding new routes.
Add Pattern Routes Into Groups	Creates routes for the nets within the selected figure group. Clear the check box to create routes for the nets within the figure group scope. For example, if you are in the edit-in-group mode, the routes will be added only within the group being edited. Tcl preference: <pre>rtPatternRouteAssistantAddRoutesIntoGroup</pre>
Allow Tapoff Jog	Adds jogs, if necessary, to ensure successful tapoffs (pin connections).

3. Click  **Apply Pattern**.

Creating and Extracting Route Patterns

You can extract the route pattern from a net and apply the same pattern (topology) to another net.

This topic describes the following:

- [Workflow for Creating and Extracting Route Patterns](#)
- [Before You Start](#)
- [Extracting Route Patterns](#)

Workflow for Creating and Extracting Route Patterns

The following table shows the high-level workflow for creating and reusing route patterns.

Step		See...
1	Review pattern route creation guidelines.	Before You Start.
2	Route a net using the Interactive Router in the Auto Welding and Auto Cloning modes.	Interactive Routing.
3	Extract the route pattern.	Extracting Route Patterns.

Before You Start

Make sure you review the following:

- [Pattern Routing Guidelines](#)
- [Limitation](#)

Pattern Routing Guidelines

Use the following guidelines to ensure successful pattern routing:

- The original route pattern must be created using the Interactive Router in the [Welding](#) and the [Cloning](#) modes.
- The route pattern must be limited to one layer and one width per trunk and associated tapoffs.
- You can select a single net for pattern extraction or realization. The source and target nets must have similar topologies.
- You should define a routing area or select a figure group if the net runs across multiple devices.
- Within the route pattern, a single channel can contain only one type of welding style: backbone or fishbone. In addition, the pins on the devices must all be aligned.

Limitation

Although the route pattern you select for extraction can have net shielding, the newly created route pattern will not have net shielding.

Extracting Route Patterns

To extract a pattern from a net:

1. In the design, define the [routing area](#), or select the figure group from which you will extract the routing pattern.

To select the figure group, do one of the following:

- On the canvas, select the group.
- In the Pattern Route Assistant, type the name of the group in the **Group** box.

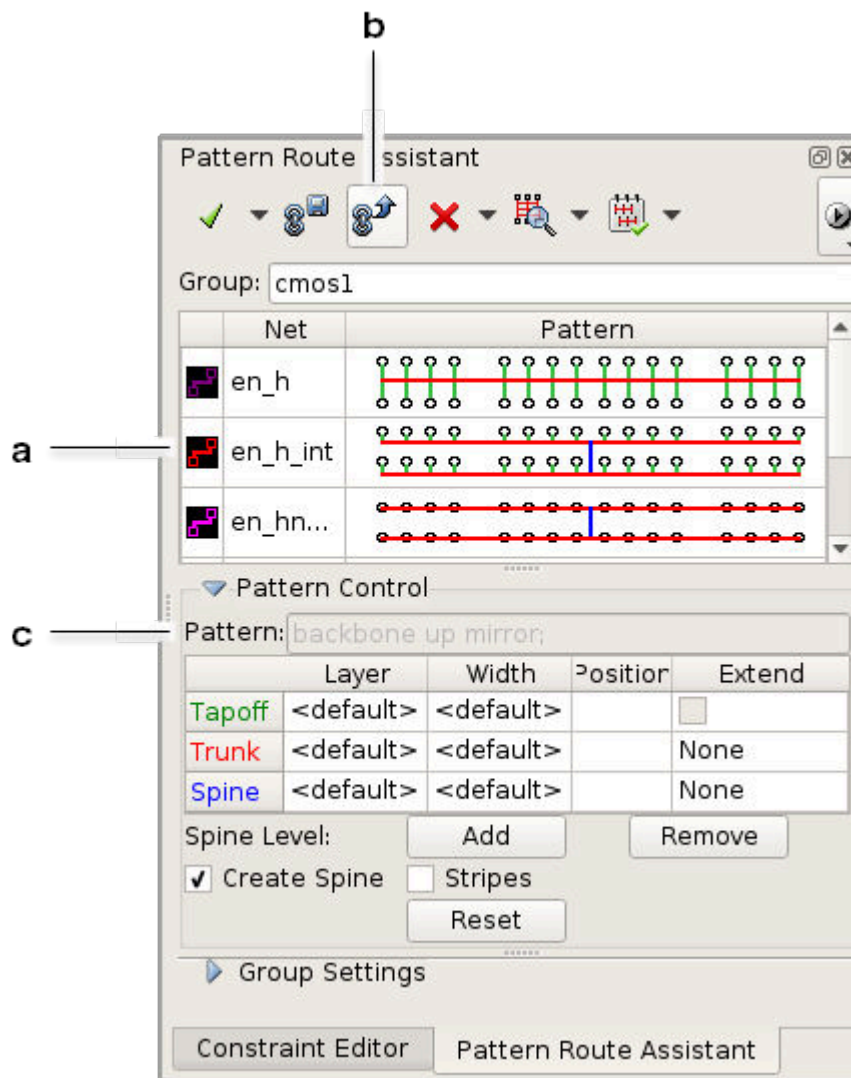
2. In the Pattern Route Assistant, display the **Pattern** description box. To do this, on the assistant toolbar, click the assistant menu , and choose **Show Pattern Description**.

3. On the Pattern Route Assistant toolbar:

- a. Select one or more nets.

- b. Click **Extract Constraint** .

The route pattern description appears in the Pattern box (c). See Route Pattern Description Reference.



Caution:

The **Extract** operation overwrites any existing routing pattern description in the **Pattern** box.

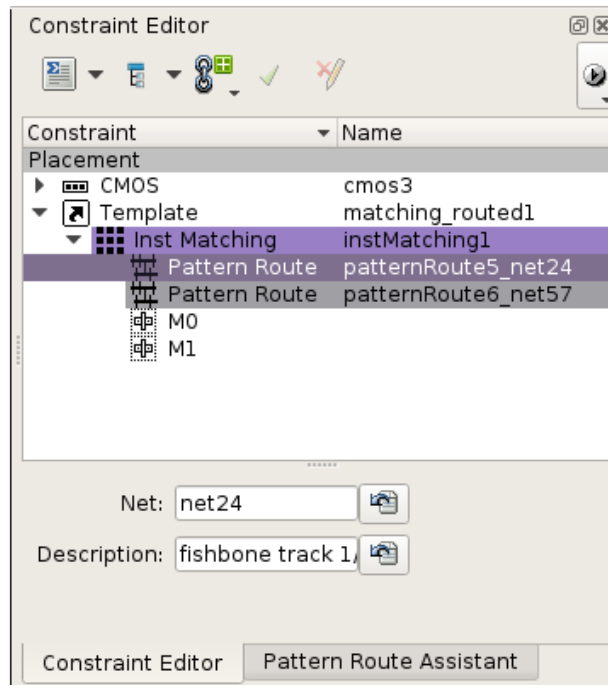
Caution:

Net shielding is not extracted. If needed, you can add shielding after the pattern routing operation is finished. For instructions, see [Adding Post-Route Shielding](#).

Note:

Route color information is not extracted. When you apply the pattern and create new routes, the routes will inherit the colors of the new tracks. See [Color Propagation](#).

4. To save the route pattern constraint, click **Save Constraint**  on the Pattern Route Assistant toolbar. The constraint is saved for the specific net in that design. You can view the constraint in the Constraint Editor.



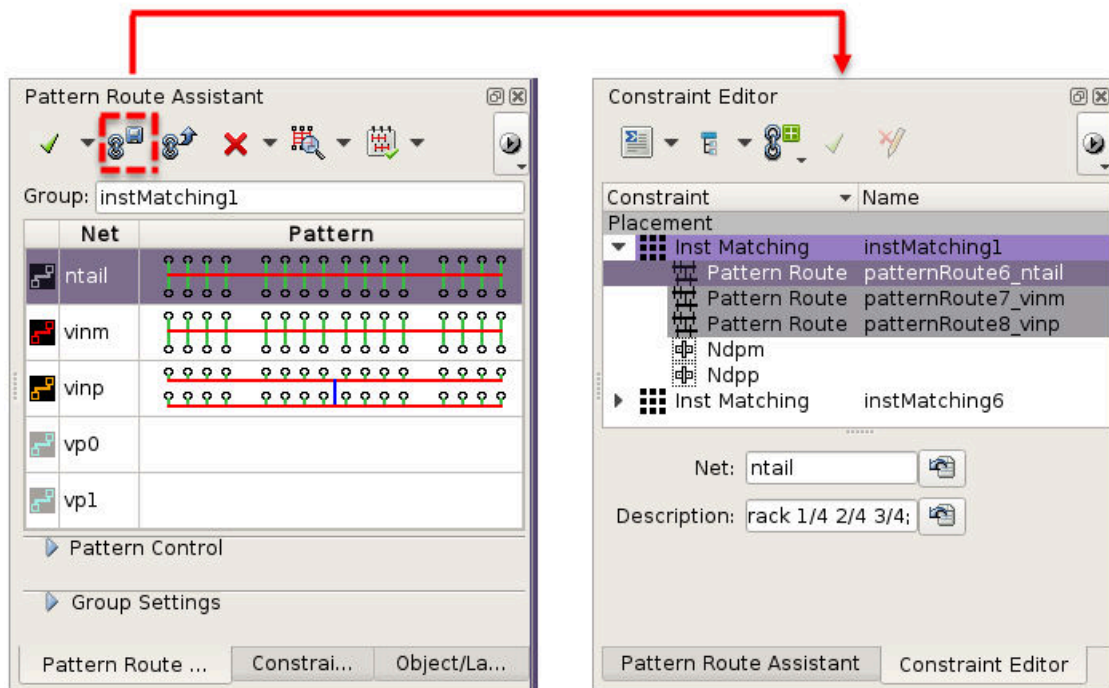
You can reuse the route patterns that you have extracted and saved. To do this, you copy and paste the route pattern description. See [Route Pattern Description](#).

Saving and Viewing Pattern Route Constraints

The pattern data and the net information are automatically stored as a route pattern constraint when you click **Apply Pattern** in the Pattern Route Assistant. See [Creating Patterned Routes](#). This topic describes how you can save a route pattern constraint without creating new routes.

After you [Setting Up Pattern Routing](#):

1. On the Pattern Route Assistant toolbar, click  **Save Constraint**.
2. To view the constraint:
 - a. Display the Constraint Editor.
 - b. Expand the **InstMatching** constraint.
 - c. Select the **Pattern Route** constraint of interest.
3. (Optional) Edit the constraint:
 - a. Specify a different **Net**.
 - b. Modify the route pattern description. See [Route Pattern Description](#).



Deleting Routes in Patterns

To delete routes that were created by the Pattern Route Assistant:

1. On the Pattern Route Assistant toolbar, display the **Delete Net Shapes** menu .
2. Choose one of the following commands in the **Delete Net Shapes** menu .

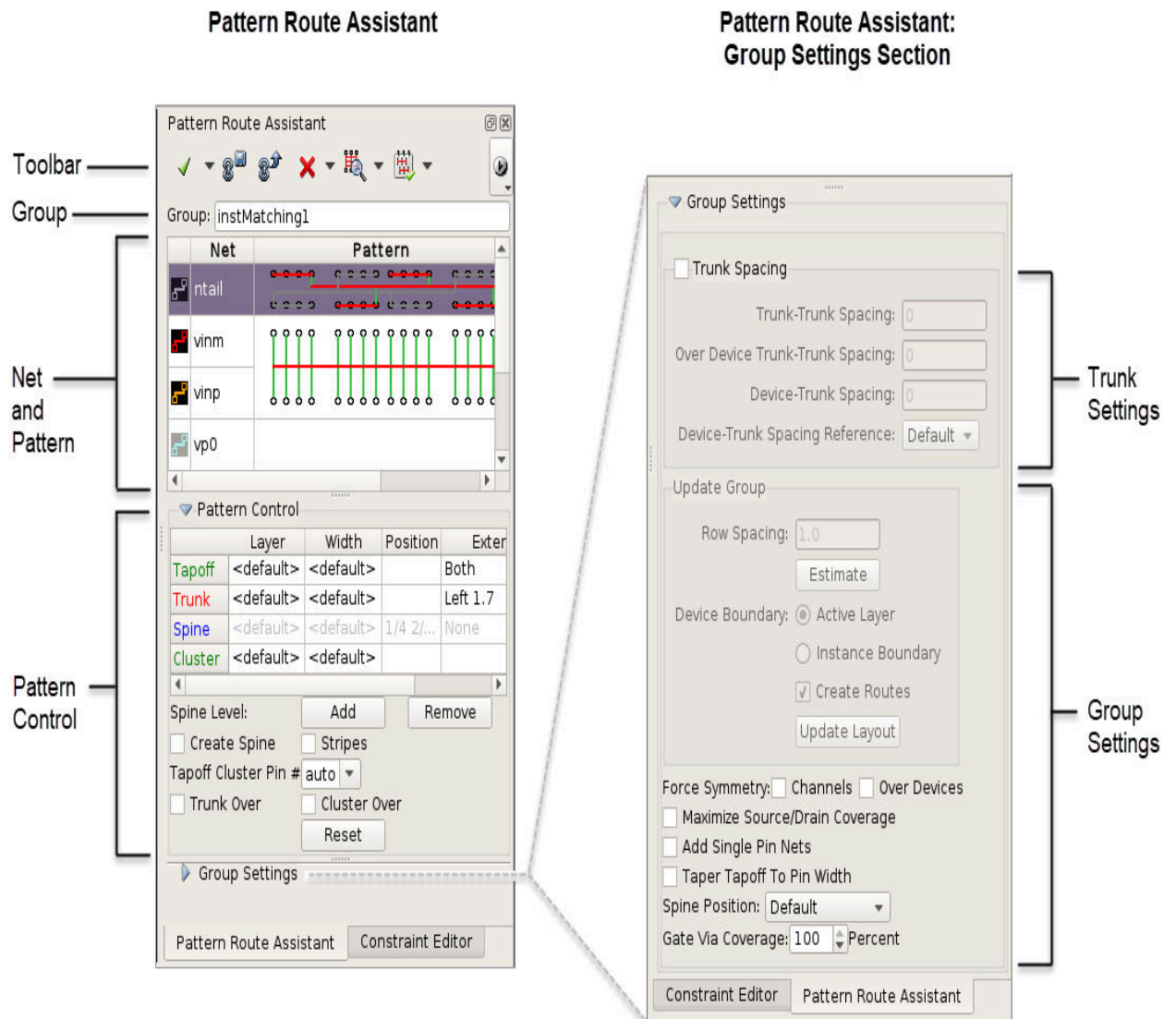
Command	Description
Delete Net Shapes	Deletes route and via shapes within the selected figure group.
Delete Net Shapes in Current Design	Deletes route and via shapes across the entire design. You must select this option if you want to delete route and via shapes at the top level. For example, clearing the Add Pattern Routes Into Groups check box might have created route shapes at the top level. To delete these top-level shapes, you must select Delete Net Shapes in Current Design .
Delete Constraint(s) with Shapes	Deletes the route shapes and the constraints that are associated with the route pattern.

3. Click **Delete Net Shapes** .

Pattern Route Assistant Reference

This topic provides a quick reference to the Pattern Route Assistant components:

- [Toolbar](#)
- [Group](#)
- [Net and Pattern](#)
- [Pattern Control](#)
- [Group Settings](#)
- [Assistant Menu](#)
- [Starter Route Patterns](#)
- [Route Pattern Description](#)
- [Via Options Dialog Box](#)



Toolbar

But ton	Tool tip	Description
	Apply Pattern	Saves the chosen route pattern and routes the selected net based on the pattern.

But ton	Tool tip	Description
	Save Constr aint	Saves the route pattern without creating routes.
	Extract Constr aint	Captures (or extracts) the specified route pattern.
	Delete Net Sha pes	Deletes the routes. See Deleting Routes in Patterns .
	Prev iew Patt ern	Displays the preview pattern route on the layout canvas. The net's flight line color is used to render the preview routes.
	Group Plan ner	Determines the topology, layer, and position for each net in a figure group and displays the information in the Console. The Group Planner is only applicable to fishbone, backbone, and backbone mirror patterns.
	Assist ant menu	See Assistant Menu .

Group

Name of the figure group that you selected for pattern routing.

Net and Pattern

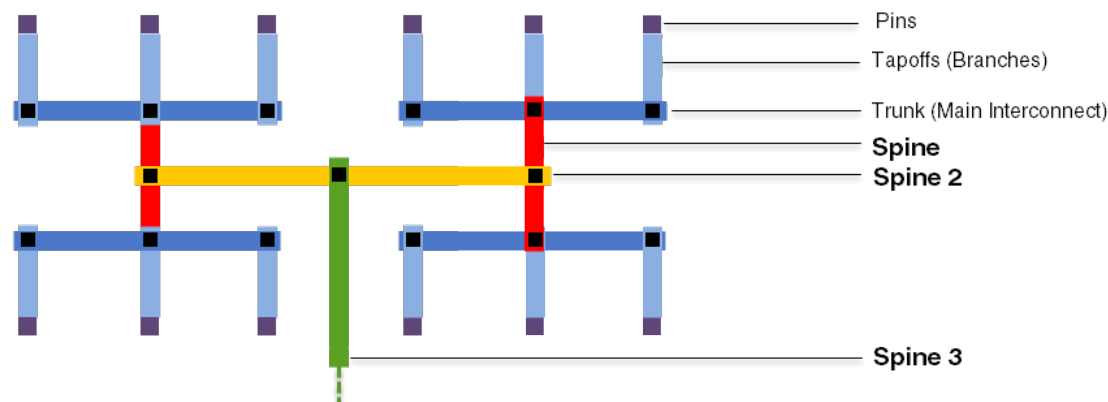
Column Name	Description
Net	Name of the net. The chosen route pattern will be applied to the selected net. Tcl preference: <code>rtPatternRouteAssistantNet</code>
Pattern	List of starter route patterns. See Starter Route Patterns .

Pattern Control

Optional layer, width, position, and extension settings that you can adjust to customize the chosen route pattern. See [Adjusting Pattern Component Attributes in the Assistant](#).

Attribute	Description
Pattern	Displays the text-based description of the selected pattern. The information is displayed only if you select Show Pattern Description in the Assistant Menu . See also Route Pattern Description .
Layer	Layer of the tapoff, trunk, spine, or cluster connection. The default layer is derived from the technology database. To route stacking, type the names of the layers to use for stacked routing.
Width	Width of the tapoff, trunk, spine, or cluster connection. The default width is derived from the technology database.
Position	Position of the tapoff, trunk, or spine. You express the position using fractions. For example, 1/2 indicates that the object is centered (it is halfway along the length of the object). Position does not apply to Cluster. Although you can type a fractional value to set the position, you might want to visually adjust the positions on the layout canvas first. See Previewing Route Patterns .
Extend	Lengthening of the tapoffs, trunks, or spines. Extend does not apply to Cluster.

Spine Level (Spine n)
Adds additional levels of spines to connect existing spines. Click **Add** to insert a new row, **Spine n**, to the attributes table. Click **Remove** to delete the most recently added **Spine n** from the attributes table.
In the following example, Spine connects trunks, Spine 2 connects Spines, Spine 3 connects Spine 2s, and so on.



Create Spine
Adds spines to connect trunks or other spines.



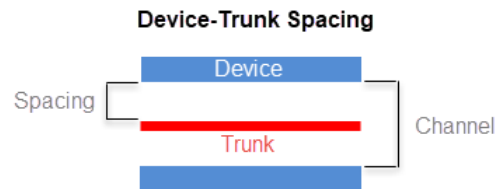
Attribute	Description																								
Stripes	The number of wire stripes in trunks and spines. ▼ Pattern Control <table><tr><th></th><th>Layer</th><th>Width</th><th>Position</th><th>Extend</th><th>Stripes</th></tr><tr><td>Tapoff</td><td><default></td><td><default></td><td></td><td>☐</td><td></td></tr><tr><td>Trunk</td><td><default></td><td><default></td><td></td><td>None</td><td>1</td></tr><tr><td>Spine</td><td><default></td><td><default></td><td>1/4 2/4 ...</td><td>None</td><td>1</td></tr></table> Spine Level: ☒ Create Spine Add Remove ☒ Stripes Reset Tapoff Cluster Pin # (Cluster route pattern only) Specifies the number of pins in the cluster: • auto: (Default) Allows the router to determine the number of pins that are grouped in a cluster. Different clusters might contain different number of pins. • <integer>: Specifies the number of pins in the clusters. The router will try to group the same number of pins in every cluster. If the number of neighboring pins exceeds the value you specify, the router will create multiple clusters and group any remaining pins. In the following example, # Pins in Tapoff Cluster is 3. The router will create as many 3-pin clusters as possible.  If the number of neighboring pins is less than the value you specify, the router still groups the pins. In the following example, the # Pins in Tapoff Cluster is 10. The router still creates a cluster of 9 pins.  Reset Reverts to the last-saved Pattern Route constraint.		Layer	Width	Position	Extend	Stripes	Tapoff	<default>	<default>		☐		Trunk	<default>	<default>		None	1	Spine	<default>	<default>	1/4 2/4 ...	None	1
	Layer	Width	Position	Extend	Stripes																				
Tapoff	<default>	<default>		☐																					
Trunk	<default>	<default>		None	1																				
Spine	<default>	<default>	1/4 2/4 ...	None	1																				

Group Settings

The Group Settings section consists of the following:

Trunk Spacing

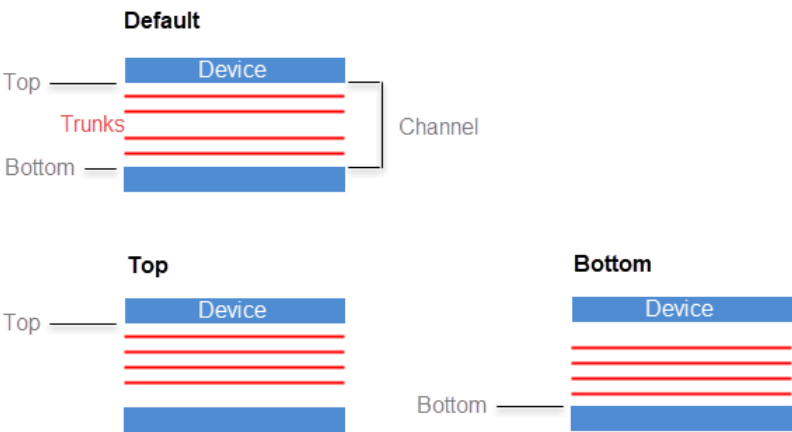
Option	Description
Trunk Spacing	Lets you control the spacing between adjacent trunks, and the spacing between devices and trunks. Tcl preference: <code>rtPatternRouteAssistantTrunkTrunkSpacingEnable</code>
Trunk-Trunk Spacing	Edge-to-edge distance between adjacent trunks in channels only. Tcl preference: <code>rtPatternRouteAssistantTrunkTrunkSpacing</code>
Over Device Trunk-Trunk Spacing	Edge-to-edge distance between adjacent trunks that are placed on top of the devices instead of in channels. Tcl preference: <code>rtPatternRouteAssistantOverDeviceTrunkTrunkSpacing</code>
Device-Trunk Spacing	Edge-to-edge distance between the device and the adjacent trunk in the channel.



Tcl preference:
`rtPatternRouteAssistantTrunkDeviceSpacing`

Option	Description
Device-Trunk Spacing Reference	Side of the device channel from which Device-Trunk Spacing is determined: • Default: Both the top and bottom sides of the channel. • Top: Top side of the device channel. • Bottom: Bottom side of the device channel.

Device-Trunk Spacing Reference



Tcl preference:
`rtPatternRouteAssistantTrunkDeviceS
pacingRef`

Update Group

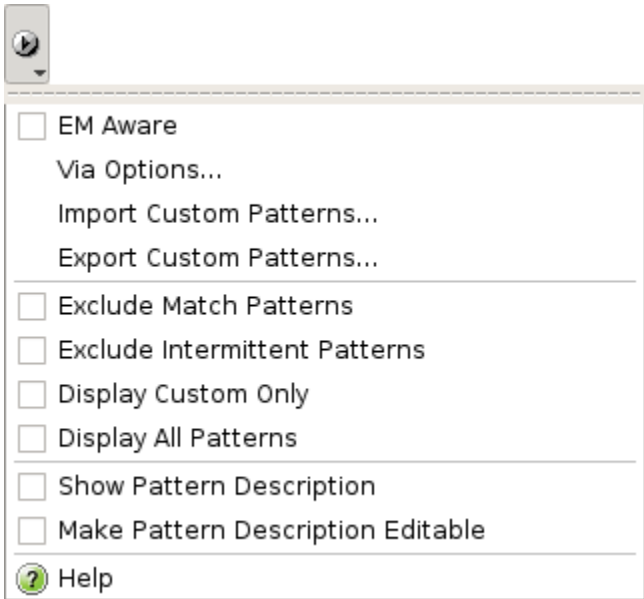
Spacing between rows, command to update the layout based on the row spacing value, option to create the routes when you update the layout.

Option	Description
Row Spacing	Distance between adjacent rows, in fins or microns. You can specify more than one row spacing value. If the number of rows is greater than the number of row spacing values, the values are repeated. For example, if a figure group contains five rows, and you specify two row spacing values (5 7), then the values will repeat: 5 7 5 7 5. When specifying multiple values, add a character space between the values. Tcl preference: <code>rtPatternRouteAssistantRowSpacing,</code> <code>rtPatternRouteAssistantRowSpacingType,</code> <code>rtPatternRouteAssistantRowSpacingNumFins</code>
Estimate	Recalculates the estimated row spacing required.
Device Boundary	Specifies the channel definition to use for pattern routing: • Instance Boundary: The channel is between cell boundaries. • Active Layer: The channel is between poly-diffusion overlaps.
Create Routes	Creates routes after updating the group placement. Tcl preference: <code>rtPatternRouteAssistantCreateRoutesDuringRowSpacingUpdate</code>
Update Layout	Updates the layout based on the calculated or edited row spacing. If you selected Create Routes, the assistant saves the route pattern constraints and creates the routes based on the route plans.

Other Group Settings

Component	Description
Force Symmetry	Considers the number of trunks to be routed in an area and ensures that routes are created symmetrically in channels, over devices, or both. If the option is off, routes might not be symmetric. The first route in the channel is always at the center. If a second route is added, it will be placed above or below the center route, resulting in an asymmetric pattern in the channel. Tcl preference: <code>rtPatternRouteAssistantForceGroupSymmetry</code>
Maximize Source/Drain Coverage	Option to maximize the cut array (number of cuts) in vias that connect routes to source/drain pins in devices. Tcl preference: <code>rtPatternRouteAssistantOverDeviceViaMaximization</code>
Add Single-Pin Nets	Includes nets that have only one pin in the Group Dashboard dialog box. If you have single-pin nets, you can select this option and use the Extend option to extend trunks to connect to the single pins. If you do not select the option, single-pin nets are not listed in the Group Dashboard, and they are not routed. Tcl preference: <code>rtPatternRouteAssistantAddSinglePinNets</code>
Taper Tapoff to Pin Width	Adjusts the tapoff width of the route to match the width of the pin. If you do not select this option, the tapoff width is used from the Pattern Control table only if the width is specified. If the tapoff width is not specified in the Pattern Control table, the default layer width is used. Tcl preference: <code>rtPatternRouteAssistantTaperTapoffToPinWidth</code>
Spine Position	Determines how spines are placed if no track regions exist in the design for the spine layer: • Default: Considers all possible spine positions and honors the layer pitch requirements. • Gate Only: Considers only the spine positions that overlap with the gates on poly layers. • SD Only: Considers only the spine positions that overlap with source and drain pins. • Terminal Aware: Considers spine positions that overlap with gates, source pins, and drain pins first, and then evaluates the spacing between terminals. Tcl preference: <code>rtPatternRouteAssistantSpinePosition</code>
Gate Via Coverage	Percent of gate pin area to use for vias. Tcl preference: <code>rtPatternRouteAssistantGateCoverageOfContacts</code>

Assistant Menu

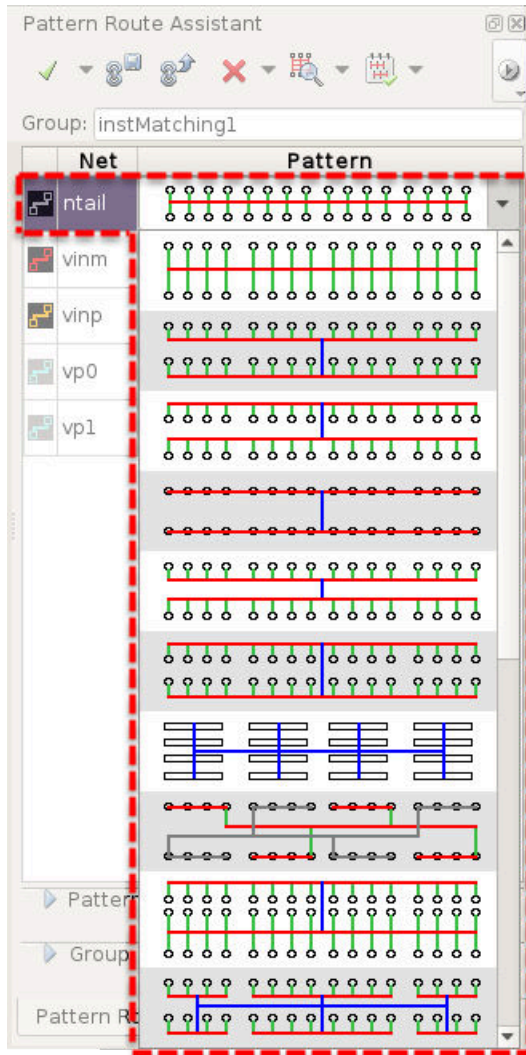


Component	Description
EM Aware	Turns on EM-aware pattern routing. See Preparing for EM-Aware Pattern Routing .
Via Options	Opens the Via Options Dialog Box .
Import Custom Patterns	
Export Custom Patterns	
Exclude Match Patterns	Hides the match-style patterns from the pattern catalog. For information about match patterns, see intermittent / match in Route Pattern Description Reference.
Exclude Intermittent Patterns	Hides the intermittent-style patterns from the pattern catalog. For information about intermittent patterns, see intermittent / match in Route Pattern Description Reference.
Display Custom Only	
Display All Patterns	
Show Pattern Description	Displays the route pattern description of the selected pattern at the top of the Pattern Control section.

Component	Description
Make Pattern Description Editable	Enables the pattern description field so that you can edit the description. See Route Pattern Description Reference for the description syntax.

Starter Route Patterns

The pattern catalog consists of a list of factory-supplied starter route patterns that you can use to create new routes. The starter route patterns cover basic and common route scenarios.



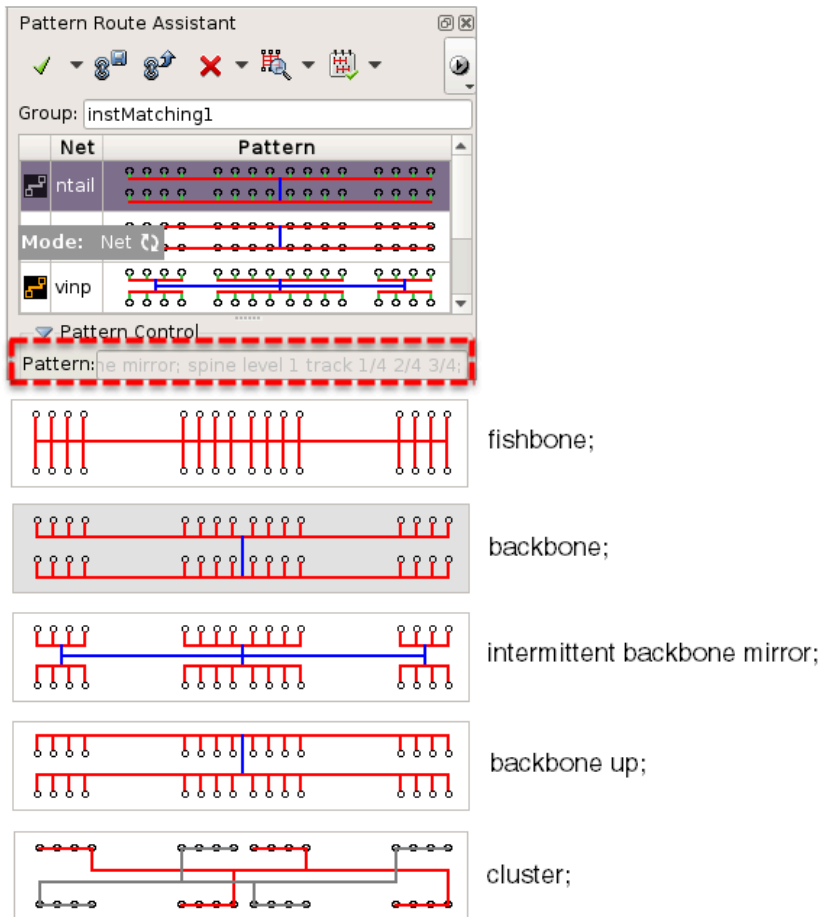
You can visually scan the list and choose the most applicable pattern for a net. Alternatively, you can let the Pattern Route Assistant choose the most applicable pattern for all the nets in a selected figure group. See [Running the Group Planner](#).

However the route patterns are chosen, you have the option of customizing the patterns to meet your design requirements. See [Adjusting Pattern Component Attributes in the Assistant](#) and [Specifying Group Settings](#).

To add custom route patterns that will cover special design requirements, see [Adding Custom Patterns in the Pattern Catalog](#).

Route Pattern Description

Each route pattern in the pattern catalog has a corresponding text description, called route pattern description. Some examples are as follows:



In a typical scenario, you would choose the most applicable route pattern from the [Starter Route Patterns](#) and, if necessary, modify the pattern attributes to customize the pattern. In case you need to make further refinements, you can edit the route pattern description.

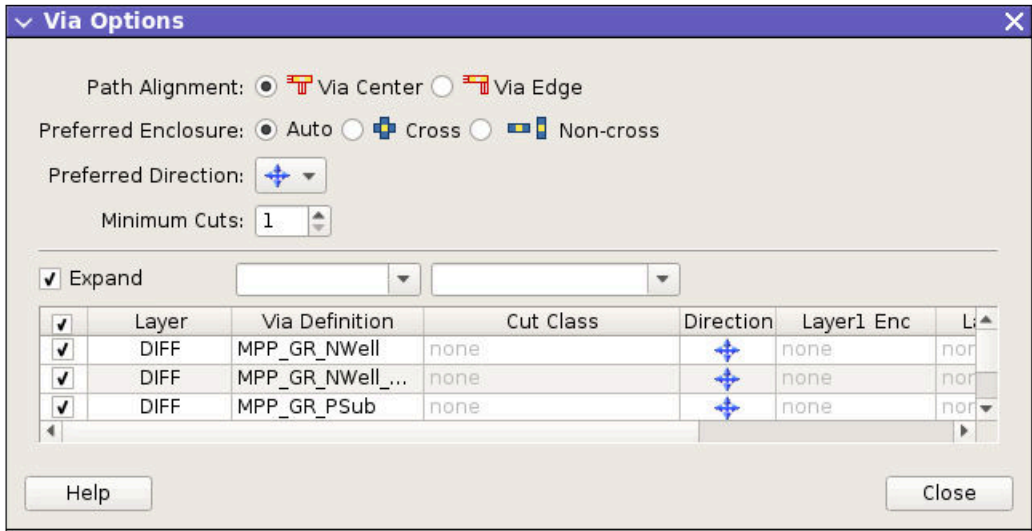
Because visually scanning a route pattern in the [Starter Route Patterns](#) is easier and faster than reading a list of text descriptions, the route pattern description is hidden by default. To display the route pattern description, in the Route Pattern Assistant, choose

Show Pattern Description from the assistant menu .

To edit the description, choose **Make Pattern Description Editable** from the assistant menu .

For details about route pattern descriptions, such as the description syntax, see Route Pattern Description Reference.

Via Options Dialog Box



Option	Description
Path Alignment	Aligns the connecting path and via using one of the following methods: • Via Center: Aligns the connecting path's centerline with the center of the via. • Via Edge: Aligns the edge of the connecting path with the edge of the via. Tcl preferences: <code>rtAllowOffsetVia</code>
Preferred Enclosure	Specifies how the via's metal layers are positioned: • Auto: The router determines the positions. • Cross: The layers are perpendicular. • Non-cross: The layers are parallel. Tcl preferences: <code>rtViaPreferredEnclosure</code>
Preferred Direction	Specifies the via orientation: • : The router determines the orientation. • : The preferred orientation is vertical. • : The preferred orientation is horizontal. • : The orientation will be vertical only. • : The orientation will be horizontal only. Tcl preferences: <code>rtPrefBarDoubleCutViaOrient</code>, <code>rtStrictFollowPrefBarDoubleCutViaOrient</code>

Option	Description
Minimum Cuts	The minimum number of cuts in the via. Tcl preferences: <code>rtMinimumCut</code>

10

Trunk Routing (Manual Method)

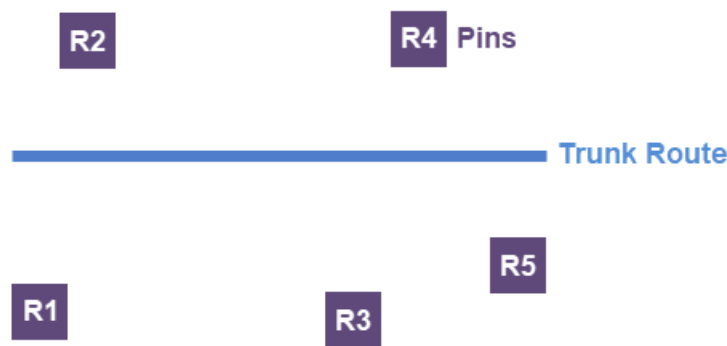
You can use the **Generate Trunk** command to create trunk-style routes in the design layout.

This chapter describes the following:

- [About the Generate Trunk Command](#)
- [Before You Start Trunk Routing](#)
- [Adding Trunk Routes](#)
- [Trunk Toolbar and Trunk Generator Dialog Box Reference](#)

About the Generate Trunk Command

The **Generate Trunk** command lets you create trunk-style routing in your designs. The available command options provide precise control of the trunk placement.



You use one of the following Tcl commands to create or delete trunk routes:

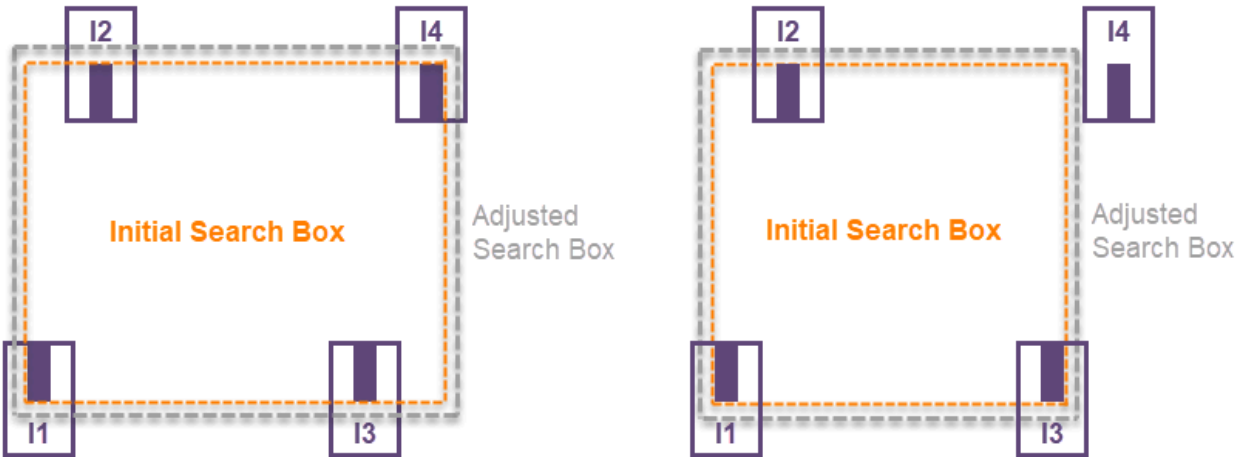
Tcl command	Description
irt::generateTrunk	Displays the Trunk toolbar. You can set various options using the graphical user interface, including the option to add shielding.

Tcl command	Description
rt::generateTrunk	The batch command version of irt::generateTrunk. The command arguments match the options available in the graphical user interface and has additional fine-tuning settings.
rt::deleteTrunkAndShield	Deletes trunk and shield shapes created by the Generate Trunk command.
rt::deleteTrunkBox	Deletes search boxes created by the Generate Trunk command.

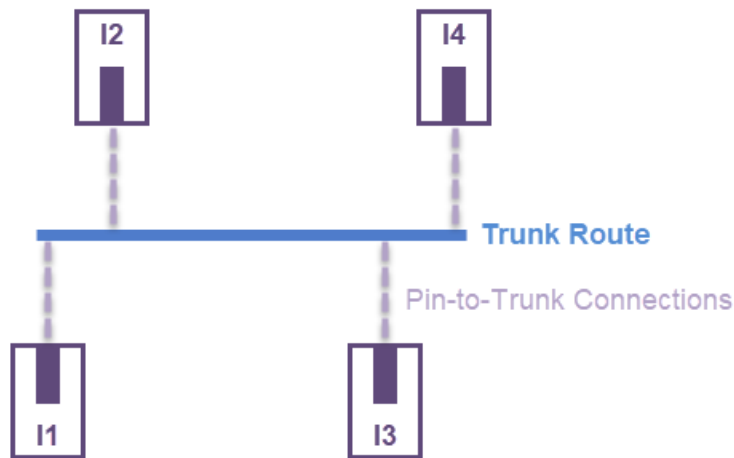
To determine trunk placement, the **Generate Trunk** command uses a search box to first estimate the optimal placement area. The search box is initially determined by the **Include** and **Exclude** instance options. However, you can adjust the size and position of the search box using the various command options to fine-tune trunk placement. The search box is displayed as an overlay on the layout canvas.

Include All Instances

Exclude: I4



After trunk routing, you can run the Auto Router to connect the trunk routes to the target pins. See [Auto Routing](#).

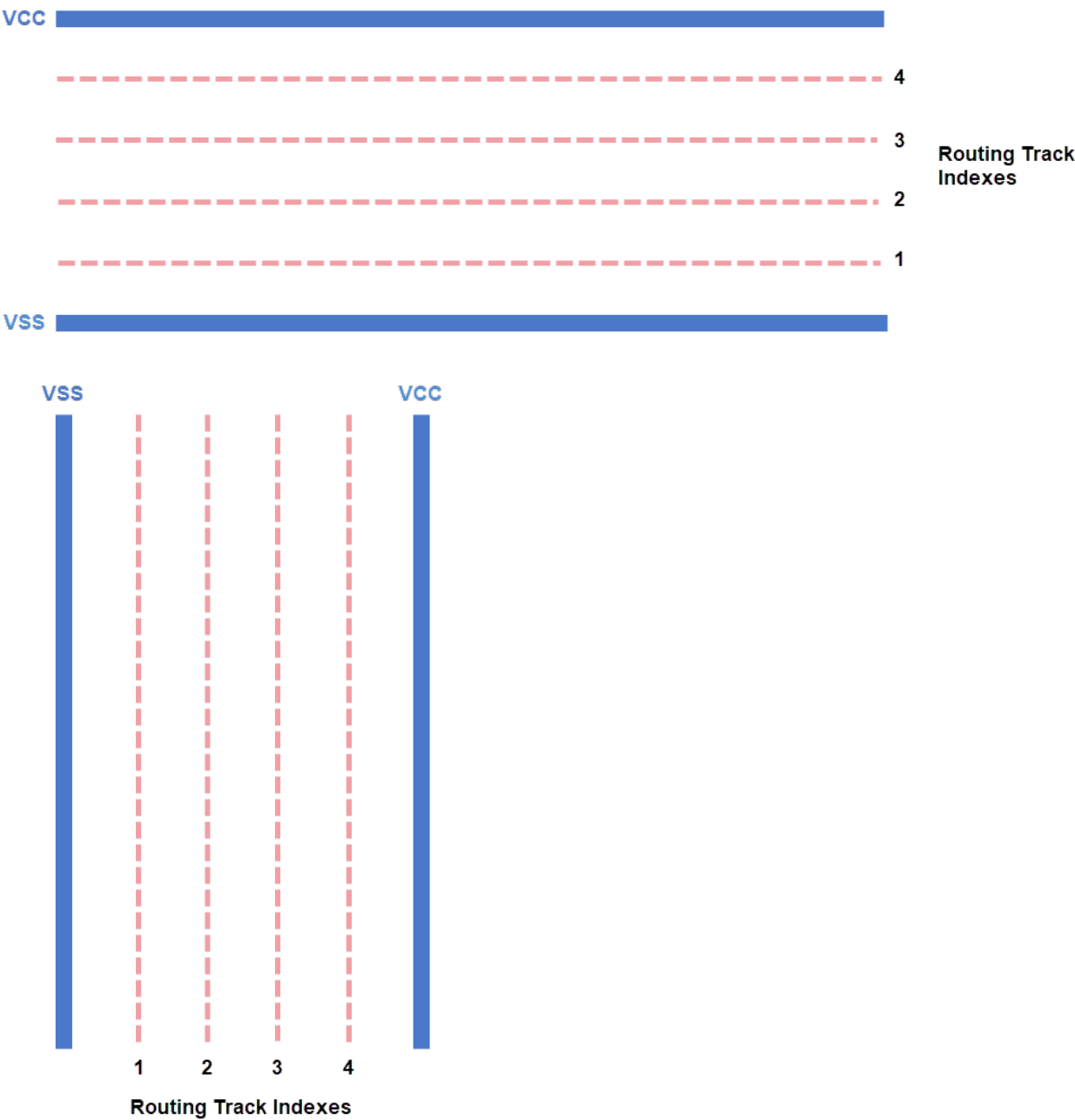


Before You Start Trunk Routing

Because trunk routes are placed on tracks, the design must contain routing tracks. For instructions on creating tracks, see [Setting Up Tracks for Routing](#).

For trunk routing, you have the option of using any defined track or specific tracks. You can identify the specific tracks using track indexes.

A track index is an integer assigned to a track between a pair of power and ground rails. The track indexes always start from the bottom-most or left-most track and increases upward, regardless of the order of the power and ground rails.



Note:
The track index sequence for horizontal trunk routing is different from the track index sequence for pattern routing. See the rules and syntax for specifying tracks for pattern routing.

Adding Trunk Routes

You use the `irt::generateTrunk` to add trunk routes.

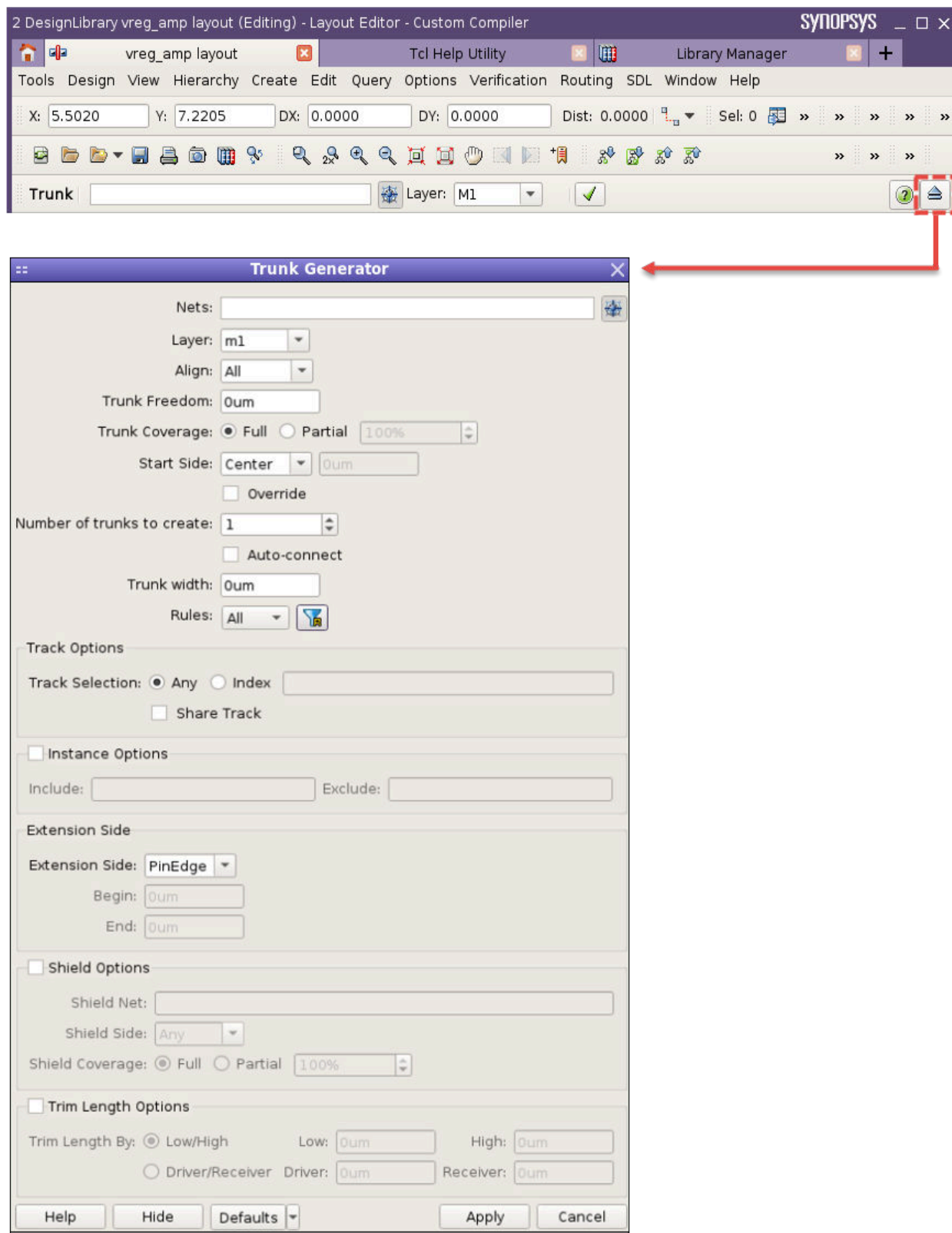
If you want to create a trunk route that consist of different nets, use the `rt::generateTrunk` Tcl command with the `-groupMode` argument. The `-groupMode` argument also adds shielding to the group.

To add trunk routes in your design:

1. In the Layout Editor, open the Console assistant. See [Displaying the Console Assistant](#).
2. In the Console, type `irt::generateTrunk` and press Enter.

The **Trunk** toolbar appears.

3. On the **Trunk** toolbar, click **Eject**  to open the **Trunk Generator** dialog box.

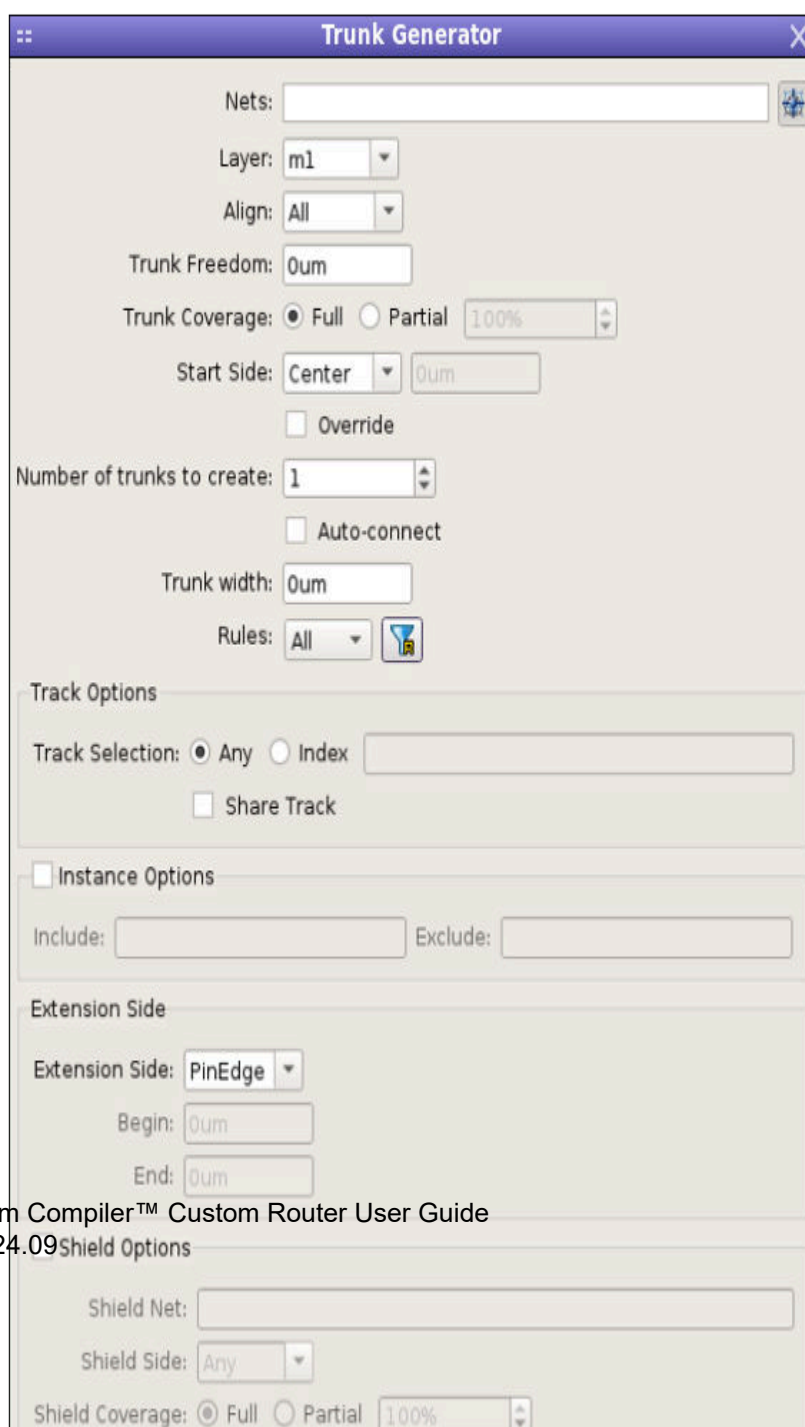
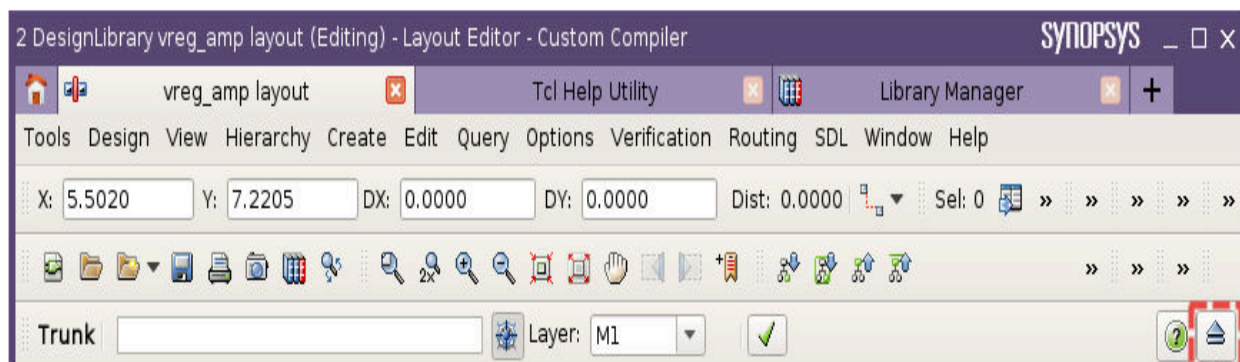


4. Set the options that determine where and how the trunk routes are placed. See [Trunk Toolbar and Trunk Generator Dialog Box Reference](#).
5. Click **OK**.

Trunk routes are created in the design.

Trunk Toolbar and Trunk Generator Dialog Box Reference

This topic provides a quick reference of the **Trunk** toolbar and the **Trunk Generator** dialog box. For instructions on creating new trunk routes, see [Adding Trunk Routes](#).

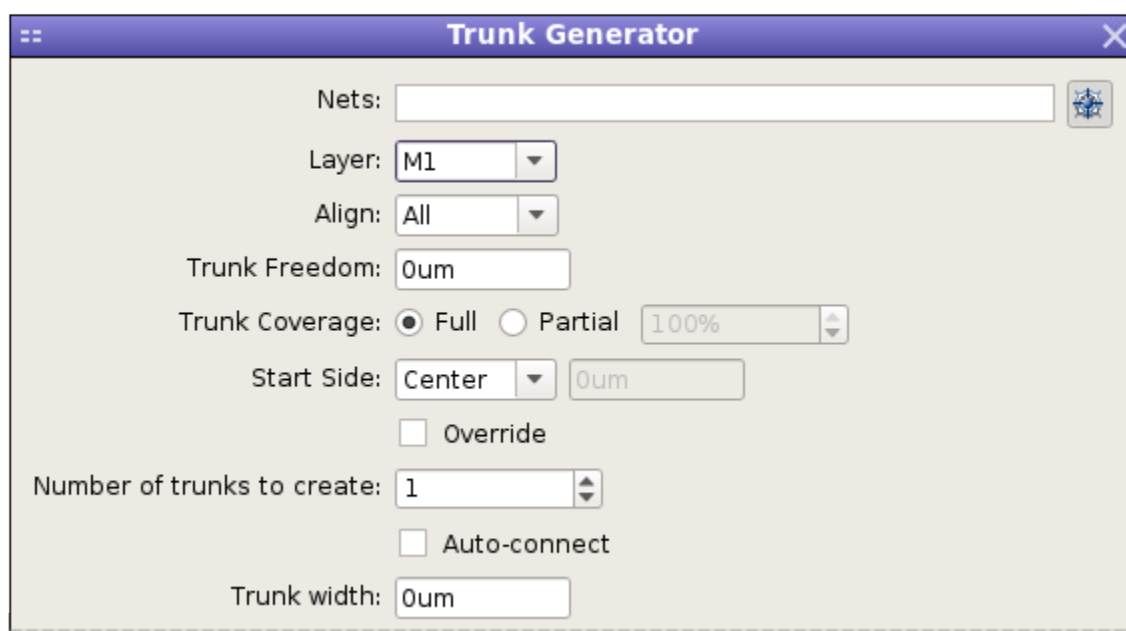


The **Trunk** toolbar and the **Generate Trunk** dialog box have some options in common. The **Generate Trunk** dialog box consists of the following:

- [General Options](#)
- [Track Options](#)
- [Instance Options](#)
- [Extension Side](#)
- [Shield Options](#)
- [Trim Length Options](#)

General Options

Specifies the nets, layers, types of pins, and other general options for generating trunk routes.



Option	Description
Nets	Nets to be routed. When specifying multiple nets, use a character space between net names.
Layer	Layer to use for trunk routing.

Option	Description
Align	Type of pins to consider for trunk route generation: • Driver pins • Receiver pins • InstPins (instance pins) • TopPin (top-level pins)
Trunk Freedom	Length and width, in μm , to increase the search box for trunk placement. See About the Generate Trunk Command .
Trunk Coverage	Required trunk length that is created without violations (DRC-clean). Trunk routing is considered successful if the percent of the routes meet the Trunk Coverage you specify.
Start Side	Starting point where the trunk generation looks for a valid location for trunk generation. • Center: Center of the search box. • Gravity: Average of the X (for horizontal trunks) or Y (for vertical trunks) center-of-pin coordinates within the search box. • High: Right side (for vertical trunks) or top (for horizontal trunks) of the search box. • HighPin: Right-most (vertical trunk) or top-most (horizontal trunk) pin within the search box. • Low: Left side (for vertical trunks) or Bottom (for horizontal trunks) of the search box. • LowPin: Left-most (vertical trunk) or bottom-most (horizontal trunk) pin within the search box. • Majority: Coordinate at which a maximum number of X (horizontal trunks) or Y (vertical trunks) pin centers exist. If more than one such coordinate exist, their median value is used. • Variable: At the location you specify. For horizontal trunks, the value you specify is an Y coordinate. For vertical trunks, the value you specify is a X coordinate. For example, for horizontal trunks, Start Side at 3 means the starting point to search for a valid location is at $Y = 3 \mu\text{m}$.
Override	Replaces existing trunks routes, associated shielding, and via shapes on the specified layer.
Number of Trunks to Create	Creates the specified number of trunks.
Auto-connect	Creates vias automatically when the newly created trunks overlap with pins or existing trunks that are on adjacent layers.
Trunk Width	Specifies the width of the trunks.

Track Options

Specifies the tracks to use during trunk routing.



Option	Description
Track Selection	Tracks to use during trunk route generation. • Any: Any defined track. • Index: Only the tracks you specify. Type the track indexes in the order of priority. Use a character space between index values. If the specified track is not within the search box, the search box is expanded to include the nearest track with the specific index. If the number of trunk routes that must be created is greater than the number of tracks you specified, the additional trunk routes are created on the adjacent tracks. For example, if two trunk routes must be created, but you only specified one track (index 1), the second trunk route is placed on the next track (index 2). You might want to specify track indexes in designs that contain power and ground mesh. For more information about track indexes, see Before You Start Trunk Routing.
Share Track	Forces trunks across multiple nets to use the same track. To use track sharing, you must specify two or more nets. In addition, the search boxes of the nets cannot overlap.

Instance Options

Specifies the instances to be considered or excluded during trunk routing.

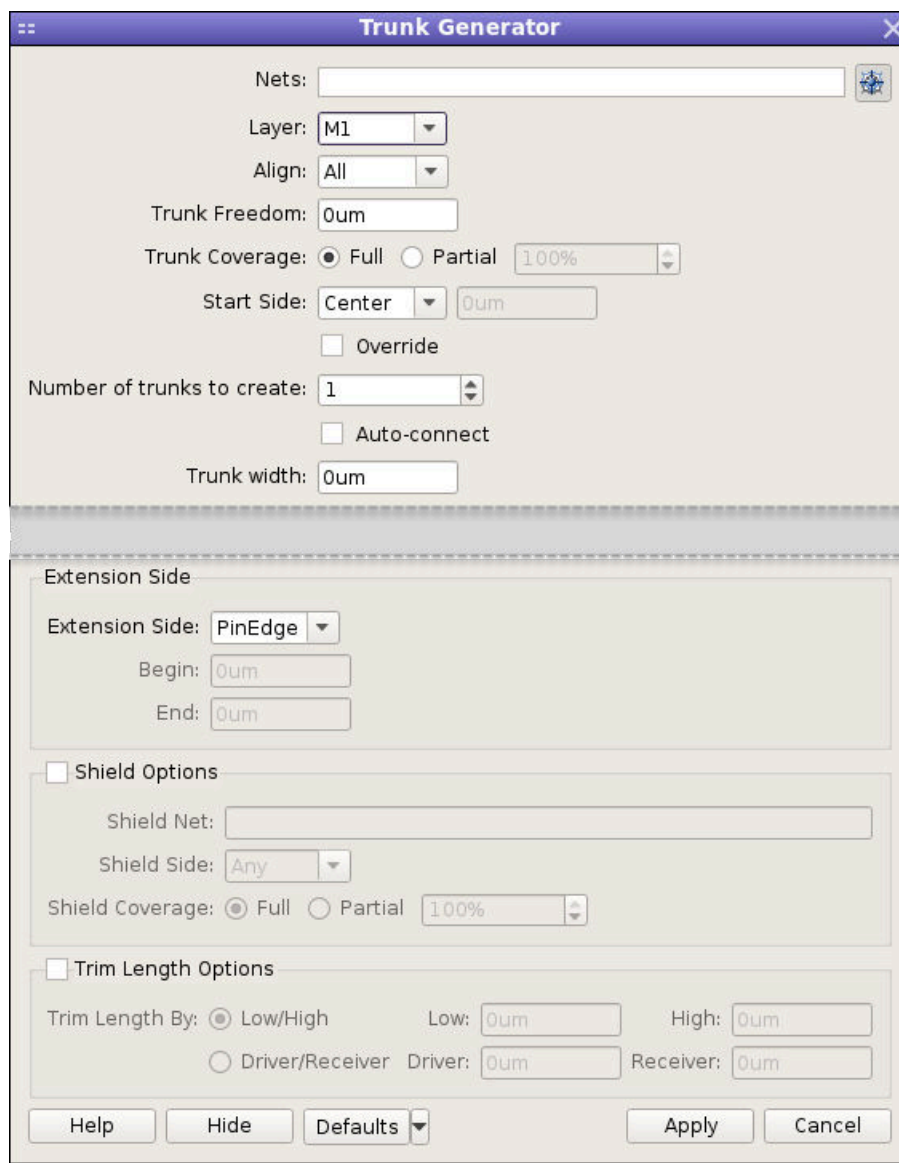
The screenshot shows the 'Trunk Generator' dialog box with the following settings:

- Nets:** (Empty text field)
- Layer:** M1 (Dropdown menu)
- Align:** All (Dropdown menu)
- Trunk Freedom:** 0um (Text field)
- Trunk Coverage:** Full (Selected radio button), Partial (Unselected radio button), 100% (Slider)
- Start Side:** Center (Dropdown menu), 0um (Text field)
- ☐ **Override**
- Number of trunks to create:** 1 (Spin box)
- ☐ **Auto-connect**
- Trunk width:** 0um (Text field)
- Track Options** (Group box):
 - Track Selection:** Any (Selected radio button), Index (Unselected radio button), (Empty text field)
 - ☐ **Share Track**
- ☐ **Instance Options** (Group box):
 - Include:** (Empty text field)
 - Exclude:** (Empty text field)

Option	Description
Include	List of instances to be considered.
Exclude	List of instances to be excluded from consideration.

Extension Side

Extends the trunk route on the specified side or based on coordinates you provide.



The Trunk Generator dialog box is divided into two main sections. The top section contains settings for the trunk itself, including a text field for 'Nets', a dropdown for 'Layer' (set to M1), a dropdown for 'Align' (set to All), a text field for 'Trunk Freedom' (set to 0um), radio buttons for 'Trunk Coverage' (Full is selected), a percentage spinner (set to 100%), a dropdown for 'Start Side' (set to Center), a text field for 'Start' (set to 0um), a checkbox for 'Override', a spinner for 'Number of trunks to create' (set to 1), a checkbox for 'Auto-connect', and a text field for 'Trunk width' (set to 0um). The bottom section is titled 'Extension Side' and contains a dropdown for 'Extension Side' (set to PinEdge), text fields for 'Begin' and 'End' (both set to 0um), a checkbox for 'Shield Options', a text field for 'Shield Net', a dropdown for 'Shield Side' (set to Any), radio buttons for 'Shield Coverage' (Full is selected), a percentage spinner (set to 100%), a checkbox for 'Trim Length Options', and a group of text fields for 'Trim Length By' (Low/High is selected) with values for Low, High, Driver, and Receiver (all set to 0um). At the bottom are buttons for 'Help', 'Hide', 'Defaults', 'Apply', and 'Cancel'.

Trunk Generator

Nets:

Layer:

Align:

Trunk Freedom:

Trunk Coverage: ☒ Full ☐ Partial

Start Side:

☐ Override

Number of trunks to create:

☐ Auto-connect

Trunk width:

Extension Side

Extension Side:

Begin:

End:

☐ Shield Options

Shield Net:

Shield Side:

Shield Coverage: ☒ Full ☐ Partial

☐ Trim Length Options

Trim Length By: ☒ Low/High ☐ Driver/Receiver

Low: High:

Driver: Receiver:

Help Hide Defaults Apply Cancel

Option	Description
Extension Side	Extends the trunk beyond the pin based on the following selection: • PinEdge: (Default) To the outer edge of the pin. • Left: To the left of the left-most pin. • Right: To the right of the right-most pin. • Top: Above the top-most pin. • Bottom: Below the bottom-most pin. • Both: In both directions (top and bottom, or left and right). • Variable: By the length you specify based on the Begin and End values.
Begin	Starts the trunk route at the specified location. The Begin and End settings have higher precedence than Extension Side. For horizontal trunks, the value you specify is an X coordinate. For vertical trunks, the value you specify is a Y coordinate. For example, for horizontal trunks, Begin at 3 means start the trunk route at X = 3 μ m.
End	Ends the extension at the specified location. The Begin and End settings have higher precedence than Extension Side. For horizontal trunks, the value you specify is an X coordinate. For vertical trunks, the value you specify is a Y coordinate. For example, for vertical trunks, End at 17 means end the trunk route at Y = 17 μ m.

Shield Options

Adds shielding during trunk routing.

Trunk Generator

Nets:

Layer:

M1

Align:

All

Trunk Freedom:

0um

Trunk Coverage:

Full Partial

100%

Start Side:

Center

0um

Override

Number of trunks to create:

1

Auto-connect

Trunk width:

0um

Shield Options

Shield Net:

Shield Side:

Any

Shield Coverage:

Full Partial

100%

Trim Length Options

Trim Length By:

Low/High

Low:

0um

High:

0um

Driver/Receiver

Driver:

0um

Receiver:

0um

Help

Hide

Defaults

Apply

Cancel

Option	Description
Shield Net	List of shielding nets to use when adding shields.
Shield Side	Side of the trunk route to add shielding: • Any: Any side of the trunk. • Low: Low side of the trunk (left side of vertical trunks, and bottom side of horizontal trunks). • High: High side of the trunk (right side of vertical trunks, top side of horizontal trunks). • Both: Both sides of the trunk (left and right of vertical trunks, top and bottom of horizontal trunks).

Option	Description
Shield Coverage	Percent of trunks that should have shielding.

Trim Length Options

Increases or decreases the trunk length on the specified side and by the specified value. A value > 0 decreases the length, and a value < 0 increases the length.

The screenshot shows the "Trunk Generator" dialog box with the following settings:

- Nets:** (Empty text field)
- Layer:** M1
- Align:** All
- Trunk Freedom:** 0um
- Trunk Coverage:** ☒ Full ☐ Partial 100%
- Start Side:** Center 0um
- ☐ Override
- Number of trunks to create:** 1
- ☐ Auto-connect
- Trunk width:** 0um
- Shield Options:**
 - ☐ Shield Options
 - Shield Net:** (Empty text field)
 - Shield Side:** Any
 - Shield Coverage:** ☒ Full ☐ Partial 100%
- Trim Length Options:**
 - ☐ Trim Length Options
 - Trim Length By:** ☒ Low/High ☐ Driver/Receiver
 - Low:** 0um **High:** 0um
 - Driver:** 0um **Receiver:** 0um

Buttons at the bottom: Help, Hide, Defaults, Apply, Cancel.

Trim Length by:	Description
Low/High	Increases or decreases the trunk length on the following side: • Low: Low side of the trunk (left side of vertical trunks, and bottom side of horizontal trunks). • High: High side of the trunk (right side of vertical trunks, top side of horizontal trunks).
Driver/Receiver	Increases or decreases the trunk length on the following side: • Driver: Close to the driver pins. • Receiver: Close to the receiver pins.

11

Trunk Routing (Automatic Method)

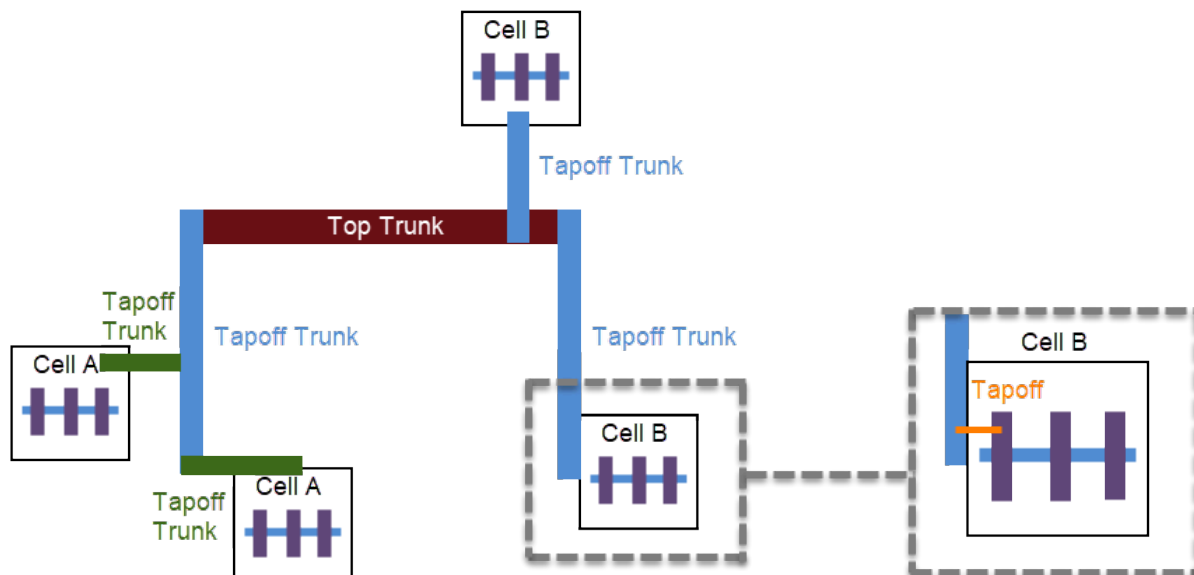
You can use the `rt::autoGenerateTrunk` command to create trunk-style routes automatically in the design layout.

This chapter describes the following:

- [About Automatic Trunk Routing](#)
- [Top-Level Trunk Layer](#)
- [Top-Level Trunk Placement](#)

About Automatic Trunk Routing

You use the `rt::autoGenerateTrunk` Tcl command to plan and create trunk routes automatically. You then use the Auto Router to add tapoffs to connect the trunks to the pins.



This topic describes the following:

- [Constraints](#)
- [Top-Level Trunks](#)
- [Routing Area](#)

Constraints

Trunk routing applies the following routing constraints:

- [Layers Constraint](#) (valid routing layers, width, spacing, and via)
- [Net Priority Constraint](#)
- [Net Shielding Constraint](#) (parallel shielding)

The command creates trunks at the 0x level based on 0x and 1x pins. The trunks are created on valid routing layers that are defined by the [Layers Constraint](#). The command does not create trunks if nets have existing trunks or if the pins are too close and would result in extremely short trunks.

Top-Level Trunks

A top-level trunk is created first and is used as a reference for all lower-layer trunks. Only one top-level trunk is created per net.

Routing Area

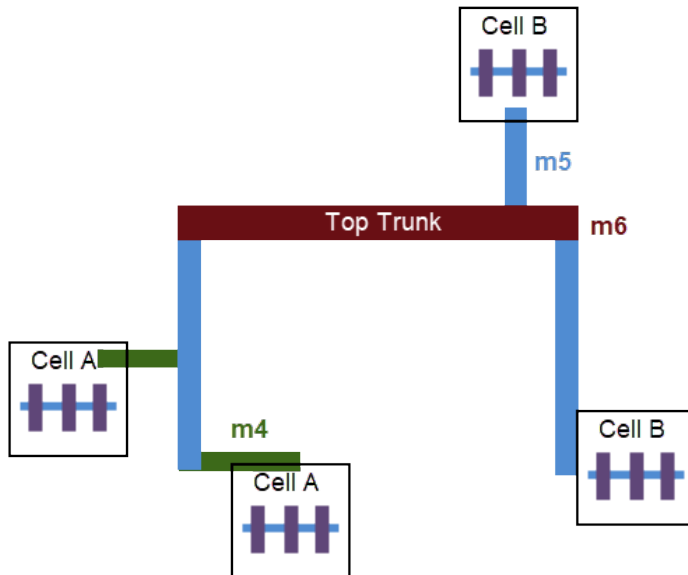
If you define a routing area, and the `rt::autoGenerateTrunk` will add trunks for pins within the routing area only.

Top-Level Trunk Layer

During automatic trunk routing, the `rt::autoGenerateTrunk` Tcl command creates a top-level trunk for each net. You can specify the layer to use for top-level trunks.

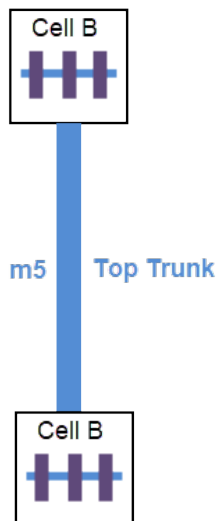
You use the `-topTrunkLayer` argument to specify the layer to use when placing top-level trunk. All tapoff trunks are placed on adjacent layers. In the following example, the command specifies using the m6 layer for the top trunk layer. As a result, the tapoff layers will be placed on the m5 and m4 layers.

```
rt::autoGenerateTrunk -design [ed] -topTrunkLayer m6
```



If the specified layer direction does not match the pin arrangement, the command moves to the next layer and creates the top-level trunk. In the following example, `-topTrunkLayer` specifies using the m6 layer, which is horizontal. However, because the pins require a vertical trunk, an m5 vertical top-level trunk is created.

```
rt::autoGenerateTrunk -design [ed] -topTrunkLayer m6
```



Top-Level Trunk Placement

During automatic trunk routing, the `rt::autoGenerateTrunk` Tcl command creates top-level trunk routes based on the `-startSide` and `-align` arguments.

This topic describes the following:

- [-startSide](#)
- [-align](#)
- [Examples](#)

-startSide

The `-startSide` argument specifies the starting point where the trunk generation looks for a valid location for top-level trunk generation:

- `center`: Center of the pin bounding box.
- `gravity`: Average of the X (vertical trunks) or Y (horizontal trunks) of terminals, where the X or Y are the center of gravity.
- `high`: Right or top side of pin bounding box.
- `low`: Left or bottom side of pin bounding box.
- `auto`: Uses the wire length and pin alignment to determine trunk placement.

-align

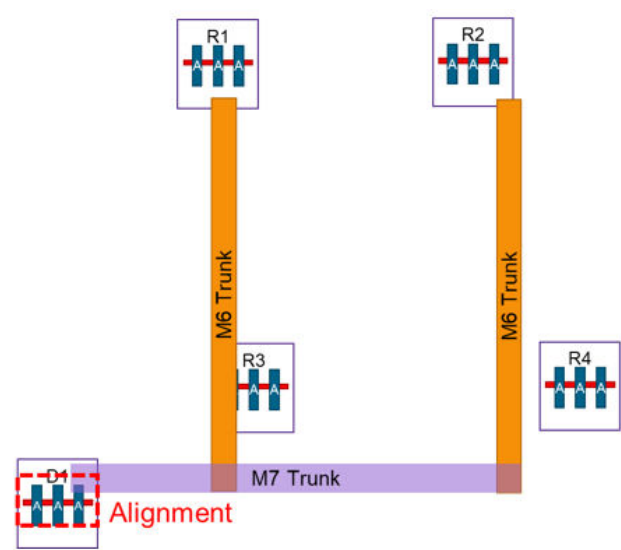
The `-startSide` argument aligns top-level trunks to the specified pins: { `all` | `driver` | `receiver` }.

- `all`: All pins.
- `driver`: Driver pins only.
- `receiver`: Receiver pins only.

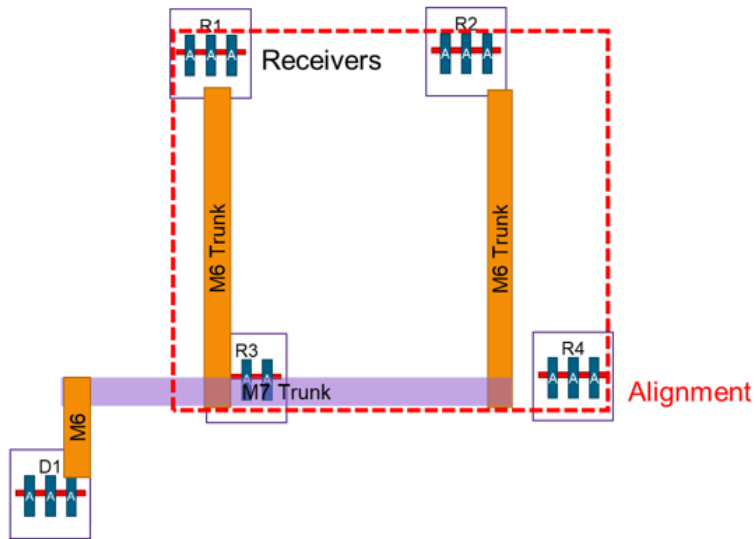
Examples

Argument	Setting
<code>-startSide</code>	<code>auto</code>
<code>-alignment</code>	<code>driver</code>

Argument	Setting
-topTrunkLayer	m7



Argument	Setting
-startSide	low
-alignment	receiver
-topTrunkLayer	m7



12

Power Routing

Several Tcl commands can be used together to create power routing structures in your design.

This chapter describes the following:

- [About Power Routing](#)
- [Before You Start](#)

About Power Routing

You can use available Tcl commands to create power routing structures in your designs, such as power grids (or power mesh). The power grid placements are based on defined tracks and are created within the PR boundary or a region that you specify.

This topic describes the following:

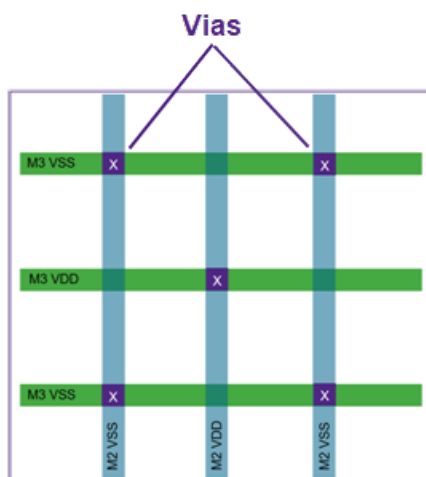
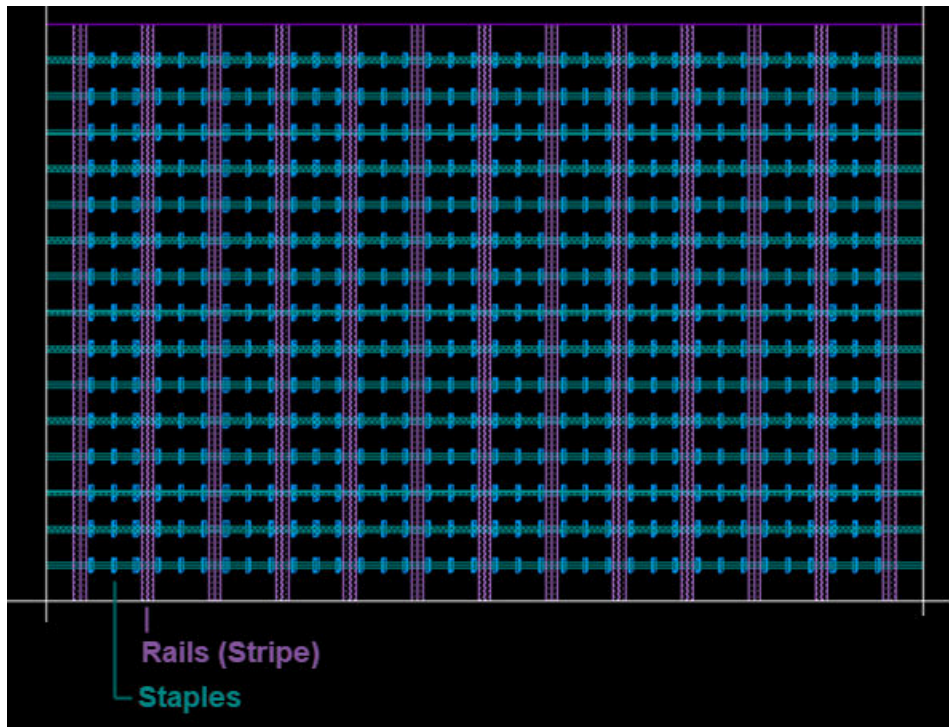
- [Power Structure Components](#)
- [Power Mesh Creation Commands](#)

Power Structure Components

Power structures typically form a grid (or mesh).

A power grid structure consists of the following components:

Power Mesh Component	Description
Power rails or power stripes	Shapes that connect VDD and VSS to standard cells.
Power staples	Short metal shapes that connect adjacent VDD or VSS rails to minimize IR drop.
Power vias	Vias that connect power rails on adjacent layers.



Power Mesh Creation Commands

You use one of the following Tcl commands to create power mesh structures in your design:

Tcl command	Description
<code>rt::createPowerStripes</code>	Creates power rails (or stripes).
<code>rt::createPowerStaples</code>	Creates power staples.
<code>rt::insertVias</code>	Adds power vias.
<code>rt::connectCellRails</code>	Connects cell rails.

Before You Start

Because power routing is based on defined tracks, you must make sure the design contains routing tracks. This topic lists resources that describe track creation and editing.

See the following topics on defining routing tracks:

- Creating and Editing Track Region PCells
- Track Pattern Assistant
- Track Region Assistant

To create power routes, use the Tcl commands in [About Power Routing](#).

13

Hierarchical Routing

The `rt::routeHierarchy` Tcl command routes the specified design from the bottom level to the top level. Because the command is aware of the instances at the top level, the routing results do not cause shorts or design violations.

This chapter describes the following:

- [About Hierarchical Routing](#)
- [Before You Start](#)
- [Routing Using `rt::routeHierarchy`](#)
- [Viewing Hierarchical Routing Reports](#)

About Hierarchical Routing

You can use the `rt::routeHierarchy` Tcl command to route the design hierarchy from the bottom level to the top level. Hierarchy routing is best suited for memory, custom digital, or other designs in which trunks, power, timing-critical, or other routes are added first at the top level, and then you route lower-level cells. Because the `rt::routeHierarchy` Tcl command is aware of the routes or obstacles at the top or upper levels as it routes, shorts and rule violations can be avoided.

This topic describes the following:

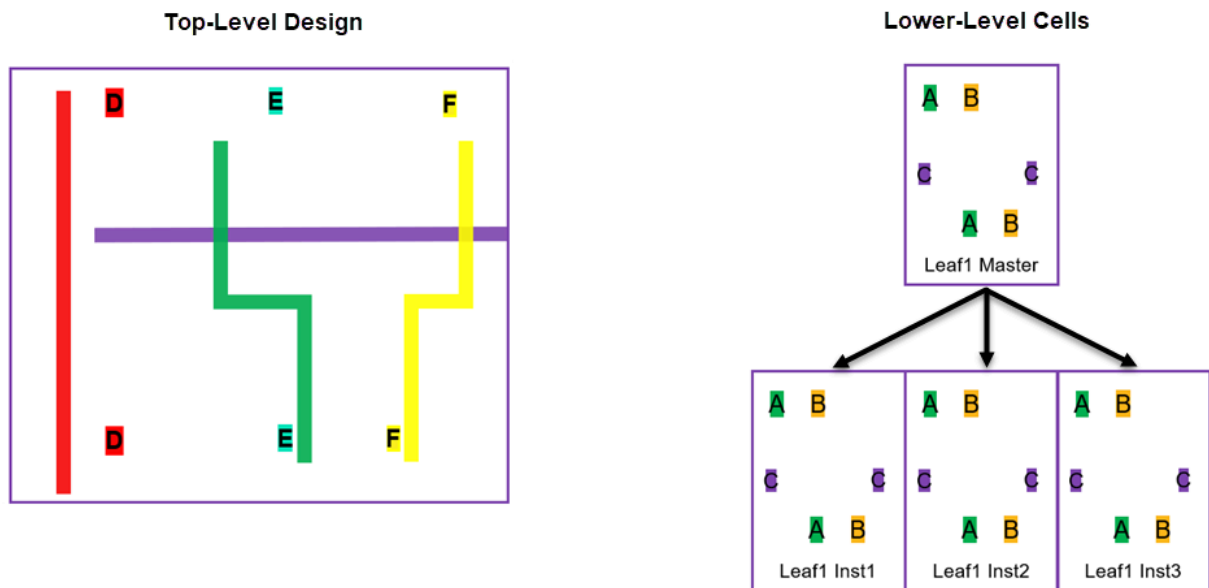
- [Bottom-to-Top Routing](#)
- [Routing Order](#)

Bottom-to-Top Routing

The `rt::routeHierarchy` Tcl command routes the design hierarchy from the bottom level to the top level. This topic describes the bottom-to-top routing process.

In some applications, preroutes are already added in the top-level design, and they cannot be moved. Later, you need to route the remaining pins across the design hierarchy.

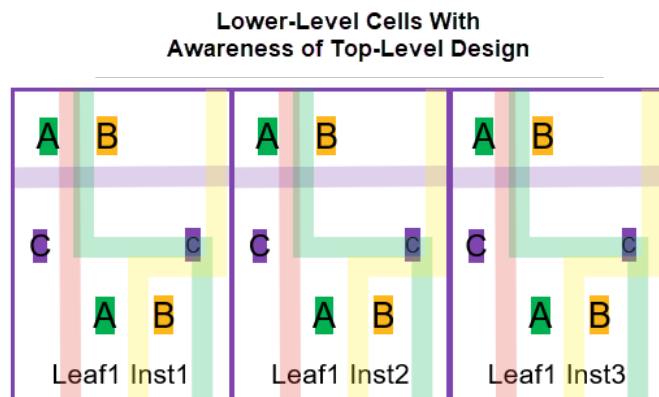
In the following example, four routes have been added previously in the top-level design. You now need to route the remaining pins in the top-level design and the lower-level cells. The lower-level cells are identical instances of a master cell.



To route the remaining pins, the `rt::routeHierarchy` Tcl command:

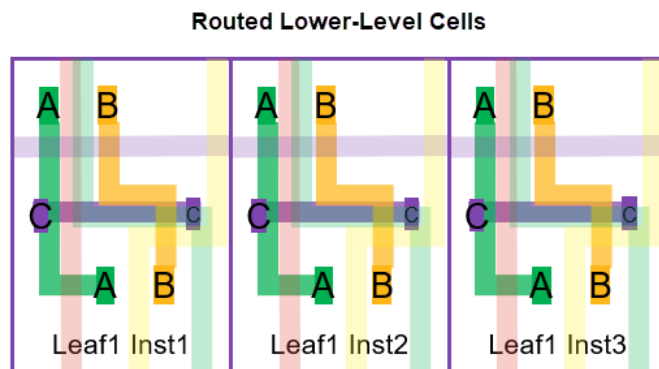
1. Detects existing shapes in the top-level design before it determines the optimal route solutions in the lower-level cells.

In the example, the command sees (or is aware of) the positions of the four top-level routes before it begins to assess potential route solutions in the Leaf1 instances.



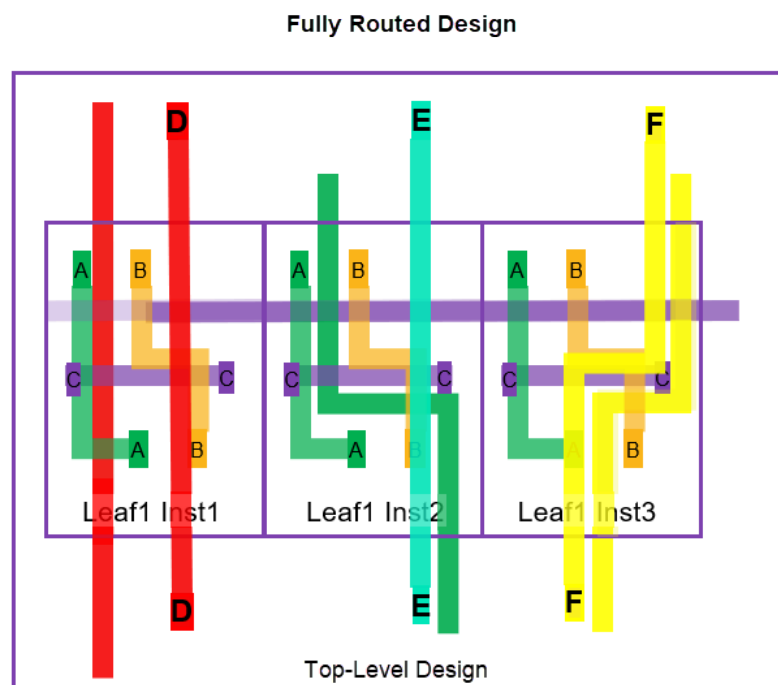
2. Using the Auto Router, creates routes in the lower-level cells with awareness of the top-level shapes. The same routing solution is applied to all identical cell instances.

In the example, nets A, B, and C are routed the same way in each Leaf1 instance. The newly added routes are added with awareness of the four top-level routes, and are thus violation-free and do not cause shorts.



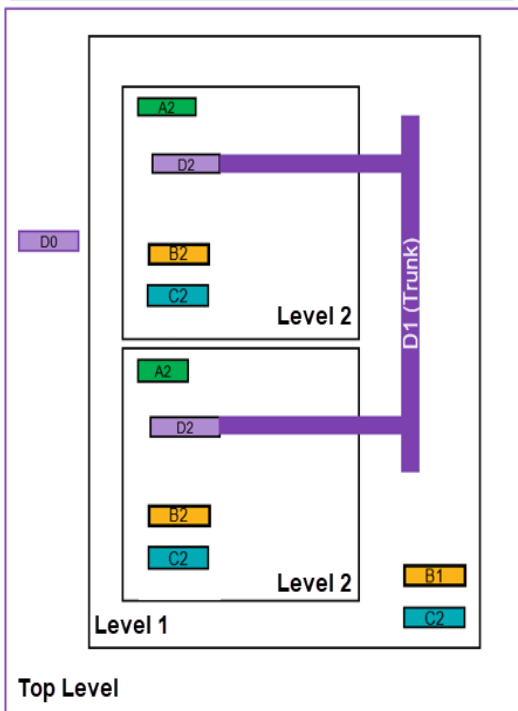
- Creates routes in the top-level design.

In the example, nets D, E, and F in the top-level design are routed after completing the routes in the lower-level cells.

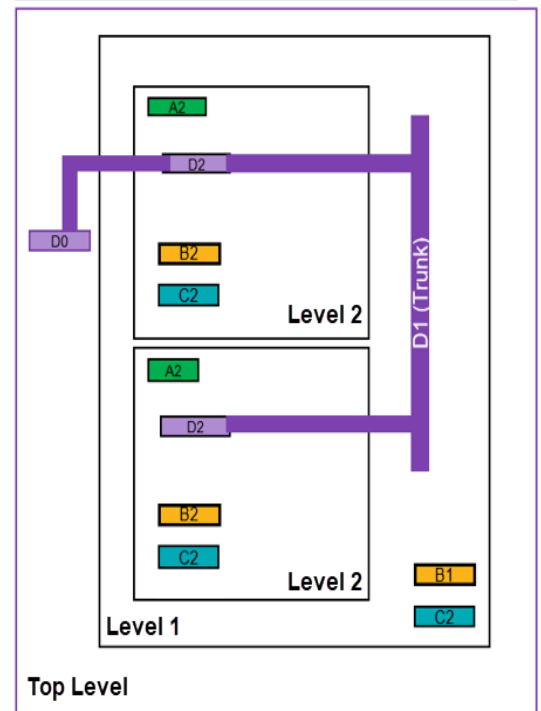


When trunks are present in the design, the same bottom-to-top process applies. In the following example, the two D2 pins in Level 2 connect to the D1 trunk before the D0 pin at the Top Level connects to the D2 pin in Level 2.

Pin-to-Trunk Routing in Lower Levels



Pin-to-Trunk Routing in Top-Level Design



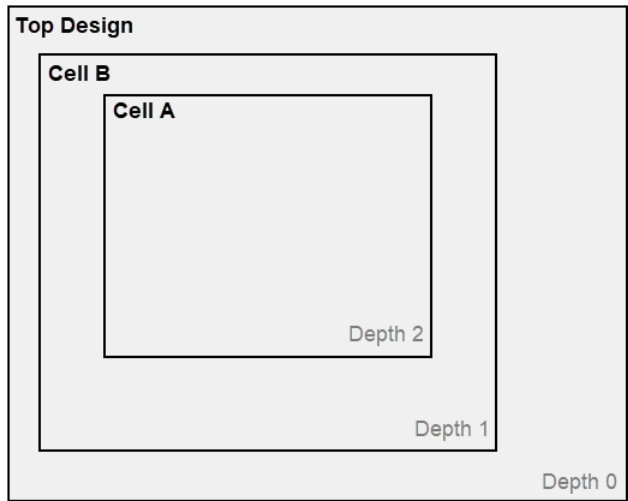
Routing Order

When using the `rt::routeHierarchy` Tcl command, you can specify multiple nets that span the design hierarchy. This topic describes the order the command uses during the routing operation.

When routing multiple nets across the design hierarchy, the `rt::routeHierarchy` Tcl command uses the following routing order:

- Bottom level to top level
- Alphabetical, if the nets are at the same hierarchy level
- Bottom-most depth, if a cell is at more than one depth

In the following example, cell A is at depths 1 and 2, so depth 2 is used when determining the order.



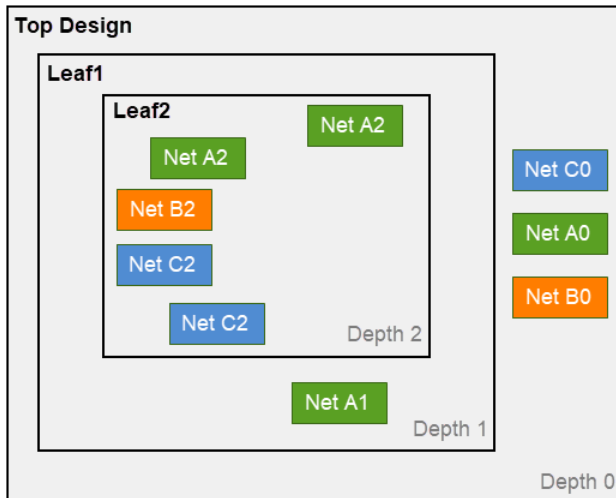
Note that the order of the nets you specify in the command is not considered.

In the following example, the command specifies routing the A0, A1, and A2 nets.

```
rt::routeHierarchical -design <topDesign> -nets <A0, A2, A1>
```

During the hierarchical routing operation, the order in which the nets are routed is as follows:

Routing Order		Net
1	Leaf2	A2
2	Leaf1	A1
3	Top Design	A0



Before You Start

For optimum results, make sure you complete the prerequisite tasks before running the `rt::routeHierarchy` Tcl command.

Before you start hierarchical routing:

1. Open the top-level design.

When you do so, all the designs that are linked to the top-level design should open also.

To ensure that the correct context is established, all the designs that are linked to the top-level design should be closed before you open the top-level design.

2. With the top-level design open, set the following Tcl preferences:

- `db::setPrefValue leStopLevel -value 32`
- `db::setPrefValue leEnableCE -value true`
- `db::setPrefValue leCEDepth -value 32`

The Tcl preference settings in the top-level design are applied to the lower-level designs during hierarchical routing.

Routing Using `rt::routeHierarchy`

You use the `rt::routeHierarchy` Tcl command to route the design hierarchy from the bottom level to the top level. This topic describes how to use the command.

To start hierarchical routing:

- In the Console or in a Tcl script, type the following Tcl command and set the applicable arguments:

```
rt::routeHierarchy -topDesign <oaDesign> -nets <{{<oaNet>}}>
```

You can specify multiple nets in the command. The nets can be from all levels of the design hierarchy. For example, you can specify nets at the top level and nets at lower levels of the hierarchy.

The order of the nets you specify in the command does not reflect the actual routing order. See [Routing Order](#).

See Also

- [About Hierarchical Routing](#)
- [Before You Start](#)
- [Viewing Hierarchical Routing Reports](#)

Viewing Hierarchical Routing Reports

The `rt::routeHierarchy` Tcl command generates a report before it starts routing and another report after it finishes routing. This topic describes the contents of the two reports.

Viewing Preroute Reports

At the beginning of the routing operation, the `rt::routeHierarchy` Tcl command generates a preroute summary report in the Console. The report lists the designs to be routed and the order in which they will be routed.

```
Information: *****
Information: * Design                               Depth  Nets Asked To Route *|
Information: * designLib/level1_cell1/layout         2              1 *
Information: * designLib/level2_cell1/layout         1              1 *
Information: * designLib/level3_cell1/layout         0              1 *
Information: *****
```

Report Column	Description
Design	Library/cell/view of the design to be routed.
Depth	Hierarchy depth of the design.
Nets Asked to Route	Number of nets specified in the <code>-nets</code> argument.

Viewing Postroute Reports

After routing, the `rt::routeHierarchy` Tcl command generates a postroute report in the Console. The report lists the designs that were routed, the order in which they were routed, the number of nets that were successfully routed, and the number of nets that were not routed.

```
Information: *****
Information: * Design          Depth  Nets Asked To Route  Attempted  Routed  Unrouted  Elapsed Time *
Information: * designLib/level1_cell/layout    2          1          1          1          0    0h:00m:01s *
Information: * designLib/level2_cell/layout    1          1          1          1          0    0h:00m:01s *
Information: * designLib/level3_cell/layout    0          1          1          1          0    0h:00m:02s *
Information: *****
```

Report Column	Description
Design	Library/cell/view of the design to be routed.
Depth	Hierarchy depth of the design.
Nets Asked to Route	Number of nets specified in the <code>-nets</code> argument.
Attempted	Number of nets that the command attempted to route in the design.
Routed	Number of nets that were successfully routed.
Unrouted	Number of nets that were not routed.
Elapsed Time	Length of time it took to route the design.

14

Adding Post-Route Shielding

This chapter describes how to add shielding after you finish routing.

This chapter contains the following topics:

- [About Post-Route Shielding](#)
- [Before You Add Post-Route Shielding](#)
- [Adding Shielding](#)
- [Adding Point-to-Point \(P2P\) Shielding](#)
- [Deleting Shielding](#)
- [Shielding Styles and Behavior](#)

About Post-Route Shielding

If you enabled the shielding options, shielding is automatically added based on your specifications during the auto routing operations. However, you can choose to disable shielding during routing and add shielding after the routing operation is finished.

Post-route shields are wrapped around existing routes without modifying the route. The supported shield types are parallel, tandem and coaxial. The shield types are described in [Net Shielding Constraint](#).

You can add post-route shielding to all selected nets. You can also add point-to-point (P2P) shielding after routing.

Parallel shielding, jumpers for parallel, and bus/diff-pair shielding can snap to active tracks or manufacturing grids.

Before You Add Post-Route Shielding

Before you add post-route shielding:

1. Open the Layout Editor and add the shielding constraints for the prerouted nets using the Constraint Editor. For more information about constraints, see [Setting Routing Constraints](#).
2. In the **Layout Design Options** dialog box (**Connectivity** tab), select **Connectivity Engine** and set the **Depth** to 0. See Enabling the Connectivity Engine.
3. (Optional) Turn on the DRD (design-rule driven) tool. See Checking for Violations as You Edit (DRD).

Adding Shielding

The **Routing > Add Shield** command lets you add shielding for prerouted nets. This topic describes the following:

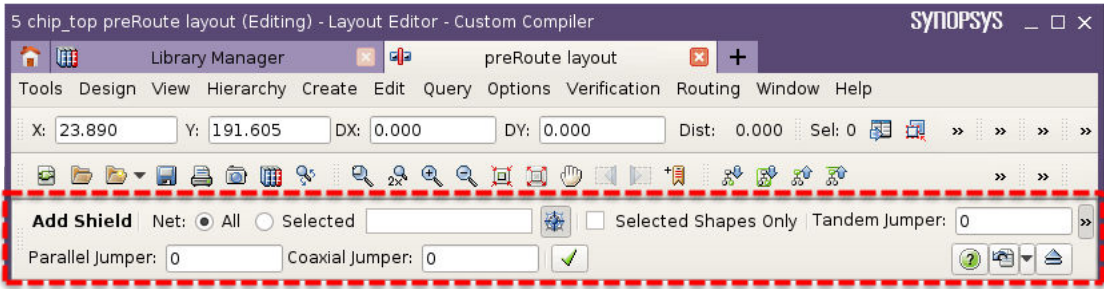
- [Enabling the Add Shield Command](#)
- [Setting Options in the Add Shield Toolbar](#)
- [Setting Options in the Add Shield Dialog Box](#)
- [Starting the Shielding Operation](#)

To add point-to-point shielding, see [Adding Point-to-Point \(P2P\) Shielding](#). To delete shielding, see [Deleting Shielding](#).

Enabling the Add Shield Command

To enable the **Add Shield** command:

1. In the Layout Editor, make sure the Custom Router is enabled. For instructions, see [Getting Started](#).
2. Choose **Routing > Add Shield**. The **Add Shield** toolbar appears.



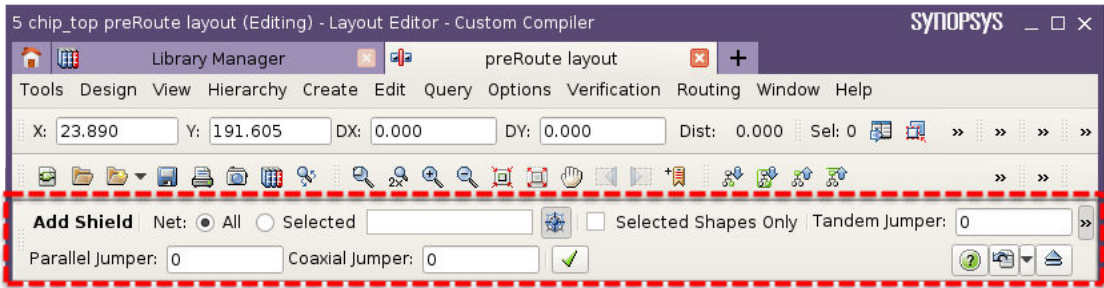
Note:

The `rtAutoShieldDialogMode` preference controls whether the dialog box or the toolbar opens by default. When the value is set to true, the dialog box opens.

Setting Options in the Add Shield Toolbar

This section describes the **Add Shield** toolbar options.

To use the `thirt::autoShield Tcl` command, see the *Custom Compiler Tcl Reference Manual*.



Option	Description
Net	• All: Adds shields to all nets. • Selected: Adds shields to the selected net. Selected net names appear in the list box.
	Opens the Design Navigator, where you can select the nets.
Selected Shapes Only	Adds shielding to route shapes that you select on the layout canvas.
Tandem Jumper Parallel Jumper Coaxial Jumper	You can specify values to add jumpers or jumper vias for the different types of shields: Tandem , Parallel , and Coaxial . If the value is not set or is 0, no jumper or jumper via is added. For the descriptions of the different shield types, see Shielding Styles and Behavior .

Shield vias are automatically added when you add shielding. By default, the shield via arrays and the corresponding signal via arrays match in structure and appearance. If you want the arrays to be different, set the `rtSyncShieldViasWithSignalVias` Tcl preference to `false`. For information about the Tcl preference, see the *Custom Compiler Tcl Reference Manual*. For more information about shield vias, see [Shield Vias](#).

Setting Options in the Add Shield Dialog Box

This section describes the options in the **Add Shield** dialog box. Except for **Selected Shapes Only**, the options in the **Add Shield** dialog box are identical to those in the **Auto Route** dialog box **Shield** tab. For the description of the options, see the [Auto Route - Shield Tab](#).

The **Add Shield** dialog box contains the following sections and options:

- Nets:** ☒ All ☐ Selected
- ☐ **Selected Shapes Only**
- Shield Nets:** [Text Field]
- Taper Halo:** [Text Field: 0]
- Enclose:** ☒ Pin ☒ Via
 - ☒ Allow Wrong Way ☐ Connect to Aligned Pin
 - ☒ Sharing ☐ Use Via Array for Coaxial
- Jumper Spacing (um)**
 - Tandem:** [Text Field: 0] **Parallel:** [Text Field: 0] **Coaxial:** [Text Field: 0]
- Minimum Length (um)**
 - Signal Segment:** [Text Field: 0] **Connected Shield:** [Text Field: 0]
- ☐ **Connect to Supply**
 - Min Shield Length:** [Text Field: 0] um
 - ☐ **Connect at Mesh Overlap**
 - Layer Offset:** ☐ Above ☒ Either ☐ Below
 - ☐ **Connect with Route**
 - Tie Spacing:** [Text Field: 0] um
 - Search Distance:** [Text Field: 0] um
- ☐ **Report Coverage**
- Rules:** [Dropdown: All] [Icon]
- Buttons:** Help, Hide, Defaults, Apply, Cancel

Starting the Shielding Operation

To start the shielding operation:

- If you are on the **Add Shield** toolbar, click .
- If you are in the **Add Shield** dialog box, click **Apply**.

Adding Point-to-Point (P2P) Shielding

The **Routing > P2P Shield** command is a semi-automatic command that lets you manually create partial, point-to-point shielding for a routed net. You can create the shield from a starting point to a target point between routing objects (path, via, pins). This feature is useful for post-editing; for example, where the shielded signal may need to be shielded partially.

This topic describes the following:

- [Before You Start](#)
- [Enabling the P2P Shield Command](#)
- [Adding Shielding](#)
- [Setting Options in the P2P Shield Toolbar](#)
- [Setting Options in the P2P Shield Dialog Box](#)

Note:

To fully shield a net, Synopsys recommends using the **Add Shield** command described in [Adding Shielding](#).

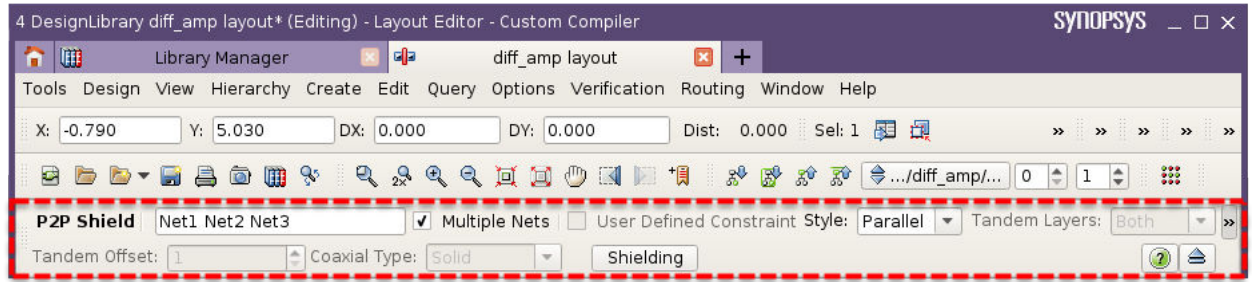
Before You Start

Make sure you have set the Net Shielding constraint. See [Setting Routing Constraints](#).

Enabling the P2P Shield Command

To enable the P2P Shield command:

1. In the Layout Editor, make sure the Custom Router is enabled. For instructions, see [Getting Started](#).
2. Choose **Routing > P2P Shield**. The **P2P Shield** toolbar appears.



Note:

The `rtCreateP2PShieldDialogMode` Tcl preference controls whether the dialog box or the toolbar opens by default. When the value is set to true, the dialog box opens. For information, see the *Custom Compiler Tcl Reference Manual*.

Adding Shielding

To add point-to-point shielding:

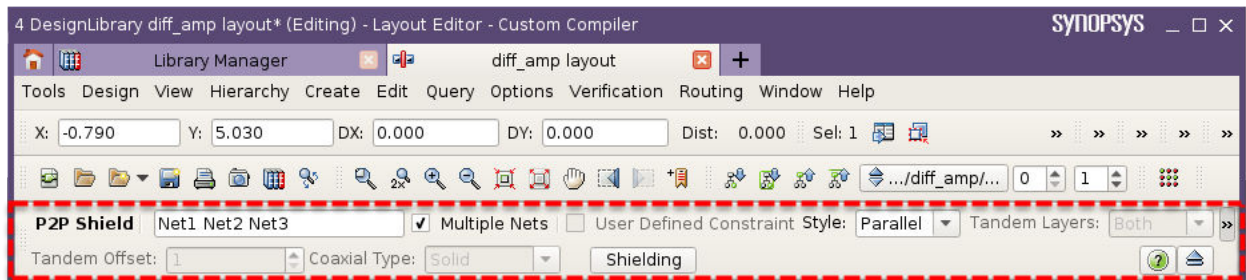
1. Set the **P2P Shield** options using one of the following methods:
 - [Setting Options in the P2P Shield Toolbar](#)
 - [Setting Options in the P2P Shield Dialog Box](#)
2. If you are adding shielding to a bus route that has bits on different nets:
 - a. On the **P2P Shielding** toolbar, select **Multiple Nets**.
 - b. In the Design Navigator, select the nets. The net names appear on the **P2P Shielding** toolbar.
3. On the layout canvas, locate the path or paths you want to shield.
4. Click the path where you want P2P Shield to start. If you are adding shielding to bus bits, click one of the bits to indicate the starting point.
5. Move the pointer toward the ending point.

As you move the pointer, a line is displayed between the starting point and where the pointer is located.
6. Click the path where you want P2P Shield to end.

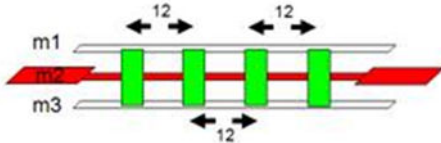
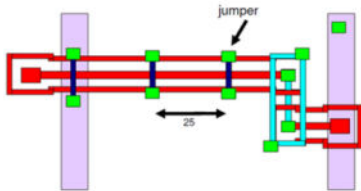
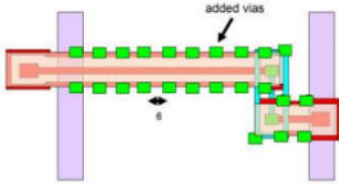
The router adds the shielding between the starting point and the ending point. If you selected **Multiple Nets**, shielding is added to all of the selected bits.

Setting Options in the P2P Shield Toolbar

This section describes the **P2P Shield** toolbar options.



Option	Description
P2P Shield	Routes the selected nets. Selected net names appear in the list box.
Multiple Nets	Allows you to apply shielding to multiple nets that belong to the same bus route.
User Defined Constraint	If specified, the P2P shield will use the constraints predefined in the Constraint Manager and the Constraint Editor.

Option	Description
Style	Specifies the type of shielding: • Tandem: adds a shield line one layer above the signal line, and another shield line one layer below the signal.  • Parallel: adds a same-layer shield line on each side of the signal line.  • Coaxial: a combination of the parallel and tandem shielding. 
Tandem Layers	When the Tandem shielding style is selected, choose the tandem layers to shield: • Both • Above • Below
Tandem Offset	When the Tandem shielding style is selected, choose the layer offset (1 or 2).
Coaxial Type	Allows the user to shield with a different coaxial shield type on a single net. The types are: • Solid • Split • Center
Shielding	Opens the Shielding Options dialog box. The options in the dialog box are identical to those in the Auto Route dialog box Shield tab. For information, see the Auto Route - Shield Tab.

Shield vias are automatically added when you add shielding. By default, the shield via arrays and the corresponding signal via arrays match in structure and appearance. If you want the arrays to be different, set the `rtSyncShieldViasWithSignalVias` Tcl preference to `false`. For information about the Tcl preference, see the *Custom Compiler Tcl Reference Manual*. For more information about shield vias, see [Shield Vias](#).

Setting Options in the P2P Shield Dialog Box

This section shows the **P2P Shield** dialog box.

The options in the **P2P Shield** dialog box are a combination of the options in the **P2P Shield** toolbar and **Auto Route** dialog box **Shield** tab. For information, see [Setting Options in the P2P Shield Toolbar](#) and the Auto Route - [Shield Tab](#).

Chapter 14: Adding Post-Route Shielding

Adding Point-to-Point (P2P) Shielding

P2P Shield

Select net to P2P Shield:

Shield Style

User Defined Constraint

Style: Parallel Tandem Layers: Both

Coaxial Type: Solid Tandem Offset: 1

Shield Nets:

Taper Halo: 0

Enclose:

Pin

Via

Allow Wrong Way

Connect to Aligned Pin

Sharing

Use Via Array for Coaxial

Jumper Spacing (um)

Tandem: 0 Parallel: 0 Coaxial: 0

Minimum Length (um)

Signal Segment: 0 Connected Shield: 0

Connect to Supply

Min Shield Length: 0 um

Connect at Mesh Overlap

Layer Offset: Above Either Below

Connect with Route

Tie Spacing: 0 um

Search Distance: 0 um

Report Coverage

Rules: All

Help

Hide

Defaults

Cancel

Deleting Shielding

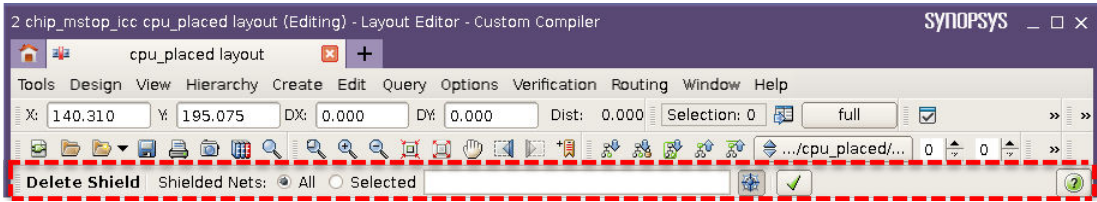
You can use the **Delete Shield** command to delete shield lines for either all shielded signal nets or just selected signal nets. This feature can be useful for post-editing. For example, in the case where you want to re-edit a signal net, but the shield lines are in the way, you could delete the shields for the net, perform your edits, and then use the **Add Shield** command to reapply shields.

Note:

Existing shield ties are removed along with the shields.

To delete shielding:

1. In the Layout Editor, choose **Routing > Delete Shield**. The **Delete Shield** toolbar appears.



2. Set following options:

Option	Description
Shielded Nets	◦ All: Deletes all shields from all the shielded signal nets. ◦ Selected: Deletes shields from only the selected shielded signal nets. The selected net names appear in the list box.
	Opens the Design Navigator.

3. On the **Delete Shield** toolbar, click .

Shielding Styles and Behavior

The following sections describe the behaviors for the different shielding types:

- [Coaxial Shielding](#)
- [Parallel Shielding](#)
- [Tandem Shielding](#)
- [Differential Pair and Bus Shielding](#)

- [Group Shielding](#)
- [Shared Shielding](#)
- [Maximizing Shared Shielding](#)
- [Shield Vias](#)

Coaxial Shielding

Coaxial shielding is based on the shield widths, shield gaps, and signal path widths on the shield layers of their shielded signal paths. The widest shield width and the largest gap is used to create shields on all shield layers.

There are three types of coaxial shields:

- *Solid coaxial shield:* Creates solid wide-width shields on both lower and upper layers that cover the signal path and two parallel shields. Because of the wide width, it can trigger wide width spacing constraint easier. There are jumper vias that are created to tie parallel and tandem shields if coaxial jumper length is specified. If the solid coaxial shield is shrunk because of maximum width constraint on the tandem layer, extra jumper paths are also created.



- *Split coaxial shield:* Because the solid-style coaxial shielding can easily trigger wide width clearance violations, you may want to choose the split style instead. The split shielding creates individual shields to sandwich its signal line and two parallel shield lines on both lower and upper layers. The largest gap of the three layers is used to create split coaxial shielding. Jumper paths and jumper vias are created on all eight surrounding shield lines together if coaxial jumper length is specified. When specifying split coaxial shielding, the split coaxial shields should remain split, but can result in wider shield widths and larger shield gaps.



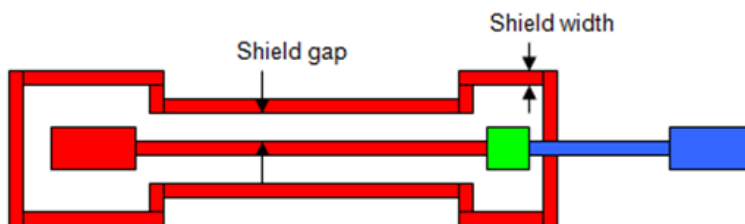
- *Center-Only coaxial shield:* The center-only coaxial shielding offers minimum coaxial shield coverage. It shields four sides of a signal path with two parallel shields and

above/below tandem shields. Jumper paths and jumper vias are created to tie four shield lines together if jumper length is specified.



Parallel Shielding

Parallel shielding puts a same-layer shield line on either side of the signal line. Optionally shields can be around a via or a pin.



Tandem Shielding

Tandem shielding shields on the layers above and below the signal line.



Tandem shielding has the following layer variations:

- *Both*: Shield wires are created for both above and under with respect to the protecting signal wire shapes.
- *Above*: Shield wires are created for only above with respect to the protecting signal wire shapes.
- *Under*: Shield wires are created for only under with respect to the protecting signal wire shapes.

The tandem layer offset controls are:

- *N-2 and N+2 layers*: The user can set tandem shield to be routed on N-2 and N+2 layers. This applies only to the tandem shield. The default value is 1, which is N-1 and N+1 layers.
- *Shield cut off around pin*: The tandem shield can be cut off around the pin. The default value is 0.0.

Differential Pair and Bus Shielding

Shielding can be applied to bus bits and differential pair routes.

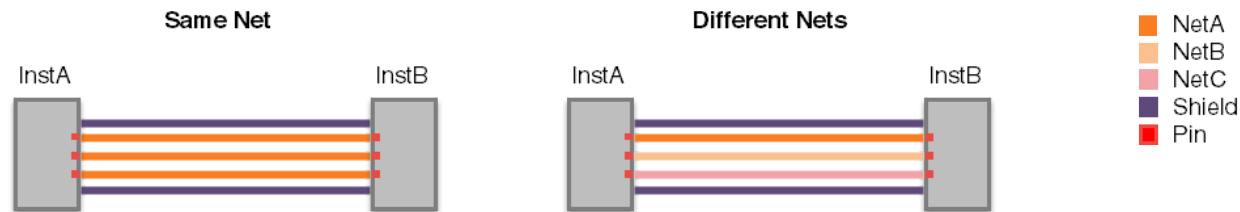
For differential pair (diff-pair) and bus shielding, the shields are determined by the **Shield Placement** parameter in the [Differential Group Constraint](#), [Bus Constraint](#), and shield setting on individual nets:

Shield Placement Setting	Description
outside	Places shields on the outside of the trunk or group.  Net Shield
interleave	Inserts shared shielding between adjacent wires.  Net Shield
doubleInterleave	Shields each wire individually.  Net Shield
halfInterleave	Inserts shared shielding every two wires.  Net Shield

If no shielding is needed for the diff-pair and bus trunks, you do not need to set [Net Shielding Constraints](#) on bit nets, regardless of the value of the **Shield Placement** parameter.

Group Shielding

Using the **Add Shield** command, you can add post-route shielding to the outer sides of a group of existing wires that connect macro boundary pins. The group of wires can belong to the same net or to different nets.



The group of wires must be created using one of the following methods:

- [Auto Routing](#)
- **Create > Interconnect** (wires are on the same net or on different nets)
- **Edit > Wire > Stripe Wire** (wires are on the same net)

The wires in the group must belong to a net group. You can create net groups using the Design Navigator or the `le::createNetGroup` Tcl command.

To prepare for group shielding, set the following Net Shielding constraint parameters:

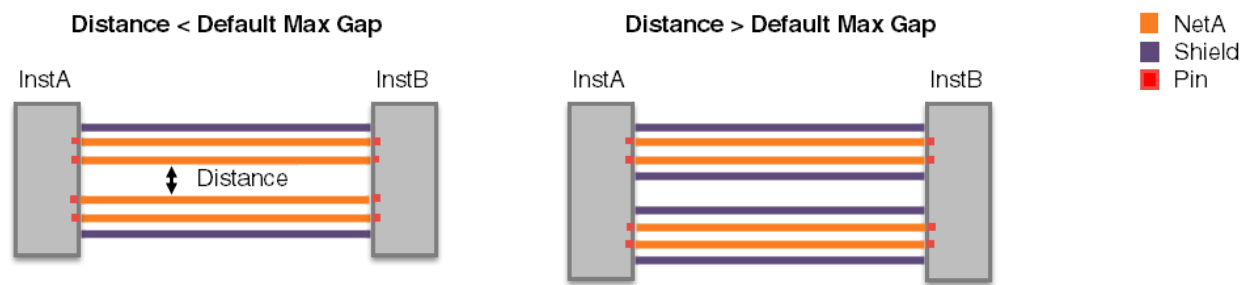
Net Shielding Constraint Parameter	Setting
Shield Net	Power or ground net, typically VSS or VDD
Style	Parallel or Coaxial
Tandem Layers	Above, Below, or Both
Tandem Offset	1
Parallel Layers	(Parallel Style only) Single or Multiple
Coaxial Type	(Coaxial Style only) Split
Group Shield	☒

Note that the Enclose Via and Enclose Pin settings are ignored. For information about the constraint parameters, see [Net Shielding Constraint](#).

The shielding results depend on the various shielding constraint settings you choose:

Style	Parallel/Tandem Layers	Description
Parallel	Single	Shielding is added on the same layer as the wires and flank the outer most wires in the group. Shield Wires Shield m3 
Parallel	Multiple	Shielding is added as follows: Shield Wires Shield m4  m3  m2 
Split Coaxial	Both	Shielding is added on the same layer as the wires and flank the outer most wires in the group. m4  m3  m2 
Split Coaxial Above	Above	Shielding is added on the same layer as the wires and flank the outer most wires in the group. m3  m2 
Split Coaxial Below	Below	Shielding is added on the same layer as the wires and flank the outer most wires in the group. m4  m3 

The Net Shielding constraint parameter, **Default Max Gap**, can also affect the shielding results. If the actual distance between the two inner-most wires is greater than the distance specified by **Default Max Gap**, shielding is added between the two wires.



Shared Shielding

Shared shielding can be applied during auto routing, point-to-point interactive routing, or post-route shielding.

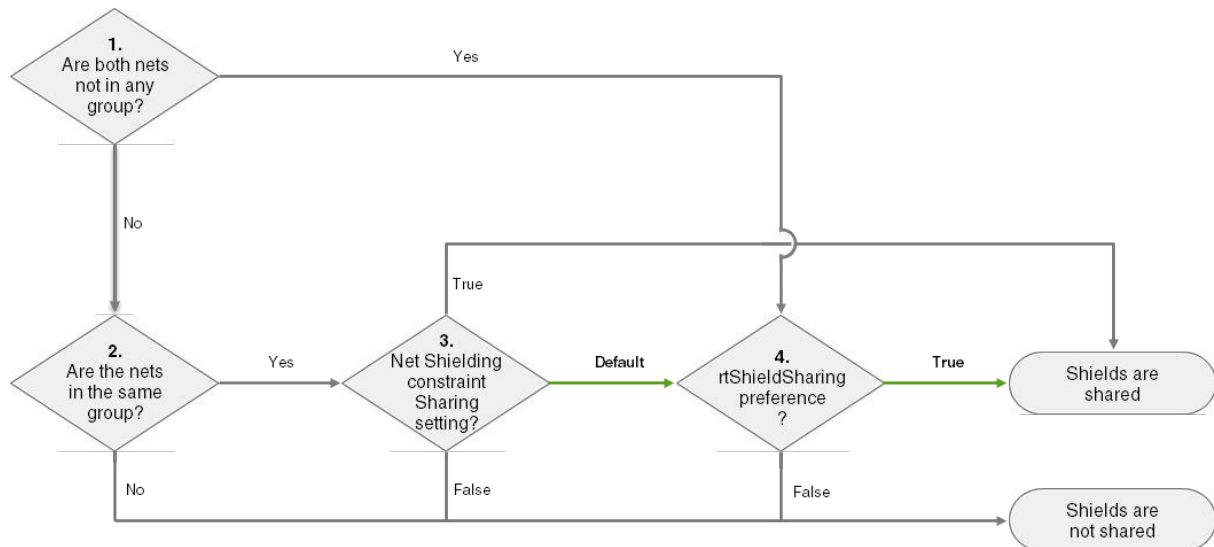
You can enable shared shielding using one of the following methods:

- **Sharing** option of the Net Shielding constraint
- `rtShieldSharing` preference

During a routing operation, the router determines shared shielding as follows:

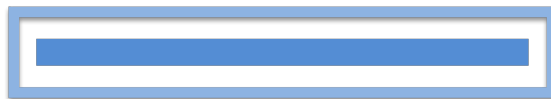
1. If both a shielded net and its neighboring net do not belong in any groups, the router checks the `rtShieldSharing` preference setting.
 - If the preference is set to `True` (default), shields are shared, as shown by the green path in the flowchart.
 - If the preference is set to `False`, shields are not shared.
2. Shared shielding is applicable only to nets that are in the same group. Nets that are not in the same group cannot share shields.
3. If nets are in the same group, the router checks the Net Shielding constraint.
 - If **Sharing** is set to **Default** (or not set), the router will use the `rtShieldSharing` preference setting, as shown by the green path in the flowchart. See step 4.
 - If the **Sharing** is set to **True**, shields are shared.
 - If the **Sharing** is set to **False**, shields are not shared.
4. The router checks the `rtShieldSharing` preference setting.
 - If the preference is set to `True` (default), shields are shared, as shown by the green path in the flowchart.
 - If the preference is set to `False`, shields are not shared.

Shared shielding by constraint and preference:



When shared shielding is enabled, the router permits the following:

- If there is enough space between two signal wires, shields are added for the individual wires.



- If there is not enough space between two signal wires, shields are shared between the two wires.



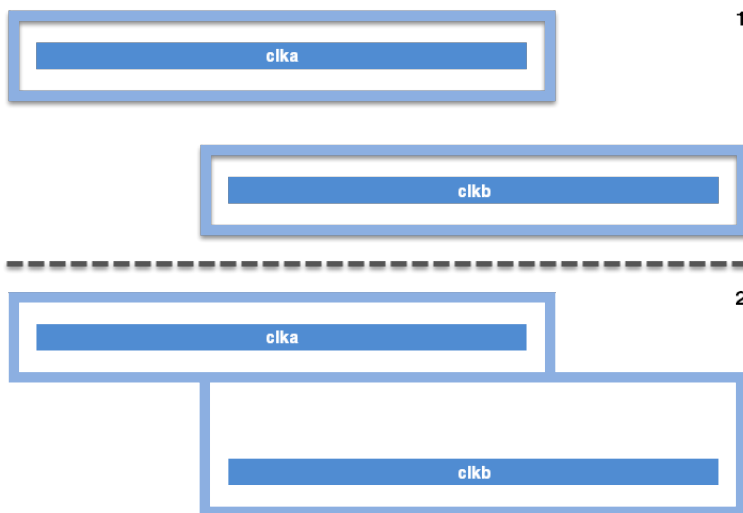
- If there is not enough space between a power grid rail and the adjacent signal wire, the power grid rail is used as shield for the signal wire.



To route wires close to power straps and use the power straps or power grid rails as shared shielding, set the `rtUsePowerStrapAsShield` preference value to true. This preference is applicable for auto routing only.

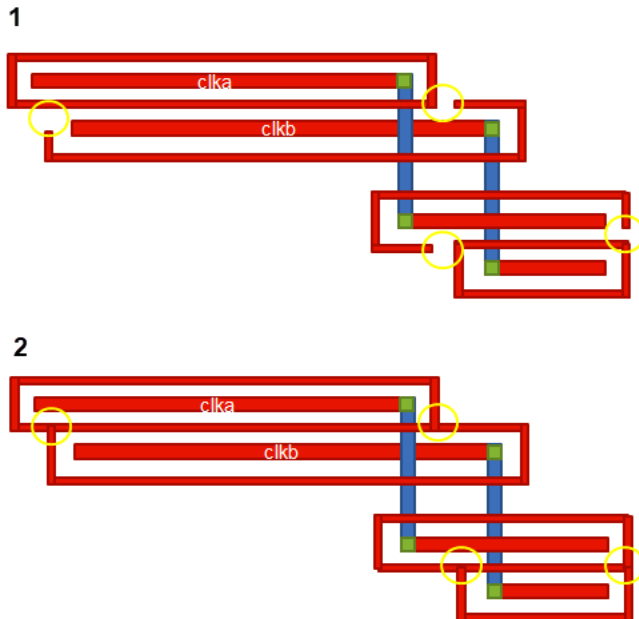
Shared shielding respects the minimum shield gap specification (the distance between the wire and shield). However, the gap can be larger than specified or is not uniform. To create more uniform shields, Synopsys recommends that you specify routing grids.

Shared shielding off (1), shared shielding on with no routing grid (2)



Enabling shared shielding can result in optimal shielding results, as the following diagrams demonstrate. The first diagram shows that when shared shielding is turned off, some gaps can occur in shielding. The second example in the figure shows that when shared shielding is turned on, the signal wires are fully shielded.

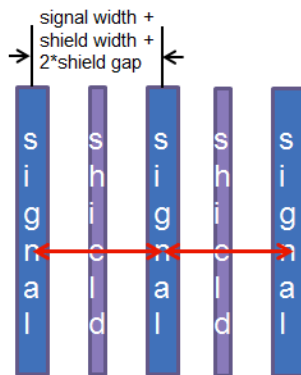
Shared shielding off (1), shared shielding on (2)



Maximizing Shared Shielding

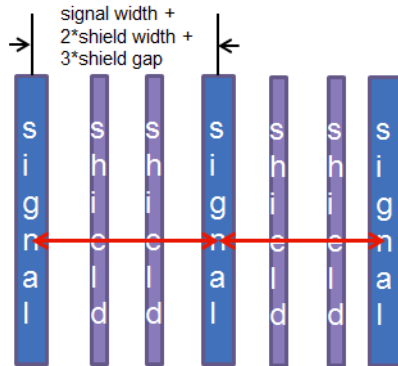
To maximize the possibility of a shield sharing result, the routing grid can be defined by user when routing the shared shielding nets.

Routing grid for shared shielding nets = signal width + shield width + 2*shield gap.



To maximize the possibility of NO shield sharing result, the routing grid can be defined by user when routing the non-shared shielding nets.

Routing grid for non-shared shielding nets = signal width + 2*shield width + 3*shield gap.



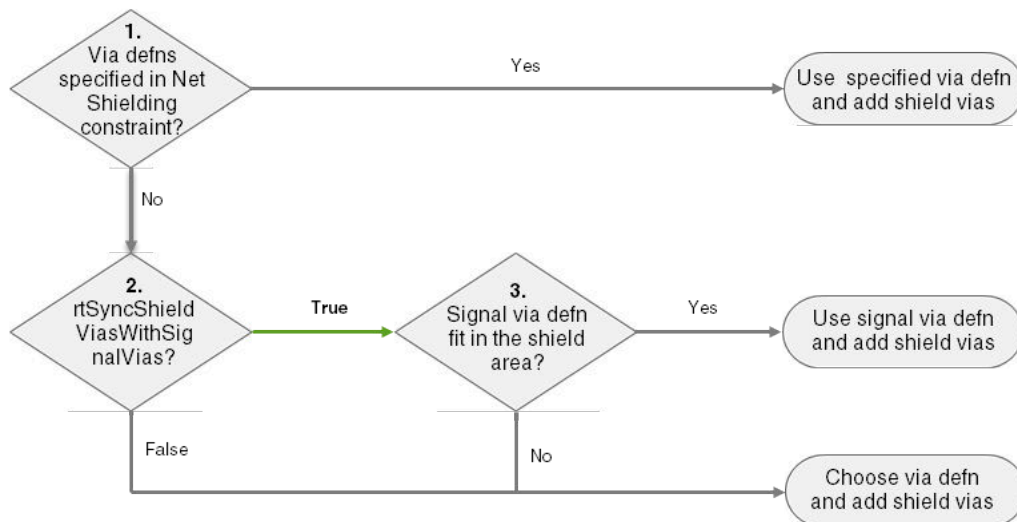
Shield Vias

Shielding can be applied to vias during auto routing, interactive routing, or post-route shielding.

When adding shield vias, the router uses the following process to determine which via definitions to use:

1. The router checks the Net Shielding constraint and uses the via definitions you specified to add shield vias.
2. If you did not specify via definitions in the Net Shielding constraint, the router checks the `rtSyncShieldViasWithSignalVias` Tcl preference.
 - If the preference is set to `true` (default, shown by the green path in the flowchart), the router tries to use the corresponding signal's via definitions. See step 3.
 - If the preference is set to `false`, the router chooses the via definition and adds the vias.
3. When the `rtSyncShieldViasWithSignalVias` preference is set to `true`, the router checks if the corresponding signal via's size fits within the area where the shield vias should be placed.
 - If the signal via's size fits, the router proceeds to use the signal's via definition to add the shield vias.
 - If the signal via's size is too large, the router chooses a different via definition to add the vias.

Process for determining the via definitions to use:



The router will not add the shield vias if one or both of the following occurs:

- Neighboring shapes encroach on the space required to add the vias.
- The shield's via area is too small to accommodate any via definition.

15

Viewing and Modifying Route Attributes

This chapter describes how to view and edit route shape properties.

This chapter contains the following topics:

- [About Routes and Route Shapes](#)
- [Viewing and Editing Route Shape Properties](#)

About Routes and Route Shapes

When placing routes, the Custom Router creates one or more of the following objects on the canvas:

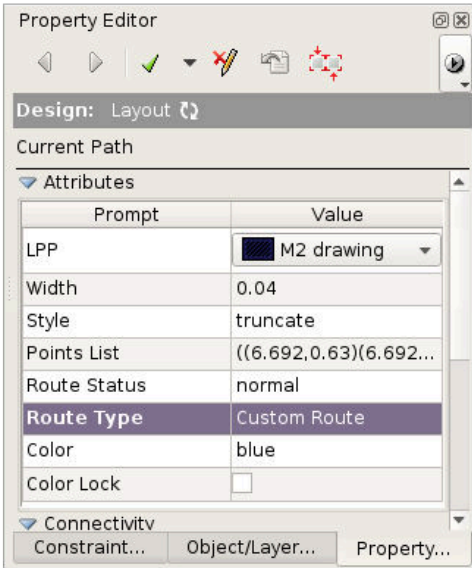
- Paths and path segments
- Vias
- Shielding

Each route object has a set of properties. For example, the properties of a path can tell you its layer-purpose pair, its width, its color, the type of signal it is supposed to carry, its voltage range, and so on.

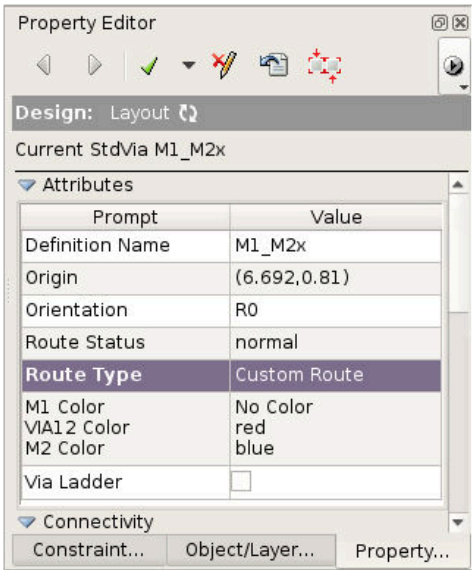
Any object that the Custom Router creates has the **Custom Route** route type property. The route type identifier allows the Custom Router to delete only the shapes it created. The identifier is also useful in co-design flows to distinguish routes created within the Custom Compiler environment from those created by the Fusion Compiler or IC Compiler II tool.

You can view the properties of route objects in the Property Editor.

Path Properties



Via Properties

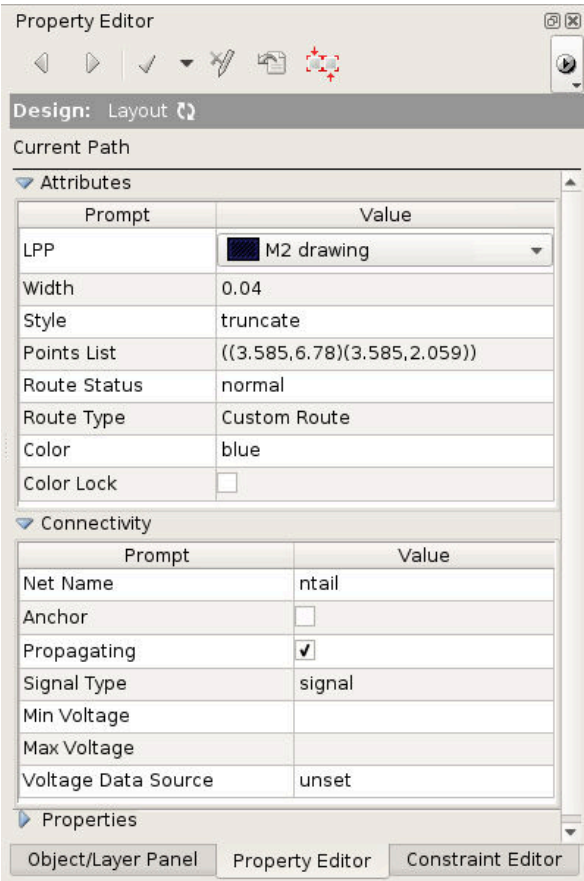


Viewing and Editing Route Shape Properties

This topic describes how to view and edit route shape properties in the Property Editor. For instructions on viewing and editing via shape properties, see [Modifying Via Attributes and Properties](#).

To view the route shape attributes:

1. In the Layout Editor, display the Property Editor.
2. On the canvas, select a route shape. The shape's properties appear in the Property Editor.



3. In the Property Editor, view or edit the **Attributes** values:

Attribute	Description
LPP	Route shape's layer-purpose pair.
Width	Width of the route shape.
Style	Route shape's end style: ◦ truncate ◦ extend ◦ round ◦ variable See Choosing End Styles.
Points List	Coordinates of the shape's vertices.

Attribute	Description
Route Status	Indicator of whether the route can be modified: ◦ normal: The route or via can be modified. ◦ fixed: The route or via can be modified manually using the Edit commands or by direct manipulation on the canvas. The route or via cannot be modified by an application such as the router. ◦ locked: The route cannot be modified.
Route Type	Indicator of how the route or via will be treated by the Fusion Compiler or IC Compiler II router: ◦ User Entered ◦ Detail Routing ◦ Global Routing ◦ Partial Parallel Route ◦ P/G Ring ◦ P/G Strap ◦ P/G Macro to IO Pin ◦ P/G Std Cell to IO Pin ◦ Clock Ring ◦ Clock Strap ◦ Clock Zero Skew Route ◦ Bus ◦ Fixed Shield ◦ Dynamic Shield ◦ Fill Track ◦ User Trunk For information about these route types, see the Fusion Compiler user documentation or the IC Compiler II user documentation. Custom Route indicates the route shape is created by the Auto Router, the Interactive Router, or the Pattern Router process. This attribute value is used by the Delete Custom Route Shape command. See Deleting Routes.
Color	The color assignment of the path segment. This attribute is available only if the object is on a colored layer.
Color Lock	The indicator that the color assignment cannot be changed. This attribute is available only if the color assignment can be edited. See Locking and Changing Colors.

4. In the Property Editor, view or edit the **Connectivity** values:

Attribute	Description
Net Name	Name of the net to which the route shape belongs.
Anchor	Indicator of whether the route shape is explicitly assigned a net. See Connectivity Anchor.
Propagating	Indicator of whether the route shape propagates connectivity. See Propagating Connectivity.
Signal Type	Type of signal that the route carries: ◦ signal ◦ power ◦ ground ◦ clock ◦ tieOff ◦ tieHi ◦ tieLo ◦ analog ◦ scan ◦ reset
Min Voltage	Net's minimum voltage value. Editing the value overrides the minimum voltage value obtained from the simulation and changes the Voltage Data Source setting to override . See Annotating Voltage Data for VDRC.
Max Voltage	Net's maximum voltage value. Editing the value overrides the maximum voltage value obtained from the simulation and changes the Voltage Data Source setting to override . See Annotating Voltage Data for VDRC.
Voltage Data Source	Source of the voltage data: ◦ unset: No voltage values are set from any source. ◦ override: The voltage values are added or modified in the layout. ◦ intent: The voltage values are added or modified in the schematic. ◦ simulation: The voltage values are obtained from simulation analysis. See Annotating Voltage Data for VDRC.

5. When you are finished editing the attributes and connectivity values, click **Apply**  on the Property Editor toolbar.

16

Deleting Routes

This chapter describes how to delete routes that were added by the Custom Router.

This chapter contains the following topics:

- [Ways You Can Delete Routes](#)
- [Before You Start](#)
- [Deleting Routes Created by the Custom Router](#)
- [Deleting Routes Created by Any Method](#)

Ways You Can Delete Routes

You can:

- Delete only the route shapes created by the Custom Router. See [Deleting Routes Created by the Custom Router](#).
- Delete all route shapes of selected nets regardless of how the routes were created. See [Deleting Routes Created by Any Method](#).

Whichever method you use, pin shapes and locked shapes cannot be deleted and will remain in the design.

To delete routes created by the Pattern Route Assistant, see [Deleting Routes in Patterns](#).

Before You Start

Make sure:

- Your design is open in the Layout Editor. See [Opening the Layout Editor and Enabling the Custom Router](#).
- (If you want to delete only the routes created by the Custom Router) The Custom Router is enabled. See [Opening the Layout Editor and Enabling the Custom Router](#).
- Plan which routes you want to remove.

Deleting Routes Created by the Custom Router

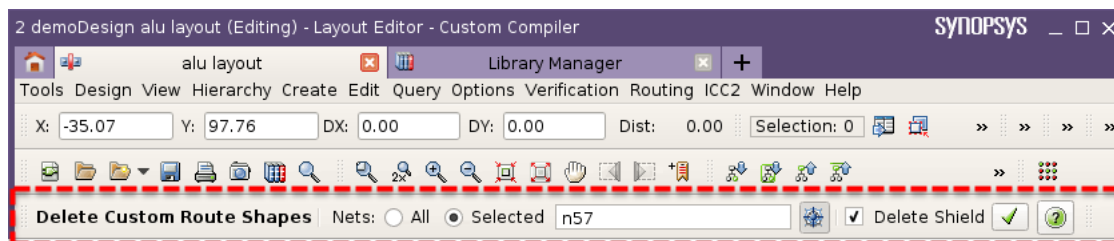
The **Route > Delete Custom Route Shapes** command removes only the routes that were created by the Auto Router, Interactive Router, and Pattern Route processes. This method facilitates reroutes and other large-scale route modifications.

During custom route deletion:

- You can remove routes on all nets or specified nets.
- Only routes that have the **normal** Route Status and **Custom Route** Route Type are removed.
- Shields surrounding the routes can also be removed.
- Via shapes, pins, and locked shapes remain in the design.

To delete route shapes created by the Custom Router:

1. In the Layout Editor, choose **Routing > Delete Custom Route Shapes**. The **Delete Custom Route Shapes** toolbar appears.



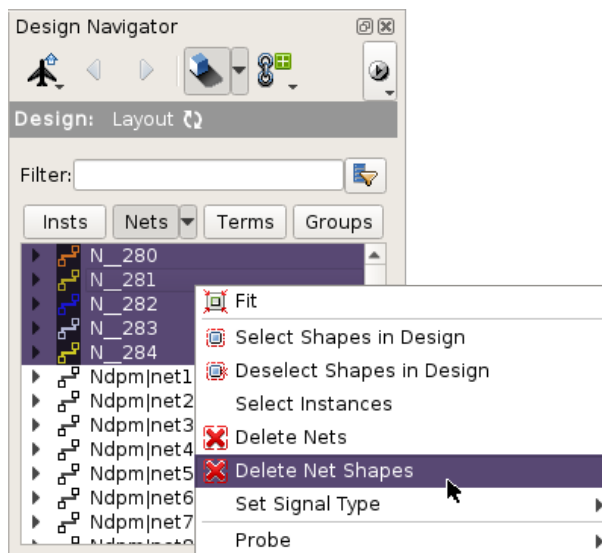
2. On the **Delete Custom Route Shapes** toolbar, select one of the following in the **Nets** area:
 - **All**: Deletes routes on all nets in the design.
 - **Selected**: Deletes routes on specified nets. You can type the net names in the box or select the nets in the Design Navigator.
3. (Optional) Select **Delete Shield** to remove any accompanying shielding. (The option is selected by default.)
4. Click .

Deleting Routes Created by Any Method

The **Delete Net Shapes** command in the Design Navigator removes routes of selected nets regardless of how the route shapes were created. Pins and locked shapes remain in the design.

To delete routes that were created by any method:

1. In the Layout Editor, display the Design Navigator.
2. In the Design Navigator:
 - a. Select the nets that you want to delete.
 - b. Right-click the selections and choose **Delete Net Shapes**.



The route shapes of the selected nets are deleted.

17

Checking Route Congestion

This chapter explains how to check route congestion in the design using the Congestion Checker.

This chapter contains the following topics:

- [Rule Filter](#)
- [About the Congestion Checker](#)
- [Starting the Congestion Checker](#)
- [Viewing the Results on the Canvas](#)
- [Viewing the Congestion Report in the Console](#)

Rule Filter

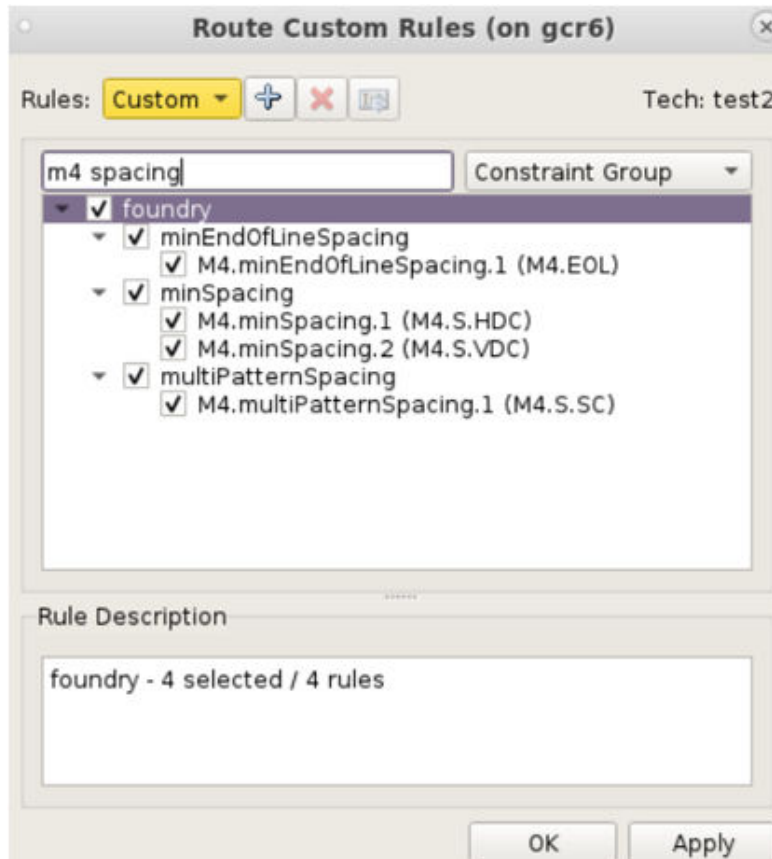
In the Rule Filter, the rules are searched through the rule filter line-edit and the search is case-insensitive. The keywords are searched in the rules tree. When there are multiple keywords specified, an AND operation is performed. The keywords are checked inside each item of the tree, including the rule ID section from the constraint.

The following keywords are added for the search:

- *lppspec*: When the first keyword is lppspec, then the search is limited to the rules applicable only inside certain region IDs. These region IDs are specified in the technology on the constraint group with an attribute lppspec. The regionrule keyword can also be used in place of lppspec.
- *failedComment*: When the first keyword is failedComment, the router shows all the constraints whose comment was not treated as rule ID.
- *noComment*: When the first keyword is noComment, all the constraints without a comment is shown.

By default, only router supported rules are displayed. The following keywords can be used to see other sets of rules. The tree-based rules box is not editable when these keywords are used.

- *allRules*: All the rules, including the supported and unsupported rules in the technology file are displayed.
- *unsupportedRules*: The rules which are not supported either by the router or by Via Param Engine are displayed.
- *allSupportedRules*: All the supported rules including the router supported rules and Via Param Engine rules are displayed.



For Example:

- To view all minSpacing rules, type minSpacing.
- To view all minSpacing rules on m0 layer, type minSpacing m0 or m0 minSpacing.
- To view rule ID M0_S_1, type m0_s_1.
- To view m0 spacing rules which are applicable only inside finboundary region ID, type lppspec m0 spacing finboundary.

About the Congestion Checker

Designing with FinFET technology has become increasingly challenging. As integrated circuits reduce in size, the component density grows dramatically. The increasing density can cause signal route congestion, thus resulting in design-rule errors and degrading performance.

You can use the Custom Compiler Congestion Checker to predict and manage the congestion in your design. Addressing congestion issues early in the process ensures that your designs will route with reduced errors and optimize performance.

This topic describes the following:

- [Congestion Calculations](#)
- [Global Routing Cells](#)
- [Analysis Results](#)

Congestion Calculations

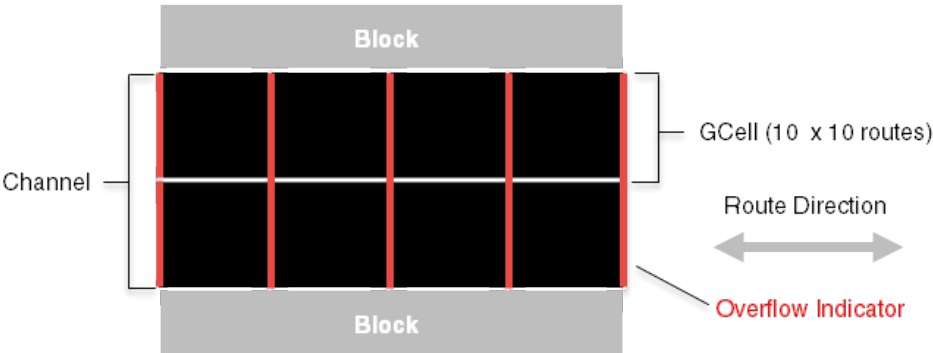
The Congestion Checker estimates congestion (or channel overflow) based on the following:

- Available space (supply in a channel)
- Number of routes that need to use the space (demand for the channel)

If the number of routes exceeds the amount of space available, the checker will report an overflow. For example, if an area in a channel can permit up to two wires (supply = 2), but three wires are placed in the channel (demand = 3), then the checker will report an overflow of 1.

Global Routing Cells

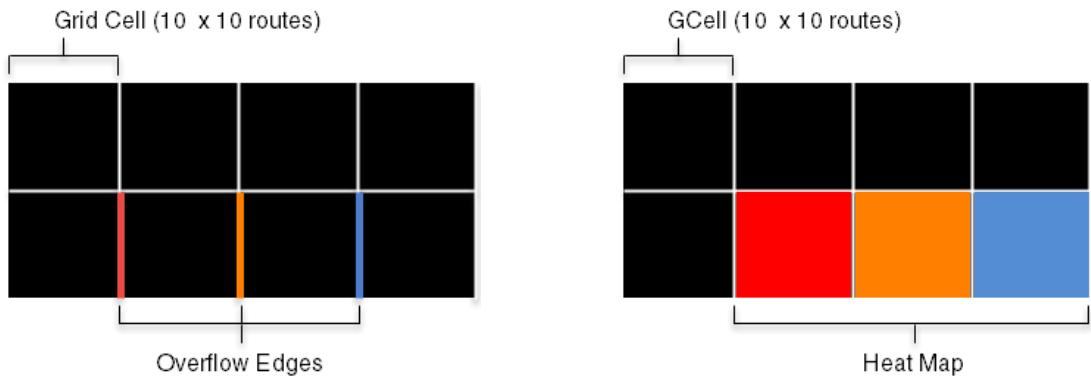
The Congestion Checker divides a design into subsections called global routing cells (GCells). During congestion analysis, the Congestion Checker calculates the overflow at the GCell edges that are perpendicular to the routes. Colored horizontal lines indicate overflow in the vertical route direction. Colored vertical lines, as shown below, indicate overflow in the horizontal route direction.



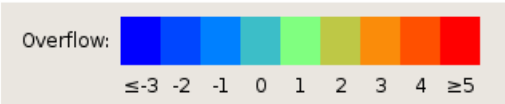
Analysis Results

The analysis results are displayed on the canvas, overlaying the design. You can choose to display the results using one of the following methods:

- **Overflow Edges:** The edges of the GCells are colored to indicate the congestion level. The colored edges are perpendicular to the direction of the routes.
- **Heat Map:** Each GCell is colored to indicate the congestion level.



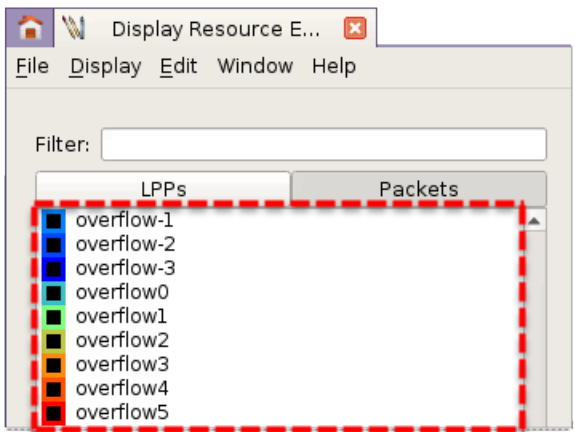
Each color is assigned an overflow value:



Overflow Value	Description
-3, -2, or -1	The channel can accommodate more wires. For example, -1 means the channel can accommodate one more wire.

Overflow Value	Description
0	The number of wires in the channel equals the number of wires permitted.
1, 2, 3, 4, or >= 5	The channel contains excess wires. For example, 3 indicates three more wires are placed in the channel than optimally permitted.

You can change the color assignments by modifying the overflow display packet. See [Creating and Editing Display Packets and Packet Aliases](#).



You can change other congestion viewing options and filter the results. See [Viewing the Results on the Canvas](#).

Starting the Congestion Checker

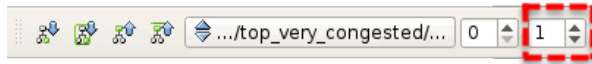
This topic describes how to set up congestion analysis using the **Check Congestion** dialog box. For information on using Tcl commands and preferences, see the [Layout Editing Tcl commands and preferences](#).

To set up the congestion analysis:

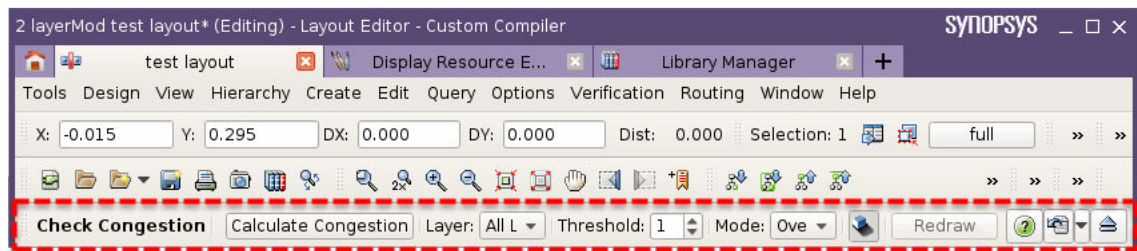
1. (Optional) Define a routing area if you want to check congestion within a specified area in the design. For instructions, see [Defining a Routing Area](#).

During analysis, the Congestion Checker will add a patterned mask over the region outside of the defined routing area to provide a visual distinction between the defined area and the rest of the design. The mask is not part of the design and is removed after the congestion checking is finished.

2. On the **Hierarchy** toolbar, set the Stop Level to 1 or greater, depending on the levels you want to include during the congestion check.



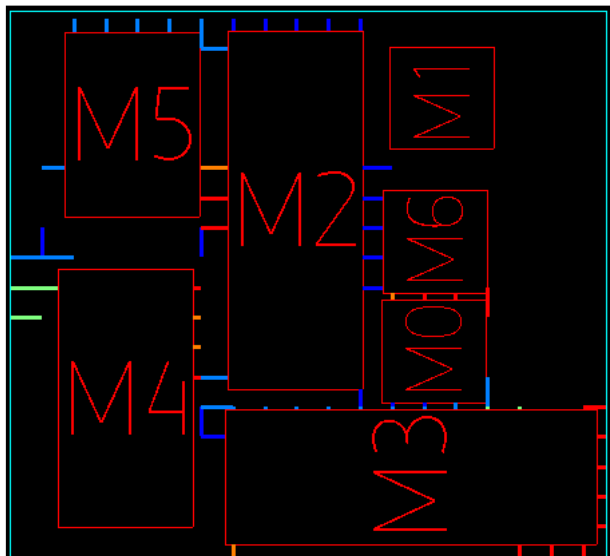
3. Choose **Routing > Check Congestion**. The **Check Congestion** toolbar appears.



4. Click **Calculate Congestion**. A progress bar appears and indicates the status of the congestion checking operation.

At any time during analysis, you can press Ctrl+C to interrupt the analysis.

When the analysis is finished, the Congestion Checker automatically displays a map of the congestion analysis results on the canvas, overlaying the design. By default, colored lines between blocks show the degree of congestion in the channels. If you defined a routing area, the Congestion checker adds a patterned mask over the region outside of the defined area. For the description of the congestion levels and the color codes, see [Analysis Results](#).



Proceed to the following:

- [Viewing the Results on the Canvas](#): The results on the canvas provides a visual overview of the congestion issues in the design.
- [Viewing the Congestion Report in the Console](#): The Congestion Report in the console presents detailed statistics of the route congestion analysis.

Viewing the Results on the Canvas

This topic describes the following:

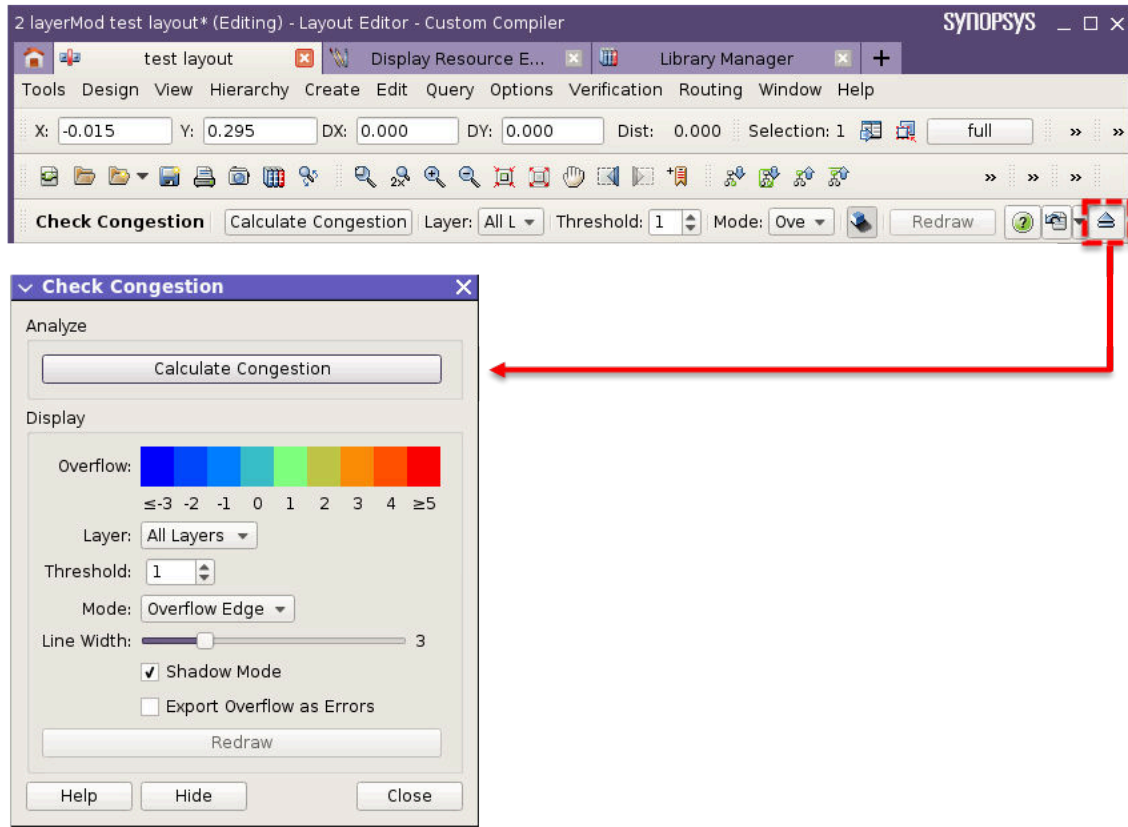
- [Ways to Modify the Display Settings](#)
- [Modifying Display Settings](#)
- [Displaying Congestion Statistics in Tooltips](#)

Ways to Modify the Display Settings

The congestion analysis results on the canvas use the default display settings. You can modify the display settings to filter the results or change how the data is displayed.

Use the options on the **Check Congestion** toolbar to modify the congestion display.

Alternatively, click **Eject**  on the toolbar and set the options in the **Check Congestion** dialog box. The dialog box contains additional settings that are not on the toolbar.



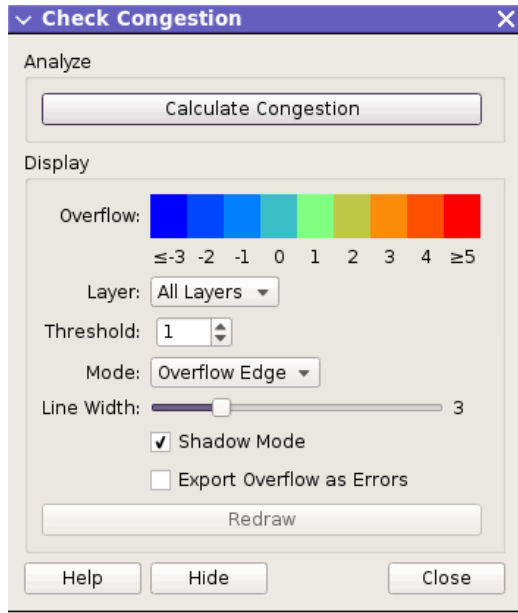
Modifying Display Settings

This topic describes how to modify the congestion data displayed on the canvas using the **Check Congestion** toolbar and dialog box. Although the instructions show the dialog box only, the same options are available on the toolbar.

To make the modifications using Tcl commands and preferences, see the Layout Editing Tclcommands and preferences.

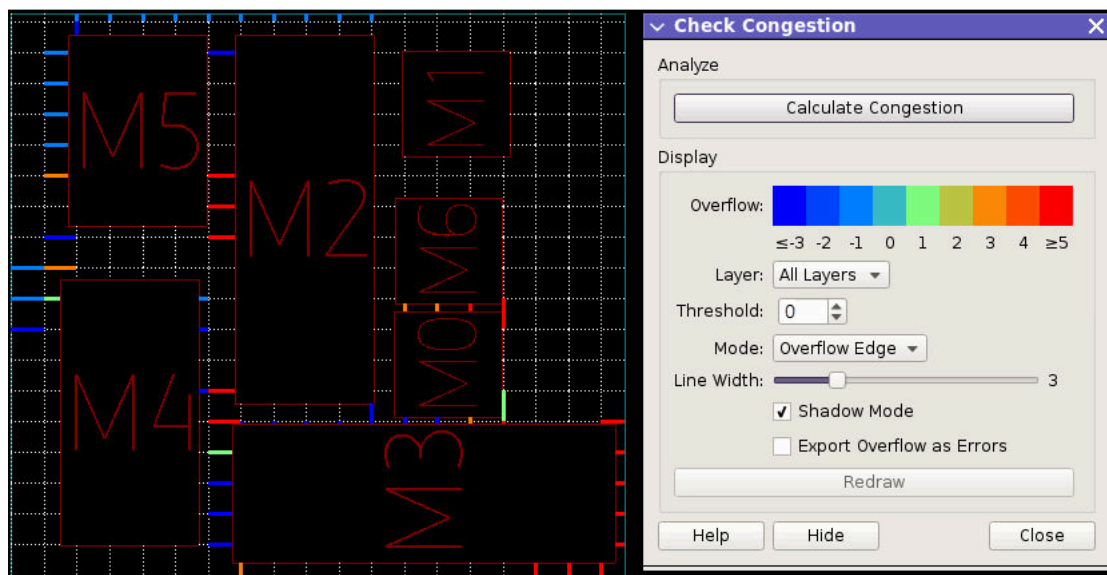
On the **Check Congestion** toolbar or in the dialog box:

1. In the **Layer** list, select the routing layer you want to view.

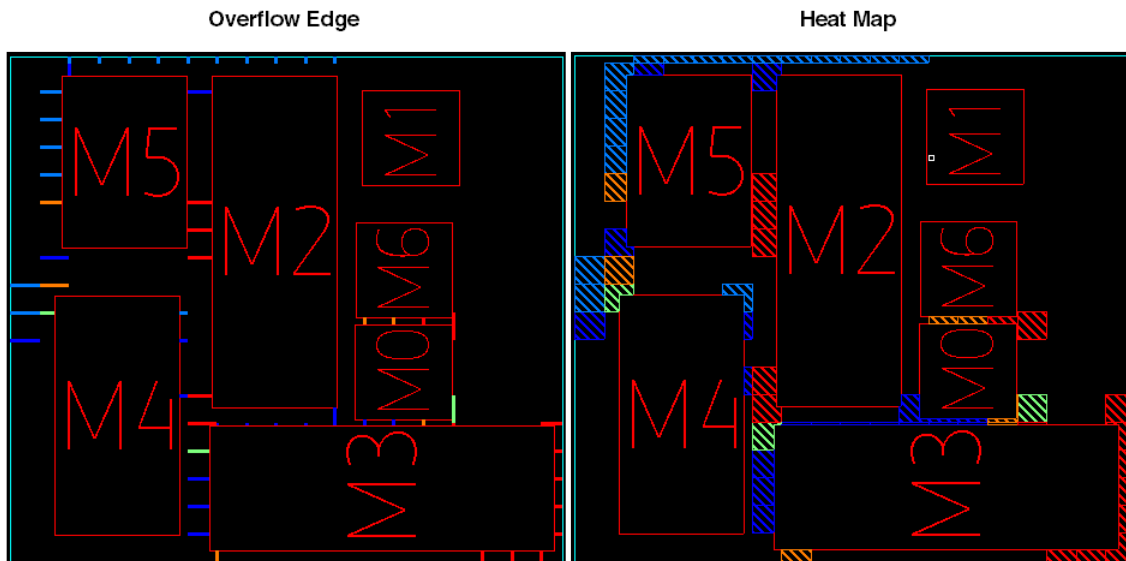


2. In the **Threshold** list, select the threshold overflow value. The Congestion Checker displays only the congestion lines whose overflow value is greater than or equal to the threshold value you specify.

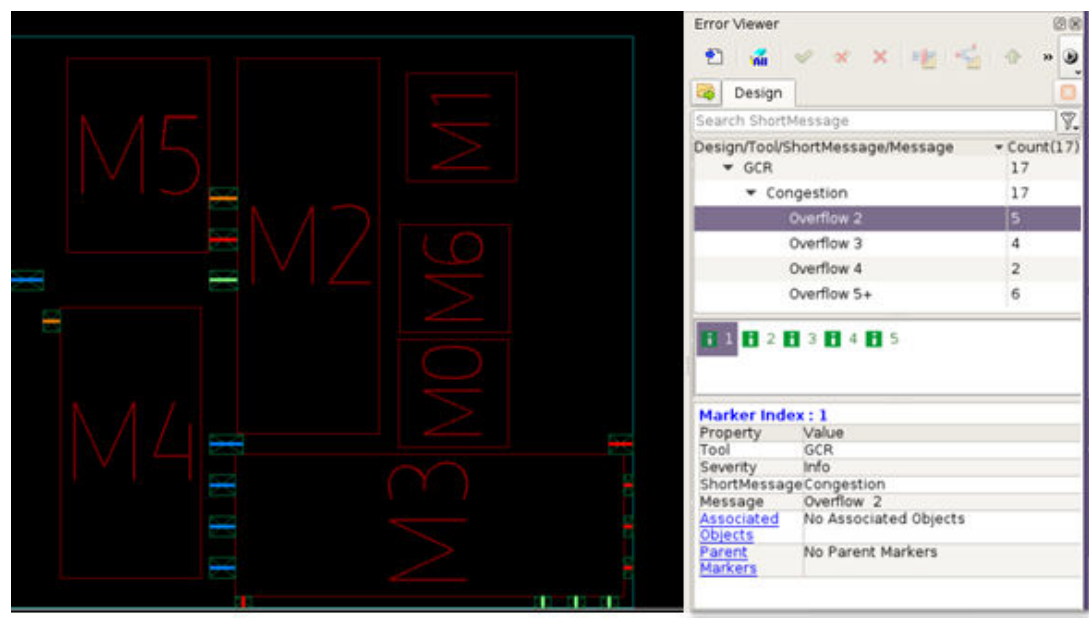
Set **Threshold** to 0 if you want to view the GCells. When the threshold is 0, you are also displaying edges with no congestion, as the following example shows.



3. In the **Mode** list, choose how you want to view the overflow information: **Overflow Edge** (default) or **Heat Map**.



4. (Dialog box only) Move the **Line Width** slider to the right to increase the width of the congestion line. If you are viewing the Heat Map, increasing the Line Width increases the width of the stipple lines.
5. Select **Shadow Mode** to increase the visual contrast between the congestion lines and the underlying design.
6. (Dialog box only) Select **Export Overflow as Errors** if you want to view and find specific overflow information using the Error Viewer. Select this option to facilitate locating congested areas in large or highly congested designs.

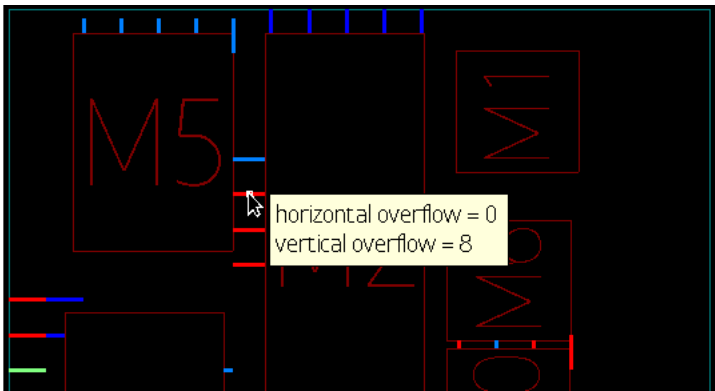


7. To apply the modified display options, click **Redraw**. The canvas updates based on your display specifications. The modified display options are also applied if you click **Calculate Congestion** when recalculating the overflow.

To resolve congested areas, widen the channel to permit more routes, and then re-run the checker.

Displaying Congestion Statistics in Tooltips

You can rest the pointer on a congestion line or heat map block. A tooltip appears and displays the number of excess wires in the channel.



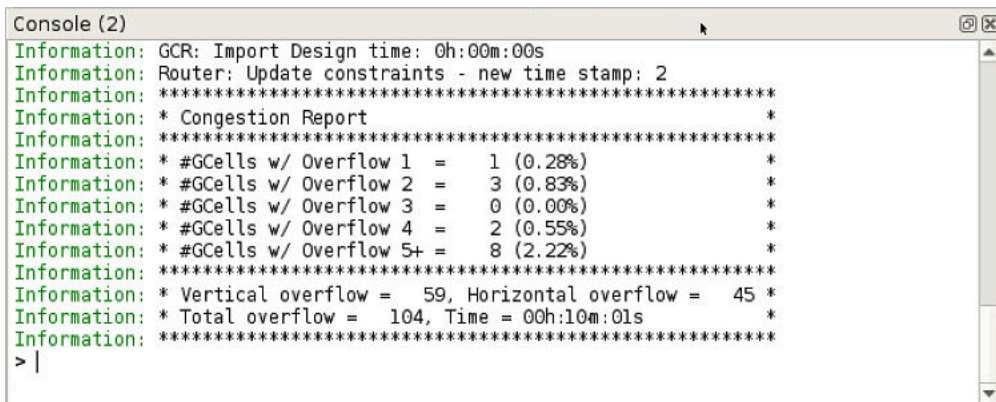
To increase the font size of the text in the tooltips:

1. In the Layout Editor, choose **Options > General**. The **General Options** dialog box opens.
2. In the **Display** section, increase the **Cursor Text Size**.
3. Click **Apply** or **OK**.

Viewing the Congestion Report in the Console

The Congestion Checker displays the Congestion Report in the Console. The Congestion Report contains the following statistics:

- *#GCells w/ Overflow n ($m\%$)*: The number of [Global Routing Cells](#) (and the percent of GCells) that have n excess wires in a channel. In the example shown, 1 GCell (which is 0.28% of GCells) has 1 more wire than optimal in the channels.
- *Vertical overflow*: The number of excess wires that are placed vertically. In the example shown, 59 excess wires are placed vertically.
- *Horizontal overflow*: The number of excess wires that are placed horizontally. In the example shown, 45 excess wires are placed horizontally.
- *Total overflow*: The total number of excess wires (Vertical overflow + Horizontal overflow). In the example shown, the total overflow is 104.
- *Time*: The length of time to check the route congestion. In the example shown, it took 10 minutes and 1 second to run the Congestion Checker.

A screenshot of a console window titled "Console (2)". The window displays the output of a congestion checker. The text is as follows:

```
Information: GCR: Import Design time: 0h:00m:00s
Information: Router: Update constraints - new time stamp: 2
Information: *****
Information: * Congestion Report *
Information: *****
Information: * #GCells w/ Overflow 1 = 1 (0.28%) *
Information: * #GCells w/ Overflow 2 = 3 (0.83%) *
Information: * #GCells w/ Overflow 3 = 0 (0.00%) *
Information: * #GCells w/ Overflow 4 = 2 (0.55%) *
Information: * #GCells w/ Overflow 5+ = 8 (2.22%) *
Information: *****
Information: * Vertical overflow = 59, Horizontal overflow = 45 *
Information: * Total overflow = 104, Time = 00h:10m:01s *
Information: *****
> |
```

18

Custom Rules

The Custom Rules feature allows you to select or deselect required router custom rules from a technology file associated with a design for routing.

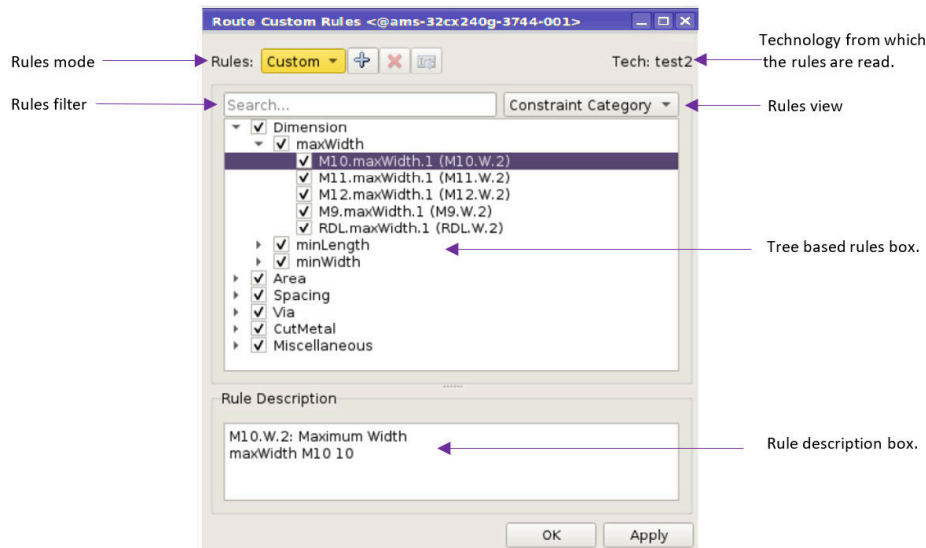
- [About Custom Rules](#)
- [Overview of Rule Selection Canvas](#)
- [Custom Rules Dialog box](#)

About Custom Rules

Router reads custom rules from a technology file associated with a design for routing. The technology file might not be optimized for a router and might impact router results. For this reason you might need to filter or skip some custom router rules from the technology file. Designers can make cautious decision to disable required custom router rules from the technology file to improve the router pattern or completion.

The Router Custom Rules feature provides the functionality to modify the required constraints. The **Route Custom Rules** dialog box lists all the constraints supported by the router from the open access technology file. All the constraints are enabled by default, and you can disable the required constraints from the technology file. This enabling or disabling of the rules applies to all router: Interactive Router, Auto Router, Pattern Router, and Topology Router. To select or deselect a constraint in the route custom rules dialog box, set the `rtEnableRules` preference to the user-defined rule set.

Figure 1 Route Custom Rules GUI



Overview of Rule Selection Canvas

The rules for a selected rules mode is listed in a tree-structure. All the constraints are the leaf items in the rule-tree display. All other levels in the tree represent rule categorization based on the rule view list.

The selection or deselection is not allowed in the tree-based rule box when the rule mode is a preset. The selection or deselection within the tree-based rule box is allowed only for the user-defined rules mode. When the rule filter is used, only those constraints or groups that match the filter criteria are shown.

For Example:

- minSpacingCG group has minSpacing constraints on m1, m2, m3, and m4 layers. When you deselect minSpacingCG, all the rules under minSpacingCG are deselected. All the minSpacing constraints on m1, m2, m3, and m4 layers are deselected. If the rule filter is specified as "m1 minspacing" and you deselect minSpacingCG, the tool deselects only minSpacing constraints on the m1 layer. minSpacing constraints on m2, m3, and m4 layers remain unaffected.

The selected rules are saved in the preference `rtEnableRulesList` as `ruleSet:name` `selectedConstraintNameList`.

When the number of deselected rules is less than the number of selected rules, the deselected rules are saved in the preference prefixed with hyphen (-), for example, `-deselectedConstraintNameList`.

Example 1 Only *minViaSpacing* rules on *v1* layer are enabled and ignore all other rules.

```
db::setPrefValue rtEnablesRuleList -value "ruleSet:custom  
v1.minViaSpacing"
```

Example 2 *minViaSpacing* rules are ignored.

```
db::setPrefValue rtEnablesRuleList -value " ruleSet:custom  
-minViaSpacing"
```

Example 3 *minViaSpacing* rules and *m1* layer *allowedSpacingRange* rules are ignored.

```
db::setPrefValue rtEnablesRuleList -value " ruleSet:custom -  
minViaSpacing m1.allowedSpacingRange"
```

Example 4 First and third *allowedSpacingRange* rule on *m1* layer are ignored.

```
db::setPrefValue rtEnablesRuleList -value " ruleSet:custom -  
m1.allowedSpacingRange.1 m1.allowedSpacingRange.3"
```

Example 5 *allowedSpacingRange* rules are ignored.

```
db::setPrefValue rtEnablesRuleList -value " ruleSet:custom -  
allowedSpacingRange"
```

Example 6 Two rulesets are created. For first ruleset *test1*, *minViaSpacing* rules and *m1* layer *allowedSpacingRange* rules are ignored. For second ruleset *test2*, *allowedEndToEndSpacing* rules are ignored.

```
db::setPrefValue rtEnablesRuleList -value "ruleSet:test1 - minViaSpacing  
m1.allowedSpacingRange ruleSet:test2 - allowedEndToEndSpacing"
```

Example 7 Two rulesets are created. For first ruleset *test1*, only metal layer *minSpacing* constraint is enabled and rest of the rules are ignored. For second ruleset *test2*, all the rules which contain width string in their name are enabled and the rest of the rules are ignored.

```
db::set::setPrefValue rtEnablesRuleList -value "ruleSet:test1  
metal.minSpacing ruleSet:test2 *width*"
```

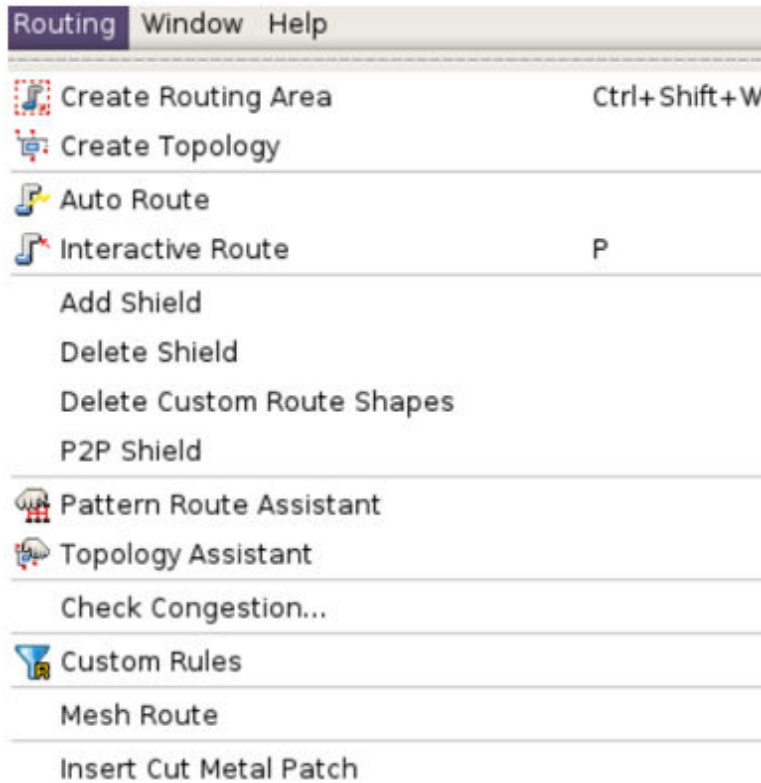
Example 8 Two rulesets are created. For first ruleset *test1*, *allowedSpacingRange* is disabled when top constraint group is foundry. For second ruleset *test2*, *minWidth* and *minSpacing* are enabled when top constraint group is foundry2.

```
db::setPrefValue rtEnablesRuleList -value " ruleSet:test1  
-foundryCG:foudry allowedSpacingRange ruleSet:test2 foundryCG:foudry2  
minWidth minSpacing"
```

Custom Rules Dialog box

The Custom Rules dialog box can open using the following methods:

- Choose **Routing > Custom Rules**.



- Open from the router Command Options Toolbar and Command Options dialog box. *Interactive Router*, *Auto Router*, `irt::checkRoute` commands lists the supported router rules. These commands include the **Rules** box and a rule filter button. Click the rule filter button to open the **Custom Rules** dialog box.



Using Rules Mode

By default, the router reads all the supported constraints from the technology file. In addition to select or deselect rules, you can apply a set of rules through a router rules mode by using the `rtEnableRules` preference. The valid values of the preference is **All** and **LVS**. The default is **All**.

Note that interactive router can create shorts.

You can add rule set by defining your own presets, See the [User-Defined Rule Modes](#) for more information.

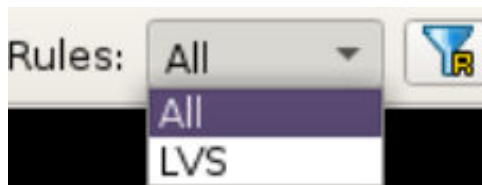
The following are the **Rules** mode for the **Custom Rules** in router:

- **All**: Reads and applies all the rules from the technology file.
- **LVS**: Reads only the shape construction rules from the technology file, and skips reading rest of the rules.

The router does not apply any spacing rules to other shapes, ignores non default rules for layer spacing, and considers all other non default rules (non default rules width and layers, and bus, matching, shield, and via constraints, and so on.). Note that no shorts are created in this mode.

Note:

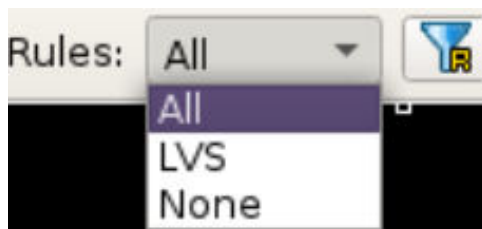
Non default rules are the design specific rules from the constraint editor.



- **None**: Reads only the shape construction rules from the technology file, and skips reading rest of the rules. This mode is supported for interactive router, pattern router, trunk generator, and mesh generator.

The router does not read any other shape rules. In this mode:

- Shorts might be created.
- Router database is not initialized for interactive router.
- Interactive router operates only on open access database.

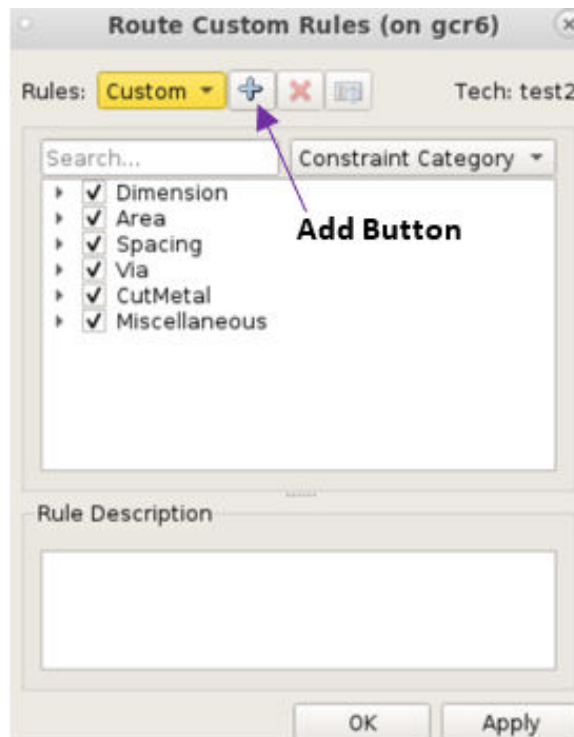


User-Defined Rule Modes

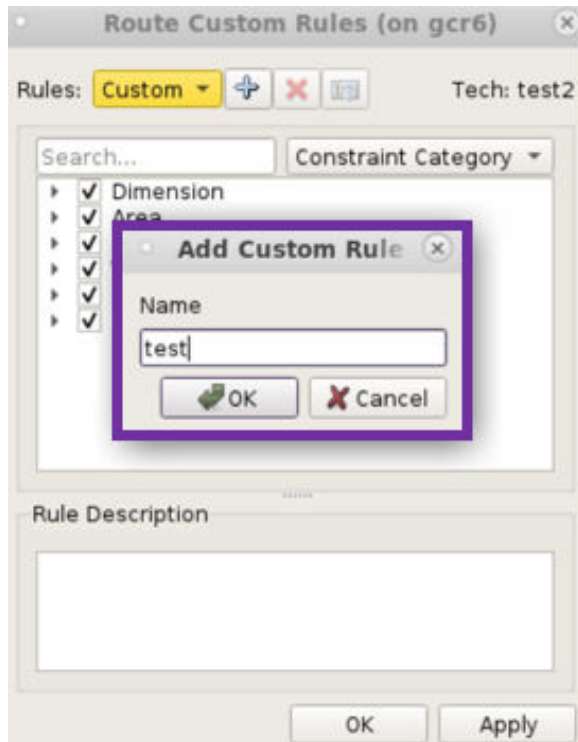
You cannot modify the existing rule mode. However, you can define new rule modes and select or deselect the user-defined rule mode, as needed. The name of user-defined rule modes cannot be the same as the presets provided by the tool.

To add the user-defined rule mode, perform the following steps:

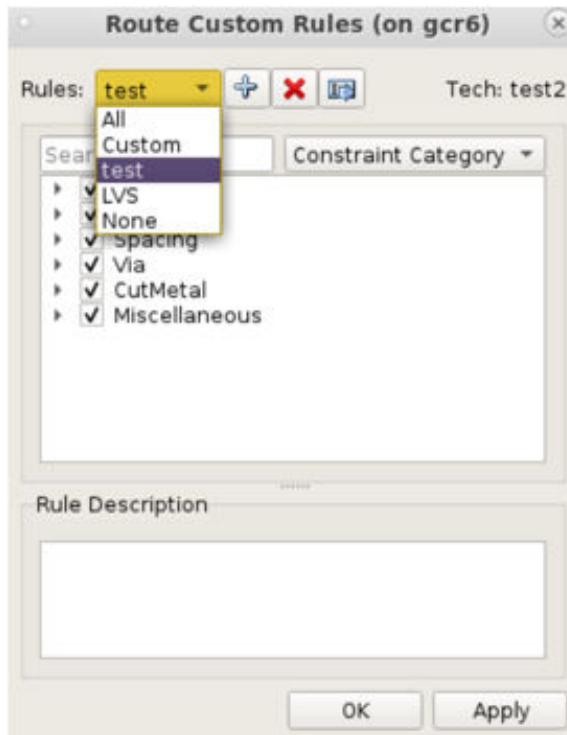
1. Click the **Add Rule Set** button () in the **Route Custom Rules** dialog box. **Add Custom Rule** pop-up appears.



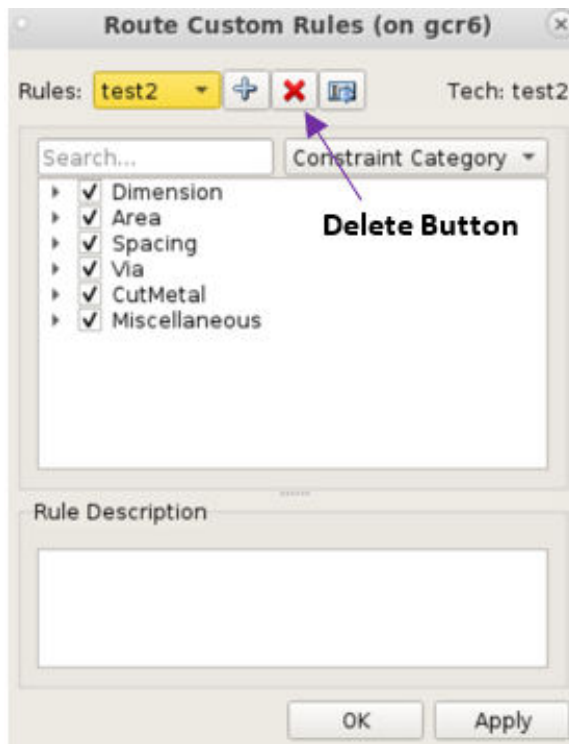
2. Type the name in the **Name** field and click **OK** to add the new user-defined rule mode.



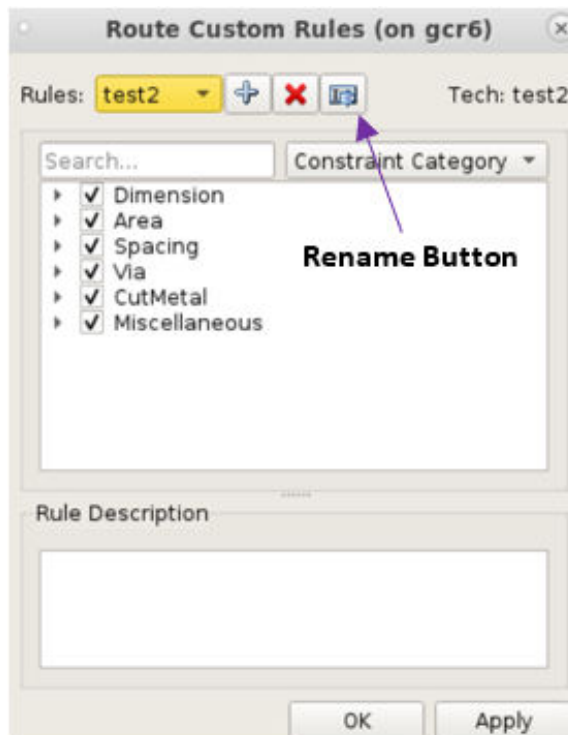
The new user-defined rule mode is created and is the active rule mode in the **Rules** list.



To delete the user-defined rule mode, select the user-defined rule from the **Rules** list and click the **Delete Rule Set** button in the **Route Custom Rules** dialog box.



To rename the user-defined rule mode, select the user-defined rule from the **Rules** list and click the **Rename Rule Set** button in the **Route Custom Rules** dialog box.



The user-defined rules mode are stored in the preference `rtEnableRulesList`. The format for `rtEnableRulesList` preference is space separated list of *ruleSet:name value*.

For Example,

Consider two rules mode: `noMinClearance` and `test`. In the `noMinClearance` rule mode, the tool ignores all `minClearance` rules. In the `test` rule mode, the tool ignores all rules except `minSpacing` and `allowedWidthRange` rules. Then, the value of the `rtEnableRulesList` preference is `ruleSet:noMinClearance -minClearance ruleSet:test minSpacing allowedWidthRange`.

In the **Route Custom Rules** dialog box (tree-based structure), right-click to use the following options to apply the settings to all rules at once:

- **Expand All**
- **Collapse All**
- **Select All**
- **Deselect All**

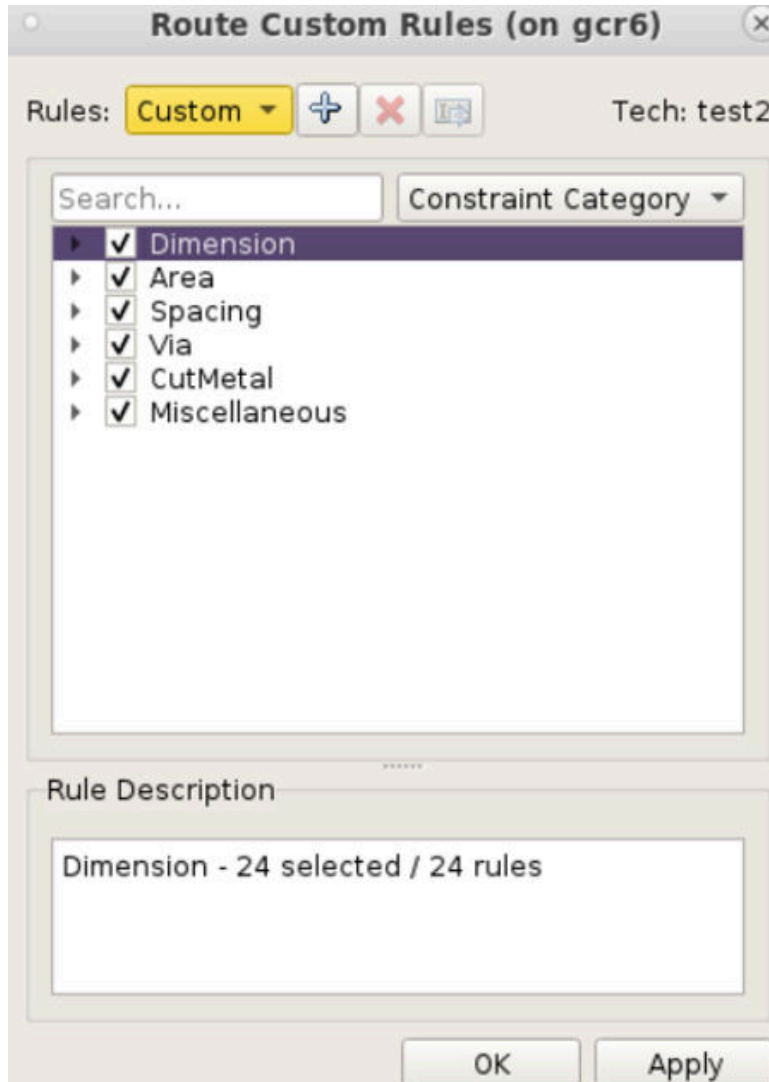
Select All and **Deselect All** options are not shown when the rules mode is a preset.

Rules View

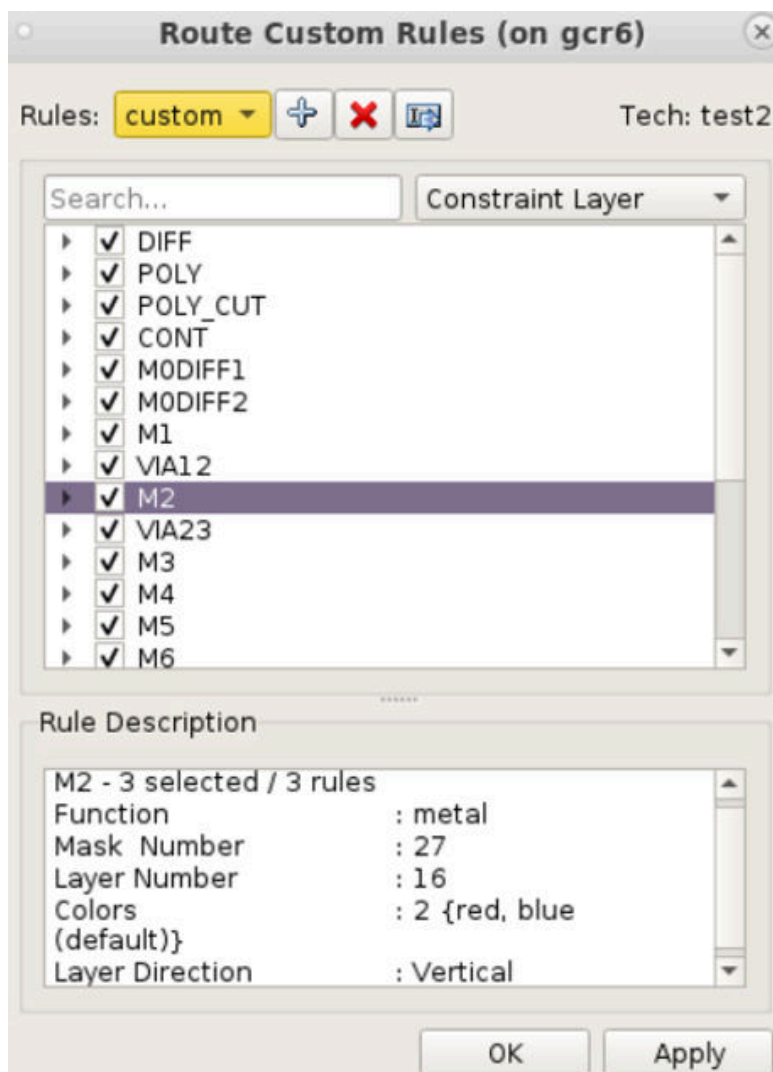
The router supported rules are read from the technology file and displayed as a tree structure in the **Route Custom Rules** dialog box. The rules name is taken from the open access constraint name. If the open access constraint comment is used and the comment is considered as DRM rule ID, the comment is shown along with the rule name. The index is appended to the rule name if multiple rules are on the same layer for a given constraint name. For example: When there are three minSpacing rules on the m1 layer, the three rules are represented as m1.minSpacing.1, m1.minSpacing.2, m1.minSpacing.3.

The rules are displayed inside the custom rule dialog box in following ways: **Constraint Category**, **Constraint Layer**, and **Constraint Group** views. The views can be selected from the view list. The view type is stored in the preference `rtEnableRulesViewType`.

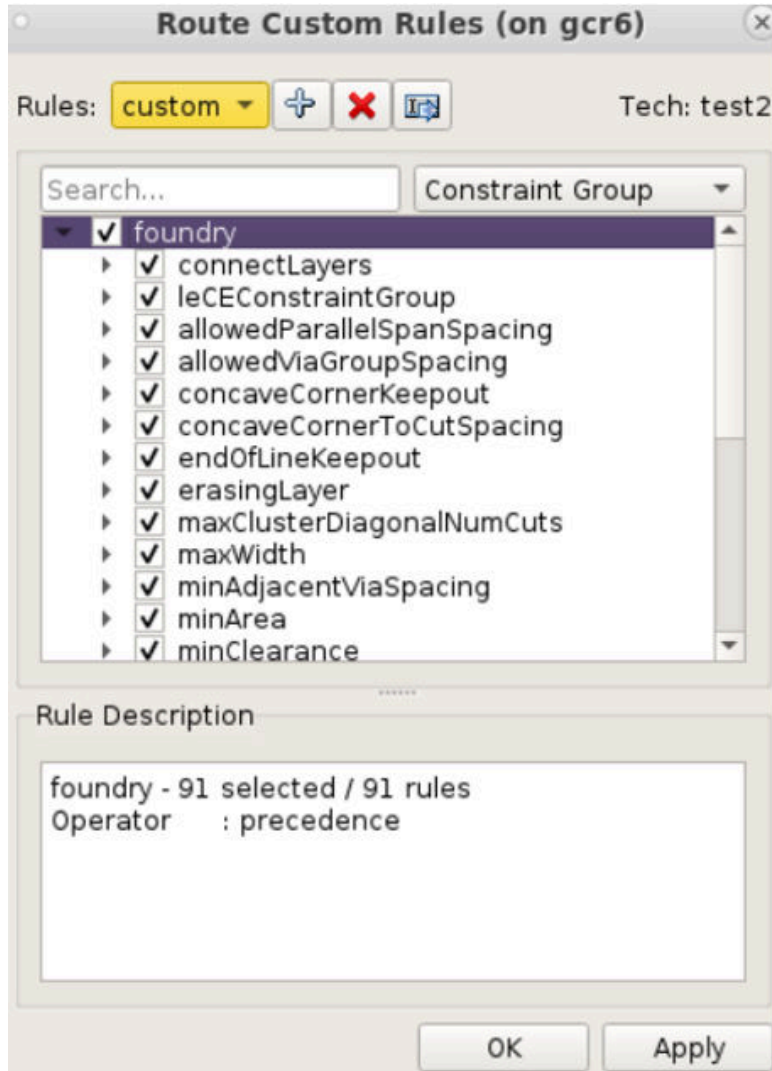
- **Constraint Category:** The constraints are grouped according to the categories of the rules. Valid categories are **Dimension**, **Area**, **Spacing**, **Via**, **Extension**, **CutMetal**, and **Miscellaneous**.



- **Constraint Layer:** The constraints are grouped according to the constraint layers. All the constraints on the same layer are grouped. Layers are displayed in increasing order of their mask number. The constraints with multiple layers (for example: minClearance) are grouped under each layer.



- **Constraint Group:** The constraints are displayed as in the technology file and follow the constraint group order in the technology file. Region specific rules are listed in a special group named *RegionRules*.



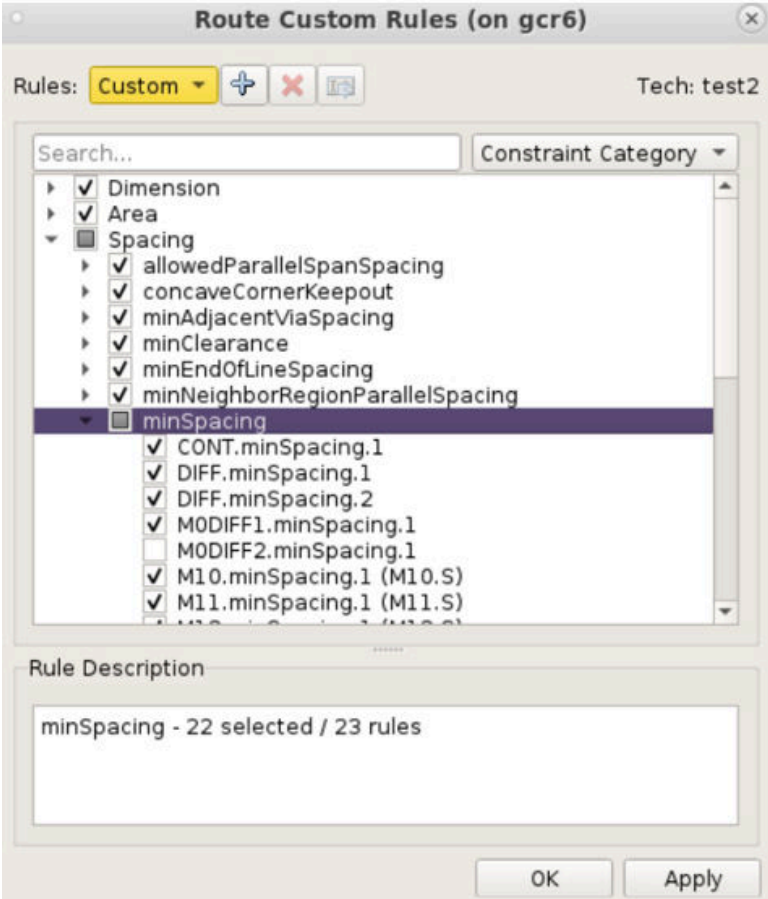
Rule Description

The **Rule Description** is available in the **Route Custom Rules** dialog box, for the selected rule. It displays the rule comment, region information, name, layer, value, and constraint parameters.

Select a rule container to display the container information. The **Rule Description** box shows how many rules are enabled and available in the selected container. In the **Constraint Layer** view, it displays layer properties such as layer function, mask

number, and number of colors. In the **Constraint Group** view, it displays constraint group properties such as constraint group operator, and LPP specifications.

Figure 2 Rule description



19

Mesh Routing

The Mesh Routing feature creates a power mesh on multiple layers and nets and generates mesh-like shapes.

This chapter describes the following topics:

- [Generating Mesh Routes](#)
- [Storing Mesh Route Creation Information](#)
- [Deleting Mesh Routes](#)

Generating Mesh Routes

The Mesh Route Generation feature uses Generate Mesh Route Tcl command to provide a GUI that allows you to generate mesh route shapes.

This topic describes the following:

- [About Mesh Route Generation](#)
- [Mesh Route Dialog Box Reference](#)
- [Recording Transaction History](#)
- [Avoiding Custom Pullback From Being Impaired](#)
- [Computing Full Design Rule and Half Design Rule](#)
- [Creating an Instance](#)
- [Creating Mesh Route With Different Intra-Group and Inter-Group Pitch](#)

About Mesh Route Generation

The Mesh Route Generation feature is an interactive command that uses Power Stripes Creation (`rt::createPowerStripes`), Power Staples Creation

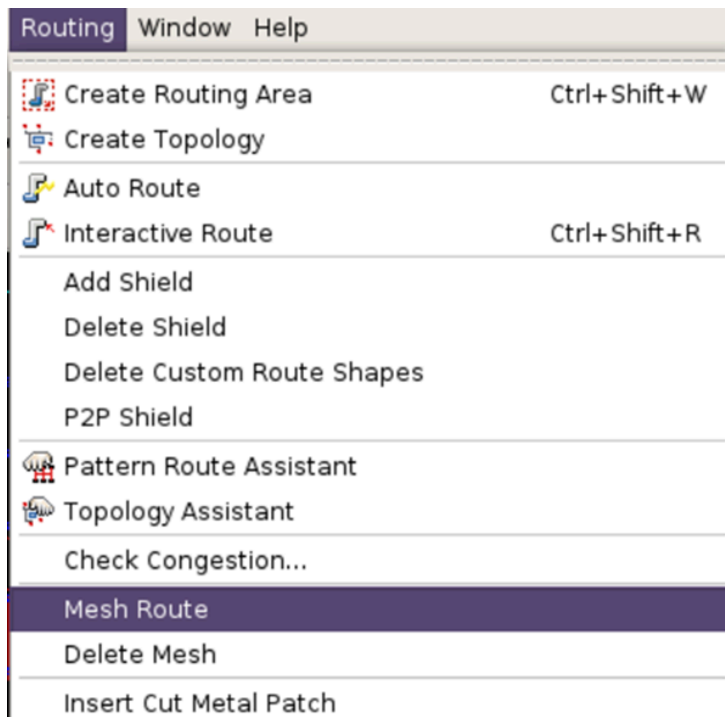
(`rt::createPowerStaples`), **Vias Insertion** (`rt::insertVias`), and **Power Hook Up** (`rt::powerHookUp`) Tcl commands to provide a GUI that achieves the following goals:

- creates power mesh on multiple layers and nets and generates mesh-like shapes
- provides an option to insert vias to connect different layers
- provides hookup on nets

You can customize the options to create power mesh with the following:

- Values such as pitch and width, or to follow the track signal type settings
- Staples or Stripes on certain layers
- Directions of the Shapes on the layer
- Signal type Nets for general usage and additional options such as removing floating shapes
- Region specification options and customized Rectangle and Polygon regions

You can access the interactive Mesh Route command from the **Routing** menu or by invoking the `irt::generateMeshRoute` command from the console.



Mesh Route Dialog Box Reference

The **Mesh Route** dialog box contains settings that determine the Mesh Route methods and specify route features that meet your design requirements.

This topic describes the following:

- [Mesh Route Dialog Box](#)

Mesh Route Dialog Box

The **Mesh Route** dialog box appears after you activate the `irt::generateMeshRoute` command. Alternatively, you can also access the dialog box from **Routing > Mesh Route** menu from toolbar.

The **Mesh Route** dialog box contains the following:

Overall Options

At the top of the **Mesh Route** dialog box, you can set the following options.

Mesh Route <@beckett>

Area: ☒ PR Boundary ☐ Rectangle ☐ Polygon ☐ Shape ☐ Track Region

Draw/Select

Stage: ☒ Mesh ☒ Insert Vias ☐ Hookup

Rules: All

Option Set:

Load From: Select Mesh

Mesh

Via

Hookup

Options

Nets:

Replace

Append

Valid	Layer	Direction	Nets	PG-Track	Width	Pitch	InterGrp Pitch	Offset	Pullback	Staple	Staple Pitch	Staple Length	Pin
☐	POLY	V		☐	0.01	0.31	0	0	0.0	☐	1.91	0.1	☐
☐	MOPOLY	Both		☐	0	0	0	0	0.0	☐	0	0	☐
☐	MODIFF1	Both		☐	0.022	0.16	0	0	0.0	☐	1.07	0.22	☐
☐	MODIFF2	V		☐	0.022	0.16	0	0	0.0	☐	1.07	0.22	☐
☐	M1	H		☐	0.022	0.23	0	0	0.0	☐	1.49	0.22	☐
☐	M2	V		☐	0.022	0.23	0	0	0.0	☐	1.49	0.22	☐
☐	M3	H		☐	0.022	0.23	0	0	0.0	☐	1.49	0.22	☐
☐	M4	V		☐	0.022	0.23	0	0	0.0	☐	1.49	0.22	☐
☐	M5	H		☐	0.044	0.46	0	0	0.0	☐	2.98	0.44	☐
☐	M6	V		☐	0.044	0.46	0	0	0.0	☐	2.98	0.44	☐
☐	M7	H		☐	0.044	0.46	0	0	0.0	☐	2.98	0.44	☐
☐	M8	V		☐	0.044	0.46	0	0	0.0	☐	2.98	0.44	☐
☐	M9	H		☐	0.5	5	0	0	0.0	☐	32.5	5	☐

Help

Hide

Defaults

OK

Apply

Cancel

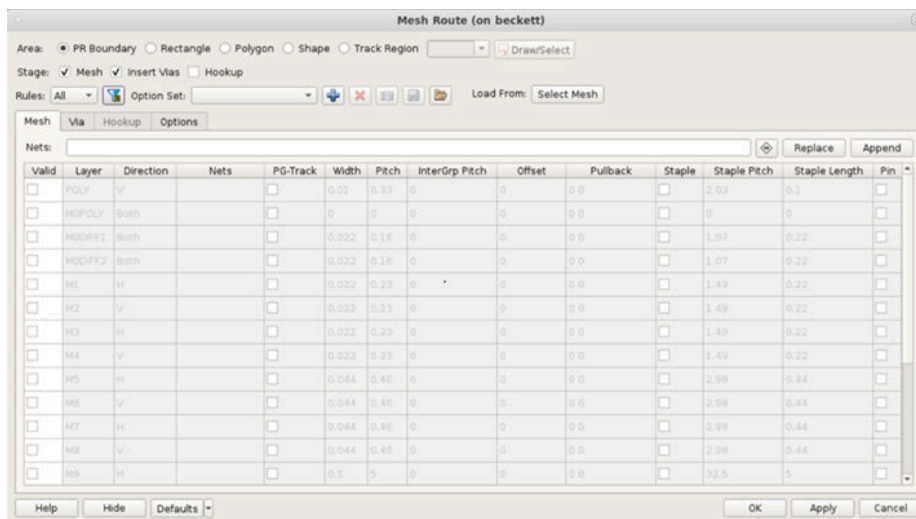
Option	Description
Area	Defines the area in the design or cell to create the power grid routing shapes or objects. • PR Boundary: Performs the command on the PR Boundary region. • Rectangle and Polygon: Performs the command on the customized and specified region that can be drawn on the Layout Editor before executing the command. Note: You can view and edit the drawn shape of these two options on the Layout Editor. When you select either one of these two modes, the Draw/Select button is enabled. You can click the button to draw the region. The Mesh Route dialog box is hidden and appears back again after the drawing operation is completed. • Track Region: Performs the command on the local track region by name. • Shape: Performs the command to select a shape or boundary representing a region to create the mesh route. You can select a shape or a boundary at the top-level. The allowed types are: ◦ Shape: Rectangle, Polygon ◦ Boundary: Alignment Boundary/ Snap Boundary / PR boundary The Draw/Select button is enabled. You can click the button to select the required shape. The Mesh Route dialog box is hidden and appears back again after the selection operation is completed.
Stage	Defines what power grid steps or stages should be performed. • Mesh: Creates the power grid paths. Runs the <code>rt::createPowerStripes</code> and / or <code>rt::createPowerStaple</code> commands. • Insert Vias: Adds Vias between the overlapping power grid paths. Runs the <code>rt::insertVias</code> command. Note: The Insert Vias stage is allowed only with the Mesh stage. If Mesh is disabled, Insert Vias is also disabled. • Hookup: calls the router to complete connections from cell input pins that are netted with power or ground to the nearest power grid mesh shapes (used for diode input pin connections). Runs the <code>rt::powerHookUp</code> command Note: You can select or deselect the stages, as required. If a stage is not selected, then the corresponding tab is disabled in the Mesh Route dialog box.

Option	Description
Rules	Allows you to control the rules that are enforced from the oaTech for better routing pattern results. • Option Set: Provides quick access to multiple presets. Click the buttons next to Option Set to create new, delete, rename, save, and load option sets. • Load From: Allows to select mesh routes created at different locations of the design.
Help	Displays the Custom Compiler help.
Hide	Hides the Create Mesh dialog box.

Mesh Tab

The **Mesh** tab allows you to create stripes or staples for the specified layers.

It is enabled when you select the **Stage: Mesh** option in the **Mesh Route** dialog box. It contains a table and an inline group for selected nets, which consists of a net input and two push buttons.



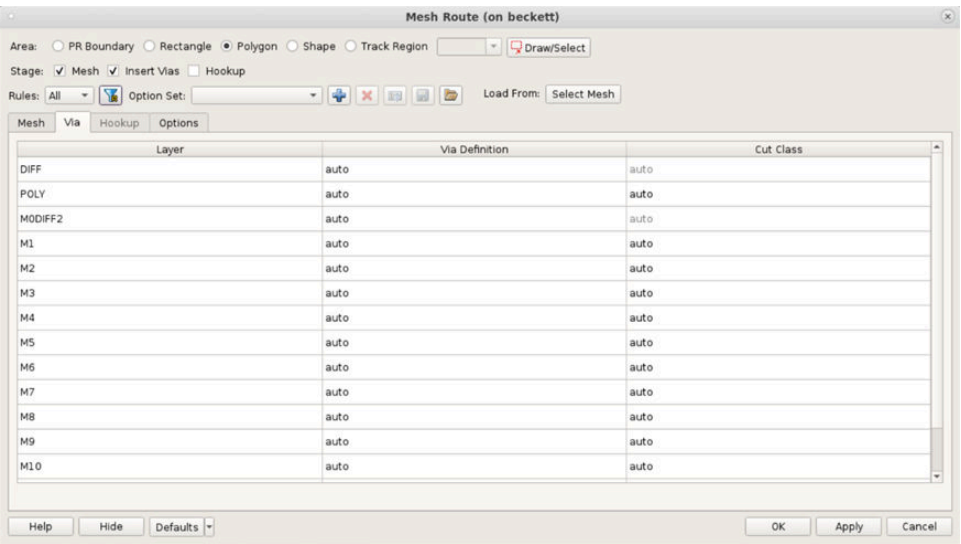
Option	Description
Nets	Allows you to input existing nets with many pattern combinations. The Design Navigator import is auto filled when the dialog is initialized. The input selection is imported from Design Navigator by clicking the  button. Note: Nets are mandatory for stripe and staple creation when Mesh stage is enabled. If no nets are provided in the Nets option under the Mesh tab, an error message appears in the Custom Compiler tool.
Import Nets	Replaces Nets (text input value) with the Design Navigator net selection. The selected scope is determined by the SDL session availability. If you open an SDL environment, the selected scope is set to Schematic, which allows both schematic and mask layout nets to be importable. Else, you open a pure mask layout and the selected scope is set to mask layout.
Replace	Replaces the data of the Nets column with the Nets input in the Net input widget.
Append	Appends the data of the Nets column with the Nets input in Net input widget.
Valid	Enables or disables the required layer.
Direction	Specifies the routing direction. Valid values are H, V, Both. The Both option means the Tcl command is performed twice on the same layer, first horizontal and then vertical.
Nets	Specifies the nets to perform mesh route on the layer. If the signal type of the input nets is “signal” then the signal type is temporarily converted into power and performs Tcl command and restores the signal type. You can also input non existing nets to the text field and is supported by the <code>-create_net</code> option when performing the Tcl command.
PG Track	Uses the power-ground tracks enabled in the design when the option is selected.
Width	Specifies the custom width value of the shapes created in the required layer. The default is <code>minWidth</code> of that layer generated from tech file.

Option	Description
Pitch	Specifies the custom pitch value of the shapes created in the required layer. The default is the five times of “min width + min spacing” of that layer generated from tech file. $\text{DefaultPitch} = [5 * (\text{minWidth} + \text{minSpacing})]$
InterGrp Pitch	Specifies the custom inter-group pitch value of the shapes created on a layer.
Offset	Specifies the custom offset value of the shapes created on a layer. A list of completion keywords to set up the offset: • HalfWidth: Half of the width value • FullWidth: width value • HalfPitch: Half of the pitch value • FullPitch: pitch value You can also input floating point values for more customizable offset options. Note: The width and pitch values are fetched from the Width and Pitch columns from the table.
Pullback	Specifies the pullback value for the shapes on a layer. The format is either one or two floating point numbers. You can also specify the following keywords: • FullDR • HalfDR • FullDR HalfDR • HalfDR FullDR
Staple	Creates power stripes when Staple is deselected and creates power staples when Staple is selected.
Staple Pitch	Specifies the pitch value of the staples created on a layer. $\text{DefaultStaplePitch} = [5 * (\text{DefaultPitch} + \text{minSpacing}) + \text{DefaultStapleLength}]$
Staple Length	Specifies the length value of the staples created on a layer. $\text{DefaultStapleLength} = (10 * \text{minWidth})$
Note: The Staple Pitch and the Staple Length options can be set only if Staple is selected.	
Pin	Create pins with the mesh stripes or staples as pin shapes.

Via Tab

The **Via** tab inserts vias to connect different layers based on the **Nets** specified in the **Mesh** tab.

It is enabled when you select the **Stage: Insert Vias** option in the **Mesh Route** dialog box.



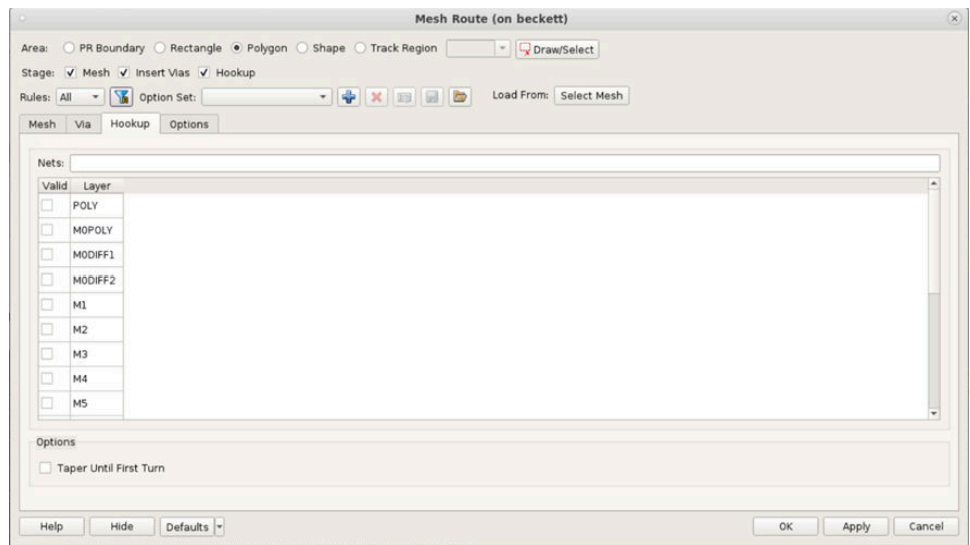
To have a consistent setting across all the routing operations, the Via information is pre-filled and synced with that information from Auto Router when the GUI is opened. After the GUI is closed, any changes done to the information are written back to the information used by Auto Router as well.

Option	Description
Layer	Specifies the lower layer of the Via. A standard Via has three layers: • lower(bottom) layer • cut layer • upper(top) layer
Via Definition	Specifies a name referring to a definition describing some information of the Via such as two physical layers representing the bottom (layer1) and top (layer2) of a Via, a set of values encapsulated by a <code>ViaParam</code> object that defines the defaults for any standard Via, and so on.
Cut Class	Specifies a name referring to an object describing the size of the cuts of Via such as cut width, cut length, and others.

Hookup Tab

The **Hookup** tab performs a power hookup on the specified nets and layers.

It is enabled when you select the **Stage: Hookup** option in the **Mesh Route** dialog box.



The specified nets follow the following order:

- If the **Nets** textbox is specified, then the list is used when performing hookup.
- If the **Nets** textbox is not specified, then the **Nets** under the **Mesh** tab along with the corresponding layers enabled in both **Mesh** and **Hookup** tabs are used.

Note:

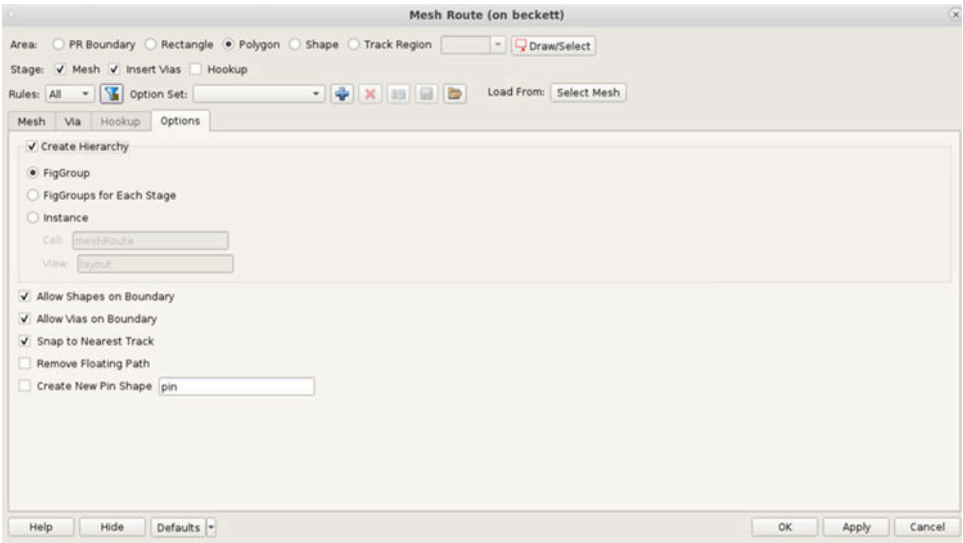
In this case the **Stage: Mesh** also needs to be enabled.

- If none of the nets are specified the command errors out.

Option	Description
Nets	Allows you to enter specific nets for hookup. Note: If no Nets are specified, then the Nets specified for selected layers in the Mesh tab are used for hookup.
Valid	Enables or disables the required layer.
Layer	Specifies the layers that are used to perform a hookup.

Options Tab

The **Options** tab includes several settings for mesh route such as different hierarchy options (no figGroup, one figGroup, figGroup for each stage and instance), and the location for Via and Shapes on boundary.

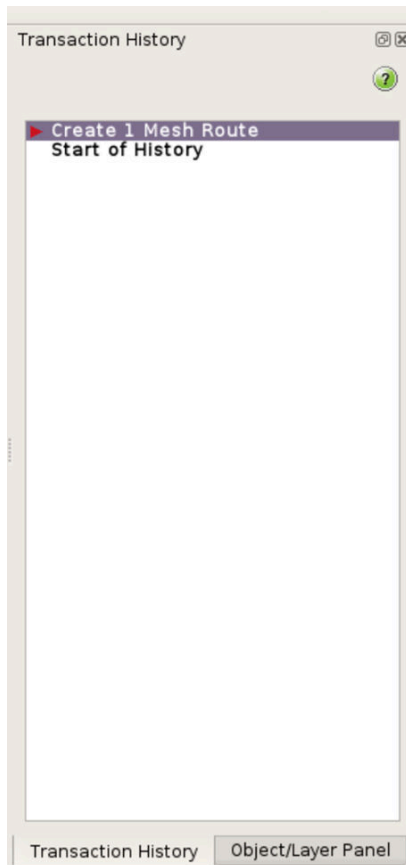


Option	Description
Create Hierarchy	Controls whether to collect the shapes for a specific purpose of grouping or not. If the option is not selected, then the mesh route results are created as individual shapes. By default, the option is selected.
FigGroup	Specifies the mesh route results under a single FigGroup called <code>MeshRoute_<Number></code> .
FigGroups for Each Stage	Specifies the mesh route results created with a figGroup for each stage called <code>MeshRoute_<Number>_<Stage Name></code> , and the figGroups for each stage are created under a single figGroup called <code>MeshRoute_<Number></code> .
Instance	Specifies the mesh route that is created with a user specified cellview and an instance called <code>MeshRoute_Inst_<Number></code> . You can specify the Cell and View prefix name for the cellview. The default for Cell is <code>meshRoute</code> and View is <code>layout</code> .
Note: FigGroup is created only if the shapes are created during execution.	
Allow Shapes on Boundary	Allows to create shapes(stripes or staples) if the centerline is on or inside a <code>PrBoundary</code> or a region. By default, the option is selected.

Option	Description
Allow Vias on Boundary	Controls the via creation on the boundary of the working area. For the Via that is close to multiple edges of the boundary, the condition on every nearby boundary edge needs to be satisfied. • ON: Allows to have Vias with the center on or within the boundary. The center is the center of the merged box of every layer of the Via. • OFF: Allows to have Vias whose bounding box is completely inside the boundary. The bounding box of a Via is the merged box of every layer of the Via. By default, the option is selected.
Snap to Nearest Track	Snaps to the closest track, when the option is selected. By default, the option is selected.
Remove Floating Path	Removes floating shapes that are not connected, when the option is selected. The option is not available when the Stage: Via option is not selected.
Create New Pin Shape	Allows to create the mesh stripes or staples as pin shapes or create extra pin shapes overlapping stripes or staples using the specified purpose names when the option is selected. You can specify one or more purposes, separated by space, in the text input. The first valid LPP is used to create the extra pin shapes, the first valid (layer_of_stripe_or_staple, purpose_specified_in_the_text_input). By default, the option is not selected. Note: • The stripes or staples in one mesh creation is processed together. One mesh creation leads to 1 or more new pins. • The stripes or staples of the same net is put into the same pin. • The stripes or staples can be on different layers. • The pin is added to the first term of the net. A term is created if there is no term. • The pin signal type is the <code>net_sigType</code> and the pin placement status is <code>oa::oacPlacedPlacementStatus</code> • A new pin is created for each mesh route when you create mesh routes many times in different location of the design. • The Vias are not in the pin and Hookup is not considered in this process.

Recording Transaction History

A transaction called **Create1 Mesh Route** is created if any Stripes or Vias are created upon execution of the `irt::generateMeshRoute` command. If none of the shapes are created during the execution of the command, the current transaction is canceled and no new transaction is created.



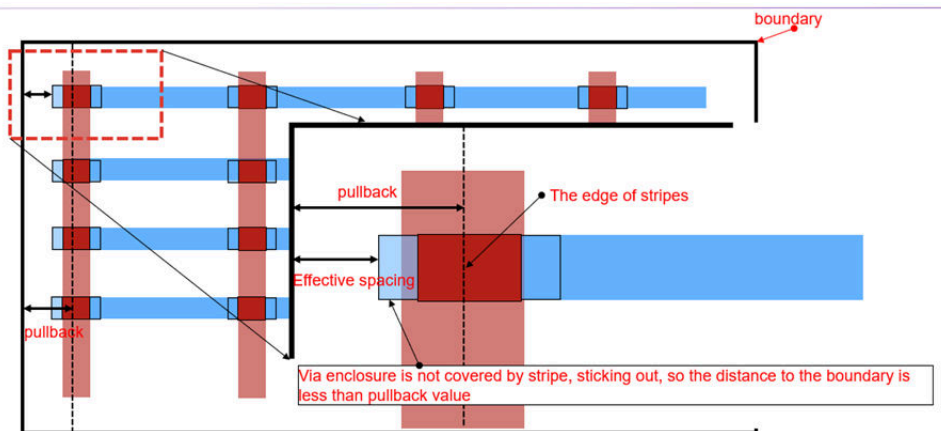
Avoiding Custom Pullback From Being Impaired

Custom Pullback value is the spacing reserved to avoid end-to-end spacing violation for the stripes in the later stages.

The upper or the lower layer of the Via around the ends of the stripes is not covered by stripes in the stripe direction. The distance from the boundary to the Via layer in the direction is shorter than the specified pullback value. This impairs the intention of pullback. It is not a DRC violation, but can cause violation in the later stages.

Note:

The stripe direction is not always the direction of the preferred layer. Instead, it is the direction of pullback.



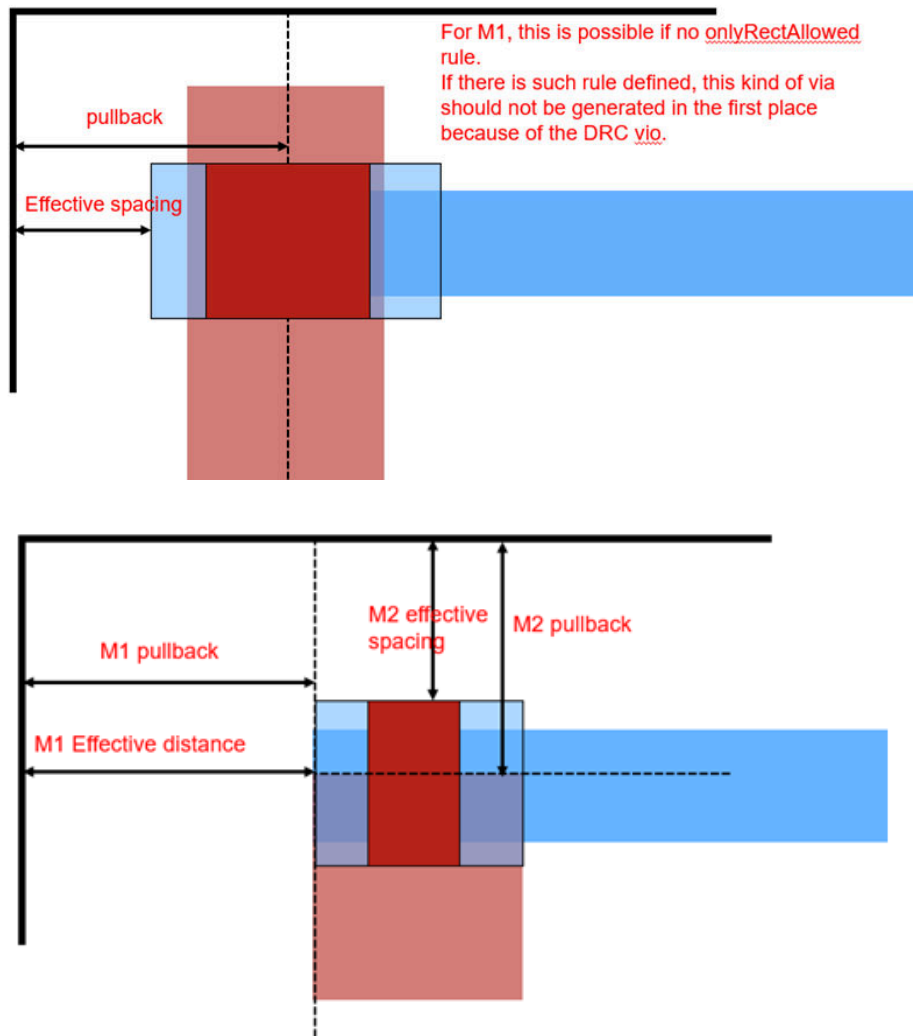
Mesh Route does not allow Vias that impair the pullback to be generated when **Ignore DRC** is **OFF**.

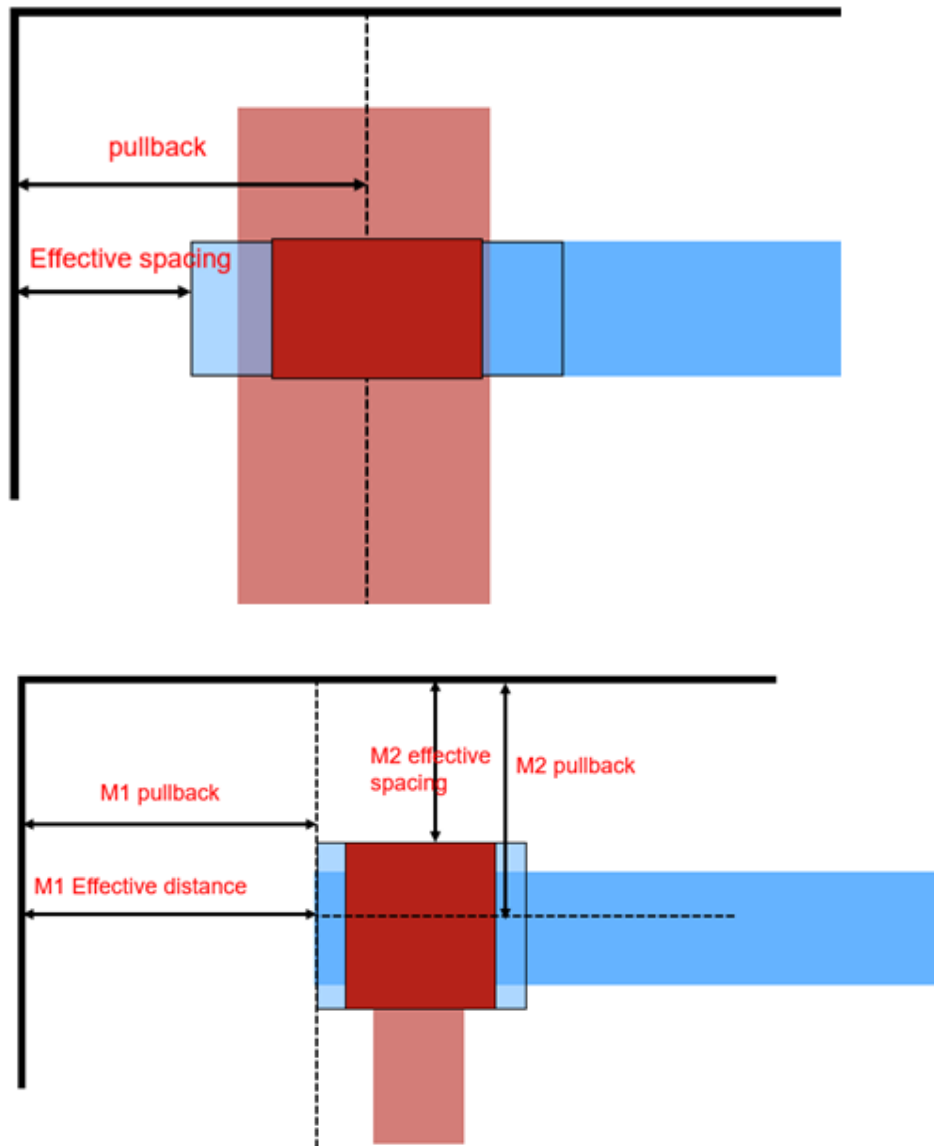
The following section describes the behavior:

- Apply a check to avoid that the intention of pullback to be impaired. If the pullback of one layer is 0, then this check is not applied.
- Check the same Via layer (upper or lower) in the stripe direction as the stripe layer. If shapes of the Via layer impairs the pullback, the tool does not allow the Via to be generated.

This check has higher priority than **Allow Vias On Boundary**. This means a Via overlapping the boundary, with **Allow Vias On Boundary** ON, is not allowed if it impairs pullback.

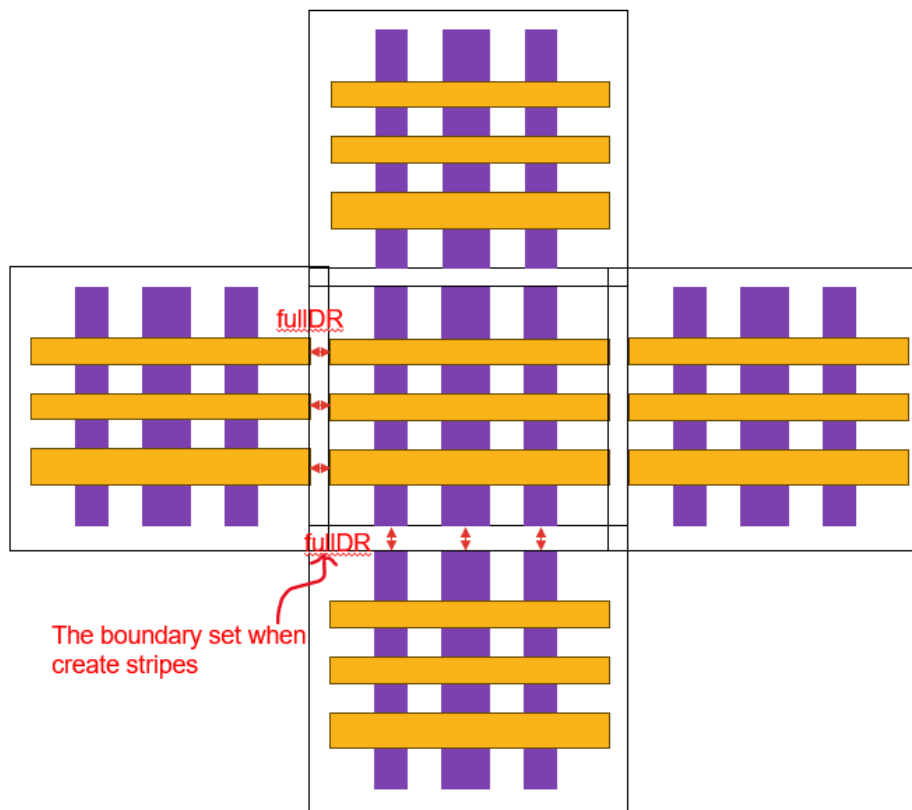
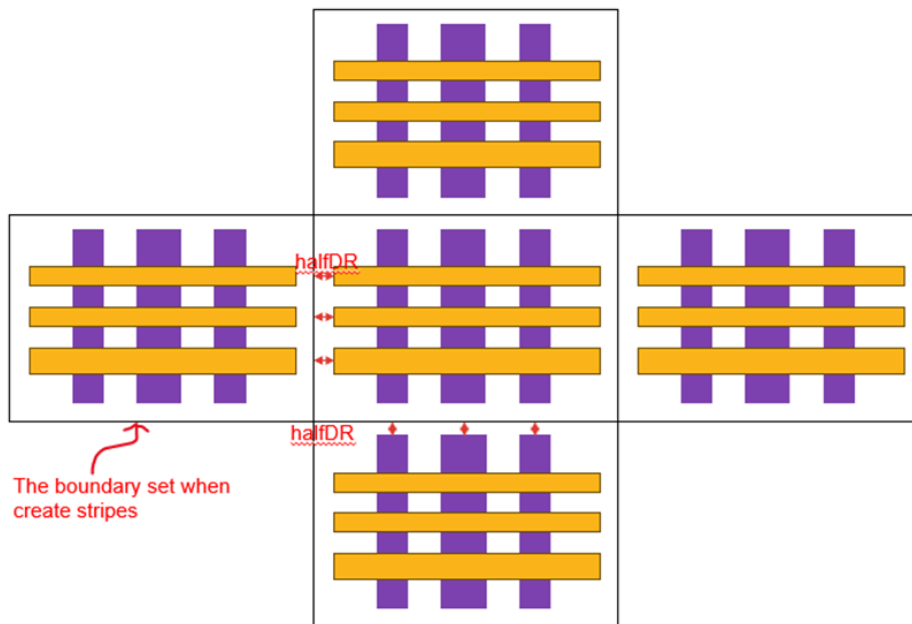
Examples: The following Vias are forbidden where Blue = M1 and Red = M2.





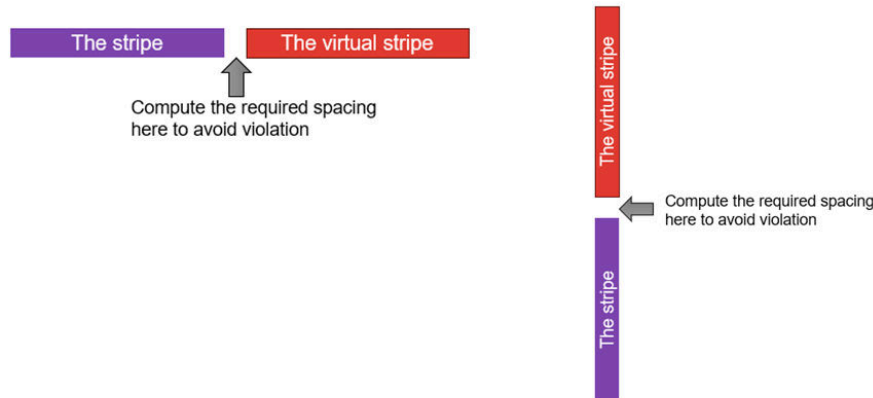
Computing Full Design Rule and Half Design Rule

Full Design Rule (FullDR) or Half Design Rule (HalfDR) is a spacing reserved for stripes to avoid violation caused by ends of stripes when two same mesh routes are abutted. Router DRC checker is used to compute the value, leveraging the router supported mechanism.



For every stripe, the value is computed as follows:

- Place a stripe and a virtual stripe beside one another.
- Compute the required spacing, based on rules/mechanisms that the router supports. Other existing shapes are ignored.



FullDR is the maximum spacing computed for each stripe.

$$\text{HalfDR} = \text{FullDR}/2$$

For one mesh route creation, there is one FullDR/HalfDR value for `pullbackLow` and one for `pullbackHigh`.

If a floating point number is provided as input, then both `pullbackLow` and `pullbackHigh` are set to the specified value. If two floating point numbers are provided as input, then the first floating point represents `pullbackLow` and the second floating point represents `pullbackHigh`.

For example:

- Pullback 2.5 means `pullbackLow` is 2.5 and `pullbackHigh` is 2.5.
- Pullback -1.5 3.6 means `pullbackLow` is -1.5 and `pullbackHigh` is 3.6.

Creating an Instance

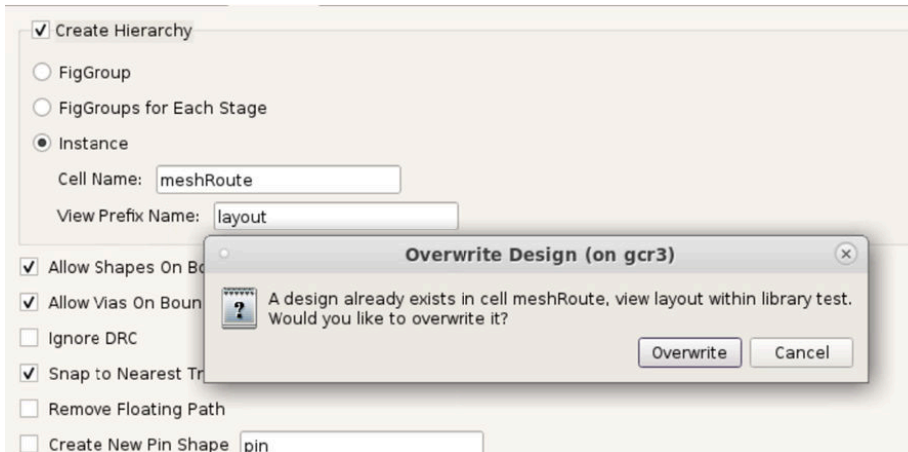
The instance creation feature converts the mesh route shapes into a cellview and creates an instance inside the current design.

To create an instance:

1. Choose **Mesh Route > Options > Instance**.
2. Specify the **Cell Name** and the **View Prefix Name**.

3. Click **Overwrite** to confirm overwriting the cellview, if the cellview exists in the library.
4. Click **Cancel** to rename the cellview.

The **Option** tab remains active.



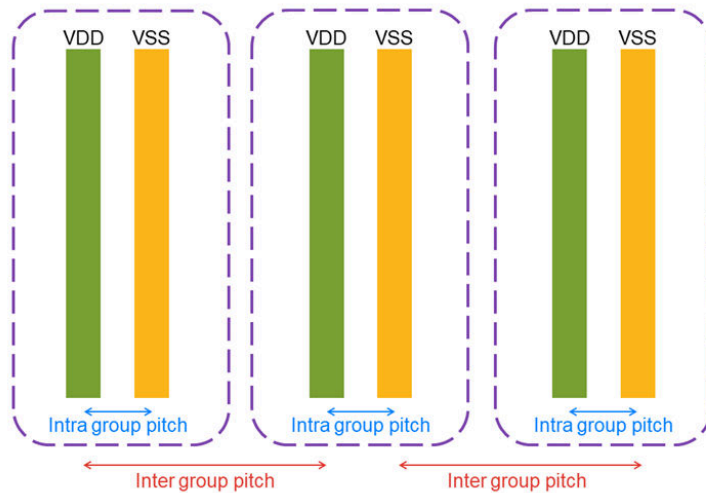
Creating Mesh Route With Different Intra-Group and Inter-Group Pitch

In the **Mesh** tab of the **Mesh Route** dialog box, there are two columns to indicate the pitch specification namely, **Pitch** (intra-group pitch) and **InterGrp Pitch**(inter-group pitch).

InterGrp Pitch is set to 0, then the entire creation is treated as one group, and only **Pitch** (intra-group pitch) impacts the pitch value.

InterGrp Pitch is set to a positive number and more than one net is available, then the entire creation is divided into multiple groups where the distance between each group is defined by **InterGrp Pitch**. Inside every group, the net pattern specified in the **Nets** column is separated by the distance defined by **Pitch** (intra-group pitch).

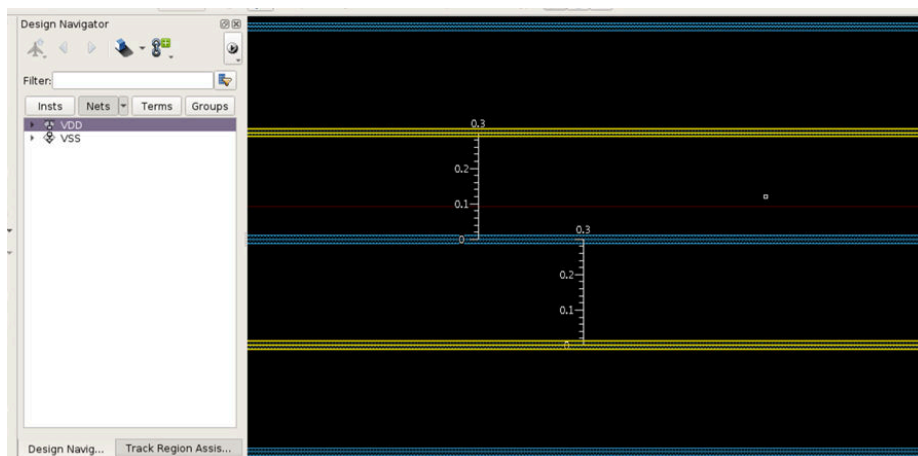
The feature to specify different pitch is available only in the Mesh Route interactive command. A preprocessing is done by calculating the corresponding offset and pitch value and decompose the net pattern when executing Power Routing Tcl command.



The following is an example of the execution Tcl command:

When inter group pitch is not specified:

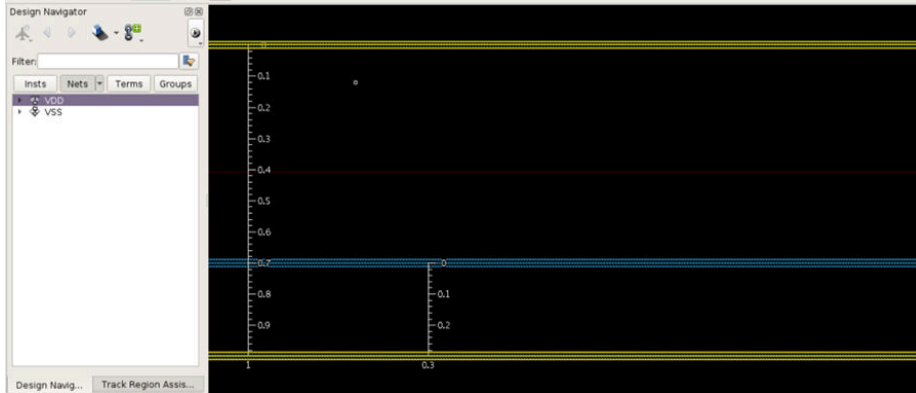
```
rt::createPowerStripes -design oa:0x7fffc72b541a -nets {VDD VSS} -layer
M1 -create_net -width 0.06 -pitch 0.3 -offset 0 -keepBoundary true
-pullBackLow 0 -pullBackHigh 0 -regionPoints {{7.269 30.449} {7.269
16.402} {23.457 16.402} {23.457 30.449} {7.269 30.449}} -rules all
-orient horizontal
```



When inter group pitch is specified:

```
rt::createPowerStripes -design oa:0x7fffc72b541a -nets {VDD} -layer
M1 -create_net -width 0.06 -pitch 1 -offset 0 -keepBoundary true
-pullBackLow 0 -pullBackHigh 0 -regionPoints {{7.269 30.449} {7.269
16.402} {23.457 16.402} {23.457 30.449} {7.269 30.449}} -rules all
-orient horizontal
```

```
rt::createPowerStripes -design oa:0x7fffc72b541a -nets {VSS} -layer  
M1 -create_net -width 0.06 -pitch 1 -offset 0.3 -keepBoundary true  
-pullBackLow 0 -pullBackHigh 0 -regionPoints {{7.269 30.449} {7.269  
16.402} {23.457 16.402} {23.457 30.449} {7.269 30.449}} -rules all  
-orient horizontal
```



Storing Mesh Route Creation Information

The Mesh Route Creation Information contains information about the settings that are used to create a mesh route.

This topic describes the following:

- [About Mesh Route Creation Information](#)

About Mesh Route Creation Information

During mesh route creation, you enter settings such as the layers used to create, nets, pitch, width, offset, stripe or staple for each layer. All these settings combined together is the mesh route creation information. Using this information, you can re-create the mesh route in the same design.

The Mesh Route Generation interactive command (`itr::generateMeshRoute`) stores the mesh route creation information every time a mesh route is created.

Mesh Route Creation Information Storage

The mesh route creation information that allows you to re-create a mesh route is stored in the design along with some additional information that can assist in a more complicated flow.

It is not possible to store and list all settings because in addition there are new settings that also needs to be stored. However, the guideline is that the information should allow you to re-create the mesh route in the same design.

Here is the basic information list that is stored in the design as part of mesh route creation information:

- Layer setting(every layer): Layer, Direction, Nets, PG-Track, Width, Pitch, InterGroupPitch, Offset, Pullback, Staple, StaplePitch, StapleLength, Pin
- Via setting: Value of `rtLayerViaDef`, `rtLayerCutClass`
- Hookup setting: Nets applied to hookup, the selected layers, the value of `rtTaperUntilFirstTurn`
- Option setting: All settings in the options tab, except the cell/view name for creating instance and the value of `rtEnableRules`
- Area setting: User specified region (PR boundary, drawn boundary, Shape), PR boundary(or design bbox if no PR boundary) / Routing window, local track region name and boundary if mesh route is created by track region mode

Structure of Mesh Route Creation Information

The mesh route objects created every time are considered as one mesh route. It includes stripes, staples, vias, pins, extra pin shapes and not `figGroup`. The mesh route objects are all put into one top-level `oaGroup` and does not depend on the `figGroup`.

Every `oaGroup` has a unique name: `synopsysCustomMeshGroup+ "_" + Serial_Number`

The `itr::generateMeshRoute` command attempts 2000 times to find a unique serial number to create the `oaGroup`. If it is unable to find, it errors out.

For example:

If you choose to create an instance, the instance is put into the `oaGroup` and the mesh route objects in the master are put into an `oaGroup`. In this case, the same mesh route information is stored on the `oaGroup` at 0x and 1x.

Then the mesh route information is stored in the `oaGroup` and the property name is `synopsysCustomMeshInfo`.

GUI changes

In the **Mesh Route** dialog box, the **Load From** button enables to retrieve the saved mesh route information during the following scenarios:

- Use mesh route information of several mesh routes created at different locations of the design that is saved to options set, in other designs.
- Create a new mesh route elsewhere based on the structure of a mesh route that is created at a specific location.
- View the setting of a created mesh route.

Note:

The following UI elements are not populated with the stored mesh route information:

- **Area**
- **Option Set**
- **Instance - Cell and View**



Deleting Mesh Routes

The Mesh Route Deletion feature uses Delete Custom Route Shapes and Delete Mesh Tcl commands to provide a GUI that allows you to delete mesh route shapes by specifying nets and mode.

This topic describes the following:

- [About Mesh Route Deletion](#)

About Mesh Route Deletion

The Mesh Route shapes can be deleted using the Delete Custom Route Shapes (`irt::deleteCustomRouteShapes`) interactive command or using a dedicated interactive command Delete Mesh (`irt::deleteMesh`) for deleting mesh route shapes.

The shapes created by Power Routing Commands attach different property names to be deleted when using the Mesh Route Generation (`irt::generateMeshRoute`) interactive command.

Here is the list of properties attached to different Power Routing commands:

Note:

These properties are not created if the environment variable `_SYNOPSIS_CUSTOM_ROUTE_DISABLE_CR_PG_PROPERTY` exists.

- `rt::createPowerStripes` (Power Stripes Creation): `CR_PG_Stripes`
- `rt::createPowerStaples` (Power Staples Creation): `CR_PG_Staples`
- `rt::insertVias` (Vias Insertion): `CR_PG_Via`

This property is attached only if one of the two connected shapes has `CR_PG_Stripes` or `CR_PG_Staples` property.

For example, the Via in the following figure is attached to `CR_PG_Via` property because the connected shapes has the `CR_PG_Stripes` property.



- `rt::powerHookUp` (Power Hook Up): `CR_PG_Hook_Up`

Since mesh route generation leverages these power routing commands to create mesh route, these properties are also applied when using Mesh Route.

In the **Only Via** mode, only vias with the `CR_PG_Via` property are removed, whereas **Mesh and Vias** delete shapes and vias with the `CR_PG_Stripes`, `CR_PG_Staples`, `CR_PG_Via` or `CR_PG_Hook_Up` properties.

Delete Custom Route Shapes Toolbar

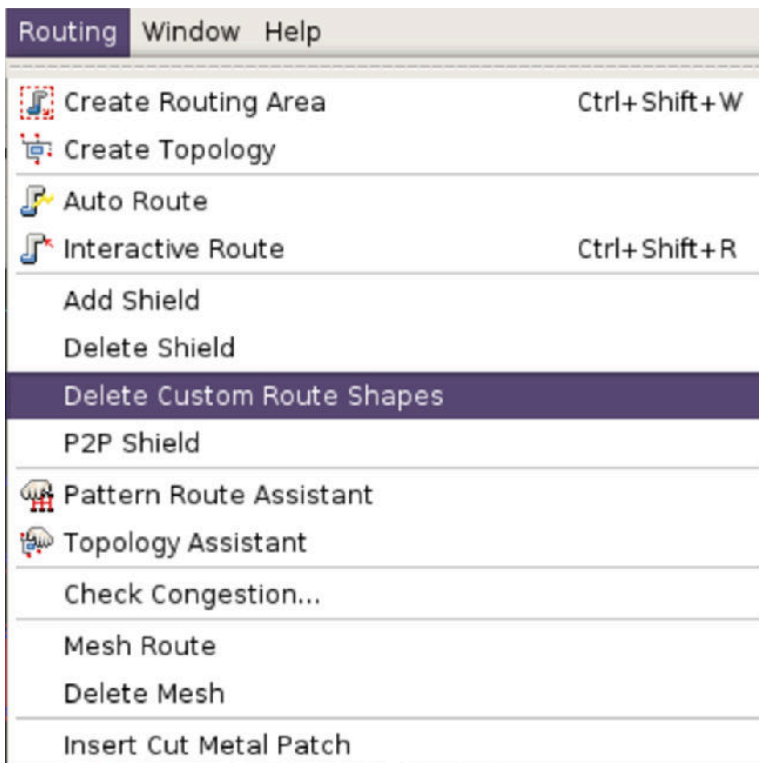
A command options toolbar appears when the `irt::deleteCustomRouteShapes` interactive command is initialized.



Option	Description
All	Deletes all the nets.
Selected	Deletes custom route shapes on the selected nets.
Include Shield	Deletes any power ground shielding shapes that may be on signal nets.

Option	Description
Only Mesh Route	Deletes only the mesh route created shapes and vias. If selected, the custom route shapes are limited to Mesh Route objects. If not, all the custom route shapes are deleted. Valid values are: • Only Vias: deletes only mesh route vias. • Mesh and Vias: deletes all the shapes created by mesh route.

Alternatively, you can access the interactive command from the **Routing** menu or by invoking the `irt::deleteCustomRouteShapes` command from the console.



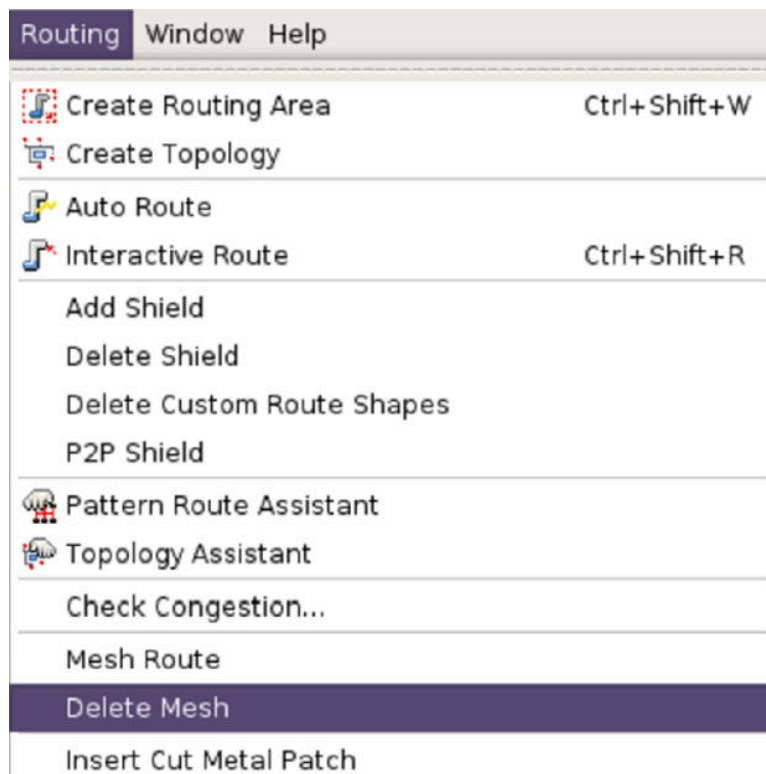
Delete Mesh Toolbar

A command options toolbar appears when the `irt::deleteMesh` interactive command is initialized.



Option	Description
All	Deletes all the nets.
Selected	Deletes mesh shapes that belongs to the specified nets.
Mode	Allows you to control the kind of the objects to be deleted from the mesh route. Valid values are: • Only Vias: deletes only mesh route vias. • Mesh and Vias: deletes all the shapes created by Mesh Route

Alternatively, you can access the interactive command from the **Routing** menu or by invoking the `irt::deleteMesh` command from the console.



20

Analyzing and Troubleshooting Routing Results

This chapter describes how to check routing results and how to debug errors.

This chapter contains the following topics:

- [Analyzing Routing Results Workflow](#)
- [Viewing the Routing Summary](#)
- [Viewing Detailed Net Information](#)
- [Troubleshooting Unrouted Nets](#)
- [Checking and Fixing Track-Related Issues](#)
- [Running the Postroute Checker](#)

If error messages appear in the Console during routing, see [Troubleshooting Error Messages](#).

Analyzing Routing Results Workflow

After the routing process is finished, you can run different types of checks to improve the quality of the results and troubleshoot errors.

1. Check the routing summary information in the Console. See [Viewing the Routing Summary](#).
2. Create a detailed net report for unrouted nets. Review the detailed net information to check the quality of the results and to look for unmet constraints. See [Viewing Detailed Net Information](#).
3. Troubleshoot unrouted nets. For information, see [Troubleshooting Unrouted Nets](#).
4. Check and fix off-track route objects. For information, see [Checking and Fixing Track-Related Issues](#).
5. Turn on the Design-Rule Driven (DRD) feature to visually check for violations. Use the Error Viewer to look for any errors and determine the cause. See [Checking for Violations as You Edit \(DRD\) and Error Viewer Assistant](#).

Note that if one of the rules in the technology is causing the problem, you will need to update the technology database and re-run the router on the design.

6. (Digital and block-level designs) Run the Postroute Checker to determine the quality of the routing results. See [Running the Postroute Checker](#).
7. Run LVS on the design. See Comparing Layout Versus Schematic (LVS).
8. When you are finished, save the design.

Viewing the Routing Summary

The Custom Compiler provides the following information in the Console before, during, and after routing:

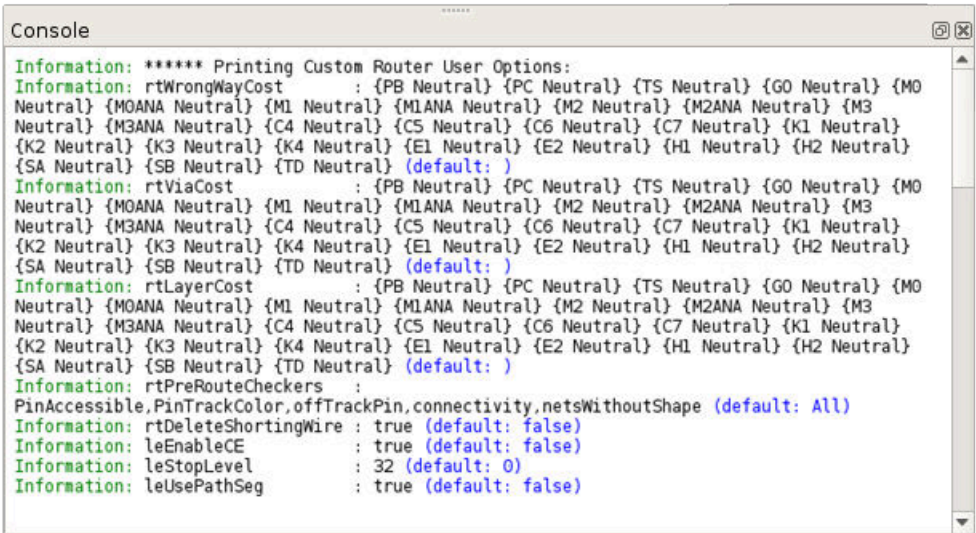
- [Preroute Information](#)
- [Routing Stage Information](#)
- [Post-Route Summary](#)

Preroute Information

The preroute information is gathered just before the routing operation starts. The information consists of:

- The list of options that have non-default values and the values that you specified during setup
- Preroute net statistics

The list of options is displayed in the Custom Router User Options section in the Console. In the example shown, the Custom Router User Options section lists the following options and the corresponding non-default values: rtWrongWayCost, rtViaCost, rtLayerCost, rtPreRouteCheckers, rtDeleteShortingWire, leEnableCE, leStopLevel, and LeUsePathSeg.



The net statistics are a series of messages that report preliminary information about the routing request. Each statistic is associated with a message number:

Net Statistic	Description
GCR-284: Number of nets to route	The number of nets that you selected for routing.
GCR-285: Number of nets rejected by router	The number of nets that will not be routed. Additional messages will appear in the Console to provide explanations: • The net contains only one pin within the design boundary or the routing area (GCR-33). • The net is a supply net (GCR-119, GCR-288).
GCR-286: Number of nets attempted by router	The remaining nets that the Custom Router will try to route.

In the following example, 596 nets were selected for routing. 56 of the nets will not be routed, so the Custom Router will try to route the remaining 540 nets. For the nets that are rejected, the Custom Router provides explanations (GCR-33) to help you troubleshoot the nets.

Chapter 20: Analyzing and Troubleshooting Routing Results

Viewing the Routing Summary

```

Console
Information: Number of nets asked to route = 596. (GCR-284)
Information: Number of nets attempted to route = 540. (GCR-285)
Information: Number of nets rejected to route = 56. (GCR-286)
Warning: (WARNING) Net N768 has no pins. Net must have more than one pin within design
prBoundary or within user specified routing area to be routed. This net will be ignored.
(GCR-033)
Warning: (WARNING) Net N767 has no pins. Net must have more than one pin within design
prBoundary or within user specified routing area to be routed. This net will be ignored.
(GCR-033)
Warning: (WARNING) Net N766 has no pins. Net must have more than one pin within design
prBoundary or within user specified routing area to be routed. This net will be ignored.
(GCR-033)
Warning: (WARNING) Net N760 has no pins. Net must have more than one pin within design
prBoundary or within user specified routing area to be routed. This net will be ignored.
(GCR-033)

```

In the following example, two nets were selected for routing. One of the nets will not be routed, so the Custom Router will route only one net. For the net that will not be routed, the Custom Router provides an explanation (GCR-288) to help you troubleshoot the net.

```

Console
Information: Number of nets asked to route = 2. (GCR-284)
Information: Number of nets attempted to route = 1. (GCR-285)
Information: Number of nets rejected to route = 1. (GCR-286)
Warning: Supply net VSS is ignored for routing. (GCR-288)
Information: *****
Information: * Nets to Route   Routed   Unrouted *
Information: *           1           1           0 *
Information: *****

```

Routing Stage Information

The routing operation consists of multiple stages:

- **Design Load:** Loading the design data.
- **Sync Design:** Reloading the design data if you have modified the design.
- **Custom Route:** Auto routing the design.
- **Bus Route:** Bus routing is in progress.
- **Add Shield:** Adding shielding to the routes.
- **Delete Shield:** Deleting shielding from the routes.

The Custom Router will report the following information for each stage:

Information Reported for Each Stage	Description
GCR-282: [<Name of step>] Elapsed time: real = XXX, cpu = XXX, [sys = XXX, usr = XXX]	The message consists of: • Name of step: The name of the routing stage (for example, Load Design). • Elapsed time: The length of time it took the stage to finish. The following elapse times are reported: real, cpu, sys, and usr. • real: The actual length of time it took to complete the stage. • cpu: The length of time during which the CPU was running the stage. • sys: The length of time it took to run the routing code. • usr: The length of time it took to run the code in the operating system kernel.
GCR-283: [<Name of step>] Memory : Total = XXX, Peak = XXX	The message consists of: • Name of step: The name of the routing stage (for example, Load Design). • Memory: Total: The amount of memory used during the routing stage: Total, and Peak. • Total: The total amount of memory that was used for the stage. • Peak: The maximum (peak) memory that was used during the stage.

In the following example, the elapsed time and memory information are reported for the following routing stages: Load Design, and Custom Route.

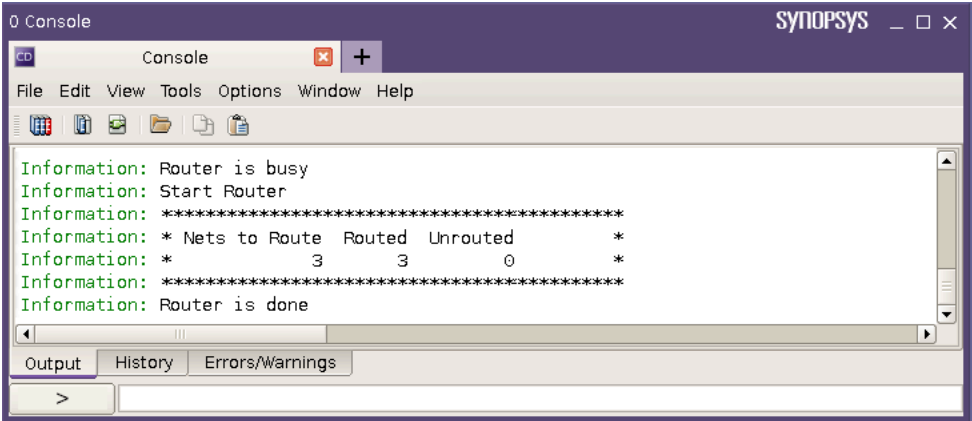
```

Console
Information: [Load Design] Elapsed time: real = 0h:00m:00s, cpu = 0h:00m:00s, [sys = 0h:00m:00s, usr = 0h:00m:00s] (GCR-282)
Information: [Load Design] Memory: total = 2.39 GB, peak = 2.39 GB. (GCR-283)
Information: [Custom Route] Elapsed time: real = 0h:00m:17s, cpu = 0h:00m:17s, [sys = 0h:00m:00s, usr = 0h:00m:17s] (GCR-282)
Information: [Custom Route] Memory: total = 2.53 GB, peak = 2.59 GB. (GCR-283)

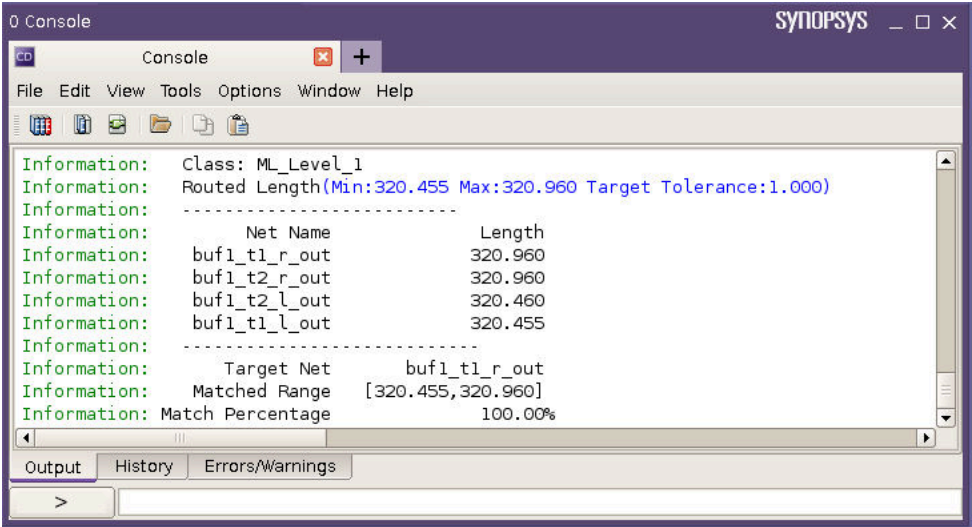
```

Post-Route Summary

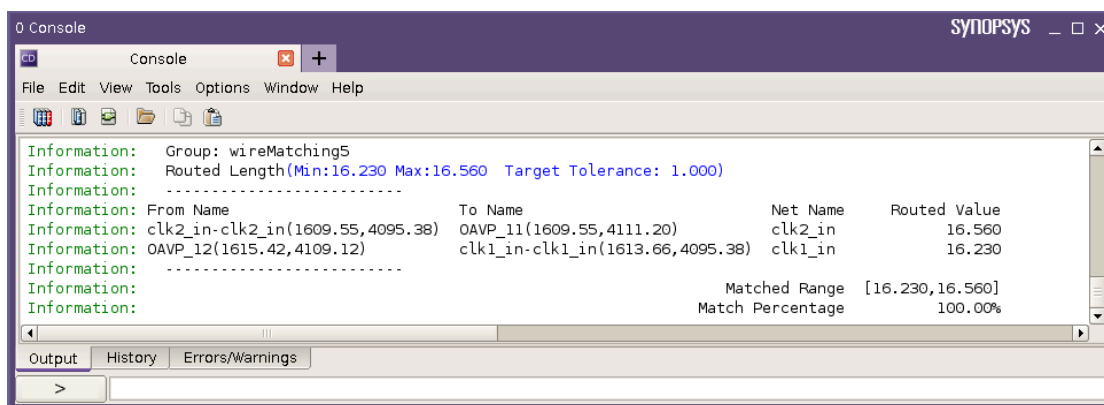
The following example shows post-route summary information that is displayed in the Console: the routing time, nets that were selected for routing, the number of routes that were routed, and the number of nets that were not routed. You can compare the post-route summary with the [Preroute Information](#) for confirmation.



The post-route summary also displays the wire matching results of a regular net group. In the example shown, the wire matching results are displayed for the following nets: buf1_t1_r_out, buf1_t2_r_out, buf1_t2_l_out, and buf1_t1_l_out. The Match Percentage is 100%, because the route lengths are all within the tolerance value relative to the nets within the same constraint.



The following example displays the wire matching results of two oaRoutes: clk2_in and clk1_in. The Match Percentage is 100%, because the route lengths are all within the tolerance value relative to the nets within the same constraint.



Use the summary as a starting point for troubleshooting unrouted nets. For example, after obtaining the names of the unrouted nets, you can proceed to create a report for each unrouted net. For more information, see [Viewing Detailed Net Information](#).

Viewing Detailed Net Information

You can create a net report for any routed and unrouted net to review layer usage and routing length information. You can use one of the following methods to view the detailed net report in the Console:

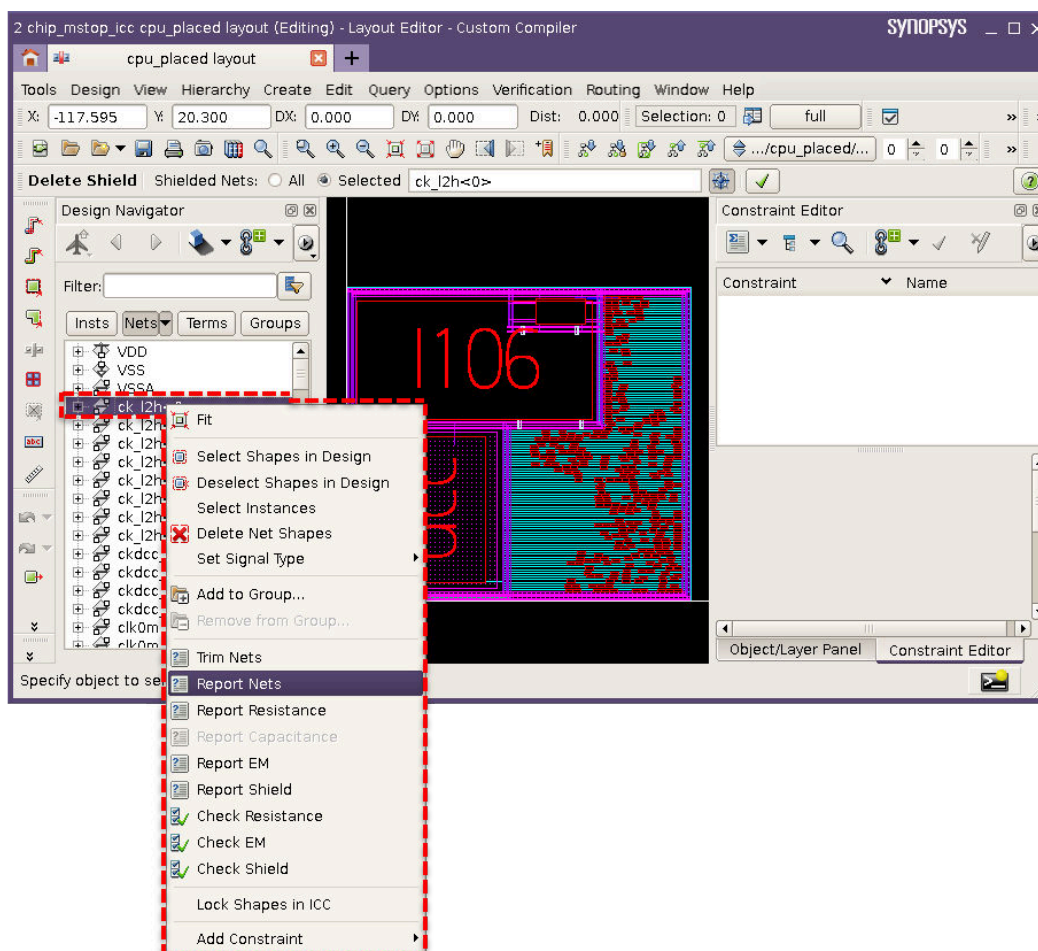
- [Creating a Net Report Using the Design Navigator](#)
- [Creating a Net Report Using `le::reportNets`](#)

To view the routing summary, see [Viewing the Routing Summary](#). To troubleshoot the routing results, see [Troubleshooting Unrouted Nets](#).

Creating a Net Report Using the Design Navigator

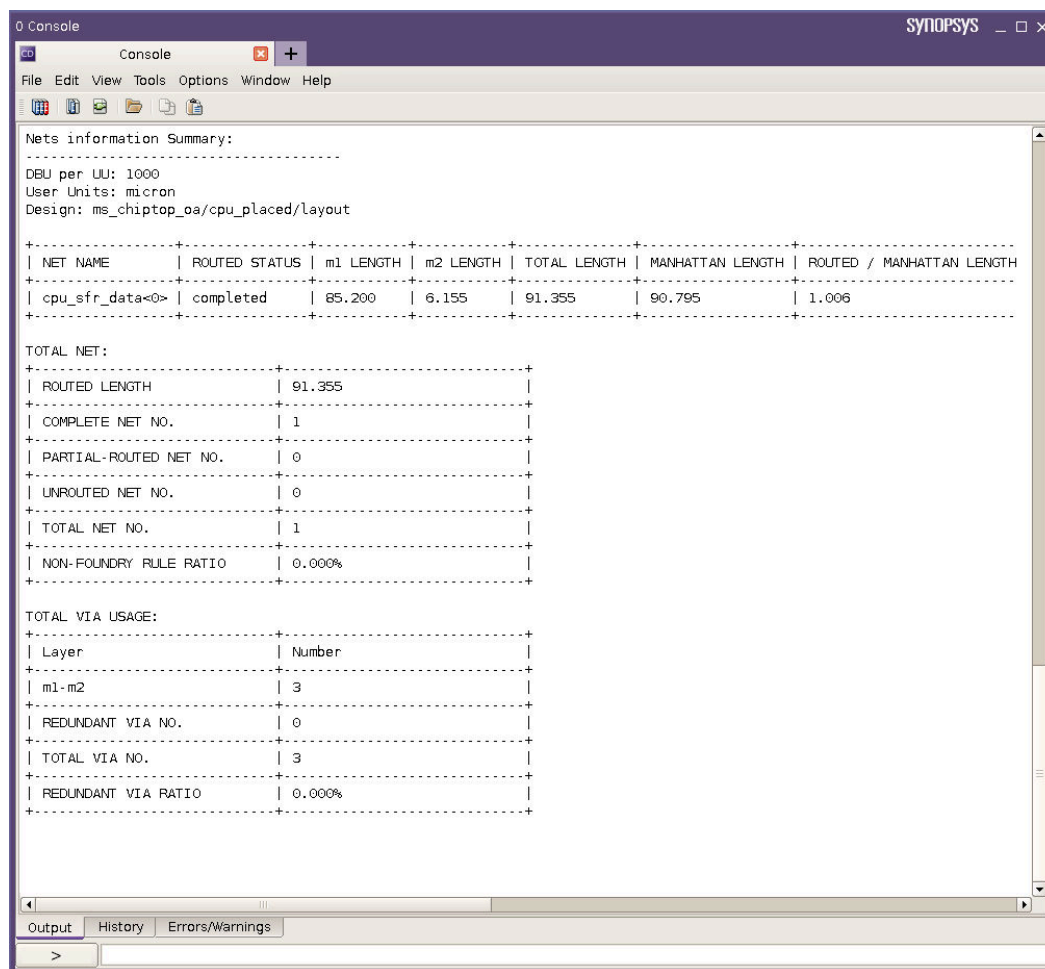
To create a net report in the Design Navigator:

1. Display the Design Navigator.
2. In the Design Navigator, make sure the **Nets** tab is displayed. The list of nets appears.
3. Right-click the net, and then choose **Report Nets**.



The net report appears in the Console window and displays the following information:

- For unrouted nets, the Console displays the Manhattan distance for each net.
- For routed nets, the Console displays the Manhattan distance, layer usage information, total net length, and quality of routes ratio.



Creating a Net Report Using `le::reportNets`

To use the `le::reportNets` Tcl command to create a net report:

In the Custom Compiler Console, type the `le::reportNets` command and the arguments. For information about the command, see the *Custom Compiler Tcl Reference Manual*.

The screenshot shows the Synopsys Console window with the following content:

```

0 Console
Console
File Edit View Tools Options Window Help
Nets information Summary:
-----
DBU per UU: 1000
User Units: micron
Design: ms_chiptop_oa/cpu_placed/layout

+-----+-----+-----+-----+-----+-----+-----+
| NET NAME      | ROUTED STATUS | m1 LENGTH | m2 LENGTH | TOTAL LENGTH | MANHATTAN LENGTH | ROUTED / MANHATTAN LENGTH |
+-----+-----+-----+-----+-----+-----+-----+
| cpu_sfr_data<0> | completed    | 85.200    | 6.155     | 91.355       | 90.795           | 1.006                     |
+-----+-----+-----+-----+-----+-----+-----+

TOTAL NET:
+-----+-----+
| ROUTED LENGTH      | 91.355 |
+-----+-----+
| COMPLETE NET NO.   | 1      |
+-----+-----+
| PARTIAL-ROUTED NET NO. | 0      |
+-----+-----+
| UNROUTED NET NO.   | 0      |
+-----+-----+
| TOTAL NET NO.      | 1      |
+-----+-----+
| NON-FOUNDY RULE RATIO | 0.000% |
+-----+-----+

TOTAL VIA USAGE:
+-----+-----+
| Layer      | Number |
+-----+-----+
| m1-m2      | 3      |
+-----+-----+
| REDUNDANT VIA NO. | 0      |
+-----+-----+
| TOTAL VIA NO.    | 3      |
+-----+-----+
| REDUNDANT VIA RATIO | 0.000% |
+-----+-----+

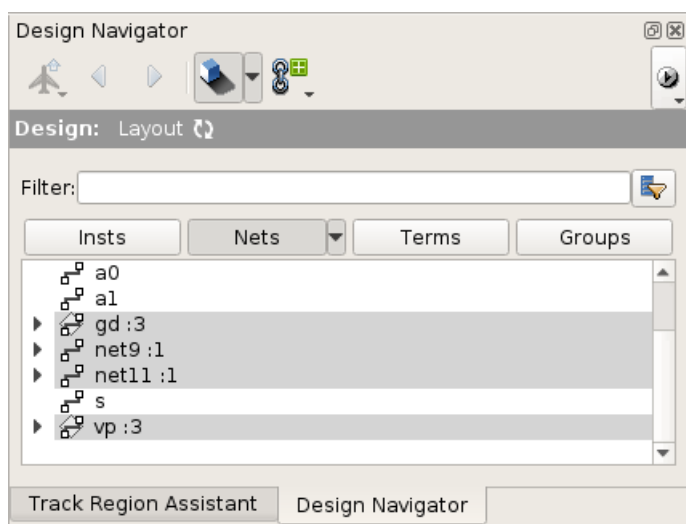
Output History Errors/Warnings
> le::reportNets [db::getNets cpu_sfr_data<0> -of [db::getAttr editDesign -of [de::getActiveContext]]]

```

Troubleshooting Unrouted Nets

Use one or more of the following methods to troubleshoot unrouted nets:

- In the Console, look at the Data Checker messages. The Data Checker runs automatically when you start the Auto Router and the Interactive Router. The Data Checker might have found problems and reported errors or warnings. For information about the Data Checker, see [Optimizing Routing Results](#) and [Checking for Constraint Compliance](#).
- In the Design Navigator **Nets** tab, look for net names that have a gray background, indicating they are unrouted nets. Set the `heDesignNavigatorShowExtendedFlightlineInfo` Tcl preference to true to display the number of unconnected pins and `instPins` next to the net name. For example, the `gd` net has 3 unconnected pins.



Select an unrouted net and visually inspect the route on the canvas. Look for blocked pins, and check for non-default rules that might prevent the router from accessing the pin. (To better view the routes on the canvas, enable the **Shadow Mode** in the Design Navigator to dim the surrounding design.

- If the unrouted net is within a defined routing area, check and make sure that all source and destination points of selected nets have to be within the defined area. The router will not be able to complete the routing if a source or destination point is missing.
- Create an interconnect to manually route a wire from the pin to see if there is a short or a design rule checking (DRC) error. Also check and make sure the route has two pins at either end.
- Verify that the two pins of the route are on a routable layer for that net.

Checking and Fixing Track-Related Issues

Throughout the design process, route and instance objects can be accidentally or intentionally placed or moved off tracks. Other track-related attribute mismatches can also occur.

This topic describes the following:

- [Before You Start](#)
- [Checking and Reporting Track-Related Issues](#)
- [Fixing Track-Related Issues](#)
- [Examples](#)

Before You Start

Before you start to check or fix off-track or other track-related attribute-mismatch issues, make sure the tracks of interest are visible and active. See the following:

- [Displaying and Hiding Track Regions on the Canvas](#)
- [Making Track Regions, Track Groups, and Individual Tracks Active](#)
- [Viewing Track Patterns](#)

Checking and Reporting Track-Related Issues

You can run the `le::checkTrackSnapping` Tcl command to check the following:

Check	Applicable Objects	Description
Track type mismatch	• Vias • Paths, path segments • Rectangles	The route object track type does not match the type of track.
Signal type mismatch	• Vias • Paths, path segments • Rectangles	The route object signal type does not match the signal type of the track.
Width mismatch	• Vias • Paths, path segments • Rectangles	The width of the route object does not match the width of the track.
Color mismatch	• Vias • Paths, path segments • Rectangles	The color of the route object does not match the color of the track.

Check	Applicable Objects	Description
Offtrack centerline	- Vias - Paths, path segments - Rectangles	The centerline of the route object is not aligned with the track.
Orientation mismatch	Instances	The orientation of the object is incorrect.
snapTrackPatterns mismatch	Instances	The snapTrackPatterns CDF parameter attributes do not match any defined tracks. See Setup for Snapping to Tracks.
Pin mismatch	Instances	The metal pin shape of the object does not snap to any tracks during instance snapping. To consider the metal pin shapes in instances snapping, enable the <code>leInstanceSnappingMetalShapes</code> Tcl preference. See Snapping Instances.
Off-track reference point	Instances	The reference point of the object is not on track.

The input to the `le::checkTrackSnapping` command is the list of objects that you want to check:

```
le::checkTrackSnapping [INPUT : vias, paths, path segments, rectangles, instances]
```

The output, `leOffTrackInfo`, contains the mismatch information found by the command:

```
le::checkTrackSnapping [OUTPUT: leOffTrackInfo]
```

The checks can be run on the entire design, a specified region, or specific objects. You can also specify the types of checks you want to run, and whether you want to add error markers on the canvas for the reported objects. See [Examples](#).

Fixing Track-Related Issues

To fix track-related issues, you can run the `le::fixTrackSnapping` Tcl command. The input to the command is the list of objects that you want to check, including the output of the `le::checkTrackSnapping` command:

```
le::fixTrackSnapping [INPUT: Vias, Paths, Path Segments, Rectangles, Instances, leOffTrackInfo]
```

The output, `leOffTrackInfo`, lists the objects that are fixed. The command can only fix specified objects that have unlocked Route Status and Placement Status.

```
le::fixTrackSnapping [OUTPUT: leOffTrackInfo]
```

See [Examples](#).

Examples

Entire Design

Check and fix track-related issues for the entire design:

```
set offTrackObjs [le::checkTrackSnapping]
le::fixTrackSnapping $offTrackObjs
```

Add error markers to the reported objects in the entire design:

```
le::checkTrackSnapping -createMarker true
```

Specific Objects

Check and fix M1 paths:

```
set M1Shapes [db::getShapes -lpp M1 -of [ed] -filter {%type=~Path/}]
set offTrackM1Shapes [le::checkTrackSnapping $M1Shapes]
le::fixTrackSnapping $offTrackM1Shapes
```

Specific Checks

Check signal type mismatches:

```
db::foreach offTrackInfo [le::checkTrackSnapping [db::getShapes -of [ed]]
{
  set mismatchInfo [db::getattr offTrackInfo.error.description]
  if { $mismatchInfo != "sigType mismatch" } {
    le::fixTrackSnapping $offTrackInfo
  }
}
```

Instances

Check and fix instances that did not snap to tracks.

```
set offTrackInsts [le::checkTrackSnapping [db::getInsts -of [ed]]]
le::fixTrackSnapping $offTrackInsts
```

Running the Postroute Checker

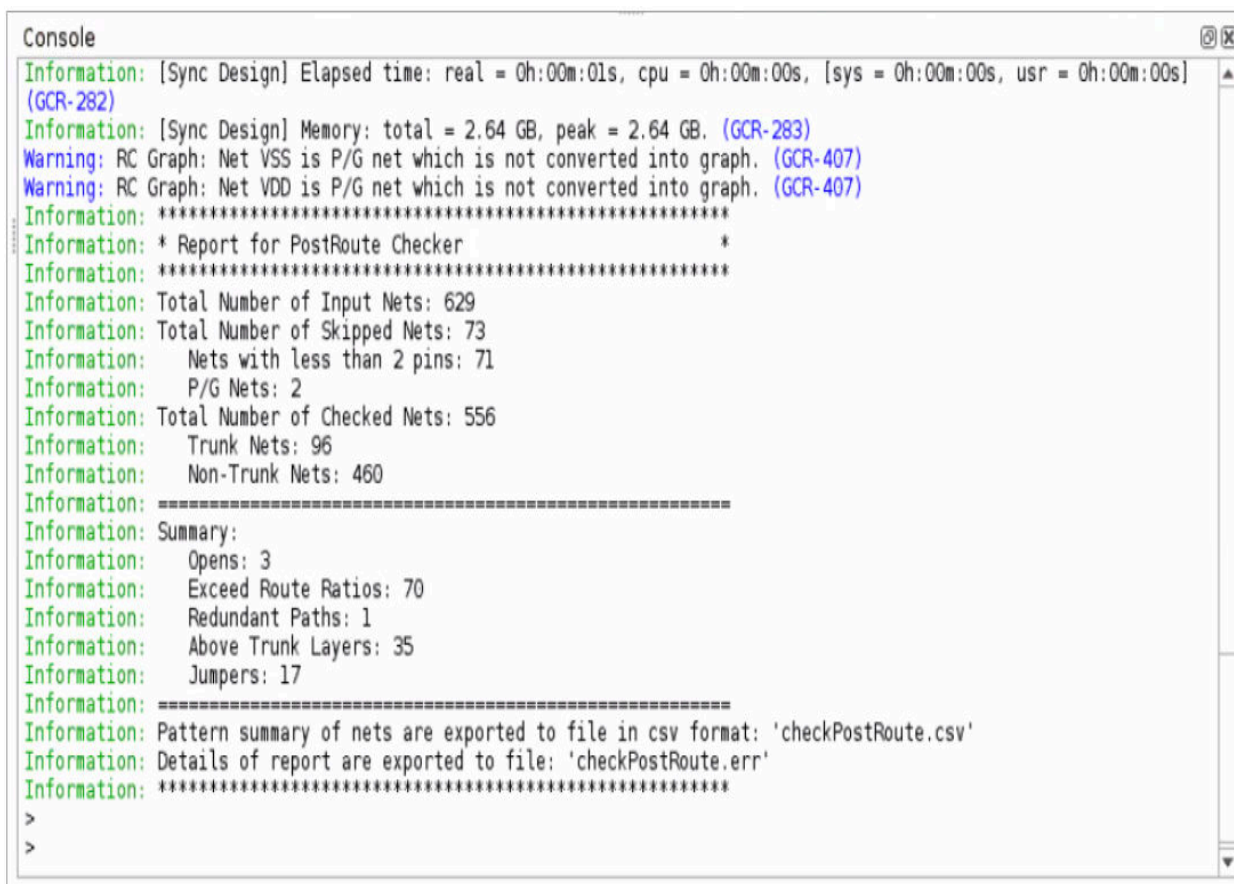
To determine the quality of the routing results, you can run the Postroute Checker on 0x route shapes. The generated reports can further assist with fixes and improve the results. The postroute checking is primarily for digital and block-level designs.

To run the Postroute Checker:

- Use the following Tcl command:

```
rt::checkPostRoute
```

When the checks are finished, a summary report appears in the Console.



```

Console
Information: [Sync Design] Elapsed time: real = 0h:00m:01s, cpu = 0h:00m:00s, [sys = 0h:00m:00s, usr = 0h:00m:00s]
(GCR-282)
Information: [Sync Design] Memory: total = 2.64 GB, peak = 2.64 GB. (GCR-283)
Warning: RC Graph: Net VSS is P/G net which is not converted into graph. (GCR-407)
Warning: RC Graph: Net VDD is P/G net which is not converted into graph. (GCR-407)
Information: *****
Information: * Report for PostRoute Checker *
Information: *****
Information: Total Number of Input Nets: 629
Information: Total Number of Skipped Nets: 73
Information: Nets with less than 2 pins: 71
Information: P/G Nets: 2
Information: Total Number of Checked Nets: 556
Information: Trunk Nets: 96
Information: Non-Trunk Nets: 460
Information: =====
Information: Summary:
Information: Opens: 3
Information: Exceed Route Ratios: 70
Information: Redundant Paths: 1
Information: Above Trunk Layers: 35
Information: Jumpers: 17
Information: =====
Information: Pattern summary of nets are exported to file in csv format: 'checkPostRoute.csv'
Information: Details of report are exported to file: 'checkPostRoute.err'
Information: *****
>
>
  
```

In addition, a checkPostRoute.csv file is generated. The file contains detailed statistics about each net that was checked: route ratio, actual length, optimal length, number of layers not within the range of non-default-rule layers, number of trunk segments, number of open pins, and so on.

A

Routing Constraint Reference

By default, the constraint parameter values are set to the values in the technology library. You can override the values in the technology library by supplying the desired values in the Constraint Editor. These design-specific constraints (intents) also supplement the process rules.

This chapter describes the design-specific routing constraints (intents) parameters that you can view or modify in the Constraint Editor:

- [Antenna Defaults Constraint](#)
- [Bus Constraint](#)
- [Color Matching Constraint](#)
- [Differential Group Constraint](#)
- [Driver Taper Halo Constraint](#)
- [Grids Constraint](#)
- [Hier Net Spacing Constraint](#)
- [Inter Group Spacing Constraint](#)
- [Layers Constraint](#)
- [Length Constraint](#)
- [Net Priority Constraint](#)
- [Net Shielding Constraint](#)
- [Pattern Route Constraint](#)
- [Shield Overhang Constraint](#)
- [Taper Halo Constraint](#)
- [Via Reliability Constraint](#)

- [Vias Constraint](#)
- [Wire Matching Constraint](#)

Design-specific constraints you create in the schematic are automatically transferred to the layout when you use the Schematic-Driven Layout (SDL) feature. See [Tools for Visually Assisted Layout Processes](#).

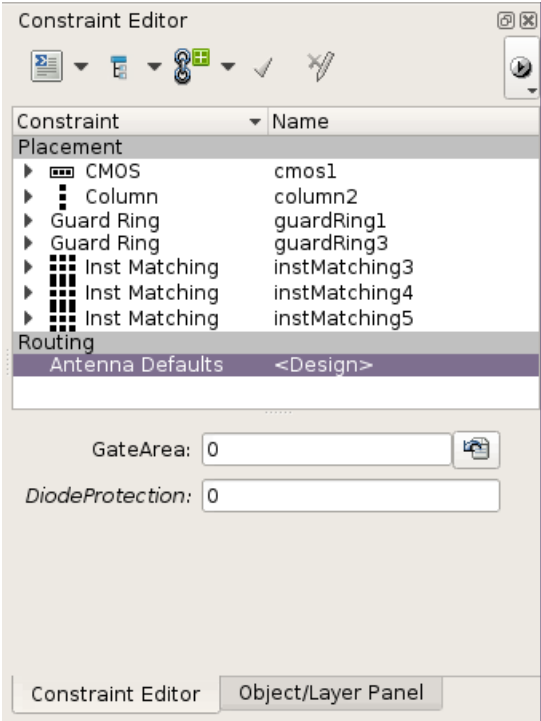
For instructions on how to create, view, or modify the design-specific constraints, see [Setting Routing Constraints](#). To modify constraint values in the technology library, review the technology database.

For information about the Resistance constraint, see [Setting Resistance Constraints](#).

Antenna Defaults Constraint

Antenna Defaults constraints lets you get and set default antenna data for instance terminals that do not already have the information stored in the design.

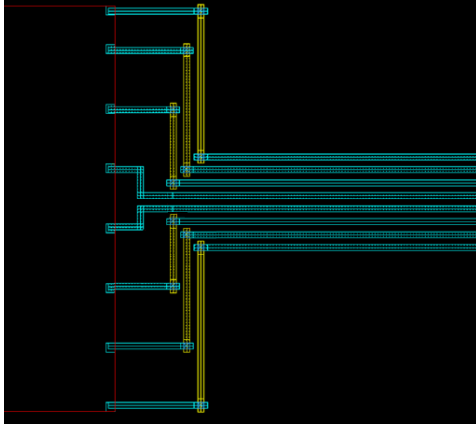
This topic describes the Antenna Defaults constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).



Parameter	Description
GateArea	The default gate area value for instance terminals.
DiodeProtection	The default diode protection value for instance terminals.

Bus Constraint

You can set Bus constraints to route an ordered list of nets as a single bus structure (or trunk). The constraints are applied per layer to the trunk shape and override any settings on the individual bits.

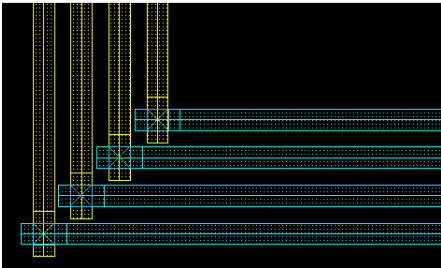
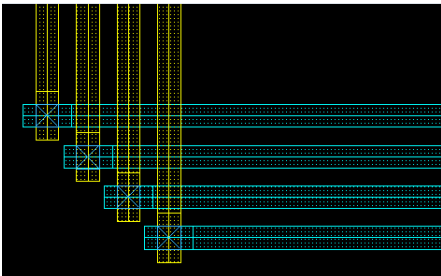


This topic describes the Bus constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).

To create Bus constraints using the `cm::createBus` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

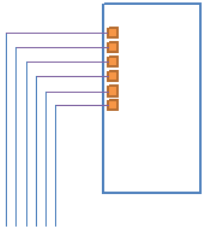
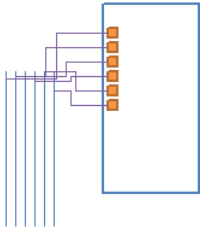


Parameter	Description
Default Gap	The minimum spacing between the outermost bits of the bus trunk and adjacent shapes.

Parameter	Description
Corner Type	How the bus trunk should be routed at corners: • default: Uses the setting in the Auto Router dialog box Bus/Diff Pair Tab. • auto: Automatically determines the optimal corner type for bit or pin alignment to reduce routing congestion. • river: Routes the bits so that they do not overlap at the corner.  • cross: Routes the bits so that they overlap at the corner. 

In the **Other Options** section, you can modify the shielding and other specifications.

Parameter	Description
Shield Placement	The shield requirements for the bus trunk: • default: Uses the setting in the Auto Router dialog box Bus/Diff Pair Tab. • outside: Places shields on the outside of the bus trunk. • interleave: Inserts shared shielding between two adjacent bits. • doubleInterleave: Shields each bit individually. • halfInterleave: Inserts shared shielding every two bits. For details, see Bus/Diff Pair Tab.
Layers	The metal layer or layers to use for routing: • default: Specifies the default range of layers. • range: Lets you specify a range of metal layers. Choose a layer in the Min and Max lists.
Min	The lowest metal layer in the range of layers for routing. This parameter is available if you set Layers to Range.

Parameter	Description
Max	The highest metal layer in the range of layers for routing. This parameter is available if you set Layers to Range .
Optimize Order	Permits the router to use the reverse net order to optimize routing.
Extend Trunk	Enables the router to adjust the trunk structure to align to the pins. The following figure shows the results of selecting Extend Trunk (left) and clearing the Extend Trunk check box (right).
 Trunk Extended to Pins  Trunk Not Extended to Pins	

In the **Layers** section, you can modify the spacing, width, and routing grid specifications.

Parameter	Description
Layer	<i>(Read only)</i> The name of the layer. You cannot change the name of the layer.
Spacing	The minimum spacing of the routing layer. The default value is obtained from the minSpacing constraints in the technology.
Width	The minimum width to be used for the layer. The default value is obtained from the minWidth constraints in the technology.
On Grid	The indicator that the bus trunk must remain on the grid for the specific routing layer.

Color Matching Constraint

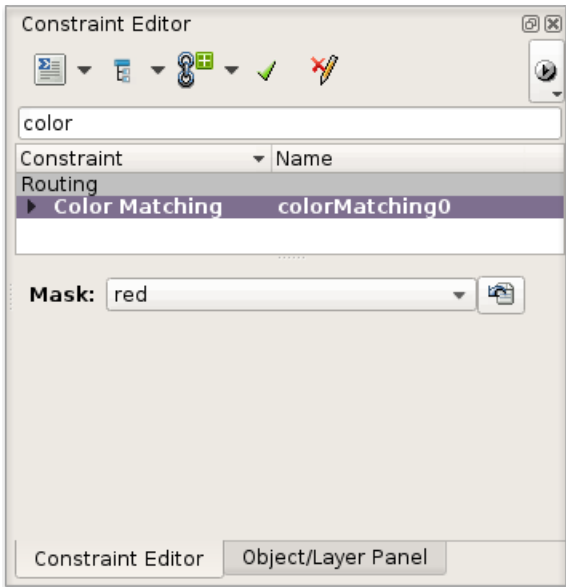
The Color Matching constraints let you specify one of the following:

- All selected nets must be on a specific mask.
- All selected nets must be on the same color mask on a per layer basis regardless of which color mask is used.

This topic describes the Color Matching constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).

Note:

Color Matching constraints that you set in the Schematic Editor are also available for viewing and editing in the Layout Editor.



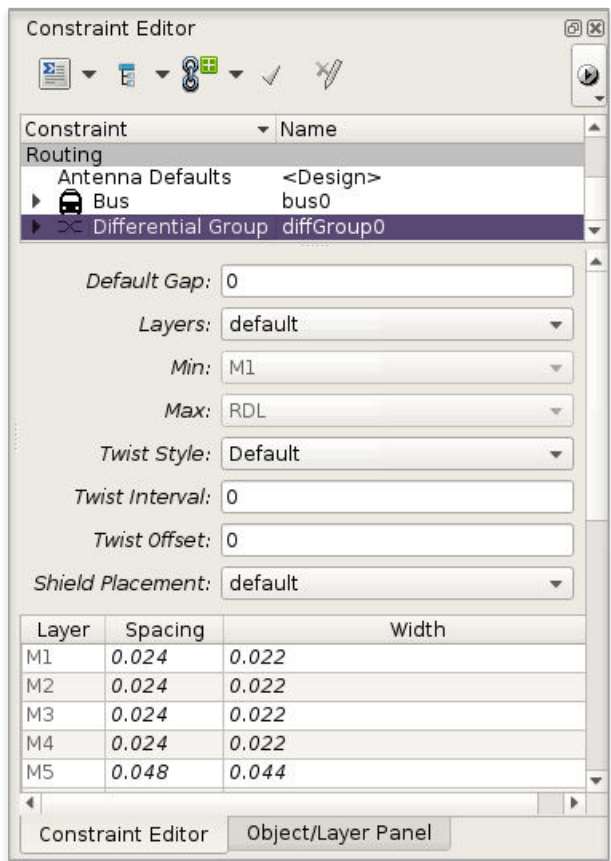
Parameter	Description
mask	The color assignment to use for all nets in the group.

Differential Group Constraint

You create Differential Group constraints for a group of two or three nets. The Custom Router pairs the closest pins, one from each net, and routes from a common gather point between the pins, keeping the resulting routing of the nets together for as long as possible.

This topic describes the Differential Group constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).

To create Differential Group constraints using the `cm::createDifferentialGroup` Tcl command, see the *Custom Compiler Tcl Reference Manual*.



Parameter	Description
Default Gap	The gap spacing between the outer most bit nets and other shapes in the design.
Layers	The layers on which the differential group is routed. You can specify the Default or a Range of layers. If you choose Range, you must specify the Min and Max layers.

Parameter	Description
Min	The lowest metal layer in the range of layers for routing. This parameter is available if you set Layers to Range .
Max	The highest metal layer in the range of layers for routing. This parameter is available if you set Layers to Range .
Twist Style	The way the nets in the group should twist: • Default: Uses the Twist Jumper Options setting in the Auto Route dialog box Bus/Diff Pair Tab. • None: Does not twist the wires. • Orthogonal: Applies 90° twists to the wires. • Diagonal: Applies 45° twists to the wires.
Twist Interval	The distance between each twist. This parameter is not available if you choose None in the Twist Style list.
Twist Offset	The distance between the first twist and the connecting pins. This parameter is not available if you choose None in the Twist Style list.
Shield Placement	The method in which the main bus trunk of the diff pair is to be shielded: • Default: Uses the Intra Shield Placement setting in the Auto Route dialog box Bus/Diff Pair Tab. • outside: Places shields on the outside of the group. • interleave: Inserts shared shielding between two adjacent routes. • doubleInterleave: Shields each route in the group individually. • halfInterleave: Inserts shared shielding every two routes. For details, see Bus/Diff Pair Tab .

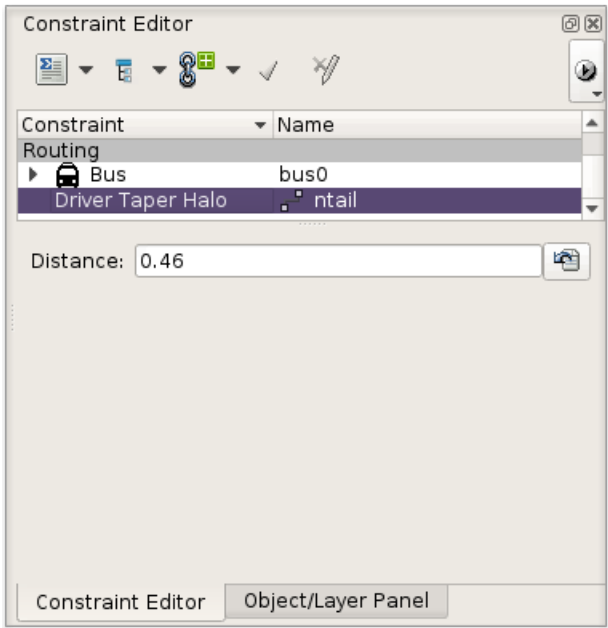
In the **Layers** table, you can modify the spacing and width specifications for each layer.

Parameter	Description
Layer	<i>(Read Only)</i> The name of the routing layer. You cannot change the name of the layer.
Spacing	The minimum spacing for the routing layer in trunk.
Width	The minimum width for routing layer in trunk.

Driver Taper Halo Constraint

You can set Driver Taper Halo constraints to taper wire up or down to connect to driver pins on a net. Only driver terminals and the instance terminals are affected by the constraint.

This topic describes the Driver Taper Halo constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).



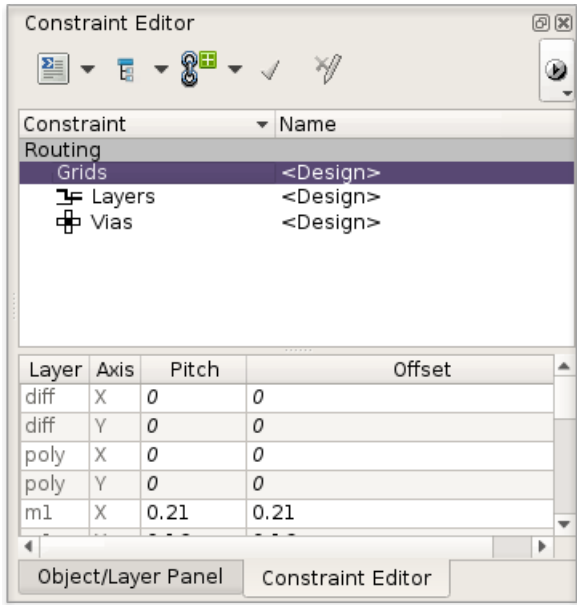
Parameter	Description
Distance	Distance between the edge of the pin and the location at which the tapering begins in the wire.

A diagram illustrating the Driver Taper Halo constraint. It shows an orange rectangle labeled 'Wire' on the left. To its right is a blue square labeled 'Pin'. A dashed rectangle surrounds the pin. A double-headed arrow points from the right edge of the orange wire to the left edge of the dashed rectangle, indicating the 'Distance' between the wire's edge and the start of the tapering region. There are also vertical double-headed arrows within the dashed rectangle.

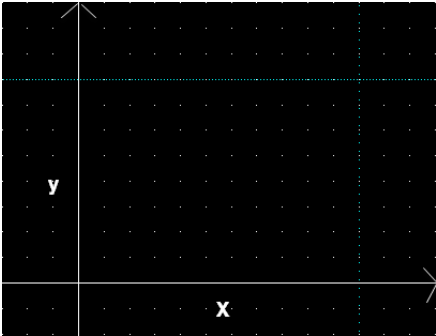
Grids Constraint

You can set Grids constraints to modify characteristics of the routing grid.

This topic describes the Grids constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).



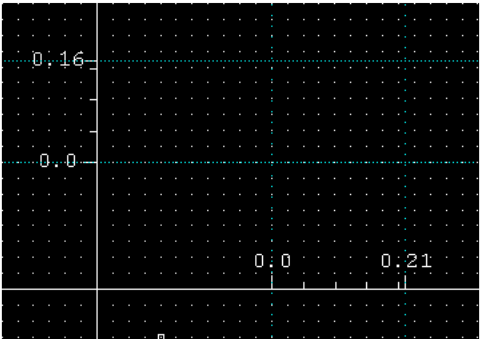
Parameter	Description
Layer	(Read Only) The name of the routing layer. You cannot change the name of the layer.
Axis	(Read Only) The grid axis: X or Y. The y-axis grid lines are perpendicular to the x-axis. The x-axis grid lines are perpendicular to the y-axis.



Parameter	Description
-----------	-------------

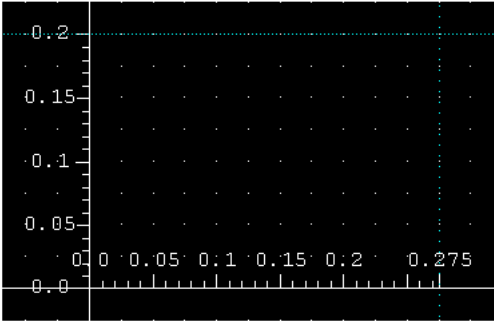
Pitch The distance between grid lines. The Pitch value must be > 0 .

Layer	Axis	Pitch	Offset
m1	X	0.21	0.275
m1	Y	0.16	0.2



Offset The distance between the origin and the first grid line along the x- and y-axis.

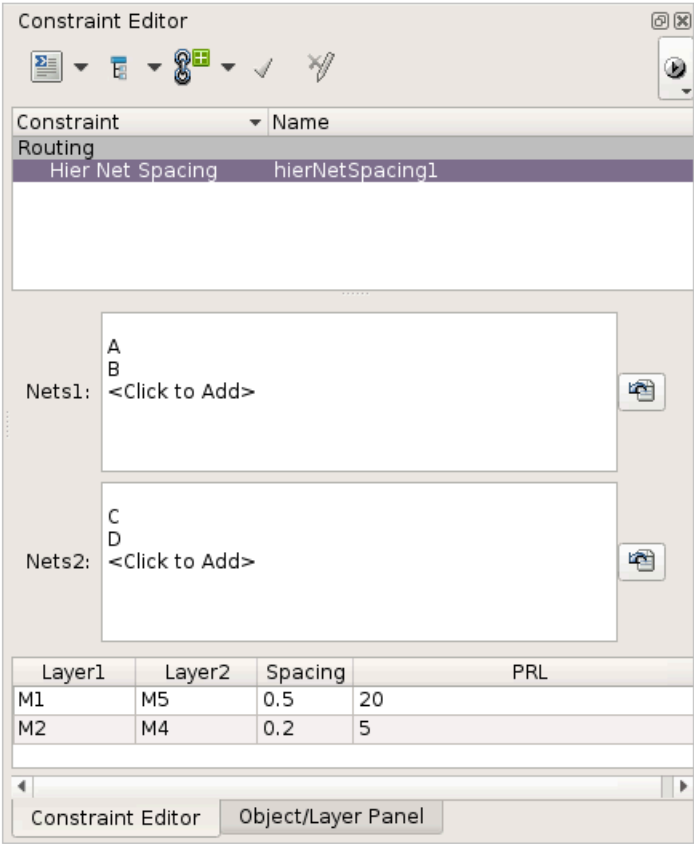
Layer	Axis	Pitch	Offset
m1	X	0.21	0.275
m1	Y	0.16	0.2



Hier Net Spacing Constraint

You can set hierarchical net spacing (Hier Net Spacing) constraints to define the minimum spacing between routes that are running parallel on different hierarchy levels in the design.

This topic describes the Hier Net Spacing constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).

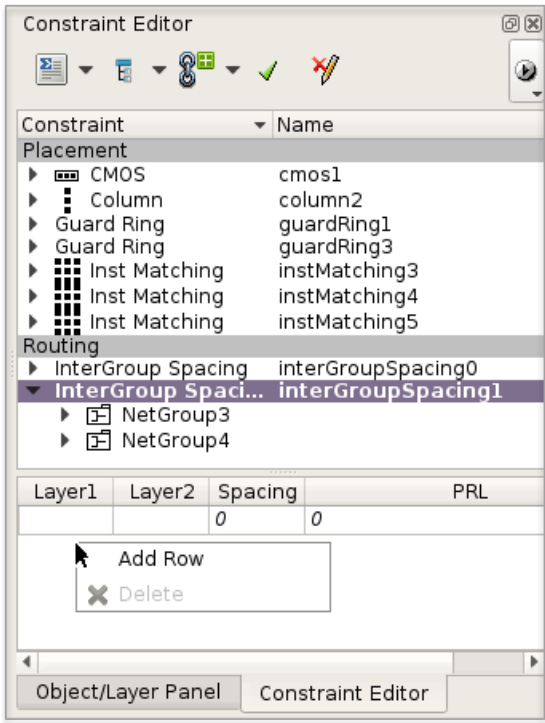


Parameter	Description
Nets1 Nets2	List of net names in net group 1 and net group 2. The net names can be occurrence paths, such as I1/I3/net3. Routes in net group 1 must be a specified distance from routes in net group 2.
Layer1 Nets2	Valid routing layers of net group 1 and net group 2.
Spacing	Minimum spacing, in user units, between the two layers.
PRL	(Parallel run length) Minimum length, in user units, that the routes must run in parallel for the constraint to apply.

Inter Group Spacing Constraint

You can set Inter Group Spacing constraints to define the minimum spacing and clearance between two nets.

This topic describes the Inter Group Spacing constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).



Parameter	Description
Layer1/2	The names of the layer1 and layer2.
Spacing	The minimum spacing between Layer1 and Layer2.
PRL	Parallel run length, the minimum length of the parallel routes before the intergroup spacing applies.

Layers Constraint

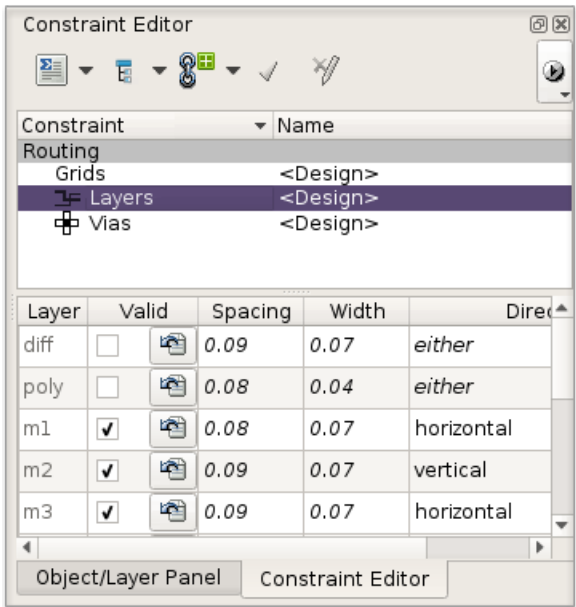
The technology contains the following rules per layer: minimum spacing, minimum width, and default routing direction. For each design, you can set the Layers constraint to enforce stricter requirements.

This section describes the design-specific Layers constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).

To set the various layer constraints in the technology, review the technology database. To override the values in the Constraint Editor for the current session only, use the settings in the **Auto Route** dialog box [Layer Tab](#).

Note:

Layers constraints that you set in the Schematic Editor are also available for viewing and editing in the Layout Editor.



Parameter	Description
Layer	(Read only) The name of the layer. You cannot change the name of the layer.
Valid	The indicator that the router can use the layer during routing. The default setting is obtained from the validRoutingLayers constraints in the technology. Select Valid to permit the router to use the layer. Clear the Valid check box to prevent the router from using the layer. If more than one poly layer appears in the Layer list, select only one of the poly layers for routing. An error results if you select more than one poly layer for routing.
Spacing	The minimum spacing value to be used for the layer. The default value is obtained from the minSpacing constraints in the technology.

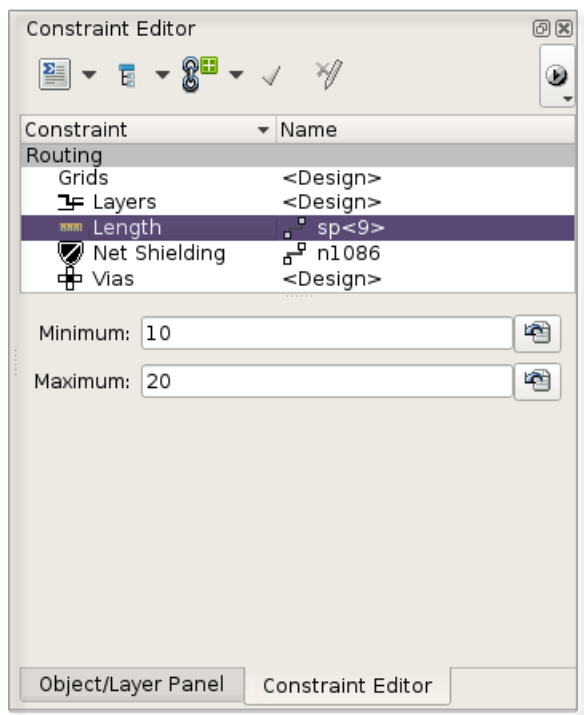
Parameter	Description
Width	The minimum width to be used for the layer. The default value is obtained from the minWidth constraints in the technology.
Direction	Preferred routing direction for the layer. The default value is obtained from the preferredRoutingDirections constraints in the technology.

Length Constraint

You can set the Length constraint to limit the length of a net within a range.

This topic describes the Length constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).

To create Length constraints using the `cm::createLength` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

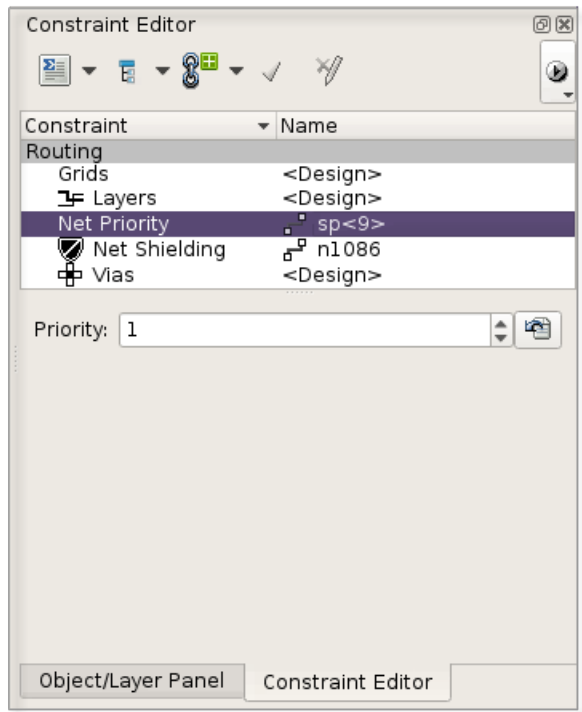


Parameter	Description
Minimum	The minimum length of the net.
Maximum	The maximum length of the net.

Net Priority Constraint

You can set the Net Priority constraint to set the routing priority for a selected net or a group of nets.

This topic describes the Net Priority constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).



Parameter	Description
Priority	Net routing priority. Valid values are -128 to 128, and positive values have higher priority.

Net Shielding Constraint

Shielding is needed around sensitive nets to reduce noise and cross talk. You set the Net Shielding constraint to add shielding to selected nets.

This topic describes the Net Shielding constraint parameters. The parameters specify the type of shielding to add to selected nets. To create or modify the constraint, see [Setting Routing Constraints](#).

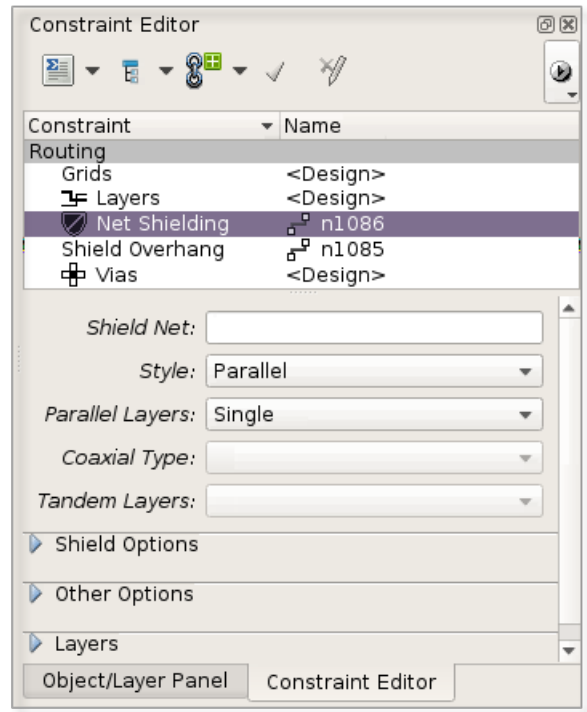
To add Net Shielding constraints using the `cm::createNetShielding` Tcl command, see the *Custom Compiler Tcl Reference Manual*.

To set additional shielding options, see the **Auto Route** dialog box [Shield Tab](#).

See also [Shielding Styles and Behavior](#).

Note:

Net Shielding constraints that you set in the Schematic Editor are also available for viewing and editing in the Layout Editor.



Parameter	Description
Shield Net	The net you want to use as shield. Typically, VSS and VDD are shield nets. Note: The value you specify in the Net Shielding constraint overrides the value you specify in the Auto Route dialog box (auto routing) and the Shielding Options dialog box (interactive routing).
Style	The type of shielding to apply to the wire: • Parallel: The shields are on the same layer as the wire and flank the wire on the left and right sides. Select Parallel if you want to add shielding to a group of nets. m3  • Tandem: The shields are on a layer above and below the wire. m4  m3  m2  • Coaxial: A combination of parallel and tandem shields surround the wire on four sides. m4  m3  m2  For more information, see Shielding Styles and Behavior.
Parallel Layers	Style of parallel group shielding: • Single: Shielding is added on the same layer as the wires and flank the outer most wires in the group. Shield Wires Shield m3  • Multiple: Shielding is added on same layer, as well as the layer above and below the wires. Shield Wires Shield m4  m3  m2 

Parameter	Description
Coaxial Type	The type of coaxial shielding: • Solid: The shields above and below the wire are solid pieces.  m4 m3 m2 • Split: The shields above and below the wire are split into smaller pieces to prevent width violations.  m4 m3 m2 • Center Only: The shields above and below the wire provide minimal coverage for the wire.  m4 m3 m2 For more information, see Coaxial Shielding.
Tandem Layers	The layers that are hosting the shields: • Both: Above and below the wire. • Above: Above the wire only. • Below: Below the wire only.

The **Shield Options** section contains the following:

Shield Options

Minimum Segment:

0

Top Offset:

1

Bottom Offset:

1

Default Width:

0.07

Default Gap:

0.08

Default Max Gap:

Parameter	Description
Minimum Segment	The minimum segment length to add shielding.

Parameter	Description
Top Offset	Shield layer offset from the wire. For example, if the wire is on m3 and Top Offset is 2, then the shield above the wire is on m5. Default: 1 Top Offset = 2 m5  m4 m3  Wire
Bottom Offset	Shield layer offset from the wire. For example, if the wire is on m3 and Bottom Offset is 1, then the shield below the wire is on m2. Default: 1 Bottom Offset = 1 m3  Wire m2 
Default Width	The default width of the shielding material on all the layers.
Default Gap	The distance between the wire and the shield.

Parameter	Description
Default Max Gap	The distance depends on the shielding application: <i>Shared shielding</i>: The maximum distance between the wire and the shared shielding. The shielding must be placed within the specified distance to be considered a shared shield. Shared Shielding  Wire Shield Wire Max Gap Max Gap <i>Group shielding</i>: The maximum distance between two inner-most wires in the group. If the actual distance is greater than the Default Max Gap, a shield will be added between the two wires. Group Shielding  Wires Wires Max Gap

The **Other Options** section contains the following:

Other Options

Second Shield Net:

Enclose Via:

Default

Enclose Pin:

Default

Vias:

Group Shield:

☐

Sharing:

Parameter	Description
Second Shield Net	The second net you want to use as shield. Note: The value you specify in the Net Shielding constraint overrides the value you specify in the Auto Route dialog box (auto routing) and the Shielding Options dialog box (interactive routing).
Enclose Via	The option to enclose vias with shielding. Select one of the following: • Default: Uses the <code>rtShieldEncloseVia</code> preference setting. • False: Does not enclose vias with shielding. • True: Encloses vias with shielding.
Enclose Pin	The option to enclose pins with shielding. Select one of the following: • Default: Uses the <code>rtShieldEnclosePin</code> preference setting. • False: Does not enclose pins with shielding. • True: Encloses pins with shielding.
Vias	The via definitions to use to connect the shielding segments. Click  , and then select one or more via definitions from the Select Via Definitions list. Although custom vias are available for selection, they will not be used. If you want to use the corresponding signal's via definitions for shield via creation, do not use this field. For information, see Shield Vias.
Group Shield	Adds shielding to the outer sides of a group of nets that connect macro boundary pins. This option is applicable when you are Adding Post-Route Shielding using the Add Shield command only. For more information, see Group Shielding. If you select Group Shield, you must choose the Parallel or Coaxial shielding style.
Sharing	The option to permit shield to be shared among signal nets in the same group that have parallel and split coaxial shield styles. Shield Sharing does not apply to other shield styles. For more information, see Shared Shielding. Select one of the following: • Default: Uses the <code>rtShieldSharing</code> preference setting. • False: Does not share shield. • True: Shares shield only in cases where the parallel and split coaxial shield styles are used.

The **Layers** section lets you specify the following parameters:

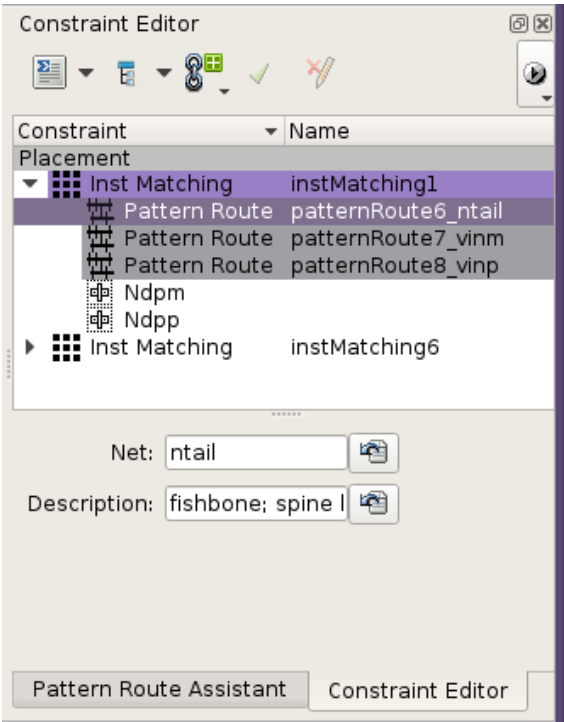
Layers				
Layer	Enabled	Gap	Width	Max Gap
DIFF	☒	0.04	0.04	
POLY	☒	0.056	0.01	
M1	☒	0.03	0.022	
M2	☒	0.03	0.022	
M3	☒	0.03	0.022	
M4	☒	0.03	0.022	
M5	☒	0.088	0.044	
M6	☒	0.088	0.044	
M7	☒	0.088	0.044	

Parameter	Description
Layer	The name of the layer. The field is read only. You cannot modify the layer names.
Enabled	The option to turn on or turn off shielding on the specific layer. The router does not add shielding on layers where you have turned off shielding.
Gap	The distance between the wire and the shield on the layer.
Width	The minimum width of the shielding material on the layer.
Max Gap	The maximum distance between the signal and the shared shielding on the layer. The shielding must be placed within the specified distance to be considered a shared shield.

Pattern Route Constraint

You can save and reuse route patterns within a design. When you realize and save a route pattern, the constraints of the pattern are saved, and you can later review or modify the constraint parameters.

This section describes the Pattern Route constraint parameters. To create and realize route patterns, see [Pattern Routing](#). To save the route pattern as a template using the Template Manager, see the *Custom Compiler SDL and Visually Assisted Automation Guide*.

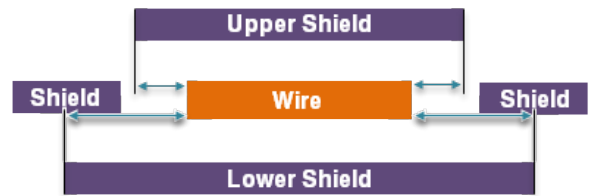


Parameter	Description
Net	The net for which the route pattern applies.
Description	The description of the pattern, in the RPD (routing pattern directive) text. For details, see Route Pattern Description Reference.

Shield Overhang Constraint

To reduce parasitic capacitance, you can create constraints to permit overhanging shields in center-only [Coaxial Shielding](#). Each shield layer above and below the signal layer overhangs both sides of the wire by the same width. However, the upper and lower shields can have different overhanging widths.

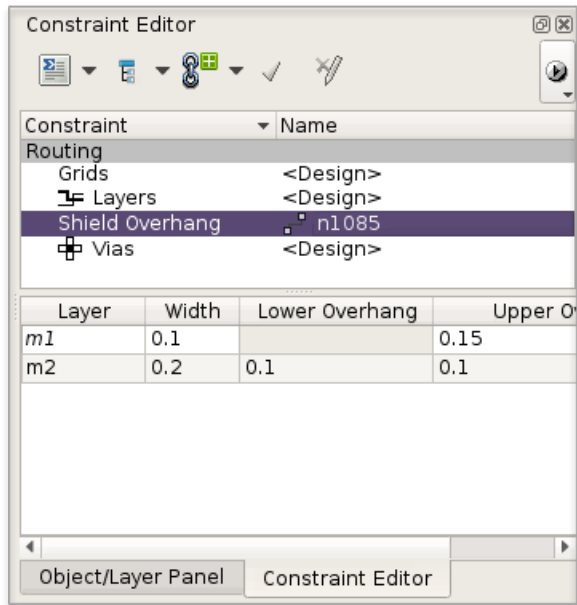
Cross-section view of a wire and center-only coaxial shields



Each Shield Overhang constraint applies to a specific signal (metal) layer. When you set the constraint, you must specify the following:

- Signal layer
- Width of the wire
- Overhanging distance, measured from the edge of the wire

This topic describes the Shield Overhang constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#). To set shielding options, see [Net Shielding Constraint](#) and the **Auto Route** dialog box [Shield Tab](#).

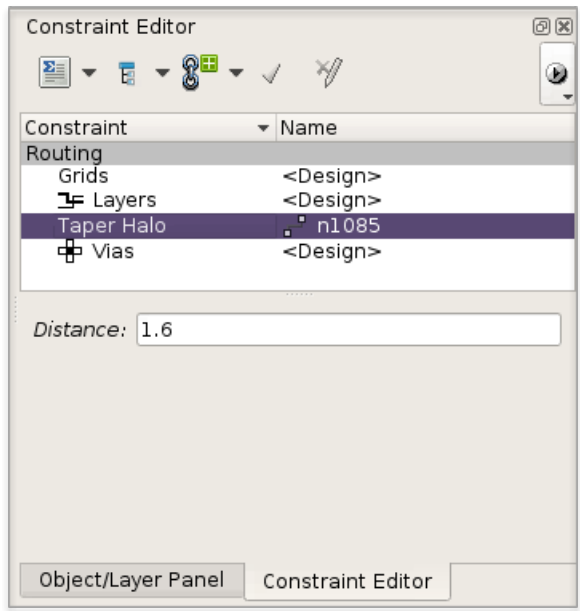


Parameter	Description
Layer	The layer on which the wire is located. To add a layer, right-click anywhere in the table and choose Add Row .
Width	The width of the wire.
Lower Overhang	The distance beyond the edge of the wire that the lower shield must extend. Default: 0 The diagram shows an orange rectangle labeled "Wire" positioned above a longer purple rectangle labeled "Lower Shield". Two horizontal double-headed arrows indicate the distance from the left and right edges of the wire to the corresponding edges of the shield, representing the lower overhang.
Upper Overhang	The distance beyond the edge of the wire that the upper shield must extend. Default: 0 The diagram shows a purple rectangle labeled "Upper Shield" positioned above an orange rectangle labeled "Wire". Two horizontal double-headed arrows indicate the distance from the left and right edges of the shield to the corresponding edges of the wire, representing the upper overhang.

Taper Halo Constraint

You can set Taper Halo constraints to taper wire up or down to connect to pins of different widths on a net.

This topic describes the Taper Halo constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).



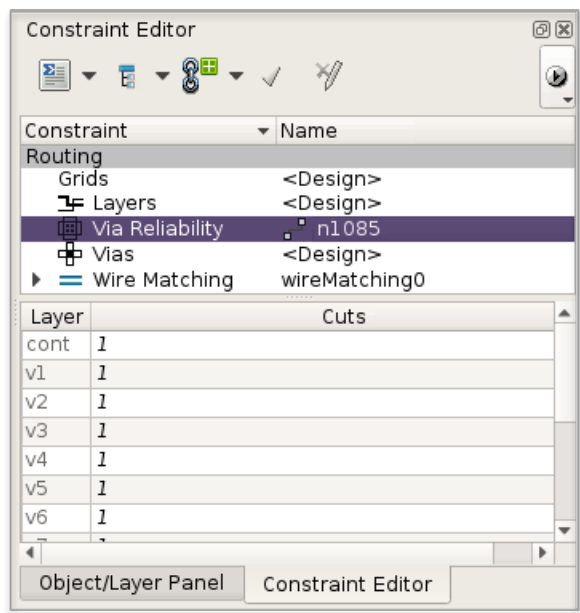
Parameter	Description
Distance	Distance between the edge of the pin and the location at which the tapering begins in the wire.

A diagram illustrating the Taper Halo constraint. It shows an orange rectangle labeled 'Wire' on the left. To its right is a blue square labeled 'Pin'. A dashed rectangle surrounds the pin. A horizontal double-headed arrow is positioned between the right edge of the wire and the left edge of the dashed rectangle, indicating the 'Distance' parameter. Vertical double-headed arrows are also shown within the dashed rectangle, indicating its width.

Via Reliability Constraint

You can set Via Reliability constraints to control the minimum number of via cuts for a specific layer. The constraint creates a fixed number of rows for each layer. The order of the rows is the increasing mask number of the cut layer.

This topic describes the Via Reliability constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).



Parameter	Description
Cuts	Minimum number of Cuts for each layer. To route with redundant vias (or double-cut vias), set the Cuts value to 2 or greater for each of the required layers.

Vias Constraint

You can set a Vias constraint to specify how the router should use the vias during routing. You can force the router to use vias in the order you specify, or you can let the router use the optimal via as it proceeds.

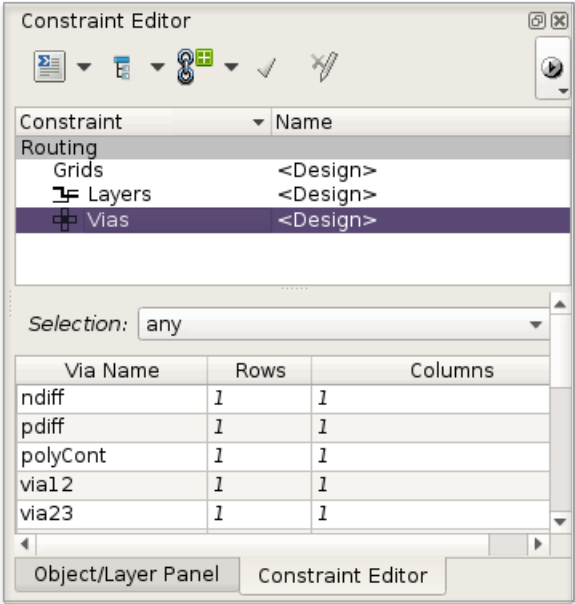
Caution:

You should provide standard via models that the Custom Router can manipulate during the routing process. However, the router is allowed to rotate individual metal shapes within a via model (but not the cut layers). Also

the router could create additional metal extensions to fulfill Minimum Area requirements.

This topic describes the Vias constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#).

For information about controlling the minimum number of via cuts per layer, see [Via Reliability Constraint](#).



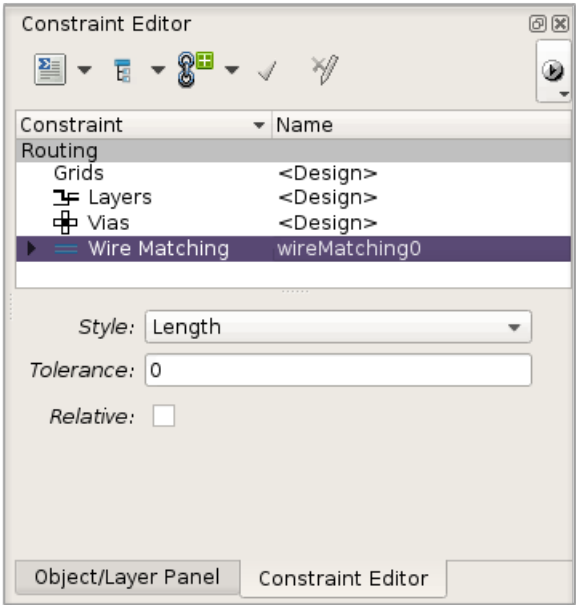
Parameter	Description
Selection	The priority to use during routing: • first: Forces the router to use the vias in the order shown in the table. Note that you can use a Tcl script to set the via priority layer by layer. For information on how to write Tcl scripts, see Introduction to Custom Compiler Tcl. • any: Uses the optimal via during routing. The router ignores the via order in the table.
Via Name	The name of the via in the net. Click the name to select the desired via from the list. You can set the order by selecting the desired vias in the order shown in the table.
Rows	The number of rows of the via array, if applicable.
Columns	The number of columns of the via array, if applicable.

Wire Matching Constraint

You can set Wire Matching constraints so that the total wire (interconnect) length or resistance of the nets falls within the specified range.

This topic describes the Wire Matching constraint parameters. To create or modify the constraint, see [Setting Routing Constraints](#). To specify the lengthening style, see [Matching Tab](#).

To create Wire Matching constraints using the `cm::createWireMatching` Tcl command, see the *Custom Compiler Tcl Reference Manual*.



Parameter	Description
Style	The type of matching to use: • Length • Resistance • Length Per Layer
Tolerance	The amount of variation permitted, in micrometers or ohms, depending the Style you selected.

Parameter	Description
Relative	The option to regard the Tolerance value as an absolute measurement or a percent. Select Relative to regard the Tolerance value as a percent. For example, if you select Relative, a Tolerance value of 5 is regarded as 5%. Clear the Relative check box to regard the Tolerance value as an absolute measurement. For example, if you clear the check box, a Tolerance value of 5 is regarded as 5 nm or 5 ohms, depending on the Style you selected.

B

Interactive Routing Cursor Movement and Behavior

This appendix provides examples of how basic cursor movements are interpreted by the router.

This appendix describes interactive routing cursor movements in the following cases:

- [Basic Interactive Cursor Movements](#)
- [Interactions with Obstructions](#)
- [Routing Over and Beyond Objects](#)
- [Routing Around Obstacles in Trackless Designs](#)
- [Adding Points Along the Route](#)
- [Adding Vias](#)
- [Unroutable Objects](#)
- [Jogs and Via Costs in Routing](#)
- [Self-Intersecting Routing Patterns and Same-Net Obstructions](#)

Basic Interactive Cursor Movements

When you begin to move the cursor on the canvas after choosing the **Routing > Interactive Route** command, the router displays a DRC-correct route from the first point you add on the canvas to the current cursor position. As you continue to move the cursor, the router computes and displays the best possible legal route, based on layer constraints, preferences, and options.

Figure 3 shows an example of tie breaking for routes of equal cost.

Figure 3 Tie Breaking Route

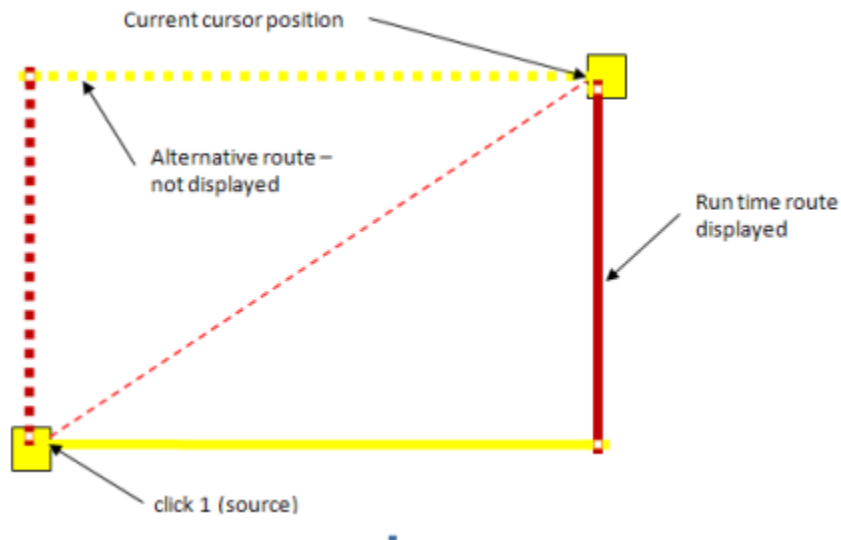
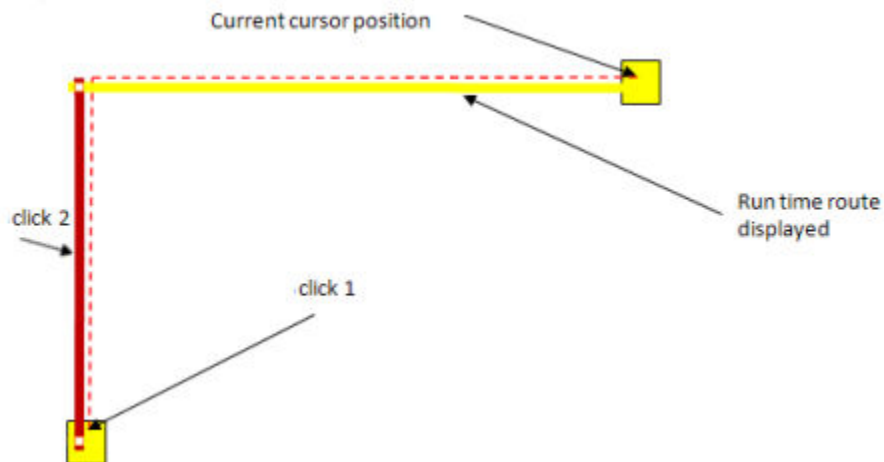


Figure 4 shows how you could insert an intermediate point for the situation displayed in Figure 3, if you wanted the alternative route.

Figure 4 Alternate Routes



Interactions with Obstructions

Figure 5 through Figure 9 demonstrate examples of interactions between design obstructions and generated routes.

Figure 5 *Interaction with Blockage*

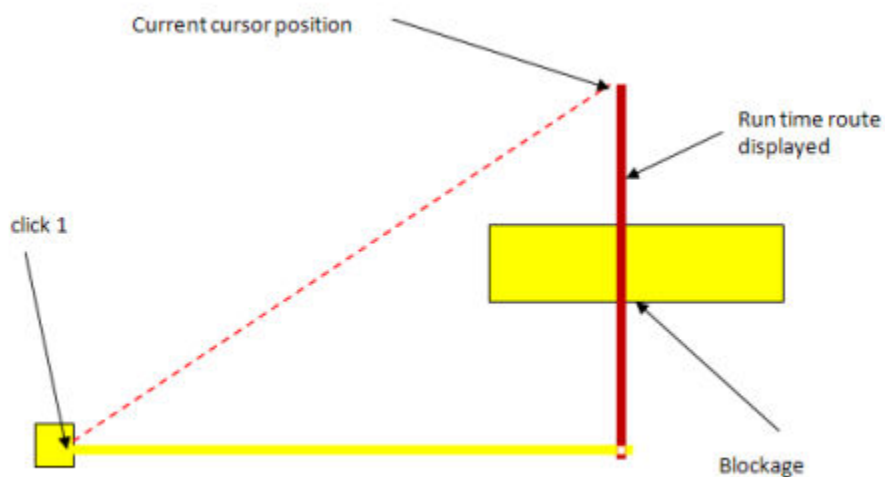


Figure 6 *Interaction with Blockage*

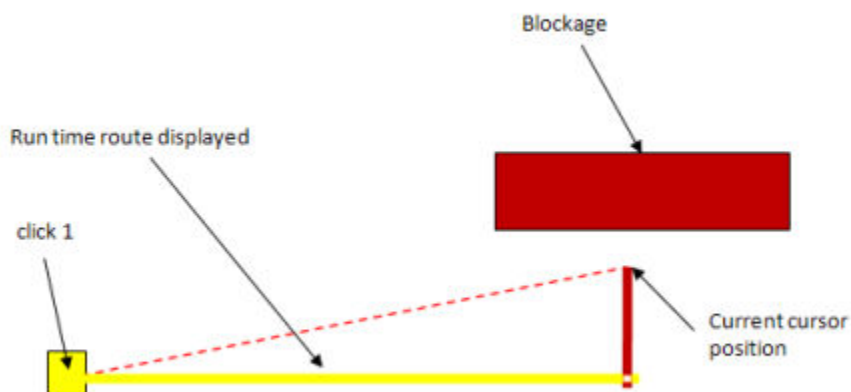


Figure 7 shows how the path followed by the mouse does not affect the displayed run-time pool route. However, as the cursor moves from the situation in Figure 6 to the situation in Figure 7, the displayed route will jump from the route displayed in Figure 6 to the route displayed in Figure 7.

Figure 7 *Interaction with Blockage*

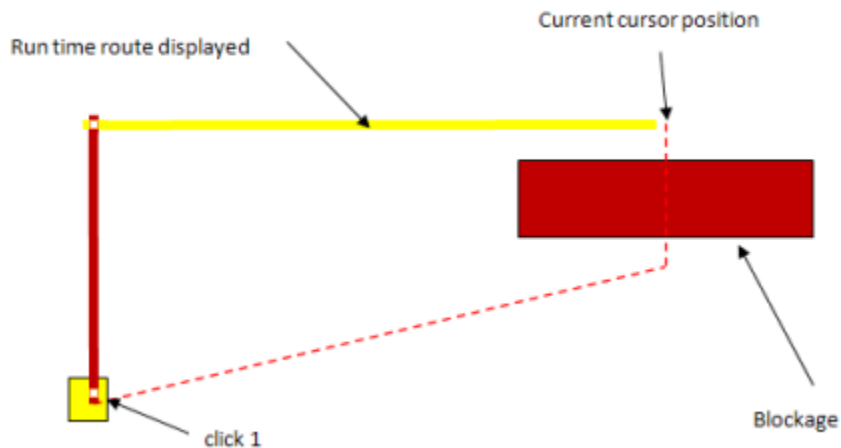


Figure 8 *Interaction with Blockage*

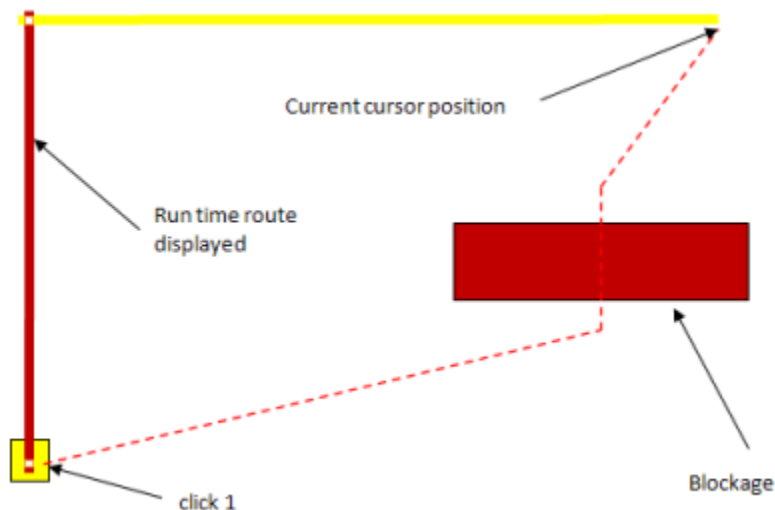
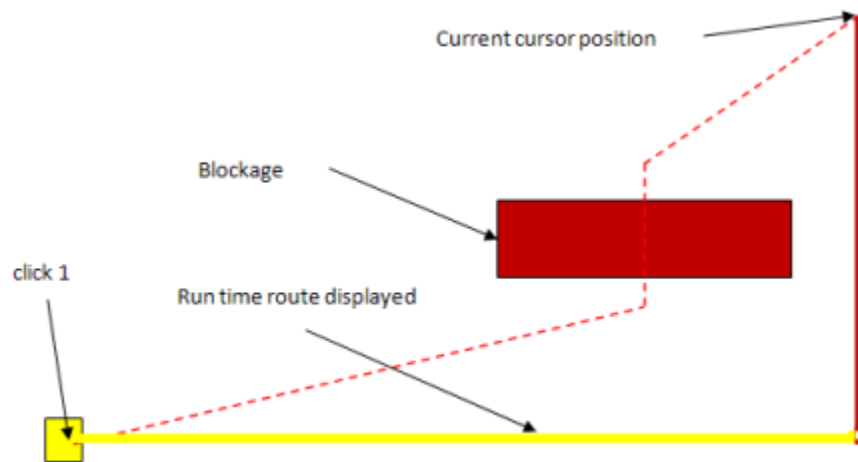


Figure 9 Interaction with Blockage



Routing Over and Beyond Objects

When routing over an object that is compatible for connectivity (for example, connectivity aware mode is off or object is not netted), the connection will be made to the active object under consideration (Figure 10).

When the cursor moves off the object's geometry, the objects will be considered to be an obstruction to be avoided (Figure 11).

Figure 10 Over Compatible Geometry

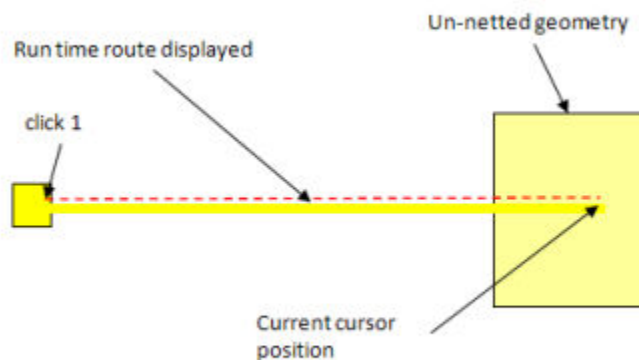
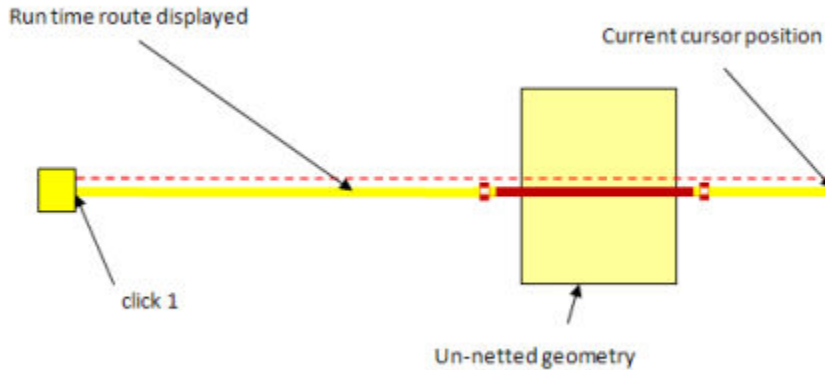


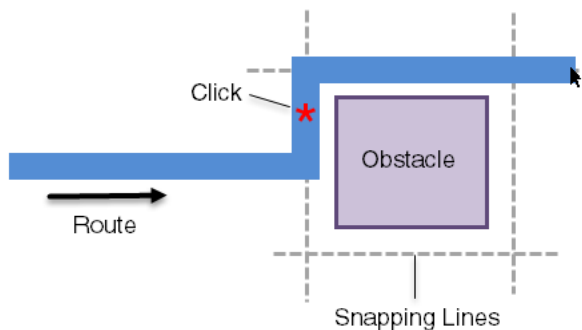
Figure 11 Over Compatible Geometry and Beyond



Routing Around Obstacles in Trackless Designs

In trackless designs, the Custom Router provides invisible snapping lines so that routes can be placed close to obstacles without violating spacing and other design rules. By eliminating time-consuming guesswork near the obstacles, the snapping lines can facilitate and accelerate routing tasks.

During interactive routing, as a route approaches an obstacle, it is snapped to the nearest snapping line surrounding the obstacle. As the route moves around the obstacle, a click along the first side ensures that the router retains its computation. As the route continues, the snapping lines ensure that the route maintains a legal distance.

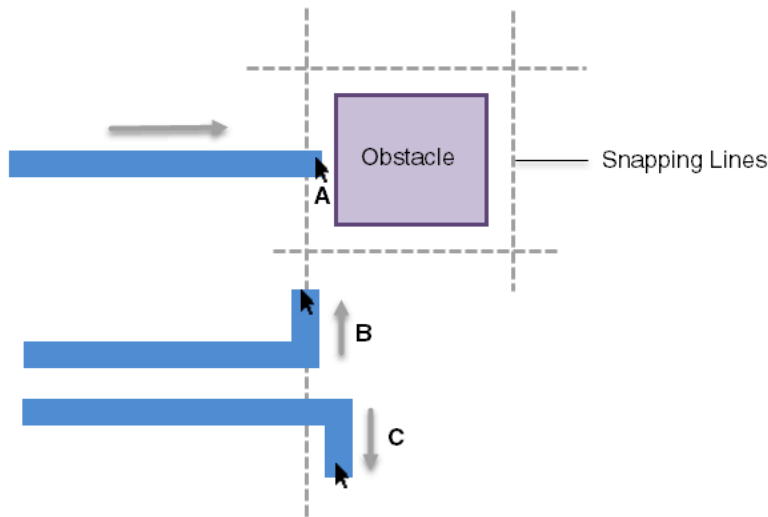


Routes are snapped to snapping lines when:

- The route pointer is between the snapping line and the edge of an obstruction (**A**).
- The route pointer is not between the snapping line and the edge of an obstruction, but it is moving toward the obstruction (**B**).

Snapping does not occur when both of the following conditions are met (**C**):

- The route pointer is not between the snapping line and the edge of the obstruction.
- The route is moving away from the obstruction.



Adding Points Along the Route

An *intermediate add-point event* in follow-the-cursor interactive routing mode is an add-point event that does not start or end the current interactive routing operation.

An add point event can be triggered by one of the following:

- A single click on the canvas
- Hitting the space bar (standard binding)
- A right-click on the canvas (click-and-menu interaction)

For follow-the-cursor interactive routing, an intermediate add point event results in the *freezing* of the route computed so far. The route however will not be committed to the database at this point; it will still reside within the run-time pool memory. The *frozen* route is displayed at the time of the add point event. You can undo this action by removing the last added point (using the Remove Point command or press Backspace). After the add-point event, the editing mode will continue to be interactive follow-the-cursor routing. The new routing source for the continuation of the routing operation will be the last segment of the route *frozen* so far. You can start the next leg of the routing operation from any point along the last *frozen* segment ([Figure 12](#)).

Figure 12 Add Point Event

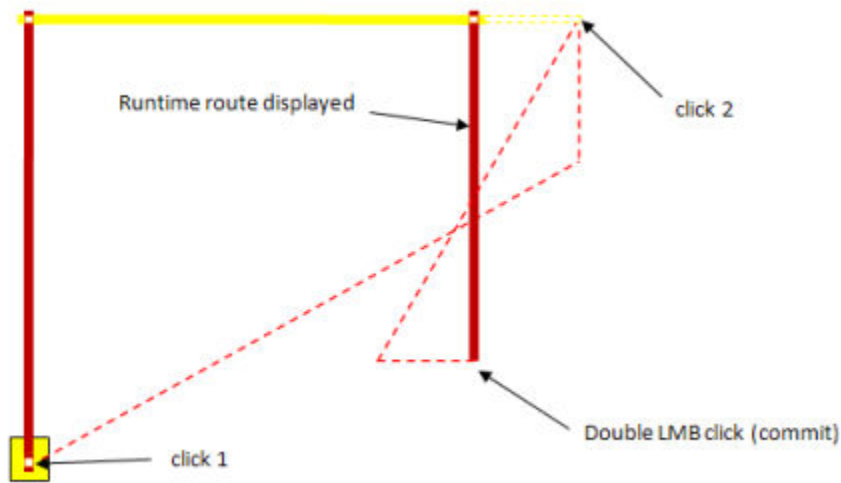
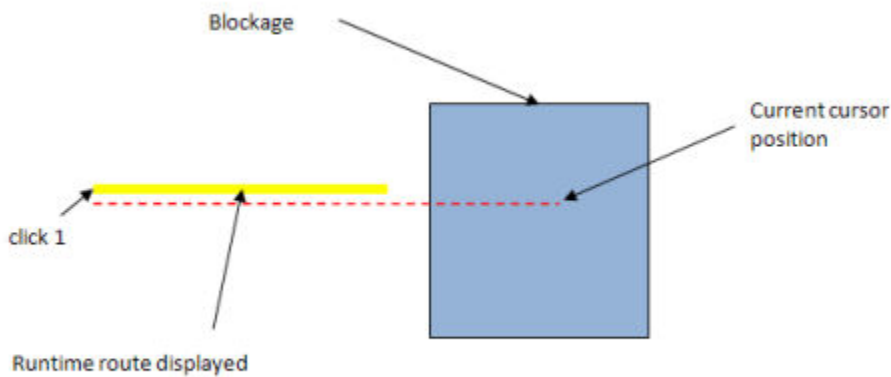
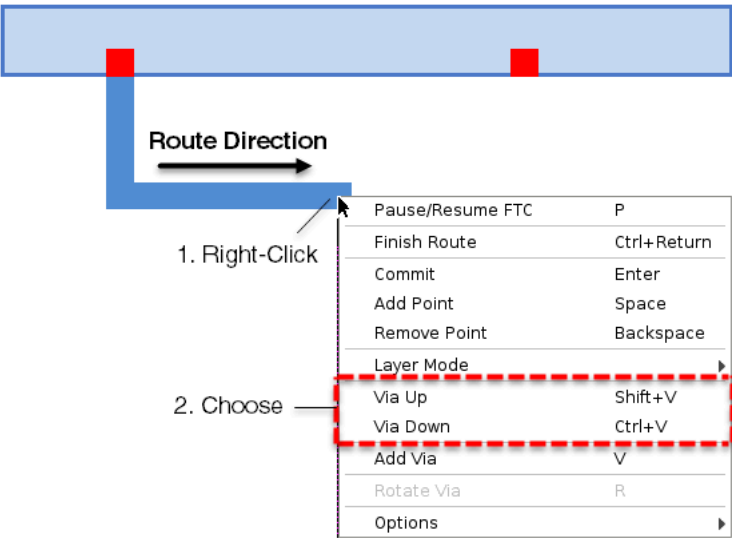


Figure 13 No Legal Solution

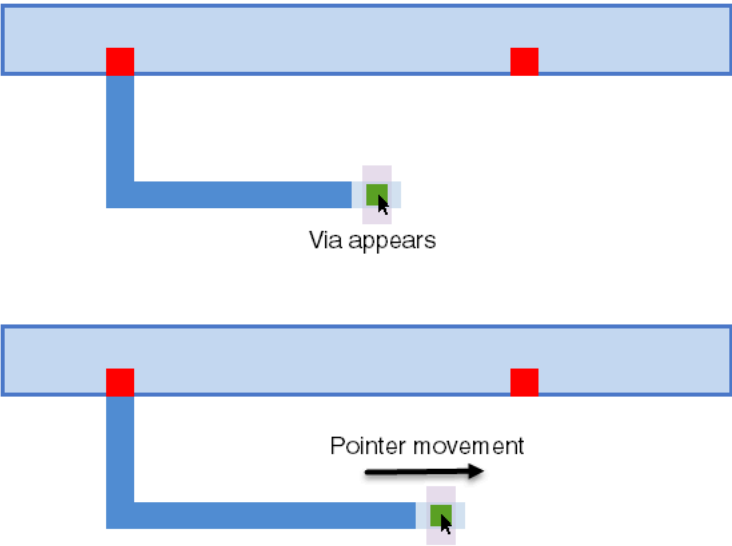


Adding Vias

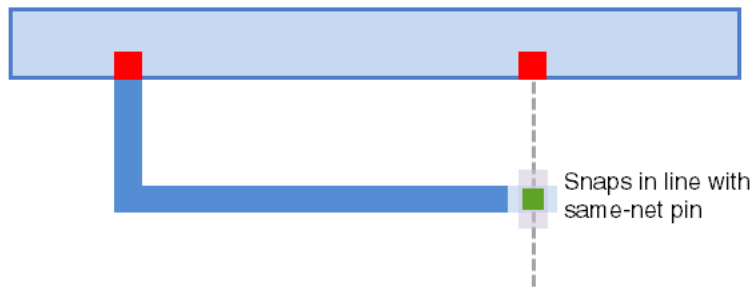
During follow-the-cursor single-layer interactive routing, you can right-click and choose **Via Up** or **Via Down** to add a via above or below the current layer. See [Adding Vias](#).



As soon as you choose **Via Up** or **Via Down**, a standard via appears under the pointer, at the end of the current route. The via moves with the pointer.



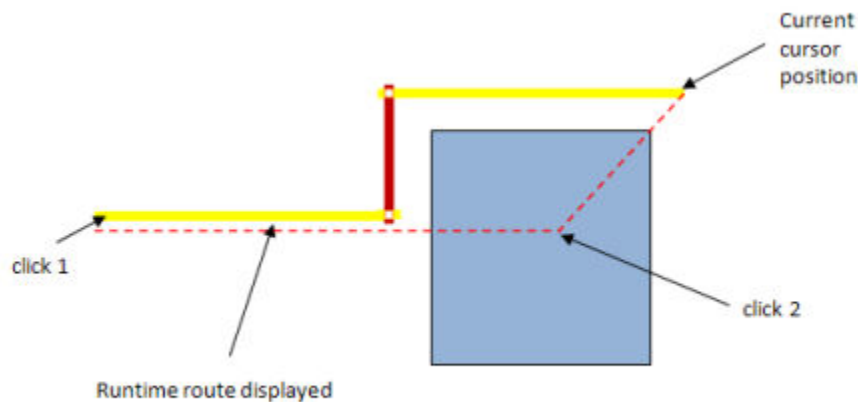
As you move the via close to a same-net pin that is on the same layer, the via snaps to a location directly in line with the pin. The snapping eliminates any position guesswork and facilitates the routing and via placement process.



To add the via, click the canvas, or right-click and choose **Add Point**. See [Adding Vias](#).

Unroutable Objects

The router does not add routes to unroutable objects. For example, in the follow-the-cursor mode, If you click to enter a point at an unroutable object, the router will add a route around the unroutable object to bypass it.



Jogs and Via Costs in Routing

[Figure 14](#) demonstrates the intended behavior when the router is in the follow-the-cursor mode with default preference costing parameter values. If the cursor is moved in such a way as to require a one track jump in the route, a wrong way jog on the same layer will be used rather than switching to the adjacent layer. However if the jump/jog is more than one track, as shown in [Figure 15](#), the router will route that segment using an adjacent layer rather than a two+ track long wrong-way jog. If you would prefer to see no routing layer change in the routing pattern, you can specify the behavior by selecting either the **Single-Layer** or **Multi-Layer** option from the **Route** toolbar or menu ([Figure 16](#)).

Figure 14 *One-track Routing Jog*

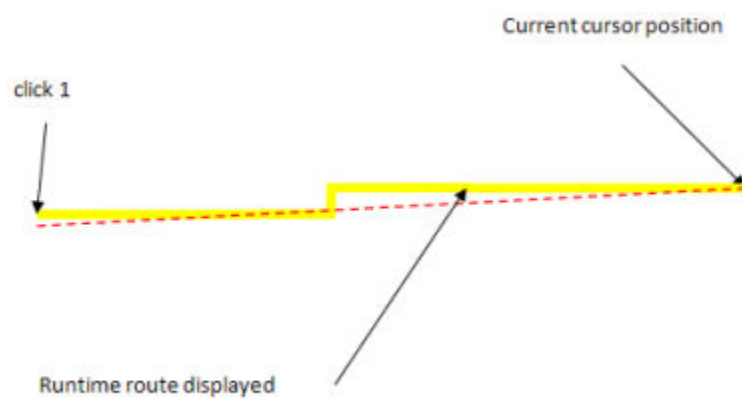


Figure 15 *Multi-track/Multi-Layer Routing Jog*

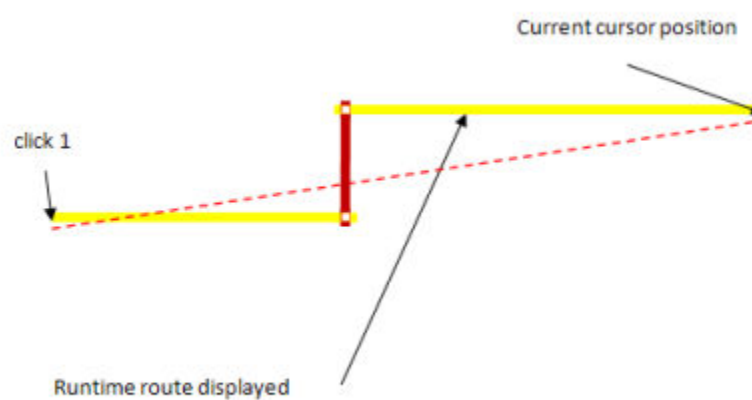
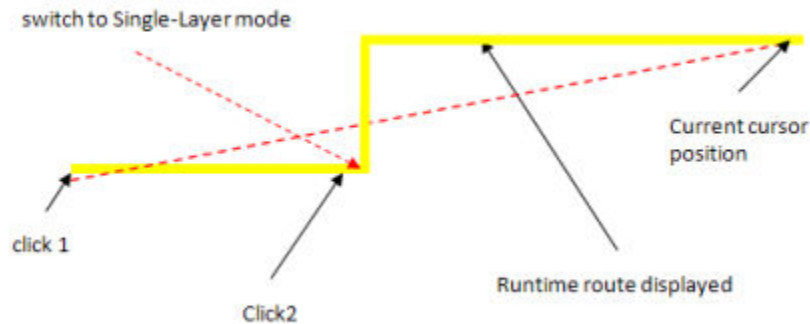
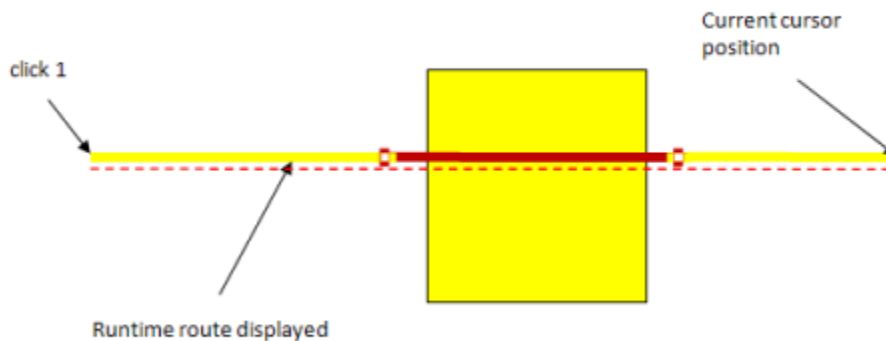


Figure 16 Single-Layer Multi-track Routing Jog



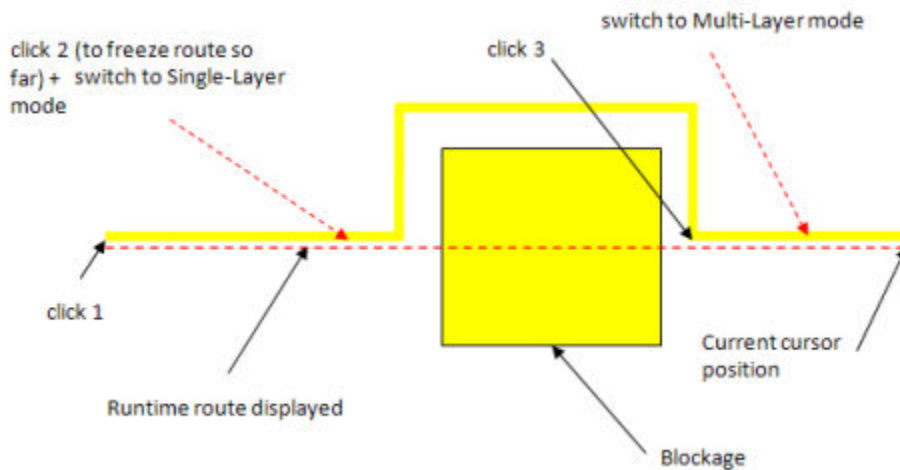
Having minimum parameter values for via and jog costs set by default means that the routing engine, in the follow-the-cursor mode, will almost always perform a via up or down operation to avoid an obstruction, if permissible. [Figure 17](#) illustrates the default behavior in the follow-the-cursor multi-layer routing mode for the mouse movement. The router will automatically create the wrong way wire on M3 in order to clear the M2 obstruction.

Figure 17 Wrong Way Routes to Avoid Obstruction



In order to realize the routing illustrated in [Figure 18](#), you have to use the menu to switch to the Single-layer mode while the route is being created, and then switch back to Multi-layer mode.

Figure 18 Same layer Jogs



Note:

In [Figure 18](#), click 3 is necessary to force the routing to hug the blockage geometry in this specific example.

If the routing automatically generated in the example in [Figure 19](#) is not the desired pattern, you can generate the pattern in [Figure 20](#) through the actions shown. Similarly, the pattern illustrated in [Figure 20](#) can also be realized easily through cursor movements and clicks.

Figure 19 Same Layer Jogs

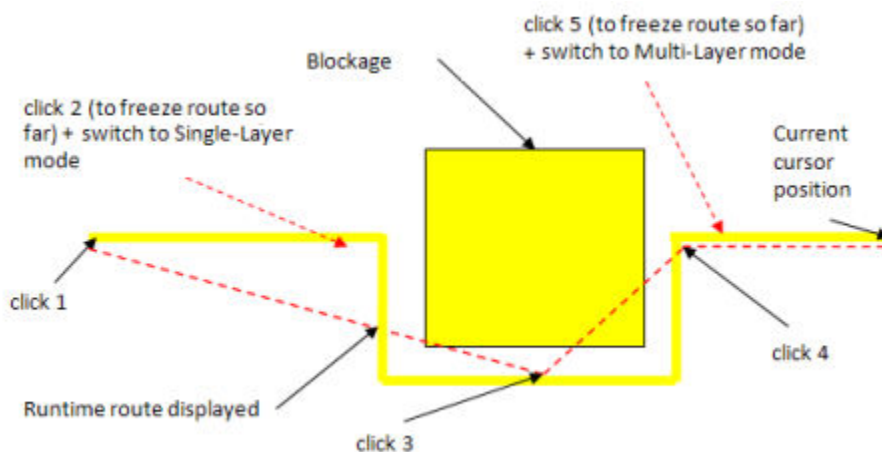
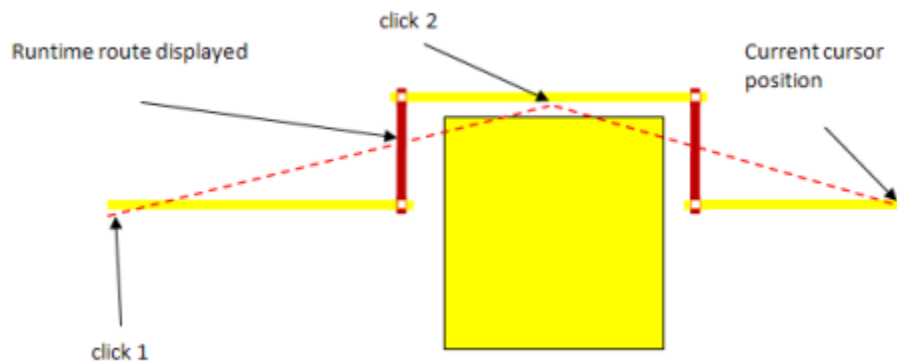


Figure 20 Wrong-way Jogs Minimized



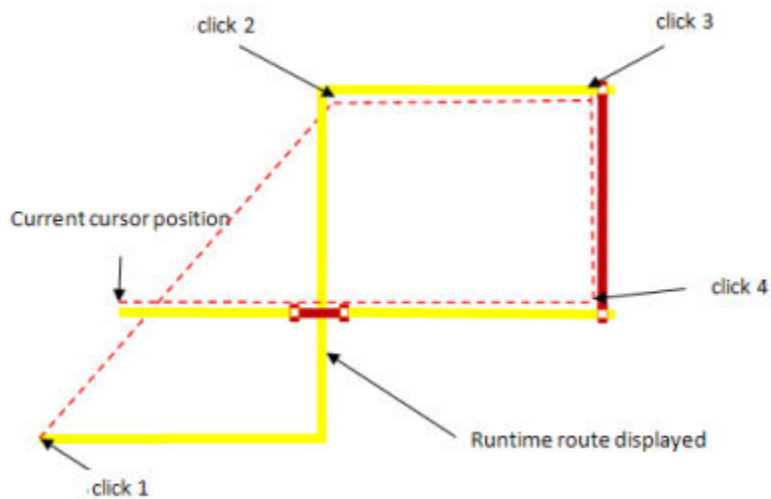
Self-Intersecting Routing Patterns and Same-Net Obstructions

The router treats same net wiring in the runtime pool (prior to a commit) as obstructions to be avoided.

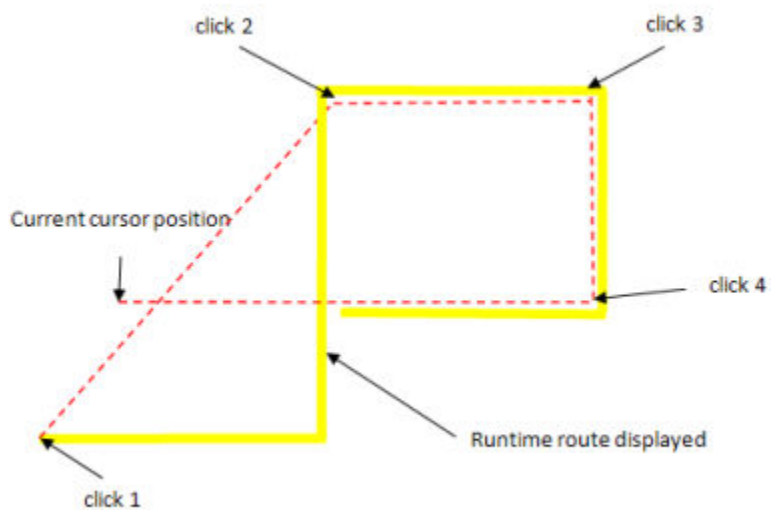
Note:

It is possible to tap into same net wiring when the wiring is not in the run time tool; that is, same net wires, vias, and pins were not created during the current routing operation. The following example shows same-net wiring being treated as a routing obstruction in follow-the-cursor multi-layer mode and how, in this case, the pattern would be routed by the routing engine

Appendix B: Interactive Routing Cursor Movement and Behavior Self-Intersecting Routing Patterns and Same-Net Obstructions



The following example shows how treating same-net wiring in the runtime pool as obstructions results in a truncated wiring pattern for the example depicted (follow-the-cursor single-layer mode).



C

Troubleshooting Error Messages

This appendix presents the list of Custom Router error messages, potential causes of the errors, and ways to resolve the errors.

This appendix contains the following topics:

- [Where the Error Messages Are Displayed](#)
- [Types of Error Messages](#)
- [Custom Router Error Messages](#)

Where the Error Messages Are Displayed

The Custom Router displays error messages in the following locations:

- [Console](#)
- [Text Viewer](#)

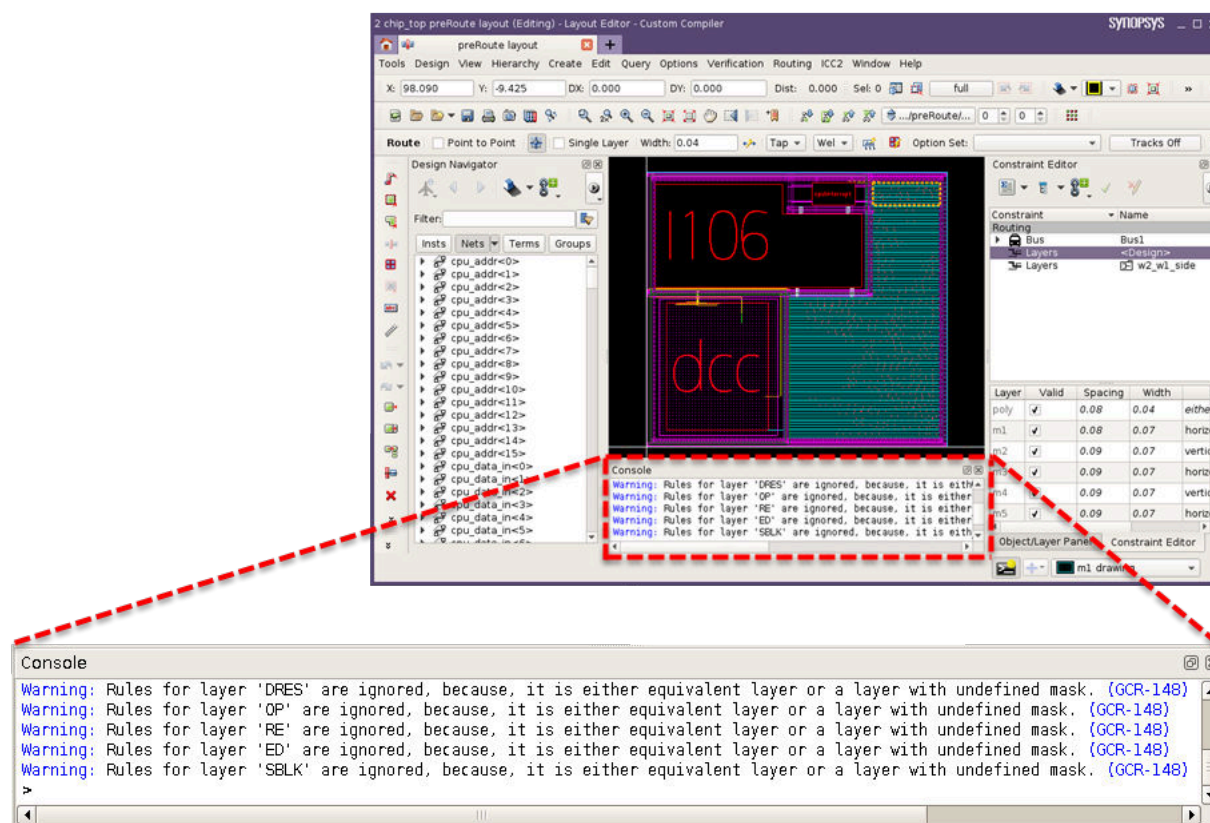
Console

Messages can appear in the Console when you:

- Enable either the Auto Router or the Interactive Router
- Start Auto Routing or Interactive Routing, and while routing is in operation

For information about the Console tab and the Console pane, see the *Custom Compiler Environment User Guide*.

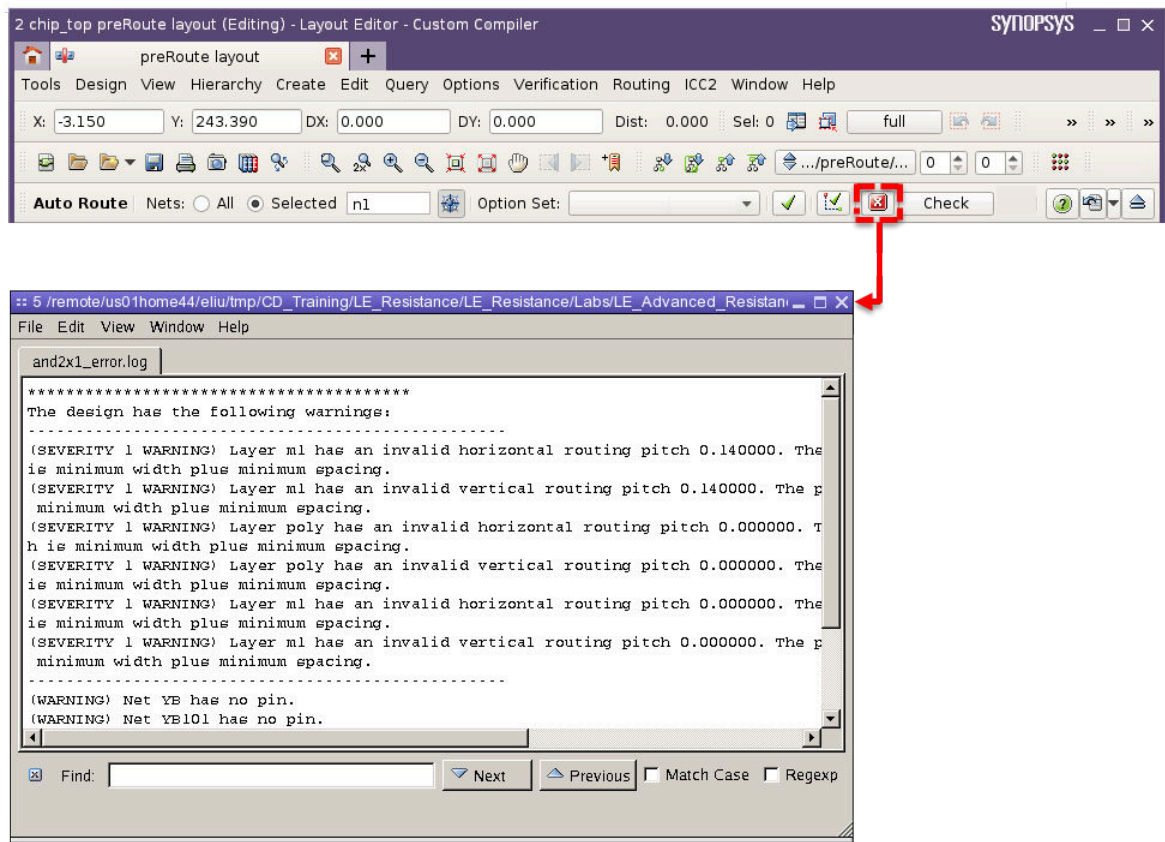
Appendix C: Troubleshooting Error Messages Where the Error Messages Are Displayed



For the descriptions of the message types, see [Types of Error Messages](#). For the list of error messages, their causes, and potential solutions, see [Custom Router Error Messages](#).

Text Viewer

Messages appear in the Text Viewer when you run the Data Checker in preparation for auto routing. For information about the Text Viewer, see the *Custom Compiler Environment User Guide*.



For the descriptions of the message types, see [Types of Error Messages](#). For the list of error messages, their causes, and potential solutions, see [Custom Router Error Messages](#).

Types of Error Messages

The types of messages are as follows:

Type of Message	Description
Error	One or more problems have occurred and the operation is unable to continue. You must correct the problems so that the operation can continue or finish successfully. In some cases, you might have to restart the operation.
Warning	One or more conditions might produce unexpected results or cause problems in the future. Although the operation was able to continue, you might want to follow the recommended corrective actions to change the results or prevent problems in the future.

Type of Message	Description
Information	Informational messages provide notifications only. Corrective actions are typically not necessary. For example, an informational message might let you know that an operation is in progress or has finished.

For the list of error messages, their causes, and potential solutions, see [Custom Router Error Messages](#).

Custom Router Error Messages

Error messages let you know of specific issues that occur during the routing operation.

This topic lists the Custom Router error messages:

- [GCR Error Messages](#)
- [IROUTE Error Messages](#)
- [PATTERN_ROUTE_ASSISTANT Error Messages](#)

Each error message table lists:

- The error and warning messages you might encounter when using the Custom Router. The messages are listed alphanumerically by message ID.
- The type of message: error or warning. See [Types of Error Messages](#).
- The causes of the errors or warnings.
- Ways to resolve the errors or warnings.

If you are still experiencing problems after trying the solutions, or if an error not on the list is displayed, contact [Synopsys Support](#).

Note:

Because Information messages are self-explanatory and do not require corrective actions, they are not listed in the following table.

GCR Error Messages

Error messages that have the GCR- prefixes pertain to the Auto Router. This topic lists the GCR- error messages, possible causes, and possible solutions

ID	Type	Message	Causes	Solutions
GCR-001	Error	Could not execute command '%s' because router is not initialized.	The Custom Compiler is not initialized, so the command cannot be executed.	In the Layout Editor, choose one of the following to initialize the Custom Router: • Routing > Auto Route • Routing > Interactive Route For detailed instructions, see Enabling the Auto Router and Enabling the Interactive Router.
GCR-002	Error	%s' execution failed with message: '%s'	An internal Custom Router command has failed. The reason will be displayed.	Contact Synopsys Support.
GCR-004	Error	The router could not be initialized because there are no layers with material %s in the database.	When reading the technology, there were no layers with the specified material defined.	Make sure the routes are on metal layers. Make sure the layers are defined in the technology database.
GCR-006	Error	The license feature %s could not be checked out because %s.	Either the feature is not installed, or the license for the feature is in use by other users.	Make sure the feature you want to use is installed correctly, or wait until other users have finished using the feature.
GCR-010	Error	The router could not be initialized.	There is a problem with the data and the Custom Router could not initialize.	Contact Synopsys Support.
GCR-011	Error	The router could not be initialized because an Open Access exception occurred: %s.	There is a problem with the OpenAccess database and the Custom Router could not initialize. The OpenAccess exception text is provided for reference.	Verify that your database has the correct permissions. Verify that network drives are connected and accessible on the system you are using. If the data appears correct on disk, Contact Synopsys Support.
GCR-012	Error	The specified path style %s is not valid because %s.	The string you entered does not match the valid path styles. Valid styles are truncate, extend, round, and variable.	Check your script for the correct spelling. Note that there are fewer valid names for path style than for path segment end styles. If the problem persists, contact Synopsys Support.

ID	Type	Message	Causes	Solutions
GCR-013	Error	The specified path segment end style %s is not valid because %s.	The string you entered does not match the valid path styles. Valid styles are truncate, extend, variable, chamfer, and custom.	Check your script for the correct spelling. If the problem persists, contact Synopsys Support.
GCR-014	Error	The router could not create a path segment because %s.	An invalid set of end points, an incorrect width, or an incorrect extension value is preventing the router from creating path segments.	Contact Synopsys Support.
GCR-015	Error	The router could not create a path because %s.	An invalid set of points (with duplicate points or collinear points), an incorrect width, or an incorrect extension value is preventing the router from creating path segments.	Contact Synopsys Support.
GCR-016	Error	The router could not create a %s via because %s.	The router tried to use an incorrect via definition name, the router could not access the via definition when trying to create the via, or the coordinates were illegal.	Contact Synopsys Support.
GCR-018	Error	The router could not be initialized because a general exception occurred.	A general exception error is preventing the router from initializing.	Contact Synopsys Support.
GCR-019	Warning	%s %s has no PR boundary defined.	The design does not have a boundary.	Create a boundary for the design. See Creating and Editing Boundaries.
GCR-020	Warning	%s rule for layer %s has a wire width of 0, which is not allowed.	In the technology database, the <code>minWidth</code> setting for the layer is set to 0.	In the technology database, set <code>minWidth</code> > 0 for the layer.
GCR-021	Warning	Layer %s has no minimum width constraint.	In the technology database, the <code>minWidth</code> constraint is not defined for the layer.	In the technology database, set <code>minWidth</code> > 0 for the layer.

ID	Type	Message	Causes	Solutions
GCR-022	Warning	Layer %s has a maximum width of %d which is less than the minimum width of %d.	The layer's <code>maxWidth</code> value cannot be less than its <code>minWidth</code> value.	In the technology database, set <code>minWidth</code> > <code>maxWidth</code> value.
GCR-023	Warning	Layer %s has no minimum spacing constraint.	In the technology database, the <code>minSpacing</code> constraint is not defined for the layer.	In the technology database, define the <code>minSpacing</code> constraint and set its value to > 0 for the layer.
GCR-024	Warning	%s rule for layer %s has a minimum spacing with value 0, which is not allowed.	In the technology database, the <code>minSpacing</code> setting for the layer is set to 0.	In the technology database, set <code>minSpacing</code> > 0 for the layer.
GCR-025	Warning	Layer %s has a minimum spacing of value 0 at 2D spacing table index (%d, %d), which is not allowed.	The system is expecting a <code>minSpacing</code> value > 0 at the location in the 2D table.	In the technology database, set <code>minSpacing</code> > 0 at the location in the 2D table.
GCR-026	Warning	Layer %s has a minimum spacing of value 0 at 1D spacing table index %d, which is not allowed.	The system is expecting a <code>minSpacing</code> value > 0 at the location in the 1D table.	In the technology database, set <code>minSpacing</code> > 0 at the location in the 1D table.
GCR-027	Warning	Layer %s has an invalid %s routing pitch %f. The pitch cannot be smaller than %f, which is minimum width plus minimum spacing.	An invalid routing pitch value is specified in the technology database.	In the technology database, set the routing pitch to a value that is <code>minWidth</code> + <code>minSpacing</code> .
GCR-028	Warning	Layer %s has a manufacturing grid of 0, which is not allowed.	The <code>mfgGridResolution</code> in the technology database for this layer is 0.	In the Controls section of the technology database, make sure <code>mfgGridResolution</code> > 0.
GCR-029	Warning	Layer %s has 0 size minimum number cut table, which is not allowed.	In the Constraint Editor, the specified layer in the Via Reliability constraint does not have a Cut value.	In the Constraint Editor, check the Via Reliability Constraint and specify the minimum number of Cuts for the layer. Check the <code>minNumCut</code> constraint in the Layer Constraints section of the technology database.

ID	Type	Message	Causes	Solutions
GCR-030	Warning	Layer %s has 0 value in minimum number cut table at index %d, which is not allowed.	In the Constraint Editor, the specified layer in the Via Reliability constraint has a Cut value of 0.	In the Constraint Editor, check the Via Reliability Constraint and make sure the minimum number of Cuts for the layer is greater than 0. Make sure the <code>minNumCut</code> constraint is defined and is greater than 0 in the Layer Constraints section of the technology database.
GCR-031	Warning	Instance %s does not have a valid placement status.	The Placement Status of the instance is Unplaced.	Place the instance in the design. In the Design Navigator, set the Placement Status for the instance.
GCR-032	Warning	Pin %s does not have a valid placement status.	The pin belongs to an instance that has a Placement status of Unplaced.	Place the pin in the design. In the Design Navigator, set the Placement Status for the instance.
GCR-033	Warning	Net %s has no pin. Net must have more than one pin within design prBoundary or within user-specified routing area to be routed. This net will be ignored.	The net is missing pins, so routes cannot be created. Routes require pins to form connections.	Add pins on the specified net. See Creating Pins and Terminals.
GCR-034	Warning	%s %s has shielding gap %f smaller than minimum spacing %f.	The distance between the signal and shield is less than the default minimum spacing value.	In the Constraint Editor, increase the Default Gap value in the Net Shielding Constraint .
GCR-035	Warning	%s %s has shielding width %f smaller than minimum width %f.	The shield width is narrower than the default minimum width value.	In the Constraint Editor, increase the Width value in the Net Shielding Constraint .
GCR-036	Warning	%s has net match length or resistance constraint with max value %f less than minimum value %f.	The Maximum value is less than the Minimum value for the Length or Resistance constraint.	Correct the Minimum and Maximum values for the Length Constraint or Defining Net-Level Resistance Constraints.

ID	Type	Message	Causes	Solutions
GCR-037	Warning	The via stack limit cannot be less than 1.	The <code>viaStackLimit</code> constraint is less than 1 in the technology database.	In the technology database, set <code>viaStackLimit</code> to 1 or greater.
GCR-038	Error	The design has no via definitions that are suitable for routing.	The vias used in the design are not defined in the technology database.	Define the vias in the technology database. For instructions, see Via Definitions Section.
GCR-039	Error	%s has no valid routing layers.	No valid routing layers are selected in the Layers constraint.	In the Constraint Editor, check the Layers Constraint and select the valid routing layers.
GCR-040	Warning	%s has no vertical routing layers.	None of the eligible routing layers has vertical direction.	In the Constraint Editor, check the layer directions in the Layers Constraint . Check the Layer Constraint section in your technology database.
GCR-041	Warning	%s has no horizontal routing layers.	None of the eligible routing layers has horizontal direction.	In the Constraint Editor, check the layer directions in the Layers Constraint . Check the Layer Constraint section in your technology database.
GCR-043	Warning	%s use via %s with cut %s number %d, which is not allowed.	Either the number of rows or number of columns in the via is 0, which is not allowed.	Make sure your vias have at least one cut per row and column.
GCR-044	Warning	%s use via with minimum cut number %d at layer %s, which is not allowed.	In the Via Reliability constraint, the number of Cuts value for the specified layer is less than 1.	In the Constraint Editor, set the number of Cuts value to 1 or greater in the Via Reliability Constraint .
GCR-045	Warning	Instance %s has no master.	The instance does not have a master. The master could be unbound or missing.	Create a master cell for the instance. See Creating and Editing Instances .
GCR-046	Warning	Instance %s is placed outside of design boundary.	The instance has Placed status but is incorrectly placed outside the design boundary.	Move the instance within the boundary, or enlarge the design boundary to enclose the instance.

ID	Type	Message	Causes	Solutions
GCR-047	Warning	Pin %s is placed outside of design boundary.	The pin is incorrectly placed.	Move the pin within the boundary, check the placement of the instance, or enlarge the design boundary to enclose the instance and all of its pins.
GCR-048	Warning	Via constraint for %s contains via %s which has no valid via def. This will not be used by router.	A via you selected in the Vias constraint has a missing or invalid via definition. The router is unable to use this via.	In the Constraint Editor, check the via selections in the Vias Constraint . Make sure that selected vias have valid via definitions or remove them from the Vias Constraint.
GCR-049	Warning	%s shorts with %s on layer %s at %s.	Two pins short circuited on the specified layer.	Change the placement of one instance so the pins no longer cause short circuitry.
GCR-050	Warning	Design %s has no net.	The design does not have any nets. You can only create routes in designs that have nets.	Create nets in the design. For instructions on creating nets, see Working With Nets .
GCR-053	Warning	Net %s shorts with net %s on layer %s at %s.	Two shapes short circuited on the specified layer.	Check the overlapping shapes and move one of them to fix the short circuitry.
GCR-054	Warning	Layer %s doesn't have a standard via available to move up.	No standard via definition was found for the specified layer.	In the Constraint Editor, select a standard via for the layer in the Vias Constraint .
GCR-057	Warning	Layer %s has %s routing pitch %f bigger than %f, which is 10x(width+spacing).	The given route pitch is very large compared to the width and spacing rules.	In the Constraint Editor, check the Pitch value in the Grids Constraint .
GCR-058	Error	Cannot define Group %s. Net is either NULL or pins have different net names.	The attempt to define a group for an oaRoute or topology_edge failed because a pin has a null net or a different net from the other pin.	Check the definition of your topology routing for the correct end points.

ID	Type	Message	Causes	Solutions
GCR-059	Warning	Net %s has shielding gap %f but the clearance for width %f is %f in layer %s spacing table. Loss of shielding can occur.	The shielding gap conflicts with the layer width and layer spacing table.	In the Constraint Editor, check the Gap parameters in the Net Shielding Constraint .
GCR-061	Error	Shielding net %s but it does not contain any valid shield rules.	The net was specified for shielding in the command net list, but the net does not have a valid shield constraint associated with it.	In the Design Navigator, check that you have selected the correct nets. In the Constraint Editor, check the Net Shielding Constraint . Make sure the constraint is applied to the correct nets.
GCR-064	Error	Unable to write detailed antenna report because file %s could not be opened.	The attempt to write the report file failed. The disk might be full, the directory might have not write access, or the file might already exist.	Check the disk space. Check and make sure you have write access to the directory. Check that the file with the same name does not already exist.
GCR-066	Warning	Shielding net %s has no pin. No shield vias can be added.	The net selected to shield your connections does not have any valid place to insert the shield vias.	Contact Synopsys Support.
GCR-067	Warning	Net %s has %f shield gap, but coaxial shielding width %f requires %f clearance in layer %s spacing table. Please change shield constraint to split-coaxial.	A coaxial shield requires more space for the specified width.	In the Constraint Editor, select the Split-Coaxial shielding Style in the Net Shielding Constraint .
GCR-068	Warning	Width %.10g is not an even multiple of the layer manufacturing grid. Router will modify width to %.10g.	The width must be an even multiple of the manufacturing grid to avoid errors in precision.	The router automatically adjusts the width to the value shown.

ID	Type	Message	Causes	Solutions
GCR-069	Warning	Cannot complete Interactive Router solution at the specified point. Topology constraints on the net prevent the creation of the topology that would result from completing a route at the specified point with the specified start. If applicable, ensure that the route start point is within the Steiner geometry.	The Interactive Router cannot end at the specified point because of the given topology specifications.	If attempting to begin or end the route at a steiner, make sure that the specified point is within the steiner geometry.
GCR-070	Warning	Cannot start Interactive Router from the specified point. Topology constraints on the net prevent the creation of the topology that would result from completing a route at the specified point.	The Interactive Router cannot begin at the specified point because of the given topology specifications.	If attempting to begin or end the route at a steiner, make sure that the specified point is within the steiner geometry.
GCR-071	Warning	Cannot start Interactive Router from the specified point. No object was selected or the selected object is not a routable object.	The Interactive Router did not detect a valid object at the selected point. Either there was no object or the object is not valid for routing.	Click an appropriate object on the canvas, such as a pin, steiner, or wire and try again.
GCR-072	Warning	Cannot complete Interactive Router operation. Cannot find a legal snapping point to nearest routable target.	The Interactive Router did not detect a valid target point for routing.	Click close to an appropriate object, such as a pin, steiner, or wire and try again.
GCR-073	Warning	Cannot complete Interactive Router operation. Cannot find a nearby routable target.	The Interactive Router did not detect a valid target point for routing.	Click close to an appropriate object, such as a pin, steiner, or wire and try again.
GCR-074	Warning	Add Point event ignored. Specified point has previously been added.	You clicked twice in the same area.	Do not click more than once in a small area.

ID	Type	Message	Causes	Solutions
GCR-075	Warning	Cannot complete Interactive Router operation at specified point. The selected object is not a routable target.	You have selected a non-routable object for interactive routing.	Click an appropriate object, such as a pin, steiner, or wire and try again.
GCR-076	Warning	Cannot complete Interactive Router operation. Cannot find a legal snapping point on specified routable target.	The target you have selected does not have a legal snapping point. The target might be off grid, or might be blocked by other nearby objects.	Make sure the target is on grid. Check for nearby obstacles before adding routes.
GCR-077	Warning	Cannot provide an Interactive Router solution from the starting point. The starting pin/wire appears to be covered by a routing obstruction. Suggest attempting the route from a different point.	One or more obstacles might be blocking the starting pin or wire.	Start the route from a different pin or wire.
GCR-078	Warning	Cannot provide an Interactive Router solution to the targeted pin/wire which appears to be covered by a routing obstruction. Suggest attempting the route to a different point.	One or more obstacles might be blocking the target pin or wire.	Try to end your route at a different pin or wire.
GCR-079	Warning	Cannot provide an Interactive Router solution to the targeted pin/wire. Verify obstructions near the target for constraints. Alternatively, verify track patterns at the targeted object.	One or more obstacles might be blocking access to the target pin or wire. Track patterns may not permit routing to the target.	Check nearby objects and review your track pattern definitions for conflicts.

ID	Type	Message	Causes	Solutions
GCR-080	Warning	Cannot provide an Interactive Router solution to the targeted pin/wire within resource limits. Routing failure occurs somewhere between the routing source and target. Verify for obstructions. Alternatively, verify that source and target are not separated by unroutable layers. Alternatively ensure that P2P guidance is orthogonal.	One or more obstacles might be preventing the router from creating a route between the specified source and target objects.	Check for obstacles along the desired route. You might have to create a different route. Check that the source and target shapes are not separated by unroutable layers. Make sure the routes are in orthogonal directions only.
GCR-081	Warning	Cannot provide an Interactive Router solution from the designated pin/wire. Verify obstructions near the source for constraints.\nAlternatively, verify track patterns at the starting object.	One or more obstacles might be preventing the router from creating a route between the specified source and target objects.	Check for obstacles along the desired route. You might have to create a different route. Check that the source and target shapes are not separated by unroutable layers. Make sure the routes are in orthogonal directions only.
GCR-082	Warning	Cannot complete Interactive Router operation. Provide additional intermediate points by left clicking as you move the cursor to the target. Alternatively, verify: valid routing layers, Non Default Rules, track patterns, pin accessibility. Alternatively, try setting the Ignore Samenet DRCs option.	Invalid routing layers, constraint violations, track patterns, and obstacles might be preventing the router from creating a route between the specified source and target objects.	Check for obstacles along the desired route. Check for constraint violations and fix them. Check the track pattern definition for conflicts.
GCR-083	Warning	Cannot provide an Interactive Router solution from the starting point. The starting pin/wire location appears to conflict with local track pattern constraints.	A conflict exists between the starting point and the track pattern constraints.	Review your track pattern definitions.

ID	Type	Message	Causes	Solutions
GCR-085	Warning	Interactive solution could only be found by ignoring routing blockages. This may result in DRCs. The routing blockages preventing a clean route appear to be located near the targeted pin/wire.	Blockages near the target pin or wire are causing the router to ignore the routing blockages to route successfully.	Review your DRC report carefully. The Interactive Router might be creating violations when it ignores a routing blockage.
GCR-086	Warning	Interactive solution could only be found by ignoring routing blockages. This may result in DRCs. The routing blockages preventing the generation of a clean route appear to be located near the source pin/wire.	Blockages near the origin pin or wire are causing the router to ignore the routing blockages to route successfully.	Review your DRC report carefully. The Interactive Router might be creating violations when it ignores a routing blockage.
GCR-087	Warning	Interactive solution could only be found by ignoring routing blockages or constraints like via redundancy. This may result in DRCs.	Blockages or constraints near the origin or target pin or wire are causing the router to fail. The router has chosen to ignore them to complete the route.	Review your DRC report carefully. The Interactive Router might be creating violations when it ignores a routing blockage.
GCR-088	Warning	Point (%d,%d) is outside of the defined routing area. Point ignored.	The router is ignoring the point outside the defined routing area.	Click inside the routing area and continue routing.
GCR-089	Error	No contiguous path found between clicked points.	There is no valid contiguous path between the two points that were clicked.	Check the area being routed and click within an open space.
GCR-090	Error	Clicked shapes have no net definition.	You clicked shapes that do not have assigned nets. The router cannot use these shapes as a starting or ending point.	Use the Property Editor Assistant to verify the shapes of the objects you want to use during routing.
GCR-091	Error	P2P shielder doesn't currently support shielding on bit nets.	The point-to-point shielding tool cannot shield individual bit nets of a bus or bundle.	Apply the shield to the entire bus or bundle. See Adding Point-to-Point (P2P) Shielding .

ID	Type	Message	Causes	Solutions
GCR-092	Warning	Cannot complete Interactive Router operation. There appears to be no possible routing path between the source and target. Verify: valid routing layers, Non Default Rules, track patterns, pin accessibility.	One or more obstacles, unroutable layers between source and target shapes, or spacing violations are preventing the router from creating routes.	Check for obstacles, unroutable layers, and spacing constraints.
GCR-094	Warning	Width %.10g on original Interactive Router tandem route start layer has been increased to %.10g to match the maximum tandem routing width required.	The router had to adjust the width on a tandem route to match the required maximum width.	Note the change in width.
GCR-095	Warning	Cannot complete Interactive Router multi-wire routing. There appears to be no possible routing path between the sources and targets. Verify: valid routing layers, Non Default Rules, pin accessibility. Alternatively ensure that P2P guide points select accessible tapoff locations and adequate open corridor areas. Routed wire connections: %u, unrouted wire connections: %u.	One or more obstacles, unroutable layers between source and target shapes, or spacing violations are preventing the router from creating routes.	Check for obstacles, unroutable layers, and spacing constraints. Also, for point-to-point routing, select accessible tapoff locations and corridors that have sufficient space.
GCR-096	Warning	Tandem solution failed. However, a base route was found. To allow partial tandem solution, set the Allow Tandem Breaks option to true (preference: rtTandemAllowBreaks)	Your settings required that tandem shield be inserted without any breaks. There was at least one section where the tandem shield was blocked by routing, blockages, or other obstacles on the layer above or below.	The router will be able to create partial tandem shields if you set the rtTandemAllowBreaks preference to true. If partial tandem shields are okay, set the Tcl preference as indicated.

ID	Type	Message	Causes	Solutions
GCR-097	Error	Shielding net %s but no shield net specified.	The net was specified for shielding in the command net list, but the net to use to create the shielding was not specified.	Check that the net to be used for shielding is specified on the Add Shield toolbar, P2P Shield toolbar, and the Auto Route dialog box. See Setting Options in the Add Shield Toolbar and the Auto Route Shield Tab .
GCR-098	Warning	Tandem vias at corners were not inserted in order to respect stacking via constraints.	The <code>viaStackLimit</code> constraint does not permit another via to be inserted at some location, so some shield vias were omitted.	Check set <code>viaStackLimit</code> constraint. Reroute the connection being shielded to an area with more room for vias to be inserted.
GCR-099	Warning	There is no PR boundary or routing area defined in the design. Router will use boundary (%f %f) (%f %f) for routing.	The design does not have a PR (design) boundary or a routing area defined, so the router computed the boundary box of all the shapes and pins.	If this is not desired, be sure to define a PR (design) boundary or routing area before starting the router. To define PR boundaries, see Creating and Editing Boundaries . See also Defining a Routing Area .
GCR-100	Warning	Cannot switch to tandem routing mode in the middle of a routing operation. Routing operation has been aborted.	You attempted to switch to the Tandem routing mode in the middle of the routing operation. The router could not continue.	Turn on the Tandem mode before starting a route operation.
GCR-111	Warning	(WARNING) Skip bit net %s for creating bus virtual pin, because only one steiner is found on it.	The router found only one steiner on a bus net bit, so it could not create a bus virtual pin.	Check the bus net bit steiner definitions.
GCR-114	Warning	Bitnets of bus %s have different corridors assigned on layer %s, all corridors on layer %s are merged into one big corridor.	The router combined one or more corridors into a single larger corridor on the specified layer.	Check your corridor definitions and define a single corridor per layer.

ID	Type	Message	Causes	Solutions
GCR-115	Warning	Detected width, spacing, rule, or constraint errors. See details in file '%s'.	One or more errors were detected while processing the constraints.	Review the errors in the specified output file and make corrections if possible. Contact Synopsys Support for assistance.
GCR-116	Warning	Shield constraint on '%s' is ignored. There is no shield net '%s' in database.	The net requested for shielding was not found in the database.	Check that the shield net name is spelled correctly and review your design data.
GCR-117	Warning	The Shielder could not create the shielding via at (%f %f) (%f %f) via due to DRC Error.	When adding shielding, the router encountered a DRC error and did not create a via at the noted coordinate.	Visually inspect the area specified for other objects that might interfere with the shielding tool.
GCR-118	Warning	Cannot shield net %s. It is not fully routed.	Shielding is only supported on fully routed nets.	Make sure the net is fully routed before adding shielding.
GCR-119	Warning	(WARNING) Supply net %s has no top level shape and would be ignored for routing.	The router cannot use a supply net without one or more top-level shapes that can be used as connection points.	Check the design data and the supply net definitions. Make sure that you have top-level shapes that can be used as connection points.
GCR-121	Warning	Shielding net %s has no pin in specified edit window. No shield vias can be added.	The viewport area does not contain any pins from the supply net, so shield vias do not have connection points.	Try zooming out so that the viewport area includes the supply shapes and shield vias have connection points.
GCR-122	Warning	Net %s unable to process oaRoute with end type of %s.	The router does not support a topology route that ends at a via.	Add a steiner or use an instance terminal for your route endpoint.
GCR-123	Warning	Detected %d feedthrough connections and %d nearby pin-pin connections.	Informational message only.	If the numbers seem unusual, review the design data.
GCR-124	Warning	Bus/Diffpair trunk %s has 3 or more terminal groups. P2P can only handle 2 terminal groups. All terminal groups are routed.	During point-to-point interactive routing, the router can support two terminal groups. However, the design contains at least three terminal groups.	Review the bus or diffpair route, making sure there are no more than two terminal groups defined, then try to route again.

ID	Type	Message	Causes	Solutions
GCR-125	Warning	Bus/Diffpair trunk %s has predefined corridor. P2P corridor is suppressed.	The bus or diffpair route already had a route-guide corridor defined. The router cannot accept the mouse-click point-to-point corridor in this case.	Check that the bus or diffpair route-guide corridor is defined as expected.
GCR-138	Warning	Could not process via %s because its viaDef %s is not bound.	Some via definitions were invalid or unbound and could not be loaded by the router.	Review your technology and library data to look for unbound vias masters. Files might have been moved or might not be readable. Correct if possible.
GCR-139	Error	TrackPattern %s with color %d exceeds maximum color of %d.	The router only supports up to three colors currently (upper and lower metal layer plus cut layer).	Contact Synopsys Support.
GCR-140	Error	Cannot use one-loop lengthening without Wire Matching or Length constraint.	Single-loop matching requires matching constraint (wireMatching or Length) assigned to the initial routed nets.	In the Constraint Editor, set the Wire Matching Constraint or Length Constraint to use the single-loop preference.
GCR-141	Error	Lengthen bounding box must be specified when using One-loop lengthening.	You must specify lengthening area (rtMatchBox) to run single-loop lengthening.	Define the wire-lengthening area using the <code>rtSingleLoopMatchUseBox Tcl</code> preference.
GCR-142	Warning	Net %s: Wire Matching or Length constraint must be specified to use One-loop lengthening.	Single-loop lengthening requires a matching constraint (wireMatching or Length) assigned to the net.	In the Constraint Editor, set the Wire Matching Constraint or Length Constraint to use the single-loop preference.
GCR-143	Warning	Net %s: Routing is incomplete. One-loop lengthening does not support partially routed nets.	Because single-loop matching is a post-route operation, all nets must be fully routed before using single-loop lengthening.	Make sure the nets are fully routed before attempting single-loop lengthening.

ID	Type	Message	Causes	Solutions
GCR-144	Warning	Net %s: Shapes on net must be located within One-loop lengthening bounding box.	The defined lengthening area (<code>rtMatchBox</code>) does not contain any shapes from the original routing, so there is no place to insert a single-loop detour wire for matching.	Redefine the lengthening area by updating the <code>rtMatchBox Tcl</code> preference.
GCR-145	Warning	Net %s: One-loop lengthening does not have enough room to satisfy constraint within the specified bounding box.	Matching algorithm has exhaustively searched all possible single-loop detour solutions based on the criteria you provided (loop min and max spacing, offset layer, on grid) within the defined lengthening area and cannot find a satisfactory solution.	Enlarge the wire-lengthening area if possible. Check the other constraint values to make sure they are compatible with single-loop lengthening.
GCR-146	Warning	Net:%s route_fromto_only preference is <on>. Skipping non-oaRoute [%s-%s][%s-%s].	You have requested to route only connections defined as routes. The specified nonroute connection will be skipped.	If the connection must be routed, define it as a route using the topology commands.
GCR-147	Warning	Net %s will be protected and will not be routable by the batch router due to the presence of diagonal segment shapes. It will however be possible to modify the net using the Interactive Router.	Diagonal paths or path segments have prevented the Auto Router from creating routes.	Use the Interactive Router to update your routes, or remove the diagonal path segments and try again.
GCR-148	Warning	Rules for layer '%s' are ignored, because, it is either equivalent layer or a layer with undefined mask.	If you have equivalent layers defined, the rules can only be processed for one of the layers.	Check your technology database and make sure rules are not set for more than one layer of an equivalent set.
GCR-149	Error	The router could not be initialized because there are more than 2 poly layers (%s) defined in the tech file.	The router currently supports no more than two poly layers.	Check your technology database and make sure only two poly layers are defined for routing.

ID	Type	Message	Causes	Solutions
GCR-150	Error	The router could not be initialized because the Custom Router does not support routing with two poly layers (two poly layers "%s" are selected for %s).	The router can only route using one poly layer when there are two poly layers defined.	Check your technology database and make sure only two poly layers are defined for routing. Use the Object/Layer Panel to deselect one of the two poly layers.
GCR-151	Warning	Cannot build the design channel map required to do device row cloning for the Interactive Router. The design appears to be too large. The current limit for the number of shapes to be processed is: %d. To increase this limit, set the <code>rtMaxShapesForCloning</code> preference to a higher value.	The router has encountered more shapes than the maximum number that your preference value permits.	Set the <code>rtMaxShapesForCloning</code> Tcl preference if possible, or define a smaller window to reduce the shape count.
GCR-152	Warning	Not all possible Interactive Router clone candidates were actually cloned. The limit of %d clones per Interactive Router operation has been reached. To increase this limit, set the <code>rtMaxCloning</code> preference to a higher value.	The router has reached the maximum number of clones based on the <code>rtMaxCloning</code> Tcl preference value.	Increase the <code>rtMaxCloning</code> value if possible.
GCR-154	Warning	Only one poly layer could be selected.	You selected more than one poly layer for routing. The router only supports one poly layer.	Check your selected layers and make sure only one poly layer is selected, then rerun your command.
GCR-156	Warning	Backup layers must be continuous for net %s.	Backup layers must be continuous.	Check the backup layers for the specified net and make sure they are continuous.
GCR-157	Warning	Backup layers must be subset of valid routing layers for net %s.	Backup layers must be a subset of the valid routing layers. Layers not in the valid routing layers set for the net cannot be used as backup layers.	Check the valid routing layers for the specified net. Make sure that only layers listed in this set are used for backup.

ID	Type	Message	Causes	Solutions
GCR-158	Warning	The net %s must have at least 4 routing layers to setup backup layers.	To specify backup layers, a net's valid routing layer set must include at least four routing layers.	Check the valid routing layers for the net and make sure that at least four layers are specified. Alternatively, do not use the backup feature.
GCR-159	Warning	Track Regions on layer '%s' overlap and create infeasible track region setup, this region '%s' will be ignored.	The tracks in track regions overlap on the same layer and create an impossible setup on the specified layer. The router will ignore the region as specified.	Check the track regions at the global and local level and make sure they do not conflict or overlap unexpectedly for same layer.
GCR-160	Warning	There are more than 64,000 shapes in a small area (%f, %f, %f, %f). The router will ignore some of the shapes.	The router detected a large number of shapes within the specified area. The router will ignore some of the shapes and continue.	Check that all pins and instances are properly placed. Check for overlapping placements. Check blockages in instance masters.
GCR-161	Error	The router could not be initialized because the routing area is outside the PR boundary (%f, %f, %f, %f).	The routing area you have defined is outside of the PR (design) boundary. The router cannot proceed.	Check the routing area definition and create an area which is within the PR (design) boundary. See Defining a Routing Area .
GCR-162	Error	The router could not be initialized because the routing area is null.	The routing area you have defined is effectively null (either the x or y dimension is zero or too small to be useful).	Check the routing area definition and define an area that is larger than the track width and spacing rules. See Defining a Routing Area .
GCR-163	Error	The router could not be initialized because there are more than %d metal and cut layers defined.	The router only supports a limited number of cut and metal layers. Your data contains more layers than the router can support.	Check the technology database and make sure that the total number of cut and metal layers is within the noted limit.
GCR-164	Error	The router could not be initialized because there are no routing layers defined in the tech data. The functions section of layerRules must have metal layers with their mask numbers.	The router did not find any metal or cut layers with their valid mask numbers in the technology library.	In the layer rules section of the technology library, check the functions subsection and make sure that the metal and cut layers are properly defined with their mask numbers.

ID	Type	Message	Causes	Solutions
GCR-165	Error	The router could not be initialized because there are no valid via definitions in the tech data.	The router did not find any valid via definitions in the technology database.	Check your technology database and make sure that the via definitions are properly defined.
GCR-166	Warning	There is no user-function 'CustomViaParameterize Func' available for custom vias, custom vias will not be modified by router.	There is no CustomViaParameterize Func user function available for custom vias.	Remove custom vias from the valid routing via definition, or provide the required function to use custom vias.
GCR-167	Warning	Found preroute shapes with MPT color or tech layer with default color. Provide active color tracks in layer %s to enable color routing methodology.	The preroute contains shapes with multiple-patterning color or layer with the default color.	Define active color tracks for the specified layer to enable color routing.
GCR-168	Error	RPDL error: '%s'.	The router encountered a syntax error while attempting to parse the RPDL extraction data. The message indicates the exact error.	Examine the RPDL file and fix the indicated syntax error. Contact Synopsys Support for assistance.
GCR-170	Error	The routing area defined in %s contains an illegal value %s.	The router encountered an illegal string value while processing the routing area using preferences.	The routing area value must consist of numbers only. You can define a rectangle by specifying the lower-left and upper-right coordinates, or you can define a polygon that contains four or more vertices in sequence.
GCR-171	Warning	Rules for layer %s are ignored because it is an equivalent layer.	If you have equivalent layers defined, the rules can only be processed for one of the layers.	Check your technology library and ensure rules are not set for more than one layer of an equivalent set.
GCR-172	Warning	Pin centered at (%.10g,%.10g) on layer %s is off-track or off-colored.	Full colored routing for multi-patterning layer requires pins to be on track with mask color matching track color.	Snap pin onto candidate track. Change the pin color to match the track color.

ID	Type	Message	Causes	Solutions
GCR-175	Warning	River style routing is not supported. The river constraint will be ignored for net %s.	The Custom Router does not support river style routing. The router detected a river constraint in your data.	Remove the river constraint and try routing again.
GCR-176	Warning	%s style shielding is not supported. This shielding type will be ignored for net %s.	The Custom Router does not support the specified shielding style. The constraint will be ignored.	Change your constraint to use parallel shielding or delete the constraint.
GCR-177	Warning	Min resistance constraints are not supported. The Custom Router will not attempt to follow the resistance constraint for net %s.	The Custom Router does not support min resistance constraints. The constraint will be ignored.	You can continue routing and the router will ignore the constraint. You can delete the constraint to eliminate the warning message.
GCR-178	Warning	The router detected a wire resistance matching constraint, which is not supported. The constraint will be ignored for net %s.	The Custom Router does not support wire matching for resistance. The constraint will be ignored.	You can continue routing and the router will ignore the constraint. You can delete the constraint to eliminate the warning message.
GCR-179	Warning	The router detected a wire length matching-per-layer constraint where there are also more than two layers defined in the NDR layer constraint for net %s. The routing quality may not be acceptable.	The Custom Router supports wire length matching on a per layer basis, but it might not be able to produce good results when three or more layers are used to route the given net.	For quality results, create a non-default-rule specifying just two layers for any net which has this type of matching per layer constraint.
GCR-180	Warning	Topology constraints may only be used for bus routing. Other topology definitions have been ignored for net %s.	The Custom Router detected topology constraints and data for a net which is not part of a bus trunk. These definitions will be ignored.	Limit your topology node and edge definitions to nets that are included in a bus trunk.
GCR-182	Warning	Wire striping routing is not supported. The wire striping constraint will be ignored for net %s.	The Custom Router does not support wire striping during routing. The router detected a wire striping constraint in your data.	Remove the wire striping constraint and try routing again.

ID	Type	Message	Causes	Solutions
GCR-188	Error	The specified group '%s' doesn't exist.	The group you specified using the -group option cannot be found in the current design.	Make sure the specified group name is correct.
GCR-189	Error	Could not extract the routing pattern for net %s. A trunk cannot span across multiple layers.	A trunk in the pattern spans across multiple layers. The trunk must be on the same layer.	Make sure the trunk stays on the same layer, or reroute so that the trunk covers smaller clusters and can remain on the same layer.
GCR-190	Error	Could not extract the routing pattern for net %s. Some of the pins are not connected to a trunk from a tapoff.	The Custom Router can only extract patterns from fully connected routes. Routes that are not connected to pins can cause the extraction process to fail.	Make sure all pins are connected to trunks from tapoffs.
GCR-191	Error	Could not extract the routing pattern for net %s. The pattern was likely produced by the Auto Router.	The route was produced by the Auto Router. The route pattern must be produced by the Interactive Router in the Auto Welding Mode and the Auto Cloning mode.	Reroute using the Interactive Router in the Auto Welding Mode and the Auto Cloning mode.
GCR-192	Error	Custom Router is not enabled.	You are attempting to use the Interactive Router, but you have not yet enabled the Custom Router.	In the Layout Editor, choose Tools > Routing , and then try using the Interactive Router.
GCR-193	Warning	Pattern Route: The tapoff was explicitly set to %s which is a horizontal layer. The net might not be routed on track.	The trunk is running through a horizontal channel, so the tapoffs are placed vertically.	Use vertical layers for tapoffs.
GCR-194	Warning	Pattern Route: No tracks are available on %s to route tapoffs. The net might not be routed on track.	The layer used for tapoffs does not have tracks.	Check the layer definition. Add track definition if needed.

ID	Type	Message	Causes	Solutions
GCR-195	Error	Multiple placement groups are within the routing area. Only a single group is permitted. Try editing the design while in the Edit in Group mode, or make sure the routing area has only one placement group.	Only one placement group is permitted within a routing area.	You must be in the Edit in Group mode when applying route patterns. Alternatively, make sure the routing area has only one placement group before you apply the route pattern.
GCR-196	Error	The routing area must contain a placement group.	The routing area must contain a placement group for the route pattern to apply.	Make sure the routing area contains a placement group before applying the route pattern.
GCR-197	Error	The design contains multiple placement groups. Define only a single placement group in the routing area, or try editing while in the Edit in Group mode.	You are attempting to apply a route pattern, but there are multiple placement groups in the design.	Try editing the design while in the Edit in Group mode, or make sure only one placement group is within the routing area.
GCR-210	Warning	User requested %s width of %f is incompatible with the allowedWidthRange constraint for layer %s. For that layer, a width of %f will be used for routing if applicable(%s)	The Custom Router cannot use the layer width you specified, because it is outside of the allowedWidthRange values in the technology database. Instead, the router will attempt to use the closest compatible width.	Review the results to make sure they are acceptable. If the results are not acceptable, make sure the layer width value is within the allowedWidthRange in the technology database. In the Constraint Editor, check the Width parameter in the Layers constraint for the net. On the Route toolbar (Interactive Router), check the Width and Auto Adjust Width settings.
GCR-211	Warning	Auto-adjust-width (%s) of %f is incompatible with the allowedWidthRange constraint for layer %s. For that layer, a width of %f will be used for routing if applicable(%s)	The Interactive Router cannot use the starting pin width, because it is outside of the allowedWidthRange values in the technology database. Instead, the router will use the closest compatible width.	Review the results to make sure they are acceptable. If the results are not acceptable, make sure the pin widths are within the allowedWidthRange.

ID	Type	Message	Causes	Solutions
GCR-212	Warning	Auto-taper (%s) of %f is incompatible with the allowedWidthRange constraint for layer %s. For that layer, a width of %f will be used for routing if applicable(%s)	The Interactive Router cannot use the starting and ending pin widths, because they are outside of the allowedWidthRange values in the technology database. Instead, the router will use the closes compatible widths.	Review the results to make sure they are acceptable. If they are not acceptable, make sure the pin widths are within the allowedWidthRange.
GCR-214	Warning	User requested width of %f for layer %s will be promoted to %f in order to use a non-reserved track.	The routing width you specified in the Layers constraint (Constraint Editor) or on the Route toolbar is increased to the Non-Reserved track width value so that the route can use the track.	Review the results to make sure they are acceptable. If the results are not acceptable, check the width settings in the Layers constraint and on the Route toolbar.
GCR-215	Warning	Auto-adjust-width of %f for layer %s will be promoted to %f in order to use a non-reserved track.	The routing width derived from the starting pin (or wire) is increased to the Non-Reserved track width value so that the route can use the track.	Review the results to make sure they are acceptable. If the results are not acceptable, check the track and pin dimensions.
GCR-216	Warning	Auto-taper-width of %f for layer %s will be promoted to %f in order to use a non-reserved track.	The requested routing width derived from the starting pin or ending pin is increased to the Non-Reserved track width value so that the route can use the track.	Review the results to make sure they are acceptable. If they are not acceptable, check the track and pin dimensions.
GCR-218	Warning	Track width of %f on layer %s is incompatible with allowedWidthRange (%s) constraint. For that layer, a width of %f will be used for routing if applicable (%s).	The non-reserved track width assigned to one or more of the tracks is outside of the allowedWidthRange values in the technology database. The allowedWidthRange values will be used.	Review the results to make sure they are acceptable. If the results are not acceptable, review the track definitions.
GCR-222	Warning	Auto-adjust-width of %f for layer %s is incompatible with a non-reserved track of width %f. The track width will be used.	The routing width derived from the starting pin (or wire) is decreased to the Non-Reserved track width value so that the route can use the track.	Review the results to make sure they are acceptable. If the results are not acceptable, review the track definitions or pin dimensions.

ID	Type	Message	Causes	Solutions
GCR-223	Warning	Auto-taper-width of %f for layer %s is incompatible with a non-reserved track of width %f. The track width will be used.	The routing width derived from the starting or ending pin will be decreased to the Non-Reserved track width value so that the route can use the track.	Review the results to make sure they are acceptable. If the results are not acceptable, review the track definitions or pin dimensions.
GCR-225	Warning	Routing width of %f on layer %s is not compatible with any neighboring tracks. Verify track settings if applicable.	The routing width (unknown source) is larger than the Non-Reserved track width value of a neighboring track. The router will ignore this or all tracks.	Review the track definitions.
GCR-237	Warning	Pins on {%s} layer(s) are ignored by the router because they are missing in the <code>rtImportPartialLayers</code> preference.	The specified pin layers are missing from the <code>rtImportPartialLayers</code> Tcl preference setting.	Review the <code>rtImportPartialLayers</code> Tcl preference setting. Add the pin layers as needed.
GCR-239	Error	The grouping rate (%d%%) is too low and not suitable for group planning flow.	Routing was aborted. The percent of nets that can be grouped is low, so the Custom Router was unable to find optimal group-routing solutions.	Select bus-like nets that can be routed together. Alternatively, turn off Auto Group before you start routing.
GCR-240	Error	Net %s is connected to standard cell pins and not suitable for group planning flow.	Routing was aborted because some of the nets you selected contain standard cell pins. Auto Group does not support standard cell pins.	Remove nets that contain standard cell pins from your net selection and try routing again.
GCR-244	Warning	Could not save Pattern Route Constraint for net %s in group %s. %s.	The constraint database cannot be edited.	Check that the design is writable and try again.
GCR-245	Error	Channel %s is fragmented and blocked on layer %s. Routing is aborted.	Too many small shapes are blocking the channel.	Look for and remove dangling wires in the channels.

ID	Type	Message	Causes	Solutions
GCR-246	Error	The percent of nets (%.2g%%) that have violations exceed the maximum allowed (%d%%).	More than 20% of nets have violations. The violations are slowing the Auto Router performance.	Use the Congestion Checker to locate and fix congested areas, and try routing again. See Checking Route Congestion .
GCR-247	Error	The percent of opened nets (%.2g%%) exceed the maximum allowed (%d%%).	One or more pins are blocked, lack of routing solution, or incorrect routing constraint settings might be preventing some nets from being routed.	Check for pin blockages, manually route open nets, check routing constraint settings, and then try routing again. Alternatively, you can turn off auto net grouping and try routing again.
GCR-248	Warning	Ignoring net '%s' connected to a standard cell pin.	The nets contain standard cell pins.	Deselect routing nets that contain standard cell pins, and try routing again.
GCR-249	Error	There are no nets for planning.	You did not select any nets for routing.	Be sure to select valid nets before you start routing.
GCR-250	Error	The percent of detailed routes (%.2g%%) that follow the global route guidance is below the maximum allowed (%.2g%%). Routing is aborted.	Obstacles in the design and incorrect routing constraint settings might be preventing the detailed routes from following the global route guidance.	Check the design for obstacles, check the routing constraints and other rules, and then try routing again. Alternatively, you can turn off auto net grouping and try routing again.
GCR-260	Error	User defined group '%s' is invalid since %s.	- The group contains only one net. The number of nets should be greater than one. - The same net appears in two groups. - Pin count differs across the nets in the group. They should be the same. - Pin layers differs across the nets in the group. They should be the same.	- Remove the group, or add more nets in the group. - Regroup the nets and make sure the net does not appear in more than one group. - Remove the nets that have a different pin count. - Check the layers across the nets and make sure they are the same.

ID	Type	Message	Causes	Solutions
GCR-264	Error	Cannot create topology. The specified object name '%s' does not exist.	You modified or deleted some pins after exporting the topology data. The topology file you are re-importing no longer matches the information in the design.	Review the design and make sure the removals are intentional.
GCR-266	Warning	Skipping object '%s' during trunk topology creation. The specified object name '%s' does not exist.	You removed pins or topology components from the design.	Review the design and make sure the removals are intentional.
GCR-267	Error	Cannot create trunk topology. Fewer than two objects were specified.	Some pins were removed so that one or more trunk topologies no longer exist. Trunk topologies require two or more pins.	Review the design and make sure the removals are intentional.
GCR-273	Error	Could not generate trunk, as the specified shield net '%s' is not a power/ground net.	The specified net shielding is not a power and ground (PG) net.	Make sure you specify a power and ground (PG) net for shielding.
GCR-274	Error	Could not generate trunk, as the specified shield net '%s' doesn't exist, and it is not 'power' or 'pwr/gnd'.	The specified net shielding does not exist.	Make sure you specify an existing power and ground (PG) net for shielding.
GCR-275	Error	Could not generate trunk, as there is no '%s' net in the current design.	You have specified <code>rt::generateTrunk -shieldNetName power</code> , but the design does not have power and ground nets.	Make sure the design has power and ground (PG) nets if you want to add shielding when creating trunk routes.
GCR-276	Error	Could not generate trunk, as there is no pin in the design or no pin matches specified options.	The design does not have pins, or none of the pins match the specified net and options.	Make sure the design has at least one pin that matches the specified net and options.
GCR-278	Error	'%s' ViaDef '%s' layer '%s' is not defined, so, this via definition will be ignored during routing. Check if it's routing layer function is defined correctly in technology	The layer function is not defined in the technology database.	Check the technology database associated with the design. If necessary, add the layer function.

ID	Type	Message	Causes	Solutions
GCR-280	Warning	Wires that are causing shorts are removed from %d nets, resulting in %d more nets being partially routed.	The <code>rtDeleteShortingWires</code> Tcl preference is <code>true</code> , so wires that are causing short circuitry are removed, resulting in partially routed nets.	If you do not want to remove the wires that are causing short circuitry, set the <code>rtDeleteShortingWires</code> Tcl preference to <code>false</code> .
GCR-281	Warning	Net %s is partially routed because of short removal.	The <code>rtDeleteShortingWires</code> Tcl preference is <code>true</code> , so wires that are causing short circuitry are removed, resulting in partially routed nets.	If you do not want to remove wires that are causing short circuitry, set <code>rtDeleteShortingWires</code> to <code>false</code> .
GCR-288	Warning	Supply net %s is ignored during routing.	- No power/ground mesh is defined for the supply net. The supply net should consist of 0x shapes from the supply net. Small 0x supply pin shapes do not count. - None of the pins needed to tie to the supply net is within reasonable distance from the closest supply network shape.	- Make sure you define power/ground mesh for the supply net and the supply net contains 0x shapes. - Make sure you define power/ground mesh for the supply net and the supply net contains 0x shapes.
GCR-289	Error	Failed to create trunk for net %s because no tracks are available on %s within the search box.	No routing tracks are found within the search box (the region that encloses all pins inside the routing area).	Make sure routing tracks are defined within the search box. In the Track Region Assistant or the Track Pattern Assistant, make sure the tracks are active.
GCR-290	Warning	Skipping object '%s' during trunk topology creation. The specified object '%s' does not belong to net '%s'.	Some pins or instTerms have net names that do not exist.	Check the pin or instTerm net assignments and try again.
GCR-291	Error	Cannot create topology. The specified object '%s' does not belong to net '%s'.	Some pins or instTerms have net names that do not exist.	Check the pin or instTerm net assignments and try again.

ID	Type	Message	Causes	Solutions
GCR-292	Error	Pin Access Failure: Pin (%s net) is not accessible at the point (%1.3f,%1.3f,%1.3f,%1.3f) because of %s DRC on the layer %s	The pin is not accessible for one or more of the following reasons: "The pin is not accessible through vias. "The pin is completely blocked. "The pin width is not within the specified <code>allowedWidthRange</code> (Layer constraint). See also Design Errors (Auto Router) or Design Errors (Interactive Router) for examples.	Use the Error Viewer Assistant to locate the pin. Review the DRC violation at the pin and fix the issue.
GCR-293	Error	Off-Track Pins: Pin Completely off track. Center of the pins are off track	The design has pins that are not on routing tracks or have widths different from the tracks. See also Design Errors (Auto Router) or Design Errors (Interactive Router) for examples.	Use the Error Viewer Assistant to locate the pins, and then place the center of the pins on the tracks.
GCR-294	Error	Pin and Track Color: The color of the %s track is:%s and the color of the pin shape(%s net) is:%s	The pin color does not match the track color. See also Design Errors (Auto Router) or Design Errors (Interactive Router) for examples.	Use the Error Viewer Assistant to locate the pins and fix the color mismatch problems.
GCR-295	Error	Pin Shorts: The shape with %s net is creating short with %s net shape on %s layer	Incorrectly positioned preroute pin shapes are causing short circuits. See also Design Errors (Auto Router) or Design Errors (Interactive Router) for examples.	Use the Error Viewer Assistant to locate the pins and fix the problem.
GCR-296	Warning	Single Pin in Routing Area: The net %s has only one shape in the routing area	Pins inside the routing area require connections to pins that are outside of the routing area.	Use the Error Viewer Assistant to locate the pins in the routing area. To fix the problem, remove the pin from the routing area or have more than one pin of on a net in the routing area.

ID	Type	Message	Causes	Solutions
GCR-297	Warning	Track Region and Layer Direction: The %s layer direction is %s and the track direction is %s	The track direction and the layer direction are not in the same direction.	Review the violation message in the Error Viewer Assistant to identify the layer for which the track regions are in the incorrect direction. If necessary, change the track direction.
GCR-298	Error	Topology segment of net '%s' is blocked around %s %s.	The Topology Router cannot find an available channel around the reported coordinate.	Inspect the area around the reported edge. Modify the topology and layout as needed to accommodate route placements.
GCR-299	Warning	Unnetted Shapes: Shape on Layer %s is not associated with the net name	Some shapes in the design are not assigned to nets.	Use the Error Viewer Assistant to locate and assign nets to the shapes.
GCR-300	Error	Could not generate trunks on the same track, as there is no common track for the search boxes of specified nets.	The design does not have routing tracks, or tracks are not found within search boxes of the specified nets.	Create tracks in the design, and make sure the tracks can be found within the search boxes of specified nets.
GCR-301	Warning	Pin Off Grid Point: Pin is not on grid point	Both ends of pins are not on grid points.	Use the Error Viewer Assistant to locate the pin. Adjust the pin so that both ends are on grid points.
GCR-302	Error	Missing Via Definitions: The via definition is missing in the tech between the layers %s	The via definition might be missing in the technology file between some layers.	Update the technology file with the missing via definition.
GCR-303	Error	Missing width Constraints: The width constraint is missing in the tech for the layer %s	Width constraints might be missing for some layers.	Use the Error Viewer Assistant to identify the layers with missing width constraints. Add the width constraints in the technology database.
GCR-304	Error	Missing Spacing Constraints: The spacing constraint is missing in the tech for the layer %s	Layer constraints might be missing for some layers.	Use the Error Viewer Assistant to identify the layers with missing spacing constraints. Add the spacing constraints in the technology database.

ID	Type	Message	Causes	Solutions
GCR-305	Warning	Nets Without Shapes: The net %s is not associated with any shape	Some nets in the design do not have shapes associated with them.	Use the Error Viewer Assistant to identify the nets that do not have shapes. Delete the nets, or add shapes to the nets.
GCR-312	Error	Could not delete tapoff, as the trunk topology has only two tapoffs.	You are attempting to delete a tapoff from a trunk that only has two tapoffs.	Delete the entire trunk, or keep the tapoff.
GCR-313	Error	There is no trunk shape with tag name '%s' in the design.	You are using a trunk route tag that does not exist in the design.	Check the tag string and try again.
GCR-322	Error	Could not generate trunk, as the specified layer %s is invalid.	The layer you specified for trunk routing does not exist or cannot be used for trunk routing.	Check the layer specification in the Trunk Generation dialog box or in the <code>rt::generateTrunk Tcl</code> command and try again.
GCR-361	Warning	Could not generate trunk, as there is no track matches the specified track index.	You specified one or more track indexes that do not exist.	Check the track indexes you specified in the Trunk Generation dialog box or in the <code>rt::generateTrunk Tcl</code> command and try again.
GCR-362	Warning	Could not fulfill the grow of trunk for net %s.	Obstacles are preventing the trunk from increasing or decreasing length based on the specified Trim value, or the trunk is unable to decrease its length any further without violating the <code>minArea</code> , <code>minLength</code> , and other rules.	Move the obstacle and try again.
GCR-363	Error	Could not generate trunk, as no net is specified.	You did not specify a net for the Generate Trunk operation.	Check the Net setting in the Trunk Generation dialog box or in the <code>rt::generateTrunk Tcl</code> command and try again.
GCR-401	Warning	RC Graph: Found floating shapes in graph %s, shape detail : %s.	Dangling path segments are found in the routing graph.	Remove dangling path segments or redundant connections on the nets or at the specified coordinates.

ID	Type	Message	Causes	Solutions
GCR-402	Warning	RC Graph: Found loop in graph %s.	Loop connections are found in the routing graph.	Check whether the loop connections can be ignored.
GCR-403	Warning	RC Graph: No RC graph is found with specified net, %s.	Routing graph does not exist for the specific net.	Use the <code>rt::rc::convertGraph</code> Tcl command to convert routing graph by specifying the net.
GCR-406	Warning	RC Graph: Some of specified nets are not from the same design.	You specified nets that are not in the design. Therefore, no routing was created for those nets.	Check the nets you specified in the <code>rt::checkPostRoute</code> Tcl command.
GCR-417	Warning	RC Graph: Failed to route incomplete nets.	Virtual connection is turned on, but the connections were not created for the specified nets.	Review the design and determine why the nets are not routed. For example, check for blockages, routing area definition, routing constraint settings, and so on.
GCR-418	Warning	Skip net %s because Pattern Route patterns haven't been realized.	Some placement groups with pattern route constraints are not realized before routing the inter-group portion that uses topologies.	Route the placement groups first, and then try again.
GCR-420	Warning	Unable to route topology for net '%s' because topology segment does not touch pin correctly at %s.	Topology path endpoint does not touch pin shapes.	Move the endpoint onto pin shapes.
GCR-421	Warning	Skip net '%s' because topology segment '%s' is out of prBoundary or routing area.	Topology segments are outside of the PR boundary or the routing area you defined.	Move the reported segment inside of the PR boundary or routing area. Alternatively, adjust the PR boundary or routing area.
GCR-428	Warning	Fail to meet EM/R constraint for net 'net_name' due to insufficient resource near topology segment: {coordinate}.	During R-aware routing, the Topology Router cannot find a channel that can fit the wire with the required width.	Widen the channel if possible. Otherwise, after a few attempts, the Topology Router will create a narrower wire in the channel, even if the width will cause violations.

ID	Type	Message	Causes	Solutions
GCR-429	Warning	Max pin-to-pin resistance constraints are only supported by topology routing with R-aware option. Auto Router will not try to satisfy max resistance constraint for net %s.	The Auto Router does not support maximum pin-to-pin resistance constraints.	Because the Topology Router supports maximum pin-to-pin resistance constraints, use topology routing with the R Aware option to route the net with maximum resistance constraint. See Generating Routes Based on Topologies .
GCR-432	Warning	Skip net '%s' because of unexpected overlapping between topology segments at '%s'.	Topology path segments are on the same layer and overlapping with each other.	Remove or move the overlapping shapes, or change routing layers.
GCR-433	Warning	Skip net '%s' because of duplicated topology segments at '%s'.	Two or more topology segments are on the same layer and have the same startpoints and endpoints.	Remove duplicate topologies.
GCR-434	Warning	Skip net '%s' because topology path is short and fully enclosed by pin shapes at '%s'.	The bounding box of a topology path is small and the topology path is fully enclosed by pin shapes.	Remove the small topology if it is not needed.
GCR-435	Warning	Skip net '%s' because floating topology is found at '%s'.	One or two topology segment ends do not touch a pin shape or a topology segment.	Correct the connection. You can re-create, move, or delete the topology segment.
GCR-436	Error	Options '%s, %s' are mutually exclusive. Do not specify both in the command '%s'.	You selected two options that cannot be set simultaneously.	Deselect one of the options and try again.
GCR-437	Error	'%s' cannot be run because the specified options are not an acceptable combination.	You selected a combination of options that cannot be specified together.	Deselect one or more of the options and try again.
GCR-438	Warning	Skip net '%s' because there is no pin of net grouped by rows.	Either no pin or none of the net pins is within existing rows.	Check and see if any net pin is within the existing rows. Adjust the pins as needed, and try again.

IROUTE Error Messages

Error messages that have the IROUTE- prefixes pertain to the Interactive Router. This topic lists the IROUTE- error messages, possible causes, and possible solutions

ID	Type	Message	Causes	Solutions
IROUTE-002	Error	Interactive Router works only in the Layout Editor.	You are attempting to enable the Interactive Router, but you are not in the Layout Editor.	Make sure you enable the Interactive Router in the Layout Editor.
IROUTE-003	Error	Interactive Router works only when the Custom Router is enabled.	You must have the Custom Router license to enable to the Interactive Router.	Make sure you have the Custom Router license, and then choose Tools > Routing in the Layout Editor.
IROUTE-004	Warning	minWidth on %s is not defined in the technology.	The <code>minWidth</code> constraint is not defined in the technology database.	Define <code>minWidth</code> in the technology database and try again.
IROUTE-005	Warning	minSpacing on %s is not defined in the technology.	The <code>minSpacing</code> constraint is not defined in the technology database.	Define the <code>minSpacing</code> constraint in the technology database and try again.
IROUTE-006	Warning	Start point is blocked by obstacle: %s %s %s on %s layer with rule %s.	The starting pin is blocked by one or more obstacles.	Remove or adjust the obstacles and try again. Alternatively, turn on Ignore Rules . See the Interactive Router Keyboard Shortcuts.
IROUTE-007	Error	Cannot start the route because the active layer %s is not a valid routing layer.	You are trying to starting routing on a non-valid routing layer.	In the Object/Layer Panel, select a valid routing layer and try again.
IROUTE-008	Warning	Cannot change the width because Auto Adjust Width is enabled.	You are trying to manually change the path width using Ctrl+scroll, but Auto Adjust Width is turned on.	If you want to manually adjust the path width, turn off Auto Adjust Width . See Route Toolbar .
IROUTE-009	Warning	Cannot change the width of the first segment when Taper Start is enabled.	You are trying to manually change the path width using Ctrl+scroll, but Taper Start is turned on.	If you want to manually adjust the path width, turn off Taper . See Route Toolbar .

ID	Type	Message	Causes	Solutions
IROUTE-010	Warning	Cannot change the width of the last segment when Taper End is enabled.	You are trying to manually change the path width using Ctrl+scroll, but Auto Adjust Width is turned on.	If you want to manually adjust the path width, turn off Auto Adjust Width . See Route Toolbar .
IROUTE-011	Warning	Route could not be found. %s. Committing partial route.	(rtAllowPartialSolution Tcl preference is true) While in the Point to Point mode, the Interactive Router was unable to generate a rule-compliant route between the specified starting and ending points, caused by the issue stated in the message. Only a partial route solution was produced.	Review the reason in the message to resolve the issue and try again.
IROUTE-012	Warning	Redo Route cannot work because there is no available reference route.	You cannot use Redo Route there is no last route.	Create a route first, and then use the Redo command.
IROUTE-013	Warning	Redo Route cannot be used with Cloning mode. Disable Cloning to use Redo Route.	Redo Route cannot work when Cloning is turned on.	Turn off Cloning and try again.
IROUTE-014	Warning	Redo Route cannot work. Click a pin first, then invoke Redo Route.	Redo Route requires that you select (or click) a pin to start from.	Click a pin to start the Redo Route operation. See Replicating the Last Route .
IROUTE-015	Warning	Redo Route cannot work. The current pin layer %s does not match reference route start layer %s.	The layer of the selected starting pin must be the same as the starting pin layer of the original route.	Select a starting pin that is on the same layer as the original starting pin.
IROUTE-017	Warning	Could not find appropriate pin for net %s, so ignoring this net for bus routing.	- No pin is selected. - The pin does not align with the leader bit.	- Make sure you are starting from pins and that you have selected the leader pin. - Make sure the selected pin aligns with the leader bit.

ID	Type	Message	Causes	Solutions
IROUTE-018	Error	Cannot export the key bindings to %s file because writing error occurred: %s.	The key binding file does not exist, or the file does not have write permission.	Make sure the target location has write permission. Check the directory to make sure the file exists.
IROUTE-019	Error	Cannot import the key bindings from %s file because reading error occurred: %s.	The key binding file does not exist.	Make sure the file is present in the directory specified by <code>SYNOPSIS_CUSTOM_PRESET_IR_KEY_BINDING</code> .
IROUTE-020	Error	Error encountered in registering the key binding from file %s: %s. Restoring the earlier key bindings.	- You have assigned the same key binding for different commands or actions. - The key binding is invalid. - The command or action you are assigning to the key binding is invalid.	To troubleshoot, export the key bindings so that you can view the original key assignments. Compare the original assignments with the new assignments for invalid key bindings or actions. Correct the assignments. See Reconfiguring Keyboard Shortcuts.
IROUTE-021	Warning	Cannot drop via at %, %s because the closest shape (on %s layer) is not a same-net shape (shape net: %s, route net: %s)	You are trying to add a via to connect two metal shapes that are not on the same net.	Make sure you are adding vias between two shapes that are on the same net.
IROUTE-022	Warning	Cannot drop via at %, %s because the closest shape (on %s layer) is a netted shape (shape net: %s) and route is unnetted.	You are trying to add a via between a netted shape and an unnetted shape.	Make sure you are adding vias between two netted shapes or two unnetted shapes.
IROUTE-023	Warning	Cannot drop via at %, %s because no shapes could be found to drop via.	You cannot add a via between two shapes on adjacent layers. The second shape is missing.	Make sure a second shape is available before using the Auto Via Drop option.
IROUTE-024	Warning	Cannot Layer Up and add via because the route is already on the highest routable layer.	You are trying to add a via above the top-most routable layer.	Continue routing on the current layer, or Layer Down and continue routing on lower layers.

ID	Type	Message	Causes	Solutions
IROUTE-025	Warning	Cannot Layer Down and add via because the route is already on the lowest routable layer.	You are trying to add a via below the bottom-most routable layer.	Continue routing on the current layer, or Layer up add continue routing on higher layers.
IROUTE-026	Warning	Cannot add via because the route cannot reach the specified layer %s from %s.	The router could not find valid via definitions to use, so no via was added.	Check the technology database and make sure valid via definitions are available for your design.
IROUTE-027	Warning	Cannot create the specified net "%s" because of %s.	The net name you specified does not exist or cannot be used.	Check the net name and try again.
IROUTE-031	Warning	Route could not be found. %s.	While in the Point to Point mode, the Interactive Router was unable to generate a rule-compliant route between the specified starting and ending points, caused by the issue stated in the message.	Review the reason in the message to resolve the issue and try again.

PATTERN_ROUTE_ASSISTANT Error Messages

Error messages that have the PATTERN_ROUTE_ASSISTANT- prefixes pertain to the Pattern Router. This topic lists the PATTERN_ROUTE_ASSISTANT- error messages, possible causes, and possible solutions

ID	Type	Message	Causes	Solutions
PATTERN_ROUTE_ASSISTANT-001	Error	The specified net %s is not in the current edit group %s.	The specified net does not belong to the current edit group.	Select the correct net or edit group.
PATTERN_ROUTE_ASSISTANT-002	Error	The specified group or current edit group %s is a stack MOS group, which is not allowed.	The Pattern Route assistant does not work with stack MOS groups.	Select non-stack-MOS groups and try again.
PATTERN_ROUTE_ASSISTANT-003	Error	The specified width %f is less than the minimum width %f for layer %s.	The specified width is less than the minimum width for the layer.	Make sure the width is greater than the minimum width.

ID	Type	Message	Causes	Solutions
PATTERN_ROUTE_ASSISTANT-004	Error	The specified width %f is greater than the maximum width %f for layer %s.	The specified width is greater than the maximum width for the layer.	Make sure the width is less than the maximum width.
PATTERN_ROUTE_ASSISTANT-005	Error	Unable to update the placement group because SDL is not enabled in the Layout Editor.	You can only update the placement group when SDL is enabled.	To enable SDL, in the Layout Editor, choose Tools > SDL .
PATTERN_ROUTE_ASSISTANT-006	Error	Could not route the group %s because it has no pattern route constraint.	You are attempting to route a group that does not have any pattern route constraints.	Specify the route patterns to use before routing the group.
PATTERN_ROUTE_ASSISTANT-007	Error	Could not estimate the row spacing for the group %s because it has no pattern route constraint.	You are attempting to route a group that does not have any pattern route constraints, so row spacing estimates are not possible.	Specify the route patterns to use before routing the group.
PATTERN_ROUTE_ASSISTANT-008	Error	Net %s tapoffs conflict with those from net %s when using %s.	The tapoffs from the nets conflict (produces short circuits) with tapoffs from other nets. Their respective pins are facing each other.	Try using a different route pattern or offsetting the tapoffs if possible. For example, you can cluster the tapoffs or split the fishbone into two backbones.
PATTERN_ROUTE_ASSISTANT-009	Error	Could not route the group %s because there is no net selected in the Design Navigator.	You enabled Route Only Selected Nets , but you did not select any net in the Design Navigator.	Select at least one net from the Design Navigator or turn off Route Only Selected Nets .
PATTERN_ROUTE_ASSISTANT-010	Error	Could not route the group %s because none of selected net has pattern route constraint.	Selected nets have no pattern constraint defined for this group.	Define pattern constraint for the selected nets.
PATTERN_ROUTE_ASSISTANT-011	Error	Could not route the group %s because none of selected net is inside this group scope.	Selected nets are not inside the group.	Select the nets that are inside the group.
PATTERN_ROUTE_ASSISTANT-012	Error	Could not plan the group %s because it is empty.	The selected group does not contain any net, so planning is not possible.	Select a different group and try again.

ID	Type	Message	Causes	Solutions
PATTERN_ROUTE_ASSISTANT-013	Error	Could not route net %s because it is already partially routed. You might want to check the "Delete Existing Shapes" box before applying the pattern.	Some pins inside the group are already routed.	Delete the routes before routing.